Lectures on Complex Geometry

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Graduate course at USTC

Course Guide

These notes are written for graduate students who have completed one semester of one-variable complex analysis, basic differential geometry, and a small amount of algebraic topology. The principal reference is J.-P. Demailly's *Complex Analytic and Differential Geometry* (2012); the references at the end of the book supply the additional results used in Hodge theory, several complex variables, and modern pluripotential theory. The presentation keeps the compact rhythm of the author's earlier notes, but adds motivation, coordinate calculations, explicit hypothesis checks, and detailed proofs of the results used later in the course.

The semester has 16 weeks, with two 90-minute sessions each week. Because the course does not assume a prior course in several complex variables, Lectures 2–8 form a deliberately slower analytic foundation: Cauchy theory on polydiscs, power series, normal families, local mapping theorems, Hartogs extension, elementary $\bar{\partial}$ solvability, subharmonic and plurisubharmonic functions, pseudoconvexity, and Stein manifolds. Later topics are introduced only after the local analytic tool they use has appeared. Every lecture opens with its notation and prerequisite ledgers, followed by a problem-driven motivation box and a timed plan, and ends with a summary and exercises. Every lecture also has an optional handout. The handouts explore examples, refinements, or later developments, but no subsequent core argument depends on them.

How to use the notes for independent study

Each lecture starts on a new page and opens with two ledgers: *Notation for this lecture* declares the recurring symbols before they are used, and *Prerequisite results used below* states the hypotheses and conclusions of the external results invoked in the lecture. A motivation box then identifies the concrete problem that the new concepts are designed to solve. These are followed by the core exposition, extended calculations, a self-study workshop, exercises, and complete solutions near the end of the book. A productive two-day cycle is the following.

(1) Before reading, copy the workshop's prerequisite ledger and mark every item that you can define without looking it up. Return to the indicated earlier lecture if more than one item is missing.

(2) Read the main lecture once for structure. On the second reading, reproduce every displayed calculation; in complex geometry a proof is often hidden in a type, sign, or transition-function check.

(3) Close the notes and answer the checkpoint questions. Compare with the solutions immediately following the checkpoint, but rewrite any missed argument in your own notation rather than merely rereading it.

(4) Attempt the numbered exercises without the solution chapter. Use the first sentence of a solution as a hint only after a serious attempt, and record which hypothesis or previous theorem unlocked the problem.

(5) Finish by testing the exit criterion aloud. The criterion asks for a reconstruction, not recognition: one should be able to state the hypotheses, carry out the model calculation, and explain where the conclusion would fail if a key assumption were removed.

The workshops deliberately repeat a small amount of notation. That repetition is a diagnostic tool: if a symbol has changed meaning between lectures, or if a theorem is being used on the wrong bidegree, the discrepancy should become visible before it enters a long proof. The convention ledger for d^{c}, curvature, weighted norms, and adjoints remains in force throughout.

Remark (Convention: the normalization of d^{c} used throughout)

We fix

$$\mathrm{d}^{\mathrm{c}} = \frac{1}{4\pi \mathrm{i}}(\partial - \bar{\partial}), \qquad \mathrm{dd}^{\mathrm{c}} = \frac{\mathrm{i}}{2\pi}\partial\bar{\partial}.$$

Thus $\mathrm{dd}^{\mathrm{c}} \log |z|^2 = \delta_0$ on $\mathbb{C}$, and for a meromorphic function f, $\mathrm{dd}^{\mathrm{c}} \log |f|^2 = [\mathrm{div}(f)]$. Other sources often differ by a factor of 2 or 2π; convert the normalization before importing any formula.

Learning objectives

By the end of the course, a student should be able to:

(1) compute fluently with (p, q)-forms, Chern connections, and curvature on complex manifolds;
(2) move correctly among the languages of functions, distributions, and currents, including their positivity notions;
(3) explain the links among pseudoconvexity, Stein geometry, psh functions, and solvability of $\bar{\partial}$;
(4) calculate basic cohomology groups using sheaves, the Dolbeault resolution, and Hodge theory;
(5) derive vanishing theorems and weighted L^2 estimates from the Bochner–Kodaira–Nakano identity;
(6) track the hypotheses, approximation steps, and limiting arguments used with positive line bundles, positive currents, Lelong numbers, and singular metrics.

Sixteen-week schedule

Week	First meeting	Second meeting
1	1. Complex manifolds, almost complex structures, and holomorphicity	2. Holomorphic functions on polydiscs: Cauchy formula, power series, and $\bar{\partial}$
2	3. Cauchy estimates, normal families, and local mapping theorems	4. Hartogs phenomena and the elementary $\bar{\partial}$ method
3	5. Subharmonic functions and regularization	6. Plurisubharmonic functions and the Levi form
4	7. Pseudoconvexity, exhaustion, and the Levi problem	8. Basic sheaf theory: local data, stalks, and exactness
5	9. Čech cohomology, Stein manifolds, and Cartan's theorems	10. Weierstrass theory, coherence, and analytic sets
6	11. Complex spaces, normalization, and divisors	12. Hermitian and Kähler metrics; the Fubini–Study metric
7	13. Elliptic operators, harmonic forms, and Hodge theory	14. Kähler identities and the $\partial\bar{\partial}$-lemma
8	15. Dolbeault, Čech, and sheaf cohomology	16. Serre duality, Frölicher, and examples

9	17. Vector bundles, connections, and Chern–Weil	18. Chern connections, line bundles, and c_1
10	19. Griffiths/Nakano positivity and curvature formulas	20. Positive line bundles, Kodaira embedding, projectivity
11	21. Currents, positivity, and Poincaré–Lelong	22. Closed positive currents, potentials, and compactness
12	23. Bedford–Taylor Monge–Ampère theory	24. Lelong numbers, Jensen's formula, and Siu's theorem
13	25. The Bochner–Kodaira–Nakano identity	26. Kodaira–Akizuki–Nakano vanishing
14	27. Hörmander's weighted L^2 estimate	28. Complete Kähler metrics, singular weights, multiplier ideals
15	29. Introduction to Ohsawa–Takegoshi extension	30. Calabi–Yau and the continuity method
16	31. Nef/big/psef cones and regularization	32. A capstone calculation and a map of the subject

Reading map to Demailly

The principal reference uses a different order and, in several places, a different normalization. The following map makes it possible to read the two texts in parallel. The lecture notes retain the convention ledger above even when the reference uses $\mathrm{dd^c} = \mathrm{i}\partial\bar\partial/\pi$ or suppresses a constant.

Lectures	Sections of [1] most closely aligned with the course
1–8	Chapter I, §§1–7: holomorphic functions, differential calculus, subharmonic and psh functions, holomorphic convexity, pseudoconvexity, and Stein manifolds. Lectures 2–4 expand the terse review in §3 for readers without prior several-complex-variable coursework.
9–11	Chapter II, §§1–9: sheaves, local analytic rings, coherence, analytic spaces, normalization, meromorphic maps, and divisors.
12	Chapter VI, §§4–5: Hermitian and Kähler manifolds, local potentials, and the basic Kähler examples. Blow-ups and modifications from Chapter II, §10 are retained as optional reading.
13–16	Chapter IV, §§1–11, and Chapter VI, §§1–12: sheaf cohomology, double complexes, Hodge theory, Kähler identities, curves, and the Frölicher sequence.
17–20	Chapter V, §§1–16, and Chapter VII, §§11–14: Hermitian bundles, Chern connections, exact sequences, projective bundles, ampleness, and Kodaira embedding.
21–24	Chapter III, §§1–8: positivity cones, closed positive currents, Monge–Ampère products, generalized Lelong numbers, Jensen–Lelong, and Siu semicontinuity.
25–29	Chapters VII–VIII: the Bochner–Kodaira–Nakano identity, curvature positivity, vanishing, complete metrics, weighted L^2 estimates, and extension.
30–32	Demailly's positivity and Monge–Ampère chapters provide the background; the Calabi–Yau, regularization, and non-pluripolar developments use the additional references listed in the bibliography.

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Chapter 1

Complex Manifolds and Several Complex Variables

1.1 Lecture 1: Complex manifolds, almost complex structures, and holomorphicity

Week 1 material • 90 minutes

Notation for this lecture

X and Y denote complex manifolds and $n = \dim_{\mathbb{C}} X$ is the complex dimension. We write $z_j = x_j + \mathrm{i} y_j$ on $\mathbb{C}^n$ and $w_a = s_a + \mathrm{i} t_a$ on $\mathbb{C}^m$. The standard complex structure J_0 is multiplication by i on the underlying real tangent space. The real tangent bundle is TX, and $J_X : TX \to TX$ is the complex-structure endomorphism, characterized by $J_X^2 = -\mathrm{id}$, where id is the identity; J_Y has the analogous meaning. We write $\mathrm{d}f_x$ for the real differential of a smooth map f at x, and $T^{1,0}X$ for the $+\mathrm{i}$ eigensubbundle of $TX \otimes_{\mathbb{R}} \mathbb{C}$. On one-forms, $J_X^* \alpha := \alpha \circ J_X$. For a smooth function f, ∂f and $\bar{\partial} f$ denote the $(1,0)$ and $(0,1)$ components of $\mathrm{d}f$. The symbols $\mathbb{C}^n$ and $\mathbb{P}^n$ mean complex affine and projective n-space, $\mathcal{O}(X)$ is the ring of holomorphic functions on X, and $[z_0 : \cdots : z_n]$ denotes homogeneous coordinates on $\mathbb{P}^n$. Finally, $\mathrm{End}(TX)$ is the bundle of real-linear endomorphisms of TX, and $b_k = \dim_{\mathbb{R}} H^k(X, \mathbb{R})$ is the k-th Betti number.

Prerequisite results used below

Holomorphic inverse function theorem. Let $U \subset \mathbb{C}^n$ be open, let $F : U \to \mathbb{C}^n$ be holomorphic, and let $a \in U$. If $\det(\partial F_i / \partial z_j)(a) \neq 0$, then there are neighborhoods $a \in U_0 \subset U$ and $F(a) \in V_0$ such that $F : U_0 \to V_0$ is biholomorphic; its inverse is holomorphic.

Holomorphic implicit function theorem. Let $U \subset \mathbb{C}^n \times \mathbb{C}^r$ be open, write its coordinates as (z, w), and let $F = (F_1, \dots, F_r) : U \to \mathbb{C}^r$ be holomorphic. Suppose $F(a, b) = 0$ and $\det(\partial F_i / \partial w_j)(a, b) \neq 0$. Then, after shrinking to neighborhoods $a \in U_z$ and $b \in U_w$, there is a unique holomorphic map $g : U_z \to U_w$ such that

$$F(z, w) = 0 \quad \Longleftrightarrow \quad w = g(z) \qquad ((z, w) \in U_z \times U_w).$$

More generally, if $F : \Omega \subset \mathbb{C}^N \to \mathbb{C}^r$ has rank r along $F^{-1}(0)$, then that zero set is locally biholomorphic to an open subset of $\mathbb{C}^{N-r}$.

Quotient-manifold criterion. If a discrete group Γ acts freely and properly discontinuously by biholomorphisms on a complex manifold M, then M/Γ has a unique complex-manifold structure for which the quotient map is a local biholomorphism.

Why these ideas are needed

Holomorphicity is first presented by formulas in coordinates, but geometry should not depend on which chart is open on the page. The first task is therefore to extract from the Cauchy–Riemann equations a coordinate-free object. The equation $\mathrm{d}h \circ J_0 = J_0 \circ \mathrm{d}h$ says exactly that a change of holomorphic coordinates preserves multiplication by i; it is the reason the local operators J_0 glue to an intrinsic J_X on the tangent bundle. Once holomorphic maps are recognized by $\mathrm{d}f \circ J_X = J_Y \circ \mathrm{d}f$, the same splitting isolates the part of $\mathrm{d}f$ that obstructs holomorphicity. This is the conceptual origin of $\bar\partial$, not merely a convenient notation for half of the real partial derivatives.

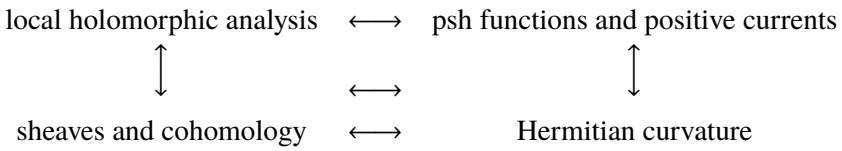
Plan for the lecture. 0–10 min: the central questions of the course; 10–25: atlases and holomorphic transition maps; 25–45: Cauchy–Riemann equations and the intrinsic almost complex structure; 45–57: holomorphic maps as complex-linear differentials; 57–67: motivation for ∂ and $\bar\partial$; 67–84: basic examples; 84–90: coordinate checks.

1.1.1 Course introduction: what complex geometry studies

Complex geometry begins with manifolds that locally look like open subsets of $\mathbb{C}^n$, but its characteristic questions are global. Four examples will guide the semester.

(1) *Existence:* when does a topological or differentiable manifold carry a complex or Kähler structure?
(2) *Functions and embeddings:* when are there enough global holomorphic functions or sections to describe the manifold inside $\mathbb{C}^N$ or $\mathbb{P}^N$?
(3) *Curvature and topology:* which cohomological restrictions follow from positivity, and when does a cohomology class contain a positive representative?
(4) *Singular limits:* how can divisors, degenerating metrics, and weak limits be treated without losing positivity?

The subject answers these questions by moving among four languages:



The arrows are the actual content of the course. The Dolbeault resolution turns $\bar\partial$ into sheaf cohomology; Chern connections turn metrics into curvature classes; Poincaré–Lelong turns zeros into currents; the Bochner–Kodaira–Nakano identity turns curvature into estimates; and those estimates produce vanishing, embeddings, and extension theorems.

Remark (A model problem)

Let $L \to X$ be a holomorphic line bundle on a compact complex manifold. Asking whether powers of L embed X into projective space forces us to use nearly the whole course: choose a Hermitian metric, read positivity from its Chern curvature, solve a weighted $\bar\partial$ equation to manufacture peak sections, use cohomology to globalize the construction, and finally check separation of points and tangent vectors. Kodaira's embedding theorem in Lecture 20 completes this circle.

How the semester is organized. Lectures 1–8 develop several-complex-variable tools from Cauchy's formula through Hartogs extension, psh functions, pseudoconvexity, and Stein theory. Lectures 9–11

introduce sheaves and analytic spaces. Lectures 12–16 pass from Kähler metrics to Hodge theory, Dolbeault cohomology, and Serre duality. Lectures 17–20 develop holomorphic vector bundles, Chern curvature, and positivity. Lectures 21–24 turn to positive currents, Monge–Ampère products, and Lelong numbers. Lectures 25–29 form the analytic climax: Bochner identities, vanishing, weighted L^2 estimates, and extension. Lectures 30–32 show how these tools enter nonlinear equations and modern positivity.

How to read a proof in this course. Before following a calculation, write a short ledger:

(1) the type and degree of every form or current;
(2) the normalization of d^{c}, curvature, and volume;
(3) the regularity and compactness or completeness hypotheses;
(4) the exact positivity statement and the vector space on which it is tested;
(5) the theorem that changes a local statement into a global one.

Most incorrect complex-geometric arguments fail at one of these five entries rather than in a long calculation.

1.1.2 Complex manifolds

A complex manifold carries two compatible structures: it is a smooth manifold of real dimension $2n$, and it has complex coordinates whose transition maps are holomorphic. The second condition replaces arbitrary smooth changes of coordinates by maps satisfying the Cauchy–Riemann equations. Consequently, surprisingly little local data can force a global conclusion.

Definition 1.1: Complex manifold

A complex manifold of complex dimension n is a Hausdorff, second-countable topological space X equipped with charts

$$\varphi_\alpha : U_\alpha \longrightarrow V_\alpha \subset \mathbb{C}^n$$

whose domains cover X, such that every φ_α is a homeomorphism and every nonempty transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is holomorphic.

Hausdorffness gives uniqueness of limits. Second countability supplies partitions of unity, metrizability, and the usual functional-analytic tools. Neither condition is cosmetic.

1.1.3 From the Cauchy–Riemann equations to an intrinsic J

The definition by charts is indispensable for constructing examples, but it hides a piece of linear algebra that is present on every tangent space. We recover it first on Euclidean space. Write coordinates on $\mathbb{C}^n$ as $z_j = x_j + \mathrm{i}y_j$ and define the standard real-linear endomorphism $J_0 : T\mathbb{C}^n \to T\mathbb{C}^n$ by

$$J_0 \frac{\partial}{\partial x_j} = \frac{\partial}{\partial y_j}, \qquad J_0 \frac{\partial}{\partial y_j} = -\frac{\partial}{\partial x_j}.$$

Thus $J_0^2 = -\mathrm{id}$. Geometrically, J_0 is multiplication by i written on the underlying real tangent space.

Proposition 1.2: Cauchy–Riemann equations in invariant form

Let $U \subset \mathbb{C}^n$ be open and let $h : U \to \mathbb{C}^m$ be smooth. Then h is holomorphic if and only if

$$\mathrm{d}h_z \circ J_0 = J_0 \circ \mathrm{d}h_z \qquad \text{for every } z \in U.$$

In other words, the real differential of a holomorphic map is complex linear.

Proof. Write the target coordinates as $w_a = s_a + \mathrm{i}t_a$ and $h^a = u^a + \mathrm{i}v^a$. On the vector $\partial/\partial x_j$ the two sides are

$$\mathrm{d}h\left(J_0 \frac{\partial}{\partial x_j}\right) = \sum_{a=1}^{m} \left(\frac{\partial u^a}{\partial y_j}\frac{\partial}{\partial s_a} + \frac{\partial v^a}{\partial y_j}\frac{\partial}{\partial t_a} \right),$$
$$J_0\left(\mathrm{d}h \frac{\partial}{\partial x_j}\right) = \sum_{a=1}^{m} \left(-\frac{\partial v^a}{\partial x_j}\frac{\partial}{\partial s_a} + \frac{\partial u^a}{\partial x_j}\frac{\partial}{\partial t_a} \right).$$

Equality is therefore equivalent to

$$\frac{\partial u^a}{\partial x_j} = \frac{\partial v^a}{\partial y_j}, \qquad \frac{\partial u^a}{\partial y_j} = -\frac{\partial v^a}{\partial x_j} \qquad (1 \le a \le m,\ 1 \le j \le n),$$

which are precisely the Cauchy–Riemann equations. Checking the $\partial/\partial y_j$ vectors gives no new equations, because $\partial/\partial y_j = J_0(\partial/\partial x_j)$ and $J_0^2 = -\mathrm{id}$. □

Let $\varphi : U \to \varphi(U) \subset \mathbb{C}^n$ be a holomorphic chart on X. Transport J_0 back to U by setting

$$J_X^{(\varphi)}\big|_x := (\mathrm{d}\varphi_x)^{-1} \circ J_0 \circ \mathrm{d}\varphi_x. \tag{1.1}$$

At first sight this definition seems to depend on the chosen coordinates. This is exactly where the Cauchy–Riemann equations enter.

Proposition 1.3: Coordinate independence of the complex structure

The endomorphism in (1.1) is unchanged when φ is replaced by another holomorphic chart. Consequently the local endomorphisms glue to a smooth bundle map $J_X : TX \to TX$ satisfying $J_X^2 = -\mathrm{id}$.

Proof. Let ψ be another chart and put $g = \psi \circ \varphi^{-1}$. On the overlap, g is holomorphic, so the preceding proposition gives $\mathrm{d}g \circ J_0 = J_0 \circ \mathrm{d}g$. The chain rule gives $\mathrm{d}\psi = \mathrm{d}g \circ \mathrm{d}\varphi$, and hence

$$\begin{aligned}(\mathrm{d}\psi)^{-1} J_0 \mathrm{d}\psi &= (\mathrm{d}\varphi)^{-1}(\mathrm{d}g)^{-1} J_0 (\mathrm{d}g)\mathrm{d}\varphi \\ &= (\mathrm{d}\varphi)^{-1} J_0 \mathrm{d}\varphi.\end{aligned}$$

Thus the two coordinate definitions agree. Smoothness and $J_X^2 = -\mathrm{id}$ may now be checked in any chart. □

Definition 1.4: Almost complex structure and integrability

An *almost complex structure* on a smooth real $2n$-manifold M is a smooth bundle endomorphism $J : TM \to TM$ with $J^2 = -\mathrm{id}$. It is *integrable* if every point has real coordinates that combine into complex coordinates in which J is J_0; equivalently, it comes from a complex-manifold atlas.

The adjective "almost" is essential: the pointwise relation $J^2 = -\mathrm{id}$ is algebraic, whereas the existence of compatible holomorphic coordinates is a differential condition. For smooth vector fields A and B, define the Nijenhuis tensor by

$$N_J(A,B) = [JA, JB] - J[JA, B] - J[A, JB] - [A, B],$$

where $[\,,\,]$ is the Lie bracket. The Newlander–Nirenberg theorem says that a smooth almost complex structure is integrable exactly when $N_J = 0$. We use this theorem only as orientation here; all complex structures constructed in this course will initially be given by explicit holomorphic charts.

If $z = (z_1, \dots, z_n)$ and $w = (w_1, \dots, w_n)$ are two holomorphic coordinate systems, their transition map is holomorphic and its complex Jacobian is invertible. The coordinate vector fields transform by

$$\frac{\partial}{\partial z_k} = \sum_j \frac{\partial w_j}{\partial z_k} \frac{\partial}{\partial w_j},$$

which defines the holomorphic tangent bundle $T^{1,0}X$. Intrinsically it is the $+\mathrm{i}$ eigensubbundle of J_X on $TX \otimes_{\mathbb{R}} \mathbb{C}$. Its dual has local frame $\mathrm{d}z_1, \dots, \mathrm{d}z_n$.

Definition 1.5: Holomorphic and biholomorphic maps

Let X and Y be complex manifolds. A continuous map $f : X \to Y$ is *holomorphic* if every coordinate expression $\psi \circ f \circ \varphi^{-1}$ is holomorphic. It is *biholomorphic* if it is bijective and both f and f^{-1} are holomorphic.

Proposition 1.6: Intrinsic criterion for a holomorphic map

A smooth map $f : X \to Y$ is holomorphic if and only if its differential intertwines the two complex structures:

$$\mathrm{d}f_x \circ J_X\big|_x = J_Y\big|_{f(x)} \circ \mathrm{d}f_x \qquad (x \in X). \tag{1.2}$$

Proof. Choose charts φ near x and ψ near $f(x)$, and set $h = \psi \circ f \circ \varphi^{-1}$. Substituting $J_X = (\mathrm{d}\varphi)^{-1} J_0 \mathrm{d}\varphi$ and $J_Y = (\mathrm{d}\psi)^{-1} J_0 \mathrm{d}\psi$ into (1.2), then applying the chain rule, shows that (1.2) is equivalent to $\mathrm{d}h \circ J_0 = J_0 \circ \mathrm{d}h$. By the Cauchy–Riemann proposition, this is equivalent to the coordinate expression h being holomorphic. □

Thus the phrase "J commutes with a holomorphic map" always means that the *differential* intertwines the source and target structures. The two J's act on different tangent bundles, so the correctly typed formula is $\mathrm{d}f \circ J_X = J_Y \circ \mathrm{d}f$, not $f \circ J_X = J_Y \circ f$.

Warning

The inverse of a bijective holomorphic map is holomorphic, but the proof uses the open mapping theorem or the local inverse theorem; it does not follow from "continuous bijection." Also remember that complex dimension n means real dimension $2n$. Losing this factor is a frequent source of errors in codimension and transversality arguments.

1.1.4 Why the Dolbeault operators appear

The exterior derivative alone cannot single out holomorphic functions: the equation $\mathrm{d}f = 0$ says that f is locally constant, which is far stronger than holomorphicity. The Cauchy–Riemann equations tell us to split $\mathrm{d}f$ into the part compatible with J and the part that measures failure of holomorphicity. Define the Wirtinger derivatives and complex one-forms by

$$\frac{\partial}{\partial z_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} - \mathrm{i}\frac{\partial}{\partial y_j}\right), \qquad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} + \mathrm{i}\frac{\partial}{\partial y_j}\right),$$

$$\mathrm{d}z_j = \mathrm{d}x_j + \mathrm{i}\,\mathrm{d}y_j, \qquad \mathrm{d}\bar{z}_j = \mathrm{d}x_j - \mathrm{i}\,\mathrm{d}y_j.$$

For a smooth complex-valued function f this gives the identity

$$\mathrm{d}f = \underbrace{\sum_j \frac{\partial f}{\partial z_j}\,\mathrm{d}z_j}_{\partial f} + \underbrace{\sum_j \frac{\partial f}{\partial \bar{z}_j}\,\mathrm{d}\bar{z}_j}_{\bar{\partial} f}. \tag{1.3}$$

Definition 1.7: Dolbeault operators on functions

The two components in (1.3) are denoted by ∂f and $\bar\partial f$. They are the $(1,0)$ and $(0,1)$ components of $\mathrm{d}f$, respectively. The operator $\bar\partial$ is the *holomorphicity detector*:

$$f \text{ is holomorphic} \iff \bar\partial f = 0.$$

This construction is coordinate independent. Indeed, if $J_X^*\alpha := \alpha \circ J_X$ is the induced action on complex-valued one-forms, then $\mathrm{d}z_j$ and $\mathrm{d}\bar z_j$ are respectively the $+\mathrm{i}$ and $-\mathrm{i}$ eigenvectors of J_X^*. Since J_X itself is coordinate independent, its two eigensubbundles and the two projections of $\mathrm{d}f$ are intrinsic. Lecture 2 extends these projections from functions to all differential forms. The payoff is much larger than a shorter notation for the Cauchy–Riemann equations: $\bar\partial^2 = 0$ creates a complex whose failure of global exactness measures the global holomorphic geometry of X.

1.1.5 Four families of examples

Example 1.8: Domains and analytic submanifolds

Every domain $\Omega \subset \mathbb{C}^n$ is a complex manifold. If $F : \Omega \to \mathbb{C}^r$ is holomorphic and has rank r along $X = F^{-1}(0)$, the holomorphic implicit-function theorem makes X a complex submanifold of dimension $n - r$. For example,

$$Q = \{[z_0 : z_1 : z_2] \in \mathbb{P}^2 : z_0^2 + z_1^2 + z_2^2 = 0\}$$

is a smooth projective curve.

Example 1.9: Complex projective space

Set $\mathbb{P}^n = (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^*$. On $U_j = \{[z] : z_j \neq 0\}$ use

$$\varphi_j([z_0 : \cdots : z_n]) = \left(\frac{z_0}{z_j}, \ldots, \widehat{\frac{z_j}{z_j}}, \ldots, \frac{z_n}{z_j}\right).$$

The transition functions are rational with nonvanishing denominators on overlaps, hence holomorphic. The quotient is compact because the unit sphere maps onto it.

Example 1.10: Complex tori

If $\Lambda \subset \mathbb{C}^n$ is a lattice of rank $2n$, then $X = \mathbb{C}^n/\Lambda$ is a compact complex manifold. Translation invariance lets $\mathrm{d}z_1, \ldots, \mathrm{d}z_n$ descend to global holomorphic one-forms. For $n = 1$ this is the analytic model of an elliptic curve.

Example 1.11: Hopf surfaces

Fix $0 < |q| < 1$ and let $\mathbb{Z}$ act on $\mathbb{C}^2 \setminus \{0\}$ by $m \cdot z = q^m z$. The quotient X_q is a compact complex surface, but it is not Kähler. Indeed, it is diffeomorphic to $S^1 \times S^3$, so $b_1 = 1$, whereas the first Betti number of a compact Kähler manifold is even. Lecture 14 will close this argument.

Proposition 1.12: Holomorphic functions on compact manifolds

If X is compact and connected, every $f \in \mathcal{O}(X)$ is constant.

Proof. The function $|f|$ attains its maximum. In a chart about a maximum point, the classical maximum-modulus principle makes f constant on the corresponding connected component. Hence $\{x : f(x) = f(x_0)\}$ is nonempty, open, and closed, so it is all of X. □

Remark (Geometric interpretation)

The scarcity of global functions explains why line bundles, meromorphic functions, sheaves, and cohomology are indispensable. A compact complex manifold may have only constant holomorphic functions while carrying many holomorphic sections of nontrivial line bundles.

1.1.6 Reading geometry from transition functions

The definition by atlases becomes useful only after one learns to extract bundles and maps from the transition functions. Suppose z and w are holomorphic coordinates on an overlap. The matrices

$$g_{wz} = \left(\frac{\partial w_j}{\partial z_k}\right)$$

satisfy the cocycle identities $g_{uz} = g_{uw}g_{wz}$ and $g_{zz} = I$. They therefore glue the trivial bundles $U_z \times \mathbb{C}^n$ to $T^{1,0}X$. Taking inverse transposes glues the cotangent bundle, taking determinants glues the canonical bundle $K_X = \Lambda^n T^{*1,0}X$, and applying any matrix representation constructs the associated tensor bundles. This is the first instance of a general principle: geometric objects are local data satisfying an overlap rule.

For projective space this calculation is concrete. On $U_0 \cap U_1 \subset \mathbb{P}^n$ write

$$w_1 = \frac{z_0}{z_1} = \frac{1}{u_1}, \qquad w_j = \frac{z_j}{z_1} = \frac{u_j}{u_1} \quad (j \geq 2),$$

where $u_j = z_j/z_0$. The transition map is holomorphic precisely where $u_1 \neq 0$, and its inverse has the same form. Differentiating it displays the nonconstant transition matrix of $T^{1,0}\mathbb{P}^n$; this is already evidence that the tangent bundle is not globally a product.

Proposition 1.13: Coordinate criterion for immersions and submersions

Let $f : X^n \to Y^m$ be holomorphic. If $\operatorname{rank}_{\mathbb{C}} \mathrm{d}f_x = r$, then there are holomorphic coordinates centered at x and $f(x)$ in which

$$f(z_1, \ldots, z_n) = (z_1, \ldots, z_r, F_{r+1}(z), \ldots, F_m(z)).$$

When $r = m$, the fibers are complex submanifolds of dimension $n - m$; when $r = n$, f is a local holomorphic embedding.

Proof. Choose a nonzero $r \times r$ minor of the complex Jacobian and apply the holomorphic inverse-function theorem to the corresponding source variables together with the remaining source coordinates. In the new coordinates the first r output components are coordinate projections. If the rank is locally constant, the remaining output components have zero derivatives along the fiber directions and hence depend only on the first r coordinates. The two extreme cases give the assertions. □

1.1.7 Quotients, compactness, and two global tests

A free properly discontinuous action of a discrete group Γ on a complex manifold M by biholomorphisms gives a complex manifold M/Γ: choose a small open set U disjoint from all its nontrivial

translates and use $U \to M/\Gamma$ as a chart. This produces tori and Hopf manifolds. Proper discontinuity is essential; a dense translation subgroup of $\mathbb{C}$ produces a quotient whose topology is not even Hausdorff.

Compactness is often best checked before holomorphic structure. The sphere $S^{2n+1} \subset \mathbb{C}^{n+1}$ surjects continuously onto $\mathbb{P}^n$, so projective space is compact. A closed fundamental parallelepiped surjects onto $\mathbb{C}^n/\Lambda$, so a complex torus is compact. For a Hopf manifold, a compact annulus $\{|q| \le |z| \le 1\}$ meets every orbit, with its two boundary components identified.

Two quick tests recur throughout the course. First, if a proposed holomorphic function on a compact connected manifold is nonconstant, the maximum principle finds an error. Second, if a proposed coordinate construction changes complex dimension, the complex rank theorem finds an error. Applied early, these tests prevent local formulas from being mistaken for global geometry.

Remark (A useful distinction)

A holomorphic bijection between complex manifolds is biholomorphic, but a bijective holomorphic map between singular analytic spaces need not be an isomorphism. The normalization map of the cusp is the standard counterexample and will appear in Lectures 10–11. Smoothness is doing real work in the manifold statement.

1.1.8 Complete verification of the basic examples

The examples above are useful only after every clause in the definition has been checked. We now carry out those checks. In each case the topology is the quotient or subspace topology, and the displayed coordinates will also prove that it is Hausdorff and second countable.

Open sets and a smooth affine hypersurface

If $\Omega \subset \mathbb{C}^n$ is open, the identity map is one chart, so there are no transition maps to check. A less tautological example is

$$X = \{(z, w) \in \mathbb{C}^2 : zw = 1\}.$$

The projection $p_z : X \to \mathbb{C}^*$ is bijective and has inverse

$$\sigma_z(\zeta) = (\zeta, \zeta^{-1}).$$

Both maps are holomorphic, so $X \simeq \mathbb{C}^*$ as a complex manifold. This also follows from the implicit-function theorem: for $F(z, w) = zw - 1$,

$$\mathrm{d}F = w\,\mathrm{d}z + z\,\mathrm{d}w,$$

and z and w cannot both vanish on X. At $(z_0, w_0) \in X$, differentiating $zw = 1$ gives the concrete tangent-space calculation

$$T^{1,0}_{(z_0,w_0)}X = \{(a, b) \in \mathbb{C}^2 : w_0 a + z_0 b = 0\} = \mathbb{C}\,(z_0, -w_0).$$

Now verify the projective conic

$$Q = \{[z_0 : z_1 : z_2] \in \mathbb{P}^2 : z_0^2 + z_1^2 + z_2^2 = 0\}.$$

On U_0 put $x = z_1/z_0$ and $y = z_2/z_0$. Then

$$Q \cap U_0 = \{1 + x^2 + y^2 = 0\}, \qquad \mathrm{d}(1 + x^2 + y^2) = 2x\,\mathrm{d}x + 2y\,\mathrm{d}y.$$

The two coefficients cannot vanish on this zero set. If $y_0 \neq 0$, projection to x is a local coordinate and implicit differentiation gives

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y};$$

if $x_0 \neq 0$, projection to y is a local coordinate. The same calculation on U_1 and U_2 covers Q. Thus Q is a one-dimensional complex submanifold. Equivalently, the homogeneous gradient $(2z_0, 2z_1, 2z_2)$ never vanishes on projective space.

The atlas of projective space

For $U_j = \{z_j \neq 0\}$ define φ_j as above. It is bijective because its inverse inserts 1 in the j-th position:

$$(u_0, \dots, \widehat{u_j}, \dots, u_n) \longmapsto [u_0 : \cdots : u_{j-1} : 1 : u_{j+1} : \cdots : u_n].$$

For example, on $U_0 \cap U_1$, write

$$u_k = \frac{z_k}{z_0} \quad (k \geq 1), \qquad v_0 = \frac{z_0}{z_1}, \quad v_k = \frac{z_k}{z_1} \quad (k \neq 1).$$

Then

$$v_0 = \frac{1}{u_1}, \qquad v_k = \frac{u_k}{u_1} \quad (k \geq 2).$$

The overlap in u-coordinates is $\{u_1 \neq 0\}$, so every denominator is nonzero. The inverse transition has the identical form with u and v interchanged. Hence the maps are biholomorphic and define a complex atlas.

The quotient is Hausdorff: each complex line has a unique orbit under positive real scaling meeting the unit sphere, and the remaining S^1-action on the sphere is by a compact group. It is second countable because it is covered by the finitely many second-countable sets $U_j \simeq \mathbb{C}^n$. Finally, the continuous map $S^{2n+1} \to \mathbb{P}^n$ is surjective, so $\mathbb{P}^n$ is compact.

For $\mathbb{P}^1$, the coordinates $u = z_1/z_0$ and $v = z_0/z_1$ satisfy $v = u^{-1}$. Identifying $\mathbb{C} \cup \{\infty\}$ with the unit sphere by inverse stereographic projection,

$$u \longmapsto \frac{1}{1 + |u|^2}\big(2\Re u, 2\Im u, |u|^2 - 1\big),$$

and sending ∞ to the north pole gives the Riemann-sphere model. Near infinity the coordinate is $v = 1/u$, exactly the second projective chart.

A complete coordinate-change check for J

The overlap map $v = u^{-1}$ also gives a concrete test of the intrinsic construction of J. Fix $u \neq 0$. Its complex derivative is $\lambda = -u^{-2} = a + \mathrm{i}b$, so its real differential, in the ordered bases (∂_x, ∂_y) and (∂_s, ∂_t), is

$$A_\lambda = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

In either coordinate system the standard complex structure has matrix

$$[J_0] = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$A_\lambda [J_0] = \begin{pmatrix} -b & -a \\ a & -b \end{pmatrix} = [J_0] A_\lambda.$$

Since $\lambda \neq 0$, the matrix A_λ is invertible, and therefore $A_\lambda^{-1}[J_0]A_\lambda = [J_0]$. This is the coordinate-independence calculation on the overlap of the two charts of $\mathbb{P}^1$.

The same calculation also shows what would go wrong for a merely smooth change of coordinates. A general real Jacobian need not commute with $[J_0]$; transporting J_0 through it can produce a different endomorphism. Holomorphicity of transition maps is exactly the hypothesis that prevents this ambiguity.

Three function-level Dolbeault calculations

For $z = x + \mathrm{i}y$, the identities $\mathrm{d}f = (\partial f/\partial z)\mathrm{d}z + (\partial f/\partial \bar{z})\mathrm{d}\bar{z}$ give

f	∂f	$\bar{\partial} f$
z^2	$2z\,\mathrm{d}z$	0
$\bar{z}$	0	$\mathrm{d}\bar{z}$
$\lvert z\rvert^2 = z\bar{z}$	$\bar{z}\,\mathrm{d}z$	$z\,\mathrm{d}\bar{z}$.

The first line records holomorphicity, the second isolates purely antiholomorphic variation, and the third shows that a real-valued nonconstant function normally has both components. Because the two summands are the eigenspace projections for J^*, these conclusions persist after any holomorphic change of coordinates.

The local charts of a complex torus

Let $\Lambda \subset \mathbb{C}^n$ be a lattice and $\pi : \mathbb{C}^n \to \mathbb{C}^n/\Lambda$ the quotient map. Discreteness gives

$$\rho := \frac{1}{2}\inf\{|\lambda| : 0 \neq \lambda \in \Lambda\} > 0.$$

For any $a \in \mathbb{C}^n$, the translates of $B_\rho(a)$ are pairwise disjoint. Therefore $\pi|_{B_\rho(a)}$ is a homeomorphism onto its image and defines a chart

$$\phi_a : \pi(B_\rho(a)) \longrightarrow B_\rho(0), \qquad \phi_a([z]) = z - a,$$

where the unique representative in $B_\rho(a)$ is used. If two such charts overlap, their representatives differ by a fixed $\lambda \in \Lambda$, and hence

$$\phi_b \circ \phi_a^{-1}(\zeta) = \zeta + a - b - \lambda.$$

This is an affine holomorphic map with Jacobian I_n. It follows at once that the forms $\mathrm{d}z_j$ are invariant under every deck translation and descend to the quotient. A closed fundamental parallelepiped is compact and maps surjectively to the quotient, proving compactness.

The quotient charts of a Hopf surface

Let $M = \mathbb{C}^2 \setminus \{0\}$ and let $\gamma(z) = qz$, $0 < |q| < 1$. The action of $\mathbb{Z}$ is free: $q^m z = z$ with $z \neq 0$ implies $q^m = 1$, hence $m = 0$. It is properly discontinuous. Indeed, if $K \Subset M$, choose $0 < r \leq |z| \leq R$ on K; then $q^m K \cap K \neq \varnothing$ forces

$$\frac{r}{R} \leq |q|^m \leq \frac{R}{r},$$

which holds for only finitely many integers m. Shrink a ball B about any point until its nontrivial translates are disjoint. The restriction $B \to \pi(B)$ is a quotient chart. On an overlap, changing the lift replaces z by $q^m z$, so every transition map is the linear biholomorphism $z \mapsto q^m z$.

Every orbit meets the compact annulus

$$A = \{z \in \mathbb{C}^2 : |q| \le |z| \le 1\};$$

therefore $A \to M/\langle\gamma\rangle$ is surjective and the quotient is compact. This proves directly that the Hopf quotient is a compact complex surface. Its non-Kählerness uses the later Hodge-theoretic fact that a compact Kähler manifold has even first Betti number.

Remark (What has actually been verified)

For every example we exhibited local coordinates, wrote every nontrivial coordinate change, checked that its denominator does not vanish on the overlap, and supplied the topological argument needed for the quotient. Merely writing a quotient or a set of equations would not prove that it is a complex manifold.

1.1.9 Self-study workshop: certifying an atlas rather than recognizing a familiar space

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. You should know the holomorphic inverse and implicit function theorems in one and several variables, the quotient topology, and the difference between a continuous bijection and a homeomorphism. The goal is to turn the definition of a complex manifold into a checklist that works on unfamiliar examples.

Reconstruct the passage from coordinates to J. Begin on $\mathbb{C}^n$ by writing the matrix of J_0 in the real basis $(\partial_{x_1}, \partial_{y_1}, \dots, \partial_{x_n}, \partial_{y_n})$. For a smooth map h, calculate both $\mathrm{d}h \circ J_0$ and $J_0 \circ \mathrm{d}h$ on every ∂_{x_j}; the equality is exactly the pair of Cauchy–Riemann equations for each component of h. Next transport J_0 through a chart φ and repeat the overlap calculation with $g = \psi \circ \varphi^{-1}$. Do not skip the line $(\mathrm{d}g)^{-1} J_0 \mathrm{d}g = J_0$: it is the precise reason the result is independent of the chart. Finally, decompose $\mathrm{d}f$ into the two eigenspaces of J^* and identify $\bar\partial f$ as the component that vanishes for holomorphic f.

Checkpoint (the intrinsic meaning of the Cauchy–Riemann equations)

(1) Which equality fails if a transition map is smooth but not holomorphic?
(2) What sign replaces complex linearity for an antiholomorphic map $h : \mathbb{C}^n \to \mathbb{C}^m$?
(3) Why does the equation $\bar\partial f = 0$ have the same meaning in every holomorphic coordinate system?

Solution (checkpoint). (1) Its real differential need not satisfy $\mathrm{d}g\, J_0 = J_0\, \mathrm{d}g$, so the two transported endomorphisms need not agree. (2) One has $\mathrm{d}h\, J_0 = -J_0\, \mathrm{d}h$; for example complex conjugation reverses multiplication by i. (3) The splitting into the $\pm\mathrm{i}$ eigensubbundles of J^* is intrinsic. Holomorphic coordinate changes preserve those subbundles, so the $(0,1)$ projection of $\mathrm{d}f$ vanishes in one holomorphic chart exactly when it vanishes in every such chart. ◇

Reconstruct the proof in four passes. Suppose a space is presented either as a zero set or as a quotient. For a zero set $X = F^{-1}(0) \subset \mathbb{C}^N$, first compute the complex Jacobian of F. Constant rank r produces $N - r$ local coordinates; one must say which coordinates are free and which are holomorphic functions of them. Next check that the subspace topology is Hausdorff and second countable. These properties are inherited from $\mathbb{C}^N$.

For a quotient M/Γ, the analytic and topological tasks separate. Freeness prevents orbifold points. Proper discontinuity supplies an open set U with $\gamma U \cap U = \varnothing$ for $\gamma \neq 1$, so the quotient map restricts to a homeomorphism $U \to \pi(U)$. If two quotient charts overlap, two lifts differ by one deck transformation; hence the transition map is that transformation in local coordinates. Finally, Hausdorffness must be proved from properness, and compactness, when claimed, from a compact set meeting every orbit.

As a small new test, let $\Gamma = \langle z \mapsto qz \rangle$ act on $\mathbb{C}^*$, where $0 < |q| < 1$. Given $a \in \mathbb{C}^*$, choose a disc $D(a, \varepsilon)$ whose radial range is too narrow to meet any $q^m D(a, \varepsilon)$ for $m \neq 0$. The quotient charts have transitions $z \mapsto q^m z$. The annulus $\{|q| \leq |z| \leq 1\}$ meets every orbit, so the quotient is a compact Riemann surface. The logarithm identifies its universal cover with $\mathbb{C}$ and its deck group with the lattice generated by $2\pi\mathrm{i}$ and $\log q$; thus it is also a one-dimensional complex torus.

Checkpoint (definitions under pressure)

(1) Why does nonvanishing of $\mathrm{d}F$ prove smoothness of $\{F = 0\}$ but not compactness?
(2) On $U_0 \cap U_1 \subset \mathbb{P}^n$, which condition guarantees that $v_k = u_k/u_1$ is holomorphic?
(3) Why is a fixed point of a group action an obstruction to the quotient being a manifold with quotient charts?

Solution (checkpoint). (1) The implicit function theorem is local and says nothing about sequences escaping to infinity; compactness is global. (2) The overlap is exactly $\{u_1 \neq 0\}$, so the denominator never vanishes there. (3) At a fixed point no neighborhood has disjoint nontrivial translates. The quotient map is not locally one-to-one, and the local model is generally a branched or orbifold quotient rather than an open subset of $\mathbb{C}^n$. ◇

Exit criterion

You can derive $\mathrm{d}h \circ J_0 = J_0 \circ \mathrm{d}h$ from the Cauchy–Riemann equations, prove that the transported J_X is chart independent, and explain why $\bar{\partial} f$ is an intrinsic test of holomorphicity. Given a new hypersurface or a discrete quotient, you can also write its chart map and inverse, compute every transition map, and separately justify Hausdorffness, second countability, dimension, and any compactness assertion.

Summary. The Cauchy–Riemann equations are equivalent to complex linearity of the real differential. This identity makes the complex structure J_X independent of holomorphic coordinates, characterizes holomorphic maps by $\mathrm{d}f \circ J_X = J_Y \circ \mathrm{d}f$, and motivates $\bar{\partial}$ as the intrinsic obstruction to holomorphicity. Projective space, complex tori, analytic submanifolds, and Hopf surfaces will be test objects throughout the course.

Exercise 1.14: The two charts of the projective line

Write the transition function for the two standard charts of $\mathbb{P}^1$ and prove that $\mathbb{P}^1$ is biholomorphic to the Riemann sphere.

Exercise 1.15: Tangent spaces

Let $X = \{F = 0\} \subset \mathbb{C}^n$ and assume $\mathrm{d}F_x$ has full rank. Prove that $T_x^{1,0} X = \ker(\mathrm{d}F_x : \mathbb{C}^n \to \mathbb{C}^r)$.

Exercise 1.16: Cauchy–Riemann equations as complex linearity

Let $h : U \subset \mathbb{C}^n \to \mathbb{C}^m$ be smooth and write $h^a = u^a + \mathrm{i}v^a$. Compute the matrices of $\mathrm{d}h \circ J_0$ and $J_0 \circ \mathrm{d}h$ in real coordinates, and prove directly that their equality is equivalent to all the Cauchy–Riemann equations for h.

Exercise 1.17: $\bar\partial$ detects antiholomorphic variation

On $\mathbb{C}$, compute ∂f and $\bar\partial f$ for $f(z) = z^2$, $f(z) = \bar z$, and $f(z) = |z|^2$. Explain why the results agree with the intrinsic J^*-eigenspace description rather than depending on the coordinate z.

Optional handout: Almost complex structures and integrability

This handout is optional. No later argument in the core lectures depends on it.

A smooth manifold M^{2n} is *almost complex* if it carries an endomorphism $J : TM \to TM$ with $J^2 = -\mathrm{id}$. Complexifying the tangent bundle gives the $\pm\mathrm{i}$ eigenspace decomposition

$$T_M \otimes_{\mathbb{R}} \mathbb{C} = T_J^{1,0}M \oplus T_J^{0,1}M.$$

This is already enough to define the type of a differential form. What is missing is a coordinate system in which J is multiplication by i.

Definition 1.18: Nijenhuis tensor

For real vector fields X, Y, define

$$N_J(X,Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y].$$

Although the formula contains Lie brackets, the derivative terms cancel and N_J is C^∞-bilinear; hence it is a tensor.

If J comes from holomorphic coordinates, coordinate vector fields commute and a direct calculation gives $N_J = 0$. Equivalently, $N_J = 0$ says that sections of $T_J^{0,1}M$ are closed under the Lie bracket. The difficult converse is the Newlander–Nirenberg theorem: a sufficiently regular almost complex structure is induced by complex coordinates exactly when $N_J = 0$. Its proof solves an overdetermined nonlinear first-order system by converting it to a $\bar\partial$ problem and iterating with regularity estimates.

Remark (Complex dimension one)

On a real surface, every almost complex structure is integrable. There is no room for the $(0,2)$ obstruction represented by N_J. Thus an oriented Riemannian surface becomes a Riemann surface after rotating each tangent plane by 90 degrees.

Exercise 1.19: A nonintegrable model

On $\mathbb{C}^2$ define $T^{0,1}$ to be spanned by $L_1 = \partial_{\bar z_1}$ and $L_2 = \partial_{\bar z_2} + \bar z_1 \partial_{z_1}$. Compute $[L_1, L_2]$ and show that this almost complex structure is not integrable.

1.2 Lecture 2: Holomorphic functions on polydiscs: Cauchy formula, power series, and $\bar\partial$

Week 1 material • 90 minutes

Notation for this lecture

$\Delta(a,r) = \{\zeta \in \mathbb{C} : |\zeta - a| < r\}$ is a disc and $P(a;r) = \prod_{j=1}^n \Delta(a_j, r_j)$ is a polydisc; $\Delta = \Delta(0,1)$. Its distinguished boundary is $\Gamma(a;r) = \prod_j \partial\Delta(a_j, r_j)$. A multi-index is $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$; we use $|\alpha| = \sum_j \alpha_j$, $\alpha! = \prod_j \alpha_j!$, $(z-a)^\alpha = \prod_j (z_j - a_j)^{\alpha_j}$, $r^\alpha = \prod_j r_j^{\alpha_j}$, and $\partial^\alpha = \partial_{z_1}^{\alpha_1} \cdots \partial_{z_n}^{\alpha_n}$. The space of holomorphic functions on Ω is $\mathcal{O}(\Omega)$.
$\mathcal{A}^{p,q}(X)$ is the space of smooth complex differential forms of bidegree (p,q). Multi-indices $I = (i_1 < \cdots < i_p)$ and $J = (j_1 < \cdots < j_q)$ abbreviate $\mathrm{d}z_I$ and $\mathrm{d}\bar z_J$. The exterior derivative is $\mathrm{d} = \partial + \bar\partial$; ∂ raises the first bidegree and $\bar\partial$ raises the second. Throughout, $\mathrm{d}^{\mathrm{c}} = (4\pi\mathrm{i})^{-1}(\partial - \bar\partial)$ and $\mathrm{d}\mathrm{d}^{\mathrm{c}} = (\mathrm{i}/2\pi)\partial\bar\partial$.

Prerequisite results used below

One-variable Cauchy formula. If g is holomorphic on a neighborhood of $\overline{\Delta(a,r)}$, then for $z \in \Delta(a,r)$,

$$g^{(k)}(z) = \frac{k!}{2\pi\mathrm{i}} \int_{|\zeta - a| = r} \frac{g(\zeta)}{(\zeta - z)^{k+1}}\, \mathrm{d}\zeta.$$

Uniform convergence tools. Uniformly convergent series of continuous functions may be integrated term by term. A uniformly convergent series of holomorphic functions whose derivative series converges uniformly on compact subsets may be differentiated term by term. Fubini's theorem permits repeated integration of a continuous function on a product of compact circles.
Complexified eigenspaces. If $J^2 = -I$, then the complexified cotangent space splits into the $+\mathrm{i}$ and $-\mathrm{i}$ eigenspaces of J^*; exterior powers give the (p,q) decomposition.

Why these ideas are needed

Lecture 1 defined holomorphicity through coordinates and the equation $\bar\partial f = 0$, but a definition does not yet give a usable calculus. In one variable, virtually every strong theorem begins with Cauchy's integral formula. On a polydisc the formula can be iterated one variable at a time, producing a joint power series, explicit coefficient bounds, and automatic smoothness. This supplies the analytic estimates needed in every later compactness or extension argument. Once the function theory is secure, the same intrinsic splitting by J^* extends ∂ and $\bar\partial$ to forms; that is the language in which psh functions, curvature, and cohomology will be written.

Plan for the lecture. 0–12 min: polydiscs and multi-indices; 12–35: iterated Cauchy formula; 35–55: Taylor expansion and coefficient estimates; 55–68: equivalent tests for holomorphicity; 68–82: (p,q)-forms and $\mathrm{d} = \partial + \bar\partial$; 82–90: worked examples.

In one complex variable, Cauchy's formula is the engine behind power series, derivative estimates, the identity theorem, and compactness. The first important fact in several variables is that the same engine still works, but the natural boundary is a product of circles rather than the topological boundary of a ball. For $a = (a_1, \dots, a_n)$ and $r = (r_1, \dots, r_n)$ with every $r_j > 0$, define

$$P(a;r) = \{z \in \mathbb{C}^n : |z_j - a_j| < r_j \text{ for } 1 \le j \le n\}.$$

Its distinguished boundary is the n-torus $\Gamma(a;r) = \{|z_j - a_j| = r_j \text{ for every } j\}$. It is this torus, not all of $\partial P(a;r)$, that appears in the Cauchy integral.

Theorem 1.20: Cauchy formula on a polydisc

Let f be holomorphic on a neighborhood of the closed polydisc $\overline{P(a;r)}$. For $z \in P(a;r)$,

$$f(z) = \frac{1}{(2\pi \mathrm{i})^n} \int_{|\zeta_1 - a_1| = r_1} \cdots \int_{|\zeta_n - a_n| = r_n} \frac{f(\zeta_1, \dots, \zeta_n)}{\prod_{j=1}^{n} (\zeta_j - z_j)} \, \mathrm{d}\zeta_n \cdots \mathrm{d}\zeta_1.$$

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$,

$$\partial^\alpha f(z) = \frac{\alpha!}{(2\pi \mathrm{i})^n} \int_{\Gamma(a;r)} \frac{f(\zeta)}{\prod_{j=1}^{n} (\zeta_j - z_j)^{\alpha_j + 1}} \, \mathrm{d}\zeta_n \cdots \mathrm{d}\zeta_1,$$

whenever the circles surround z.

Proof. Fix $z_2, \dots, z_n$ and apply the one-variable Cauchy formula in z_1. The resulting integrand is holomorphic in z_2; apply the formula in that variable, and continue. Continuity on the compact product of circles permits repeated integration by Fubini's theorem. Differentiating the kernel under the integral sign is legitimate on every smaller polydisc, because its distance from each circle is positive. Since

$$\frac{\partial^{\alpha_j}}{\partial z_j^{\alpha_j}} \frac{1}{\zeta_j - z_j} = \frac{\alpha_j!}{(\zeta_j - z_j)^{\alpha_j + 1}},$$

multiplying the one-variable factors gives the derivative formula. □

Corollary 1.21: Taylor expansion and Cauchy estimates

Under the hypotheses above,

$$f(z) = \sum_{\alpha \in \mathbb{N}^n} c_\alpha (z - a)^\alpha, \qquad c_\alpha = \frac{\partial^\alpha f(a)}{\alpha!},$$

and the series converges absolutely and uniformly on every smaller closed polydisc. If $M = \sup_{\overline{P(a;r)}} |f|$, then

$$|\partial^\alpha f(a)| \le \alpha! \, M \, r^{-\alpha}, \qquad r^{-\alpha} = \prod_{j=1}^{n} r_j^{-\alpha_j}.$$

Proof. For $|z_j - a_j| < r_j$, expand each kernel as the geometric series

$$\frac{1}{\zeta_j - z_j} = \sum_{k \ge 0} \frac{(z_j - a_j)^k}{(\zeta_j - a_j)^{k+1}}.$$

On a smaller polydisc the product of these series converges uniformly on $\Gamma(a;r)$, so it may be integrated term by term. The coefficient formula follows. Taking absolute values in the derivative formula gives the estimate because the length $2\pi r_j$ of each circle cancels one power of r_j in the denominator. □

The corollary has two conceptual consequences. First, holomorphic functions are automatically C^∞ and real analytic. Second, the derivatives in different variables commute because they are coefficients of one convergent power series; no extra regularity hypothesis is required.

Theorem 1.22: Three practical tests for holomorphicity

For a complex-valued function f on a domain $\Omega \subset \mathbb{C}^n$, consider:

(1) f is holomorphic in the sense of complex Fréchet differentiability;

(2) f is C^1 and $\partial f/\partial \bar{z}_j = 0$ for every j;
(3) near every point, f is represented by a convergent power series in $z_1, \dots, z_n$ with no $\bar{z}$ variables.

Then the three conditions are equivalent. If f is merely separately holomorphic in every variable, it is also jointly holomorphic; the nontrivial removal of the regularity assumption is proved in Lecture 4.

Proof. Complex differentiability gives the Cauchy–Riemann equations by restricting the differential to each coordinate line, so (1) implies (2). For a C^1 function satisfying (2), the one-variable Cauchy–Pompeiu formula in each variable has no interior $\bar{\partial}$ term. Iteration gives the polydisc Cauchy formula, hence the power series in (3). A normally convergent power series may be differentiated term by term; its differential is complex linear, proving (3) implies (1). □

Lecture 1 split $\mathrm{d}f$ using the eigenspaces of J^*. We now extend that split to forms. In local coordinates,

$$T^*X \otimes_{\mathbb{R}} \mathbb{C} = T^{*1,0}X \oplus T^{*0,1}X, \qquad \mathrm{d}z_j = \mathrm{d}x_j + \mathrm{i}\,\mathrm{d}y_j, \quad \mathrm{d}\bar{z}_j = \mathrm{d}x_j - \mathrm{i}\,\mathrm{d}y_j.$$

Therefore

$$\mathcal{A}^k(X, \mathbb{C}) = \bigoplus_{p+q=k} \mathcal{A}^{p,q}(X), \qquad \alpha = \sum_{|I|=p, |J|=q} \alpha_{I\bar{J}}\, \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J.$$

Definition 1.23: Dolbeault operators

The exterior derivative decomposes by type as $\mathrm{d} = \partial + \bar{\partial}$, where

$$\partial : \mathcal{A}^{p,q} \to \mathcal{A}^{p+1,q}, \qquad \bar{\partial} : \mathcal{A}^{p,q} \to \mathcal{A}^{p,q+1}.$$

For a function f,

$$\partial f = \sum_j \frac{\partial f}{\partial z_j}\mathrm{d}z_j, \quad \bar{\partial} f = \sum_j \frac{\partial f}{\partial \bar{z}_j}\mathrm{d}\bar{z}_j, \quad \frac{\partial}{\partial z_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} - \mathrm{i}\frac{\partial}{\partial y_j}\right).$$

The identity $\mathrm{d}^2 = 0$ and uniqueness of type give

$$\partial^2 = 0, \qquad \bar{\partial}^2 = 0, \qquad \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

A function is holomorphic exactly when $\bar{\partial} f = 0$.

Warning

Conjugation exchanges type but does not reverse wedge order: if $\alpha \in \mathcal{A}^{p,q}$, then $\bar{\alpha} \in \mathcal{A}^{q,p}$ and $\overline{\alpha \wedge \beta} = \bar{\alpha} \wedge \bar{\beta}$. A second common error is to omit the factor $1/2$ in $\partial/\partial z$.

The local solvability of $\bar{\partial}$, which is subtler than the algebraic identity $\bar{\partial}^2 = 0$, is deferred to Lecture 4 so that we can first see the one-variable Cauchy–Green operator from which it is built.

Remark (Convention)

With $\mathrm{d}^{\mathrm{c}} = (4\pi\mathrm{i})^{-1}(\partial - \bar{\partial})$,

$$\mathrm{d}\mathrm{d}^{\mathrm{c}} = \frac{\mathrm{i}}{2\pi}\partial\bar{\partial}, \qquad \mathrm{d}\mathrm{d}^{\mathrm{c}}|z|^2 = \frac{\mathrm{i}}{2\pi}\sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j.$$

For real u, $\mathrm{d}^{\mathrm{c}}u$ is a real one-form. Test every imported formula first on $u = |z|^2$ and then on

$u = \log |z|^2$.

Proposition 1.24: Holomorphic pullback preserves type

If $f : X \to Y$ is holomorphic, then $f^*\mathcal{A}^{p,q}(Y) \subset \mathcal{A}^{p,q}(X)$ and $f^*\partial = \partial f^*$, $f^*\bar\partial = \bar\partial f^*$.

Proof. Locally $f^*(\mathrm{d}w_j) = \sum_k (\partial f_j/\partial z_k)\mathrm{d}z_k$, with no $\mathrm{d}\bar z_k$ term; conjugation gives the other type. Pullback commutes with d, and comparison of type components gives the last two identities. □

1.2.1 Type decomposition as an eigenspace calculation

The almost-complex structure acts on one-forms by $(J^*\alpha)(v) = \alpha(Jv)$. On the complexified cotangent space its eigenvalues are i on $T^{*1,0}X$ and $-\mathrm{i}$ on $T^{*0,1}X$. Hence the projections of a complex one-form α are

$$\alpha^{1,0} = \frac{1}{2}(\alpha - \mathrm{i}J^*\alpha), \qquad \alpha^{0,1} = \frac{1}{2}(\alpha + \mathrm{i}J^*\alpha).$$

Taking exterior powers gives the full (p,q) decomposition. This makes clear that type is intrinsic even though the symbols $\mathrm{d}z_I \wedge \mathrm{d}\bar z_J$ use a chart.

For a smooth function on $\mathbb{C}^n$,

$$\mathrm{d}f = \sum_j f_{z_j}\mathrm{d}z_j + \sum_j f_{\bar z_j}\mathrm{d}\bar z_j.$$

The Cauchy–Riemann equations are therefore the single invariant equation $\bar\partial f = 0$. For the test function $f(z_1, z_2) = z_1\bar z_2 + |z_2|^2$ one obtains

$$\partial f = \bar z_2 \mathrm{d}z_1 + \bar z_2 \mathrm{d}z_2, \qquad \bar\partial f = z_1 \mathrm{d}\bar z_2 + z_2 \mathrm{d}\bar z_2.$$

Differentiating once more verifies directly that $\partial\bar\partial f = -\bar\partial\partial f$; doing this calculation once is the safest way to remember the wedge sign.

For a general (p,q)-form, the coefficient formula is

$$\bar\partial\left(\sum_{I,J} \alpha_{I\bar J}\mathrm{d}z_I \wedge \mathrm{d}\bar z_J\right) = \sum_{I,J,k} \frac{\partial \alpha_{I\bar J}}{\partial \bar z_k}\mathrm{d}\bar z_k \wedge \mathrm{d}z_I \wedge \mathrm{d}\bar z_J.$$

One may move $\mathrm{d}\bar z_k$ to the customary position after the holomorphic differentials, but this contributes $(-1)^p$. Many apparent disagreements between coordinate formulas are only this ordering convention.

1.2.2 From the local lemma to a global obstruction

The local Dolbeault lemma says that every closed (p,q)-form with $q > 0$ has local primitives. Let $\{U_i\}$ be a cover and choose $\alpha|_{U_i} = \bar\partial\beta_i$. On overlaps,

$$\bar\partial(\beta_i - \beta_j) = 0.$$

Thus the failure of the local primitives to glue is itself holomorphic data. Changing β_i changes this overlap family by a coboundary. Sheaf cohomology, introduced in Lecture 9, is designed to record exactly this ambiguity.

The first nontrivial example is a compact Riemann surface. Locally every smooth $(0,1)$-form is a $\bar\partial$ of a function, but globally the equation may fail. On a complex torus $X = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, the form

$\mathrm{d}\bar{z}$ descends and is $\bar{\partial}$-closed. If $\mathrm{d}\bar{z} = \bar{\partial}u$ globally, pairing with the holomorphic form $\mathrm{d}z$ and applying Stokes would give

$$0 = \int_X \mathrm{d}z \wedge \bar{\partial}u = \int_X \mathrm{d}z \wedge \mathrm{d}\bar{z},$$

which is impossible. Local exactness and global exactness are therefore genuinely different statements.

Remark (Normalization audit)

With $\mathrm{d}^{\mathrm{c}} = (4\pi\mathrm{i})^{-1}(\partial - \bar{\partial})$,

$$\mathrm{d}^{\mathrm{c}}u = \frac{1}{4\pi}\sum_j (u_{x_j}\mathrm{d}y_j - u_{y_j}\mathrm{d}x_j), \qquad \mathrm{dd}^{\mathrm{c}}u = \frac{\mathrm{i}}{2\pi}\partial\bar{\partial}u.$$

Both operators take real functions to real forms. The upright symbols emphasize that they are differential operators rather than scalar variables.

1.2.3 Coordinate calculations with type

Let $z = x + \mathrm{i}y$. Solving $\mathrm{d}z = \mathrm{d}x + \mathrm{i}\mathrm{d}y$ and $\mathrm{d}\bar{z} = \mathrm{d}x - \mathrm{i}\mathrm{d}y$ for the real forms gives

$$\mathrm{d}x = \frac{1}{2}(\mathrm{d}z + \mathrm{d}\bar{z}), \qquad \mathrm{d}y = \frac{1}{2\mathrm{i}}(\mathrm{d}z - \mathrm{d}\bar{z}).$$

The dual formulas are

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - \mathrm{i}\frac{\partial}{\partial y}\right), \qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathrm{i}\frac{\partial}{\partial y}\right).$$

Consequently, for $f(x, y) = x^2 + y^2 = z\bar{z}$,

$$\partial f = \bar{z}\,\mathrm{d}z, \qquad \bar{\partial} f = z\,\mathrm{d}\bar{z},$$

and a second differentiation gives

$$\partial\bar{\partial} f = \mathrm{d}z \wedge \mathrm{d}\bar{z}, \qquad \bar{\partial}\partial f = \mathrm{d}\bar{z} \wedge \mathrm{d}z = -\mathrm{d}z \wedge \mathrm{d}\bar{z}.$$

This single calculation displays both $\partial\bar{\partial} + \bar{\partial}\partial = 0$ and the sign caused by exchanging two one-forms.

For a general $(1,1)$-form on $\mathbb{C}^2$,

$$\alpha = \sum_{j,k=1}^{2} a_{j\bar{k}}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k,$$

the two components of its derivative are

$$\partial\alpha = \sum_{\ell,j,k} \frac{\partial a_{j\bar{k}}}{\partial z_\ell}\mathrm{d}z_\ell \wedge \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k,$$

$$\bar{\partial}\alpha = \sum_{\ell,j,k} \frac{\partial a_{j\bar{k}}}{\partial \bar{z}_\ell}\mathrm{d}\bar{z}_\ell \wedge \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k.$$

The first has type $(2,1)$ and the second type $(1,2)$. Since these summands lie in distinct subbundles, $\mathrm{d}\alpha = 0$ is equivalent to the two equations $\partial\alpha = 0$ and $\bar{\partial}\alpha = 0$.

A complete $\bar\partial$ solution in one variable

On a disc, solve

$$\bar\partial u = (z^2 + \bar z)\,\mathrm{d}\bar z.$$

Since $\partial u/\partial \bar z = z^2 + \bar z$, one may integrate while treating z as a constant:

$$u(z) = z^2\bar z + \frac{1}{2}\bar z^2 + h(z),$$

where h is arbitrary and holomorphic. Direct differentiation verifies the equation. The nonuniqueness is exactly $\ker \bar\partial = \mathcal{O}$.

In several variables compatibility appears. For

$$f = f_1\,\mathrm{d}\bar z_1 + f_2\,\mathrm{d}\bar z_2,$$

one computes

$$\bar\partial f = \left(\frac{\partial f_2}{\partial \bar z_1} - \frac{\partial f_1}{\partial \bar z_2}\right)\mathrm{d}\bar z_1 \wedge \mathrm{d}\bar z_2.$$

Thus $\bar\partial u = f$ can hold only if the coefficient in parentheses vanishes. For the concrete closed form

$$f = \bar z_2\,\mathrm{d}\bar z_1 + \bar z_1\,\mathrm{d}\bar z_2$$

the condition holds, and $u = \bar z_1 \bar z_2$ is a solution. By contrast, $f = \bar z_2\,\mathrm{d}\bar z_1$ is not closed and cannot be a $\bar\partial$ of a function.

The real-coordinate form of d^{c}

With the convention $\mathrm{d}^{\mathrm{c}} = (4\pi\mathrm{i})^{-1}(\partial - \bar\partial)$ and real u,

$$\begin{aligned}\partial u - \bar\partial u &= \frac{1}{2}(u_x - \mathrm{i}u_y)(\mathrm{d}x + \mathrm{i}\mathrm{d}y) - \frac{1}{2}(u_x + \mathrm{i}u_y)(\mathrm{d}x - \mathrm{i}\mathrm{d}y)\\ &= \mathrm{i}(u_x\,\mathrm{d}y - u_y\,\mathrm{d}x).\end{aligned}$$

Hence

$$\boxed{\mathrm{d}^{\mathrm{c}}u = \frac{1}{4\pi}(u_x\,\mathrm{d}y - u_y\,\mathrm{d}x)},$$

which is visibly real. Applying d gives

$$\mathrm{d}\mathrm{d}^{\mathrm{c}}u = \frac{1}{4\pi}(u_{xx} + u_{yy})\,\mathrm{d}x \wedge \mathrm{d}y = \frac{\mathrm{i}}{2\pi}u_{z\bar z}\,\mathrm{d}z \wedge \mathrm{d}\bar z.$$

For $u = |z|^2$, this becomes $\mathrm{d}\mathrm{d}^{\mathrm{c}}|z|^2 = \pi^{-1}\mathrm{d}x \wedge \mathrm{d}y$.

A pullback calculation

Let $F : \mathbb{C} \to \mathbb{C}^2$, $F(\zeta) = (\zeta^2, e^\zeta)$. Then

$$F^*(\mathrm{d}z_1) = 2\zeta\,\mathrm{d}\zeta, \qquad F^*(\mathrm{d}z_2) = e^\zeta\,\mathrm{d}\zeta, \qquad F^*(\mathrm{d}\bar z_j) = \overline{F^*(\mathrm{d}z_j)}.$$

There is no $\mathrm{d}\bar\zeta$ term in the pullback of a $(1,0)$-form. For instance,

$$F^*(\mathrm{d}z_1 \wedge \mathrm{d}\bar z_2) = 2\zeta e^{\bar\zeta}\,\mathrm{d}\zeta \wedge \mathrm{d}\bar\zeta,$$

which remains of type $(1,1)$. If instead $F(\zeta) = (\bar\zeta, 0)$, then $F^*(\mathrm{d}z_1) = \mathrm{d}\bar\zeta$; this is the explicit failure of type preservation for an antiholomorphic map.

1.2.4 Self-study workshop: reconstructing the polydisc Cauchy calculus

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall the one-variable Cauchy formula, the geometric series, and the definitions of $\partial/\partial z_j$ and $\partial/\partial \bar{z}_j$. Before calculating, write the multi-index conventions $\alpha!$, r^α, and $(z-a)^\alpha$.

A derivation to reproduce. For $f \in \mathcal{O}(P(0;r))$, iterated Cauchy integration gives

$$c_\alpha = \frac{1}{(2\pi \mathrm{i})^n} \int_{\Gamma(0;r)} \frac{f(\zeta)}{\zeta_1^{\alpha_1+1} \cdots \zeta_n^{\alpha_n+1}} \, \mathrm{d}\zeta_n \cdots \mathrm{d}\zeta_1.$$

Parametrizing $\zeta_j = r_j \mathrm{e}^{\mathrm{i}\theta_j}$ and taking absolute values yields $|c_\alpha| \le M r^{-\alpha}$. On $P(0;\rho)$ with $\rho_j < r_j$,

$$\sum_\alpha |c_\alpha z^\alpha| \le M \prod_{j=1}^n \sum_{k \ge 0} (\rho_j / r_j)^k < \infty.$$

This proves absolute and uniform convergence on smaller polydiscs without appealing to a formal manipulation of power series.

Checkpoint (one function in three languages)

Let $f(z_1, z_2) = 1/(1 - z_1 z_2)$ on Δ^2.

(1) Write its power series and prove normal convergence on $\{|z_1|, |z_2| \le \rho\}$ for every $\rho < 1$.
(2) Compute ∂f and $\bar{\partial} f$.
(3) Which Taylor coefficients c_{jk} vanish, and why is that visible from the integral formula?

Solution (checkpoint). The geometric series gives $f = \sum_{k \ge 0} z_1^k z_2^k$; on the stated compact set it is dominated by $\sum \rho^{2k}$. Moreover

$$\partial f = \frac{z_2 \mathrm{d}z_1 + z_1 \mathrm{d}z_2}{(1 - z_1 z_2)^2}, \qquad \bar{\partial} f = 0.$$

Thus $c_{jk} = 0$ unless $j = k$. In the coefficient integral, independent rotations of the two circle variables make the integral zero unless the exponents match the single monomial powers appearing in the geometric expansion. ◇

Exit criterion

You can derive the polydisc Cauchy formula by iteration, extract a coefficient, prove normal convergence on a smaller polydisc, and translate $\bar{\partial} f = 0$ between the Cauchy–Riemann, power-series, and differential-form descriptions.

Summary. Cauchy's formula on a polydisc gives convergent multivariable Taylor series and coefficient estimates. Holomorphicity may be recognized by complex differentiability, the equations $\bar{\partial} f = 0$, or a local power series. Type decomposition then extends the function-level operator $\bar{\partial}$ to all differential forms.

Exercise 1.25: Coefficients on a bidisc

Let f be holomorphic on a neighborhood of $\{|z_1| \le 2, |z_2| \le 3\}$ and satisfy $|f| \le 12$ there. Bound the coefficient of $z_1^4 z_2^2$ in its Taylor series at the origin, and bound $|\partial_{z_1}^4 \partial_{z_2}^2 f(0)|$.

Exercise 1.26: A power series domain

Determine where $\sum_{j,k\geq 0} z_1^j z_2^k / 2^{j+k}$ converges absolutely, compute its sum, and explain why compact subsets of that domain permit termwise differentiation.

Exercise 1.27: Sign check

For $\alpha \in \mathcal{A}^{p,q}$ and $\beta \in \mathcal{A}^{r,s}$, prove $\bar{\partial}(\alpha \wedge \beta) = \bar{\partial}\alpha \wedge \beta + (-1)^{p+q}\alpha \wedge \bar{\partial}\beta$.

Exercise 1.28: Reality of d^{c}

For a real function u on $\mathbb{C}$, express $\mathrm{d}^{\mathrm{c}}u$ in x, y coordinates and verify that it is real.

Optional handout: Why the distinguished boundary is enough

This handout is optional. No later argument in the core lectures depends on it.

The topological boundary of a bidisc contains the two solid faces $\{|z_1| = r_1, |z_2| \leq r_2\}$ and $\{|z_1| \leq r_1, |z_2| = r_2\}$. The iterated Cauchy formula integrates only over their intersection $\Gamma(0; r)$, a two-dimensional torus. This is possible because holomorphicity in z_1 first reconstructs the function from one circle and holomorphicity in z_2 reconstructs the resulting coefficients from the other.

For $0 < \rho_j < r_j$, successive applications of the one-variable maximum principle give

$$\sup_{P(0;\rho)} |f| \leq \sup_{\Gamma(0;r)} |f|.$$

Thus the distinguished boundary controls the entire polydisc even though it is only a small subset of the topological boundary. The same iteration shows that every derivative on $P(0;\rho)$ is controlled by the supremum on $\Gamma(0;r)$, with a constant depending on the gaps $r_j - \rho_j$.

This product geometry is also why polydiscs are the natural local sets for several complex variables: the ball is geometrically symmetric, but the polydisc makes the one-variable analytic mechanism completely transparent.

1.3 Lecture 3: Cauchy estimates, normal families, and local mapping theorems

Week 2 material • 90 minutes

Notation for this lecture

$\Omega \subset \mathbb{C}^n$ is a connected open set, $K \Subset \Omega$ means that K is compact and contained in Ω, and $P(a;r)$ is the polydisc defined in Lecture 2. For $f \in \mathcal{O}(\Omega)$, $\|f\|_K = \sup_K |f|$. The notation $f_\nu \to f$ in $\mathcal{O}(\Omega)$ means uniform convergence on every compact subset. A family $\mathcal{F} \subset \mathcal{O}(\Omega)$ is *locally uniformly bounded* if every point has a neighborhood U with $\sup_{f\in\mathcal{F}} \|f\|_U < \infty$, and it is *normal* if every sequence has a compactly convergent subsequence. For a holomorphic map $F = (F_1, \dots, F_m)$, $J_{\mathbb{C}}F = (\partial F_i/\partial z_j)$ is its complex Jacobian matrix and $\operatorname{rank}_{\mathbb{C}} J_{\mathbb{C}}F$ is its complex rank.

Prerequisite results used below

Cauchy estimate on a polydisc. If f is holomorphic near $\overline{P(a;r)}$ and $M = \sup_{P(a;r)} |f|$, then $|\partial^\alpha f(a)| \le \alpha! M r^{-\alpha}$.
Arzelà–Ascoli. On a compact metric space, a uniformly bounded and equicontinuous sequence of complex-valued functions has a uniformly convergent subsequence.
Contraction mapping theorem. A contraction of a complete metric space has a unique fixed point, and the fixed point depends continuously on parameters when the contraction constant is uniform.

Why these ideas are needed

Power series prove local existence, but later arguments need stability: what happens to holomorphic functions under limits, and when can equations be solved in holomorphic coordinates? Cauchy estimates answer both questions. They upgrade a uniform bound to bounds for every derivative, yielding Montel compactness and passage to limits without losing holomorphicity. The same quantitative control makes a map with invertible complex Jacobian a small perturbation of its linear part; a contraction argument then produces a holomorphic inverse. The implicit theorem is the geometric payoff: nonsingular holomorphic equations cut out complex submanifolds.

Plan for the lecture. 0–18 min: Cauchy estimates as quantitative control; 18–34: identity and maximum principles; 34–52: locally uniform convergence; 52–66: Montel's theorem; 66–78: holomorphic inverse theorem; 78–90: implicit theorem and submanifold examples.

The qualitative statement "holomorphic functions are analytic" becomes useful only after it is quantified. If $P(a;r) \Subset \Omega$ and $M = \sup_{P(a;r)} |f|$, Lecture 2 gives

$$|\partial^\alpha f(a)| \le \alpha! M r^{-\alpha}. \tag{3.1}$$

Thus a bound on the function on a slightly larger set controls every derivative on a smaller set. This is the basic compactness mechanism of complex analysis.

Theorem 1.29: Identity principle in several variables

Let $\Omega \subset \mathbb{C}^n$ be connected and $f \in \mathcal{O}(\Omega)$. If f vanishes on a nonempty open subset, then $f \equiv 0$. More generally, if all derivatives of f at one point vanish, then $f \equiv 0$.

Proof. The Taylor expansion makes f zero on a polydisc about the given point. The set of points having a zero neighborhood is open. It is also closed: if such points approach $b \in \Omega$, every derivative vanishes along the sequence and hence at b by continuity; the Taylor series at b is zero. Connectedness finishes the proof. □

Proposition 1.30: Maximum modulus and open mapping

If $f \in \mathcal{O}(\Omega)$ is nonconstant, then $|f|$ has no local maximum and f is an open map.

Proof. If $|f|$ has a local maximum at a, restrict f to every complex line through a. The one-variable maximum principle makes every such restriction locally constant; the Taylor series then has no nonconstant term. The identity principle gives global constancy. For openness, choose a complex line on which the first nonzero homogeneous Taylor term of $f - f(a)$ does not vanish. The one-variable restriction is nonconstant and maps a small disc to a neighborhood of $f(a)$. □

Theorem 1.31: Weierstrass convergence theorem

If $f_\nu \in \mathcal{O}(\Omega)$ converges uniformly on compact subsets to f, then $f \in \mathcal{O}(\Omega)$ and $\partial^\alpha f_\nu \to \partial^\alpha f$ uniformly on compact subsets for every multi-index α.

Proof. Choose nested polydiscs $K \subset P' \Subset P \Subset \Omega$. Uniform convergence on $\overline{P}$ allows passage to the limit in the Cauchy integral formula, proving that f satisfies the same formula. The derivative formula applied to $f_\nu - f$ gives

$$\sup_K |\partial^\alpha (f_\nu - f)| \le C_{K,P,\alpha} \sup_{\overline{P}} |f_\nu - f| \longrightarrow 0.$$

□

Theorem 1.32: Montel theorem

A locally uniformly bounded family $\mathcal{F} \subset \mathcal{O}(\Omega)$ is normal: every sequence in $\mathcal{F}$ has a subsequence converging uniformly on compact subsets to a holomorphic function.

Proof. On $K \Subset P \Subset \Omega$, the first-derivative case of (3.1) gives a uniform Lipschitz bound on K. Thus the family is uniformly bounded and equicontinuous there, so Arzelà–Ascoli gives a uniformly convergent subsequence. Apply this successively to a countable exhaustion $K_1 \Subset K_2 \Subset \cdots$ and take the diagonal subsequence. The Weierstrass convergence theorem makes the limit holomorphic. □

The next two theorems explain why nonsingular holomorphic equations define complex manifolds. For $F = (F_1, \dots, F_m)$, write $J_{\mathbb{C}}F(a) = (\partial F_i/\partial z_j)(a)$ for its complex Jacobian matrix.

Theorem 1.33: Holomorphic inverse function theorem

Let $F : \Omega \subset \mathbb{C}^n \to \mathbb{C}^n$ be holomorphic and suppose $\det J_{\mathbb{C}}F(a) \neq 0$. Then there are neighborhoods U of a and V of $F(a)$ such that $F : U \to V$ is biholomorphic.

Proof. After translations and composition with the inverse of $J_{\mathbb{C}}F(a)$, assume $a = F(a) = 0$ and $J_{\mathbb{C}}F(0) = I$. Write $F(z) = z + R(z)$. Shrink a closed ball so that $\|J_{\mathbb{C}}R(z)\| \le 1/2$. For w small, the map $\Phi_w(z) = w - R(z)$ sends that ball to itself and is a contraction. Its unique fixed point $G(w)$ satisfies $F(G(w)) = w$. The real inverse theorem gives smoothness of G, and differentiating the identity gives $\mathrm{d}G = (\mathrm{d}F \circ G)^{-1}$. The right side is complex linear, so Lecture 1's J-criterion makes G holomorphic. □

Theorem 1.34: Holomorphic implicit function theorem

Let $F = (F_1, \dots, F_r) : \Omega \subset \mathbb{C}^n \to \mathbb{C}^r$ be holomorphic, let $F(a) = 0$, and suppose the $r \times r$ matrix $(\partial F_i/\partial z_j(a))_{1 \le i,j \le r}$ is invertible. Writing $z = (z', z'') \in \mathbb{C}^r \times \mathbb{C}^{n-r}$, there are neighborhoods of a'' and a unique holomorphic map g such that near a,

$$F(z', z'') = 0 \quad \Longleftrightarrow \quad z' = g(z'').$$

Consequently $F^{-1}(0)$ is a complex submanifold of dimension $n - r$ near a.

Proof. Apply the inverse theorem to $\Psi(z', z'') = (F(z', z''), z'')$. Its complex Jacobian is block triangular with invertible diagonal blocks. In the inverse coordinates $(w, z'') = \Psi(z)$, the zero set is simply $\{w = 0\}$; transforming back gives the graph $z' = g(z'')$. Uniqueness follows from injectivity of Ψ. □

Example 1.35: A smooth hypersurface and a singular one

For $F(z_1, z_2) = z_1^2 + z_2^2 - 1$, the gradient never vanishes on $F^{-1}(0)$, so the zero set is a one-dimensional complex submanifold. For $G(z_1, z_2) = z_1 z_2$, the gradient vanishes at the origin and $G^{-1}(0)$ is the union of the two coordinate axes; it is not a manifold there. Rank failure does not always imply singularity, but it warns that the chosen equations no longer prove regularity.

1.3.1 Cauchy estimates on nested polydiscs

Suppose $K \subset P(a; \rho) \Subset P(a; r) \Subset \Omega$ and put $\delta_j = r_j - \rho_j > 0$. Centering a polydisc of radius δ_j at each $z \in P(a; \rho)$ gives the uniform estimate

$$\sup_{P(a;\rho)} |\partial^\alpha f| \le \alpha! \, \|f\|_{P(a;r)} \prod_j \delta_j^{-\alpha_j}. \tag{3.2}$$

The estimate depends on the positive collar between the two polydiscs. Derivatives become expensive as the collar shrinks; this is why compact convergence, rather than pointwise convergence, is the natural topology on $\mathcal{O}(\Omega)$.

1.3.2 Two contrasting sequences

On the unit disc, $f_\nu(z) = z^\nu$ converges to 0 uniformly on every smaller disc but not uniformly on the whole open unit disc. Formula (3.2) shows that every fixed derivative also tends to zero on smaller discs. In contrast, $g_\nu(z) = \nu z$ has no compactly convergent subsequence near any nonzero point; the family is not locally uniformly bounded. These examples distinguish pointwise, locally uniform, and global uniform convergence.

1.3.3 Why rank is counted over $\mathbb{C}$

For a holomorphic map, Lecture 1 gives $\mathrm{d}F \circ J_0 = J_0 \circ \mathrm{d}F$. Hence the real rank of $\mathrm{d}F$ is twice the complex rank of $J_{\mathbb{C}}F$. If $F : \mathbb{C}^n \to \mathbb{C}^r$ has complex rank r at a point, its real differential has rank $2r$, and the real dimension $2n - 2r$ supplied by the smooth implicit theorem is exactly complex dimension $n - r$. The holomorphic inverse theorem adds the conclusion that the resulting coordinates are holomorphic.

1.3.4 An explicit inverse-theorem estimate

Assume $F(0) = 0$ and $J_{\mathbb{C}}F(0) = I$. Choose $R > 0$ so that $\|J_{\mathbb{C}}F(z) - I\| \le 1/2$ for $|z| \le R$. Put $Q(z) = F(z) - z$. The mean-value estimate on the underlying real space gives

$$|Q(z) - Q(w)| \le \frac{1}{2}|z - w|.$$

If $|y| \le R/2$, then $\Phi_y(z) = y - Q(z)$ satisfies

$$|\Phi_y(z)| \le |y| + |Q(z) - Q(0)| \le R/2 + |z|/2 \le R$$

on the closed ball $\overline{B}_R$, and it is a contraction with constant $1/2$. The fixed point $z = G(y)$ is the unique solution of $F(z) = y$ in that ball. This calculation records exactly how small the target neighborhood must be.

1.3.5 Local coordinates on a quadric

Let $Q = \{z \in \mathbb{C}^3 : z_1^2 + z_2^2 + z_3^2 = 1\}$. At $p = (1, 0, 0)$, the derivative with respect to z_1 is 2. The implicit theorem gives

$$z_1 = g(z_2, z_3) = \sqrt{1 - z_2^2 - z_3^2},$$

where the square root is the branch with value 1 at the origin. Expanding,

$$g = 1 - \frac{1}{2}(z_2^2 + z_3^2) + O(|(z_2, z_3)|^4).$$

Thus (z_2, z_3) are explicit holomorphic coordinates near p. At every point of Q, at least one coordinate is nonzero, so some partial derivative of the defining function is nonzero; these charts cover Q.

1.3.6 Zero sets need not be discrete

In one variable, a nonzero holomorphic function has isolated zeros. In several variables, $f(z_1, z_2) = z_1$ vanishes on the hypersurface $\{z_1 = 0\}$. The identity principle therefore requires vanishing on an open subset of the ambient domain, not merely on a set with an accumulation point. What survives is restriction to complex lines: if the restriction to a line is not identically zero, its zeros on that line are isolated.

1.3.7 Self-study workshop: using one collar to control all derivatives

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Record two nested polydiscs $P(a; \rho) \Subset P(a; r)$ and their coordinate gaps $\delta_j = r_j - \rho_j$. Recall Cauchy's derivative formula, Arzelà–Ascoli, and the contraction mapping theorem.

The proof chain. A uniform bound on the outer polydisc gives the derivative estimate (3.2) on the inner one. The first derivatives give equicontinuity; Arzelà–Ascoli gives a subsequence on one compact set; a diagonal process handles an exhaustion. Cauchy's formula passes to the limit and proves that the limit is holomorphic. For a local inverse, normalize the derivative to the identity and use the same smallness of the derivative of the remainder to obtain a contraction.

Checkpoint (regular or singular level set)

For each function below, determine the points at which its zero set is proved smooth by the implicit theorem, and give local coordinates there:

$$F(z_1, z_2) = z_1 z_2 - 1, \qquad G(z_1, z_2) = z_1^2 - z_2^3.$$

What happens at the origin for the second example?

Solution (checkpoint). On $F^{-1}(0)$ neither coordinate is zero. Since $\partial F/\partial z_1 = z_2$ and $\partial F/\partial z_2 = z_1$, either coordinate may be solved as a holomorphic function of the other; for example $z_2 = 1/z_1$. For G, the gradient $(2z_1, -3z_2^2)$ is nonzero away from the origin, so the curve is smooth there. At $(0, 0)$ the Jacobian criterion gives no chart, and the parametrization $(z_1, z_2) = (t^3, t^2)$ exhibits a cusp. It is not a complex submanifold at the origin. ◇

Exit criterion

You can derive a nested-polydisc derivative estimate, explain the diagonal argument in Montel's theorem, normalize an inverse-function problem, and use a named nonzero Jacobian minor to produce explicit local coordinates on a level set.

Summary. Cauchy estimates turn local bounds into derivative bounds. They imply stability under compact convergence and, together with Arzelà–Ascoli, Montel compactness. The holomorphic inverse and implicit function theorems convert an invertible complex Jacobian into holomorphic coordinates and regular level sets.

Exercise 1.36: A quantitative derivative bound

Suppose $f \in \mathcal{O}(\Delta^2)$ and $|f| \le 1$. Prove that for $|z_1|, |z_2| \le 1/2$, $|\partial_{z_1}^2 \partial_{z_2} f(z)| \le 16$. State exactly which larger polydisc you use around z.

Exercise 1.37: An implicit curve

Near $(1, 0)$ solve $z_1^2 + z_2^2 = 1$ for z_1 as a holomorphic function of z_2. Compute the first two derivatives at 0 and explain why the same graph cannot be continued as a single-valued function on all of $\mathbb{C} \setminus \{\pm 1\}$.

Exercise 1.38: Normality with a moving pole

For $f_a(z) = 1/(1 - az_1)$ with $|a| \le 1/2$, prove directly that the family is normal on Δ^n and identify the limit of any subsequence for which $a_\nu \to a$.

Optional handout: Hurwitz's theorem and stability of zeros

This handout is optional. No later argument in the core lectures depends on it.

Let f_ν be nowhere-zero holomorphic functions on a connected domain and suppose $f_\nu \to f$ uniformly on compact subsets. Hurwitz's theorem says that either f is nowhere zero or $f \equiv 0$. To see the mechanism, assume $f(a) = 0$ but f is not identically zero. Restrict to a generic complex line through a on which the first nonzero homogeneous Taylor term does not vanish. On a small circle in that line, f has no zeros. Uniform convergence and Rouché's theorem force f_ν to have the same positive number of zeros inside, a contradiction.

The conclusion is false for pointwise convergence. It is also deliberately weaker than continuity of individual zero sets: in several variables a zero set may split, develop singularities, or change topology. What compact convergence preserves is the analytic constraint detected on every transverse one-dimensional slice.

1.4 Lecture 4: Hartogs phenomena and the elementary $\bar\partial$ method

Week 2 material • 90 minutes

Notation for this lecture

$\Delta_r = \{z \in \mathbb{C} : |z| < r\}$, $\Delta = \Delta_1$, and $A_r = \{z \in \mathbb{C} : r < |z| < 1\}$. The standard Hartogs figure is $H_r = (\Delta_r \times \Delta) \cup (\Delta \times A_r) \subset \Delta^2$. A function is *separately holomorphic* if it is holomorphic in each one-variable slice with all other coordinates fixed. The symbol $C_c^\infty(\mathbb{C})$ denotes smooth compactly supported functions. For $g \in C_c^\infty(\mathbb{C})$,

$$(Tg)(z) = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{g(\zeta)}{\zeta - z}\, \mathrm{d}\zeta \wedge \mathrm{d}\bar\zeta$$

is the Cauchy–Green operator. The spaces $\mathcal{A}^{p,q}$ and the operators $\partial, \bar\partial$ have the meanings declared in Lecture 2.

Prerequisite results used below

Laurent coefficients. If h is holomorphic on an annulus containing $|\zeta| = \rho$, then $h(z) = \sum_{k\in\mathbb{Z}} a_k z^k$ there, with

$$a_k = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=\rho} \frac{h(\zeta)}{\zeta^{k+1}}\, \mathrm{d}\zeta.$$

If h extends holomorphically across the inner disc, every a_k with $k < 0$ vanishes.

Baire category theorem. A complete metric space cannot be a countable union of closed sets with empty interior. We use it only to obtain a uniform bound for a family of one-variable slices.

Stokes–Cauchy calculation. On a planar domain, apply Stokes' theorem to $f(\zeta)(\zeta - z)^{-1}\mathrm{d}\zeta$ after deleting a small disc about z; the small boundary tends to $2\pi \mathrm{i} f(z)$.

Why these ideas are needed

Several complex variables first becomes visibly different from one-variable theory in Hartogs' theorem: separate holomorphicity automatically becomes joint, and a compact hole may be filled without any boundedness assumption near the hole. The Laurent coefficient proof is elementary enough to expose the mechanism. It also points to a broader question: instead of extending a function across a special hole, can we correct a nonholomorphic function by solving an equation? Cauchy–Pompeiu identifies $\bar\partial f$ as the defect, and the Cauchy–Green integral supplies a right inverse. This is the first model of the $\bar\partial$ method that later drives Stein theory, Hörmander estimates, vanishing, and extension.

Plan for the lecture. 0–15 min: separate versus joint holomorphicity; 15–42: Hartogs figure and extension theorem; 42–55: geometric meaning of the missing compact set; 55–70: Cauchy–Pompeiu and the Cauchy–Green operator; 70–84: local $\bar\partial$ lemma for forms; 84–90: compatibility checks.

Separate holomorphicity has no real-variable analogue. A function may be given one variable at a time, yet the conclusion is a joint power series in all variables.

Theorem 1.39: Hartogs theorem on separate holomorphicity

Let $\Omega \subset \mathbb{C}^n$ be a domain. If $f : \Omega \to \mathbb{C}$ is holomorphic in each variable separately while the others are fixed, then f is jointly holomorphic.

Proof. We record the analytic heart of the proof. First suppose f is locally bounded. Choose a closed polydisc $\overline{P(a;r)} \Subset \Omega$. Repeated one-variable Cauchy formulas give the integral in Theorem 1.20; local boundedness justifies Fubini and differentiation under the integral. Hence f has a joint power series.

It remains to obtain local boundedness. On a slightly smaller product of discs, apply the Baire category theorem to the closed sets

$$E_N = \{z' : \sup_{|z_n - a_n| \le r_n} |f(z', z_n)| \le N\}.$$

Their union is the whole base because each one-variable slice is continuous on a compact disc. Each E_N is closed: it is the intersection, over $|z_n - a_n| \le r_n$, of the closed sets $\{z' : |f(z', z_n)| \le N\}$. The Baire category theorem therefore gives an E_N containing a base polydisc. Cauchy's formula in the remaining variables propagates this bound to a neighborhood. Induction on n then reduces the problem to the locally bounded case. □

The more striking Hartogs phenomenon is that a compact hole in dimension at least two may be invisible to holomorphic functions. Fix $0 < r < 1$ and define the Hartogs figure

$$H_r = (\Delta_r \times \Delta) \cup (\Delta \times A_r), \qquad A_r = \{\zeta \in \mathbb{C} : r < |\zeta| < 1\}.$$

Theorem 1.40: Hartogs extension on the standard figure

Every $f \in \mathcal{O}(H_r)$ extends uniquely to a holomorphic function on the bidisc Δ^2.

Proof. Choose ρ with $r < \rho < 1$. For fixed $z_1 \in \Delta$, expand in the annular variable:

$$f(z_1, z_2) = \sum_{k \in \mathbb{Z}} a_k(z_1) z_2^k, \qquad a_k(z_1) = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=\rho} \frac{f(z_1, \zeta)}{\zeta^{k+1}} \, \mathrm{d}\zeta.$$

Each a_k is holomorphic on Δ by differentiation under the integral. When $|z_1| < r$, the function is holomorphic for $|z_2| < 1$, so its Laurent coefficients $a_k(z_1)$ vanish for every $k < 0$. The identity principle makes those coefficients zero for every $z_1 \in \Delta$. Thus only nonnegative powers remain, and the resulting series converges normally on smaller bidiscs. It defines the required extension. Uniqueness follows from the identity principle because H_r is open. □

The proof shows exactly where dimension two enters: one variable supplies an annulus on which to compute Laurent coefficients, while the other variable carries the identity theorem that kills the negative coefficients. Punctures and compact holes therefore behave very differently from the one-variable case.

We now return to the operator introduced in Lecture 1. Let $\mathrm{d}A(\zeta) = \mathrm{d}\xi \, \mathrm{d}\eta$ for $\zeta = \xi + \mathrm{i}\eta$. Our orientation convention is

$$\mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta} = -2\mathrm{i} \, \mathrm{d}A(\zeta).$$

For a C^1 function on a bounded planar domain D with positively oriented, piecewise smooth boundary, the Cauchy–Pompeiu formula is

$$f(z) = \frac{1}{2\pi \mathrm{i}} \int_{\partial D} \frac{f(\zeta)}{\zeta - z} \, \mathrm{d}\zeta + \frac{1}{2\pi \mathrm{i}} \int_D \frac{\partial f / \partial \bar{\zeta}}{\zeta - z} \, \mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta}. \tag{4.1}$$

Equivalently, its area-measure form is

$$f(z) = \frac{1}{2\pi \mathrm{i}} \int_{\partial D} \frac{f(\zeta)}{\zeta - z} \, \mathrm{d}\zeta - \frac{1}{\pi} \int_D \frac{\partial f / \partial \bar{\zeta}}{\zeta - z} \, \mathrm{d}A(\zeta).$$

Thus the sign in the second display is not a new convention: it is exactly the first display after substituting $\mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta} = -2\mathrm{i} \, \mathrm{d}A$. The interior integral measures the failure of f to be holomorphic.

Proposition 1.41: The Cauchy–Green right inverse

For $g \in C_c^\infty(\mathbb{C})$, set

$$(Tg)(z) = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{g(\zeta)}{\zeta - z}\, \mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta}.$$

Then Tg is smooth and $\bar{\partial}(Tg) = g\, \mathrm{d}\bar{z}$.

Proof. We separate the two points that are often hidden in the phrase "fundamental solution": the integral is smooth, and its $\bar{\partial}$ derivative is the prescribed form.

First convert the form integral to ordinary area measure. Since $\mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta} = -2\mathrm{i}\, \mathrm{d}A(\zeta)$,

$$(Tg)(z) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{g(\zeta)}{z - \zeta}\, \mathrm{d}A(\zeta) = (E * g)(z), \qquad E(z) = \frac{1}{\pi z}. \tag{4.2}$$

The kernel E is locally integrable because $\int_{|z|<\varepsilon} |z|^{-1} \mathrm{d}A = 2\pi\varepsilon$. Choose $\chi \in C_c^\infty(\mathbb{C})$ equal to 1 near the origin and write $E = \chi E + (1-\chi)E$. The second summand is smooth. The first is an L^1 function of compact support, so for every real multi-index α,

$$\partial_z^\alpha \big((\chi E) * g \big) = (\chi E) * (\partial^\alpha g).$$

The right-hand side is continuous by dominated convergence; repeating the argument for every α proves $Tg \in C^\infty(\mathbb{C})$.

It remains to compute the derivative of the kernel, including its normalization. For a test function $\varphi \in C_c^\infty(\mathbb{C})$, take a disc D containing its support. Formula (4.1), applied to φ at $z = 0$, has zero boundary term and gives

$$\varphi(0) = -\frac{1}{\pi} \int_{\mathbb{C}} \frac{\partial \varphi / \partial \bar{z}}{z}\, \mathrm{d}A(z) = \left\langle \frac{\partial E}{\partial \bar{z}}, \varphi \right\rangle.$$

Here the last equality is the definition of the distributional derivative. Hence

$$\frac{\partial}{\partial \bar{z}} \left(\frac{1}{\pi z} \right) = \delta_0. \tag{4.3}$$

Differentiation commutes with convolution of a distribution with a test function, and therefore

$$\frac{\partial (Tg)}{\partial \bar{z}} = \left(\frac{\partial E}{\partial \bar{z}} \right) * g = \delta_0 * g = g.$$

Finally, for a scalar function u one has $\bar{\partial} u = (\partial u / \partial \bar{z}) \mathrm{d}\bar{z}$; consequently $\bar{\partial}(Tg) = g\, \mathrm{d}\bar{z}$, as claimed. □

Proposition 1.42: Cauchy–Green operator with parameters

Let

$$P = \prod_{j=1}^{n} \Delta(a_j, r_j) \subset \mathbb{C}^n, \qquad g \in C^\infty(P),$$

fix $1 \le k \le n$, and choose $\psi \in C_c^\infty(\Delta(a_k, r_k))$. Define

$$(T_{k,\psi} g)(z) = \frac{1}{2\pi \mathrm{i}} \int_{\Delta(a_k, r_k)} \frac{\psi(\tau)\, g(z_1, \dots, z_{k-1}, \tau, z_{k+1}, \dots, z_n)}{\tau - z_k}\, \mathrm{d}\tau \wedge \mathrm{d}\bar{\tau}. \tag{4.4}$$

Then $T_{k,\psi} g$ is smooth on P. If $j \neq k$, differentiation in the j-th variable passes through the

integral:

$$\frac{\partial}{\partial \bar{z}_j} T_{k,\psi} g = T_{k,\psi}\left(\frac{\partial g}{\partial \bar{z}_j}\right). \tag{4.5}$$

In the variable being integrated,

$$\frac{\partial}{\partial \bar{z}_k} T_{k,\psi} g = \psi(z_k) g(z). \tag{4.6}$$

Thus the operator preserves holomorphic dependence on every parameter z_j, $j \neq k$, and solves the $\bar{\partial}$ equation in z_k wherever $\psi = 1$.

Proof. For fixed values of the parameter variables, formula (4.4) is the Cauchy–Green operator applied to the compactly supported smooth function

$$\tau \longmapsto \psi(\tau) g(z_1, \ldots, z_{k-1}, \tau, z_{k+1}, \ldots, z_n),$$

extended by zero outside $\Delta(a_k, r_k)$. Proposition 1.41 therefore gives (4.6) and smoothness in z_k.

Let $K \Subset P$ be compact. For parameters in K, the τ-support remains inside the fixed compact set $\operatorname{supp}\psi$, while every parameter derivative of g is bounded on a neighborhood of $K \times \operatorname{supp}\psi$. The singular factor $|\tau - z_k|^{-1}$ is locally integrable, uniformly for z_k in compact subsets. Dominated convergence therefore permits every derivative in a parameter variable to pass under the integral. This proves (4.5), and iteration gives all mixed derivatives. Hence the function is smooth in all variables. □

Lemma 1.43: Coordinate bookkeeping for $\bar{\partial}$

On a coordinate domain, let coefficientwise differentiation of a form be denoted by $\partial\alpha/\partial\bar{z}_j$. Then

$$\bar{\partial}\alpha = \sum_{j=1}^{n} \mathrm{d}\bar{z}_j \wedge \frac{\partial \alpha}{\partial \bar{z}_j}. \tag{4.7}$$

Suppose that a form α involves only $\mathrm{d}\bar{z}_1, \ldots, \mathrm{d}\bar{z}_k$ and that $\bar{\partial}\alpha = 0$. Every coefficient of α is then holomorphic in the remaining variables $z_{k+1}, \ldots, z_n$. Moreover, if $k \geq 1$, there is a unique splitting

$$\alpha = \mathrm{d}\bar{z}_k \wedge \gamma + \alpha', \tag{4.8}$$

where neither γ nor α' involves $\mathrm{d}\bar{z}_k$.

Proof. Formula (4.7) follows by applying $\bar{\partial}$ to the coefficient functions; the coordinate differentials are constant. Write α in increasing antiholomorphic multi-indices. If $j > k$, the coefficient of $\mathrm{d}\bar{z}_j \wedge \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ in (4.7), up to the sign needed to restore the chosen order, is $\partial\alpha_{I\bar{J}}/\partial\bar{z}_j$. No other summand can contribute to that same multi-index because j does not occur in J. Closedness makes this derivative zero. The holomorphicity test then gives joint holomorphic dependence on $z_{k+1}, \ldots, z_n$. Finally, separate the terms whose antiholomorphic multi-index contains k from those whose multi-index does not. Moving $\mathrm{d}\bar{z}_k$ to the front fixes the sign and gives the unique forms γ and α' in (4.8). □

Theorem 1.44: Dolbeault–Grothendieck lemma

Let X be a complex manifold and let $\alpha \in \mathcal{A}^{p,q}(X)$ with $q \geq 1$. If $\bar{\partial}\alpha = 0$, then every point $x \in X$ has a coordinate neighborhood U and a form $\beta \in \mathcal{A}^{p,q-1}(U)$ such that

$$\alpha|_U = \bar{\partial}\beta.$$

Equivalently, the Dolbeault complex is locally exact in every positive antiholomorphic degree.

This result is also called the local Dolbeault lemma; the name Dolbeault–Grothendieck lemma emphasizes the local exactness statement used in the Dolbeault resolution.

Proof. We first prove an explicit statement on polydiscs. Let

$$P' = P(a; r') \Subset P(a; r) = P$$

be concentric polydiscs, so $r'_j < r_j$ for every j. For $0 \le k \le n$, let (S_k) be the following assertion:

If $\alpha \in \mathcal{A}^{p,q}(P)$ is $\bar\partial$-closed, $q \ge 1$, and involves only $\mathrm{d}\bar z_1, \dots, \mathrm{d}\bar z_k$, then there is $\beta \in \mathcal{A}^{p,q-1}(P')$ with $\bar\partial\beta = \alpha|_{P'}$.

The assertion (S_0) holds because a (p,q)-form of positive antiholomorphic degree that involves no antiholomorphic differential is zero. Assume now that $k \ge 1$ and that (S_{k-1}) is known.

Step 1: identify the holomorphic parameters. By Lemma 1.43, all coefficients of α are holomorphic in $z_{k+1}, \dots, z_n$. Split

$$\alpha = \mathrm{d}\bar z_k \wedge \gamma + \alpha',$$

where γ and α' involve only $\mathrm{d}\bar z_1, \dots, \mathrm{d}\bar z_{k-1}$. Their coefficients are coefficients of α, up to sign, and hence have the same holomorphic parameter dependence.

Step 2: choose the region on which cancellation will be exact. Choose ρ with $r'_k < \rho < r_k$ and a cutoff $\psi \in C^\infty_c(\Delta(a_k, r_k))$ that equals 1 on a neighborhood of $\overline{\Delta(a_k, \rho)}$. Put

$$P^\flat = \Delta(a_1, r_1) \times \cdots \times \Delta(a_k, \rho) \times \cdots \times \Delta(a_n, r_n).$$

Then $P' \Subset P^\flat \subset P$.

Step 3: solve in the k-th variable. Apply $T_{k,\psi}$ coefficientwise to γ and call the resulting $(p, q-1)$-form G. It still involves only $\mathrm{d}\bar z_1, \dots, \mathrm{d}\bar z_{k-1}$. Proposition 1.42 gives, on $P^\flat$,

$$\frac{\partial G}{\partial \bar z_k} = \gamma, \qquad \frac{\partial G}{\partial \bar z_j} = 0 \quad (j > k). \tag{4.9}$$

The second identity is the point of keeping track of the other variables as holomorphic parameters.

Step 4: remove one antiholomorphic differential. Using (4.7) and (4.9), on $P^\flat$ we have

$$\bar\partial G = \sum_{j<k} \mathrm{d}\bar z_j \wedge \frac{\partial G}{\partial \bar z_j} + \mathrm{d}\bar z_k \wedge \gamma.$$

Consequently

$$\widetilde\alpha := \alpha - \bar\partial G = \alpha' - \sum_{j<k} \mathrm{d}\bar z_j \wedge \frac{\partial G}{\partial \bar z_j}$$

involves only $\mathrm{d}\bar z_1, \dots, \mathrm{d}\bar z_{k-1}$. It is closed because

$$\bar\partial\widetilde\alpha = \bar\partial\alpha - \bar\partial^2 G = 0.$$

Step 5: apply the induction hypothesis. The pair $P' \Subset P^\flat$ is again a pair of concentric polydiscs. Assertion (S_{k-1}) therefore gives $\widetilde\beta \in \mathcal{A}^{p,q-1}(P')$ with $\bar\partial\widetilde\beta = \widetilde\alpha|_{P'}$. Set

$$\beta = G|_{P'} + \widetilde\beta.$$

Then

$$\bar\partial\beta = \bar\partial G + \widetilde\alpha = \alpha \qquad \text{on } P'.$$

This proves (S_k), and induction gives (S_n).

It remains to pass from coordinates to the manifold. Fix $x \in X$ and choose a holomorphic chart $\phi : V_0 \to \phi(V_0) \subset \mathbb{C}^n$ with $a = \phi(x)$. Choose concentric polydiscs $P' \Subset P \Subset \phi(V_0)$ centered at a. The pullback by ϕ^{-1} carries α to a closed (p,q)-form on P. Assertion (S_n) supplies a primitive on P'. Pulling it back by ϕ gives the required form on the coordinate neighborhood $U = \phi^{-1}(P')$, because biholomorphic pullback preserves bidegree and commutes with $\bar\partial$. □

Remark (What the proof uses)

The restriction $q \geq 1$ enters in the base case (S_0). The polydisc shrinks because the cutoff equals one only on a smaller disc, so (4.6) cancels the $\mathrm{d}\bar z_k$ component only there. Different cutoffs can produce different primitives; for $q = 1$ any two differ by a holomorphic p-form. Finally, the argument is local. It gives no assertion that the local primitives glue; the failure to glue is measured by Dolbeault cohomology.

Example 1.45: Compatibility before solving

On $\mathbb{C}^2$, a $(0,1)$-form $\alpha = a_1\mathrm{d}\bar z_1 + a_2\mathrm{d}\bar z_2$ satisfies

$$\bar\partial\alpha = \left(\frac{\partial a_2}{\partial \bar z_1} - \frac{\partial a_1}{\partial \bar z_2}\right)\mathrm{d}\bar z_1 \wedge \mathrm{d}\bar z_2.$$

Thus $\bar\partial u = \alpha$ requires equality of the two mixed derivatives. For $\alpha = \bar z_2\mathrm{d}\bar z_1 + \bar z_1\mathrm{d}\bar z_2$, the condition holds and $u = \bar z_1\bar z_2$ is an explicit solution.

1.4.1 Deriving Cauchy–Pompeiu with the sign visible

Let D be a smoothly bounded planar domain and let $D_\varepsilon = D \setminus \overline{\Delta(z,\varepsilon)}$. Since $\bar\partial_\zeta(\zeta - z)^{-1} = 0$ on D_ε,

$$\mathrm{d}\left(\frac{f(\zeta)}{\zeta - z}\,\mathrm{d}\zeta\right) = -\frac{\partial f/\partial\bar\zeta}{\zeta - z}\,\mathrm{d}\zeta \wedge \mathrm{d}\bar\zeta.$$

Stokes' theorem gives contributions from ∂D and the negatively oriented inner circle. On the inner circle, $f(\zeta) = f(z) + o(1)$ and $\int (\zeta - z)^{-1}\mathrm{d}\zeta = 2\pi\mathrm{i}$. Moving this term to the other side yields formula (4.1). Writing the deleted circle and its orientation prevents the most common sign error.

1.4.2 Hartogs extension in higher dimension

For $n \geq 2$, split coordinates as $(z', z_n) \in \mathbb{C}^{n-1} \times \mathbb{C}$. If a domain contains

$$(\Delta_r^{n-1} \times \Delta) \cup (\Delta^{n-1} \times A_r),$$

expand in z_n on the annulus. Every Laurent coefficient is holomorphic in z'. The negative coefficients vanish on the open set Δ_r^{n-1} and hence vanish on all of Δ^{n-1} by the several-variable identity principle. The same proof therefore fills the entire missing polydisc.

1.4.3 Local solvability versus global obstruction

The Dolbeault–Grothendieck lemma says that $\bar\partial$-closed forms of positive antiholomorphic degree have local primitives. It does not say that the primitives glue globally. On a complex torus $X = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, the form $\mathrm{d}\bar z$ descends and is $\bar\partial$-closed. If it were $\bar\partial u$ globally, then Stokes' theorem would give

$$0 = \int_X \mathrm{d}z \wedge \bar\partial u = \int_X \mathrm{d}z \wedge \mathrm{d}\bar z,$$

a contradiction. The difference between local and global solvability is the reason Dolbeault cohomology will later be introduced.

1.4.4 Two explicit Hartogs calculations

On $\Delta \times A_r$, consider

$$f(z_1, z_2) = \frac{1}{1 - z_1 z_2} = \sum_{k \geq 0} z_1^k z_2^k.$$

There are no negative Laurent powers, so the series already displays the extension to Δ^2. In contrast, $1/z_2$ has a negative Laurent coefficient and cannot extend. The reason it is not a counterexample to Hartogs' theorem is that it is not holomorphic on the other arm $\Delta_r \times \Delta$, where $z_2 = 0$ occurs.

For a less immediate example, suppose on the annular arm

$$f(z_1, z_2) = \sum_{k \in \mathbb{Z}} a_k(z_1) z_2^k.$$

The values on $\Delta_r \times \Delta$ force $a_{-1}, a_{-2}, \ldots$ to vanish for $|z_1| < r$. Each coefficient is holomorphic for $|z_1| < 1$, so the identity theorem forces global vanishing. Nothing is being assumed about the size of f near the missing set.

1.4.5 Solving a $\bar{\partial}$ equation by inspection

Treat z as constant when differentiating with respect to $\bar{z}$. Thus

$$\bar{\partial}\left(z^2 \bar{z} + \frac{1}{2}\bar{z}^2\right) = (z^2 + \bar{z}) \mathrm{d}\bar{z}.$$

Every other solution differs by a holomorphic function, because $\ker(\bar{\partial} : \mathcal{A}^{0,0} \to \mathcal{A}^{0,1}) = \mathcal{O}$. In two variables,

$$\bar{\partial}(\bar{z}_1 \bar{z}_2) = \bar{z}_2 \mathrm{d}\bar{z}_1 + \bar{z}_1 \mathrm{d}\bar{z}_2.$$

The coefficient compatibility in the main text is exactly the equality of mixed $\bar{z}$ derivatives.

1.4.6 Following the Dolbeault induction in $\mathbb{C}^2$

The abstract induction in Theorem 1.44 can be read as an explicit two-stage algorithm. Let

$$P = \Delta(a_1, r_1) \times \Delta(a_2, r_2), \qquad \alpha = a_1^\circ \, \mathrm{d}\bar{z}_1 + a_2^\circ \, \mathrm{d}\bar{z}_2 \in \mathcal{A}^{0,1}(P),$$

where the superscript $\circ$ distinguishes the coefficient functions from the centers of the discs. Direct calculation gives

$$\bar{\partial}\alpha = \left(\frac{\partial a_2^\circ}{\partial \bar{z}_1} - \frac{\partial a_1^\circ}{\partial \bar{z}_2}\right) \mathrm{d}\bar{z}_1 \wedge \mathrm{d}\bar{z}_2.$$

Thus closedness is the compatibility condition

$$\frac{\partial a_2^\circ}{\partial \bar{z}_1} = \frac{\partial a_1^\circ}{\partial \bar{z}_2}. \tag{4.10}$$

Fix a smaller concentric polydisc $P' \Subset P$.

First stage: remove the $\mathrm{d}\bar{z}_2$ component. Choose a cutoff ψ_2 in the second disc that equals one on a slightly larger disc containing the second projection of P', and set

$$G_2 = T_{2,\psi_2} a_2^\circ.$$

On the corresponding intermediate polydisc, $\partial G_2/\partial\bar z_2 = a_2^\circ$. Hence

$$\alpha - \bar\partial G_2 = b_1\,\mathrm{d}\bar z_1, \qquad b_1 = a_1^\circ - \frac{\partial G_2}{\partial \bar z_1}.$$

The new coefficient is holomorphic in the variable already eliminated. Indeed, using (4.10), commutation of mixed derivatives, and the defining equation for G_2, we obtain

$$\frac{\partial b_1}{\partial \bar z_2} = \frac{\partial a_1^\circ}{\partial \bar z_2} - \frac{\partial}{\partial \bar z_1}\frac{\partial G_2}{\partial \bar z_2} = \frac{\partial a_1^\circ}{\partial \bar z_2} - \frac{\partial a_2^\circ}{\partial \bar z_1} = 0.$$

Second stage: remove the $\mathrm{d}\bar z_1$ *component.* Choose a cutoff ψ_1 in the first disc, equal to one near the first projection of P', and put

$$G_1 = T_{1,\psi_1} b_1.$$

The parameter identity (4.5) and the preceding holomorphicity give

$$\frac{\partial G_1}{\partial \bar z_1} = b_1, \qquad \frac{\partial G_1}{\partial \bar z_2} = T_{1,\psi_1}\left(\frac{\partial b_1}{\partial \bar z_2}\right) = 0$$

on P'. Therefore

$$\beta = G_2 + G_1 = T_{2,\psi_2} a_2^\circ + T_{1,\psi_1}\left(a_1^\circ - \frac{\partial}{\partial \bar z_1} T_{2,\psi_2} a_2^\circ\right)$$

satisfies $\bar\partial\beta = \alpha$ on P'. For $a_1^\circ = \bar z_2$ and $a_2^\circ = \bar z_1$, this construction differs from the inspection solution $\bar z_1\bar z_2$ by a holomorphic function, as it must.

1.4.7 A cutoff creates an error that $\bar\partial$ can repair

Suppose a holomorphic function is known near a closed set and choose a smooth cutoff χ equal to one near that set. The product χf is globally smooth but usually not holomorphic:

$$\bar\partial(\chi f) = f\,\bar\partial\chi.$$

If one solves $\bar\partial u = f\,\bar\partial\chi$, then $F = \chi f - u$ is holomorphic. This simple "extend, measure the defect, solve, subtract" pattern is the prototype for later extension and vanishing proofs. The hard part in global applications is obtaining a solution with the right support or norm estimate.

1.4.8 Self-study workshop: seeing the Hartogs mechanism rather than memorizing the theorem

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall Laurent coefficient integrals, the identity principle, Cauchy–Pompeiu, and the bidegree of $\bar\partial$. Draw the two arms of H_r before writing a formula.

The coefficient argument. The annular arm lets us compute all Laurent coefficients in z_2 for every $z_1 \in \Delta$. The solid arm says the negative coefficients vanish when $z_1 \in \Delta_r$. Holomorphic dependence of the coefficients on z_1 and the identity theorem propagate that vanishing to the whole base. Normal convergence of the remaining Taylor series then supplies the extension.

Checkpoint (locate every hypothesis)

Answer the following without citing Hartogs' theorem as a black box.
(1) Where is the annulus used?
(2) Where is the second complex variable used?
(3) Why is $1/z_2$ not a counterexample?
(4) Why must a form equal to $\bar\partial u$ first be $\bar\partial$-closed?

Solution (checkpoint). The annulus supplies the Laurent coefficient integrals in z_2. The second variable makes those coefficients holomorphic functions on a connected disc; their vanishing on Δ_r then propagates by the identity theorem. The function $1/z_2$ is not defined holomorphically on the solid arm, which contains points with $z_2 = 0$. Finally, $\bar\partial u = \alpha$ implies $\bar\partial\alpha = \bar\partial^2 u = 0$. ⋄

Exit criterion

You can reconstruct the Laurent coefficient proof of Hartogs extension, derive the sign in Cauchy–Pompeiu by deleting a small disc, solve a simple $\bar\partial$ equation, and state the closedness compatibility condition before seeking a primitive.

Summary. Separate holomorphicity forces joint holomorphicity, and a Hartogs figure has no holomorphic memory of its compact hole. The Cauchy–Pompeiu formula identifies $\bar\partial f$ as the obstruction term, while the Cauchy–Green operator gives a local right inverse and leads to the local Dolbeault lemma.

Exercise 1.46: Filling a Hartogs hole

Let f be holomorphic on H_r and suppose on $\Delta \times A_r$ that $f(z_1, z_2) = \sum_{k\in\mathbb{Z}} a_k(z_1) z_2^k$. Prove directly that every a_k with $k < 0$ vanishes, carefully naming the open set on which it first vanishes.

Exercise 1.47: A solvable and an unsolvable form

On $\mathbb{C}^2$, decide which of

$$\alpha = \bar z_2 \mathrm{d}\bar z_1 + \bar z_1 \mathrm{d}\bar z_2, \qquad \gamma = \bar z_2 \mathrm{d}\bar z_1$$

can equal $\bar\partial u$. For the solvable one, find all solutions.

Exercise 1.48: Cauchy–Pompeiu check

Use (4.1) with $f(z) = \bar z$ on a disc centered at the origin to recover the value $\bar z$ from its boundary and interior terms.

Optional handout: The Bochner–Martinelli representation

This handout is optional. No later argument in the core lectures depends on it.

On $\mathbb{C}^n \setminus \{0\}$, the Bochner–Martinelli kernel is the $(n, n-1)$-form

$$K_{\mathrm{BM}}(\zeta, z) = \frac{(n-1)!}{(2\pi\mathrm{i})^n} \frac{\sum_{j=1}^n (-1)^{j-1} (\bar\zeta_j - \bar z_j)\, \mathrm{d}\bar\zeta_1 \wedge \cdots \widehat{\mathrm{d}\bar\zeta_j} \cdots \wedge \mathrm{d}\bar\zeta_n \wedge \mathrm{d}\zeta_1 \wedge \cdots \wedge \mathrm{d}\zeta_n}{|\zeta - z|^{2n}}.$$

For a smoothly bounded domain D and a C^1 function f near $\overline{D}$, its integral representation has the schematic form

$$f(z) = \int_{\partial D} f(\zeta) K_{\mathrm{BM}}(\zeta, z) - \int_D \bar\partial f(\zeta) \wedge K_{\mathrm{BM}}(\zeta, z),$$

with the displayed convention determining the signs. When $n = 1$, this is precisely Cauchy–Pompeiu. For $n > 1$ the kernel is not holomorphic in z, so it is not a complete replacement for the one-variable Cauchy kernel. Its value is instead that it exposes the same boundary-versus-$\bar{\partial}$-defect structure in every dimension.

Chapter 2

Subharmonic Functions, Pseudoconvexity, and Stein Manifolds

2.1 Lecture 5: Subharmonic functions, mean values, and regularization

Week 3 material • 90 minutes

Notation for this lecture

$\Omega \subset \mathbb{C}$ is a plane domain, $D(a,r)$ is the disc of center a and radius r, and $u : \Omega \to [-\infty, \infty)$ is upper semicontinuous unless more regularity is stated. Its circular mean is $M_u(a,r) = (2\pi)^{-1}\int_0^{2\pi} u(a + re^{i\theta})\,d\theta$. $\Delta = \partial_x^2 + \partial_y^2$ is the Euclidean Laplacian and $dA = dx\,dy$ is planar area measure. With the convention of Lecture 1,

$$dd^c u = \frac{1}{4\pi}(\Delta u)\,dA$$

for $u \in C^2(\Omega)$; the same identity holds distributionally for $u \in L^1_{\mathrm{loc}}(\Omega)$. $C_c^\infty(\Omega)$ is the space of compactly supported smooth test functions. A distribution T is a continuous linear functional on this space, written $\langle T, \chi\rangle$; its derivative is defined by $\langle \partial_j T, \chi\rangle = -\langle T, \partial_j \chi\rangle$. Weak convergence $T_\nu \rightharpoonup T$ means convergence on every test function. The positive distribution $\mu_u = dd^c u$ is the Riesz measure when u is subharmonic. The relation $D \Subset \Omega$ means that $\overline{D}$ is compact in Ω. For such a disc D, $G_D(z,\zeta)$ is the Green kernel normalized by $dd^c_z G_D(z,\zeta) = \delta_\zeta$ and zero boundary values. ρ_ε denotes a nonnegative radial mollifier of total mass one, and $u_\varepsilon = u * \rho_\varepsilon$ wherever the convolution is defined.

Prerequisite results used below

Mean-value theorem for harmonic functions. If h is harmonic on a neighborhood of $\overline{D(a,r)}$, then $h(a) = M_h(a,r)$.

Green representation. If $D \Subset \mathbb{C}$ has a Green kernel G_D, $u \in C^2(\overline{D})$, and H_u is the harmonic function with boundary values $u|_{\partial D}$, then, with the normalization of this text,

$$u(z) = H_u(z) + \int_D G_D(z,\zeta)\,dd^c u(\zeta).$$

Thus the sign of $dd^c u$ controls comparison with harmonic functions.

Why these ideas are needed

Holomorphic functions produce natural potentials such as $\log|f|$, but these have logarithmic poles and are not smooth. We need a class large enough to contain them, stable enough to survive decreasing limits, and rigid enough to retain the maximum principle. Subharmonic functions meet all three demands. Their mean-value, distributional, and regularized descriptions serve different purposes: the first controls point values, the second records positive Laplacian mass, and the third justifies calculations by smooth approximation. Learning to pass correctly among these descriptions is the one-variable model for later pluripotential theory.

Plan for the lecture. 0–15 min: the minimal distribution toolkit; 15–32: the definition and the smooth Laplacian criterion; 32–55: local integrability, Weyl's lemma, and the distributional criterion; 55–72: locality and the maximum principle; 72–90: circular means, Riesz measures, and regularization.

2.1.1 The minimal distribution toolkit

The functions needed in complex analysis are often not twice differentiable: $\log|z|$ is the fundamental example. A distribution records a function through integration against smooth probes. If $u \in L^1_{\mathrm{loc}}(\Omega)$, define

$$\langle T_u, \chi\rangle = \int_\Omega u\chi \, dA, \qquad \chi \in C_c^\infty(\Omega).$$

Its derivative is moved onto the test function by integration by parts:

$$\langle \partial_x T_u, \chi\rangle = -\int_\Omega u\, \partial_x\chi \, dA, \qquad \langle \Delta T_u, \chi\rangle = \int_\Omega u\, \Delta\chi \, dA.$$

The signs are chosen so that classical and distributional derivatives agree whenever u is smooth. A distribution T is positive if $\langle T, \chi\rangle \geq 0$ for every nonnegative test function χ. A basic theorem of distribution theory says that every positive distribution is a positive Radon measure. Thus the inequality $\Delta u \geq 0$ still has a concrete meaning when u is nonsmooth: its Laplacian is a positive measure.

Example 2.1: A jump creates a Dirac mass

For the Heaviside function $H = 1_{[0,\infty)}$ on $\mathbb{R}$,

$$\langle H', \chi\rangle = -\int_0^\infty \chi'(x)\, dx = \chi(0),$$

so $H' = \delta_0$, where $\delta_0(\chi) = \chi(0)$. Distributional differentiation therefore remembers concentration at a singular point instead of discarding it.

Definition 2.2: Convolution, mollifier, and inner domain

Let f, g be functions on $\mathbb{R}^m$ such that the following integral is defined; it is enough, for example, that $f \in L^1_{\mathrm{loc}}$ and g is bounded with compact support. Their *convolution* is

$$(f * g)(x) = \int_{\mathbb{R}^m} f(x-y)g(y)\, dy.$$

A *mollifier* is a function $\rho \in C_c^\infty(\mathbb{R}^m)$ satisfying $\rho \geq 0$ and $\int \rho = 1$. We take $\operatorname{supp}\rho \subset B(0,1)$ and set

$$\rho_\varepsilon(x) = \varepsilon^{-m}\rho(x/\varepsilon), \qquad \Omega_\varepsilon = \{x \in \Omega : \operatorname{dist}(x, \mathbb{R}^m \setminus \Omega) > \varepsilon\}.$$

For $u \in L^1_{\mathrm{loc}}(\Omega)$, its mollification is

$$u_\varepsilon(x) = (u * \rho_\varepsilon)(x) = \int_{B(0,1)} u(x - \varepsilon y)\rho(y)\,\mathrm{d}y, \qquad x \in \Omega_\varepsilon. \tag{5.1}$$

When $m = 2$ we identify $\mathbb{R}^2$ with $\mathbb{C}$ and write $\mathrm{d}y = \mathrm{d}A(y)$.

Lemma 2.3: Basic properties of mollification

For $u \in L^1_{\mathrm{loc}}(\Omega)$, the function u_ε is smooth on Ω_ε, converges to u in L^1_{loc} as $\varepsilon \downarrow 0$, and satisfies

$$\partial^\alpha u_\varepsilon = u * (\partial^\alpha \rho_\varepsilon)$$

for every real multi-index α. More generally, if D is a distribution, then differentiation commutes with mollification:

$$\partial^\alpha (D * \rho_\varepsilon) = D * (\partial^\alpha \rho_\varepsilon) = (\partial^\alpha D) * \rho_\varepsilon. \tag{5.2}$$

Proof. On a compact subset of Ω_ε, formula (5.1) pairs u only with translates of the fixed compactly supported smooth kernel. Difference quotients may therefore be passed to the kernel, which gives smoothness and the first derivative formula; iteration gives all multi-indices. The distributional identity is the same calculation written using test functions. Finally, translations are continuous in L^1 on compact sets. Since

$$u_\varepsilon - u = \int_{B(0,1)} \big(u(\,\cdot - \varepsilon y) - u\big)\rho(y)\,\mathrm{d}y,$$

Minkowski's inequality and dominated convergence give $u_\varepsilon \to u$ in L^1_{loc}. □

For $\Omega \subset \mathbb{C}$, a smooth function is subharmonic when $\Delta u \geq 0$. To admit logarithmic poles such as $\log|z - a|$, the definition must use upper semicontinuity and distributions.

Definition 2.4: Subharmonic function

A function $u : \Omega \to [-\infty, \infty)$ is *subharmonic* if $u \not\equiv -\infty$, u is upper semicontinuous, and whenever $\overline{D(a,r)} \subset \Omega$,

$$u(a) \leq \frac{1}{2\pi} \int_0^{2\pi} u(a + r\mathrm{e}^{\mathrm{i}\theta})\,\mathrm{d}\theta.$$

The integral is understood as an extended Lebesgue integral. Upper semicontinuity bounds the positive part on the circle, so the only possible infinite value is $-\infty$; the inequality rules that out when $u(a) > -\infty$.

Upper semicontinuity means that $\{u < c\}$ is open for every $c \in \mathbb{R}$, or equivalently that $\limsup_{z \to a} u(z) \leq u(a)$. It prevents us from changing an almost-everywhere representative by raising its value at an isolated point.

Proposition 2.5: The C^2 Laplacian criterion

Let $u \in C^2(\Omega)$. Then u is subharmonic if and only if $\Delta u \geq 0$ pointwise on Ω.

Proof. Fix $a \in \Omega$ and define, for $0 < r < \operatorname{dist}(a, \partial\Omega)$,

$$M_u(a,r) = \frac{1}{2\pi} \int_0^{2\pi} u(a + r\mathrm{e}^{\mathrm{i}\theta})\,\mathrm{d}\theta.$$

Differentiation under the integral, followed by the divergence theorem, gives

$$\frac{\partial}{\partial r} M_u(a,r) = \frac{1}{2\pi r} \int_{D(a,r)} \Delta u \, \mathrm{d}A. \tag{5.3}$$

If $\Delta u \geq 0$, then $M_u(a,r)$ is nondecreasing. Since $M_u(a,r) \to u(a)$ as $r \downarrow 0$, the submean inequality follows.

Conversely, suppose that u satisfies the submean inequality and that $\Delta u(a) < 0$ at some point a. Continuity of Δu gives $r_0 > 0$ such that $\overline{D(a,r_0)} \subset \Omega$ and $\Delta u < 0$ on this disc. Formula (5.3) then gives $\partial_r M_u(a,r) < 0$ for $0 < r \leq r_0$. Hence $M_u(a,r) < \lim_{s \downarrow 0} M_u(a,s) = u(a)$, contrary to the submean inequality. Thus $\Delta u \geq 0$ everywhere. □

Proposition 2.6: Local integrability

Every subharmonic function belongs to $L^1_{\mathrm{loc}}(\Omega)$.

Proof. First suppose that $u(b) > -\infty$ and $\overline{D(b,R)} \subset \Omega$. Upper semicontinuity gives an upper bound $u \leq C$ on this closed disc; increasing C if necessary, assume $C \geq 0$. For every $0 < r < R$, the submean inequality implies

$$\frac{1}{2\pi} \int_0^{2\pi} u^-(b + r\mathrm{e}^{\mathrm{i}\theta}) \, \mathrm{d}\theta \leq C - u(b), \qquad u^- := \max\{-u, 0\}.$$

Indeed, the average of u is at least $u(b)$, while $u^+ \leq C$. Integration in polar coordinates shows that $u^- \in L^1(D(b,R))$; the upper bound gives the same conclusion for u^+. Thus u is integrable on every slightly smaller disc centered at a point where it is finite.

Since $u \not\equiv -\infty$, such a point b exists. To propagate the conclusion, let

$$E = \{a \in \Omega : u \text{ is integrable on some neighborhood of } a\}.$$

The preceding argument shows that E is nonempty, and it is open by definition. It is also closed in Ω. Indeed, let a lie in the closure of E and choose $R > 0$ with $\overline{D(a,2R)} \subset \Omega$. Select $b \in E \cap D(a, R/4)$. Because u is integrable near b, it is finite almost everywhere there; hence one can choose a finite point $c \in D(a, R/2)$ arbitrarily close to b. The first part of the proof, applied at c inside $D(c, 3R/2) \subset D(a, 2R)$, shows that u is integrable on $D(a,R)$. Thus $a \in E$. The domain Ω is connected, so $E = \Omega$. □

Lemma 2.7: Weyl's lemma

Let $U \subset \mathbb{R}^m$ be open and let T be a distribution on U. If $\Delta T = 0$ in distributions, then there is a function $H \in C^\infty(U)$ such that $T = T_H$ and $\Delta H = 0$ pointwise.

Proof. Let ρ_ε be a radial mollifier. On U_ε the function $T_\varepsilon = T * \rho_\varepsilon$ is smooth and

$$\Delta T_\varepsilon = (\Delta T) * \rho_\varepsilon = 0.$$

Fix $U' \Subset U$ and choose $\delta > 0$ smaller than the distance from U' to ∂U. For sufficiently small ε, the function T_ε is harmonic on the δ-neighborhood of U'. The mean-value property and radiality of ρ_δ give $T_\varepsilon * \rho_\delta = T_\varepsilon$ on U'. Letting $\varepsilon \downarrow 0$ in distributions yields $T * \rho_\delta = T$ on U'. The left-hand side is a smooth harmonic function. Since U' was arbitrary, these local representatives agree and define the desired function H. □

Lemma 2.8: The planar fundamental solution

For $\zeta \in \mathbb{C}$, the function

$$E_\zeta(z) = \frac{1}{2\pi} \log|z - \zeta|$$

belongs to $L^1_{\mathrm{loc}}(\mathbb{C})$ and satisfies

$$\Delta E_\zeta = \delta_\zeta$$

in the sense of distributions. Here δ_ζ is the Dirac measure at ζ, defined by $\langle \delta_\zeta, \chi \rangle = \chi(\zeta)$ for $\chi \in C^\infty_c(\mathbb{C})$.

Proof. Translation reduces the proof to $\zeta = 0$. The singularity is locally integrable because, for every $R > 0$,

$$\int_{D(0,R)} |\log|z||\, \mathrm{d}A(z) = 2\pi \int_0^R r|\log r|\, \mathrm{d}r < \infty.$$

It remains to compute the distributional Laplacian. Fix $\chi \in C^\infty_c(\mathbb{C})$ and choose $R > 0$ so large that χ vanishes on a neighborhood of $\partial D(0,R)$. For $0 < \varepsilon < R$, set $A_{\varepsilon,R} = D(0,R) \setminus \overline{D(0,\varepsilon)}$. The function $\log|z|$ is harmonic on this annulus. Green's second identity therefore gives

$$\int_{A_{\varepsilon,R}} \log|z|\, \Delta\chi(z)\, \mathrm{d}A(z) = \int_{\partial A_{\varepsilon,R}} (\log|z|\, \partial_n \chi - \chi\, \partial_n \log|z|)\, \mathrm{d}s,$$

where n is the outward unit normal to the annulus and $\mathrm{d}s$ is arc-length measure. The outer boundary contributes zero. On $|z| = \varepsilon$, the outward normal to the annulus is $-\partial_r$, so $\partial_n \log|z| = -1/\varepsilon$ and $\mathrm{d}s = \varepsilon\, \mathrm{d}\theta$. Consequently,

$$\int_{A_{\varepsilon,R}} \log|z|\, \Delta\chi(z)\, \mathrm{d}A(z) = \int_0^{2\pi} \left(\chi(\varepsilon e^{i\theta}) - \varepsilon \log\varepsilon\, \partial_r \chi(\varepsilon e^{i\theta}) \right) \mathrm{d}\theta.$$

The first term on the right tends to $2\pi\chi(0)$. The absolute value of the second term is at most $2\pi\varepsilon|\log\varepsilon|\, \|\nabla\chi\|_{L^\infty}$, hence tends to zero. On the left, local integrability of $\log|z|$ allows $\varepsilon \downarrow 0$. Thus

$$\frac{1}{2\pi} \int_{\mathbb{C}} \log|z|\, \Delta\chi(z)\, \mathrm{d}A(z) = \chi(0),$$

which is precisely $\Delta E_0 = \delta_0$. Translating the identity proves the statement for every ζ. □

Theorem 2.9: Equivalent characterizations of subharmonicity

For $u \in L^1_{\mathrm{loc}}(\Omega)$, the following are equivalent.

(1) The almost-everywhere class of u contains a unique subharmonic representative.
(2) $\Delta u \geq 0$ in distributions: $\int u \Delta\chi \geq 0$ for every nonnegative $\chi \in C^\infty_c(\Omega)$.
(3) For any fixed radial mollifier from Definition 2.2, the convolutions u_ε are smooth and subharmonic on Ω_ε and decrease to the unique subharmonic representative of the class of u as $\varepsilon \downarrow 0$.

Proof. We prove the implications in an order that also constructs the representative.

From the submean inequality to the distribution inequality. Let u be the subharmonic representative and choose a radial mollifier $\rho(y) = \widehat{\rho}(|y|)$. Lemma 2.3 gives smoothness. If $\overline{D(a, r+\varepsilon)} \subset \Omega$,

apply the submean inequality at each point $a - w$, multiply by $\rho_\varepsilon(w)$, and use Fubini's theorem:

$$\begin{aligned} u_\varepsilon(a) &= \int u(a-w)\rho_\varepsilon(w)\,\mathrm{d}A(w) \\ &\le \frac{1}{2\pi}\int_0^{2\pi}\int u(a+re^{i\theta}-w)\rho_\varepsilon(w)\,\mathrm{d}A(w)\,\mathrm{d}\theta \\ &= \frac{1}{2\pi}\int_0^{2\pi} u_\varepsilon(a+re^{i\theta})\,\mathrm{d}\theta. \end{aligned}$$

Thus u_ε is smooth and subharmonic on Ω_ε. Proposition 2.5 therefore gives $\Delta u_\varepsilon \ge 0$. For a nonnegative $\chi \in C_c^\infty(\Omega)$, choose ε small enough that $\operatorname{supp}\chi \subset \Omega_\varepsilon$. Therefore

$$\int_\Omega u\,\Delta\chi\,\mathrm{d}A = \lim_{\varepsilon\downarrow 0}\int_\Omega u_\varepsilon\Delta\chi\,\mathrm{d}A = \lim_{\varepsilon\downarrow 0}\int_\Omega (\Delta u_\varepsilon)\chi\,\mathrm{d}A \ge 0.$$

The first equality uses L^1_{loc} convergence. Hence (1)⇒(2).

From the distribution inequality to a subharmonic representative. Assume (2). The positive distribution $\mu = \Delta u$ is a positive Radon measure. Choose concentric discs $D \Subset D_1 \Subset \Omega$. Since μ is locally finite, $\mu(D_1) < \infty$. Define the logarithmic potential

$$v(z) = \frac{1}{2\pi}\int_{D_1} \log|z-\zeta|\,\mathrm{d}\mu(\zeta).$$

This extended integral is well defined in $[-\infty,\infty)$: for each fixed z, the positive part of $\log|z-\zeta|$ is bounded as ζ ranges over the bounded disc D_1. We prove carefully that v is locally integrable. Fix a compact set $K \subset \mathbb{C}$ and choose $R_K > 1$ such that $|z-\zeta| < R_K$ for all $z \in K$ and $\zeta \in D_1$. Translation and polar coordinates give, uniformly for $\zeta \in D_1$,

$$\begin{aligned} \int_K |\log|z-\zeta||\,\mathrm{d}A(z) &= \int_{K-\zeta} |\log|w||\,\mathrm{d}A(w) \\ &\le 2\pi\int_0^{R_K} r|\log r|\,\mathrm{d}r =: C_K < \infty. \end{aligned}$$

Here $K-\zeta := \{z-\zeta : z \in K\}$ is the translate of K by $-\zeta$. Since the integrand is nonnegative, Tonelli's theorem applies without any prior integrability assumption and yields

$$\int_K\int_{D_1} |\log|z-\zeta||\,\mathrm{d}\mu(\zeta)\,\mathrm{d}A(z) \le C_K\mu(D_1) < \infty.$$

It follows that the inner integral is finite for almost every $z \in K$; at such points,

$$|v(z)| \le \frac{1}{2\pi}\int_{D_1} |\log|z-\zeta||\,\mathrm{d}\mu(\zeta).$$

After integration over K, the preceding estimate gives

$$\|v\|_{L^1(K)} \le \frac{C_K}{2\pi}\mu(D_1) < \infty.$$

Thus $v \in L^1_{\mathrm{loc}}(\mathbb{C})$; in particular, the set where the extended integral equals $-\infty$ has area zero.

We now verify the equation for v. Lemma 2.8 states that, for every $\chi \in C_c^\infty(\mathbb{C})$,

$$\frac{1}{2\pi}\int_{\mathbb{C}} \log|z-\zeta|\,\Delta\chi(z)\,\mathrm{d}A(z) = \chi(\zeta).$$

If $\chi \in C_c^\infty(D_1)$, the same estimate, with $K = \operatorname{supp}\chi$, and the boundedness of $\Delta\chi$ show that the absolute value of the double integrand below is integrable. Fubini's theorem therefore gives

$$\begin{aligned}\langle \Delta v, \chi\rangle &= \frac{1}{2\pi}\int_{D_1}\left(\int_{\mathbb{C}} \log|z-\zeta|\,\Delta\chi(z)\,\mathrm{d}A(z)\right)\mathrm{d}\mu(\zeta)\\ &= \int_{D_1}\chi(\zeta)\,\mathrm{d}\mu(\zeta) = \langle \mu, \chi\rangle.\end{aligned}$$

This proves $\Delta v = \mu$ on D_1.

For completeness, v is subharmonic in the original mean-value sense. Put

$$k_N(z,\zeta) = \max\{\log|z-\zeta|, -N\}, \qquad v_N(z) = \frac{1}{2\pi}\int_{D_1} k_N(z,\zeta)\,\mathrm{d}\mu(\zeta).$$

We first verify the assertion about the kernel that will be integrated. For $b = a - \zeta$, the mean-value property applied to a nonsingular logarithm gives

$$\frac{1}{2\pi}\int_0^{2\pi} \log|a + r\mathrm{e}^{\mathrm{i}\theta} - \zeta|\,\mathrm{d}\theta = \log\max\{r, |a-\zeta|\}.$$

Indeed, if $|b| > r$, the left-hand side is the circular mean of the harmonic function $z \mapsto \log|b+z|$ on $D(0,r)$ and equals $\log|b|$. If $|b| < r$, factor out r and apply the mean-value property to $z \mapsto \log|1 + (b/r)z|$ on the unit disc; the remaining mean is zero. To pass to $|b| = r$, rotate and write $t = |b|/r \uparrow 1$. For $1/2 \le t \le 1$,

$$|\mathrm{e}^{\mathrm{i}\theta} + t|^2 = (1-t)^2 + 4t\cos^2(\theta/2) \ge 2\cos^2(\theta/2).$$

Together with $|\mathrm{e}^{\mathrm{i}\theta}+t| \le 2$, this gives an integrable majorant for $|\log|\mathrm{e}^{\mathrm{i}\theta}+t||$, because $\log|\cos(\theta/2)|$ is integrable. Dominated convergence proves the boundary case. The displayed circle-average identity proves directly that $\log|\,\cdot - \zeta|$ satisfies the submean inequality.

For fixed ζ, the function $k_N(\,\cdot\,,\zeta)$ is the maximum of this subharmonic function and the constant $-N$, and hence also satisfies the submean inequality. Indeed, if f and g satisfy the submean inequality on a circle, then

$$\max\{f(a), g(a)\} \le \max\left\{\frac{1}{2\pi}\int f, \frac{1}{2\pi}\int g\right\} \le \frac{1}{2\pi}\int \max\{f,g\}.$$

We record separately why integration in ζ preserves continuity. Define $g_N(r) = \max\{\log r, -N\}$ for $r > 0$ and $g_N(0) = -N$. Its slope is zero on $[0, \mathrm{e}^{-N}]$ and at most e^N on $[\mathrm{e}^{-N}, \infty)$; hence g_N is e^N-Lipschitz. The reverse triangle inequality gives

$$|k_N(z,\zeta) - k_N(z',\zeta)| \le \mathrm{e}^N|z-z'|.$$

Since $\mu(D_1) < \infty$, integration yields the explicit estimate

$$|v_N(z) - v_N(z')| \le \frac{\mathrm{e}^N\mu(D_1)}{2\pi}|z-z'|.$$

Thus v_N is Lipschitz continuous.

To check the submean inequality, fix $a \in \mathbb{C}$ and $r > 0$. For every $\zeta \in D_1$,

$$k_N(a,\zeta) \le \frac{1}{2\pi}\int_0^{2\pi} k_N(a + r\mathrm{e}^{\mathrm{i}\theta}, \zeta)\,\mathrm{d}\theta.$$

On the compact set $\partial D(a,r)\times\overline{D_1}$, the kernel k_N is bounded above and is bounded below by $-N$. Its absolute value is therefore integrable with respect to $\mathrm{d}\theta\,\mathrm{d}\mu(\zeta)$. Integrating the preceding inequality in ζ and applying Fubini's theorem gives

$$v_N(a) \le \frac{1}{2\pi}\int_0^{2\pi} v_N(a+re^{i\theta})\,\mathrm{d}\theta.$$

This proves directly that v_N is a continuous subharmonic function.

Finally, $k_N \downarrow \log|z-\zeta|$. Applying monotone convergence to the nonnegative functions $k_1 - k_N$ shows that $v_N \downarrow v$, with the value $-\infty$ allowed. A decreasing limit of upper-semicontinuous functions is upper semicontinuous, and the submean inequality passes to the limit by the same argument on the circle, applied to $v_1 - v_N$. Because $v \in L^1_{\mathrm{loc}}$, the limit is not identically $-\infty$. Thus v is subharmonic.

The distribution $h = u - v$ satisfies $\Delta h = 0$ on D_1. Weyl's lemma, Lemma 2.7, gives a smooth harmonic function H representing h. Hence

$$u^\sharp = H + v$$

is a subharmonic representative of u on D.

We now prove the uniqueness that is needed both here and in the statement of the theorem. Suppose that w_1 and w_2 are subharmonic functions on an open set $U \subset \mathbb{C}$ and that $w_1 = w_2$ almost everywhere on U. Fix $a \in U$ and choose $R > 0$ with $\overline{D(a,R)} \subset U$. Let $N = \{z \in D(a,R) : w_1(z) \ne w_2(z)\}$; then N has area zero. Polar coordinates centered at a give

$$0 = \int_{D(a,R)} 1_N\,\mathrm{d}A = \int_0^R r\left(\int_0^{2\pi} 1_N(a+re^{i\theta})\,\mathrm{d}\theta\right)\mathrm{d}r.$$

The integrand is nonnegative. Hence, for almost every $r \in (0,R)$, the two functions agree for almost every point of the circle $\partial D(a,r)$. Each w_k is locally integrable, so a second application of polar-coordinate Fubini shows that $\theta \mapsto w_k(a+re^{i\theta})$ is integrable for almost every r, for $k = 1,2$. After intersecting these full-measure sets of radii, the circular means are equal. We may therefore choose radii $r_j \downarrow 0$ such that

$$M_{w_1}(a,r_j) = M_{w_2}(a,r_j) \qquad \text{for every } j.$$

Every subharmonic function w satisfies

$$w(a) = \lim_{r\downarrow 0} M_w(a,r)\ :$$

the submean inequality gives the lower bound, while upper semicontinuity gives the upper bound. Applying this identity to the chosen radii gives $w_1(a) = w_2(a)$. Since a was arbitrary, $w_1 = w_2$ throughout U. Thus a subharmonic representative of an almost-everywhere class is unique. In particular, the local representatives constructed above agree on overlaps and therefore glue, proving (2)⇒(1), including the uniqueness asserted in (1).

Monotonicity of radial regularization. We give all steps because radiality is used twice, for two different purposes. First fix $a \in \Omega$ and choose $R > 0$ such that $\overline{D(a,R)} \subset \Omega$. Choose a disc D_1 such that $\overline{D(a,R)} \subset D_1 \Subset \Omega$. The local decomposition constructed above writes $u^\sharp = H + v$ on $D(a,R)$, where H is harmonic and

$$v(z) = \frac{1}{2\pi}\int_{D_1} \log|z-\zeta|\,\mathrm{d}\mu(\zeta).$$

For $0 < r < R$, the mean-value property for H and the circle-average identity for the logarithm give

$$M_{u^\sharp}(a,r) = H(a) + \frac{1}{2\pi}\int_{D_1} \log\max\{r,|a-\zeta|\}\,\mathrm{d}\mu(\zeta).$$

Here Fubini's theorem is legitimate. Indeed, the positive parts of $\log|a + re^{i\theta} - \zeta|$ are bounded uniformly for $\zeta \in D_1$, and the circle-average identity, together with $\log\max\{r, |a-\zeta|\} \geq \log r$, gives a uniform bound for the integrals of the negative parts. Since $\mu(D_1) < \infty$, the absolute values are integrable with respect to $\mathrm{d}\theta\,\mathrm{d}\mu(\zeta)$. The displayed formula shows that $r \mapsto M_{u^\sharp}(a,r)$ is nondecreasing, because μ is positive.

We now use radiality of the mollifier. Since u and $u^\sharp$ agree almost everywhere, their convolutions with ρ_ε are equal. Write $\rho(y) = \widehat{\rho}(|y|)$ and define

$$w(s) = 2\pi s\widehat{\rho}(s), \qquad 0 \leq s \leq 1.$$

The conditions $\rho \geq 0$ and $\int_{\mathbb{C}} \rho\,\mathrm{d}A = 1$ say precisely that $w \geq 0$ and $\int_0^1 w(s)\,\mathrm{d}s = 1$. Polar coordinates, followed by the change $\theta \mapsto \theta + \pi$, yield

$$u_\varepsilon(a) = \int_0^1 M_{u^\sharp}(a, \varepsilon s) w(s)\,\mathrm{d}s. \tag{5.4}$$

If $0 < \varepsilon_1 < \varepsilon_2$ and $\overline{D(a,\varepsilon_2)} \subset \Omega$, the monotonicity of circular means gives $u_{\varepsilon_1}(a) \leq u_{\varepsilon_2}(a)$. Thus $u_\varepsilon(a)$ decreases as $\varepsilon \downarrow 0$.

It remains to identify the limit. The submean inequality gives $u^\sharp(a) \leq M_{u^\sharp}(a, \varepsilon s)$ for $s > 0$, so (5.4) gives $u^\sharp(a) \leq u_\varepsilon(a)$. If $u^\sharp(a)$ is finite, upper semicontinuity says that for every $\eta > 0$ there is $\delta > 0$ such that

$$u^\sharp(z) \leq u^\sharp(a) + \eta \quad \text{whenever } |z - a| < \delta.$$

For $0 < \varepsilon < \delta$, the convolution samples only this disc, and hence $u_\varepsilon(a) \leq u^\sharp(a) + \eta$. If $u^\sharp(a) = -\infty$, the same argument is applied with an arbitrary real upper bound A in place of $u^\sharp(a) + \eta$; it gives $u_\varepsilon(a) \leq A$ for all sufficiently small ε. Therefore $u_\varepsilon(a) \downarrow u^\sharp(a)$ at every point a. By the uniqueness just proved, $u^\sharp$ is the unique subharmonic representative named in (1); this proves (3) from either (1) or (2).

Finally, (3)⇒(2) follows from the integration-by-parts calculation above, because $u_\varepsilon \to u$ in L^1_{loc} and $\Delta u_\varepsilon \geq 0$. This completes the equivalence. □

Proposition 2.10: Subharmonicity is a local property

Let $\Omega \subset \mathbb{C}$ be a domain and let $u : \Omega \to [-\infty, \infty)$. Then u is subharmonic on Ω if and only if every point $a \in \Omega$ has an open neighbourhood $U_a \subset \Omega$ such that $u|_{U_a}$ is subharmonic on U_a.

Proof. If u is subharmonic on Ω, its restriction to any open subset is subharmonic, so only the converse needs proof. Assume that the neighbourhoods $\{U_a\}_{a\in\Omega}$ have the stated property. Upper semicontinuity is local: for every $c \in \mathbb{R}$,

$$\{u < c\} = \bigcup_{a\in\Omega} \big(U_a \cap \{u < c\}\big)$$

is open in Ω. Proposition 2.6, applied on the neighborhoods U_a, gives $u \in L^1_{\mathrm{loc}}(\Omega)$.

It remains to check the distribution inequality globally. Let $0 \leq \chi \in C_c^\infty(\Omega)$ and put $K = \operatorname{supp}\chi$. Choose finitely many members $U_1, \ldots, U_N$ of the above cover and nonnegative functions $\eta_j \in C_c^\infty(U_j)$, $1 \leq j \leq N$, such that

$$0 \leq \eta_j \leq 1, \qquad \sum_{j=1}^N \eta_j = 1 \quad \text{near } K.$$

Such functions exist by a smooth partition of unity subordinate to the finite cover of the compact set K. Each $\eta_j\chi$ is a nonnegative test function on U_j. Since $u|_{U_j}$ is subharmonic, Theorem 2.9 gives

$$\int_\Omega u\,\Delta\chi\,\mathrm{d}A = \sum_{j=1}^N \int_{U_j} u\,\Delta(\eta_j\chi)\,\mathrm{d}A \geq 0.$$

Thus $\Delta u \geq 0$ on all of Ω. The same theorem supplies a global subharmonic representative $u^\sharp$ of the L^1_{loc} class of u. On every U_a, the two subharmonic functions $u^\sharp|_{U_a}$ and $u|_{U_a}$ agree almost everywhere; the uniqueness argument in the theorem therefore gives $u^\sharp = u$ on U_a. Since these neighbourhoods cover Ω, we have $u = u^\sharp$ everywhere, so the original function u is subharmonic. □

Proposition 2.11: Maximum principle

If a subharmonic function u on a connected domain Ω attains its finite global supremum at a point of Ω, then it is constant. If Ω is bounded and u extends upper semicontinuously to $\overline{\Omega}$, then $\sup_\Omega u \leq \sup_{\partial\Omega} u$.

Proof. Put $M = \sup_\Omega u$ and suppose $u(a) = M$. Choose $R > 0$ with $\overline{D(a,R)} \subset \Omega$. For every $0 < r < R$, the submean inequality and the pointwise bound $u \leq M$ give

$$M = u(a) \leq \frac{1}{2\pi}\int_0^{2\pi} u(a + re^{i\theta})\,d\theta \leq M.$$

Hence equality holds throughout. Define the nonnegative measurable function $g = M - u$, allowing the value $+\infty$ where $u = -\infty$. The preceding equality is equivalent to

$$\int_0^{2\pi} g(a + re^{i\theta})\,d\theta = 0 \qquad (0 < r < R).$$

Tonelli's theorem now gives

$$\int_{D(a,R)} g(z)\,dA(z) = \int_0^R \int_0^{2\pi} g(a + re^{i\theta}) r\,d\theta\,dr = 0.$$

Since $g \geq 0$, it follows that $g = 0$ almost everywhere on $D(a,R)$, or equivalently that $u = M$ almost everywhere there.

This full-measure set is dense in $D(a,R)$: every nonempty open disc has positive area and therefore meets it. For any $x \in D(a,R)$, choose points $x_j \to x$ with $u(x_j) = M$. Upper semicontinuity gives

$$u(x) \geq \limsup_{j\to\infty} u(x_j) = M.$$

The reverse inequality follows from the definition of M, so $u = M$ throughout $D(a,R)$. We have proved that the nonempty set $\{u = M\}$ is open. It is closed because its complement $\{u < M\}$ is open by upper semicontinuity. Connectedness of Ω now gives $u \equiv M$.

For the boundary statement, upper semicontinuity on the compact set $\overline{\Omega}$ makes u attain its maximum there. If that maximum were larger than $\sup_{\partial\Omega} u$, it would be attained at an interior point; the first part would make $u \equiv M$ on Ω. For every $\xi \in \partial\Omega$, choose $z_j \in \Omega$ with $z_j \to \xi$. Upper semicontinuity on $\overline{\Omega}$ would then give $u(\xi) \geq \limsup_j u(z_j) = M$, contradicting $M > \sup_{\partial\Omega} u$. □

Proposition 2.12: Circular means and logarithmic slope

Set

$$M_u(a,r) = \frac{1}{2\pi}\int_0^{2\pi} u(a + re^{i\theta})d\theta.$$

Then $M_u(a,r)$ increases with r, is convex as a function of $\log r$, and tends to $u(a)$ as $r \downarrow 0$.

Proof. For smooth u, Green's formula gives

$$\frac{d}{dr}M_u(a,r) = \frac{1}{2\pi r}\int_{D(a,r)} \Delta u\,dA \geq 0.$$

Hence the mean is nondecreasing. With $t = \log r$,

$$\frac{\mathrm{d}}{\mathrm{d}t} M_u(a, \mathrm{e}^t) = \frac{1}{2\pi} \int_{D(a,\mathrm{e}^t)} \Delta u \, \mathrm{d}A.$$

The right-hand side is nondecreasing in t, so the function of t is convex.

For a general subharmonic function, take the decreasing smooth radial mollifications from Theorem 2.9. On every compact interval of admissible radii their circular means decrease to $M_u(a, r)$. Monotonicity and the convexity inequality pass to this pointwise limit. Finally, the submean inequality gives $u(a) \le M_u(a, r)$. Given $\delta > 0$, upper semicontinuity gives $u(z) \le u(a) + \delta$ on a sufficiently small disc about a, so $M_u(a, r) \le u(a) + \delta$ for small r. Therefore $M_u(a, r) \to u(a)$ as $r \downarrow 0$. □

With the normalization fixed in Lecture 1, the Riesz measure is

$$\mu_u = \mathrm{dd}^{\mathrm{c}} u = \frac{1}{4\pi} \Delta u$$

after identifying a scalar distribution with its associated multiple of the area form. Thus $\Delta u \ge 0$ and $\mu_u \ge 0$ are the same positivity condition, but the factor 4π must be retained when the two notations are compared. For $u = \log |f|^2$, Poincaré–Lelong gives $\mu_u = \sum_a \operatorname{ord}_a(f) \delta_a$.

Example 2.13: The model singularity

Let $u(z) = \alpha \log |z - a|^2 + h(z)$, where $\alpha \ge 0$ and h is harmonic. Then u is subharmonic and

$$M_u(a, r) = 2\alpha \log r + h(a).$$

Hence the slope of the circular mean as a function of $\log r$ is 2α. This calculation anticipates the Lelong number in Lecture 24.

Warning

A distribution inequality specifies only an almost-everywhere class. To obtain a subharmonic *function*, one must select its unique subharmonic representative. Raising the value at one point does not change Δu but destroys the mean inequality.

2.1.2 Riesz decomposition and the Poisson correction

The distribution $\mu_u = \mathrm{dd}^{\mathrm{c}} u$ separates a subharmonic function into a singular potential and a harmonic remainder. Let $D \Subset \Omega$ be a disc and let $G_D(z, \zeta)$ be its Green kernel, normalized by

$$\mathrm{dd}^{\mathrm{c}}_z G_D(z, \zeta) = \delta_\zeta, \qquad G_D(\,\cdot\,, \zeta)|_{\partial D} = 0.$$

Then, after restricting to a slightly smaller disc if necessary,

$$u(z) = h(z) + \int_D G_D(z, \zeta) \, \mathrm{d}\mu_u(\zeta),$$

where h is harmonic. Indeed, the difference of the two sides has vanishing distributional Laplacian and is therefore represented by a smooth harmonic function by Weyl's lemma (Lemma 2.7). This is the local Riesz decomposition.

For the disc $D(0, R)$ one may take

$$G_D(z, \zeta) = \log \left| \frac{R(z - \zeta)}{R^2 - \bar{\zeta} z} \right|^2.$$

The denominator is nonzero on the disc, so its logarithm is harmonic in z; on $|z| = R$ the quotient has modulus one. A point mass in μ_u therefore produces a logarithmic pole, while a diffuse measure produces a distributed potential.

The same information appears in the circular means. For smooth u and $0 < r < R$, Green's formula and our normalization give

$$\frac{\mathrm{d}}{\mathrm{d}r} M_u(0,r) = \frac{2}{r}\mu_u(D(0,r)).$$

Approximation extends this identity in the integrated sense to every subharmonic function:

$$M_u(0,R) - M_u(0,r) = 2\int_r^R \frac{\mu_u(D(0,t))}{t}\,\mathrm{d}t.$$

This formula simultaneously proves monotonicity and explains convexity in $\log r$.

2.1.3 What regularization preserves

Choose a radial mollifier $\rho \geq 0$ supported in the unit disc and set $u_\varepsilon = u * \rho_\varepsilon$ on $\Omega_\varepsilon = \{z : \operatorname{dist}(z, \partial\Omega) > \varepsilon\}$. Then

$$\mathrm{dd}^{\mathrm{c}} u_\varepsilon = (\mathrm{dd}^{\mathrm{c}} u) * \rho_\varepsilon = \mu_u * \rho_\varepsilon \geq 0.$$

For any fixed nonnegative radial mollifier, polar coordinates express $u_\varepsilon(z)$ as a positive average of circular means centered at z. Because those means increase with their radius, $u_\varepsilon(z)$ increases with ε and therefore decreases to $u(z)$ as $\varepsilon \downarrow 0$. A nonradial approximate identity still converges in L^1_{loc}, but need not be monotone. The distinction is important in nonlinear pluripotential theory.

Here is a useful form of Hartogs' lemma. If u_j and u are subharmonic and $u_j \to u$ in $L^1_{\mathrm{loc}}(\Omega)$, then, for $K \Subset U \Subset \Omega$,

$$\limsup_{j\to\infty} \sup_K u_j \leq \sup_U u.$$

To see the estimate, choose a radius $r > 0$ such that $D(z,r) \subset U$ for every $z \in K$. The area submean inequality gives

$$u_j(z) \leq \frac{1}{\pi r^2}\int_{D(z,r)} u_j\,\mathrm{d}A.$$

This area version follows by integrating the circular submean inequality over $0 < s < r$ with weight $2s/r^2$. The error made by replacing u_j with u is bounded uniformly in $z \in K$ by $(\pi r^2)^{-1}\|u_j - u\|_{L^1(U)}$, which tends to zero, while the remaining average is at most $\sup_U u$. Thus L^1_{loc} convergence does not permit isolated upward spikes in the limit.

Remark (A working dictionary)

Boundary values determine the harmonic part through the Poisson kernel; the positive measure μ_u determines the failure of harmonicity; logarithmic atoms record zeros with multiplicity. Jensen's formula, the argument principle, and the one-dimensional Poincaré–Lelong formula are three expressions of this same decomposition.

2.1.4 Three calculations worth doing by hand

Jensen's formula and the normalization constant

Let u be smooth near $\overline{D(0,R)}$ and write $M(r) = M_u(0,r)$. Integrating the circular-mean identity (5.3) gives

$$\begin{aligned} M(R) - u(0) &= \int_0^R \frac{1}{2\pi r} \int_{|\zeta|<r} \Delta u(\zeta)\, \mathrm{d}A(\zeta)\, \mathrm{d}r \\ &= \frac{1}{2\pi} \int_{|\zeta|<R} \left(\int_{|\zeta|}^R \frac{\mathrm{d}r}{r} \right) \Delta u(\zeta)\, \mathrm{d}A(\zeta) \\ &= \frac{1}{2\pi} \int_{|\zeta|<R} \log \frac{R}{|\zeta|}\, \Delta u(\zeta)\, \mathrm{d}A(\zeta). \end{aligned}$$

Since $\mu_u = \mathrm{d}\mathrm{d}^{\mathrm{c}} u = (4\pi)^{-1} \Delta u\, \mathrm{d}A$, this is

$$M(R) - u(0) = 2 \int_{D(0,R)} \log \frac{R}{|\zeta|}\, \mathrm{d}\mu_u(\zeta). \tag{5.5}$$

The factor 2 is forced by our choice $\mathrm{d}\mathrm{d}^{\mathrm{c}} = (\mathrm{i}/2\pi)\partial\bar\partial$. Formula (5.5) extends to subharmonic functions by applying it to the smooth regularizations on a slightly smaller disc and passing to the limit. If $u(0) = -\infty$, both sides are interpreted in the extended sense.

For example, $\mathrm{d}\mathrm{d}^{\mathrm{c}} \log|z-a| = \frac{1}{2}\delta_a$. Hence, when $|a| < R$, (5.5) gives

$$\frac{1}{2\pi} \int_0^{2\pi} \log|R\mathrm{e}^{\mathrm{i}\theta} - a|\, \mathrm{d}\theta = \log R.$$

If $|a| > R$, the function is harmonic in $D(0,R)$ and the same average equals $\log|a|$. Together these give

$$\frac{1}{2\pi} \int_0^{2\pi} \log|R\mathrm{e}^{\mathrm{i}\theta} - a|\, \mathrm{d}\theta = \log \max\{R, |a|\}.$$

A corner in the graph becomes a circle measure

Consider the continuous subharmonic function

$$u(z) = \max\{\log|z|^2, -2\}.$$

It equals -2 on $|z| \le \mathrm{e}^{-1}$ and $2\log|z|$ on $|z| \ge \mathrm{e}^{-1}$. Both pieces are harmonic away from the joining circle, so the distributional Laplacian is supported on that circle. The inner radial derivative is 0 and the outer radial derivative is $2/r$. Applying Green's identity on the two sides of the circle, the terms containing the boundary value of u cancel and one obtains, for every $\chi \in C_c^\infty(\mathbb{C})$,

$$\langle \Delta u, \chi \rangle = \int_{|z|=\mathrm{e}^{-1}} \frac{2}{\mathrm{e}^{-1}} \chi(z)\, \mathrm{d}s(z) = 2 \int_0^{2\pi} \chi(\mathrm{e}^{-1}\mathrm{e}^{\mathrm{i}\theta})\, \mathrm{d}\theta.$$

Consequently

$$\mu_u = \mathrm{d}\mathrm{d}^{\mathrm{c}} u = \frac{1}{2\pi} (\theta \mapsto \mathrm{e}^{-1}\mathrm{e}^{\mathrm{i}\theta})_* \mathrm{d}\theta.$$

Thus μ_u is the uniform probability measure on $|z| = \mathrm{e}^{-1}$. There is no mass at the origin: truncation made the function constant near the origin and moved the entire logarithmic mass to the joining circle.

What regularization does to the Riesz measure

For a subharmonic function u and a nonnegative mollifier, differentiation commutes with convolution, so

$$\mu_{u_\varepsilon} = \mathrm{dd}^{\mathrm{c}}(u * \rho_\varepsilon) = (\mathrm{dd}^{\mathrm{c}} u) * \rho_\varepsilon = \mu_u * \rho_\varepsilon.$$

The right-hand side is a smooth nonnegative density. In the preceding example it spreads the circle measure across an annulus of thickness comparable to ε. Its total mass is unchanged as long as the whole annulus remains inside the region under consideration. This is the measure-theoretic reason that regularization preserves subharmonicity.

The domain must shrink from Ω to Ω_ε: the value $u_\varepsilon(z)$ samples u on the entire disc $D(z, \varepsilon)$. Radiality has a separate role. It writes the regularization as a weighted average of circular means, which gives $u_\varepsilon \downarrow u$ as $\varepsilon \downarrow 0$. Positivity alone preserves subharmonicity, whereas radiality supplies this monotone pointwise approximation.

2.1.5 Self-study workshop: moving among the definitions of subharmonicity

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall Green's formula, the mean-value property of harmonic functions, convolution with a radial mollifier, and the distributional derivatives introduced at the start of this lecture. The central task is to know where smoothness is used and how Weyl's lemma recovers a classical function from a distribution.

A dependency map. Proposition 2.5 settles the smooth case directly: formula (5.3) turns the sign of Δu into monotonicity of circular means, and the converse follows by shrinking to a disc on which $\Delta u < 0$. No distribution theory is needed at this stage.

For a general upper-semicontinuous function, first prove local integrability. This allows Δu to be defined by

$$\langle \Delta u, \chi \rangle = \int_\Omega u \, \Delta\chi \, \mathrm{d}A.$$

Regularization proves that the mean inequality implies $\Delta u \geq 0$. In the opposite direction, the logarithmic potential produces a function with precisely this Laplacian. Subtracting it leaves a distribution with zero Laplacian, and Weyl's lemma says that this remainder is a smooth harmonic function. Finally, the limit of circular means chooses the unique upper-semicontinuous representative.

For $u(z) = |z|^2$, one obtains $M_u(a, r) = |a|^2 + r^2$, so the mean inequality is strict for $r > 0$. For $u(z) = \log|z|$, the function is harmonic off the origin but has a positive point mass in its distributional Laplacian. Jensen's formula records exactly the jump caused by that mass.

For a nonsmooth upper-semicontinuous u, the pointwise symbol Δu may not exist. Radial convolution on a slightly smaller domain gives smooth subharmonic functions. The shrinking of the domain is essential: $u * \rho_\varepsilon$ at z samples values in $D(z, \varepsilon)$.

Checkpoint (check the two analytic bridges)

(1) Let μ be a finite positive measure and $v(z) = (2\pi)^{-1} \int \log|z - \zeta| \, \mathrm{d}\mu(\zeta)$. Write the test-function calculation proving $\Delta v = \mu$.
(2) If $h = u - v$ satisfies $\Delta h = 0$ only in the sense of distributions, what exactly does Weyl's lemma add?
(3) Why do two subharmonic representatives that agree almost everywhere agree at every point?

Solution (checkpoint). (1) For $\chi \in C_c^\infty$, Fubini's theorem and the fundamental-solution identity give

$$\begin{aligned}\langle \Delta v, \chi \rangle &= \frac{1}{2\pi} \int \left(\int \log|z-\zeta| \, \Delta\chi(z) \, \mathrm{d}A(z) \right) \mathrm{d}\mu(\zeta) \\ &= \int \chi(\zeta) \, \mathrm{d}\mu(\zeta).\end{aligned}$$

(2) Weyl's lemma produces $H \in C^\infty$ with $h = T_H$ as distributions and $\Delta H = 0$ pointwise. Thus the remainder is not merely a formal distributional class: it has a genuine harmonic representative. (3) If $u_1 = u_2$ almost everywhere, Fubini gives radii $r_j \downarrow 0$ for which their circular means agree. Since $u_k(a) = \lim_{j\to\infty} M_{u_k}(a, r_j)$ for a subharmonic function, $u_1(a) = u_2(a)$ for every a. ◇

Exit criterion

You can prove the C^2 criterion from (5.3), verify a distributional Laplacian against test functions, state Weyl's lemma, and explain how the canonical upper-semicontinuous representative is recovered.

Summary. Mean values, a positive distributional Laplacian, and decreasing smooth approximation are equivalent languages. Upper semicontinuity controls point values, distributions control limits, and convolution makes calculations possible.

Exercise 2.14: Convexity of circular means

Use the three-circles theorem, or lift $u(\mathrm{e}^w)$ to a half-plane, to prove that $M_u(0, r)$ is convex in $\log r$.

Exercise 2.15: Decreasing limits

Prove that a decreasing limit of subharmonic functions is either subharmonic or identically $-\infty$ on each connected component.

Optional handout: Riesz decomposition and Green potentials

This handout is optional. No later argument in the core lectures depends on it.

Let $D \Subset \mathbb{C}$ be a disc and u subharmonic. Its Riesz measure is $\mu_u = \mathrm{dd}^{\mathrm{c}} u$. Choose the Green function $G_D(z, \zeta)$, normalized to vanish on ∂D and to have logarithmic singularity at $z = \zeta$. Then, on a smaller disc, one can write

$$u(z) = h(z) + \int_D G_D(z, \zeta) \, \mathrm{d}\mu_u(\zeta),$$

where h is harmonic. This is the Riesz decomposition: a subharmonic function is a harmonic function plus the potential of its positive Laplacian.

To justify the formula for a general u, let $v(z) = \int_D G_D(z, \zeta) \, \mathrm{d}\mu_u(\zeta)$. The defining identity for the Green kernel and the same test-function calculation used for the logarithmic potential give $\mathrm{dd}^{\mathrm{c}} v = \mu_u$ on compactly contained subdiscs. Hence $\mathrm{dd}^{\mathrm{c}}(u - v) = 0$. Weyl's lemma (Lemma 2.7) then supplies the smooth harmonic representative h of $u - v$.

Example 2.16: Zeros of a holomorphic function

For $u = \log|f|^2$, the measure μ_u is the divisor of f. Evaluating the Riesz formula at the center of a disc yields Jensen's formula: the boundary mean of $\log|f|$ differs from $\log|f(0)|$ by the logarithmic contribution of the zeros.

Remark (Higher dimensions)

For psh functions, $\mathrm{dd}^{\mathrm{c}}u$ is matrix-valued rather than a scalar measure. Local psh potentials of closed positive $(1,1)$-currents are the several-variable successor of Riesz decomposition, while Lelong numbers extract the coefficient of the logarithmic part.

Exercise 2.17: Green function of the unit disc

Verify that $G_{\mathbb{D}}(z,\zeta) = \log\left|\frac{z-\zeta}{1-\bar{\zeta}z}\right|^2$ vanishes on the unit circle and compute its dd^{c} in the z variable.

2.2 Lecture 6: Plurisubharmonic functions and the Levi form

Week 3 material • 90 minutes

Notation for this lecture

X is a complex manifold, $\Omega \subset \mathbb{C}^n$ is a domain, and $\mathrm{PSH}(\Omega)$ is the set of plurisubharmonic functions on it. For $u \in C^2(\Omega)$, the Levi matrix is $H_u = (u_{j\bar{k}})$ with $u_{j\bar{k}} = \partial^2 u/(\partial z_j \partial \bar{z}_k)$, and its Levi form on $\xi \in \mathbb{C}^n$ is $\mathcal{L}_u(\xi) = \sum u_{j\bar{k}} \xi_j \bar{\xi}_k$. Thus u is psh exactly when $H_u \geq 0$, and strictly psh when $H_u > 0$. For a fixed real closed $(1,1)$-form θ, the notation $u \in \mathrm{PSH}(X, \theta)$ means that u is quasi-psh and $\theta + \mathrm{dd}^c u \geq 0$ as a current. A set contained locally in $\{v = -\infty\}$ for some psh $v \not\equiv -\infty$ is called pluripolar.

Prerequisite results used below

Line-restriction chain rule. For $u \in C^2(\Omega)$ and a complex affine line $\ell(\zeta) = p + \zeta\xi$,

$$\frac{\partial^2 (u \circ \ell)}{\partial \zeta \, \partial \bar{\zeta}}(0) = \sum_{j,k} u_{j\bar{k}}(p) \xi_j \bar{\xi}_k .$$

Hence subharmonicity on every complex line is equivalent, in the C^2 case, to positive semidefiniteness of the Levi matrix.

Decreasing-limit theorem for subharmonic functions. A decreasing limit of subharmonic functions on a plane domain is either subharmonic or identically $-\infty$. Applied after restriction to every complex line, the same statement gives the corresponding stability of psh functions.

Why these ideas are needed

In several complex variables, positivity of the full real Laplacian forgets which directions are complex. The replacement is to demand subharmonicity on every complex line. For smooth functions this condition is encoded by the Levi matrix $(u_{j\bar{k}})$, so it is invariant under holomorphic coordinate changes and can be compared directly with positive $(1,1)$-forms. Plurisubharmonic functions are the natural local potentials of positive currents, the admissible weights in L^2 estimates, and the exhaustion functions that characterize domains of holomorphy. The definition is therefore chosen to support all three later roles at once.

Plan for the lecture. 0–20 min: restriction to complex lines; 20–45: Levi matrix and positive currents; 45–65: stability and suprema; 65–78: qpsh and θ-psh functions; 78–90: pluripolar sets.

Definition 2.18: Positive $(1,1)$-current

A real $(1,1)$-current T is *positive* if it takes nonnegative values on every compactly supported smooth strongly positive $(n-1, n-1)$-form. In local coordinates one may write

$$T = \frac{\mathrm{i}}{2\pi} \sum_{j,k} T_{j\bar{k}} \, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k .$$

Positivity means equivalently that the matrix of distribution coefficients is positive: for every compactly supported smooth vector field $\xi = (\xi_1, \ldots, \xi_n)$, the distribution $\sum_{j,k} \xi_j \bar{\xi}_k T_{j\bar{k}}$ is positive, hence is a positive measure. When the coefficients are smooth, this reduces to positive semidefiniteness of the Hermitian matrix $(T_{j\bar{k}})$ at every point.

Definition 2.19: Plurisubharmonic function

An upper-semicontinuous function $u : \Omega \subset \mathbb{C}^n \to [-\infty, \infty)$ is *plurisubharmonic* (psh) if $u \not\equiv -\infty$ and, for every complex affine line L, its restriction to each component of $\Omega \cap L$ is subharmonic or identically $-\infty$. Write $u \in \mathrm{PSH}(\Omega)$.

Theorem 2.20: Three tests for plurisubharmonicity

Let $\Omega \subset \mathbb{C}^n$ be a domain.

(1) If $u \in C^2(\Omega)$, then u is psh exactly when its Levi matrix $H_u = (u_{j\bar{k}})$ is positive semidefinite.

(2) If $v \in L^1_{\mathrm{loc}}(\Omega)$, then $\mathrm{dd}^{\mathrm{c}} v$ is a positive $(1,1)$-current exactly when the almost-everywhere class of v contains a unique plurisubharmonic representative.

(3) Every psh function is locally a pointwise decreasing limit of smooth psh functions. Conversely, a decreasing limit of smooth psh functions is psh unless it is identically $-\infty$.

Warning

The phrase "the class of v contains a psh representative" in part (2) cannot be replaced by "an upper-semicontinuous representative is psh." On the unit ball, set $v(0) = 1$ and $v(z) = 0$ for $z \neq 0$. Then v is upper semicontinuous, $v = 0$ almost everywhere, and therefore $\mathrm{dd}^{\mathrm{c}} v = 0$ as a distribution. Nevertheless, the submean inequality fails at the origin on every complex line through 0. The positive-current condition determines an almost-everywhere class; the theorem selects its unique psh representative, which in this example is the zero function.

Proof. The smooth line test. For $\xi \in \mathbb{C}^n$, put $v(\tau) = u(z + \tau\xi)$. The chain rule gives

$$\Delta_\tau v(0) = 4 \sum_{j,k} u_{j\bar{k}}(z)\xi_j \bar{\xi}_k .$$

Thus every line restriction is subharmonic exactly when the Levi matrix is semipositive, proving (1).

From line subharmonicity to a positive current. Suppose that u is psh. We first prove in detail that u is subharmonic as a function on the real space $\mathbb{R}^{2n}$.

Let $S^{2n-1} \subset \mathbb{C}^n$ be the unit sphere and let σ be its normalized surface measure. Fix $x \in \Omega$ and $r > 0$ such that $\overline{B(x,r)} \subset \Omega$. We first record the elementary geometry that relates complex lines to this sphere. For a unit vector $\xi \in S^{2n-1}$, let $L_\xi = x + \mathbb{C}\xi$. A point of L_ξ has the form $x + \tau\xi$, and

$$|(x + \tau\xi) - x| = |\tau|\,|\xi| = |\tau|.$$

It follows that

$$L_\xi \cap \partial B(x,r) = \{x + r\mathrm{e}^{\mathrm{i}\theta}\xi : 0 \le \theta < 2\pi\},$$

which is an ordinary Euclidean circle. Two unit vectors ξ and η determine the same complex line through x exactly when $\eta = \mathrm{e}^{\mathrm{i}\alpha}\xi$ for some α; in that case the two displayed parametrizations differ only by the change of variable $\theta \mapsto \theta + \alpha$. Conversely, if $y \in \partial B(x,r)$, then $\xi = (y - x)/r$ is a unit vector and y belongs to the circle associated with L_ξ. Thus these circles cover the sphere, and two of them are either equal or disjoint. The quotient language sometimes used for this fact merely identifies ξ with all of its phase rotations $\mathrm{e}^{\mathrm{i}\alpha}\xi$; we will not need to form that quotient.

For every unit vector $\xi \in S^{2n-1}$, the function

$$\tau \longmapsto u(x + \tau\xi)$$

is subharmonic on the relevant plane disc or is identically $-\infty$. Hence

$$u(x) \le \frac{1}{2\pi} \int_0^{2\pi} u(x + r\mathrm{e}^{\mathrm{i}\theta}\xi)\, \mathrm{d}\theta. \tag{6.1}$$

If $u(x) > -\infty$, the restriction cannot be identically $-\infty$, and (6.1) is the ordinary one-variable submean inequality. If $u(x) = -\infty$, the desired spherical submean inequality is automatic: upper semicontinuity bounds the positive part of u on $\partial B(x, r)$, so its spherical average is a well-defined extended integral in $[-\infty, \infty)$.

Assume, therefore, that $u(x) > -\infty$. Upper semicontinuity gives a finite constant C such that $u \le C$ on the compact sphere $\partial B(x, r)$. Define the nonnegative function

$$q(\zeta) = C - u(x + r\zeta), \qquad \zeta \in S^{2n-1}.$$

Rewriting (6.1) gives, for every unit vector ξ,

$$\frac{1}{2\pi} \int_0^{2\pi} q(\mathrm{e}^{\mathrm{i}\theta}\xi)\, \mathrm{d}\theta \le C - u(x).$$

Integrate this inequality with respect to ξ. Tonelli's theorem applies because $q \ge 0$. Moreover, for each fixed θ, the map $\xi \mapsto \mathrm{e}^{\mathrm{i}\theta}\xi$ is unitary and hence preserves the normalized surface measure. Consequently,

$$\begin{aligned} \int_{S^{2n-1}} q(\zeta)\, \mathrm{d}\sigma(\zeta) &= \int_{S^{2n-1}} \left(\frac{1}{2\pi} \int_0^{2\pi} q(\mathrm{e}^{\mathrm{i}\theta}\xi)\, \mathrm{d}\theta \right) \mathrm{d}\sigma(\xi) \\ &\le C - u(x). \end{aligned}$$

Thus q is integrable on the sphere, and subtracting the last inequality from C gives

$$u(x) \le \int_{S^{2n-1}} u(x + r\zeta)\, \mathrm{d}\sigma(\zeta). \tag{6.2}$$

This is the Euclidean spherical submean inequality. Notice that the same complex line is represented by all vectors $\mathrm{e}^{\mathrm{i}\alpha}\xi$. Averaging over all unit vectors counts those circles repeatedly, but the repetition is harmless; the unitary-invariance calculation above is exactly what removes it.

We next prove local integrability rather than importing it from real potential theory. Choose $b \in \Omega$ with $u(b) > -\infty$ and $\overline{B(b, R)} \subset \Omega$. Upper semicontinuity gives a constant $C \ge 0$ such that $u \le C$ on this closed ball. Put $u^+ = \max\{u, 0\}$ and $u^- = \max\{-u, 0\}$. Formula (6.2) gives, for $0 < r < R$,

$$\int_{S^{2n-1}} u^-(b + r\zeta)\, \mathrm{d}\sigma(\zeta) \le C - u(b).$$

Indeed, the spherical average of u is at least $u(b)$, while the spherical average of u^+ is at most C. Polar coordinates now give

$$\begin{aligned} \int_{B(b,R)} u^-\, \mathrm{d}V &= |S^{2n-1}| \int_0^R r^{2n-1} \left(\int_{S^{2n-1}} u^-(b + r\zeta)\, \mathrm{d}\sigma(\zeta) \right) \mathrm{d}r \\ &\le |S^{2n-1}|\, (C - u(b)) \frac{R^{2n}}{2n} < \infty. \end{aligned}$$

Since $u^+ \le C$ on the ball, this proves $u \in L^1(B(b, R))$.

For completeness, this integrability propagates from b to all of Ω. Let

$$E = \{a \in \Omega : u \text{ is integrable on a neighborhood of } a\}.$$

The preceding calculation shows that E is nonempty, and it is open by definition. To see that it is closed, let a lie in the closure of E and choose $R > 0$ with $\overline{B(a, 2R)} \subset \Omega$. Choose $b \in E \cap B(a, R/4)$. Since u is integrable near b, it is finite almost everywhere on some sufficiently small ball about b contained in $B(a, R/2)$. Choose a finite point c in that ball. Then

$$\overline{B(c, 3R/2)} \subset B(a, 2R) \subset \Omega, \qquad B(a, R) \subset B(c, 3R/2).$$

Applying the preceding ball calculation at c with radius $3R/2$ shows that u is integrable on a neighborhood of $B(a,R)$, so $a \in E$. The domain is connected, and hence $E = \Omega$.

We have proved both that $u \in L^1_{\mathrm{loc}}$ and that a psh function is real subharmonic. In particular, integrating (6.2) in the radial variable gives the ball submean inequality

$$u(x) \le \frac{1}{|B(x,r)|}\int_{B(x,r)} u = \frac{2n}{r^{2n}}\int_0^r s^{2n-1}\left(\int_{S^{2n-1}} u(x+s\zeta)\,\mathrm{d}\sigma(\zeta)\right)\mathrm{d}s, \tag{6.3}$$

which will be used below.

Why mollification preserves plurisubharmonicity. Let $\rho \in C_c^\infty(\mathbb{C}^n)$ be any nonnegative mollifier supported in the unit ball, and for $\varepsilon > 0$ define on the inner domain Ω_ε from Definition 2.2

$$u_\varepsilon(x) = \int_{\mathbb{C}^n} u(x-\varepsilon y)\rho(y)\,\mathrm{d}V(y).$$

This function is smooth by Lemma 2.3. We verify the psh condition rather than merely assert it. Fix $a \in \mathbb{C}^n$, $\xi \in \mathbb{C}^n \setminus \{0\}$, and a closed disc $\overline{D(\tau_0, r)}$ such that

$$a + \tau\xi \in \Omega_\varepsilon \quad \text{for every } \tau \in \overline{D(\tau_0,r)}.$$

For each $y \in \operatorname{supp}\rho$, the function

$$\tau \longmapsto u(a - \varepsilon y + \tau\xi)$$

is the restriction of u to a complex line parallel to $a + \mathbb{C}\xi$; it is subharmonic on $D(\tau_0, r)$ or identically $-\infty$. The definition of Ω_ε guarantees that all its arguments lie in Ω. Therefore its one-variable submean inequality gives

$$u(a-\varepsilon y+\tau_0\xi) \le \frac{1}{2\pi}\int_0^{2\pi} u(a-\varepsilon y+(\tau_0+r\mathrm{e}^{\mathrm{i}\theta})\xi)\,\mathrm{d}\theta.$$

Multiply by the nonnegative weight $\rho(y)$ and integrate in y. The local integrability proved above permits Fubini's theorem, and we obtain

$$u_\varepsilon(a+\tau_0\xi) \le \frac{1}{2\pi}\int_0^{2\pi} u_\varepsilon(a+(\tau_0+r\mathrm{e}^{\mathrm{i}\theta})\xi)\,\mathrm{d}\theta.$$

Thus the restriction of u_ε to every complex line is subharmonic. Since u_ε is smooth, it is upper semicontinuous and finite; hence $u_\varepsilon \in \mathrm{PSH}(\Omega_\varepsilon)$. Equivalently, the smooth line test in part (1) says that its Levi matrix is positive semidefinite, so $\mathrm{dd^c}u_\varepsilon \ge 0$. Since $u_\varepsilon \to u$ in L^1_{loc}, for every test form η we have

$$\langle \mathrm{dd^c}u, \eta\rangle = \lim_{\varepsilon\downarrow 0}\langle \mathrm{dd^c}u_\varepsilon, \eta\rangle.$$

The weak limit of positive currents is positive, so $\mathrm{dd^c}u \ge 0$.

From a positive current to the canonical representative. Assume $v \in L^1_{\mathrm{loc}}$ and $\mathrm{dd^c}v \ge 0$. The distribution identity (5.2) gives

$$\mathrm{dd^c}v_\varepsilon = (\mathrm{dd^c}v) * \rho_\varepsilon \ge 0. \tag{6.4}$$

Indeed, the right-hand side is a positive weighted average of translates of the positive current $\mathrm{dd^c}v$. Thus v_ε is smooth and psh by part (1). It is also subharmonic as a function on $\mathbb{R}^{2n}$, because its real Laplacian is four times the trace of its Levi matrix.

We spell out the compactness argument that recovers point values. On concentric balls $K \Subset B \Subset \Omega$, the real submean inequality bounds $\sup_K v_\varepsilon$ by a constant times $\|v_\varepsilon\|_{L^1(B)}$. These norms stay bounded because $v_\varepsilon \to v$ in L^1_{loc}. Hence the upper envelope

$$v^\sharp(x) = \limsup_{\substack{y\to x\\ \varepsilon\downarrow 0}} v_\varepsilon(y) \tag{6.5}$$

is psh. Here is the submean verification. Given a complex line disc centered at x, choose $x_j \to x$ and $\varepsilon_j \downarrow 0$ that realize the limsup in (6.5), apply the submean inequality to v_{ε_j} on the corresponding translated discs, and then apply Fatou's lemma to a common local upper bound. The result is the submean inequality for $v^\sharp$. Formula (6.5) also makes $v^\sharp$ upper semicontinuous.

It remains to check that no new almost-everywhere class has appeared. Write

$$A_r v(x) = \frac{1}{|B(x,r)|} \int_{B(x,r)} v.$$

If x is a Lebesgue point of v, then $v_\varepsilon(x) \to v(x)$, so $v^\sharp(x) \geq v(x)$. Conversely, the real submean inequality gives $v_\varepsilon(y) \leq A_r v_\varepsilon(y)$ whenever the ball is admissible. For fixed r, convolution commutes with the ball average, and $A_r v$ is continuous; hence

$$\lim_{\substack{y \to x \\ \varepsilon \downarrow 0}} A_r v_\varepsilon(y) = A_r v(x).$$

Consequently $v^\sharp(x) \leq A_r v(x)$ for every sufficiently small r. Letting $r \downarrow 0$ at a Lebesgue point gives $v^\sharp(x) \leq v(x)$. Thus $v^\sharp = v$ almost everywhere, and $v^\sharp$ is the required psh representative.

We next prove uniqueness and the forward direction of part (3). Let u be psh, fix $x \in \Omega$, and choose $R > 0$ such that $\overline{B(x,R)} \subset \Omega$. With σ denoting the normalized surface measure on S^{2n-1}, define the spherical mean

$$M_u(x,r) = \int_{S^{2n-1}} u(x + r\zeta)\, \mathrm{d}\sigma(\zeta), \qquad 0 < r < R.$$

For every unit vector ξ, the circular mean

$$m_u(x,r;\xi) = \frac{1}{2\pi} \int_0^{2\pi} u(x + r\mathrm{e}^{\mathrm{i}\theta}\xi)\, \mathrm{d}\theta$$

is nondecreasing in r, because the restriction of u to $x + \mathbb{C}\xi$ is subharmonic or identically $-\infty$. Averaging in ξ and using the unitary invariance of σ, as in the proof of (6.2), gives

$$M_u(x,r) = \int_{S^{2n-1}} m_u(x,r;\xi)\, \mathrm{d}\sigma(\xi).$$

It follows that $r \mapsto M_u(x,r)$ is nondecreasing. Moreover,

$$u(x) \leq M_u(x,r) \leq \sup_{\partial B(x,r)} u,$$

where the first inequality is the submean inequality and the second follows because σ is a probability measure. Upper semicontinuity therefore gives

$$M_u(x,r) \downarrow u(x) \qquad \text{as } r \downarrow 0,$$

also when $u(x) = -\infty$: in that case, for every $A \in \mathbb{R}$, upper semicontinuity gives $u < A$ on a sufficiently small ball centered at x.

Choose a nonnegative radial mollifier $\rho(z) = \widehat{\rho}(|z|)$ supported in the unit ball and normalized by $\int_{\mathbb{C}^n} \rho\, \mathrm{d}V = 1$. Set

$$w(s) = |S^{2n-1}| s^{2n-1} \widehat{\rho}(s), \qquad 0 \leq s \leq 1.$$

Then $w \geq 0$, $\int_0^1 w(s)\, \mathrm{d}s = 1$, and polar coordinates give, whenever $0 < \varepsilon < R$,

$$(u * \rho_\varepsilon)(x) = \int_0^1 M_u(x, \varepsilon s) w(s)\, \mathrm{d}s. \tag{6.6}$$

The monotonicity of the spherical means shows that the left-hand side decreases as $\varepsilon \downarrow 0$. Since w is a probability density and the convolution samples only $B(x,\varepsilon)$, the submean inequality also gives

$$u(x) \le (u * \rho_\varepsilon)(x) \le \sup_{B(x,\varepsilon)} u.$$

The upper-semicontinuity argument above now identifies the limit, including at a point where $u(x) = -\infty$, and yields

$$u * \rho_\varepsilon(x) \downarrow u(x). \tag{6.7}$$

The convolutions are smooth and psh by the first half of the proof, so this proves the forward direction of (3).

Suppose now that u_1 and u_2 are plurisubharmonic representatives of the same almost-everywhere class on Ω. They belong to $L^1_{\mathrm{loc}}(\Omega)$ by the first half of the proof. Hence, on every relatively compact ball and for every sufficiently small $\varepsilon > 0$, equality $u_1 = u_2$ almost everywhere implies

$$u_1 * \rho_\varepsilon = u_2 * \rho_\varepsilon.$$

Letting $\varepsilon \downarrow 0$ and using (6.7) for each function gives $u_1(x) = u_2(x)$ at every $x \in \Omega$. This proves the uniqueness asserted in (2). Finally, a decreasing limit of smooth psh functions restricts on each complex line to a decreasing limit of subharmonic functions, so it is psh unless it is identically $-\infty$. □

Proposition 2.21: Stability of psh functions

The following well-defined operations preserve plurisubharmonicity: nonnegative linear combinations that are not identically $-\infty$, finite maxima, decreasing limits unless identically $-\infty$, holomorphic pullback unless a connected component is sent into the pole set, and composition $\chi \circ u$ with χ convex and increasing, where $\chi(-\infty) := \lim_{t\to-\infty} \chi(t)$. If a nonempty family of psh functions is locally uniformly bounded above, then the upper-semicontinuous regularization of its supremum is psh.

Proof. Nonnegative linear combinations and finite maxima can be checked after restriction to complex lines. For a holomorphic map F, take local decreasing smooth psh approximations $u_j \downarrow u$. The smooth chain rule shows that $u_j \circ F$ is psh, and its decreasing limit is psh unless it is identically $-\infty$ on a component. For smooth u and χ,

$$\mathrm{dd^c}(\chi \circ u) = \chi'(u)\mathrm{dd^c}u + \chi''(u)\frac{\mathrm{i}}{2\pi}\partial u \wedge \bar\partial u \ge 0;$$

decreasing smooth convex increasing approximation of χ, followed by decreasing approximation of u, gives the general case. Decreasing limits follow from monotone convergence in the submean inequality. Finally, Choquet's lemma reduces a locally bounded-above supremum to the upper regularization of an increasing sequence of finite maxima; the submean inequality then passes to the limit by Fatou's lemma. □

Example 2.22: Analytic singularities

Let $\Omega \subset \mathbb{C}^n$ be a domain. If $f_1, \dots, f_N \in \mathcal{O}(\Omega)$ are not all identically zero and $c > 0$, then

$$u = c \log\left(|f_1|^2 + \cdots + |f_N|^2\right)$$

is psh.

To prove this, restrict first to a complex affine line $\ell(\zeta) = a + \zeta\xi$, where $\xi \in \mathbb{C}^n \setminus \{0\}$. Let $D \subset \mathbb{C}$ be a connected component of $\ell^{-1}(\Omega)$ and set $h_j = f_j \circ \ell \in \mathcal{O}(D)$. If all the functions

h_j vanish identically on D, then $u \circ \ell \equiv -\infty$ there, which is allowed in the definition of a psh function. Suppose that they do not all vanish identically and put

$$S(\zeta) = \sum_{j=1}^{N} |h_j(\zeta)|^2.$$

For $\delta > 0$, introduce the everywhere positive smooth function

$$S_\delta(\zeta) = \delta + S(\zeta) = \delta + \sum_{j=1}^{N} |h_j(\zeta)|^2.$$

Direct differentiation gives

$$\Delta_\zeta \log S_\delta = 4\frac{\partial^2 \log S_\delta}{\partial\zeta\,\partial\bar{\zeta}} = 4\frac{S_\delta \sum_{j=1}^{N} |h_j'|^2 - \left|\sum_{j=1}^{N} h_j'\overline{h_j}\right|^2}{S_\delta^2} \geq 0. \tag{6.8}$$

Indeed, Cauchy–Schwarz gives

$$\left|\sum_{j=1}^{N} h_j'\overline{h_j}\right|^2 \leq \left(\sum_{j=1}^{N} |h_j'|^2\right)\left(\sum_{j=1}^{N} |h_j|^2\right) \leq S_\delta \sum_{j=1}^{N} |h_j'|^2.$$

Thus $\log S_\delta$ is smooth and subharmonic on D. As $\delta \downarrow 0$, these functions decrease pointwise to $\log S$, with value $-\infty$ at a common zero of the h_j. The limit is not identically $-\infty$, because the h_j are not all identically zero on D; hence the decreasing-limit property for subharmonic functions shows that $\log S$ is subharmonic on D. Multiplication by $c > 0$ preserves subharmonicity. Since this conclusion holds on every complex line, u is psh.
Functions of this form have *analytic singularities*; their $-\infty$ locus is the analytic set $\{f_1 = \cdots = f_N = 0\}$.

Warning

$\min(u, v)$ is usually not psh. A supremum need not be upper semicontinuous, so one must take $\sup_i^* u_i(x) = \limsup_{y\to x} \sup_i u_i(y)$. Finally, the statement $\mathrm{dd^c} u \geq 0$ for a distribution class still requires recovery of its unique plurisubharmonic representative.

Global psh functions on a compact connected complex manifold are constant. On a disconnected compact manifold they are constant on each connected component. To retain a background curvature term, use the following relative notion.

Definition 2.23: Quasi-psh and θ-psh functions

A function is *quasi-plurisubharmonic* (qpsh) if locally it is the sum of a psh function and a smooth function. For a closed real $(1,1)$-form θ, a qpsh function u is θ-psh if

$$\theta_u := \theta + \mathrm{dd^c} u \geq 0.$$

We write $u \in \mathrm{PSH}(X, \theta)$.

For example on $\mathbb{P}^1$,

$$u([z_0 : z_1]) = \log \frac{|z_1|^2}{|z_0|^2 + |z_1|^2}$$

satisfies $\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}} u = [z_1 = 0]$.

Definition 2.24: Pluripolar set

A set $E \subset X$ is *locally pluripolar* if, near every point of X, there is a psh function u whose pole set contains E. In these notes, "pluripolar" without a modifier means this local condition. Requiring one psh function on all of X is the stronger notion of *globally pluripolar*. Every proper analytic subset is locally pluripolar, but pluripolar sets may be much more complicated.

Lemma 2.25: Regularized maximum

Let $\rho \in C_c^\infty(\mathbb{R})$ be nonnegative, even, and satisfy $\int \rho = 1$. For $\varepsilon_j > 0$ set

$$M_\varepsilon(t_1, \dots, t_p) = \int_{\mathbb{R}^p} \max_j (t_j + \varepsilon_j s_j) \prod_{j=1}^{p} \rho(s_j)\, \mathrm{d}s_1 \cdots \mathrm{d}s_p.$$

Then M_ε is smooth, convex, and nondecreasing in each variable; it commutes with adding the same constant to all variables; and

$$\max_j t_j \le M_\varepsilon(t) \le \max_j t_j + C \max_j \varepsilon_j.$$

Consequently, $M_\varepsilon(u_1, \dots, u_p)$ is psh whenever the u_j are smooth psh functions.

Proof.

The function M_ε is the convolution on $\mathbb{R}^p$ of the convex function $\max_j t_j$ with the product mollifier $\prod_j \varepsilon_j^{-1} \rho(s_j/\varepsilon_j)$. Convolution with a smooth compactly supported kernel gives smoothness. Translating one coordinate upward cannot decrease the maximum, so convolution preserves coordinatewise monotonicity; averaging the convexity inequality preserves convexity. Because ρ is even, the product measure has barycenter zero, and Jensen's inequality gives $M_\varepsilon(t) \ge \max_j t_j$. If $|s_j| \le R$ on $\operatorname{supp} \rho$, then $|\max_j (t_j + \varepsilon_j s_j) - \max_j t_j| \le R \max_j \varepsilon_j$, which gives the upper bound and the stated constant. Adding c to every t_j adds c inside the maximum and hence after integration.

For $F = M_\varepsilon$ and smooth functions u_j, the chain rule gives

$$\mathrm{dd}^{\mathrm{c}} F(u) = \sum_j F_j(u)\, \mathrm{dd}^{\mathrm{c}} u_j + \frac{\mathrm{i}}{2\pi} \sum_{j,k} F_{jk}(u)\, \partial u_j \wedge \bar\partial u_k.$$

Coordinatewise monotonicity gives $F_j \ge 0$, while convexity makes the real Hessian (F_{jk}) positive semidefinite. Both terms are therefore semipositive when the u_j are psh.

□

2.2.1 The Levi form of a logarithmic norm

Let $F = (f_1, \dots, f_N) : \Omega \to \mathbb{C}^N$ be holomorphic, and assume that no connected component of Ω is contained in $F^{-1}(0)$. At a point where $F \neq 0$, put $S = \sum_\alpha |f_\alpha|^2$. Direct differentiation gives

$$\partial\bar\partial \log S = \frac{S \sum_\alpha \partial f_\alpha \wedge \overline{\partial f_\alpha} - \left(\sum_\alpha \bar f_\alpha \partial f_\alpha\right) \wedge \overline{\left(\sum_\beta \bar f_\beta \partial f_\beta\right)}}{S^2}.$$

Evaluating on a vector ξ turns the numerator into

$$|F|^2 |\mathrm{d}F(\xi)|^2 - |\langle \mathrm{d}F(\xi), F\rangle|^2 \ge 0.$$

Thus $\log|F|^2$ is psh away from $F^{-1}(0)$, and its extension with value $-\infty$ on the zero set is psh everywhere. To see the extension directly, apply the same calculation to the nowhere-zero tuple $(\sqrt{\delta}, f_1, \dots, f_N)$ and let $\log(\delta + |F|^2)$ decrease as $\delta \downarrow 0$. The hypothesis excluding an identically zero component is necessary because our definition does not count the function $-\infty$ as psh. Geometrically, the displayed form is the pullback of the Fubini–Study form under $[F] : \Omega \setminus F^{-1}(0) \to \mathbb{P}^{N-1}$.

The equality directions are also informative. The Levi form vanishes on ξ exactly when $\mathrm{d}F(\xi)$ is proportional to F. Therefore a logarithmic norm is usually only semipositive. Adding $\varepsilon|z|^2$ makes it strictly psh on a coordinate domain while retaining the same pole set.

2.2.2 Strictness, pullback, and compact manifolds

If $u \in C^2$ is psh and χ is convex increasing, then

$$\mathrm{i}\partial\bar\partial(\chi \circ u) = \chi'(u)\mathrm{i}\partial\bar\partial u + \chi''(u)\mathrm{i}\partial u \wedge \bar\partial u.$$

The second term supplies positivity in the gradient direction but says nothing along the level set of u. Thus strict convexity of χ does not automatically turn every psh function into a strictly psh one. In constructing exhaustions, one usually adds a separate strict term before composing.

For a holomorphic map $g : Y \to X$, a C^2 psh function satisfies

$$\mathcal{L}_{u\circ g}(v) = \mathcal{L}_u(\mathrm{d}g(v)).$$

Pullback preserves semipositivity, but it preserves strict positivity only when g is immersive. This is the function-level analogue of pulling back a Kähler form.

On a compact connected manifold, a global psh function is constant by the maximum principle. The relative expression $\theta + \mathrm{dd^c}u$ avoids this obstruction because positivity is imposed on the sum, rather than on $\mathrm{dd^c}u$ alone. In the principal applications, θ represents a positive or pseudoeffective class. Replacing θ by $\theta + \mathrm{dd^c}\psi$ replaces u by $u - \psi$ and leaves the current unchanged. Hence the space of θ-psh potentials depends on the Bott–Chern class of θ, up to this natural translation.

Warning

The pole set $\{u = -\infty\}$ of a psh function is globally pluripolar. A locally pluripolar set need not be represented by one global psh function unless an additional global theorem or construction is available. Likewise, a measure-zero set need not be pluripolar, and a pluripolar set may be dense. Measure theory and pluripotential smallness are different notions.

2.2.3 The line test and the Levi matrix in coordinates

Let $u \in C^2(\Omega)$ and fix $a \in \Omega$, $v \in \mathbb{C}^n$. On the complex line $\ell(\zeta) = a + \zeta v$ the chain rule gives

$$\frac{\partial^2(u \circ \ell)}{\partial\zeta\partial\bar\zeta}(0) = \sum_{j,k=1}^{n} u_{j\bar k}(a) v_j \bar v_k.$$

Since $\Delta_\zeta = 4\partial^2/(\partial\zeta\partial\bar\zeta)$, the restriction of u to every complex line is subharmonic exactly when the Hermitian matrix $H_u = (u_{j\bar k})$ is semipositive. With our normalization,

$$\mathrm{dd^c}u = \frac{\mathrm{i}}{2\pi}\sum_{j,k} u_{j\bar k}\,\mathrm{d}z_j \wedge \mathrm{d}\bar z_k,$$

so this is also exactly positivity of the $(1,1)$-form $\mathrm{dd^c}u$.

For $u(z) = |z_1|^2 + 2|z_2|^2 + 2\Re(z_1 z_2)$, the holomorphic term $2\Re(z_1 z_2)$ contributes nothing to $\partial\bar\partial u$, and

$$H_u = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} > 0.$$

This example is a useful warning: strict plurisubharmonicity is determined by mixed $z, \bar z$ derivatives, not by the full real Hessian.

Logarithms of holomorphic tuples

Let $F = (f_1, \dots, f_N)$ be holomorphic, with no connected component of the domain contained in $F^{-1}(0)$, and put $S = \sum_\alpha |f_\alpha|^2$. Away from $F^{-1}(0)$,

$$\partial S = \sum_\alpha \bar f_\alpha \, \partial f_\alpha,$$

$$\partial\bar\partial \log S = \frac{S \sum_\alpha \partial f_\alpha \wedge \overline{\partial f_\alpha} - \left(\sum_\alpha \bar f_\alpha \partial f_\alpha\right) \wedge \overline{\left(\sum_\beta \bar f_\beta \partial f_\beta\right)}}{S^2}.$$

Evaluating on a vector v turns the numerator into

$$S \sum_\alpha |\mathrm{d}f_\alpha(v)|^2 - \left|\sum_\alpha \bar f_\alpha \mathrm{d}f_\alpha(v)\right|^2 \geq 0$$

by Cauchy–Schwarz. Hence $\log \sum |f_\alpha|^2$ is psh away from the common zero set. The smooth psh functions $\log(\delta + \sum |f_\alpha|^2)$ decrease, as $\delta \downarrow 0$, to the extension with value $-\infty$ on that zero set. If all the f_α vanished identically on one component, this extension would be identically $-\infty$ there and would fall outside our definition. This calculation is the local model for analytic singularities.

Taking $F = (1, z_1, \dots, z_n)$ recovers the Fubini–Study potential. Taking $F = (z_1, \dots, z_n)$ shows that $\log |z|^2$ is psh and has Levi rank $n-1$ away from the origin: the radial vector is the null direction, while tangent directions to the sphere are positive.

A strictly psh exhaustion of the ball

On the unit ball put $\rho(z) = -\log(1 - |z|^2)$. Then

$$\rho_{j\bar k} = \frac{\delta_{jk}}{1 - |z|^2} + \frac{\bar z_j z_k}{(1 - |z|^2)^2}.$$

For $v \neq 0$,

$$\sum \rho_{j\bar k} v_j \bar v_k = \frac{|v|^2}{1 - |z|^2} + \frac{|\langle v, z\rangle|^2}{(1 - |z|^2)^2} > 0.$$

Moreover $\rho(z) \to +\infty$ as $z \to \partial B$. Thus it is a smooth strictly psh exhaustion. This one formula simultaneously proves strict pseudoconvexity and supplies the proper function used in Stein theory.

Stability operations checked directly

For smooth psh functions u, v and $a, b \geq 0$, the identities $\mathrm{d}\mathrm{d}^{\mathrm{c}}(au + bv) = a\mathrm{d}\mathrm{d}^{\mathrm{c}}u + b\mathrm{d}\mathrm{d}^{\mathrm{c}}v \geq 0$ and, for a holomorphic map F, $\mathrm{d}\mathrm{d}^{\mathrm{c}}(u \circ F) = F^*(\mathrm{d}\mathrm{d}^{\mathrm{c}}u) \geq 0$ prove the first two stability statements. For a convex increasing $\chi \in C^2(\mathbb{R})$,

$$\mathrm{d}\mathrm{d}^{\mathrm{c}}\chi(u) = \chi'(u)\mathrm{d}\mathrm{d}^{\mathrm{c}}u + \frac{\mathrm{i}}{2\pi}\chi''(u)\partial u \wedge \bar\partial u \geq 0.$$

For nonsmooth functions, use decreasing smooth approximation. This also explains the hypothesis on pullback: if a component is mapped into $\{u = -\infty\}$, the composite is identically $-\infty$ there. One should not interpret the displayed pullback of $\mathrm{dd^c}u$ literally when u is singular. The maximum of two psh functions is psh because its restriction to every complex line is the maximum of two subharmonic functions; upper-semicontinuous regularization is needed for arbitrary suprema.

There is no analogous minimum principle. Both $|z|^2$ and 1 are subharmonic on $\mathbb{C}$. However, $\min\{|z|^2, 1\}$ has a downward jump in radial derivative at $|z| = 1$ and hence a negative circle contribution to its distributional Laplacian.

2.2.4 How the smooth Levi test extends to nonsmooth psh functions

For a locally integrable function u, the notation $\mathrm{dd^c}u \geq 0$ means

$$\langle \mathrm{dd^c}u, \eta \rangle \geq 0$$

for every compactly supported strongly positive test form η of complementary bidegree. Convolving in a coordinate ball preserves positivity because

$$\mathrm{dd^c}(u * \rho_\varepsilon) = (\mathrm{dd^c}u) * \rho_\varepsilon$$

is an average of translates of the positive current $\mathrm{dd^c}u$. Hence the smooth function u_ε has a semipositive Levi matrix, and its restriction to every complex line is subharmonic.

Pass to the upper-semicontinuous envelope $u^* = \limsup^* u_\varepsilon$. Local L^1 bounds and the real submean inequality give local upper bounds for the family. On each line, Fatou's lemma then preserves the submean inequality, so u^* is psh. At a Lebesgue point x, comparison with the ball averages $A_r u(x)$ gives

$$u(x) \leq u^*(x) \leq A_r u(x),$$

and letting $r \downarrow 0$ shows $u^*(x) = u(x)$. Thus u^* represents the original L^1 class. Conversely, if u is psh, then every u_ε is psh on a smaller ball: translating a complex line and averaging its subharmonic restrictions preserves subharmonicity. Therefore $\mathrm{dd^c}u_\varepsilon \geq 0$, and letting $\varepsilon \downarrow 0$ gives $\mathrm{dd^c}u \geq 0$ weakly. Regularization is thus the bridge from the smooth Levi calculation to the nonsmooth theorem, but it produces the canonical psh representative rather than validating an arbitrary pointwise representative.

The order of operations matters. A general current cannot be pulled back to an arbitrary submanifold. For a singular psh function one restricts the function to a complex line first, or regularizes before restricting; one does not silently pull back the current $\mathrm{dd^c}u$. This prevents a common invalid proof of the line criterion.

2.2.5 Self-study workshop: turning plurisubharmonicity into Hermitian linear algebra

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. You need the subharmonic line test from Lecture 5, the $(1,1)$ calculus of Lecture 2, and the eigenvalue criterion for Hermitian matrices. Always distinguish psh from strictly psh.

One computation, three interpretations. Let $u \in C^2(\Omega)$ for $\Omega \subset \mathbb{C}^n$. At p and $v = (v_1, \ldots, v_n)$, restrict u to the complex line $\zeta \mapsto p + \zeta v$. The one-variable chain rule gives

$$\frac{\partial^2}{\partial\zeta\,\partial\bar{\zeta}} u(p + \zeta v)\Big|_{\zeta=0} = \sum_{j,k} u_{j\bar{k}}(p) v_j \bar{v}_k.$$

Therefore the line restriction is subharmonic for every v exactly when the Levi matrix $(u_{j\bar{k}})$ is positive semidefinite. Equivalently, $\mathrm{dd^c}u$ is a positive $(1,1)$-form. Strict psh means positive definite, locally with a quantitative lower bound after shrinking to a compact set.

For

$$u(z) = \log(1+|z_1|^2) + |z_2|^2 \quad \text{on } \mathbb{C}^2,$$

the Levi matrix is diagonal:

$$\begin{pmatrix} (1+|z_1|^2)^{-2} & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus u is strictly psh. By contrast, $|z_1|^2 - |z_2|^2$ fails the line test on $v = (0,1)$.

If $F = (f_1, \dots, f_N)$ is a holomorphic tuple with no common zero, then $\log \sum_\alpha |f_\alpha|^2$ is psh. The calculation is a Cauchy–Schwarz inequality: the Hessian is the squared norm of the component of $\mathrm{d}F(v)$ orthogonal to the line spanned by F. This geometric reading also identifies the zero directions.

Checkpoint (stability rules with hypotheses)

(1) If u is psh and χ is convex increasing, why is $\chi \circ u$ psh?
(2) Is the difference of two psh functions necessarily psh?
(3) Why can there be no strictly psh function on a compact connected complex manifold?

Solution (checkpoint). For smooth functions and $\chi \in C^2$,

$$\partial\bar{\partial}(\chi(u)) = \chi'(u)\partial\bar{\partial}u + \chi''(u)\partial u \wedge \bar{\partial}u,$$

and both terms are positive under the stated hypotheses. A general finite convex increasing χ admits decreasing smooth convex increasing approximations; use $\chi(-\infty) = \lim_{t\to-\infty}\chi(t)$ at pole points. Decreasing smooth approximation of u then gives the general case. Differences need not be psh, as $|z_1|^2 - |z_2|^2$ shows. A psh function on a compact connected manifold is constant by restriction to coordinate discs and the maximum principle, so it cannot be strict. ◇

Exit criterion

You can pass fluently among line restrictions, the Levi matrix, and positivity of $\mathrm{dd^c}u$, and you can prove rather than memorize the main stability operations.

Summary. Plurisubharmonicity synchronizes one-variable subharmonicity in every complex direction. The Levi matrix, positive $(1,1)$-currents, and monotone regularization are equivalent entry points. The θ-psh language permits singular potentials on compact manifolds.

Exercise 2.26: Maximum versus minimum

Prove that $\max(\log|z_1|, \log|z_2|)$ is psh and give an explicit pair of psh functions whose minimum is not psh.

Exercise 2.27: A pole on projective space

Check the formula for u in both affine charts and verify directly that $\omega_{\mathrm{FS}} + \mathrm{dd^c}u = [z_1 = 0]$.

Optional handout: Plurisubharmonic envelopes and Choquet's lemma

This handout is optional. No later argument in the core lectures depends on it.

Suppose $\mathcal{F}$ is a nonempty family of psh functions locally uniformly bounded above. The raw supremum need not be upper semicontinuous, but

$$P = \left(\sup_{u\in\mathcal{F}} u\right)^*$$

is psh. A useful reduction is Choquet's lemma: there is a sequence $u_j \in \mathcal{F}$ such that $(\sup_j u_j)^* = P$. Thus an uncountable extremal problem can be handled by a sequence.

Choose a countable basis of relatively compact balls and, on each one, select functions whose suprema approximate the supremum of the family on a dense set. Taking finite maxima produces an increasing sequence. The submean inequality passes to the upper-semicontinuous regularization, proving plurisubharmonicity.

Definition 2.28: Relative extremal function

For $K \Subset \Omega$, define the raw envelope

$$h_{K,\Omega} = \sup\{u \in \mathrm{PSH}(\Omega) : u \le 0 \text{ on } \Omega,\ u \le -1 \text{ on } K\}.$$

Its upper-semicontinuous regularization $h^*_{K,\Omega}$ is called the relative extremal function of K in Ω.

Recall that a bounded domain is *hyperconvex* if it admits a negative continuous psh exhaustion. On such a domain, Bedford–Taylor theory shows that the envelope is maximal away from the obstacle:

$$(\mathrm{dd^c} h^*_{K,\Omega})^n = 0 \qquad \text{on } \Omega \setminus K.$$

The *relative Monge–Ampère capacity* is

$$\mathrm{Cap}(K,\Omega) = \sup\left\{\int_K (\mathrm{dd^c} w)^n : w \in \mathrm{PSH}(\Omega),\ -1 \le w \le 0\right\}.$$

With our normalization, Bedford–Taylor's equilibrium theorem identifies it with $\int_\Omega (\mathrm{dd^c} h^*_{K,\Omega})^n$. This is the prototype for obstacle problems and pluripotential envelopes on compact Kähler manifolds.

Exercise 2.29: Envelope in the disc

For $K = \overline{D(0,r)} \subset \mathbb{D}$, determine $h^*_{K,\mathbb{D}}$ explicitly using radial harmonic functions on the annulus.

2.3 Lecture 7: Pseudoconvex domains, exhaustion, and the Levi problem

Week 4 material • 90 minutes

Notation for this lecture

A smoothly bounded domain is written $\Omega = \{\rho < 0\}$, where ρ is a real defining function and $\mathrm{d}\rho \neq 0$ on $\partial\Omega$. Its complex tangent space is $T^{1,0}_p\partial\Omega = \{\xi \in \mathbb{C}^n : \sum_j \rho_j(p)\xi_j = 0\}$, where $\rho_j = \partial\rho/\partial z_j$. The boundary Levi form is $\mathcal{L}_\rho(\xi) = \sum \rho_{j\bar{k}}\xi_j\bar{\xi}_k$ restricted to that space. An exhaustion is a proper function whose sublevel sets are relatively compact; a complex manifold is pseudoconvex here if it admits a continuous psh exhaustion. The unit disc is Δ, Δ_r is the radius-r disc, and $A_{r,1} = \{r < |z| < 1\}$ is an annulus.

Prerequisite results used below

Regular level-set theorem. If a smooth real function ρ satisfies $\mathrm{d}\rho(p) \neq 0$, then near p the set $\{\rho = 0\}$ is a smooth real hypersurface and $T_p\{\rho = 0\} = \ker \mathrm{d}\rho(p)$.
Strictly psh sublevel sets. If $\rho \in C^2$ is strictly psh near the boundary of $\Omega = \{\rho < 0\}$ and $\mathrm{d}\rho \neq 0$ on $\partial\Omega$, then the Levi form of $\partial\Omega$ is positive on every nonzero complex tangent vector. Conversely, a strictly pseudoconvex boundary admits a defining function that is strictly psh on a collar after replacing ρ by $\mathrm{e}^{A\rho} - 1$ for sufficiently large A.

Why these ideas are needed

Local holomorphic coordinates do not tell us whether functions defined on a domain can be continued past its boundary or whether the $\bar{\partial}$ equation has global solutions there. A boundary condition must distinguish genuine analytic barriers from accidental choices of defining function. Pseudoconvexity does this through the Levi form, while a plurisubharmonic exhaustion records the same geometry without requiring a smooth boundary. The Levi problem is the organizing question: does this apparently differential-convex condition exactly characterize domains of holomorphy? Its answer explains why pseudoconvexity becomes the standing hypothesis of later L^2 theory.

Plan for the lecture. 0–15 min: Hartogs extension and domains of holomorphy; 15–48: the Cartan–Thullen theorem; 48–65: the Levi form and psh exhaustion; 65–82: the Levi problem; 82–90: examples.

In one variable every domain can be the maximal domain of definition of a holomorphic function. In several variables, Hartogs extension can fill holes automatically. A genuine domain of holomorphy must therefore satisfy a complex convexity condition.

Definition 2.30: Domain of holomorphy

A domain $\Omega \subset \mathbb{C}^n$ is a *domain of holomorphy* if the following local nonextension property holds. For every connected open set $U \subset \mathbb{C}^n$ with $U \cap \partial\Omega \neq \varnothing$, and every connected component V of $U \cap \Omega$, there is a function $f \in \mathcal{O}(\Omega)$ for which $f|_V$ has no holomorphic extension to U. The function in this definition is initially allowed to depend on the pair (U, V). The Cartan–Thullen theorem below proves the stronger conclusion that one function $F \in \mathcal{O}(\Omega)$ can be chosen simultaneously for all such pairs. In particular, F cannot extend holomorphically to any connected domain properly containing Ω.

Definition 2.31: Holomorphic hull and holomorphic convexity

For a compact set $K \Subset \Omega$, its *holomorphic hull* is

$$\widehat{K}_{O(\Omega)} = \{z \in \Omega : |f(z)| \le \sup_K |f| \text{ for every } f \in O(\Omega)\}.$$

The domain Ω is *holomorphically convex* if $\widehat{K}_{O(\Omega)}$ is compact in Ω for every compact $K \Subset \Omega$. Thus no sequence in a holomorphic hull may approach the boundary or escape to infinity.

Definition 2.32: Levi pseudoconvexity

Suppose $\Omega = \{\rho < 0\}$ has C^2 boundary and $\mathrm{d}\rho \neq 0$ on $\partial\Omega$. At $p \in \partial\Omega$, its complex tangent space is

$$T^{1,0}_p \partial\Omega = \left\{ \xi \in \mathbb{C}^n : \sum_j \rho_j(p)\xi_j = 0 \right\}.$$

Here

$$\rho_j = \frac{\partial \rho}{\partial z_j}, \qquad \rho_{j\bar{k}} = \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k}.$$

The Levi form is

$$\mathcal{L}_{\rho,p}(\xi) = \sum_{j,k=1}^{n} \rho_{j\bar{k}}(p)\xi_j \bar{\xi}_k.$$

The boundary is *Levi pseudoconvex* at p if $\mathcal{L}_{\rho,p}(\xi) \ge 0$ for every $\xi \in T^{1,0}_p \partial\Omega$, and it is *strictly pseudoconvex* at p if the inequality is strict for every nonzero $\xi \in T^{1,0}_p \partial\Omega$. We apply either adjective to $\partial\Omega$ when the corresponding condition holds at every boundary point.

We must check that neither the complex tangent space nor Levi pseudoconvexity depends on the defining function. Let ρ and $\widetilde{\rho}$ be two C^2 defining functions near $p \in \partial\Omega$, both with the same negative-side convention:

$$\Omega = \{\rho < 0\} = \{\widetilde{\rho} < 0\}, \qquad \mathrm{d}\rho \neq 0, \quad \mathrm{d}\widetilde{\rho} \neq 0 \quad \text{on } \partial\Omega.$$

After shrinking the neighborhood of p, there is a positive C^1 function a such that

$$\widetilde{\rho} = a\rho.$$

Here is the local calculation. Since $\mathrm{d}\rho(p) \neq 0$, the real implicit function theorem provides real coordinates (y, t) in which $t = \rho$ and the boundary is $\{t = 0\}$. Because $\widetilde{\rho}(y, 0) = 0$, the fundamental theorem of calculus gives

$$\widetilde{\rho}(y,t) = t \int_0^1 \frac{\partial \widetilde{\rho}}{\partial t}(y, st)\, \mathrm{d}s = t\, a(y,t).$$

The function a is C^1. For $t < 0$ sufficiently close to zero, both $t = \rho(y,t)$ and $\widetilde{\rho}(y,t)$ are negative, so $a(y,t) > 0$. Moreover, differentiating $\widetilde{\rho} = a\rho$ on $\{t = 0\}$ gives

$$\mathrm{d}\widetilde{\rho} = a\, \mathrm{d}\rho.$$

The two differentials are nonzero there, hence a cannot vanish on the boundary. Consequently $a(p) > 0$, and, after one more shrinking, $a > 0$ in a neighborhood of p.

Taking $(1,0)$-parts at p yields

$$\partial\widetilde{\rho}(p) = a(p)\partial\rho(p).$$

Therefore

$$\{\xi : \partial\widetilde{\rho}(p)(\xi) = 0\} = \{\xi : \partial\rho(p)(\xi) = 0\},$$

which proves that $T^{1,0}_p\partial\Omega$ is independent of the chosen defining function.

It remains to check the Levi form. Fix a vector $\xi \in T^{1,0}_p\partial\Omega$ and consider the affine holomorphic disc $\gamma(\zeta) = p + \zeta\xi$. Complex tangency says

$$\frac{\partial}{\partial\zeta}(\rho\circ\gamma)(0) = 0;$$

because ρ is real-valued, the $\bar{\zeta}$-derivative vanishes as well. Together with $\rho(p) = 0$, this gives $\rho\circ\gamma(\zeta) = O(|\zeta|^2)$. On the other hand, $a\circ\gamma(\zeta) = a(p) + O(|\zeta|)$. Thus the second-order Taylor parts of $\widetilde{\rho}\circ\gamma = (a\circ\gamma)(\rho\circ\gamma)$ and $a(p)\rho\circ\gamma$ agree. Taking their mixed second derivatives at zero, we obtain

$$\mathcal{L}_{\widetilde{\rho},p}(\xi) = \frac{\partial^2(\widetilde{\rho}\circ\gamma)}{\partial\zeta\,\partial\bar{\zeta}}(0) = a(p)\frac{\partial^2(\rho\circ\gamma)}{\partial\zeta\,\partial\bar{\zeta}}(0) = a(p)\mathcal{L}_{\rho,p}(\xi).$$

Since $a(p) > 0$, semipositivity and positive definiteness on the complex tangent space are unchanged. Hence Levi pseudoconvexity and strict pseudoconvexity are intrinsic properties of the oriented boundary $\partial\Omega$. The negative-side convention matters: replacing ρ by $-\rho$ reverses the sign of the Levi form, but it also replaces the domain $\{\rho < 0\}$ by the opposite side $\{\rho > 0\}$.

Definition 2.33: Pseudoconvex complex manifold

A complex manifold X is *pseudoconvex*, or *weakly pseudoconvex*, if it admits a smooth psh exhaustion $\psi : X \to \mathbb{R}$. We will say explicitly that X *admits a smooth strictly psh exhaustion* when strict positivity holds everywhere. The term *strongly pseudoconvex manifold* is often reserved for the different, 1-convex condition in which strict positivity is required only outside a compact set. Exhaustion means that every sublevel set $\{\psi < c\}$ is relatively compact.

Why every convex domain is pseudoconvex. Let $\Omega \subset \mathbb{C}^n$ be a convex domain. We first treat the general case, without making any regularity assumption on $\partial\Omega$. If $\Omega = \mathbb{C}^n$, then $|z|^2$ is a smooth strictly psh exhaustion. Suppose from now on that $\Omega \neq \mathbb{C}^n$, and define the boundary-distance function

$$\delta(z) = \operatorname{dist}(z, \mathbb{C}^n \setminus \Omega) = \operatorname{dist}(z, \partial\Omega), \qquad z \in \Omega.$$

We claim that δ is concave. Indeed, if $x, y \in \Omega$ and $0 \le t \le 1$, then

$$B(x, \delta(x)) \subset \Omega, \qquad B(y, \delta(y)) \subset \Omega.$$

Convexity of Ω and the elementary identity for Minkowski sums of Euclidean balls give

$$\begin{aligned} B\big((1-t)x + ty, (1-t)\delta(x) + t\delta(y)\big) &= (1-t)B(x, \delta(x)) + tB(y, \delta(y)) \\ &\subset \Omega. \end{aligned}$$

Consequently,

$$\delta((1-t)x + ty) \ge (1-t)\delta(x) + t\delta(y),$$

as claimed.

The function $s \mapsto -\log s$ is convex and decreasing on $(0, \infty)$. The preceding concavity inequality therefore implies

$$-\log\delta((1-t)x + ty) \le -\log\big((1-t)\delta(x) + t\delta(y)\big) \le (1-t)(-\log\delta(x)) + t(-\log\delta(y)).$$

Thus $-\log\delta$ is convex. It follows that

$$\Phi(z) = |z|^2 - \log\delta(z)$$

is a continuous convex function on Ω. A continuous real convex function on a convex domain in $\mathbb{C}^n$ is psh: its restriction to every complex affine line is convex on the underlying real plane, and Jensen's inequality on circles gives the submean inequality. Hence Φ is psh.

We next verify properness rather than merely asserting that Φ blows up at the boundary. Choose $q \in \partial\Omega$. Since $q \notin \Omega$,

$$\delta(z) \le |z - q| \le |z| + |q|,$$

and hence

$$\Phi(z) \ge |z|^2 - \log(|z| + |q|) \longrightarrow +\infty \qquad \text{as } |z| \longrightarrow \infty.$$

Thus every sublevel $\{\Phi \le c\}$ is bounded. On the same sublevel,

$$|z|^2 - \log\delta(z) \le c \quad \Longrightarrow \quad \delta(z) \ge \exp(|z|^2 - c) \ge \mathrm{e}^{-c}.$$

It therefore stays a positive distance from $\mathbb{C}^n \setminus \Omega$, so it is compact in Ω. This proves that Φ is a continuous psh exhaustion.

Finally, our definition asks for a *smooth* psh exhaustion. The regularization argument in Step 1 of the proof of Theorem 2.36 below applies here: smooth Φ by convolution on an increasing sequence of compact sublevels, add arbitrarily small positive multiples of $|z|^2$, and glue successive approximants with the regularized maximum of Lemma 2.25. It produces a smooth strictly psh function Ψ satisfying $|\Psi - \Phi| \le 1$. Since a bounded perturbation of an exhaustion is still an exhaustion, Ψ is the smooth psh exhaustion required by Definition 2.33. Hence every convex domain is pseudoconvex.

When $\partial\Omega$ is C^2, the same conclusion is also visible directly in the boundary Levi form. Let ρ be a local defining function near $p \in \partial\Omega$, with $\Omega = \{\rho < 0\}$. Convexity says that the real tangent hyperplane at p supports Ω. If $\xi \in T^{1,0}_p\partial\Omega$, then the real two-plane $p + \mathbb{C}\xi$ lies in that tangent hyperplane. Hence, for small $\zeta \in \mathbb{C}$, $p + \zeta\xi$ is not in Ω and

$$(\rho \circ \gamma)(\zeta) \ge 0 = (\rho \circ \gamma)(0), \qquad \gamma(\zeta) = p + \zeta\xi.$$

The function $\rho \circ \gamma$ has a local minimum at zero. Its real Laplacian there is therefore nonnegative, and the chain rule gives

$$0 \le \Delta_\zeta(\rho \circ \gamma)(0) = 4\frac{\partial^2(\rho \circ \gamma)}{\partial\zeta\,\partial\bar{\zeta}}(0) = 4\mathcal{L}_{\rho,p}(\xi).$$

Thus a convex domain with C^2 boundary is Levi pseudoconvex, in agreement with the exhaustion proof above.

Example 2.34: An exhaustion from a defining function

For a bounded smoothly bounded strictly pseudoconvex domain, modify ρ so that $-\log(-\rho)$ is strictly psh near the boundary, then patch it with a small multiple of $|z|^2$. The identity

$$\partial\bar{\partial}[-\log(-\rho)] = \frac{\partial\bar{\partial}\rho}{-\rho} + \frac{\partial\rho \wedge \bar{\partial}\rho}{\rho^2}$$

shows that the first term controls complex tangential directions and the second controls the normal direction.

Warning

Levi semipositivity at the boundary is tested only in complex tangent directions. Requiring $\partial\bar\partial\rho \geq 0$ throughout a neighborhood is stronger and often needs a new defining function. Real convexity implies pseudoconvexity, but not conversely.

The passage from local nonextension to a compactness property is the first global step in the Levi problem. The forward implication comes from Cauchy estimates: a point in a holomorphic hull has at least the same Taylor radius as the compact set being hulled. The reverse implication uses holomorphic functions to separate a point from a hull and then assembles these separating functions into an interpolation series. The following proof gives both constructions; compare [1, Chapter I, Proposition 6.9 and Theorem 6.11].

Theorem 2.35: Cartan–Thullen theorem

Let $\Omega \subset \mathbb{C}^n$ be a domain. The following conditions are equivalent.

(1) Ω is a domain of holomorphy in the sense of Definition 2.30.
(2) Ω is holomorphically convex.
(3) If $(z_j)_{j\geq 1}$ is a sequence of pairwise distinct points of Ω with no accumulation point in Ω, and $(a_j)_{j\geq 1}$ is any sequence in $\mathbb{C}$, then there is an $F \in \mathcal{O}(\Omega)$ such that $F(z_j) = a_j$ for every $j \geq 1$.
(4) There is one function $F \in \mathcal{O}(\Omega)$ such that, for every connected open set $U \subset \mathbb{C}^n$ with $U \cap \partial\Omega \neq \varnothing$ and every connected component V of $U \cap \Omega$, the restriction $F|_V$ has no holomorphic extension to U.

Proof. We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (1).

Step 1: a domain of holomorphy is holomorphically convex. Write

$$\|z\|_\infty = \max_{1\leq k\leq n} |z_k|, \qquad P(z,r) = \{w \in \mathbb{C}^n : \|w - z\|_\infty < r\}.$$

For a nonempty subset $A \subset \Omega$, define its boundary distance by

$$\delta_\Omega(A) = \inf\{\|z - w\|_\infty : z \in A,\ w \in \mathbb{C}^n \setminus \Omega\}.$$

If $K \Subset \Omega$ is a nonempty compact set, then $\delta_\Omega(K) > 0$. We claim that

$$\delta_\Omega\big(\widehat{K}_{\mathcal{O}(\Omega)}\big) = \delta_\Omega(K). \tag{2.1}$$

Fix $0 < r < \delta_\Omega(K)$ and $f \in \mathcal{O}(\Omega)$. The set

$$K_r = \bigcup_{z\in K} \overline{P(z,r)}$$

is a compact subset of Ω; put $M_{f,r} = \sup_{K_r} |f|$. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$, write $|\alpha| = \alpha_1 + \cdots + \alpha_n$, $\alpha! = \alpha_1! \cdots \alpha_n!$, and $\partial^\alpha = \partial_{z_1}^{\alpha_1} \cdots \partial_{z_n}^{\alpha_n}$. The Cauchy inequalities on the polydisc give

$$\frac{|\partial^\alpha f(z)|}{\alpha!} \leq M_{f,r} r^{-|\alpha|}, \qquad z \in K. \tag{2.2}$$

For fixed α, the function $\partial^\alpha f$ belongs to $\mathcal{O}(\Omega)$. Hence the definition of the holomorphic hull extends (2.2) to every $a \in \widehat{K}_{\mathcal{O}(\Omega)}$:

$$\frac{|\partial^\alpha f(a)|}{\alpha!} \leq \sup_K \frac{|\partial^\alpha f|}{\alpha!} \leq M_{f,r} r^{-|\alpha|}. \tag{2.3}$$

If $0 < s < r$ and $w \in P(a,s)$, the series of absolute values of the Taylor series at a is bounded by

$$M_{f,r} \sum_{\alpha \in \mathbb{N}^n} (s/r)^{|\alpha|} = M_{f,r}(1 - s/r)^{-n}.$$

Thus the Taylor series of f at a converges normally on every smaller polydisc $P(a,s)$ and defines a holomorphic function on $P(a,r)$.

We now use the nonextension property, and this is the reason for its local formulation in Definition 2.30. If $P(a,r) \not\subset \Omega$, then the connected open set $U = P(a,r)$ meets $\partial\Omega$. Let V be the connected component of $U \cap \Omega$ that contains a. For every $f \in \mathcal{O}(\Omega)$, its Taylor series at a is a holomorphic function on U and agrees with f near a; the identity theorem therefore shows that it agrees with f on V. This says that every $f \in \mathcal{O}(\Omega)$ extends from V to U, contradicting condition (1). Consequently $P(a,r) \subset \Omega$, and $\delta_\Omega(a) \geq r$. Since $r < \delta_\Omega(K)$ was arbitrary,

$$\delta_\Omega\big(\widehat{K}_{\mathcal{O}(\Omega)}\big) \geq \delta_\Omega(K).$$

The opposite inequality follows from $K \subset \widehat{K}_{\mathcal{O}(\Omega)}$, proving (2.1).

It remains to turn (2.1) into compactness. The coordinate functions $z_1, \ldots, z_n$ belong to $\mathcal{O}(\Omega)$, so

$$|z_k| \leq \sup_K |z_k|, \qquad z \in \widehat{K}_{\mathcal{O}(\Omega)}, \quad 1 \leq k \leq n.$$

The hull is therefore bounded. It is closed in Ω, since it is the intersection over $f \in \mathcal{O}(\Omega)$ of the closed sets $\{|f| \leq \sup_K |f|\}$. Formula (2.1) keeps it a positive distance from $\mathbb{C}^n \setminus \Omega$, so it is also closed in $\mathbb{C}^n$. The Heine–Borel theorem now makes it compact. This proves (2).

Step 2: holomorphic convexity gives interpolation on a discrete set. We first construct a convenient exhaustion. Choose compact sets $L_1 \subset L_2 \subset \cdots \subset \Omega$ with $\bigcup_{\nu \geq 1} L_\nu = \Omega$. Starting with $K_0 = \varnothing$, suppose K_ν has been constructed. Choose a compact neighborhood $K'_\nu \Subset \Omega$ of K_ν and set

$$K_{\nu+1} = \widehat{K'_\nu \cup L_{\nu+1}}_{\mathcal{O}(\Omega)}.$$

Condition (2) makes $K_{\nu+1}$ compact. Moreover, taking a holomorphic hull twice does not enlarge it, because every $h \in \mathcal{O}(\Omega)$ satisfies

$$\sup_{\widehat{A}_{\mathcal{O}(\Omega)}} |h| = \sup_A |h|$$

for every compact $A \Subset \Omega$. Hence each K_ν is holomorphically convex, $K_\nu \Subset \operatorname{int} K_{\nu+1}$, and $\bigcup_{\nu \geq 1} K_\nu = \Omega$.

Let $(z_j)_{j \geq 1}$ and $(a_j)_{j \geq 1}$ be as in condition (3). For every j, let $\nu(j) \geq 0$ be the least integer such that

$$z_j \in K_{\nu(j)+1} \setminus K_{\nu(j)}.$$

Every compact subset of Ω contains only finitely many of the z_j; therefore $\nu(j) \to \infty$ as $j \to \infty$. When $\nu(j) \geq 1$, the point z_j does not belong to the holomorphic hull $K_{\nu(j)}$. Thus there is a $g_j \in \mathcal{O}(\Omega)$ such that, after multiplication by a nonzero constant,

$$g_j(z_j) = 1, \qquad q_j := \sup_{K_{\nu(j)}} |g_j| < 1. \tag{2.4}$$

For the finitely many indices with $\nu(j) = 0$, take $g_j \equiv 1$; the estimates on $K_0 = \varnothing$ below are then vacuous.

The next factor prevents a term chosen at stage j from changing values already interpolated. Set $P_1 \equiv 1$. For $j \geq 2$ and every $1 \leq k < j$, choose an index $\mu(k,j) \in \{1, \ldots, n\}$ for which $(z_j)_{\mu(k,j)} \neq (z_k)_{\mu(k,j)}$, and define

$$P_j(z) = \prod_{k=1}^{j-1} \frac{z_{\mu(k,j)} - (z_k)_{\mu(k,j)}}{(z_j)_{\mu(k,j)} - (z_k)_{\mu(k,j)}}.$$

Then $P_j(z_j) = 1$ and $P_j(z_k) = 0$ for $k < j$.

We choose coefficients $\lambda_j \in \mathbb{C}$ and exponents $m_j \in \mathbb{N}$ inductively. After the terms with index less than j have been fixed, put

$$\lambda_j = a_j - \sum_{k=1}^{j-1} \lambda_k P_k(z_j) g_k(z_j)^{m_k}. \tag{2.5}$$

If $\nu(j) \geq 1$, choose m_j so large that

$$\sup_{K_{\nu(j)}} |\lambda_j P_j g_j^{m_j}| \leq 2^{-j}; \tag{2.6}$$

this is possible by (2.4), since P_j is bounded on the compact set $K_{\nu(j)}$. For the finitely many indices with $\nu(j) = 0$, choose any positive exponent. Define

$$F(z) = \sum_{j=1}^{\infty} \lambda_j P_j(z) g_j(z)^{m_j}. \tag{2.7}$$

If $C \Subset \Omega$, choose N with $C \subset K_N$. For all sufficiently large j, one has $\nu(j) \geq N$, and (2.6) bounds the j-th summand by 2^{-j} on C. The Weierstrass test gives normal convergence of (2.7), so $F \in \mathcal{O}(\Omega)$. At z_j, every later summand vanishes because $P_k(z_j) = 0$ for $k > j$, while (2.5) makes the sum of the first j terms equal to a_j. Hence $F(z_j) = a_j$ for all j, proving (3).

Step 3: interpolation produces one universal nonextendible function. If $\partial\Omega = \varnothing$, then $\Omega = \mathbb{C}^n$ and condition (4) is vacuous; take $F \equiv 0$. Assume from now on that $\partial\Omega \neq \varnothing$. Let $\mathcal{B}$ be the countable family of Euclidean balls in $\mathbb{C}^n \simeq \mathbb{R}^{2n}$ whose centers and radii are rational. Consider all pairs (B, W) for which

$$B \in \mathcal{B}, \qquad B \cap \partial\Omega \neq \varnothing, \qquad W \text{ is a connected component of } B \cap \Omega.$$

There are only countably many such pairs. Indeed, each $B \cap \Omega$ is open in $\mathbb{R}^{2n}$, each of its connected components is open, and each component contains a point with rational real coordinates. Enumerate the pairs as $(B_\ell, W_\ell)_{\ell \geq 1}$.

For each ℓ, choose $p_\ell \in \partial W_\ell \cap B_\ell$. Such a point exists because W_ℓ is a nonempty proper open subset of the connected ball B_ℓ. Moreover $p_\ell \in \partial\Omega$: a boundary point of a component of the open set $B_\ell \cap \Omega$ that lies inside B_ℓ cannot lie in Ω. Define the compact exhaustion

$$C_m = \left\{ z \in \Omega : \|z\| \leq m, \quad \operatorname{dist}(z, \mathbb{C}^n \setminus \Omega) \geq \frac{1}{m} \right\}, \qquad m \geq 1,$$

where $\|z\|$ is the Euclidean norm. Every compact subset of Ω is contained in C_m for all sufficiently large m.

Choose a map $\sigma : \mathbb{N}_{\geq 1} \to \mathbb{N}_{\geq 1}$ for which every positive integer occurs infinitely often. Because $p_{\sigma(m)} \in \partial W_{\sigma(m)}$ and $p_{\sigma(m)} \notin C_m$, we may choose pairwise distinct points

$$z_m \in W_{\sigma(m)} \setminus C_m, \qquad |z_m - p_{\sigma(m)}| < \frac{1}{m}. \tag{2.8}$$

The sequence (z_m) has no accumulation point in Ω: if $C \Subset \Omega$, then $C \subset C_m$ for all large m, whereas $z_m \notin C_m$. Condition (3), applied with the prescribed values $a_m = m$, therefore gives an $F \in \mathcal{O}(\Omega)$ satisfying

$$F(z_m) = m, \qquad m \geq 1. \tag{2.9}$$

We verify that this single F has the property in (4). Let $U \subset \mathbb{C}^n$ be connected and open with $U \cap \partial\Omega \neq \varnothing$, and let V be a connected component of $U \cap \Omega$. Since V is a nonempty proper open subset of the connected set U, there is a point $p \in \partial V \cap U$. Choose a rational ball B_ℓ with $p \in B_\ell$ and

$\overline{B_\ell} \subset U$, and choose a connected component W_ℓ of $B_\ell \cap \Omega$ that is contained in V. Along the infinite subsequence for which $\sigma(m) = \ell$, (2.8) and (2.9) give

$$z_m \in W_\ell, \qquad z_m \longrightarrow p_\ell \in B_\ell, \qquad |F(z_m)| = m \longrightarrow \infty.$$

If $F|_V$ extended to a holomorphic function on U, that extension would be bounded on a small closed ball about p_ℓ and would agree with F at the points z_m for all large m, a contradiction. This proves (4).

Step 4: the universal function gives the definition. Condition (4) immediately implies condition (1), since the same function F works for every pair (U, V) in Definition 2.30. It also justifies the global formulation often used informally. If F extended to a connected domain $\Omega' \supsetneq \Omega$, a path in Ω' from a point of Ω to a point of $\Omega' \setminus \Omega$ would meet some $p \in \partial\Omega \cap \Omega'$. A sufficiently small ball $U \Subset \Omega'$ about p and the component of $U \cap \Omega$ met by the path would contradict (4). The cycle of implications is complete. □

Theorem 2.36: The Levi problem

For a domain $\Omega \subset \mathbb{C}^n$, the following are equivalent:

(1) Ω is a domain of holomorphy;
(2) Ω is holomorphically convex;
(3) Ω is pseudoconvex in the sense of Definition 2.33; equivalently, it has a continuous psh exhaustion;
(4) Ω is a Stein manifold.

For a C^2 boundary, these are also equivalent to Levi pseudoconvexity.

Condition (3) refers explicitly to Definition 2.33, which asks for a smooth psh exhaustion. The continuous formulation in the same item is equivalent for a domain in $\mathbb{C}^n$: Step 1 below regularizes a continuous psh exhaustion to a smooth strictly psh exhaustion, while every smooth psh exhaustion is continuous. Condition (4) uses the intrinsic definition given in Definition 3.12: holomorphic convexity, separation of points, and enough global holomorphic differentials to give local coordinates.

For classroom use, the theorem is best read as a dictionary with two directions. To prove that a concrete domain is a domain of holomorphy or a Stein manifold, it is usually enough to write down one psh exhaustion and verify its sublevels are relatively compact. To prove that a C^2 domain is not pseudoconvex, one may instead find a complex tangential vector on which the boundary Levi form is negative. When the boundary is not smooth, a Hartogs extension argument can replace the boundary calculation. Examples 2.38–2.41 carry out these three tests explicitly.

Proof (with Oka's boundary lemmas as quoted inputs). We first state the quoted inputs in the precise forms used below.

Oka's two boundary lemmas. Let $D \subset \mathbb{C}^n$ be a domain with C^2 boundary. At each $p \in \partial D$, choose a local defining function r_p on a neighborhood V_p, so that $D \cap V_p = \{r_p < 0\}$ and $\mathrm{d}r_p \neq 0$ on $\partial D \cap V_p$.

(O1) Suppose that

$$\sum_{j,k=1}^{n} (r_p)_{j\bar{k}}(p)\xi_j\bar{\xi}_k \geq 0 \quad \text{whenever } p \in \partial D \quad \text{and} \quad \sum_{j=1}^{n} (r_p)_j(p)\xi_j = 0. \tag{7.1}$$

Then D has a continuous psh exhaustion. More precisely, on every relatively compact boundary patch one may write $r = r_p$, modify r without changing the set $\{r < 0\}$, and choose $A > 0$ and $\varepsilon > 0$ so that

$$-\log(-r) + A|z|^2 \tag{7.2}$$

is psh on the inner collar $\{-\varepsilon < r < 0\}$ of that patch and tends to $+\infty$ at its boundary part. A locally finite regularized-maximum gluing of these collar barriers, together with a term growing at infinity, gives the stated exhaustion of D.

(O2) Suppose instead that at some $p \in \partial D$ the form in (7.1) is negative on a nonzero complex tangent vector. For $0 < s < 1$, write

$$H_s = \big[(\Delta_s \times \Delta) \cup (\Delta \times \{1 - s < |w| < 1\})\big] \times \Delta^{n-2} \subset \Delta^n. \tag{7.3}$$

Its envelope of holomorphy is the full polydisc Δ^n. There is a holomorphic copy of such an H_s lying in D near p whose filled polydisc crosses ∂D. Equivalently, there is a neighborhood U of p, with $U \not\subset D$, such that every $f \in \mathcal{O}(D)$ has a unique holomorphic extension to the strictly larger connected domain $D \cup U$.

For $n = 1$ the complex tangent space is zero-dimensional, so (O2) cannot occur; the displayed Hartogs figure is needed only when $n \geq 2$. We use (O1) and (O2) only in the final boundary argument.

The two analytic tools used in the proof of (3) $\Longrightarrow$ (2) are proved elsewhere in these notes: local smoothing of psh functions follows from Theorem 2.20 and Lemma 2.25, and the required $\bar\partial$ estimate is the complete Kähler estimate of Theorem 7.23.

Step 1: continuous and smooth psh exhaustions give the same condition. Let ρ be a continuous psh exhaustion. After composing with an increasing convex function, arrange that

$$K_j = \{\rho \leq j\} \Subset \operatorname{int} K_{j+1}, \qquad \bigcup_{j \geq 1} K_j = \Omega.$$

On a neighborhood of K_{j+1}, convolution on slightly smaller coordinate balls gives smooth psh approximants decreasing to ρ. Because ρ is continuous, the convergence is uniform on K_{j+1}. Add a positive multiple $\varepsilon_j |z|^2$ to make the approximant strictly psh, choosing ε_j so small that $\varepsilon_j \sup_{K_{j+1}} |z|^2 < 2^{-j}$. On an overlap use the regularized maximum from Lemma 2.25. If the smoothing width is smaller than the gap between the two inputs, that maximum agrees with the larger input; hence the new approximation can be made to agree with the old one on K_{j-1} and to extend it across K_{j+1}. Induction gives a globally defined smooth strictly psh function ψ with

$$|\psi - \rho| \leq 1.$$

The last inequality makes ψ proper, so it is an exhaustion. The converse needs no construction: a smooth psh exhaustion is a continuous psh exhaustion. This proves the equivalence asserted inside condition (3).

Step 2: a psh exhaustion forces holomorphic convexity. Choose an increasing convex function χ whose derivative grows sufficiently fast on every band $\{j \leq \psi \leq j + 1\}$, and put

$$\tau = \chi(\psi), \qquad \omega = \mathrm{i}\partial\bar\partial\tau = \mathrm{i}\chi'(\psi)\partial\bar\partial\psi + \mathrm{i}\chi''(\psi)\partial\psi \wedge \bar\partial\psi.$$

The function χ can be chosen inductively so that every curve crossing infinitely many bands has infinite ω-length. Thus ω is a complete Kähler metric, while τ remains a smooth strictly psh exhaustion.

Fix a compact set $K \Subset \Omega$ and put $m = \sup_K \tau$. Choose $c > m$. We show that every point a with $\tau(a) > c + 1$ can be separated from K by the absolute value of one holomorphic function. This will put the entire holomorphic hull of K inside the compact sublevel $\{\tau \leq c + 1\}$.

Choose a coordinate ball $B \ni a$ with $\overline{B} \subset \{\tau > c\}$, and choose $\eta \in C_c^\infty(B)$ equal to 1 near a. For $A > 0$ consider the psh weight

$$\Phi_A = A\tau + (n + 1) \log |z - a|^2.$$

The logarithmic term is singular only where $\bar\partial\eta = 0$. Replace it temporarily by $(n+1)\log(|z-a|^2+\varepsilon^2)$, apply Theorem 7.23 to $\bar\partial\eta \wedge \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$, and then let $\varepsilon \downarrow 0$ by weak compactness. Removing the common holomorphic volume form, we obtain a function satisfying

$$\bar\partial u_A = \bar\partial\eta$$

with finite Φ_A-weighted norm. Since $\operatorname{supp}\bar\partial\eta \subset \{\tau > c\}$ and the logarithmic factor is bounded on that support, the estimate has the form

$$\int_\Omega |u_A|^2 \mathrm{e}^{-\Phi_A}\, \mathrm{d}V_\omega \le \frac{C}{A}\,\mathrm{e}^{-Ac}, \tag{7.4}$$

where C may depend on a, B, η, but not on A.

Let U be a fixed neighborhood of K with $\sup_U \tau < c$. The point a is a positive distance from U, so the logarithmic factor is bounded above and below there. On the fixed compact set $\overline{U}$, the ω-volume and Euclidean volume are uniformly comparable. Formula (7.4) therefore gives

$$\int_U |u_A|^2\, \mathrm{d}V \le C_U \mathrm{e}^{-A(c-\sup_U \tau)}.$$

Because $\bar\partial u_A = 0$ on U, the mean-value inequality for holomorphic functions turns this into

$$\sup_K |u_A| \longrightarrow 0 \qquad (A \to \infty). \tag{7.5}$$

Near a, the function u_A is also holomorphic. Its weighted integrability contains

$$\int_{|z-a|<r} |u_A(z)|^2 |z-a|^{-2(n+1)}\, \mathrm{d}V_\omega(z) < \infty.$$

If the first nonzero homogeneous term of u_A had degree d, the radial integral would be a constant multiple of

$$\int_0^r t^{2d-3}\, \mathrm{d}t,$$

which is finite only for $d > 1$. In particular, $u_A(a) = 0$.

Now set $F_A = \eta - u_A$. It is holomorphic on all of Ω, $F_A(a) = 1$, and (7.5) gives $\sup_K |F_A| < 1/2$ for large A. Thus $a \notin \widehat{K}_{O(\Omega)}$. We have proved

$$\widehat{K}_{O(\Omega)} \subset \{\tau \le c + 1\}.$$

The hull is closed in Ω, being an intersection of closed sets of the form $\{|f| \le \sup_K |f|\}$. It is therefore a closed subset of a compact sublevel and is compact. This proves (3) $\Longrightarrow$ (2).

Step 3: holomorphic convexity produces a continuous psh exhaustion. Assume (2). We give the construction because it explains the reverse implication without hiding another equivalence theorem. Choose compact sets

$$K_1 \Subset \operatorname{int} K_2 \Subset \cdots, \qquad \bigcup_j K_j = \Omega,$$

such that every K_j is holomorphically convex. To obtain them, start with any compact exhaustion, enlarge its j-th member to a compact neighborhood of K_{j-1}, and replace that neighborhood by its holomorphic hull. Condition (2) makes the hull compact, and taking the hull twice changes nothing.

Put

$$A_j = K_{j+2} \setminus \operatorname{int} K_{j+1}.$$

The compact sets A_j and K_j are disjoint. If $a \in A_j$, then $a \notin \widehat{K}_j = K_j$; hence some $h_a \in O(\Omega)$ satisfies

$$|h_a(a)| > \sup_{K_j} |h_a|.$$

Choose a number s_a strictly between the two sides and replace h_a by h_a/s_a. Then

$$q_a := \sup_{K_j} |h_a| < 1 < |h_a(a)|.$$

The map $z \mapsto |h_a(z)|$ is continuous for the Euclidean topology and has value larger than 1 at a. Hence there are a Euclidean neighborhood W_a of a and a number $r_a > 1$ such that $|h_a| \geq r_a$ on W_a. Finitely many of these neighborhoods, $W_{a_1}, \dots, W_{a_{N_j}}$, cover the compact set A_j. For every ℓ choose an exponent m_ℓ so large that

$$q_{a_\ell}^{2m_\ell} \leq \frac{2^{-j}}{N_j}, \qquad r_{a_\ell}^{2m_\ell} \geq j,$$

and set $g_{j\ell} = h_{a_\ell}^{m_\ell}$. We then have

$$v_j := \sum_{\ell=1}^{N_j} |g_{j\ell}|^2 \leq 2^{-j} \quad \text{on } K_j, \qquad v_j \geq j \quad \text{on } A_j. \tag{7.6}$$

Indeed, the first inequality follows by summing the N_j bounds on K_j; if $z \in A_j$, then $z \in W_{a_\ell}$ for at least one ℓ, and the corresponding single summand is already at least j. This is the precise place where both compactness of A_j and the strict separation inequality are used.

For a fixed K_N, estimate (7.6) applies to every $j \geq N$. Consequently

$$\rho(z) = |z|^2 + \sum_{j=1}^{\infty} v_j(z) \tag{7.7}$$

converges uniformly on K_N. Thus ρ is a continuous strictly psh function: each partial sum is psh, local uniform limits preserve the submean inequality, and $|z|^2$ supplies strict positivity.

It remains to check properness. If $z \notin K_{N+1}$, choose the least m with $z \in K_m$. Then $m \geq N+2$ and $z \in A_{m-2}$, so (7.6) gives $\rho(z) \geq v_{m-2}(z) \geq m-2 \geq N$. Hence

$$\{\rho < N\} \subset K_{N+1}.$$

Every sublevel is relatively compact, and ρ is a continuous psh exhaustion. Step 1 now supplies the smooth psh exhaustion required in Definition 2.33. This proves (2) $\Longrightarrow$ (3).

Step 4: condition (4) introduces no additional difficulty for a domain. By Definition 3.12, a Stein manifold is holomorphically convex, so (4) $\Longrightarrow$ (2). Conversely, suppose (2). The coordinate functions $z_1, \dots, z_n$ separate any two distinct points of Ω, and their differentials form a basis of the holomorphic cotangent space at every point. Together with holomorphic convexity, these are exactly the remaining requirements in Definition 3.12. Thus (2) $\Longrightarrow$ (4).

Step 5: Cartan–Thullen and the quoted boundary lemmas finish the proof. Theorem 2.35 gives (1) $\Longleftrightarrow$ (2). Combined with Steps 1–4, this proves the equivalence of (1)–(4).

Now assume that $\partial\Omega$ is C^2. If its Levi form is semipositive, the hypothesis (7.1) of (O1) holds, so (O1) gives a continuous psh exhaustion. By Step 1 this is the smooth psh exhaustion required by condition (3). Formula (7.2) also explains the geometry: the term $\partial r \wedge \bar\partial r / r^2$ controls the complex normal direction, while the Levi form controls complex tangential directions.

Conversely, suppose one of conditions (1)–(4) holds and the Levi form has a negative complex tangential direction at p. By (O2), every $f \in \mathcal{O}(\Omega)$ extends to the same strictly larger connected domain $\Omega \cup U$. Restricting these extensions to a small connected neighborhood of p contradicts Definition 2.30, or equivalently condition (4) of Theorem 2.35. The Levi form must therefore be semipositive. This proves both directions of the boundary equivalence. □

Proposition 2.37: Every plane domain is pseudoconvex

Every domain $\Omega \subset \mathbb{C}$ is pseudoconvex in the sense of Definition 2.33. In fact, it admits a smooth strictly subharmonic exhaustion.

Proof. If $\Omega = \mathbb{C}$, the function $z \mapsto |z|^2$ is already a smooth strictly subharmonic exhaustion. Assume from now on that $\Omega \neq \mathbb{C}$. Then $\partial\Omega$ is nonempty. Put

$$\delta_\Omega(z) = \operatorname{dist}(z, \partial\Omega), \qquad u(z) = -\log \delta_\Omega(z).$$

The distance to a closed set is 1-Lipschitz, so δ_Ω is continuous. Moreover, $\delta_\Omega(z) > 0$ for every $z \in \Omega$; hence u is a finite continuous function on Ω.

We first prove that u is subharmonic. Since

$$\delta_\Omega(z) = \inf_{a \in \partial\Omega} |z - a|,$$

and $t \mapsto -\log t$ is decreasing, we have the pointwise identity

$$u(z) = \sup_{a \in \partial\Omega} \bigl(-\log |z - a|\bigr). \tag{7.8}$$

For a fixed $a \in \partial\Omega$, the function $h_a(z) = -\log|z - a|$ is harmonic on Ω, because its only singularity is at $a \notin \Omega$. Let $\overline{\Delta(z_0, s)} \Subset \Omega$. The mean-value identity for each h_a, followed by the elementary inequality "supremum of averages $\leq$ average of suprema," gives

$$\begin{aligned} u(z_0) &= \sup_{a \in \partial\Omega} \frac{1}{2\pi} \int_0^{2\pi} h_a(z_0 + s\mathrm{e}^{\mathrm{i}\theta})\, \mathrm{d}\theta \\ &\leq \frac{1}{2\pi} \int_0^{2\pi} \sup_{a \in \partial\Omega} h_a(z_0 + s\mathrm{e}^{\mathrm{i}\theta})\, \mathrm{d}\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + s\mathrm{e}^{\mathrm{i}\theta})\, \mathrm{d}\theta. \end{aligned} \tag{7.9}$$

Thus u satisfies the submean inequality on every relatively compact disc and is subharmonic.

Now define

$$\rho(z) = |z|^2 + u(z) = |z|^2 - \log \delta_\Omega(z). \tag{7.10}$$

It is continuous and strictly subharmonic: for example, $\rho - \frac{1}{2}|z|^2 = u + \frac{1}{2}|z|^2$ is subharmonic. It remains to check the exhaustion property rather than merely its boundary blow-up.

Fix one point $a_0 \in \partial\Omega$. Since $\delta_\Omega(z) \leq |z - a_0|$, we have

$$\rho(z) \geq |z|^2 - \log|z - a_0|,$$

and the right-hand side tends to $+\infty$ as $|z| \to \infty$. Consequently every sublevel $\{\rho \leq c\}$ is bounded. On the same sublevel, (7.10) also gives

$$-\log \delta_\Omega(z) \leq c - |z|^2 \leq c, \qquad \text{hence} \qquad \delta_\Omega(z) \geq \mathrm{e}^{-c}. \tag{7.11}$$

Thus the sublevel stays a positive distance from $\partial\Omega$. A convergent sequence in it can therefore neither approach the boundary nor leave Ω; continuity of ρ shows that its limit remains in the same sublevel. The sublevel is closed and bounded in $\mathbb{C}$, hence compact. Therefore ρ is a continuous strictly subharmonic exhaustion.

Finally apply the convolution and regularized-maximum construction from Step 1 in the proof of Theorem 2.36. It gives a smooth strictly subharmonic function $\widetilde{\rho}$ with $|\widetilde{\rho} - \rho| \leq 1$. This bounded error preserves properness, so $\widetilde{\rho}$ is the smooth psh exhaustion required by Definition 2.33. □

For a C^2 boundary in one complex dimension, the complex tangent space has dimension $n-1 = 0$, so the boundary Levi condition is vacuous. The proposition is stronger: it applies to arbitrary plane domains, including domains with nonsmooth or highly disconnected boundary.

Example 2.38: Ellipsoids and psh sublevel domains

Choose numbers $a_1, \dots, a_n > 0$ and consider the ellipsoid

$$E = \left\{ z \in \mathbb{C}^n : s(z) := \sum_{j=1}^{n} \frac{|z_j|^2}{a_j^2} < 1 \right\}.$$

The function

$$\Phi(z) = -\log(1 - s(z))$$

is smooth on E. Since

$$\frac{\partial s}{\partial z_j} = \frac{\bar{z}_j}{a_j^2}, \qquad \frac{\partial^2 s}{\partial z_j \partial \bar{z}_k} = \frac{\delta_{jk}}{a_j^2},$$

the chain rule gives

$$\mathcal{L}_\Phi(z;\xi) = \frac{1}{1-s(z)} \sum_{j=1}^{n} \frac{|\xi_j|^2}{a_j^2} + \frac{1}{(1-s(z))^2} \left| \sum_{j=1}^{n} \frac{\bar{z}_j \xi_j}{a_j^2} \right|^2 > 0 \qquad (\xi \neq 0).$$

Thus Φ is strictly psh. It is also an exhaustion, a point that should be checked separately from positivity: for $C \in \mathbb{R}$,

$$\{z \in E : \Phi(z) \le C\} = \{z \in \mathbb{C}^n : s(z) \le 1 - \mathrm{e}^{-C}\}.$$

This set is empty if $C < 0$ and is a compact ellipsoid contained in E if $C \ge 0$. Theorem 2.36 now shows at once that E is holomorphically convex, is a domain of holomorphy, and is Stein. On the boundary, the defining function $s - 1$ has the positive definite Levi matrix $\operatorname{diag}(a_1^{-2}, \dots, a_n^{-2})$, so the boundary formulation of the same theorem gives the same conclusion.

The calculation illustrates a reusable construction. Let v be smooth and psh on a neighborhood of the closure of a relatively compact domain $D = \{v < c\} \subset \mathbb{C}^n$, and assume $v(z) \to c$ as $z \to \partial D$. Then

$$\partial\bar{\partial}[-\log(c - v)] = \frac{\partial\bar{\partial} v}{c - v} + \frac{\partial v \wedge \bar{\partial} v}{(c - v)^2} \ge 0.$$

Moreover, $-\log(c-v)$ tends to $+\infty$ at ∂D, and the relative compactness of D then makes it a psh exhaustion. Adding $\varepsilon|z|^2$, where $\varepsilon > 0$, makes it strictly psh without destroying properness.

Example 2.39: Products and annular polycylinders

Finite products preserve pseudoconvexity. Indeed, if X_j has a smooth psh exhaustion ρ_j, replace ρ_j by e^{ρ_j} so that it is positive and still psh; explicitly,

$$\mathrm{dd^c} \mathrm{e}^{\rho_j} = \mathrm{e}^{\rho_j} \left(\mathrm{dd^c} \rho_j + \tfrac{\mathrm{i}}{2\pi} \partial\rho_j \wedge \bar{\partial}\rho_j \right) \ge 0.$$

On $X_1 \times \cdots \times X_m$ the function

$$\rho(z_1, \dots, z_m) = \sum_{j=1}^{m} \mathrm{e}^{\rho_j(z_j)}$$

is psh. A nonempty sublevel $\{\rho \le c\}$ has $c > 0$, and then every $\rho_j(z_j) \le \log c$, so the sublevel is contained in a product of compact sublevels; it is therefore compact.

For a concrete example, let $0 < r < R$ and $A_{r,R} = \{z \in \mathbb{C} : r < |z| < R\}$. Setting $t = |z|^2$,

define

$$\psi_{r,R}(z) = -\log\frac{t-r^2}{R^2-r^2} - \log\frac{R^2-t}{R^2-r^2}.$$

Both logarithmic terms are nonnegative and tend to $+\infty$ at their respective boundary circles. If F is a C^2 function of one real variable, direct differentiation gives

$$\frac{\partial^2}{\partial z\partial\bar z}F(|z|^2) = F'(t) + tF''(t).$$

For the function above this coefficient equals

$$\frac{r^2}{(t-r^2)^2} + \frac{R^2}{(R^2-t)^2} > 0.$$

Thus $\psi_{r,R}$ is strictly subharmonic. If $\psi_{r,R} \le C$, the nonnegativity of the two summands gives

$$t - r^2 \ge (R^2-r^2)\mathrm{e}^{-C}, \qquad R^2 - t \ge (R^2-r^2)\mathrm{e}^{-C};$$

the sublevel therefore stays a positive distance from both boundary circles and is compact in $A_{r,R}$. Hence $\psi_{r,R}$ is an exhaustion. Consequently every annular polycylinder

$$A_{r_1,R_1} \times \cdots \times A_{r_n,R_n}$$

is pseudoconvex, with exhaustion $\sum_j \psi_{r_j,R_j}(z_j)$. This domain has holes in each coordinate direction. Theorem 2.36 says that it is nevertheless a domain of holomorphy and a Stein manifold. It should not be confused with the spherical shell $\{r < |z| < R\} \subset \mathbb{C}^n$, which is not pseudoconvex when $n \ge 2$: at a point p of its inner boundary, the domain-side defining function $r^2 - |z|^2$ has Levi form $-|\xi|^2$ on every nonzero vector satisfying $\sum_j \bar p_j \xi_j = 0$.

Example 2.40: Complements of holomorphic hypersurfaces

Let $f \in \mathcal{O}(\mathbb{C}^n)$ be nonconstant and put

$$D_f = \mathbb{C}^n \setminus \{f = 0\}.$$

On D_f, the function $g = 1/f$ is holomorphic, so $|g|^2$ is psh. More explicitly,

$$\mathcal{L}_{|g|^2}(z;\xi) = \left|\sum_{j=1}^n \frac{\partial g}{\partial z_j}(z)\xi_j\right|^2 \ge 0.$$

Therefore

$$\rho_f(z) = |z|^2 + \frac{1}{|f(z)|^2}$$

is smooth on D_f. Its Levi form is

$$\mathcal{L}_{\rho_f}(z;\xi) = |\xi|^2 + \left|\sum_{j=1}^n \frac{\partial g}{\partial z_j}(z)\xi_j\right|^2 > 0 \qquad (\xi \ne 0),$$

so it is strictly psh. Its sublevels are compact: a nonempty sublevel $\{\rho_f \le c\}$ has $c > 0$, and on it $|z|^2 \le c$ and $|f(z)| \ge c^{-1/2}$. Thus the point can neither escape to infinity nor approach the hypersurface $\{f = 0\}$. Since the sublevel is closed in D_f, it is compact. Thus

D_f is pseudoconvex and, by Theorem 2.36, it is also holomorphically convex, a domain of holomorphy, and Stein.
Taking $f(z) = z_1 \cdots z_k$ shows, for example, that

$$(\mathbb{C}^*)^k \times \mathbb{C}^{n-k} = \mathbb{C}^n \setminus \{z_1 \cdots z_k = 0\}$$

is pseudoconvex. For instance, on $\mathbb{C}^2 \setminus \{z_1 z_2 = 1\}$ one may take

$$|z_1|^2 + |z_2|^2 + |z_1 z_2 - 1|^{-2}$$

as a smooth strictly psh exhaustion. The hypersurface assumption matters: for $n \geq 2$, the punctured space $\mathbb{C}^n \setminus \{0\}$ is not pseudoconvex. Every holomorphic function on a punctured ball extends across the missing point by the Hartogs theorem, so the punctured space is not a domain of holomorphy; the negative conclusion then follows from Theorem 2.36.

Example 2.41: The Hartogs triangle

Let

$$T = \{(z, w) \in \mathbb{C}^2 : |z| < |w| < 1\}.$$

The boundary of T is not C^2 at the origin, so the global boundary Levi test is not available. Instead consider the biholomorphism

$$\Delta \times \Delta^* \longrightarrow T, \qquad (\zeta, w) \longmapsto (\zeta w, w),$$

whose inverse is $(z, w) \mapsto (z/w, w)$. On the unit disc the function

$$b(\zeta) = -\log(1 - |\zeta|^2)$$

is strictly subharmonic because

$$\frac{\partial^2 b}{\partial \zeta \partial \bar{\zeta}} = \frac{1}{(1 - |\zeta|^2)^2} > 0.$$

It follows by holomorphic pullback that the three functions

$$-\log|w|, \qquad -\log(1 - |w|^2), \qquad -\log\bigl(1 - |z|^2/|w|^2\bigr)$$

are psh on T; the first is pluriharmonic and the other two are the pullbacks of b by $(z, w) \mapsto w$ and $(z, w) \mapsto z/w$. Their sum

$$\Psi_T(z, w) = -\log|w| - \log(1 - |w|^2) - \log\bigl(1 - |z|^2/|w|^2\bigr)$$

is in fact strictly psh: in the coordinates $(\zeta, w) = (z/w, w)$, the last two summands have diagonal Levi matrix

$$\operatorname{diag}\bigl((1 - |\zeta|^2)^{-2}, (1 - |w|^2)^{-2}\bigr).$$

We still have to verify properness. All three summands are nonnegative. If $\Psi_T(z, w) \leq C$, then

$$|w| \geq \mathrm{e}^{-C}, \qquad |w|^2 \leq 1 - \mathrm{e}^{-C}, \qquad \frac{|z|^2}{|w|^2} \leq 1 - \mathrm{e}^{-C}.$$

Thus the sublevel remains a positive distance from the three boundary regimes $w = 0$, $|w| = 1$, and $|z| = |w|$; it is closed and bounded in $\mathbb{C}^2$, hence compact in T. Therefore Ψ_T is a smooth strictly psh exhaustion, and Theorem 2.36 shows that T is a domain of holomorphy and a Stein manifold. This example demonstrates why the exhaustion definition is more robust than a boundary definition for domains with corners or singular boundary points.

2.3.1 The Hartogs phenomenon behind pseudoconvexity

The Hartogs phenomenon begins in complex dimension two. Fix $0 < r < 1$ and consider the Hartogs figure

$$H = \big(\Delta_r \times \Delta\big) \cup \big(\Delta \times A_{1-r,1}\big) \subset \mathbb{C}^2,$$

where $\Delta_s = \{\zeta \in \mathbb{C} : |\zeta| < s\}$, $\Delta = \Delta_1$, and $A_{s,t} = \{\zeta \in \mathbb{C} : s < |\zeta| < t\}$. The first set is the *vertical arm*; the second is the *annular arm*. Their intersection is nonempty, so a holomorphic function on H consists of holomorphic functions on the two arms that agree on the overlap.

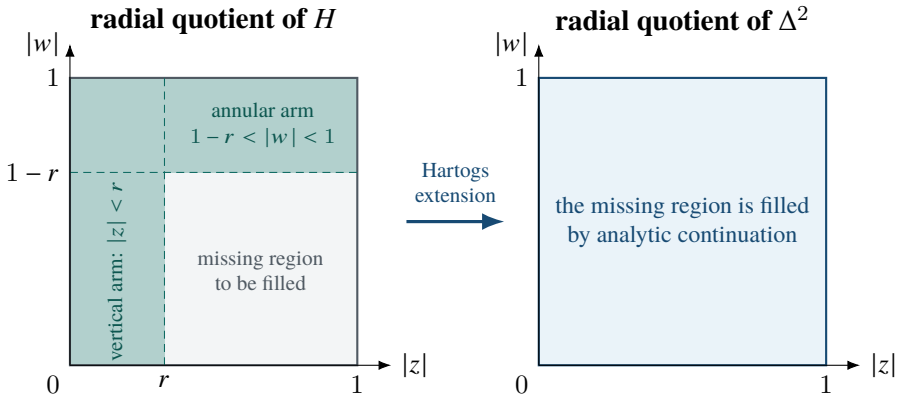


Figure 2.1: The map $q : \Delta^2 \to [0,1)^2$, $q(z,w) = (|z|,|w|)$, collapses the $S^1 \times S^1$ orbits. In this radial quotient the Hartogs figure is the shaded L-shaped set on the left. Its envelope of holomorphy is the full bidisc, represented by the filled square on the right. Thus this is a quotient picture of a four-real-dimensional domain, not a literal planar cross-section.

The picture records the two jobs performed by the two arms. The annular arm reaches every $z \in \Delta$, so it defines all Laurent coefficients $a_k(z)$ on the full z-disc. The vertical arm reaches the centre $w = 0$ when $|z| < r$, so it forces the negative coefficients to vanish there. Holomorphic dependence on z then propagates this vanishing across the whole disc and fills precisely the gray region in the left panel.

We now verify this mechanism algebraically, including the convergence needed to produce an actual extension. Let $f \in \mathcal{O}(H)$ and choose a radius ρ with $1 - r < \rho < 1$. On the annular arm, expand in the second variable:

$$f(z,w) = \sum_{k\in\mathbb{Z}} a_k(z) w^k, \qquad a_k(z) = \frac{1}{2\pi \mathrm{i}} \int_{|w|=\rho} \frac{f(z,w)}{w^{k+1}}\, \mathrm{d}w.$$

For fixed $z \in \Delta$, the integral is independent of the chosen $\rho \in (1-r, 1)$ by Cauchy's theorem on an annulus. Differentiation under the integral sign shows that each a_k is holomorphic on the full parameter disc Δ.

If $|z| < r$, then (z,w) lies in the vertical arm for every $w \in \Delta$. Consequently $w \mapsto f(z,w)$ is holomorphic through $w = 0$, and its Laurent coefficients satisfy

$$a_k(z) = 0 \qquad (k < 0, |z| < r).$$

For each fixed negative integer k, the function $a_k \in \mathcal{O}(\Delta)$ therefore vanishes on the nonempty open disc Δ_r. The identity theorem gives $a_k \equiv 0$ on Δ. This is the step in which holomorphic dependence on the parameter z fills the hole.

It remains to justify convergence in the region that was missing from H. Fix $0 < R < 1$ and $0 < S < 1$, and choose

$$\max\{S, 1-r\} < \rho < 1.$$

The compact set $\{|z| \le R, |w| = \rho\}$ lies in the annular arm, so

$$M_{R,\rho} := \max_{|z|\le R,\ |w|=\rho} |f(z,w)| < \infty.$$

Cauchy's coefficient estimate gives, for every $k \ge 0$ and $|z| \le R$,

$$|a_k(z)| \le M_{R,\rho}\rho^{-k}.$$

Hence for $|z| \le R$ and $|w| \le S$,

$$\sum_{k=0}^{\infty} |a_k(z)w^k| \le M_{R,\rho} \sum_{k=0}^{\infty} (S/\rho)^k < \infty.$$

The Weierstrass test gives normal convergence on every compact subset of Δ^2. Thus

$$\widetilde{f}(z,w) = \sum_{k=0}^{\infty} a_k(z)w^k$$

defines a holomorphic function on the full bidisc. On the annular arm it is the Laurent series of f. On the vertical arm, for each $|z| < r$, it agrees with f first on the annular overlap and then on the whole w-disc by the one-variable identity theorem. Therefore $\widetilde{f}|_H = f$. The extension is unique because two extensions agree on the nonempty open set $\Delta_r \times \Delta$ inside the connected bidisc.

For a calculation that can be carried out on the board, take

$$f(z,w) = \frac{1}{2 - z - w}.$$

Since $|z + w| < 2$ on Δ^2, this function is holomorphic there, and on H its coefficients are

$$a_k(z) = \frac{1}{(2-z)^{k+1}} \quad (k \ge 0), \qquad a_k(z) = 0 \quad (k < 0).$$

The extension constructed above is the familiar geometric series

$$\widetilde{f}(z,w) = \sum_{k=0}^{\infty} \frac{w^k}{(2-z)^{k+1}} = \frac{1}{2 - z - w}.$$

This example makes the coefficient construction visible, although the extension is already apparent from the final rational formula.

The function $1/w$ illustrates the obstruction in the opposite direction. It is holomorphic on the annular arm, where its only Laurent coefficient is $a_{-1} \equiv 1$. It cannot be the restriction of a function in $\mathcal{O}(H)$, because on the vertical arm such a function would have to be holomorphic at $w = 0$ and hence would have $a_{-1} = 0$ for $|z| < r$. More generally, no negative Laurent term can be supported only over part of the parameter disc: once its coefficient vanishes on Δ_r, the identity theorem forces it to vanish on all of Δ.

This elementary calculation explains why an inner hole is often invisible to holomorphic functions in several variables. A domain of holomorphy must prevent analytic discs from sweeping through its boundary in this way. The Levi form is the infinitesimal test for precisely that failure.

2.3.2 Boundary calculations worth knowing

For the unit ball, $\rho(z) = |z|^2 - 1$ and

$$\mathcal{L}_\rho(\xi) = |\xi|^2.$$

The boundary is strictly pseudoconvex. More generally, every bounded real convex domain with C^2 boundary is pseudoconvex because the real Hessian restricted to a complex line controls the Levi form.

The spherical annulus $\{r < |z| < 1\} \subset \mathbb{C}^n$ has two boundary components. The outer one is pseudoconvex with the outward defining function $|z|^2 - 1$. Near the inner boundary the defining function is $r^2 - |z|^2$, whose Levi form is negative on complex tangent vectors. For $n \geq 2$ the annulus is therefore not pseudoconvex, in agreement with Hartogs extension across the inner ball.

For a product domain such as a polydisc, the natural boundary has corners and the C^2 Levi test does not apply globally. Nevertheless

$$\psi(z) = -\sum_{j=1}^{n} \log(1 - |z_j|^2)$$

is a smooth strictly psh exhaustion. Exhaustion functions are therefore more robust than boundary defining functions.

Remark (Holomorphic hulls as a compactness test)

For $K \Subset \Omega$, the holomorphic hull

$$\widehat{K}_\Omega = \{z \in \Omega : |f(z)| \leq \sup_K |f| \text{ for every } f \in \mathcal{O}(\Omega)\}$$

contains everything that holomorphic functions cannot separate from K by size. Pseudoconvexity makes these hulls relatively compact. Thus the Levi problem turns a local Hessian inequality into a global statement about all holomorphic functions at once.

2.3.3 Boundary Levi forms by direct calculation

For the unit ball take $r(z) = |z|^2 - 1$. Then

$$r_j = \bar{z}_j, \qquad r_{j\bar{k}} = \delta_{jk}.$$

At $p \in \partial B$, a vector v is complex tangential precisely when $\sum_j \bar{p}_j v_j = 0$. Its boundary Levi form is

$$\mathcal{L}_r(v) = \sum_{j,k} r_{j\bar{k}} v_j \bar{v}_k = |v|^2 > 0.$$

Thus the sphere is strictly pseudoconvex.

If $\widetilde{r} = hr$ with $h > 0$ smooth, then

$$\partial\bar{\partial}(hr) = h\partial\bar{\partial}r + r\partial\bar{\partial}h + \partial h \wedge \bar{\partial}r + \partial r \wedge \bar{\partial}h.$$

On $\partial\Omega$, $r = 0$; on complex tangent vectors, $\partial r(v) = \bar{\partial} r(\bar{v}) = 0$. Therefore

$$\mathcal{L}_{\widetilde{r}}(v) = h\mathcal{L}_r(v).$$

The sign of the boundary Levi form is independent of the chosen defining function.

For the product domain Δ^2 the boundary is not smooth at the corners, so no single smooth defining function describes all of it. Nevertheless

$$\Phi(z) = -\log(1 - |z_1|^2) - \log(1 - |z_2|^2)$$

is a smooth strictly psh exhaustion. Hence pseudoconvexity is naturally phrased by exhaustions, not only by smooth boundary Levi forms.

Hartogs extension by Laurent coefficients

Let

$$H = \{(z,w) : |z| < 1,\ 1-\varepsilon < |w| < 1\} \cup \{(z,w) : |z| < \varepsilon,\ |w| < 1\}$$

be a Hartogs figure and let $f \in \mathcal{O}(H)$. On the annular part expand in w:

$$f(z,w) = \sum_{k\in\mathbb{Z}} a_k(z) w^k, \qquad a_k(z) = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=r} \frac{f(z,\zeta)}{\zeta^{k+1}}\, \mathrm{d}\zeta,$$

where $1-\varepsilon < r < 1$. The integral shows that every a_k is holomorphic in z for $|z| < 1$. When $|z| < \varepsilon$, the function is holomorphic for $|w| < 1$, so Cauchy's theorem gives $a_k(z) = 0$ for all $k < 0$. The identity theorem then gives $a_k \equiv 0$ on the connected disc $|z| < 1$ for every $k < 0$.

The remaining power series

$$\widetilde{f}(z,w) = \sum_{k\geq 0} a_k(z) w^k$$

converges normally on smaller bidiscs by Cauchy estimates and extends f to Δ^2. This is the precise mechanism behind the Hartogs phenomenon: the inner hole cannot support negative Laurent coefficients once a thin cylinder fills it for one open set of parameter values.

Why $-\log(-r)$ is psh near a pseudoconvex boundary

Assume $r < 0$ on Ω and $\mathrm{d}r \neq 0$ on the boundary. A calculation gives

$$\partial\bar\partial[-\log(-r)] = -\frac{\partial\bar\partial r}{r} + \frac{\partial r \wedge \bar\partial r}{r^2}.$$

The second term controls the complex normal direction. On tangential vectors the second term vanishes and the first has the sign of the boundary Levi form because $r < 0$. Strict positivity in the normal direction and continuity show that, after possibly replacing r by $e^{Ar}r$ with A large, this is psh in a collar of the boundary. Gluing it with a bounded interior function produces a psh exhaustion.

The logical content of the Levi problem

For a domain $\Omega \subset \mathbb{C}^n$, the chain of implications is

$$\text{psh exhaustion} \Longrightarrow \text{holomorphic convexity} \Longrightarrow \text{domain of holomorphy}.$$

The difficult implication uses L^2 solution of $\bar\partial$ to construct holomorphic functions that become large along a sequence approaching the boundary. The compatibility condition is enforced by starting with a cutoff meromorphic model and correcting its $\bar\partial$-error. Thus the theorem does not claim that an arbitrary psh exhaustion is holomorphic; it says that its Levi positivity supplies the estimate needed to manufacture enough holomorphic functions.

2.3.4 A proof ledger for the Levi problem

A full proof uses results developed later, so we record exactly what each quoted input contributes. Let ρ be a continuous psh exhaustion of $\Omega \subset \mathbb{C}^n$. Choose regular values $c_j \to +\infty$ and set

$$\Omega_j = \{\rho < c_j\} \Subset \Omega_{j+1}.$$

Regularized maxima replace ρ near each compact sublevel by a smooth strictly psh function. The Oka–Cartan bump lemma then decomposes the passage from Ω_j to Ω_{j+1} into finitely many strictly pseudoconvex bumps.

On one bump, start from a cutoff of a meromorphic model having a pole at a selected point outside Ω_j. Its failure to be holomorphic is a compactly supported form $g = \bar\partial(\chi h)$. Hörmander's estimate, with a large multiple of the psh weight, solves

$$\bar\partial u = g, \qquad \int_{\Omega_{j+1}} |u|^2 e^{-A\rho} \le C_A \int_{\Omega_{j+1}} |g|^2 e^{-A\rho}.$$

The correction $\chi h - u$ is holomorphic. The weight makes u small on Ω_j, while the pole model is large at the chosen exterior point. Repeating the construction over the finite bumps yields Runge approximation $O(\Omega_{j+1}) \to O(\Omega_j)$ with quantitative control.

A diagonal series of the resulting functions diverges along every sequence leaving compact subsets of Ω. Hence holomorphic hulls of compact sets remain compact, and Ω is a domain of holomorphy. The converse is the Cartan–Thullen theorem: failure of pseudoconvexity creates a Hartogs figure across which every holomorphic function extends. Thus the two deep analytic inputs are the bump lemma and the weighted $\bar\partial$ estimate; the remaining steps are exhaustion, cutoff correction, Runge approximation, and a diagonal argument.

2.3.5 Self-study workshop: reading pseudoconvexity from a defining function

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall the Levi matrix, regular level sets, complex tangent spaces, and proper exhaustions. The sign of a boundary Levi form is meaningful only after fixing the convention $\Omega = \{\rho < 0\}$.

The boundary calculation as an algorithm. At $p \in \partial\Omega$ with $\mathrm{d}\rho(p) \neq 0$, a $(1,0)$ vector $v = \sum v_j \partial/\partial z_j$ is complex tangential when

$$\partial\rho(v) = \sum_j \rho_j(p) v_j = 0.$$

The boundary Levi form is the restriction of $(\rho_{j\bar k})$ to this hyperplane. For the ellipsoid

$$\Omega = \{|z_1|^2 + 2|z_2|^2 < 1\}, \qquad \rho = |z_1|^2 + 2|z_2|^2 - 1,$$

we have

$$\mathcal{L}_\rho(v) = |v_1|^2 + 2|v_2|^2 > 0$$

for every nonzero vector, hence in particular on the complex tangent space. The domain is strongly pseudoconvex.

Changing the defining function does not change the boundary sign. If $\widetilde\rho = h\rho$ with $h > 0$, then on a tangential vector at the boundary all terms containing ρ or $\partial\rho(v)$ disappear, leaving

$$\mathcal{L}_{\widetilde\rho}(v) = h\,\mathcal{L}_\rho(v).$$

This is the calculation that makes pseudoconvexity intrinsic.

Near a pseudoconvex boundary, the candidate $-\log(-\rho)$ has Hessian

$$\partial\bar\partial[-\log(-\rho)] = \frac{\partial\bar\partial\rho}{-\rho} + \frac{\partial\rho \wedge \bar\partial\rho}{\rho^2}.$$

The second term controls the complex normal direction, while the first carries the tangential Levi form. A small strictly psh correction handles any remaining compact interior region when constructing an exhaustion.

Checkpoint (where each hypothesis is used)

(1) Why is the inner boundary of a spherical shell in $\mathbb{C}^n$, $n \geq 2$, not pseudoconvex with the domain-side convention?
(2) Does pseudoconvexity make the boundary Levi term in an L^2 identity zero?
(3) What extra condition turns a psh function into an exhaustion?

Solution (checkpoint). At the inner boundary a defining function for the shell has Levi form negative on complex tangential directions, so the sign fails. Pseudoconvexity makes the boundary term nonnegative, not zero. An exhaustion must be proper: every sublevel set must be relatively compact in the domain. ◇

Exit criterion

Given a smooth defining function, you can compute the complex tangent equation, restrict the Levi matrix, verify invariance under positive rescaling, and explain how a boundary calculation enters a global exhaustion.

Summary. Pseudoconvexity is the correct complex analogue of convexity. Boundary Levi forms, psh exhaustions, holomorphic convexity, and domains of holomorphy become one cluster of equivalent notions through the Levi problem.

Exercise 2.42: Ball and polydisc

Construct smooth strictly psh exhaustions of the unit ball and the polydisc. Explain how to regularize the nonsmooth maximum naturally arising for the polydisc.

Exercise 2.43: Changing a defining function

Prove that replacing ρ by $a\rho$ multiplies its Levi form by a on the complex tangent space of the boundary.

Optional handout: The $\bar{\partial}$-Neumann problem

This handout is optional. No later argument in the core lectures depends on it.

On a smoothly bounded domain $\Omega \subset \mathbb{C}^n$, solving $\bar{\partial} u = f$ with minimum L^2 norm is governed by the complex Laplacian

$$\Box_q = \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}$$

with the $\bar{\partial}$-Neumann boundary condition. If $\Box_q$ has a bounded inverse N_q on the orthogonal complement of its kernel, the canonical solution is $u = \bar{\partial}^* N_q f$.

For a defining function ρ, a smooth $(0,q)$-form v belongs to the domain of $\bar{\partial}^*$ only if contraction of v with $\bar{\partial}\rho$ vanishes on the boundary. In the Morrey–Kohn identity, integration by parts then produces the boundary term

$$\int_{\partial\Omega} \sum_{|K|=q-1} \sum_{j,k} \rho_{j\bar{k}} v_{jK} \overline{v_{kK}} \, \frac{\mathrm{d}S}{|\nabla \rho|}.$$

Equivalently one may normalize the defining function along the boundary and absorb this positive density into surface measure. Pseudoconvexity makes the term nonnegative. This is the boundary counterpart of curvature positivity in the Bochner–Kodaira–Nakano identity.

Strict pseudoconvexity gives subelliptic estimates: roughly, one controls a positive fraction of a Sobolev derivative of v by $\|\bar\partial v\|+\|\bar\partial^* v\|$. Compactness of the relevant Sobolev embedding then implies that N_q is compact. Weak pseudoconvexity can lose subellipticity and leads to subtle questions about boundary type.

Remark (Two solution mechanisms)

Hörmander's weighted estimate obtains coercivity from an interior psh weight. The $\bar\partial$-Neumann method can obtain it from the boundary Levi form. Many arguments mix the two sources of positivity.

2.4 Lecture 8: Basic sheaf theory: local data, stalks, and exactness

Week 4 material • 90 minutes

Notation for this lecture

X is a topological space, later specialized to a complex manifold, and $U, V, W \subset X$ are open sets. A presheaf or sheaf is denoted by $\mathcal{F}$; its sections on U are $\mathcal{F}(U)$, the restriction of $s \in \mathcal{F}(U)$ to $V \subset U$ is $s|_V$, and its stalk and the germ of s at x are $\mathcal{F}_x$ and s_x. On a complex manifold, $\mathcal{O}_X(U) = \mathcal{O}(U)$ is the ring of holomorphic functions on U and $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$ is the maximal ideal of germs vanishing at x. The ideal sheaf of the reduced point $\{x\}$ is denoted $\mathcal{I}_{\{x\}} \subset \mathcal{O}_X$; its stalk at x is $\mathfrak{m}_x$, whereas its stalk at any $y \neq x$ is $\mathcal{O}_{X,y}$. The symbols $\Gamma(X, \mathcal{F}) = \mathcal{F}(X)$ and $H^0(X, \mathcal{F}) = \Gamma(X, \mathcal{F})$ both denote global sections.

Prerequisite results used below

Open covers and restrictions. If $U = \bigcup_i U_i$, an assertion about functions is *local* when it can be checked on the U_i. Restricting a function first from U to V and then from V to W gives the same result as restricting directly from U to W.

Modules and exact sequences. For a ring R, an R-module is an abelian group on which elements of R act compatibly with addition and multiplication. A sequence $A \xrightarrow{a} B \xrightarrow{b} C$ is exact at B when $\operatorname{im} a = \ker b$. We will explain how these notions are localized over a complex manifold.

Identity principle. If two holomorphic functions on a connected complex manifold agree on a nonempty open set, then they agree everywhere. Equivalently, a holomorphic function vanishing on a nonempty open set is identically zero.

No prior sheaf theory is assumed. Presheaves, sheaves, germs, stalks, sheaf morphisms, sheaf images, exactness, locally free sheaves, and coherent sheaves are all defined in this lecture before they are used.

Why these ideas are needed

A holomorphic function is simultaneously a global object and a collection of local functions that agree on overlaps. The same pattern appears for holomorphic forms, sections of vector bundles, local defining equations, and local solutions of $\bar{\partial}$-equations. Writing a separate gluing argument every time would hide this common structure. A sheaf isolates it.

The central distinction is between *local solvability* and *global solvability*. A map of sheaves may be surjective because every germ has a local preimage, while the corresponding map on global sections is not surjective because the local preimages do not glue. Germs and stalks make the local statement precise; exact sequences organize it; the first Čech cocycle records the failure to glue. Lecture 9 will turn that cocycle into cohomology and only then state Cartan's theorems. Thus this lecture assumes no previous exposure to sheaf theory.

Plan for the lecture. 0–20 min: presheaves and the two sheaf axioms; 20–38: germs, stalks, and local rings; 38–58: morphisms, sheaf images, and stalkwise exactness; 58–75: locally free and coherent sheaves; 75–90: a first gluing obstruction.

2.4.1 Why complex geometry needs sheaves

Holomorphic functions cannot be discussed only one open set at a time. If $V \subset U$, a function on U restricts to one on V; conversely, functions defined on the members of an open cover glue when they

agree on overlaps. A sheaf isolates these two operations.

2.4.2 How to read the notation

Before giving the axioms, it is useful to separate the three operations that the notation compresses. Fix a point $x \in V \subset U \subset X$, where U and V are open. For any sheaf $\mathcal{F}$ there are canonical maps

$$\mathcal{F}(U) \longrightarrow \mathcal{F}(V) \longrightarrow \mathcal{F}_x, \qquad s \longmapsto s|_V \longmapsto s_x.$$

The first arrow restricts the domain; the second forgets the particular neighborhood and remembers only the behavior sufficiently close to x. There is usually no canonical arrow in the opposite direction: a germ need not extend to a prescribed larger open set. This asymmetry is the elementary source of the local-to-global problems treated by Cartan's theorems.

For a concrete instance, let $D(0,r) = \{z \in \mathbb{C} : |z| < r\}$ and $f(z) = (3-z)^{-1} \in \mathcal{O}(D(0,2))$. Restriction gives the same formula on $D(0,1)$, and the germ $f_0 \in \mathcal{O}_{\mathbb{C},0}$ can equally be represented by that restriction or by the convergent series $\sum_{k\geq 0} z^k/3^{k+1}$ on any sufficiently small disc. These are not three different functions: they are three levels of information about one local object.

Definition 2.44: Presheaf and sheaf

A *presheaf* $\mathcal{F}$ on a topological space X consists of an object $\mathcal{F}(U)$ for every open set $U \subset X$ and a restriction map

$$\rho_V^U : \mathcal{F}(U) \longrightarrow \mathcal{F}(V), \qquad s \longmapsto s|_V, \quad V \subset U,$$

such that $s|_U = s$ and $(s|_V)|_W = s|_W$ whenever $W \subset V \subset U$. The objects may be sets, abelian groups, rings, or modules, and all restriction maps must preserve the corresponding structure. The presheaf is a *sheaf* if, for every open cover $U = \bigcup_i U_i$, it also satisfies:

(1) *locality*: if $s, t \in \mathcal{F}(U)$ and $s|_{U_i} = t|_{U_i}$ for every i, then $s = t$;
(2) *gluing*: if $s_i \in \mathcal{F}(U_i)$ and $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ for all i, j, then there is $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for every i.

Locality makes the glued section unique.

Definition 2.45: Morphisms of presheaves and sheaves

Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves of sets on X. A *morphism of presheaves* $\alpha : \mathcal{F} \to \mathcal{G}$ is a family of maps

$$\alpha_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U), \qquad U \subset X \text{ open},$$

such that restriction before or after applying α gives the same answer. Explicitly, for every inclusion $V \subset U$,

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\alpha_U} & \mathcal{G}(U) \\ {\scriptstyle \rho_V^U}\downarrow & & \downarrow{\scriptstyle \rho_V^U} \\ \mathcal{F}(V) & \xrightarrow{\alpha_V} & \mathcal{G}(V) \end{array} \qquad \text{commutes.}$$

If the presheaves take values in abelian groups, rings, or modules, every α_U is required to be a homomorphism of the corresponding kind. A *morphism of sheaves* is a morphism of the underlying presheaves. Identity morphisms and composites are defined open set by open set.

The compatibility square is not cosmetic. It says that α is one operation defined on local data, rather than an unrelated map for each open set. It also produces, for every $x \in X$, an induced map on stalks

$$\alpha_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x, \qquad \alpha_x(s_x) = (\alpha_U(s))_x.$$

This formula is well defined. Indeed, if $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ represent the same germ at x, then they agree on some neighborhood $W \subset U \cap V$ of x. The compatibility square gives $\alpha_U(s)|_W = \alpha_W(s|_W) = \alpha_W(t|_W) = \alpha_V(t)|_W$, so their images have the same germ.

The model example is the *structure sheaf* $\mathcal{O}_X$, defined by $\mathcal{O}_X(U) = \mathcal{O}(U)$. Holomorphicity is local, so compatible holomorphic functions glue to a holomorphic function. The same argument gives the sheaves C^0_X of continuous functions, $\mathcal{A}_X$ of smooth functions, and Ω^p_X of holomorphic p-forms. If $E \to X$ is a holomorphic vector bundle, then $U \mapsto H^0(U, E)$ is the sheaf of holomorphic sections of E.

The word "constant" hides an instructive distinction that will reappear in the kernel of the differential.

Definition 2.46: Constant presheaf and locally constant sheaf

Fix an abelian group A, regarded as a discrete topological group. The *constant presheaf* A_{pre} is defined by

$$A_{\mathrm{pre}}(U) = A \quad (U \neq \varnothing), \qquad A_{\mathrm{pre}}(\varnothing) = 0,$$

with identity restriction maps between nonempty open sets. Thus a section has one single value on all of U.

The *locally constant sheaf* with fiber A, denoted $\underline{A}_X$, is

$$\underline{A}_X(U) = \{a : U \to A : \text{ every } x \in U \text{ has a neighborhood on which } a \text{ is constant}\},$$

with pointwise addition and the usual restriction of functions. A sheaf $\mathcal{L}$ is called *locally constant* if every $x \in X$ has an open neighborhood U and an abelian group A_x such that $\mathcal{L}|_U \simeq \underline{A_x}_U$.

Let us check the assertions implicit in this definition. Compatible locally constant functions $a_i : U_i \to A$ glue pointwise to a function $a : U \to A$. Near x, choose i with $x \in U_i$; the function a_i is constant on a smaller neighborhood of x, so the glued function is locally constant. Hence $\underline{A}_X$ is a sheaf. On a complex manifold, every open set is locally connected, so its connected components are open and the displayed definition permits one independent value on each component.

By contrast, A_{pre} is generally not a sheaf. If $U = U_1 \sqcup U_2$ and $a_1 \neq a_2$ in A, the constant sections a_j on U_j agree on the empty overlap, but no section of $A_{\mathrm{pre}}(U)$ restricts to both. The natural morphism $A_{\mathrm{pre}} \to \underline{A}_X$ sends a value to the corresponding constant function, and its induced map on every stalk is an isomorphism. Consequently

$$(A_{\mathrm{pre}})^{\#} \simeq \underline{A}_X.$$

Thus sheafification does not change the local values; it supplies the missing gluing across disconnected pieces.

Definition 2.47: Germ and stalk

Let $x \in X$. A *germ* at x is an equivalence class of pairs (U, s) with $x \in U$ and $s \in \mathcal{F}(U)$. Two pairs (U, s) and (V, t) are equivalent if there is an open neighborhood $W \subset U \cap V$ of x on which $s|_W = t|_W$. The set of germs is the *stalk*

$$\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U),$$

and the germ of s at x is denoted s_x.

Definition 2.48: Sheafification

Let $\mathcal{P}$ be a presheaf and let $\mathcal{P}_x$ denote the set of its germs at x. Its *sheafification* $\mathcal{P}^{\#}$ is defined as follows. A section $\sigma \in \mathcal{P}^{\#}(U)$ assigns to every $x \in U$ a germ $\sigma(x) \in \mathcal{P}_x$, subject to the local representability condition: for every $x \in U$, there are an open neighborhood $V \subset U$ of x and a section $t \in \mathcal{P}(V)$ such that

$$\sigma(y) = t_y \qquad \text{for every } y \in V.$$

Restrictions are obtained by restricting the function $x \mapsto \sigma(x)$. There is a natural morphism

$$\eta : \mathcal{P} \longrightarrow \mathcal{P}^{\#}, \qquad \eta_U(t)(x) = t_x.$$

The local representability condition proves the identity axiom for $\mathcal{P}^{\#}$ because equality is checked pointwise. Compatible germ-valued functions glue by taking their union, and local representatives for the pieces remain local representatives for that union; this proves the gluing axiom. Moreover, $\eta_x : \mathcal{P}_x \to \mathcal{P}^{\#}_x$ is bijective for every x. Thus sheafification changes the gluing behavior without changing any stalk. It also has the following universal property: every morphism from $\mathcal{P}$ to a sheaf $\mathcal{G}$ factors uniquely through $\eta : \mathcal{P} \to \mathcal{P}^{\#}$. For the presheaf of globally constant functions above, a locally representable germ-valued section is exactly a locally constant function, as claimed.

For example, $\mathcal{O}_{X,x}$ is the ring of holomorphic function germs at x. It is a local ring: its unique maximal ideal is

$$\mathfrak{m}_x = \{f_x \in \mathcal{O}_{X,x} : f(x) = 0\}.$$

Indeed, a germ is invertible precisely when its value at x is nonzero; after shrinking, its reciprocal is holomorphic. In a coordinate z centered at $0 \in \mathbb{C}$, the stalk is the convergent power-series ring $\mathbb{C}\{z\}$ and $\mathfrak{m}_0 = (z)$.

The stalk records exactly local information. If $s, t \in \mathcal{F}(U)$, then $s = t$ if and only if $s_x = t_x$ for every $x \in U$. The nontrivial direction follows because equality of the two germs gives, for each x, a neighborhood U_x on which the sections agree; locality for the cover $U = \bigcup_x U_x$ then gives $s = t$.

There are therefore three levels that must not be confused. A section $s \in \mathcal{F}(U)$ is defined on one specified open set; its germ s_x forgets everything away from x; the stalk $\mathcal{F}_x$ contains all germs at x, including germs that may have no representative on the original set U. Passing from sections to germs is what makes exactness a local question. Passing back from compatible germs or local sections to one global section is a gluing problem, and that reverse passage is where cohomology will enter.

Definition 2.49: $\mathcal{O}_X$-modules and their morphisms

A sheaf $\mathcal{F}$ is an $\mathcal{O}_X$-*module* if every $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module and restriction commutes with multiplication by holomorphic functions. A morphism $\alpha : \mathcal{F} \to \mathcal{G}$ of $\mathcal{O}_X$-modules is a sheaf morphism in the sense of Definition 2.45 for which every α_U is $\mathcal{O}_X(U)$-linear. The stalk map defined above is then $\mathcal{O}_{X,x}$-linear.

Two constructions used repeatedly below deserve explicit notation. If $U \subset X$ is open, the *restriction sheaf* $\mathcal{F}|_U$ is the sheaf on U defined by $(\mathcal{F}|_U)(V) = \mathcal{F}(V)$ for every open set $V \subset U$. If $i : Y \hookrightarrow X$ is a continuous inclusion and $\mathcal{G}$ is a sheaf on Y, its *direct image* is the sheaf on X defined by

$$(i_*\mathcal{G})(U) = \mathcal{G}(U \cap Y).$$

No extension of a section away from Y is implicit in the symbol i_*.

Definition 2.50: Subsheaves, quotients, and ideal products

A *subsheaf* $\mathcal{G} \subset \mathcal{F}$ of $\mathcal{O}_X$-modules is a sheaf whose sections $\mathcal{G}(U)$ form $\mathcal{O}_X(U)$-submodules of $\mathcal{F}(U)$, compatibly with restriction. The *quotient sheaf* $\mathcal{F}/\mathcal{G}$ is the sheafification of the presheaf $U \mapsto \mathcal{F}(U)/\mathcal{G}(U)$; consequently

$$(\mathcal{F}/\mathcal{G})_x \simeq \mathcal{F}_x/\mathcal{G}_x.$$

If $\mathcal{I} \subset \mathcal{O}_X$ is an ideal sheaf, then $\mathcal{I}\mathcal{F} \subset \mathcal{F}$ is the subsheaf whose local sections can locally be written as finite sums $f_1 s_1 + \cdots + f_N s_N$ with $f_j \in \mathcal{I}$ and $s_j \in \mathcal{F}$. On stalks, $(\mathcal{I}\mathcal{F})_x = \mathcal{I}_x \mathcal{F}_x$.

The *support* of a sheaf $\mathcal{F}$ is

$$\operatorname{Supp} \mathcal{F} = \{x \in X : \mathcal{F}_x \neq 0\}.$$

For a general sheaf this set need not be closed, but it is closed for every coherent analytic sheaf used below. The definition is stalkwise: saying that $\mathcal{F}$ is supported on a subset A means $\mathcal{F}_x = 0$ for $x \notin A$.

Typical $\mathcal{O}_X$-modules are $\mathcal{O}_X^{\oplus r}$, the sheaf of sections of a holomorphic vector bundle, and an ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$. For a closed analytic subspace $i : Y \hookrightarrow X$, restriction of holomorphic functions is a morphism $\mathcal{O}_X \to i_* \mathcal{O}_Y$, and its kernel is the *ideal sheaf*

$$\mathcal{I}_Y = \ker(\mathcal{O}_X \longrightarrow i_* \mathcal{O}_Y).$$

If Y is reduced, this says concretely that $\mathcal{I}_Y(U) = \{f \in \mathcal{O}_X(U) : f|_{Y \cap U} = 0\}$. The kernel definition also remembers nilpotent structure when Y is nonreduced.

The reduced point already contains the bookkeeping needed in the proof of Cartan A. Let $X = \mathbb{C}$, $Y = \{0\}$, and write $\mathcal{I}_0 = \mathcal{I}_{\{0\}}$. If $0 \in U$, then

$$\mathcal{I}_0(U) = \{f \in \mathcal{O}(U) : f(0) = 0\};$$

if $0 \notin U$, the condition is empty and $\mathcal{I}_0(U) = \mathcal{O}(U)$. Consequently $(\mathcal{I}_0)_0 = (z) \subset \mathbb{C}\{z\}$, whereas $(\mathcal{I}_0)_x = \mathcal{O}_{\mathbb{C},x}$ for $x \neq 0$. Thus an ideal sheaf of a point is not the zero sheaf away from the point; its *quotient* $\mathcal{O}_{\mathbb{C}}/\mathcal{I}_0$ is the sheaf supported at the point.

We now define kernels and images carefully. The distinction between two possible notions of image is the first genuinely sheaf-theoretic point in the lecture.

Definition 2.51: Kernel, presheaf image, sheaf image, and cokernel

Let $\alpha : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups. The *kernel* is defined sectionwise:

$$(\ker \alpha)(U) = \ker\big(\alpha_U : \mathcal{F}(U) \to \mathcal{G}(U)\big).$$

The *presheaf image* is the subpresheaf

$$(\operatorname{im}_{\mathrm{pre}} \alpha)(U) = \operatorname{im}\big(\alpha_U : \mathcal{F}(U) \to \mathcal{G}(U)\big).$$

The *sheaf image* $\operatorname{im} \alpha$ is the sheafification of this presheaf. Equivalently, it is the subsheaf of $\mathcal{G}$ given by

$$(\operatorname{im} \alpha)(U) = \left\{ g \in \mathcal{G}(U) : \begin{array}{l} \text{for every } x \in U \text{ there are an open } V_x \subset U,\ x \in V_x, \\ \text{and } f_x \in \mathcal{F}(V_x) \text{ with } \alpha_{V_x}(f_x) = g|_{V_x} \end{array} \right\}.$$

Finally, the *cokernel* is the sheafification

$$\operatorname{coker} \alpha = (U \longmapsto \mathcal{G}(U)/\operatorname{im} \alpha_U)^{\#}.$$

The same definitions apply to sheaves of modules, using submodules and quotient modules.

The displayed image condition should be read as "locally liftable," not "globally liftable." We verify that these definitions do what they should.

Proposition 2.52: Why the kernel is a sheaf and the image must be sheafified

For a sheaf morphism $\alpha : \mathcal{F} \to \mathcal{G}$ of abelian groups:

(1) the sectionwise kernel is already a sheaf;
(2) the locally liftable sections in the definition form a sheaf, canonically isomorphic to $(\operatorname{im}_{\mathrm{pre}} \alpha)^{\#}$;
(3) for every $x \in X$ there are natural identifications

$$(\ker \alpha)_x = \ker \alpha_x, \qquad (\operatorname{im} \alpha)_x = \operatorname{im} \alpha_x, \qquad (\operatorname{coker} \alpha)_x = \operatorname{coker} \alpha_x.$$

Consequently the canonical morphism $\mathcal{F}/\ker \alpha \to \operatorname{im} \alpha$ is an isomorphism.

Proof. Suppose $U = \bigcup_i U_i$ and $f_i \in (\ker \alpha)(U_i)$ agree on all overlaps. They glue in $\mathcal{F}$ to $f \in \mathcal{F}(U)$. Since $\alpha_U(f)|_{U_i} = \alpha_{U_i}(f_i) = 0$ for every i, locality in $\mathcal{G}$ gives $\alpha_U(f) = 0$. Thus f lies in the kernel, proving the gluing axiom; locality is inherited from $\mathcal{F}$.

For the image, let locally liftable sections $g_i \in \mathcal{G}(U_i)$ agree on overlaps. They glue in $\mathcal{G}$ to $g \in \mathcal{G}(U)$. Given $x \in U$, choose i with $x \in U_i$. On a smaller neighborhood of x, the section g_i, and hence g, has a lift to $\mathcal{F}$. Thus g satisfies the local-lifting condition. Locality is again inherited from $\mathcal{G}$, so these sections define a sheaf. It contains the presheaf image and has the same germs; the universal property of sheafification identifies it with $(\operatorname{im}_{\mathrm{pre}} \alpha)^{\#}$.

The kernel identity follows directly from representatives of germs. For the image identity, a germ of a locally liftable section has, after shrinking, the form $\alpha(f)$ and hence lies in $\operatorname{im} \alpha_x$; the reverse inclusion is immediate. Sheafification preserves stalks, so the cokernel identity follows from the corresponding quotient of stalks. The last morphism is an isomorphism on every stalk by the ordinary first isomorphism theorem for groups, hence is an isomorphism of sheaves. □

Here is the promised explicit failure of the sheaf axiom for a presheaf image.

Example 2.53: The presheaf image of the exponential map is not a sheaf

On $X = \mathbb{C}^*$ consider the morphism of sheaves of abelian groups

$$E : \mathcal{O}_X \longrightarrow \mathcal{O}_X^*, \qquad E_U(f) = e^f,$$

where addition is used on $\mathcal{O}_X$ and multiplication on $\mathcal{O}_X^*$. Cover $\mathbb{C}^*$ by small simply connected discs $U_i \Subset \mathbb{C}^*$. On each U_i the coordinate function $z|_{U_i}$ has a holomorphic logarithm ℓ_i, and therefore

$$z|_{U_i} = e^{\ell_i} \in (\operatorname{im}_{\mathrm{pre}} E)(U_i).$$

These sections agree on overlaps because they are all restrictions of the same function z.
If $\operatorname{im}_{\mathrm{pre}} E$ were a sheaf, its gluing axiom would give an element of $(\operatorname{im}_{\mathrm{pre}} E)(\mathbb{C}^*)$ restricting to all $z|_{U_i}$. As this presheaf is a subpresheaf of $\mathcal{O}_X^*$, locality forces that element to be z itself. Hence there would be $f \in \mathcal{O}(\mathbb{C}^*)$ with $e^f = z$. Differentiating gives $f'(z) = 1/z$, but for the unit circle γ one has

$$0 = \int_\gamma f'(z)\,\mathrm{d}z \quad \text{whereas} \quad \int_\gamma \frac{\mathrm{d}z}{z} = 2\pi\mathrm{i},$$

a contradiction. Thus the compatible local sections in the presheaf image do not glue there. After sheafification they do glue: every nowhere-zero holomorphic function has a logarithm locally, so $\operatorname{im} E = \mathcal{O}_X^*$ as sheaves.

The same ideas identify a familiar kernel. Let $\mathrm{d} : \mathcal{O}_X \to \Omega^1_X$ be the exterior derivative, viewed as a morphism of sheaves of complex vector spaces. On an open set U, the equation $\mathrm{d}f = 0$ says that f is constant on each connected component of U, equivalently locally constant. Therefore

$$\ker(\mathrm{d} : \mathcal{O}_X \to \Omega^1_X) = \underline{\mathbb{C}}_X.$$

The right side is the locally constant sheaf, not the constant presheaf: on a disconnected open set the constants on different components may be different.

The category in which a map lives matters. Multiplication by the coordinate,

$$m_z : \mathcal{O}_{\mathbb{C}} \longrightarrow \mathcal{O}_{\mathbb{C}}, \qquad f \longmapsto zf,$$

is a morphism of $\mathcal{O}_{\mathbb{C}}$-modules because $m_z(hf) = h\, m_z(f)$. Its cokernel is the one-dimensional skyscraper at 0, and hence

$$0 \longrightarrow \mathcal{O}_{\mathbb{C}} \xrightarrow{m_z} \mathcal{O}_{\mathbb{C}} \xrightarrow{f \mapsto f(0)} i_*\mathbb{C} \longrightarrow 0, \qquad i : \{0\} \hookrightarrow \mathbb{C},$$

is exact. At $x \neq 0$, the germ z_x is a unit, so $(m_z)_x$ is an isomorphism; at 0, its cokernel is $\mathbb{C}\{z\}/(z) \simeq \mathbb{C}$. By contrast, $\mathrm{d} : \mathcal{O}_{\mathbb{C}} \to \Omega^1_{\mathbb{C}}$ is a morphism of sheaves of complex vector spaces but not of $\mathcal{O}_{\mathbb{C}}$-modules, since $\mathrm{d}(hf) = h\,\mathrm{d}f + f\,\mathrm{d}h$. Cartan A and B concern the $\mathcal{O}_X$-module category.

Proposition 2.54: Stalkwise criterion for exactness

A sequence of sheaves of abelian groups or $\mathcal{O}_X$-modules

$$\mathcal{F}' \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\beta} \mathcal{F}''$$

is exact if and only if

$$\mathcal{F}'_x \xrightarrow{\alpha_x} \mathcal{F}_x \xrightarrow{\beta_x} \mathcal{F}''_x$$

is exact for every $x \in X$.

Proof. Exactness of sheaves gives exactness after taking germs. Conversely, let $s \in \mathcal{F}(U)$ satisfy $\beta_U(s) = 0$. For each $x \in U$, stalkwise exactness gives a germ $t_x \in \mathcal{F}'_x$ with $\alpha_x(t_x) = s_x$. Represent t_x on a neighborhood V_x of x. Equality of germs means that, after shrinking V_x, one has $\alpha(t_x) = s|_{V_x}$. Thus s belongs locally to the image of α, which is exactly the assertion $\ker\beta = \operatorname{im}\alpha$ as sheaves. The same argument at the two ends handles injectivity and surjectivity. □

The ideal sequence is the basic coordinate calculation. If $Y = \{z_{k+1} = \cdots = z_n = 0\} \subset U \subset \mathbb{C}^n$, then restriction sends

$$f(z_1, \ldots, z_n) \longmapsto f(z_1, \ldots, z_k, 0, \ldots, 0),$$

and its kernel is $(z_{k+1}, \ldots, z_n)$. Consequently

$$0 \longrightarrow \mathcal{I}_Y \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Y \longrightarrow 0 \tag{8.1}$$

is exact. Here $(i_*\mathcal{O}_Y)(U) = \mathcal{O}_Y(U \cap Y)$; the symbol i_* merely regards a sheaf on Y as a sheaf on X supported on Y, as defined above. Thus (8.1) is the sheaf-theoretic version of the elementary statement that a holomorphic function restricts to zero on Y exactly when it belongs locally to the defining ideal.

Definition 2.55: Locally free and coherent analytic sheaves

An $\mathcal{O}_X$-module $\mathcal{E}$ is *locally free of rank r* if every point has a neighborhood U on which $\mathcal{E}|_U \simeq \mathcal{O}_U^{\oplus r}$. These are precisely the sheaves of holomorphic sections of rank-r holomorphic vector bundles.

An $\mathcal{O}_X$-module $\mathcal{F}$ is *coherent* if it is locally generated by finitely many sections and the relations among any finite collection of local sections are again locally finitely generated. On a complex manifold this is equivalent to the existence, near every point, of a finite presentation

$$\mathcal{O}_U^{\oplus p} \xrightarrow{A} \mathcal{O}_U^{\oplus q} \longrightarrow \mathcal{F}|_U \longrightarrow 0,$$

where A is a $q \times p$ matrix of holomorphic functions.

Here is how to read the finite presentation without category-theoretic shorthand. The images $s_1, \dots, s_q$ of the standard basis of $\mathcal{O}_U^{\oplus q}$ generate every section of $\mathcal{F}$ locally. A column $(a_1, \dots, a_q)^t$ is a relation when $a_1 s_1 + \cdots + a_q s_q = 0$. Exactness says that all such relations are locally generated by the p columns of the holomorphic matrix A. Thus coherence gives one neighborhood on which both generators and their relations are finite; it is stronger than asking separately that every stalk be a finitely generated module.

Local freeness is the absence of relations after choosing a frame. Coherence allows relations but requires finitely many generators and finitely many relations on one neighborhood, not merely separately at each stalk. The sheaves $\mathcal{O}_X^{\oplus r}$, locally free sheaves, and ideal sheaves of analytic subsets are coherent; the last assertion is Oka's coherence theorem, whose local analytic input is developed in Lecture 10.

The definitions and the warning that a quotient presheaf need not already be a sheaf follow [1, Chapter II, §1]; the locally free–coherent distinction is developed in [1, Chapter II, §3]. The point of coherence is not terminological: kernels and cokernels of holomorphic bundle maps need not remain locally free, whereas coherent analytic sheaves form a category stable under the finite kernel, image, and cokernel operations used below.

If $i : \{x\} \hookrightarrow X$ and V is a vector space, the *skyscraper sheaf* i_*V has sections V on every open set containing x and zero sections on open sets disjoint from x, with the evident restriction maps. It packages data supported at one point. In particular, $i_*(\mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x)$ is the fiber of $\mathcal{F}$ at x, regarded as a sheaf on X.

Lemma 2.56: Nakayama's lemma in the form used below

Let $(A, \mathfrak{m})$ be a local ring and let M be a finitely generated A-module. If $m_1, \dots, m_r \in M$ have images spanning $M/\mathfrak{m}M$ over the residue field $A/\mathfrak{m}$, then $m_1, \dots, m_r$ generate M over A.

Proof. Let $N = Am_1 + \cdots + Am_r$. The spanning assumption says $M = N + \mathfrak{m}M$, so the finite module $Q = M/N$ satisfies $Q = \mathfrak{m}Q$. Choose generators $q_1, \dots, q_s$ of Q. There are $a_{ij} \in \mathfrak{m}$ such that

$$q_i = \sum_{j=1}^{s} a_{ij} q_j.$$

In matrix notation, $(\mathrm{id} - A)q = 0$. Since $\det(\mathrm{id} - A) \equiv 1 \pmod{\mathfrak{m}}$, the determinant is a unit in the local ring. Multiplication by the adjugate matrix gives $q = 0$. Hence $Q = 0$ and $M = N$. □

2.4.3 The first gluing obstruction

Write $\Gamma(X, \mathcal{F}) = \mathcal{F}(X)$ and $H^0(X, \mathcal{F}) = \Gamma(X, \mathcal{F})$ for the global sections. Consider a short exact sequence

$$0 \longrightarrow \mathcal{F}' \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\beta} \mathcal{F}'' \longrightarrow 0 \tag{8.2}$$

Before trying to lift a section, it is important to know exactly what applying $\Gamma(X,-)$ preserves. The induced sequence

$$0 \longrightarrow \Gamma(X,\mathcal{F}') \xrightarrow{\alpha_X} \Gamma(X,\mathcal{F}) \xrightarrow{\beta_X} \Gamma(X,\mathcal{F}'') \tag{8.3}$$

is exact at the first two terms. Indeed, injectivity of α can be checked on stalks, so α_X is injective. Moreover, if $t \in \Gamma(X,\mathcal{F})$ satisfies $\beta_X(t) = 0$, stalkwise exactness shows that t locally belongs to the image of α; the local preimages are unique because α is injective, hence they agree on overlaps and glue to a global preimage in $\Gamma(X,\mathcal{F}')$. This proves $\ker \beta_X = \operatorname{im} \alpha_X$.

There is deliberately no zero at the right of (8.3). Surjectivity of the sheaf map β gives local lifts of a global section of $\mathcal{F}''$, but those lifts may fail to glue. In categorical language, the global-section functor $\Gamma(X,-)$ is *left exact*, not generally exact. The next calculation isolates the first obstruction to right exactness.

Now let $s \in \mathcal{F}''(X)$ be a global section. Surjectivity of β as a sheaf gives only local lifts: choose an open cover $X = \bigcup_i U_i$ and sections $t_i \in \mathcal{F}(U_i)$ with $\beta(t_i) = s|_{U_i}$. On an overlap, $t_i - t_j$ lies in $\ker \beta = \operatorname{im} \alpha$, so there is a unique $a_{ij} \in \mathcal{F}'(U_i \cap U_j)$ with

$$\alpha(a_{ij}) = t_i - t_j.$$

On a triple overlap the three differences telescope, hence

$$a_{ij} + a_{jk} + a_{ki} = 0. \tag{8.4}$$

Such a family is called a Čech 1-cocycle. Replacing t_i by $t_i + \alpha(b_i)$ changes a_{ij} by $b_i - b_j$, called a coboundary. The quotient of cocycles by coboundaries is denoted $\check{H}^1(\mathfrak{U},\mathcal{F}')$ for the cover $\mathfrak{U} = \{U_i\}$.

The class of (a_{ij}) vanishes exactly when the local lifts can be corrected to glue. Indeed, if $a_{ij} = b_i - b_j$, then $t_i - \alpha(b_i) = t_j - \alpha(b_j)$ on every overlap, so the sheaf axiom produces a global lift $t \in \mathcal{F}(X)$. Conversely, a global lift gives zero differences. The intrinsic group $H^1(X,\mathcal{F}')$ records this obstruction independently of the cover; on an acyclic cover it is computed by the displayed Čech quotient. More generally $H^q(X,\mathcal{F})$ denotes sheaf cohomology in degree q. Lecture 9 constructs these groups more systematically. For Cartan B, the operational meaning is already visible: $H^1 = 0$ means that every such local lifting obstruction can be removed.

2.4.4 Self-study workshop: turning local gluing failures into cohomology classes

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall the sheaf axioms, germs and stalks, kernels and cokernels, and an open cover with ordered indices. Keep separate three levels: sections on an open set, germs at a point, and Čech cochains on intersections.

Why exactness is local but splitting is not. A sequence of sheaves

$$0 \longrightarrow \mathcal{F}' \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\beta} \mathcal{F}'' \longrightarrow 0$$

is exact precisely when the sequence of stalks is exact at every point. Surjectivity of β therefore means that a germ of a section of $\mathcal{F}''$ has a lift near each point; it does not promise one lift over the whole open set.

Let $s \in H^0(X,\mathcal{F}'')$. Choose an open cover and local lifts $t_i \in \mathcal{F}(U_i)$. On $U_i \cap U_j$,

$$\beta(t_i - t_j) = s - s = 0,$$

so exactness gives $t_i - t_j = \alpha(a_{ij})$ for a unique $a_{ij} \in \mathcal{F}'(U_i \cap U_j)$. On triple overlaps,

$$a_{ij} + a_{jk} + a_{ki} = 0.$$

Thus (a_{ij}) is a Čech 1-cocycle. Replacing t_i by $t_i + \alpha(b_i)$ changes a_{ij} by the coboundary $b_i - b_j$, so the cohomology class is independent of all choices. It vanishes exactly when the local lifts can be corrected to glue. This is the connecting map $\partial_{\mathrm{conn}} : H^0(X, \mathcal{F}'') \to H^1(X, \mathcal{F}')$.

Checkpoint (track the choices)

(1) Where is exactness at $\mathcal{F}$ used in the construction?
(2) If $a_{ij} = b_i - b_j$, which corrected lifts glue?
(3) Why does stalkwise surjectivity not imply surjectivity on global sections?

Solution (checkpoint). Exactness identifies $t_i - t_j \in \ker \beta$ with a section from $\operatorname{im} \alpha$. If $a_{ij} = b_i - b_j$, then $t_i - \alpha(b_i)$ have zero differences and glue. Global surjectivity can fail because the local lifts may have a nonzero connecting class in $H^1(X, \mathcal{F}')$. ◇

Exit criterion

Starting from a short exact sequence and a global quotient section, you can construct the connecting cocycle, prove independence of choices, and interpret its vanishing as a gluing statement.

Summary. A sheaf records sections, restrictions, and unique gluing, and a sheaf morphism is a family of maps commuting with all restrictions. The constant presheaf sheafifies to the locally constant sheaf. Kernels are computed sectionwise, whereas images and cokernels must in general be sheafified; the exponential map on $\mathbb{C}^*$ gives an explicit failure of the gluing axiom for a presheaf image. Germs and stalks retain exactly the local information, while stalkwise exactness distinguishes local lifting from global lifting. Locally free and coherent sheaves are the finite analytic objects to which the later Cartan theorems apply.

Exercise 2.57: The exponential sequence

Prove exactness of $0 \to \mathbb{Z} \to \mathcal{O}_X \xrightarrow{f \mapsto \exp(2\pi i f)} \mathcal{O}_X^* \to 1$. The key point is the local existence of a logarithm; explain separately why a global logarithm need not exist.

Exercise 2.58: Nakayama at a stalk

If $\mathcal{F}$ is coherent, explain why a basis of $\mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x$ lifts to finitely many generators of $\mathcal{F}$ on a neighborhood of x.

Optional handout: A concrete dictionary for sections, germs, and stalks

This handout is optional. No later argument in the core lectures depends on it.

The same holomorphic function appears at three different levels. If $U \subset X$ is open and $f \in \mathcal{O}_X(U)$, then f is a *section on* U; for $x \in U$, the equivalence class $f_x \in \mathcal{O}_{X,x}$ is its *germ*; and the ring $\mathcal{O}_{X,x}$ containing all such germs is the *stalk*. Shrinking U changes the representative but not the germ. Passing to a stalk therefore means "forget the chosen neighborhood but retain all information sufficiently near x."

In a coordinate z centered at $0 \in \mathbb{C}$, every germ has a convergent expansion

$$f_0 = \sum_{k \geq 0} a_k z^k \in \mathbb{C}\{z\}.$$

It is a unit precisely when $a_0 \neq 0$. Its image modulo the maximal ideal $\mathfrak{m}_0 = (z)$ is the value $a_0 = f(0)$, while its image modulo $\mathfrak{m}_0^2 = (z^2)$ remembers the value and first derivative:

$$[f_0] \longleftrightarrow (f(0), f'(0)).$$

Thus quotients by powers of a maximal ideal encode finite jets.

For the inclusion $i : \{0\} \hookrightarrow \mathbb{C}$, let $\mathcal{I}_{\{0\}}$ be the ideal sheaf of holomorphic functions vanishing at 0 and consider evaluation at the origin:

$$0 \longrightarrow \mathcal{I}_{\{0\}} \longrightarrow \mathcal{O}_{\mathbb{C}} \xrightarrow{\mathrm{ev}_0} i_*\mathbb{C} \longrightarrow 0.$$

On an open set containing 0, the last arrow sends f to $f(0)$; on an open set disjoint from 0, its target is zero. The map is surjective as a sheaf because every complex number is locally represented by a constant function. At 0 the kernel stalk is $(\mathcal{I}_{\{0\}})_0 = \mathfrak{m}_0$, while away from 0 both $(\mathcal{I}_{\{0\}})_x$ and $\mathcal{O}_{\mathbb{C},x}$ are the same local ring. Stalkwise exactness holds at every point, which is exactly the criterion for the displayed sequence of sheaves to be exact.

By contrast, the exponential map $\mathcal{O}_{\mathbb{C}^*} \to \mathcal{O}_{\mathbb{C}^*}^*$ is surjective on stalks but not on global sections: every nonzero germ has a logarithm, whereas the global function z has no holomorphic logarithm on $\mathbb{C}^*$. These two examples give a useful diagnostic:

(1) to verify a statement about sheaves, shrink around one point and compute a stalk;
(2) to verify a statement about global sections, keep the whole space and check whether the local choices glue.

Example 2.59: Multiplication by the coordinate

Consider the sheaf morphism

$$m_z : \mathcal{O}_{\mathbb{C}} \longrightarrow \mathcal{O}_{\mathbb{C}}, \qquad m_z(f) = zf.$$

Its kernel is zero. Its sheaf image is $\mathcal{I}_{\{0\}}$: on an open set containing 0, every function of the form zf vanishes at 0; conversely, if $g(0) = 0$, then g/z has a removable singularity and is holomorphic. The cokernel is the skyscraper $i_*\mathbb{C}$, so there is a short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{C}} \xrightarrow{m_z} \mathcal{O}_{\mathbb{C}} \longrightarrow i_*\mathbb{C} \longrightarrow 0.$$

The stalk calculation makes the geometry transparent. If $x \neq 0$, then z is a unit in $\mathcal{O}_{\mathbb{C},x}$ and $(m_z)_x$ is an isomorphism. At $x = 0$, its image is $\mathfrak{m}_0 = (z)$ and its cokernel is $\mathcal{O}_{\mathbb{C},0}/\mathfrak{m}_0 \simeq \mathbb{C}$. Thus the cokernel records exactly the value at the one point where multiplication by z fails to be invertible.

Exercise 2.60: Three levels of one function

Let $f(z) = z^3 + 2z$ on $\mathbb{C}$. Compute its section on a disc $D(0, r)$, its germ at 0, and its classes modulo $\mathfrak{m}_0$, $\mathfrak{m}_0^2$, and $\mathfrak{m}_0^4$. State explicitly which information is lost at each quotient.

Chapter 3

Sheaves, Analytic Spaces, and Kähler Metrics

3.1 Lecture 9: Global sections, cohomology, and coherent sheaves

Week 5 material • 90 minutes

Notation for this lecture

X denotes a topological space and, from the coherence discussion onward, a complex manifold. The symbols $\mathcal{F}', \mathcal{F}, \mathcal{F}''$ denote sheaves of abelian groups or $\mathcal{O}_X$-modules. Their global sections are $\Gamma(X, \mathcal{F}) = \mathcal{F}(X) = H^0(X, \mathcal{F})$, and $H^q(X, \mathcal{F})$ denotes sheaf cohomology. For a short exact sequence, $\partial^q_{\mathrm{conn}}$ denotes its degree-q connecting homomorphism; this symbol is kept distinct from the Dolbeault operator ∂.

For an ordered open cover $\mathfrak{U} = \{U_i\}_{i \in I}$, $C^q(\mathfrak{U}, \mathcal{F})$ is the group of Čech q-cochains, δ is the Čech coboundary, and $\check{H}^q(\mathfrak{U}, \mathcal{F})$ is the resulting cohomology. The locally constant sheaf with fiber $\mathbb{Z}$ is denoted $\underline{\mathbb{Z}}_X$.

When X is a complex manifold, $\mathcal{O}_X$ is its structure sheaf and $\mathcal{O}(X) = \Gamma(X, \mathcal{O}_X)$. For $x \in X$, the local ring is $\mathcal{O}_{X,x}$, its maximal ideal is $\mathfrak{m}_x$, and $\mathcal{I}_{\{x\}}$ is the ideal sheaf of the reduced point $\{x\}$. If $s_1, \dots, s_r$ are local sections of an $\mathcal{O}_X$-module $\mathcal{F}$, their relation sheaf is the kernel of

$$\mathcal{O}_X^{\oplus r} \longrightarrow \mathcal{F}, \qquad (a_1, \dots, a_r) \longmapsto \sum_{j=1}^{r} a_j s_j.$$

For a compact set $K \subset X$, its holomorphic hull is

$$\widehat{K}_{\mathcal{O}(X)} = \{x \in X : |f(x)| \le \sup_K |f| \text{ for every } f \in \mathcal{O}(X)\}.$$

Prerequisite results used below

Sheaf exactness. Recall from Lecture 8 that a sequence of sheaves is exact if and only if it is exact on every stalk. In particular, a surjective sheaf morphism gives local lifts of sections, not necessarily a lift on the whole open set.

Complexes. For homomorphisms $d^q : C^q \to C^{q+1}$ with $d^{q+1}d^q = 0$, the quotient $\ker d^q / \operatorname{im} d^{q-1}$ measures the failure of exactness at C^q. This elementary construction will be applied first to overlap data and then to a resolution of a sheaf.

Modules. A finite presentation of a module has the form $R^p \to R^q \to M \to 0$. In the analytic setting the matrices have holomorphic entries and the same presentation is required

on one neighborhood, not merely at one stalk.

Why these ideas are needed

The functor of global sections is the first genuinely local-to-global operation in sheaf theory. Applied to a short exact sequence, it preserves injectivity and exactness in the middle, but it need not preserve surjectivity. The reason is geometric: a quotient section has local lifts, while those lifts may disagree on overlaps. This failure is not an annoyance to be hidden; it is the phenomenon that sheaf cohomology is designed to measure.

We shall therefore introduce cohomology in the order in which one actually needs it. First we prove that $\Gamma(X,-)$ is left exact and exhibit a concrete failure of right exactness on $\mathbb{C}^*$. Next we turn the differences of local lifts into a Čech cocycle and then into the connecting homomorphism of the long exact cohomology sequence. This makes a vanishing theorem immediately meaningful: it is a theorem saying that specified local solutions can be corrected and glued globally.

Cartan's vanishing theorem is not stated for arbitrary sheaves, but for coherent $\mathcal{O}_X$-modules. We develop that class systematically before using the theorem. Coherence means that both generators and relations are finite on one neighborhood; Oka's coherence theorem makes this class stable under the kernels and cokernels that arise from holomorphic equations. The theorem and its consequences belong in the classroom narrative, while its Weierstrass-division proof is placed in Appendix A.4.

Plan for the lecture. 0–18 min: left exactness of $\Gamma(X,-)$ and its failure to be right exact; 18–38 min: the connecting cocycle and H^1; 38–52 min: the long exact sequence and the purpose of vanishing theorems; 52–66 min: coherent sheaves and Oka coherence; 66–82 min: exact-sequence properties of coherent sheaves; 82–90 min: Cartan A and B and the role of vanishing.

3.1.1 The first fact: global sections are left exact

Let

$$0 \longrightarrow \mathcal{F}' \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\beta} \mathcal{F}'' \longrightarrow 0 \tag{9.1}$$

be a short exact sequence of sheaves of abelian groups on X. Exactness is a local condition: (9.1) means that $0 \to \mathcal{F}'_x \to \mathcal{F}_x \to \mathcal{F}''_x \to 0$ is exact for every $x \in X$.

Proposition 3.1: Left exactness of the global-section functor

Applying $\Gamma(X,-)$ to (9.1) gives an exact sequence

$$0 \longrightarrow \Gamma(X,\mathcal{F}') \xrightarrow{\alpha_X} \Gamma(X,\mathcal{F}) \xrightarrow{\beta_X} \Gamma(X,\mathcal{F}''). \tag{9.2}$$

Thus $\Gamma(X,-)$ is a left exact functor.

Proof. Suppose $s' \in \Gamma(X,\mathcal{F}')$ and $\alpha_X(s') = 0$. Every germ $\alpha_x(s'_x)$ is zero. Since α_x is injective, $s'_x = 0$ for every $x \in X$, and the locality axiom for a sheaf gives $s' = 0$. Hence α_X is injective.

The inclusion $\operatorname{im}\alpha_X \subseteq \ker\beta_X$ follows from $\beta\alpha = 0$. Conversely, take $s \in \Gamma(X,\mathcal{F})$ with $\beta_X(s) = 0$. For every x, exactness of the stalk sequence gives a germ $t_x \in \mathcal{F}'_x$ satisfying $\alpha_x(t_x) = s_x$. Represent t_x by a section on a neighborhood U_x. After shrinking U_x, equality of germs gives $\alpha(t_x) = s|_{U_x}$. If $U_x \cap U_y \neq \varnothing$, then

$$\alpha(t_x - t_y) = 0.$$

Injectivity of α as a sheaf morphism gives $t_x = t_y$ on the overlap. The sections t_x therefore glue to $t \in \Gamma(X, \mathcal{F}')$ with $\alpha_X(t) = s$. Thus $\ker \beta_X = \operatorname{im} \alpha_X$. □

The proof stops exactly where it should: it does not prove that β_X is surjective. Surjectivity of β as a sheaf morphism means that every germ of a section of $\mathcal{F}''$ has a local lift. It does not say that all those local lifts agree.

Example 3.2: $\Gamma(X, -)$ is not right exact

On $X = \mathbb{C}^* = \mathbb{C} \setminus \{0\}$, consider the exponential sequence

$$0 \longrightarrow \underline{\mathbb{Z}}_X \longrightarrow \mathcal{O}_X \xrightarrow{E} \mathcal{O}_X^* \longrightarrow 1, \qquad E(f) = \exp(2\pi \mathrm{i} f). \tag{9.3}$$

Here $\mathcal{O}_X^*$ is the sheaf of nowhere-vanishing holomorphic functions under multiplication. The map $\underline{\mathbb{Z}}_X \to \mathcal{O}_X$ sends an integer-valued locally constant function to the same holomorphic function. The kernel of E is precisely $\underline{\mathbb{Z}}_X$. Moreover, every nonvanishing holomorphic germ has a logarithm on a sufficiently small disc, so E is surjective as a sheaf morphism.
Nevertheless, the global section $z \in \mathcal{O}_X^*(X)$ has no global preimage. Indeed, if $\exp(2\pi \mathrm{i} f(z)) = z$ on $\mathbb{C}^*$, differentiation would give

$$2\pi \mathrm{i} f'(z) = \frac{1}{z}.$$

The integral of the left-hand side around the unit circle is zero because f' has a global primitive, whereas the integral of $\mathrm{d}z/z$ is $2\pi \mathrm{i}$. This contradiction shows that

$$\Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(X, \mathcal{O}_X^*)$$

is not surjective. Thus the missing right exactness is already visible in the elementary problem of choosing a logarithm consistently around a loop.

3.1.2 The failure of surjectivity produces a cocycle

Return to (9.1), and let $s \in \Gamma(X, \mathcal{F}'')$. Choose an open cover $\mathfrak{U} = \{U_i\}_{i \in I}$ and local lifts $t_i \in \mathcal{F}(U_i)$ satisfying $\beta(t_i) = s|_{U_i}$. On $U_i \cap U_j$ one has

$$\beta(t_i - t_j) = 0.$$

Exactness and injectivity of α therefore give a unique section $a_{ij} \in \mathcal{F}'(U_i \cap U_j)$ such that

$$\alpha(a_{ij}) = t_i - t_j. \tag{9.4}$$

On a triple intersection the differences telescope, so

$$a_{ij} + a_{jk} + a_{ki} = 0. \tag{9.5}$$

Equation (9.5) is the cocycle condition. If the lift t_i is replaced by $t_i + \alpha(b_i)$, then a_{ij} is replaced by $a_{ij} + b_i - b_j$. Hence the obstruction is not the particular family (a_{ij}), but its class modulo such changes.

We now place this calculation in a complex. Totally order the index set I. For a sheaf $\mathcal{A}$ of abelian groups define

$$C^q(\mathfrak{U}, \mathcal{A}) = \prod_{i_0 < \cdots < i_q} \mathcal{A}(U_{i_0} \cap \cdots \cap U_{i_q}),$$

and define the Čech coboundary by

$$(\delta c)_{i_0 \ldots i_{q+1}} = \sum_{k=0}^{q+1} (-1)^k c_{i_0 \ldots \widehat{i_k} \ldots i_{q+1}}\big|_{U_{i_0} \cap \cdots \cap U_{i_{q+1}}}. \tag{9.6}$$

Every term in $\delta^2 c$ occurs twice with opposite signs, hence $\delta^2 = 0$. The groups of cocycles, coboundaries, and Čech cohomology are

$$Z^q = \ker \delta, \qquad B^q = \operatorname{im} \delta, \qquad \check{H}^q(\mathfrak{U}, \mathcal{A}) = Z^q / B^q.$$

For $q = 0$, the cocycle condition says that local sections agree on overlaps, so the sheaf axiom gives

$$\check{H}^0(\mathfrak{U}, \mathcal{A}) = \Gamma(X, \mathcal{A}).$$

For $q = 1$, a class is represented by overlap errors such as (a_{ij}). The class $[(a_{ij})]$ vanishes if and only if there are $b_i \in \mathcal{F}'(U_i)$ with $a_{ij} = b_i - b_j$. In that case the corrected lifts $t_i - \alpha(b_i)$ agree and glue to a global lift of s. Therefore

$$[(a_{ij})] = 0 \quad \Longleftrightarrow \quad s \in \operatorname{im}\big(\Gamma(X, \mathcal{F}) \to \Gamma(X, \mathcal{F}'')\big). \tag{9.7}$$

Remark (The first cohomology group is a defect of surjectivity)

For a fixed short exact sequence (9.1), the connecting map gives an injective map

$$\frac{H^0(X, \mathcal{F}'')}{\operatorname{im}\big(H^0(X, \mathcal{F}) \to H^0(X, \mathcal{F}'')\big)} \hookrightarrow H^1(X, \mathcal{F}').$$

Its image consists exactly of those H^1-classes produced by trying to lift a global section of this particular quotient $\mathcal{F}''$. Thus $H^1(X, \mathcal{F}')$ is a universal receptacle for all such first gluing obstructions; a single short exact sequence need not realize every class in it.

Remark (Why higher degrees appear)

A 1-cocycle is an error on pairwise intersections. Correcting it requires local 0-cochains. In a more complicated problem, choices made on pairwise intersections may themselves disagree on triple intersections, producing a 2-cocycle, and the same pattern continues. The long exact sequence packages all degrees at once; it is the systematic version of repeatedly lifting, taking a difference, and descending to the kernel.

3.1.3 Intrinsic cohomology and the long exact sequence

The group $\check{H}^q(\mathfrak{U}, \mathcal{A})$ initially depends on the chosen cover. The intrinsic groups $H^q(X, \mathcal{A})$ may be defined as the right derived functors of the left exact functor $\Gamma(X, -)$. Concretely, choose an injective resolution

$$0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{I}^0 \xrightarrow{d^0} \mathcal{I}^1 \xrightarrow{d^1} \mathcal{I}^2 \longrightarrow \cdots. \tag{9.8}$$

A sheaf $\mathcal{I}$ is called *injective* if every morphism from a subsheaf $\mathcal{B}' \hookrightarrow \mathcal{B}$ to $\mathcal{I}$ extends to a morphism from $\mathcal{B}$ to $\mathcal{I}$. Injective resolutions exist. Define

$$H^q(X, \mathcal{A}) = H^q\big(\Gamma(X, \mathcal{I}^\bullet), d\big) = \frac{\ker\big(\Gamma(X, \mathcal{I}^q) \to \Gamma(X, \mathcal{I}^{q+1})\big)}{\operatorname{im}\big(\Gamma(X, \mathcal{I}^{q-1}) \to \Gamma(X, \mathcal{I}^q)\big)}$$

for $q \geq 1$, and set $H^0(X, \mathcal{A}) = \Gamma(X, \mathcal{A})$. Comparison morphisms between injective resolutions show that the result is independent of the choice.

If $\mathfrak{U}$ is a *Leray cover* for $\mathcal{A}$, meaning

$$H^r(U_{i_0} \cap \cdots \cap U_{i_p}, \mathcal{A}) = 0 \quad \text{for every } r > 0$$

whenever the finite intersection is nonempty, then the Čech comparison theorem gives $\check{H}^q(\mathfrak{U}, \mathcal{A}) \simeq H^q(X, \mathcal{A})$. Thus the overlap calculation above is not a heuristic: on an acyclic cover it computes the intrinsic group.

Theorem 3.3: The long exact cohomology sequence

The short exact sequence (9.1) induces natural connecting homomorphisms $\partial^q_{\mathrm{conn}} : H^q(X,\mathcal{F}'') \to H^{q+1}(X,\mathcal{F}')$ and an exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(X,\mathcal{F}') \longrightarrow H^0(X,\mathcal{F}) \longrightarrow H^0(X,\mathcal{F}'') \xrightarrow{\partial^0_{\mathrm{conn}}} H^1(X,\mathcal{F}') \\ \longrightarrow H^1(X,\mathcal{F}) \longrightarrow H^1(X,\mathcal{F}'') \xrightarrow{\partial^1_{\mathrm{conn}}} H^2(X,\mathcal{F}') \longrightarrow \cdots . \end{aligned} \tag{9.9}$$

For $s \in H^0(X,\mathcal{F}'')$, the element $\partial^0_{\mathrm{conn}}(s)$ is represented by the overlap cocycle (9.4).

Proof (the degree-zero construction and the general mechanism). The calculation leading to (9.7) proves that the connecting class is independent of the chosen local lifts and that

$$\ker \partial^0_{\mathrm{conn}} = \mathrm{im}\big(H^0(X,\mathcal{F}) \to H^0(X,\mathcal{F}'')\big).$$

It also proves exactness at $H^1(X,\mathcal{F}')$: if a cocycle in $\mathcal{F}'$ becomes a coboundary in $\mathcal{F}$, its local primitives have images in $\mathcal{F}''$ that glue to a global section, whose connecting class is the original cocycle.

For every degree at once, take compatible injective resolutions of the three sheaves. This produces a short exact sequence of complexes of global sections. A cocycle in the quotient complex is lifted, differentiated, and then regarded as a cocycle in the kernel complex. A different lift changes the result by a coboundary. The resulting maps are the $\partial^q_{\mathrm{conn}}$, and the standard kernel–image chase proves exactness. Lecture 15 returns to this construction using fine and Dolbeault resolutions. □

3.1.4 What a vanishing theorem accomplishes

The first line of (9.9) gives a precise measurement of the failure of right exactness:

$$\mathrm{coker}\Big(H^0(X,\mathcal{F}) \longrightarrow H^0(X,\mathcal{F}'')\Big) \simeq \mathrm{im}\,\partial^0_{\mathrm{conn}} \subseteq H^1(X,\mathcal{F}'). \tag{9.10}$$

Consequently, if $H^1(X,\mathcal{F}') = 0$, every global section of $\mathcal{F}''$ lifts globally. More generally, the vanishing of $H^q(X,\mathcal{F}')$ removes degree-q gluing obstructions. A sheaf $\mathcal{A}$ is called Γ-*acyclic on* X if $H^q(X,\mathcal{A}) = 0$ for all $q \geq 1$.

This is the point of a vanishing theorem. It turns a left exact functor into an exact one on a controlled class of short exact sequences. Extension, interpolation, and the first Cousin problem will all have the same proof pattern:

$$\text{encode the data in a quotient} \longrightarrow \text{identify the coherent kernel} \longrightarrow \text{kill its } H^1.$$

The remaining question is why the word *coherent* occurs in the vanishing theorem.

3.1.5 Coherent analytic sheaves: the finite class stable under equations

Let X now be a complex manifold and let $\mathcal{F}$ be an $\mathcal{O}_X$-module.

Definition 3.4: Finite type, finite presentation, and coherence

The sheaf $\mathcal{F}$ is of *finite type* if every point has a neighborhood U and a surjective morphism

$$\mathcal{O}_U^{\oplus q} \longrightarrow \mathcal{F}|_U \longrightarrow 0$$

for some finite q. It is *locally finitely presented* if one can extend this to an exact sequence

$$\mathcal{O}_U^{\oplus p} \xrightarrow{A} \mathcal{O}_U^{\oplus q} \longrightarrow \mathcal{F}|_U \longrightarrow 0 \tag{9.11}$$

with finite p, q and a matrix A of holomorphic functions.
The sheaf $\mathcal{F}$ is *coherent* if it is of finite type and, for every open $U \subset X$ and every finite collection $s_1, \dots, s_r \in \mathcal{F}(U)$, the relation sheaf

$$\mathrm{Rel}(s_1, \dots, s_r) = \ker\bigl(\mathcal{O}_U^{\oplus r} \to \mathcal{F}|_U\bigr)$$

is of finite type. Thus coherence requires generators and the relations among any finite list of sections to vary finitely on one neighborhood.

Choosing local generators of a coherent sheaf and then finite generators of their relation sheaf gives (9.11). The converse is not a purely formal statement over an arbitrary sheaf of rings; on a complex manifold it follows from Oka's theorem below. This logical order is worth remembering: stalkwise Noetherianity alone would not supply generators on a common neighborhood.

Theorem 3.5: Oka's coherence theorem

The structure sheaf $\mathcal{O}_X$ of a complex manifold is coherent.

The proof is an induction on the dimension using Weierstrass division. Division turns an apparently infinite relation into a bounded-degree remainder problem in one fewer variable; uniform Cauchy estimates make the resulting generators valid on one neighborhood. This technical proof is not needed in the classroom development and is given separately in Appendix A.4.

3.1.6 Exact-sequence properties of coherent analytic sheaves

For a morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of $\mathcal{O}_X$-modules, the symbols $\ker \varphi$, $\operatorname{im} \varphi$, and $\operatorname{coker} \varphi$ always denote the corresponding *sheaves*. They are characterized stalkwise by

$$(\ker \varphi)_x = \ker \varphi_x, \qquad (\operatorname{im} \varphi)_x = \operatorname{im} \varphi_x, \qquad (\operatorname{coker} \varphi)_x = \operatorname{coker} \varphi_x. \tag{9.12}$$

In particular, a section of the sheaf cokernel need only lift locally to $\mathcal{G}$. This distinction is exactly the one encountered in the failure of right exactness of global sections.

The next results do not use Oka's theorem: they follow directly from the definition of coherence and hold for modules over any sheaf of rings.

Lemma 3.6: Coherence gives a finite presentation

Every coherent $\mathcal{O}_X$-module is locally finitely presented.

Proof. Fix $x \in X$. Since $\mathcal{F}$ is of finite type, after replacing X by a neighborhood U of x there are sections $s_1, \dots, s_q \in \mathcal{F}(U)$ which define a surjection

$$\pi : \mathcal{O}_U^{\oplus q} \longrightarrow \mathcal{F}|_U, \qquad (a_1, \dots, a_q) \longmapsto \sum_{j=1}^{q} a_j s_j.$$

Coherence says that $\ker \pi$ is of finite type. After shrinking once more around x, choose generators $r_1, \dots, r_p$ of $\ker \pi$. Sending the standard basis of $\mathcal{O}_U^{\oplus p}$ to these sections gives an exact sequence

$$\mathcal{O}_U^{\oplus p} \longrightarrow \mathcal{O}_U^{\oplus q} \xrightarrow{\pi} \mathcal{F}|_U \longrightarrow 0.$$

Both p and q are finite, so this is a finite presentation on one neighborhood of x. □

Proposition 3.7: The coherent-sheaf calculus

Let X be a complex manifold.

(1) If $\mathcal{H} \subset \mathcal{G}$ is a finite-type subsheaf of a coherent $\mathcal{O}_X$-module $\mathcal{G}$, then $\mathcal{H}$ is coherent.

(2) If $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism, $\mathcal{F}$ is of finite type, and $\mathcal{G}$ is coherent, then $\ker \varphi$ is of finite type.

(3) If $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of coherent $\mathcal{O}_X$-modules, then $\ker \varphi$ and $\operatorname{coker} \varphi$ are coherent.

(4) In a short exact sequence

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0, \tag{9.13}$$

if two of the three sheaves are coherent, then so is the third.

(5) The full subcategory $\mathrm{Coh}(\mathcal{O}_X)$ of coherent $\mathcal{O}_X$-modules is a weak Serre subcategory of the category of all $\mathcal{O}_X$-modules: it is closed under kernels, cokernels, and extensions. Consequently $\mathrm{Coh}(\mathcal{O}_X)$ is an abelian category, and its inclusion into the category of all $\mathcal{O}_X$-modules is exact.

Proof. For (1), take finitely many sections $h_1, \dots, h_r \in \mathcal{H}(U)$ on an open set $U \subset X$. Because $\mathcal{H} \hookrightarrow \mathcal{G}$ is injective, the relations among the h_j viewed as sections of $\mathcal{H}$ are exactly their relations viewed as sections of $\mathcal{G}$. The latter relation sheaf is of finite type because $\mathcal{G}$ is coherent. This verifies the relation condition for $\mathcal{H}$; the finite-type condition is an assumption.

For (2), fix $x \in X$. On a neighborhood U of x, choose generators $s_1, \dots, s_q$ of $\mathcal{F}$ and let

$$\pi : \mathcal{O}_U^{\oplus q} \longrightarrow \mathcal{F}|_U$$

be the associated surjection. The composite $\varphi\pi : \mathcal{O}_U^{\oplus q} \to \mathcal{G}|_U$ is determined by the finite list $\varphi(s_1), \dots, \varphi(s_q)$. Hence $\mathcal{K} := \ker(\varphi\pi)$ is of finite type by coherence of $\mathcal{G}$. Moreover,

$$\pi(\mathcal{K}) = \ker(\varphi|_U) \tag{9.14}$$

as sheaves. Indeed, the inclusion from left to right is immediate. Conversely, a local section of $\ker \varphi$ has, after shrinking, a preimage under the sheaf surjection π; that preimage lies in $\mathcal{K}$. The image of a finite-type sheaf is generated by the images of any finite set of generators, so (9.14) proves that $\ker \varphi$ is of finite type near x.

For the kernel assertion in (3), part (2) shows that $\ker \varphi$ is of finite type. It is a subsheaf of the coherent sheaf $\mathcal{F}$, so part (1) makes it coherent. Set $\mathcal{Q} = \operatorname{coker} \varphi$. This sheaf is of finite type because it is a quotient of the finite-type sheaf $\mathcal{G}$. We verify the relation condition for $\mathcal{Q}$. Fix $x \in X$ and finitely many sections $\bar{s}_1, \dots, \bar{s}_n \in \mathcal{Q}(U)$ on an open neighborhood U of x. Since $\mathcal{Q}$ is the sheaf cokernel, after shrinking U the sections $\bar{s}_i$ have lifts $s_i \in \mathcal{G}(U)$. The sheaf $\operatorname{im} \varphi$ is of finite type, being the image of the finite-type sheaf $\mathcal{F}$; shrink again and choose generators $t_1, \dots, t_m$ of $\operatorname{im} \varphi$ on U. Define

$$\Psi : \mathcal{O}_U^{\oplus(m+n)} \longrightarrow \mathcal{G}|_U, \qquad \Psi(a, b) = \sum_{j=1}^{m} a_j t_j - \sum_{i=1}^{n} b_i s_i. \tag{9.15}$$

Here $a = (a_1, \dots, a_m)$ and $b = (b_1, \dots, b_n)$ are local sections of the two free summands. The sheaf $\ker \Psi$ is of finite type because $\mathcal{G}$ is coherent. Projection to the b-coordinates maps $\ker \Psi$ onto $\mathrm{Rel}(\bar{s}_1, \dots, \bar{s}_n)$ as a sheaf. To see surjectivity, observe that a local tuple b is a relation among the $\bar{s}_i$ precisely when $\sum b_i s_i$ is a local section of $\operatorname{im} \varphi$; it can then be written locally as $\sum a_j t_j$. Thus the relation sheaf is the image of a finite-type sheaf and is itself of finite type. Therefore $\mathcal{Q}$ is coherent.

For (4), two cases follow immediately from (3): if $\mathcal{F}_1$ and $\mathcal{F}_2$ are coherent, then $\mathcal{F}_3$ is a cokernel; if $\mathcal{F}_2$ and $\mathcal{F}_3$ are coherent, then $\mathcal{F}_1$ is a kernel. Suppose now that $\mathcal{F}_1$ and $\mathcal{F}_3$ are coherent. Near a fixed point choose finitely many generators of $\mathcal{F}_3$ and local lifts of them to $\mathcal{F}_2$. Adjoin the images in $\mathcal{F}_2$ of finitely many generators of $\mathcal{F}_1$. These sections generate $\mathcal{F}_2$, so $\mathcal{F}_2$ is of finite type.

It remains to control relations. Let $s_1, \dots, s_n \in \mathcal{F}_2(U)$ and denote their images in $\mathcal{F}_3(U)$ by $\bar{s}_1, \dots, \bar{s}_n$. Coherence of $\mathcal{F}_3$ makes

$$\mathcal{K}_3 = \operatorname{Rel}(\bar{s}_1, \dots, \bar{s}_n)$$

finite type. For a local section $a = (a_1, \dots, a_n)$ of $\mathcal{K}_3$, the sum $\sum a_i s_i$ lies in the kernel of $\mathcal{F}_2 \to \mathcal{F}_3$, which is identified with $\mathcal{F}_1$. This defines a morphism $\theta : \mathcal{K}_3 \to \mathcal{F}_1$. Its kernel is exactly $\operatorname{Rel}(s_1, \dots, s_n)$. Part (2), applied to θ, shows that this kernel is of finite type. Hence $\mathcal{F}_2$ is coherent.

Finally, (3) gives closure under kernels and cokernels, while (4) gives closure under extensions; this is precisely the weak Serre property in (5). The zero sheaf is coherent, and (4) applied to $0 \to \mathcal{F} \to \mathcal{F} \oplus \mathcal{G} \to \mathcal{G} \to 0$ gives finite direct sums. Thus all kernels, cokernels, and finite biproducts needed for an abelian category remain inside $\operatorname{Coh}(\mathcal{O}_X)$. They are the same sheaves as those computed in the ambient module category, so the inclusion functor is exact. □

Oka's theorem now converts the preceding abstract calculus into the convenient finite-presentation criterion used in computations.

Corollary 3.8: Finite presentations and finite constructions

On a complex manifold, an $\mathcal{O}_X$-module is coherent if and only if it is locally finitely presented. If $\mathcal{F}$ and $\mathcal{G}$ are coherent, then so are $\operatorname{im}(\varphi)$ for every morphism $\varphi : \mathcal{F} \to \mathcal{G}$, every finite direct sum, every extension and quotient of coherent sheaves, and $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$.

Proof. One implication is Lemma 3.6. Conversely, suppose that near a point there is an exact sequence

$$\mathcal{O}_X^{\oplus p} \xrightarrow{A} \mathcal{O}_X^{\oplus q} \longrightarrow \mathcal{F} \longrightarrow 0. \tag{9.16}$$

Oka's theorem makes $\mathcal{O}_X$ coherent, and finite direct sums are coherent by Proposition 3.7(4). The image of A is a finite-type sheaf because it is an image of $\mathcal{O}_X^{\oplus p}$. It is also a subsheaf of the coherent sheaf $\mathcal{O}_X^{\oplus q}$, so it is coherent by part (1) of that proposition. The quotient $\mathcal{F}$ is then coherent by part (3).

For a morphism $\varphi : \mathcal{F} \to \mathcal{G}$, its image is the kernel of the natural map $\mathcal{G} \to \operatorname{coker} \varphi$, hence is coherent. The assertions about finite sums, extensions, and quotients have already been proved in Proposition 3.7. Finally, tensor a finite presentation $\mathcal{O}_X^{\oplus p} \to \mathcal{O}_X^{\oplus q} \to \mathcal{F} \to 0$ with $\mathcal{G}$. Right exactness of the tensor product gives

$$\mathcal{G}^{\oplus p} \longrightarrow \mathcal{G}^{\oplus q} \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} \longrightarrow 0.$$

The first two sheaves are coherent, so their cokernel is coherent. □

A further useful local test says that injectivity at one stalk persists after shrinking, provided the source is finite type and the target is coherent.

Proposition 3.9: Stalkwise injectivity persists locally

Let $\varphi : \mathcal{G} \to \mathcal{F}$ be a morphism of $\mathcal{O}_X$-modules, let $x \in X$, and assume that $\mathcal{G}$ is of finite type and $\mathcal{F}$ is coherent. If $\varphi_x : \mathcal{G}_x \to \mathcal{F}_x$ is injective, then there is an open neighborhood U of x such that $\varphi|_U : \mathcal{G}|_U \to \mathcal{F}|_U$ is injective.

Proof. Let $\mathcal{K} = \ker \varphi$. Proposition 3.7(2) shows that $\mathcal{K}$ is of finite type, while injectivity of φ_x says that $\mathcal{K}_x = 0$. Choose a neighborhood V of x on which $\mathcal{K}$ is generated by finitely many sections

$k_1, \dots, k_r$. Since $(k_j)_x = 0$, each k_j vanishes on some neighborhood $V_j \subset V$ of x. On the finite intersection $U = V_1 \cap \cdots \cap V_r$, all generators vanish, hence $\mathcal{K}|_U = 0$. Therefore $\varphi|_U$ is injective. □

Here are the basic examples, with the reason for coherence made explicit.

(1) A locally free sheaf $\mathcal{E}$ of finite rank is locally isomorphic to $\mathcal{O}_X^{\oplus r}$, hence is coherent by Oka's theorem.

(2) If $A : \mathcal{O}_X^{\oplus p} \to \mathcal{O}_X^{\oplus q}$ is a holomorphic matrix, then $\ker A$, $\operatorname{im} A$, and $\operatorname{coker} A$ are coherent. These sheaves record respectively relations, equations that are attained, and equations modulo those attained.

(3) In coordinates centered at x, the reduced-point ideal is generated by $z_1, \dots, z_n$. Hence $\mathcal{I}_{\{x\}}$ is coherent, and $\mathcal{O}_X/\mathcal{I}_{\{x\}}$ is the one-dimensional skyscraper sheaf at x.

(4) If $Y \subset X$ is a complex submanifold, coordinates in which $Y = \{z_{k+1} = \cdots = z_n = 0\}$ show that its ideal sheaf $\mathcal{I}_Y$ is coherent and give the exact sequence

$$0 \longrightarrow \mathcal{I}_Y \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Y \longrightarrow 0.$$

The coherence of the full vanishing ideal of a singular analytic set requires an additional argument and is proved in Lecture 10.

Coherent sheaves are therefore the natural coefficient objects for analytic geometry. They are broad enough to contain vector bundles, systems of holomorphic equations, their solution sheaves, and quotient structures; finite enough that local analytic estimates apply to finitely many generators; and stable under the exact sequences used by cohomology. Cartan B would be false if "coherent" were replaced by "arbitrary sheaf": for example, $\mathbb{C}^*$ is Stein but $H^1(\mathbb{C}^*, \underline{\mathbb{Z}}_{\mathbb{C}^*}) \simeq \mathbb{Z}$.

3.1.7 Reading a finite presentation stalk by stalk

Suppose that on an open set $U \subset X$ one has a presentation

$$\mathcal{O}_U^{\oplus p} \xrightarrow{A} \mathcal{O}_U^{\oplus q} \xrightarrow{\pi} \mathcal{F}|_U \longrightarrow 0. \tag{9.17}$$

Let $e_1, \dots, e_q$ be the standard frame of $\mathcal{O}_U^{\oplus q}$ and put $s_j = \pi(e_j)$. For every $x \in U$, taking stalks gives

$$\mathcal{O}_{X,x}^{\oplus p} \xrightarrow{A_x} \mathcal{O}_{X,x}^{\oplus q} \longrightarrow \mathcal{F}_x \longrightarrow 0.$$

Thus every germ $u_x \in \mathcal{F}_x$ is a finite sum $u_x = \sum_j f_{j,x}(s_j)_x$. A column $a = (a_1, \dots, a_q)^t$ is a relation among the s_j precisely when $a \in \ker \pi = \operatorname{im} A$. The columns of A therefore generate all relations, not just at the chosen point but throughout the same neighborhood U. This uniform neighborhood is the extra content that a merely finitely generated stalk does not provide.

Example 3.10: A cokernel whose rank jumps

On $X = \mathbb{C}$, multiplication by the coordinate defines

$$0 \longrightarrow \mathcal{O}_{\mathbb{C}} \xrightarrow{z} \mathcal{O}_{\mathbb{C}} \longrightarrow \mathcal{Q} \longrightarrow 0. \tag{9.18}$$

If $x \neq 0$, then z is a unit in $\mathcal{O}_{\mathbb{C},x}$, so $\mathcal{Q}_x = 0$. At the origin,

$$\mathcal{Q}_0 = \mathcal{O}_{\mathbb{C},0}/(z) \simeq \mathbb{C}.$$

Hence $\mathcal{Q}$ is not locally free near 0: its fiber rank jumps. It is nevertheless coherent, because it is the cokernel of a morphism between finite free sheaves. This is the simplest reason that one must enlarge the category of vector bundles before taking kernels and cokernels.

Example 3.11: A relation sheaf computed explicitly

On $\mathbb{C}^2$ with coordinates (z_1, z_2), consider

$$\varphi : \mathcal{O}_{\mathbb{C}^2}^{\oplus 2} \longrightarrow \mathcal{O}_{\mathbb{C}^2}, \qquad (a, b) \longmapsto a z_1 + b z_2.$$

We claim that near the origin

$$\ker \varphi = \mathcal{O}_{\mathbb{C}^2} \cdot (-z_2, z_1). \tag{9.19}$$

The displayed vector is a relation. Conversely, if $a z_1 + b z_2 = 0$, restrict to the hypersurface $z_1 = 0$. Then $b(0, z_2) z_2 = 0$, so the identity theorem gives $b(0, z_2) = 0$. Division by the coordinate gives $b = z_1 c$ for a holomorphic function c. Substitution yields $z_1(a + c z_2) = 0$, hence $a = -c z_2$. Therefore $(a, b) = c(-z_2, z_1)$, proving (9.19). Oka coherence is the general theorem asserting that finitely many such local relations suffice even when no direct division by a coordinate is available.

3.1.8 A dependency map for Cartan B

It is useful to separate the roles of the results that enter the applications below. Oka coherence is local: it says that finite holomorphic equations and their relations remain inside the coherent category. Cartan B is global: on a Stein manifold it says that every coherent sheaf is Γ-acyclic. The long exact sequence is the bridge converting that vanishing into a lifting statement.

For example, let $Y \subset X$ be a closed complex submanifold of a Stein manifold. From

$$0 \longrightarrow \mathcal{I}_Y \longrightarrow \mathcal{O}_X \longrightarrow i_* \mathcal{O}_Y \longrightarrow 0$$

one obtains

$$\mathcal{O}(X) \longrightarrow \mathcal{O}(Y) \xrightarrow{\partial^0_{\mathrm{conn}}} H^1(X, \mathcal{I}_Y).$$

Oka coherence puts $\mathcal{I}_Y$ in the class to which Cartan B applies; Cartan B makes the last group zero; exactness then proves that every holomorphic function on Y extends to X. None of these three steps can simply be omitted: the exact sequence identifies the obstruction, coherence makes the theorem applicable, and vanishing removes the obstruction.

3.1.9 Stein manifolds and Cartan's theorems

Compact complex manifolds have too few holomorphic functions. Stein manifolds lie at the other extreme: their holomorphic functions are rich enough to make them resemble affine algebraic varieties.

Definition 3.12: Stein manifold

A complex manifold X is *Stein* if:

(1) for every compact K, the holomorphic hull $\widehat{K}_{\mathcal{O}(X)} = \{x : |f(x)| \le \sup_K |f| \text{ for all } f \in \mathcal{O}(X)\}$ is compact;
(2) global holomorphic functions separate points;
(3) at each x, the differentials of global holomorphic functions span $T^{*1,0}_x X$.

The third condition has a concrete local meaning. If the global differentials span $T^{*1,0}_x X$, choose $n = \dim X$ of them that form a basis. The holomorphic inverse function theorem says that the corresponding global functions form a holomorphic coordinate system near x.

Theorem 3.13: Analytic and geometric characterizations of Stein manifolds

For a connected complex manifold X, the following are equivalent:

(1) X is Stein;
(2) X has a smooth strictly psh exhaustion;
(3) X is biholomorphic to a closed complex submanifold of some $\mathbb{C}^N$;
(4) for every coherent $\mathcal{O}_X$-module $\mathcal{F}$, germs of global sections generate every stalk $\mathcal{F}_x$, and $H^q(X, \mathcal{F}) = 0$ for every $q \geq 1$.

Proof (with the proper embedding theorem as the quoted input). The implication from a smooth strictly psh exhaustion to the defining Stein properties was proved in the Levi problem: weighted $\bar{\partial}$ solutions separate points and tangent vectors, and the boundary-growing construction gives holomorphic convexity.

Assume next that X is Stein. The Bishop–Narasimhan–Remmert proper embedding theorem constructs finitely many global functions whose joint map $X \to \mathbb{C}^N$ is a proper injective immersion. Its proof repeats the separation argument on an exhaustion, adds boundary-growing functions for properness, and then uses a finite-dimensional generic-projection theorem with dimension estimates for the double-point and tangent loci. Properness makes the image closed, and a proper injective immersion is a biholomorphism onto that closed complex submanifold.

A closed complex submanifold $Y \subset \mathbb{C}^N$ has the smooth strictly psh exhaustion $|z|^2|_Y$: strict positivity restricts to $T^{1,0}Y$, and closedness makes the restriction proper. This proves the equivalence of the first three conditions.

The two sheaf-theoretic assertions in (4), traditionally called Cartan A and B, are proved in the next theorem. Conversely, assume (4). Apply its global-generation assertion to the ideal sheaf $\mathcal{I}_{\{x\}}$. At a second point $y \neq x$ its stalk is $\mathcal{O}_{X,y}$, so among the global generators there is a function vanishing at x but not at y; this separates the two points. At x, generators of $\mathcal{I}_{\{x\}}$ map to generators of $\mathfrak{m}_x/\mathfrak{m}_x^2$, so their differentials span the cotangent space. The vanishing assertion in (4) removes the additive Cousin obstructions in the Runge construction on an exhaustion. The resulting boundary-growing global functions keep every holomorphic hull inside a compact sublevel. Hence X is holomorphically convex and satisfies all three defining conditions. All four conditions are equivalent. □

3.1.10 What the two Cartan statements will mean

All objects in the next theorem have now been defined, but it is worth translating the statement before proving it. A global section $s \in H^0(X, \mathcal{F})$ gives a germ $s_x \in \mathcal{F}_x$. Multiplying such germs by coefficients $f_x \in \mathcal{O}_{X,x}$ gives an $\mathcal{O}_{X,x}$-linear map

$$H^0(X, \mathcal{F}) \otimes_{\mathbb{C}} \mathcal{O}_{X,x} \longrightarrow \mathcal{F}_x, \qquad s \otimes f_x \longmapsto f_x s_x. \tag{9.20}$$

Theorem A says precisely that this map is surjective. Equivalently, every germ $u_x \in \mathcal{F}_x$ can be written

$$u_x = f_{1,x}(s_1)_x + \cdots + f_{N,x}(s_N)_x$$

for finitely many global sections s_j and holomorphic germs $f_{j,x}$. It does not say that u_x itself is the germ of a global section, nor that one fixed finite list generates $\mathcal{F}$ at every point of a noncompact X.

Theorem B concerns a different problem. Local lifts or local solutions may disagree on overlaps; their differences form cocycles. The group $H^q(X, \mathcal{F})$ records the degree-q obstruction after quotienting by all possible corrections. Thus its vanishing says that every such obstruction is a coboundary. The adjective *coherent* is essential in both statements: it provides locally finite generators and relations, allowing the analytic estimates to be applied to a finite system of holomorphic functions.

Theorem 3.14: Cartan's Theorems A and B

Let X be Stein and $\mathcal{F}$ a coherent $\mathcal{O}_X$-module.

Theorem A. For every $x \in X$, the stalk $\mathcal{F}_x$ is generated by germs of global sections in $H^0(X, \mathcal{F})$.

Theorem B. $H^q(X, \mathcal{F}) = 0$ for every $q \geq 1$.

Proof (using the coherent-sheaf Runge lemma). Choose a strongly pseudoconvex exhaustion $K_1 \Subset K_2 \Subset \cdots$ of X, with strongly pseudoconvex neighborhoods U_j satisfying $K_j \Subset U_j \Subset K_{j+1}$. The analytic input is the following coherent Runge package.

First, a coherent sheaf on a sufficiently small coordinate polydisc has, after shrinking, a finite free resolution

$$0 \longrightarrow \mathcal{O}^{r_m} \longrightarrow \cdots \longrightarrow \mathcal{O}^{r_0} \longrightarrow \mathcal{F} \longrightarrow 0. \tag{9.21}$$

The free sheaves are resolved by their Dolbeault complexes. Solving $\bar\partial$ coefficientwise and descending through the short exact sequences obtained from (9.21) gives local acyclicity for $\mathcal{F}$. This finite-resolution step is the extra ingredient missing if one tries to apply the Dolbeault lemma directly to an arbitrary coherent sheaf.

Second, the bump proof on a strongly pseudoconvex pair gives both local vanishing on U_j and Runge approximation: if $K \Subset U$ is holomorphically convex, sections of $\mathcal{F}$ near K can be approximated uniformly there by sections on U. For a free sheaf this is obtained by cutting off a vector of holomorphic functions and solving the resulting $\bar\partial$ error with a weight large outside the smaller set. Resolution (9.21) then carries the assertion to $\mathcal{F}$, one kernel at a time. This is the coherent-sheaf Runge lemma; its scalar analytic estimate is the one proved in Lectures 27–28.

To prove Theorem B, take a locally finite Stein-coordinate cover and a Čech q-cocycle c, $q \geq 1$. The local vanishing just described solves $c = \delta b_1$ near K_1. Suppose b_j has been constructed near K_j. Solve once near K_{j+1} and compare the two solutions; their difference is a $(q-1)$-cocycle. For $q = 1$ it is a section and Runge approximation changes the new solution so that it differs from b_j by less than 2^{-j} on K_{j-1}. For $q > 1$, repeat this correction for the difference cocycle and descend one Čech degree at a time until reaching sections. The corrected cochains form a normally convergent sequence on compact sets, because the error on each fixed K_N is bounded by the tail of $\sum_{j>N} 2^{-j}$. Its limit b satisfies $\delta b = c$. Hence all positive Čech cohomology vanishes. Using a Leray refinement gives $H^q(X, \mathcal{F}) = 0$.

For Theorem A fix $x \in X$. Apply Theorem B to

$$0 \longrightarrow \mathcal{I}_{\{x\}}\mathcal{F} \longrightarrow \mathcal{F} \longrightarrow i_*\bigl(\mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x\bigr) \longrightarrow 0,$$

where $i : \{x\} \hookrightarrow X$ and the last term is a skyscraper sheaf. The right-hand term is a finite-dimensional skyscraper sheaf, and the kernel is coherent. By Theorem B,

$$H^1(X, \mathcal{I}_{\{x\}}\mathcal{F}) = 0.$$

Therefore global sections of $\mathcal{F}$ surject onto the fiber $\mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x$. Lift a basis; Nakayama's lemma says that their germs generate $\mathcal{F}_x$. This proves Theorem A. □

Theorem A globalizes local equations; Theorem B removes gluing obstructions. Together they express the analytic analogue of affineness. Two classical problems make the distinction between additive and multiplicative gluing especially clear.

3.1.11 The first and second Cousin problems

Example 3.15: The first Cousin problem: prescribed principal parts

Let $X = \bigcup_i U_i$ be an open cover and choose meromorphic functions $m_i \in \mathcal{M}_X(U_i)$, where $\mathcal{M}_X$ denotes the sheaf of meromorphic functions. Assume that

$$g_{ij} := m_i - m_j \in \mathcal{O}_X(U_i \cap U_j). \tag{9.22}$$

Thus the poles of m_i and m_j, including their polar parts, agree on an overlap; only a holomorphic summand may differ. The class of m_i modulo holomorphic functions is called its *principal part*. The *first Cousin problem* asks for one global meromorphic function $m \in \mathcal{M}_X(X)$ such that

$$m - m_i \in \mathcal{O}_X(U_i) \qquad \text{for every } i. \tag{9.23}$$

On a triple overlap the differences telescope:

$$g_{ij} + g_{jk} + g_{ki} = 0.$$

Hence (g_{ij}) is a Čech 1-cocycle with values in the additive sheaf $\mathcal{O}_X$. If its class vanishes, write $g_{ij} = h_i - h_j$ with $h_i \in \mathcal{O}_X(U_i)$. Then

$$(m_i - h_i) - (m_j - h_j) = g_{ij} - (h_i - h_j) = 0,$$

so the functions $m_i - h_i$ glue to a solution of (9.23). Conversely, a solution m gives $h_i = m_i - m$ and therefore makes the cocycle a coboundary. Thus the obstruction is precisely

$$[g_{ij}] \in H^1(X, \mathcal{O}_X).$$

In particular, Cartan B shows that the first Cousin problem is always solvable on a Stein manifold. In one complex variable this contains the Mittag–Leffler problem of constructing a meromorphic function with prescribed polar parts at a discrete set.

Example 3.16: The second Cousin problem: prescribed zeros and poles

Again let $X = \bigcup_i U_i$, and choose nonzero meromorphic functions $f_i \in \mathcal{M}_X(U_i)$ such that

$$g_{ij} := \frac{f_i}{f_j} \in \mathcal{O}_X^*(U_i \cap U_j). \tag{9.24}$$

The condition says that f_i and f_j have the same zero and pole orders on the overlap. Such compatible local zero-and-pole data is called a *divisor*. The *second Cousin problem* asks for a global meromorphic function $f \in \mathcal{M}_X(X)$ having exactly these local zeros and poles; equivalently,

$$\frac{f}{f_i} \in \mathcal{O}_X^*(U_i) \qquad \text{for every } i. \tag{9.25}$$

Now (g_{ij}) is a multiplicative 1-cocycle: $g_{ij}g_{jk}g_{ki} = 1$. It defines a class

$$[g_{ij}] \in H^1(X, \mathcal{O}_X^*) = \mathrm{Pic}(X),$$

where $\mathrm{Pic}(X)$ denotes the group of isomorphism classes of holomorphic line bundles under tensor product. The displayed element is the class of the line bundle obtained by using the g_{ij} as transition functions. It vanishes exactly when there are $a_i \in \mathcal{O}_X^*(U_i)$ with $g_{ij} = a_i/a_j$. In that case the meromorphic functions f_i/a_i agree on overlaps and glue to the desired global f.

Unlike the first Cousin problem, Cartan B does not directly kill this obstruction, because $\mathcal{O}_X^*$ is not an $\mathcal{O}_X$-module. The exponential sequence

$$0 \longrightarrow \underline{\mathbb{Z}}_X \longrightarrow \mathcal{O}_X \xrightarrow{h \mapsto e^{2\pi i h}} \mathcal{O}_X^* \longrightarrow 1 \tag{9.26}$$

relates it to topology; here $\underline{\mathbb{Z}}_X$ is the locally constant integer sheaf defined in Lecture 8. If X is Stein, Cartan B gives $H^q(X, \mathcal{O}_X) = 0$ for $q \geq 1$, so the long exact sequence yields an isomorphism

$$c_1 : H^1(X, \mathcal{O}_X^*) \xrightarrow{\sim} H^2(X, \underline{\mathbb{Z}}_X).$$

Therefore the second Cousin problem on a Stein manifold is solvable if and only if this integral class vanishes. Steinness removes the analytic obstruction but not the topological one. For example, $(\mathbb{C}^*)^2$ is Stein and deformation retracts onto the real two-torus, whose H^2 is $\mathbb{Z}$; hence nonzero second-Cousin obstructions do occur even on Stein manifolds.

Proposition 3.17: Interpolation on a closed discrete set

Let $A = \{a_j\}$ be a closed discrete subset of a Stein manifold, and prescribe at each a_j a finite jet, of an order m_j that may depend on j. One function $f \in \mathcal{O}(X)$ realizes all the prescribed jets.

Proof. Define an ideal sheaf $\mathcal{I}$ by

$$\mathcal{I}_{a_j} = \mathfrak{m}_{a_j}^{m_j+1}, \qquad \mathcal{I}_x = \mathcal{O}_{X,x} \quad (x \notin A).$$

Closed discreteness is used here: every point has a neighborhood meeting at most one a_j, so this prescription is locally a power of one finitely generated maximal ideal, or the unit ideal. Hence $\mathcal{I}$ is coherent. The quotient $\mathcal{O}_X/\mathcal{I}$ is supported on A, and a global section of it is precisely an arbitrary collection of the prescribed finite jets. The long exact sequence of

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X/\mathcal{I} \longrightarrow 0$$

contains

$$H^0(X, \mathcal{O}_X) \longrightarrow H^0(X, \mathcal{O}_X/\mathcal{I}) \longrightarrow H^1(X, \mathcal{I}).$$

Theorem B makes the last group zero, so the first arrow is surjective. □

Remark (Geometric interpretation)

A Stein manifold is a noncompact complex manifold on which holomorphic functions can serve as coordinates. A strictly psh exhaustion provides convexity; L^2 estimates turn convexity into $\bar{\partial}$-solvability; Cartan B turns solvability into global gluing.

3.1.12 How Cartan B removes a gluing obstruction

Let $Y \subset X$ be a closed complex submanifold of a Stein manifold and let $\mathcal{I}_Y$ be its ideal sheaf. The exact sequence

$$0 \longrightarrow \mathcal{I}_Y \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

gives

$$\mathcal{O}(X) \longrightarrow \mathcal{O}(Y) \longrightarrow H^1(X, \mathcal{I}_Y).$$

Theorem B makes the last group zero, so every holomorphic function on Y extends to X. The statement is qualitative: no norm estimate is supplied. Lecture 29 adds an L^2 estimate and thereby obtains information invisible to Theorem B.

The same pattern works for interpolation. If Z is a finite nonreduced zero-dimensional subspace, the quotient $\mathcal{O}_X/\mathcal{I}_Z$ stores prescribed finite jets. Surjectivity of

$$\mathcal{O}(X) \longrightarrow H^0(X, \mathcal{O}_X/\mathcal{I}_Z)$$

produces a global holomorphic function with those jets. Choosing Z to be two reduced points separates points; choosing a double point produces prescribed differentials.

3.1.13 From global generators to an embedding

Cartan A says that a coherent sheaf is generated by global sections at every point. Apply it first to the ideal of a point and then to its square. The quotients

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \simeq T_x^{*1,0}X, \qquad \mathcal{O}_X/\mathcal{I}_{\{x\}} \simeq i_*\mathbb{C},$$

where $i : \{x\} \hookrightarrow X$ is the inclusion. These quotients show that global functions can separate tangent directions and values. On a compact piece K, finitely many such functions work simultaneously. A suitable finite map

$$F = (f_1, \ldots, f_N) : X \longrightarrow \mathbb{C}^N$$

is then immersive and injective on K.

Holomorphic convexity supplies additional global functions that become large outside successive compact sets. Adjoining them makes F proper. A proper injective holomorphic immersion is a closed holomorphic embedding. This is the mechanism behind the embedding characterization of Stein manifolds; both local generation and global compactness are needed.

The basic examples fit the definition cleanly. Every domain in $\mathbb{C}^n$ that is pseudoconvex is Stein; closed complex submanifolds of $\mathbb{C}^N$ are Stein; products and unbranched coverings of Stein manifolds are Stein. In contrast, a compact connected positive-dimensional complex manifold is not Stein: its holomorphic functions are constant and therefore cannot separate points.

Remark (A practical coherent-sheaf workflow)

To globalize local data on a Stein manifold: encode the data as a quotient of a coherent sheaf, identify the kernel, write the short exact sequence, and use Theorem B to kill the connecting obstruction. Theorem A is used when generators are needed; Theorem B is used when compatibility or extension is needed.

3.1.14 Checking the Stein axioms in model cases

The space $\mathbb{C}^n$ is holomorphically separable because a coordinate z_j separates any two distinct points. It is holomorphically convex: if $K \Subset \mathbb{C}^n$ and $x \in \widehat{K}_{\mathcal{O}}$, then applying the coordinate functions gives $|x_j| \le \sup_K |z_j|$ for every j, so the hull is closed and bounded, hence compact.

For $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$, the two global holomorphic functions z and z^{-1} give the bounds

$$\inf_K |z| \le |x| \le \sup_K |z| \qquad (x \in \widehat{K}_{\mathcal{O}}).$$

Thus the hull stays in a compact annulus. The function $|z|^2 + |z|^{-2}$ is a proper strictly subharmonic exhaustion, giving the analytic criterion as well.

If $Y \subset X$ is a closed complex submanifold of a Stein manifold, Cartan A applied to its ideal sheaf provides local defining functions from global ones. Holomorphic separability restricts from X to Y, and holomorphic convexity follows because $\widehat{K}_{\mathcal{O}(Y)} \subset \widehat{K}_{\mathcal{O}(X)} \cap Y$ after extending the finitely many functions needed near the hull. This proves that Y is Stein; the extension step is a use of Cartan B, not a purely topological fact.

The integral cocycle hidden in the second Cousin problem

The preceding introduction says that multiplicative data have an integral obstruction. Here is the overlap calculation that produces it. Pass to a sufficiently fine refinement on which every g_{ij} has a holomorphic logarithm, and choose logarithms ℓ_{ij} satisfying

$$g_{ij} = e^{2\pi i \ell_{ij}}, \qquad \ell_{ji} = -\ell_{ij}.$$

The multiplicative cocycle identity implies

$$n_{ijk} := \ell_{ij} + \ell_{jk} + \ell_{ki} \in \mathbb{Z}$$

on every triple intersection. Since the left side is holomorphic and takes values in the discrete set $\mathbb{Z}$, it is locally constant. Moreover $\delta n = 0$ because $n = \delta\ell$ and $\delta^2 = 0$. Thus (n_{ijk}) is a Čech 2-cocycle with values in the locally constant sheaf $\underline{\mathbb{Z}}_X$.

Changing a logarithm by an integer-valued 1-cochain changes n by an integral 2-coboundary. Hence the class

$$[n_{ijk}] \in H^2(X, \underline{\mathbb{Z}}_X)$$

depends only on the multiplicative class $[g_{ij}]$; it is its first Chern class. If this integral class vanishes, modify the ℓ_{ij} by integers so that $\ell_{ij} + \ell_{jk} + \ell_{ki} = 0$. They are then an additive holomorphic 1-cocycle. On a Stein manifold, Cartan B gives holomorphic functions b_i with $\ell_{ij} = b_i - b_j$. Setting $a_i = e^{2\pi i b_i}$ yields

$$g_{ij} = \frac{a_i}{a_j},$$

which is exactly the correction needed to solve the second Cousin problem. This calculation shows concretely where the topological obstruction survives after the holomorphic one has been removed.

Interpolation on a discrete set

Let $D = \{a_1, a_2, \ldots\} \subset X$ be closed and discrete, and let $\mathcal{I}_D$ be its ideal sheaf. The sequence

$$0 \longrightarrow \mathcal{I}_D \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0$$

gives

$$H^0(X, \mathcal{O}_X) \longrightarrow H^0(D, \mathcal{O}_D) \longrightarrow H^1(X, \mathcal{I}_D).$$

The ideal sheaf is coherent, so Cartan B kills the last group. Since a holomorphic function on a discrete reduced space is just an arbitrary collection of values, every sequence (c_j) is the restriction of some $f \in \mathcal{O}(X)$. The same argument with $\mathcal{I}_D^{m+1}$ prescribes finite jets of order m.

How Theorem A begins an embedding proof

Fix $x \in X^n$ and let $\mathcal{I}_{\{x\}}$ be the ideal sheaf of the reduced point $\{x\}$. Its stalk at x is the maximal ideal $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$, and $\mathfrak{m}_x/\mathfrak{m}_x^2$ is the holomorphic cotangent space at x. Theorem A gives global functions $f_1, \ldots, f_N$ whose germs generate $\mathfrak{m}_x$. Their differentials span $T_x^{*1,0}X$, so some n of them have linearly independent differentials. The map

$$F = (f_{i_1}, \ldots, f_{i_n}) : X \to \mathbb{C}^n$$

is therefore a local embedding at x by the holomorphic inverse-function theorem. Holomorphic separability supplies global functions distinguishing pairs of points. On a relatively compact set, compactness reduces these local and pairwise choices to finitely many functions, yielding a proper holomorphic embedding after an exhaustion argument. The algebraic input is generation of the stalk $\mathfrak{m}_x$ and its square $\mathfrak{m}_x^2$; the analytic input is the rank calculation.

3.1.15 How Cartan B turns local data into a global section

Let $\mathcal{F}$ be coherent on a Stein manifold and suppose local sections $s_i \in \mathcal{F}(U_i)$ are to be glued. Their differences $c_{ij} = s_i - s_j$ form a Čech one-cocycle precisely when $c_{ij} + c_{jk} + c_{ki} = 0$. Cartan B says its class in $H^1(X, \mathcal{F})$ vanishes, so there are $b_i \in \mathcal{F}(U_i)$ with

$$c_{ij} = b_i - b_j.$$

The corrected sections $s_i - b_i$ agree and therefore glue. This calculation is the last step in extension, interpolation, and Cousin problems; the deep theorem is the vanishing used to find the corrections.

For an extension problem, let $Y \subset X$ be a closed analytic subspace with ideal sheaf $\mathcal{I}_Y$. The exact sequence

$$0 \longrightarrow \mathcal{I}_Y\mathcal{F} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/\mathcal{I}_Y\mathcal{F} \longrightarrow 0$$

gives local lifts s_i of a prescribed section $\bar{s}$ on Y. Their differences lie in $\mathcal{I}_Y\mathcal{F}$ and define the obstruction $\delta(\bar{s}) \in H^1(X, \mathcal{I}_Y\mathcal{F})$. The product sheaf is coherent, and Cartan B makes this group zero, so the preceding correction produces a global lift. Every hypothesis is now visible: closed analyticity gives a coherent ideal, coherence survives products and quotients, and the Stein condition supplies the vanishing.

3.1.16 Self-study workshop: using Cartan's theorems as problem-solving tools

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall holomorphic convexity, coherent ideal sheaves, short exact sequences, and the meaning of a stalk. It is not enough to remember the slogans for Theorems A and B; one must identify the coherent sheaf to which each theorem is applied.

Three standard translations. To separate a point x from a closed analytic set A, use the coherent ideal sheaf $\mathcal{I}_A$. Theorem A says that germs of global sections generate $(\mathcal{I}_A)_x$. If $x \notin A$, this stalk is $\mathcal{O}_{X,x}$, so a suitable global $f \in \mathcal{I}_A(X)$ satisfies $f(x) \neq 0$ while $f|_A = 0$.

To prescribe a finite jet at x, use

$$0 \longrightarrow \mathcal{I}_{\{x\}}^{k+1} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X/\mathcal{I}_{\{x\}}^{k+1} \longrightarrow 0.$$

The quotient is a coherent skyscraper sheaf. Theorem B kills $H^1(X, \mathcal{I}_{\{x\}}^{k+1})$, so the restriction map on global sections is surjective. Thus an abstract vanishing theorem becomes a concrete interpolation statement.

For a first Cousin problem, local meromorphic functions f_i with holomorphic differences $f_i - f_j$ define a Čech 1-cocycle in $\mathcal{O}_X$. If X is Stein, $H^1(X, \mathcal{O}_X) = 0$, hence $f_i - f_j = g_i - g_j$ for holomorphic g_i. The functions $f_i - g_i$ glue to one global meromorphic function. Every sign in the correction can be checked on an overlap.

Checkpoint (choose the sheaf)

(1) Which sheaf controls a holomorphic function with a prescribed value and first derivative at x?
(2) Which cohomology group obstructs gluing additive Cousin data?
(3) Why is a compact connected positive-dimensional complex manifold not Stein?

Solution (checkpoint). Use $\mathcal{O}_X/\mathcal{I}_{\{x\}}^2$ and the exact sequence with kernel $\mathcal{I}_{\{x\}}^2$. Additive Cousin data define a class in $H^1(X, \mathcal{O}_X)$. A compact connected complex manifold has only constant holomorphic functions, so it cannot separate two points and therefore fails a Stein axiom. ◇

Exit criterion

For a separation, interpolation, or gluing problem, you can write the relevant short exact sequence, name the coherent kernel, identify the obstruction group, and state exactly where Theorem A or B removes the obstruction.

Summary. The functor $\Gamma(X, -)$ is left exact but generally not right exact; its failure to lift quotient sections is measured first by a connecting class in H^1 and then by the higher derived functors H^q. A vanishing theorem restores the missing surjectivity. Coherent analytic sheaves are the correct coefficient class because they have finite generators and relations and, by Oka coherence, are stable under finite kernels, images, and cokernels. On a Stein manifold Cartan B kills their positive cohomology, while Cartan A supplies global generators. The first Cousin problem is therefore always solvable on a Stein manifold; the second retains its integral H^2-obstruction.

Exercise 3.18: Open Riemann surfaces

Consult the Behnke–Stein theorem and explain why every open Riemann surface is Stein. Why is the Levi condition automatic in complex dimension one?

Exercise 3.19: Closed complex submanifolds

Assume X is Stein and $Y \subset X$ is a closed complex submanifold. Restrict a strictly psh exhaustion to prove that Y is Stein.

Optional handout: Runge approximation and the Oka viewpoint

This handout is optional. No later argument in the core lectures depends on it.

Let X be Stein. A compact set $K \subset X$ is $\mathcal{O}(X)$-convex if $K = \widehat{K}_{\mathcal{O}(X)}$. The Oka–Weil theorem says that every function holomorphic on a neighborhood of such a K can be approximated uniformly on K by functions in $\mathcal{O}(X)$. For $X = \mathbb{C}$, this is Runge's theorem without prescribed poles.

The proof has the same local-to-global shape as the rest of the course. Cover K by small coordinate sets, approximate locally, and represent the mismatch by a $\bar\partial$-closed form supported near the boundary of the cover. Solve $\bar\partial$ with a weight that is small on K and large away from it. The correction is then globally holomorphic and uniformly small on K.

Oka theory reverses the emphasis. Stein manifolds are flexible as *sources*; Oka manifolds are flexible as *targets*. A complex manifold Y is Oka if continuous maps from Stein spaces to Y can, under suitable hypotheses, be deformed to holomorphic maps with approximation and interpolation. Complex Lie groups and their homogeneous spaces are basic examples.

Example 3.20: The exponential map

A holomorphic map from a Stein manifold to $\mathbb{C}^*$ is a nowhere-zero holomorphic function. The exponential sequence separates the analytic problem of finding a logarithm from its integral topological obstruction. This is a simple model for the Oka principle: once the topological obstruction vanishes, analytic flexibility solves the holomorphic problem.

Exercise 3.21: Polynomial convexity of the ball

Show directly that the closed unit ball in $\mathbb{C}^n$ is polynomially convex by separating an exterior point with a complex linear polynomial.

3.2 Lecture 10: Weierstrass theory and analytic sets

Week 5 material • 90 minutes

Notation for this lecture

$\mathcal{O}_{\mathbb{C}^n,0}$ is the local ring of holomorphic germs at the origin and $\mathbb{C}\{z_1,\dots,z_n\}$ is the ring of convergent power series there. A unit is a germ nonzero at the base point. We distinguish $z' = (z_1,\dots,z_{n-1})$ from the chosen Weierstrass variable z_n. A nonzero germ $f \in \mathcal{O}_{\mathbb{C}^n,0}$ is *regular of order d in z_n* if the one-variable germ $z_n \mapsto f(0,z_n)$ vanishes to exact order d at 0. If $f = f_m + f_{m+1} + \cdots$ is its expansion into homogeneous polynomials with $f_m \neq 0$, then f_m is called the *initial homogeneous part* of f and $m = \operatorname{ord}_0 f$. A monic polynomial $P(z',z_n) = z_n^d + a_1(z')z_n^{d-1} + \cdots + a_d(z')$ with $a_j(0) = 0$ is a Weierstrass polynomial of degree d. For an ideal I, $(f_1,\dots,f_r)$ means the ideal it generates. If I is an ideal in a ring A, then A/I is the quotient ring or module. A ring is *Noetherian* if every ideal is finitely generated. A *domain* has no nonzero zero-divisors, and a *unique factorization domain* (UFD) is a domain in which every nonzero nonunit has a factorization into irreducibles, unique up to order and units. A *finite R-module* means an R-module generated by finitely many elements. An ideal $\mathfrak{p} \neq A$ is prime if $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. The radical $\sqrt{I} = \{a : a^N \in I \text{ for some } N \geq 1\}$ is the intersection of the prime ideals containing I; a prime minimal among them is a *minimal prime over I*.

If $A \subset X$ is an analytic subset of a complex manifold, then $\mathcal{I}_A \subset \mathcal{O}_X$ denotes the sheaf of germs vanishing on A, $\mathcal{O}_A = (\mathcal{O}_X/\mathcal{I}_A)|_A$ its structure sheaf, and $I_{A,x} = (\mathcal{I}_A)_x$. We write A_{Reg} and A_{Sing} for the regular and singular loci. The germ (A,x) is *pure of dimension d* if all its irreducible components have dimension d. An analytic germ is *irreducible* if it is not the union of two proper analytic subgerms.

For an analytic germ (A,x), $\mathcal{O}_{A,x}$ is its local ring and $\mathfrak{m}_{A,x}$ its maximal ideal. Its local dimension is the Krull dimension $\dim_x A := \dim \mathcal{O}_{A,x}$. The Zariski cotangent and tangent spaces are

$$T_x^* A := \mathfrak{m}_{A,x}/\mathfrak{m}_{A,x}^2, \qquad T_x A := \operatorname{Hom}_{\mathbb{C}}(T_x^* A, \mathbb{C}).$$

The module of Kähler differentials $\Omega^1_{A,x}$ is generated by symbols $\mathrm{d}f$, $f \in \mathcal{O}_{A,x}$, subject to $\mathbb{C}$-linearity and the Leibniz rule $\mathrm{d}(fg) = f\,\mathrm{d}g + g\,\mathrm{d}f$.

If a finite module M over a ring R has a presentation $R^p \xrightarrow{B} R^q \to M \to 0$, its k-th Fitting ideal $\operatorname{Fitt}_k(M)$ is generated by the $(q-k)\times(q-k)$ minors of B (with the usual conventions for minors of size 0 or larger than the matrix). Row and column operations show that this ideal is independent of the presentation. For a pure d-dimensional analytic set, the zero set of $\operatorname{Fitt}_d(\Omega^1_A)$ is its singular locus.

For a finite holomorphic map $\pi : A \to U$, the discriminant locus $\Delta_\pi \subset U$ is the set of points near which π is not a disjoint union of biholomorphisms. On $U \setminus \Delta_\pi$ the map is called unbranched. If $W(T) = \prod_{\nu=1}^{q}(T - w_\nu)$ is monic, its polynomial discriminant is $\prod_{\mu<\nu}(w_\mu - w_\nu)^2$; it is symmetric in the roots, hence a polynomial in the coefficients of W, and it vanishes exactly when a root is repeated.

Prerequisite results used below

Holomorphic implicit function theorem. Let $F : U \subset \mathbb{C}^{n+r} \to \mathbb{C}^r$ be holomorphic, $F(a) = 0$, and suppose some $r \times r$ minor of its complex Jacobian is invertible at a. Then $F^{-1}(0)$ is, near a, the graph of a holomorphic map in the remaining n variables.

Identity theorem for germs. If a holomorphic function on a connected domain vanishes on a nonempty open subset, it vanishes identically. Hence the local ring of germs on an irreducible

smooth germ has no zero divisors, and equality of convergent power series can be checked on any sufficiently small representative.

Argument principle and Rouché's theorem. If a holomorphic function g has no zero on the positively oriented circle $|\zeta| = R$, then

$$\frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=R} \frac{g'(\zeta)}{g(\zeta)}\, \mathrm{d}\zeta$$

is the number of zeros of g in $|\zeta| < R$, counted with multiplicity. If $|g - h| < |g|$ on the circle, then g and h have the same number of zeros inside it. We use these one-variable results for the slices $z_n \mapsto f(z', z_n)$.

Two algebra facts. If R is Noetherian, every submodule of a finite R-module is finite. Gauss's lemma says that if R is a UFD, then the polynomial ring $R[T]$ is again a UFD.

Why these ideas are needed

The implicit-function theorem describes a zero set only where its Jacobian has full rank. Singular analytic sets occur exactly where that tool fails, so a different local normal form is needed. Weierstrass preparation isolates one variable and replaces a holomorphic equation by a monic polynomial with holomorphic coefficients; division then gives a controlled remainder. These statements make finite projection and local algebra calculable. They also supply the analytic mechanism behind Oka coherence, but the technical relation-sheaf induction has been moved to Appendix A.4. The classroom narrative can therefore focus on how Weierstrass theory controls analytic sets.

Plan for the lecture. 0–22 min: local rings of convergent power series; 22–50 min: Weierstrass preparation and division; 50–62 min: Noetherianity and the role of finite remainders; 62–82 min: dimension and regular points of analytic sets; 82–90 min: computations.

The ring of germs at $0 \in \mathbb{C}^n$ is $\mathcal{O}_{\mathbb{C}^n,0} = \mathbb{C}\{z_1, \dots, z_n\}$, with maximal ideal $\mathfrak{m} = (z_1, \dots, z_n)$. It is a Noetherian local domain, but this is not a formal consequence of the polynomial case; convergence is controlled by Weierstrass theory.

Lemma 3.22: Choosing a Weierstrass direction

Let $0 \neq f \in \mathbb{C}\{z_1, \dots, z_n\}$ and put $m = \operatorname{ord}_0 f$. After an invertible complex-linear change of coordinates, f is regular of order m in the last variable. One direction can be chosen to work simultaneously for any prescribed finite collection of nonzero germs.

Proof. Write the Taylor expansion as

$$f = f_m + f_{m+1} + \cdots,$$

where each f_k is homogeneous of degree k and $f_m \neq 0$. Choose $v \in \mathbb{C}^n$ with $f_m(v) \neq 0$, and choose a linear coordinate system whose last coordinate axis is the line $\mathbb{C}v$. Along that axis,

$$f(tv) = t^m f_m(v) + t^{m+1} f_{m+1}(v) + \cdots,$$

so the restriction has exact order m. The forbidden directions form the proper algebraic cone $\{f_m = 0\}$. For finitely many germs the union of the corresponding proper cones is still a proper subset of $\mathbb{C}^n$; a direction outside that union works for all of them. □

Definition 3.23: Weierstrass polynomial

A Weierstrass polynomial of degree d in z_n is

$$P(z', z_n) = z_n^d + a_1(z')z_n^{d-1} + \cdots + a_d(z'), \qquad a_j(0) = 0.$$

Theorem 3.24: Weierstrass preparation and division

Suppose $f \in \mathbb{C}\{z', z_n\}$ is regular of order d in z_n, meaning $f(0, z_n) = z_n^d g(z_n)$ with $g(0) \neq 0$. Then uniquely $f = uP$, where u is a unit and P is a degree-d Weierstrass polynomial. For every h, there are unique q and $r \in \mathbb{C}\{z'\}[z_n]$ with $\deg_{z_n} r < d$ such that $h = qf + r$.

Proof. We first prove preparation. Choose $R > 0$ so small that $f(0, \zeta)$ has no zero for $0 < |\zeta| \leq R$. On $|\zeta| = R$ the function $f(0, \zeta)$ has a positive minimum. After shrinking the z'-polydisc, Rouché's theorem shows that every slice $\zeta \mapsto f(z', \zeta)$ has exactly d zeros $w_1(z'), \ldots, w_d(z')$ in $|\zeta| < R$, counted with multiplicity. The individual roots need not be single-valued, so we do not try to label them holomorphically. Instead define, for $1 \leq k \leq d$,

$$S_k(z') = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=R} \zeta^k \frac{\partial f/\partial \zeta}{f}(z', \zeta)\, \mathrm{d}\zeta.$$

The denominator is uniformly nonzero on the contour, hence differentiation under the integral shows that S_k is holomorphic in z'. The residue theorem gives $S_k = \sum_{\nu=1}^{d} w_\nu^k$. If e_j is the j-th elementary symmetric function and $e_0 = 1$, Newton's recursive identities are

$$j e_j = \sum_{k=1}^{j} (-1)^{k-1} e_{j-k} S_k, \qquad 1 \leq j \leq d.$$

They show inductively that every e_j is holomorphic in z'. Therefore

$$P(z', z_n) = \prod_{\nu=1}^{d} (z_n - w_\nu(z')) = z_n^d + a_1(z')z_n^{d-1} + \cdots + a_d(z')$$

has holomorphic coefficients. Since all roots equal 0 at $z' = 0$, each $a_j(0) = 0$.

The functions f and P have the same zeros, with the same multiplicities, on every vertical disc. Thus $u = f/P$ and $u^{-1} = P/f$ have removable singularities on each such disc. Joint holomorphicity follows from Cauchy's formula on a slightly larger circle on which both quotients are already holomorphic in (z', z_n). Consequently u is holomorphic and nowhere zero. The root multiset determines the monic polynomial P, so both P and u are unique.

For division it suffices first to divide by P. Shrink the polydisc so that all roots of $P(z', \cdot)$ lie in $|\zeta| < R/2$ and h is holomorphic for $|\zeta| \leq R$. For $|z_n| < R/2$, set

$$q(z', z_n) = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=R} \frac{h(z', \zeta)}{P(z', \zeta)(\zeta - z_n)}\, \mathrm{d}\zeta,$$
$$r(z', z_n) = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=R} \frac{h(z', \zeta)}{P(z', \zeta)} \frac{P(z', \zeta) - P(z', z_n)}{\zeta - z_n}\, \mathrm{d}\zeta.$$

All denominators are uniformly separated from zero on the contour, so both expressions are holomorphic. The difference quotient in the second integral is a polynomial of degree at most $d - 1$ in z_n; hence so is r. Adding the two integrands and applying Cauchy's formula gives $h = Pq + r$. If two such decompositions existed, formal division by the monic polynomial P would make their remainders equal, and then their quotients equal. Finally $f = uP$ gives $h = f(q/u) + r$. This proves preparation and division, including convergence and uniqueness. □

Corollary 3.25: Noetherianity and factoriality of the local ring

$\mathbb{C}\{z_1, \dots, z_n\}$ is Noetherian and is a unique factorization domain.

Proof. We argue by induction on n. The ring $\mathbb{C}\{z_1\}$ is a discrete valuation ring: every nonzero germ is a unit times a power of z_1. Assume $R' = \mathbb{C}\{z_1, \dots, z_{n-1}\}$ is Noetherian and let $I \subset R'\{z_n\}$ be a nonzero ideal. After a generic linear change of variables, choose $f \in I$ regular of order d in z_n and replace it by its Weierstrass polynomial. Divide every $g \in I$ by f:

$$g = qf + r_0 + r_1 z_n + \cdots + r_{d-1} z_n^{d-1}.$$

The coefficient vectors $(r_0, \dots, r_{d-1})$ form an R'-submodule $M \subset (R')^d$. Since R' is Noetherian, M is generated by finitely many remainder vectors, represented by $g_1, \dots, g_N \in I$. Subtracting an $R'\{z_n\}$-linear combination of the g_j from any $g \in I$ leaves a multiple of f. Thus $I = (f, g_1, \dots, g_N)$, proving Noetherianity.

For unique factorization, again use induction. We first record the bridge between analytic and polynomial divisibility. If $P, F \in R'[z_n]$, with P Weierstrass, and $F = Ph$ for some $h \in R'\{z_n\}$, polynomial division gives $F = PQ + R$ with $Q, R \in R'[z_n]$ and $\deg_{z_n} R < \deg_{z_n} P$. Uniqueness in Weierstrass division forces $R = 0$ and $h = Q$. Thus analytic divisibility by P already occurs in $R'[z_n]$.

Now write a nonzero nonunit as uP by preparation. By the induction hypothesis R' is a UFD, so Gauss's lemma makes $R'[z_n]$ a UFD. If $P = gh$ in $R'\{z_n\}$, preparing the two nonunits gives $g = u_1 P_1$ and $h = u_2 P_2$, where P_1, P_2 are Weierstrass polynomials. Uniqueness of preparation yields $P = P_1 P_2$ and $u_1 u_2 = 1$. Conversely, the divisibility observation above shows that an irreducible Weierstrass polynomial in $R'[z_n]$ remains irreducible in the analytic ring. Existence and uniqueness of factorization therefore reduce to those in $R'[z_n]$. □

The finite-remainder argument in this proof is also the analytic input behind Oka's coherence theorem, stated as Theorem 3.5. We use the theorem in the classroom text without interrupting the study of analytic sets with its technical induction. A complete relation-sheaf proof is given in Appendix A.4. Notice that coherence of the *full* vanishing ideal of an analytic set is still a further theorem below; it is not a formal consequence of the Noetherianity of individual stalks.

Definition 3.26: Analytic set and regular point

A closed set $A \subset X$ is analytic if locally it is the common zero set of finitely many holomorphic functions. A point $x \in A$ is regular if A is a complex submanifold near x; all other points form A_{Sing}.

Full rank of the Jacobian is a sufficient regularity test. If the chosen list of equations is redundant, rank deficiency need not mean that the set is singular: one must inspect the ideal, not merely a presentation.

Theorem 3.27: Local structure of analytic sets

An analytic set has locally finitely many irreducible components. The regular part of each irreducible component is dense and connected. At every point of a pure d-dimensional component, a generic linear projection to $\mathbb{C}^d$ is finite after the source and target are sufficiently shrunk; away from its discriminant it is an unbranched finite covering.

Proof. We separate the algebraic, finite-map, and extension steps.

Step 1: local components. At $x \in A$, the radical ideal $I_{A,x} \subset \mathcal{O}_{X,x}$ has finitely many minimal prime ideals because $\mathcal{O}_{X,x}$ is Noetherian. Their zero germs are exactly the irreducible components of (A, x). After shrinking once, representatives of these finitely many germs give all irreducible components near x.

Step 2: finite projection. Let (B, x) be a pure d-dimensional component. A generic linear map $p : \mathbb{C}^N \to \mathbb{C}^d$ has isolated fiber $B \cap p^{-1}(p(x))$ at x: one may choose successive generic linear forms so that each new level equation lowers the local dimension by one, until only x remains. Successive Weierstrass preparation in the remaining $N - d$ coordinates shows that $\mathcal{O}_{B,x}$ is a finite module over $\mathcal{O}_{\mathbb{C}^d,p(x)}$. After shrinking source and target, the restriction $p : B \to U$ is therefore finite: its fibers are finite and the inverse image of a compact subset of U is compact.

Step 3: the good locus. Let q be the generic number of points in a fiber. A generic linear function t on the remaining coordinates separates those q points and satisfies a monic degree-q polynomial over the base. Because the ground field has characteristic zero, the generic roots are distinct, so the polynomial's discriminant is not identically zero. Its zero set, enlarged if necessary to include the locus where t fails to generate the finite algebra, gives a proper analytic subset $\Delta_p \subset U$. At a point over $U \setminus \Delta_p$, the relevant root is simple. The holomorphic implicit-function theorem then writes each sheet as a graph; thus p is locally a disjoint union of biholomorphisms there. In particular, B_{Reg} is nonempty and dense.

Step 4: connectedness. The covering $p^{-1}(U \setminus \Delta_p) \to U \setminus \Delta_p$ is connected. Indeed, if its sheets split into two monodromy orbits, take for each orbit the product of the corresponding linear factors of the primitive polynomial. The coefficients are symmetric within that orbit, hence single-valued and holomorphic off Δ_p; they are locally bounded because all roots remain in one fixed polydisc. The removable-singularity theorem extends the coefficients across Δ_p. The two resulting monic factors cut B into two proper analytic subsets, contradicting irreducibility.

Thus B_{Reg} contains a connected dense subset. A space containing a connected dense subset is connected, so B_{Reg} is connected. □

Lemma 3.28: Uniform denominator description of a vanishing ideal

Let $(A, 0) \subset (\mathbb{C}^N, 0)$ be an irreducible analytic germ. After shrinking to a neighborhood U of 0, there are holomorphic functions $G_1, \dots, G_M, \delta \in \mathcal{O}(U)$, all G_j vanishing on A, such that δ does not vanish identically on A and, for every $x \in A \cap U$,

$$(\mathcal{I}_A)_x = \{f \in \mathcal{O}_{U,x} : \delta_x f \in (G_{1,x}, \dots, G_{M,x})\}. \tag{10.2}$$

Proof. Choose a finite projection $\pi : A \cap U \to U' \subset \mathbb{C}^d$ as in the preceding theorem and a generic linear function t on the remaining coordinates which separates the points of a generic fiber. If the generic degree is q, the values of t on a fiber are the roots of a monic polynomial

$$W(z', T) = T^q + a_1(z')T^{q-1} + \cdots + a_q(z').$$

The coefficients are elementary symmetric functions of the fiber values and extend holomorphically across the discriminant. Let $\delta(z')$ be the discriminant of W. On $\{\delta \neq 0\}$, Lagrange interpolation expresses every remaining coordinate z_k in the form

$$\delta(z')z_k = B_k(z', t), \qquad \deg_T B_k < q.$$

Both sides are holomorphic, so the identities extend to A. Take the G_j to be $W(z', t)$ and the finitely many functions $\delta(z')z_k - B_k(z', t)$, together with local equations needed for the chosen embedding.

We explain the membership test, which is the point of the lemma. Repeated Weierstrass division by W reduces $\delta^m f$, modulo $(G_1, \dots, G_M)$, to a polynomial $R(z', t)$ of degree $< q$; the exponent m is fixed after shrinking U, because only the finitely many coordinate substitutions above are used. If f vanishes on A, then on a fiber with $\delta \neq 0$ the polynomial $R(z', T)$ vanishes at the q distinct roots of $W(z', T)$. Its degree is $< q$, hence $R = 0$. Holomorphicity extends this identity across $\{\delta = 0\}$. Thus $\delta^m f \in (G_1, \dots, G_M)$.

Replace δ by δ^m. This proves the inclusion from left to right in (10.2). Conversely, if $\delta f = \sum a_j G_j$, then f vanishes on the dense set $A \cap \{\delta \neq 0\}$. The identity theorem on the irreducible germ gives $f|_A = 0$, proving the reverse inclusion. □

Theorem 3.29: Coherence of vanishing ideals

For every analytic subset A of a complex manifold X, the full vanishing ideal $\mathcal{I}_A$ is coherent. Consequently $\mathcal{O}_A = (\mathcal{O}_X/\mathcal{I}_A)|_A$ is a coherent sheaf of rings.

Proof. First suppose (A, x) is irreducible and choose $G_1, \dots, G_M, \delta$ as in the lemma. Formula (10.2) identifies $\mathcal{I}_A$ with the image under projection to the first component of the relation sheaf

$$\mathcal{R}(\delta, -G_1, \dots, -G_M) = \{(f, a_1, \dots, a_M) : \delta f - \textstyle\sum_j a_j G_j = 0\}.$$

Oka's theorem makes this relation sheaf coherent, and the image of a morphism of coherent sheaves is coherent. If (A, x) has irreducible components $A_1, \dots, A_s$, then

$$\mathcal{I}_A = \mathcal{I}_{A_1} \cap \cdots \cap \mathcal{I}_{A_s}.$$

A finite intersection of coherent subsheaves is the kernel of the map to the direct sum of the corresponding quotients, hence is coherent. The quotient defining $\mathcal{O}_A$ is coherent by the exact-sequence property. □

Corollary 3.30: Analyticity of the singular locus

For an analytic set A, the singular locus A_{Sing} is analytic and has smaller dimension on every irreducible component.

Proof. Suppose first that $A \subset U \subset \mathbb{C}^N$ is pure of dimension d. By the preceding theorem, after shrinking U its full vanishing ideal is generated by $g_1, \dots, g_s$. The conormal presentation is

$$\mathcal{O}_A^{s} \xrightarrow{(\partial g_i/\partial z_j)} \mathcal{O}_A^{N} \longrightarrow \Omega^1_A \longrightarrow 0.$$

At $x \in A$, the germ is smooth of dimension d exactly when the differentials of the full ideal span an $(N-d)$-dimensional conormal space. Hence

$$A_{\mathrm{Sing}} = V\big(\mathrm{Fitt}_d(\Omega^1_A)\big),$$

or concretely, it is cut out inside A by all $(N-d)\times(N-d)$ minors of the displayed Jacobian matrix. Thus it is analytic. It is proper by the density of the regular locus proved above, so its dimension is $< d$ on each irreducible component. For a non-pure set, take the union of the singular loci of the finitely many local irreducible components and all their pairwise intersections. □

Example 3.31: The cusp

The curve $A = \{y^2 = x^3\} \subset \mathbb{C}^2$ is singular at the origin, but $t \mapsto (t^2, t^3)$ is its normalization. The ring $\mathbb{C}\{t^2, t^3\} \subset \mathbb{C}\{t\}$ is not integrally closed: t is integral over it.

Warning

Being defined by r equations gives $\operatorname{codim} A \le r$, not $\operatorname{codim} A \ge r$. Equations may be dependent and some components may have larger dimension. A complete intersection requires equality and, scheme-theoretically, a regular sequence.

3.2.1 A division calculation and its geometric meaning

Let $P(x, y) = y^2 - x$. Weierstrass division in the variable y gives

$$y^4 + x = (y^2 + x)(y^2 - x) + (x^2 + x).$$

The remainder has degree less than two in y, and uniqueness shows that no hidden convergent-series terms can occur. Modulo (P), every germ therefore has a unique representative $a(x) + b(x)y$. Algebraically,

$$\mathbb{C}\{x, y\}/(y^2 - x)$$

is a free module of rank two over $\mathbb{C}\{x\}$. Geometrically, projection to the x-axis is a finite map of degree two near the origin.

For the cusp $y^2 = x^3$, the same projection is finite and the discriminant is supported at $x = 0$. Away from the discriminant the two roots vary holomorphically on simply connected open sets. At the origin the sheets join, and the parametrization $t \mapsto (t^2, t^3)$ separates the hidden parameter. Weierstrass preparation thus converts a local analytic set into a branched finite cover plus lower-dimensional exceptional data.

3.2.2 Tangent space, dimension, and the chosen equations

If the germ $(A, 0) \subset (\mathbb{C}^N, 0)$ has ideal I, its Zariski tangent space is

$$T_0 A = \{v \in \mathbb{C}^N : \mathrm{d}f_0(v) = 0 \text{ for every } f \in I\}.$$

Equivalently,

$$T_0^* A \simeq \mathfrak{m}_A/\mathfrak{m}_A^2.$$

This definition uses the full ideal and is independent of a selected generating list. For $y^2 = x^3$, the defining equation has no linear term, so $T_0 A = \mathbb{C}^2$ although A has dimension one. The tangent-space jump detects singularity.

The local dimension is instead the Krull dimension of $\mathcal{O}_{A,0}$. A finite map to $\mathbb{C}^d$ proves that the local ring is finite over a convergent power-series ring in d variables, making the analytic and algebraic notions of dimension agree. The inequality "r equations give codimension at most r" follows because each new equation can lower dimension by at most one.

Oka coherence upgrades these stalk calculations to neighborhoods. For a finite holomorphic matrix $A : \mathcal{O}^p \to \mathcal{O}^q$, the sheaves $\ker A$, $\operatorname{im} A$, and $\operatorname{coker} A$ are coherent. Rank-drop loci are therefore analytic: they are cut out intrinsically by minors, or more generally by Fitting ideals. This is the basic mechanism for proving that singular loci and degeneracy loci are analytic subsets.

Warning

The dimension of the Zariski tangent space is an upper bound for local dimension, not its definition. It is exact at a smooth point and may be much larger at a singular point. Likewise, one Jacobian matrix built from redundant equations may have smaller rank than the intrinsic conormal space.

3.2.3 Weierstrass preparation and division in examples

Consider

$$f(z, w) = e^w (w^2 - z).$$

Since $f(0,w) = e^w w^2$ has order two, preparation gives a degree-two polynomial. In this case it is visible without a theorem:

$$f = uP, \qquad u = e^w, \qquad P = w^2 - z.$$

To divide $h = w^4 + z$ by P, use

$$w^4 = (w^2 + z)(w^2 - z) + z^2.$$

Hence

$$h = (w^2 + z)P + (z^2 + z),$$

and the remainder has degree zero in w, as required. For division by $f = uP$, the quotient is $(w^2 + z)e^{-w}$ and the remainder is unchanged. Uniqueness follows from the degree condition: if $(q - q')P = r' - r$ and $q - q' \neq 0$, then the monicity of P makes the left side have w-degree at least 2, whereas the right side has w-degree less than 2. Hence both sides vanish, so $q = q'$ and $r = r'$.

The finite-map content is also explicit. On $X = \{w^2 = z\}$ the projection $(z,w) \mapsto z$ has two points over $z \neq 0$ and one double point over 0. The local algebra

$$\mathcal{O}_X \simeq \mathbb{C}\{z\}[w]/(w^2 - z)$$

is a free $\mathbb{C}\{z\}$-module with basis $1, w$ because Weierstrass division reduces every power series uniquely to $a(z) + b(z)w$.

Regularity, tangent spaces, and tangent cones

For $X = V(f_1, \dots, f_r) \subset \mathbb{C}^n$, differentiating every equation along a holomorphic curve γ in X gives

$$\{\gamma'(0) : \gamma : (\mathbb{C},0) \to (X,x) \text{ is holomorphic}\} \subseteq \bigcap_j \ker \mathrm{d}f_j(x) = T_x X.$$

The right-hand side is the Zariski tangent space. At a smooth point it agrees with the usual holomorphic tangent space; choosing a minimal set of equations and applying the implicit-function theorem gives equality with the curve velocities and dimension $\dim_x X$.

For a hypersurface germ $\{f = 0\}$, the tangent cone at the origin is the zero set of the lowest nonzero homogeneous part of the Taylor series of f. It records tangent directions with multiplicity, rather than only the vector space T_0X.

For the cusp $C = \{y^2 - x^3 = 0\}$,

$$\mathrm{d}(y^2 - x^3) = 2y\,\mathrm{d}y - 3x^2\,\mathrm{d}x,$$

which vanishes at the origin. Thus the Zariski tangent space is all of $\mathbb{C}^2$ even though the curve has dimension one. The lowest-degree homogeneous term is y^2, so the tangent cone is the double line $y = 0$. The parametrization $t \mapsto (t^2, t^3)$ has derivative zero at 0, another signal that it is not a local embedding there.

For the node $N = \{y^2 - x^2 - x^3 = 0\}$ the differential also vanishes at 0, but the lowest homogeneous part factors:

$$y^2 - x^2 = (y - x)(y + x).$$

The tangent cone is the union of two distinct lines. Locally one can choose a square root of $1 + x$ and factor

$$y^2 - x^2(1 + x) = \big(y - x\sqrt{1 + x}\big)\big(y + x\sqrt{1 + x}\big),$$

so the node has two smooth branches, unlike the irreducible cusp.

Why redundant equations can mislead

The smooth line $L = \{y = 0\} \subset \mathbb{C}^2$ can also be presented as $V(y^2, y^3)$. The Jacobian of this displayed pair vanishes everywhere on L, but L is not singular. The radical ideal is (y), and its one generator has nonzero differential. Therefore the geometric regularity test uses the ideal sheaf (or the module of Kähler differentials), not the rank of an arbitrary redundant list.

For a hypersurface $f = 0$, the singular locus is

$$V\left(f, \frac{\partial f}{\partial z_1}, \dots, \frac{\partial f}{\partial z_n}\right).$$

For the cusp this is only the origin. Away from the origin either x or y is nonzero, one partial derivative is nonzero, and the implicit-function theorem gives a one-dimensional chart. This verifies directly that the regular locus is dense.

3.2.4 Uniform estimates behind the local theorem

The contour formulas in the proof do more than establish existence. Fix a smaller parameter polydisc K and radii $0 < \rho < R$ so that every root of $P(z, \cdot)$ lies in $|w| < \rho$ and

$$\mu := \min_{z \in K,\, |\zeta| = R} |P(z, \zeta)| > 0.$$

If h is bounded on $K \times \{|w| \le R\}$, the quotient formula gives, for $|w| \le \rho$,

$$|q(z, w)| \le \frac{R}{\mu(R - \rho)} \sup |h|.$$

The difference quotient of P is bounded on the same compact set, so each coefficient of the remainder is bounded by $C_P \sup |h|$, where C_P depends only on P, K, ρ, R, not on h. Cauchy's estimates give analogous bounds for every derivative on a still smaller polydisc.

This uniformity explains why Weierstrass division is useful for sheaves. A purely stalkwise argument could choose a different neighborhood for every relation. The fixed constants above allow all divisions of a bounded family to be carried out on one common smaller polydisc. Consequently generators constructed by the finite-remainder argument generate nearby stalks as well as the stalk at the base point; that neighborhood control is exactly what coherence requires.

The finite-remainder mechanism also proves Oka's coherence theorem. Since the relation-sheaf induction is not needed for the classroom discussion of analytic sets, it has been moved to Appendix A.4; there the quotient and remainder terms, the lower-dimensional matrix, and the common-neighborhood issue are all written out explicitly.

3.2.5 Why the singular locus is analytic

Suppose locally that $A \subset \mathbb{C}^N$ has ideal generated by $f_1, \dots, f_r$. The conormal sequence gives a presentation

$$\mathcal{O}_A^r \xrightarrow{(\partial f_i / \partial z_j)} \mathcal{O}_A^N \longrightarrow \Omega_A^1 \longrightarrow 0.$$

If A is pure of dimension d, then at a regular point Ω_A^1 is free of rank d. Equivalently, some $(N - d) \times (N - d)$ Jacobian minor is a unit. Consequently

$$A_{\mathrm{Sing}} = V\big(\mathrm{Fitt}_d(\Omega_A^1)\big).$$

This formula both proves analyticity and shows why adding redundant equations does not change the answer: Fitting ideals depend on the presented module, not on the chosen presentation.

3.2.6 Self-study workshop: using Weierstrass division as finite-dimensional algebra

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall convergent power series, units in a local ring, finite modules, and the implicit function theorem. Distinguish a Weierstrass polynomial from an arbitrary polynomial in the distinguished variable.

A model quotient calculated completely. Consider

$$P(z,w) = w^2 - z^3 \in \mathbb{C}\{z,w\}.$$

It is monic of degree two in w. Weierstrass division says that every $f \in \mathbb{C}\{z,w\}$ has a unique expression

$$f = qP + a(z) + b(z)w.$$

Hence

$$\mathbb{C}\{z,w\}/(P) \simeq \mathbb{C}\{z\} \oplus \mathbb{C}\{z\}w$$

as a $\mathbb{C}\{z\}$-module. The projection of the analytic curve $V(P)$ to the z-disc is therefore finite of generic degree two. The fiber over $z \neq 0$ has the two points $w = \pm z^{3/2}$ locally after choosing a branch, while the fiber over 0 is one point with multiplicity two.

The same equation also illustrates why finite projection does not imply smoothness. The gradient

$$\left(\frac{\partial P}{\partial z}, \frac{\partial P}{\partial w}\right) = (-3z^2, 2w)$$

vanishes at the origin, so the curve is singular there. The local ring is reduced because P has no repeated irreducible factor, but it is not normal; normalization will add the parameter t with $z = t^2$, $w = t^3$.

In a proof of coherence, division replaces infinitely many coefficients in the w-direction by the finite remainder vector (a, b). Relations among analytic generators can then be handled inductively in one fewer variable. The finiteness of the remainder is the mechanism, not a decorative normal form.

Checkpoint (read geometry from the remainder)

(1) What is the rank of $\mathbb{C}\{z,w\}/(w^d + a_1 w^{d-1} + \cdots + a_d)$ over $\mathbb{C}\{z\}$?
(2) Why is uniqueness in division essential for this assertion?
(3) For $P = w^2 - z$, is $V(P)$ singular at the origin?

Solution (checkpoint). The quotient is free of rank d with basis $1, w, \ldots, w^{d-1}$. Uniqueness proves that these residue classes are independent as well as generating. For $w^2 - z$, the gradient is $(-1, 2w)$, which never vanishes, so the curve is smooth despite the two-to-one projection away from the branch point. ◇

Exit criterion

You can divide by a Weierstrass polynomial, identify the finite module and its rank, and separately test smoothness, reducedness, and finiteness of projection.

Summary. Weierstrass division lowers the dimension of local analytic problems and replaces an arbitrary holomorphic function by a finite remainder. This mechanism underlies Oka coherence, whose proof is recorded in the appendix, and makes finite projections and local dimension accessible. An analytic set must be understood through its full ideal sheaf, not one convenient list of equations.

Exercise 3.32: Node and cusp

Compare $xy = 0$ and $y^2 = x^3$ at the origin: irreducible components, Zariski tangent spaces, and normalizations.

Exercise 3.33: A division calculation

Divide $h(x, y) = y^4 + x$ by $P(x, y) = y^2 - x$ and write the quotient and remainder.

Optional handout: Intersection multiplicity and local algebra

This handout is optional. No later argument in the core lectures depends on it.

Let $f, g \in \mathbb{C}\{x, y\}$ define plane curves with an isolated intersection at the origin. Their local intersection multiplicity is

$$i_0(f, g) = \dim_{\mathbb{C}} \frac{\mathbb{C}\{x, y\}}{(f, g)}.$$

The dimension is finite exactly when the two germs have no common irreducible component. This definition is invariant under local biholomorphisms, symmetric in f, g, and stable under small perturbation when all nearby intersection points are counted.

Example 3.34: Tangency

For $f = y$ and $g = y - x^m$,

$$i_0(f, g) = \dim_{\mathbb{C}} \mathbb{C}\{x\}/(x^m) = m.$$

Thus multiplicity records the order of tangency, not merely the set-theoretic fact that the curves meet at one point.

If f is irreducible with parametrization $\gamma(t)$, then $i_0(f, g) = \operatorname{ord}_t(g \circ \gamma)$. For the cusp $f = y^2 - x^3$ with $\gamma(t) = (t^2, t^3)$, its intersection with $y = 0$ has multiplicity 3, while its intersection with $x = 0$ has multiplicity 2.

For a finite-length R-module M, its length $\ell_R(M)$ is the number of factors in a composition series whose successive quotients are simple modules; when $R/\mathfrak{m} \simeq \mathbb{C}$, this equals $\dim_{\mathbb{C}} M$. For a local ring $(R, \mathfrak{m})$ of dimension d, the Hilbert–Samuel function $\ell_R(R/\mathfrak{m}^{k+1})$ agrees for large k with a polynomial of degree d. Its leading term is $e(R)k^d/d!$; the integer $e(R)$ is the multiplicity of the analytic germ. Smooth points have multiplicity one. Lecture 24 relates this algebraic number to the Lelong number of the integration current.

Exercise 3.35: A monomial intersection

Compute $\dim_{\mathbb{C}} \mathbb{C}\{x, y\}/(x^a, y^b)$ and interpret the answer as an intersection multiplicity.

3.3 Lecture 11: Complex analytic spaces, normalization, and divisors

Week 6 material • 90 minutes

Notation for this lecture

For a complex analytic space X, $\mathcal{O}_{X,x}$ is its local ring, X_{Reg} and X_{Sing} are its regular and singular loci, and $\mathcal{M}_X$ denotes the sheaf of meromorphic functions. A space is *reduced* if $\mathcal{O}_{X,x}$ has no nonzero nilpotents, and *normal* if every local ring is a domain integrally closed in its fraction field. For a reduced ring R, its *total ring of fractions* $Q(R)$ is the localization obtained by inverting all non-zero-divisors; when R is a domain, this is its fraction field. A ring element a is a *non-zero-divisor* if multiplication $r \mapsto ar$ is injective. The height $\operatorname{ht}\mathfrak{p}$ of a prime ideal is the supremum of the lengths of strict chains of prime ideals ending at $\mathfrak{p}$. A sequence $a_1, \dots, a_r$ in a local ring is *regular* if a_j is a non-zero-divisor in $R/(a_1, \dots, a_{j-1})$ for every j. An analytic subspace is a *local complete intersection* if its ideal is locally generated by a regular sequence whose length equals the codimension. $\nu : X^\nu \to X$ denotes normalization. A prime divisor is an irreducible codimension-one analytic subset D, and $\operatorname{ord}_D(f)$ is the order of a meromorphic function along it. A Weil divisor is a locally finite formal sum $\sum_D m_D D$; a Cartier divisor is locally represented by nonzero meromorphic functions modulo holomorphic units. Its associated line bundle is $\mathcal{O}_X(D)$.
A meromorphic function on a reduced space is *weakly holomorphic* if it is holomorphic on the regular locus and locally bounded near the singular locus. $\widetilde{\mathcal{O}}_X$ denotes the sheaf of weakly holomorphic functions. For irreducible reduced spaces, a holomorphic map $F : Y \to X$ is *dominant* if its image is dense in X; equivalently, pullback embeds the meromorphic function field of X into that of Y. A holomorphic map is *finite* if it is proper and has finite fibers; locally this is equivalent to its target structure sheaf making the direct image of the source structure sheaf a finite module. For a holomorphic map $\nu : Y \to X$, the direct-image sheaf is defined by $(\nu_*\mathcal{O}_Y)(U) = \mathcal{O}_Y(\nu^{-1}(U))$. The symbol $F^\sharp$ denotes the pullback morphism on structure sheaves induced by a holomorphic map F.

Prerequisite results used below

Integral-dependence criterion. An element b of a field extension of a domain A is integral over A if it satisfies a monic equation $b^d + a_1 b^{d-1} + \cdots + a_d = 0$ with $a_j \in A$. The integral closure $\widetilde{A}$ is the set of all such elements in the fraction field of A.
Finite algebras and analytic spaces. If $\mathcal{A}$ is a coherent $\mathcal{O}_X$-algebra that is locally finite as an $\mathcal{O}_X$-module, then locally choosing module generators and their multiplication relations realizes $\mathcal{A}$ as the structure sheaf of a complex space finite over X. This is the analytic analogue of constructing Spec of a finite algebra; it will be used explicitly in the normalization proof.

Why these ideas are needed

Manifolds exclude crossings, cusps, nilpotent structure, and many natural limits of geometric constructions. Complex analytic spaces retain local holomorphic equations while allowing these singularities. The local ring distinguishes phenomena that the underlying point set cannot see. Normalization then asks for the best finite normal model of a singular space, separating hidden branches without discarding holomorphic functions. Divisors focus attention on codimension one because zeros and poles are measured there. Together these concepts provide the minimum language needed to state later birational and extension arguments accurately.

Plan for the lecture. 0–20 min: locally ringed spaces; 20–40: nonreduced structures; 40–58: normalization; 58–78: Weil and Cartier divisors and meromorphic functions; 78–90: Hartogs extension and examples.

Definition 3.36: Complex analytic space

A complex analytic space is a locally ringed space $(X, \mathcal{O}_X)$ locally isomorphic to

$$(A, \mathcal{O}_\Omega/\mathcal{I})|_A, \qquad A = V(\mathcal{I}) \subset \Omega \subset \mathbb{C}^N,$$

where $\mathcal{I}$ is coherent. The space is reduced if $\mathcal{O}_X$ has no nonzero nilpotents.

The double line is $(\mathbb{C}^2, \mathcal{O}_{\mathbb{C}^2}/(y^2))$. Its underlying set is $\{y = 0\}$, but $y \neq 0$ while $y^2 = 0$. This infinitesimal normal information is invisible in the set of points.

Definition 3.37: Normal space and normalization

A reduced complex space is normal if every local ring $\mathcal{O}_{X,x}$ is a domain integrally closed in its fraction field. A *normalization* of a reduced space X is a finite holomorphic map $\nu : X^\nu \to X$ from a normal space such that ν is biholomorphic over the normal locus of X and $\nu_*\mathcal{O}_{X^\nu}$ is the integral closure of $\mathcal{O}_X$ in $\mathcal{M}_X$.

The definition does not by itself prove that X^ν exists. The next theorem is the analytic existence statement; the proof also explains why the map is finite rather than merely bijective over the regular locus.

Lemma 3.38: Weakly holomorphic functions and universal denominators

Let X be reduced. Near every point there is a holomorphic function h whose zero set is nowhere dense and such that

$$h\widetilde{\mathcal{O}}_X \subset \mathcal{O}_X,$$

where $\widetilde{\mathcal{O}}_X$ is the sheaf of weakly holomorphic functions. On an irreducible germ, $\widetilde{\mathcal{O}}_{X,x}$ is exactly the integral closure of $\mathcal{O}_{X,x}$ in its meromorphic function field and is a finite $\mathcal{O}_{X,x}$-module.

Proof. First suppose that (X, x) is irreducible of local dimension d. Choose a finite projection $\pi : X \to B \subset \mathbb{C}^d$ and a linear function t separating the points of a generic fiber. Away from the discriminant, a weakly holomorphic function f has values $f_1, \dots, f_q$ on the q sheets. Lagrange interpolation gives

$$f = b_0(z') + b_1(z')t + \cdots + b_{q-1}(z')t^{q-1}.$$

The denominator of the interpolation matrix is the Vandermonde determinant. Its square is the discriminant $\delta(z')$ of the monic polynomial of t. Since f is locally bounded, Cramer's rule shows that δb_j is locally bounded near the discriminant; the removable-singularity theorem on the base extends it holomorphically. Hence δf is holomorphic on X. For finitely many local components, multiply the corresponding discriminants and functions separating the components. This gives one function h valid for all nearby stalks.

If $g \in \mathcal{M}_{X,x}$ satisfies a monic equation over $\mathcal{O}_{X,x}$, the usual root bound for a monic polynomial makes g locally bounded away from its initial denominator. On X_{Reg} this bound rules out every polar hypersurface, and the one-variable removable-singularity theorem on transverse discs extends g holomorphically. The same bound near X_{Sing} then shows that g is weakly holomorphic. Conversely, the elementary symmetric functions of the values $f_1, \dots, f_q$ are bounded and extend holomorphically

to B. Therefore f satisfies the monic equation

$$T^q - \sigma_1 T^{q-1} + \cdots + (-1)^q \sigma_q = 0$$

with holomorphic coefficients. This proves that weak holomorphicity and integral dependence agree. Finally multiplication by h embeds $\widetilde{\mathcal{O}}_{X,x}$ into the Noetherian module $\mathcal{O}_{X,x}$; its image is an ideal and hence finitely generated. Multiplication by h is an isomorphism onto that image as an $\mathcal{O}_{X,x}$-module, proving finiteness. □

Theorem 3.39: Oka normalization theorem

Every reduced complex analytic space X has a normalization $\nu : X^\nu \to X$. It is unique up to a unique isomorphism over X. If X is irreducible and $F : Y \to X$ is a dominant holomorphic map from a normal irreducible space, then F factors uniquely through ν.

Proof. The construction is local on X. At a reducible germ, normalize its finitely many irreducible components and take their disjoint union; hence it suffices to work on an irreducible neighborhood U. The lemma makes $\widetilde{\mathcal{O}}_X|_U$ a finite $\mathcal{O}_U$-algebra. Choose module generators $1, f_1, \dots, f_m$ and a universal denominator h, so that $f_j = g_j/h$ on the regular locus. Each f_j satisfies a monic polynomial over $\mathcal{O}_U$. Map the dense open set $U_{\mathrm{Reg}} \cap \{h \neq 0\}$ into $U \times \mathbb{C}^m$ by

$$x \longmapsto (x, f_1(x), \dots, f_m(x))$$

and take its analytic closure. Writing w_j for the coordinates on $\mathbb{C}^m$, the equations $hw_j = g_j$, the monic equations for w_j, and the multiplication table of the finite algebra show that this closure is analytic, has finite fibers over U, and is proper after U is shrunk. Its structure sheaf is precisely $\widetilde{\mathcal{O}}_X|_U$.

Call the resulting finite space U^ν. A weakly holomorphic function on U^ν restricts over U_{Reg} to an element of $\widetilde{\mathcal{O}}_X(U)$, hence is a holomorphic combination of the chosen generators and extends to U^ν. Thus U^ν is normal. On overlaps the two constructions have the same integral-closure algebra, so they are uniquely isomorphic and glue to X^ν.

For the universal property, pullback embeds the meromorphic function field of X in that of Y. Every element integral over $\mathcal{O}_X$ remains integral over $\mathcal{O}_Y$ after pullback and is therefore holomorphic because Y is normal. Hence $F^\sharp : \mathcal{O}_X \to F_*\mathcal{O}_Y$ extends uniquely to the integral closure $\nu_*\mathcal{O}_{X^\nu}$. This algebra map gives the required lift $Y \to X^\nu$. Uniqueness holds on the dense locus where ν is an isomorphism, and therefore everywhere by the identity theorem. □

For curves, normal means smooth. In higher dimension a normal space may be singular, but its singular set has codimension at least two. Indeed, a height-one localization of a Noetherian normal domain is a one-dimensional normal local domain, hence a discrete valuation ring and therefore regular. Since Lecture 10 proved that the singular locus is analytic, it can have no codimension-one component.

Theorem 3.40: Hartogs–Riemann extension

If X is normal and $A \subset X$ is analytic with $\operatorname{codim}_X A \geq 2$, then $\mathcal{O}_X(U) \to \mathcal{O}_X(U \setminus A)$ is an isomorphism.

Proof. Injectivity follows from the identity theorem because $U \setminus A$ is dense. For surjectivity, let $f \in \mathcal{O}_X(U \setminus A)$ and work near $x \in A$. On the smooth locus of X, choose a two-dimensional complex slice transverse to A. The usual Hartogs figure argument extends f across $A \cap X_{\mathrm{Reg}}$, and the Cauchy integral depends holomorphically on the remaining coordinates. We may therefore regard f as holomorphic on $U \setminus X_{\mathrm{Sing}}$.

Put $n = \dim_x X$. Choose a finite local projection $\pi : X \to B \subset \mathbb{C}^n$ and a linear function t which separates the points of a generic fiber. Outside the discriminant, denote the values of f on the sheets by

$f_1, \dots, f_q$. Their elementary symmetric functions σ_k are holomorphic away from $\pi(A \cup X_{\mathrm{Sing}})$. They also extend across the generic points of the discriminant: there X is smooth and f is holomorphic, and the σ_k are the coefficients of multiplication by f on the finite local algebra. The only set left on the smooth base has codimension at least two, because a finite map preserves dimension. The ordinary Hartogs theorem on B therefore extends every σ_k.

Lagrange interpolation in t gives a common discriminant denominator, exactly as in the preceding lemma, so f defines a meromorphic germ on X. It satisfies

$$f^q - \pi^*\sigma_1 f^{q-1} + \cdots + (-1)^q \pi^*\sigma_q = 0.$$

Thus f is integral over $\mathcal{O}_{X,x}$. Normality gives $f \in \mathcal{O}_{X,x}$. The local extensions agree on the dense complement of A and hence glue over U. □

Definition 3.41: Weil and Cartier divisors

On a pure-dimensional normal space, a Weil divisor is a locally finite integral combination of irreducible codimension-one analytic subsets. A Cartier divisor is represented by local meromorphic functions f_i with $f_i/f_j \in \mathcal{O}^*(U_i \cap U_j)$. A divisor $\mathrm{div}(f)$ is principal.

The local rings of a smooth manifold are UFDs, so Weil divisors are locally principal and hence Cartier. This may fail on a singular space.

Proposition 3.42: Equations give an upper bound on codimension

Let X be irreducible and $f_1, \dots, f_p \in \mathcal{O}(X)$. Every irreducible component of $V(f_1, \dots, f_p)$ has codimension at most p. If each successive function is a non-zero-divisor in the corresponding quotient, codimension rises by one at every step and the zero set is a local complete intersection.

Proof. Work at the generic point of an irreducible component Z of the zero set. The analytic local ring is Noetherian. Krull's principal ideal theorem says that every minimal prime over a principal ideal has height at most one. Applying it successively to (f_1), (f_1, f_2), and so on shows

$$\mathrm{ht}(f_1, \dots, f_p) \le p,$$

which is exactly $\mathrm{codim}_X Z \le p$.

If f_j is a non-zero-divisor modulo $(f_1, \dots, f_{j-1})$, multiplication by f_j is injective there. Every minimal prime containing the enlarged ideal then has height one more than a minimal prime of the preceding quotient; the dimension drops by exactly one. Induction gives codimension p, and the local ideal is generated by the regular sequence $f_1, \dots, f_p$. This is the local complete-intersection condition. □

Definition 3.43: Meromorphic function

The sheaf $\mathcal{M}_X$ is obtained by inverting local non-zero-divisors. Locally a meromorphic function is g/h, but it need not be one quotient on an entire open set. On a manifold, zeros and poles have pure codimension one; the indeterminacy set has codimension at least two.

Remark (Detecting the pole set)

For a meromorphic function f, let $\mathcal{J}_x = \{u \in \mathcal{O}_{X,x} : u f_x \in \mathcal{O}_{X,x}\}$. Its pole set is

$$\{x : \mathcal{J}_x \neq \mathcal{O}_{X,x}\} = \mathrm{Supp}(\mathcal{O}_X/\mathcal{J}).$$

This ideal-theoretic description is intrinsic and continues to work when no single quotient

represents f on the whole open set.

3.3.1 Normalization as a universal repair

For an irreducible reduced germ with local ring A, let $\widetilde{A}$ be the integral closure of A in its fraction field. Finiteness of normalization in complex analytic geometry turns $\widetilde{A}$ into the local ring of a finite analytic map $\nu : X^\nu \to X$. The construction is universal: every dominant holomorphic map from a normal irreducible space to X factors uniquely through X^ν.

For the cusp,

$$A = \mathbb{C}\{t^2, t^3\} \subset \mathbb{C}\{t\} = \widetilde{A}.$$

The element t is integral because it satisfies $T^2 - t^2 = 0$ with coefficient $t^2 \in A$. The normalization map $t \mapsto (t^2, t^3)$ is bijective, but its inverse is not holomorphic at the origin. This is why a bijective finite holomorphic map of analytic spaces need not be an isomorphism.

For the node $xy = 0$, normalization separates the two branches:

$$\mathbb{C}\{x, y\}/(xy) \longrightarrow \mathbb{C}\{x\} \oplus \mathbb{C}\{y\}, \qquad f(x, y) \longmapsto (f(x, 0), f(0, y)).$$

The image consists of pairs with equal constant term. Normalization therefore may separate points, unlike the cuspidal example, where it only repairs the local ring.

3.3.2 Divisors, valuations, and line bundles

On a normal irreducible space, each prime divisor D defines an order $\operatorname{ord}_D(f)$ for a nonzero meromorphic function: compute the vanishing order at a generic smooth point of D. Hartogs extension across codimension at least two shows that this generic value determines the codimension-one behavior globally. Thus

$$\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g), \qquad \operatorname{div}(f/g) = \operatorname{div}(f) - \operatorname{div}(g).$$

A Cartier divisor represented by f_i on U_i defines the invertible subsheaf

$$\mathcal{O}_X(D)|_{U_i} := \mathcal{O}_{U_i} f_i^{-1} \subset \mathcal{M}_X.$$

Writing $e_i = f_i^{-1}$ for its local holomorphic frame, one has $e_i = (f_j/f_i)e_j$ on $U_i \cap U_j$. The meromorphic section $s_D = 1$ has local coefficient f_i in the frame e_i, so $\operatorname{div}(s_D) = D$. Conversely, a nonzero meromorphic section of a line bundle determines a Cartier divisor. On a manifold every Weil divisor is Cartier because regular local rings are factorial; on a normal singular space the two notions can differ.

Remark (Why codimension two disappears)

Holomorphic functions on a normal space extend across codimension-two analytic sets, and Weil divisors only record codimension one. Nevertheless codimension-two sets are not geometrically irrelevant: they can be indeterminacy loci of meromorphic maps or singular loci of sheaves. They disappear only from these two particular invariants.

3.3.3 Normalization of the cusp from its local ring

Let $A = \mathbb{C}\{x, y\}/(y^2 - x^3)$. The parametrization

$$\nu : \mathbb{C} \longrightarrow C, \qquad t \longmapsto (t^2, t^3)$$

induces the injection

$$A \hookrightarrow \mathbb{C}\{t\}, \qquad x \mapsto t^2, \quad y \mapsto t^3.$$

The element t is integral over A because it satisfies the monic equation $T^2 - x = 0$. It is not in A: every element of $\mathbb{C}\{t^2, t^3\}$ has zero coefficient of t^1. Since $t = y/x$ in the common fraction field away from the origin, adjoining t gives

$$\overline{A} = A[t] = \mathbb{C}\{t\}.$$

The map ν is finite because $\mathbb{C}\{t\} = A + At$ is generated by $1, t$ as an A-module. It is bijective near the origin, but its inverse is not holomorphic at the cusp: a holomorphic inverse would put t in A. This is the concrete reason a finite bijective map of analytic spaces need not be an isomorphism.

A Weil divisor that is not Cartier

Consider the normal quadric cone

$$X = \{(x, y, z) \in \mathbb{C}^3 : xy = z^2\}$$

and the irreducible codimension-one set $D = \{x = z = 0\} \subset X$. The finite double cover

$$\pi : \mathbb{C}^2 \to X, \qquad (s, t) \longmapsto (s^2, t^2, st)$$

is the quotient by $(s, t) \mapsto (-s, -t)$. The set-theoretic inverse image of D is $\{s = 0\}$.

If D were Cartier at the vertex, a local equation g would pull back to a holomorphic function whose divisor is exactly $\{s = 0\}$ with multiplicity one. Up to a unit this pullback would be s. But $g \circ \pi$ is invariant under $(s, t) \mapsto (-s, -t)$, whereas a unit times s changes sign at the level of its lowest nonzero homogeneous term. An invariant function vanishing only on $s = 0$ has even generic vanishing order there. This contradiction shows that D is Weil but not Cartier at the vertex. Twice the divisor is Cartier because $x = s^2$ has divisor $2D$.

Cartier divisors and their line bundles

Let a Cartier divisor be represented by meromorphic functions f_i with $f_i/f_j \in \mathcal{O}^*(U_i \cap U_j)$. Define

$$\mathcal{O}_X(D)|_{U_i} := \mathcal{O}_{U_i}\, f_i^{-1} \subset \mathcal{M}_X.$$

This definition is independent of i: on an overlap, f_i^{-1} and f_j^{-1} differ by the holomorphic unit f_j/f_i. If $e_i = f_i^{-1}$ denotes the resulting holomorphic frame, then

$$e_i = \frac{f_j}{f_i} e_j.$$

The constant meromorphic function 1 is a global meromorphic section s_D of this line bundle. Since $1 = f_i e_i$ on U_i, the local coefficient of s_D is f_i, and therefore $\operatorname{div}(s_D) = D$. This convention fixes both the sign of D and the direction of the transition functions.

If $D = \operatorname{div}(f)$ is principal, choose every $f_i = f|_{U_i}$. Then $g_{ij} = 1$, so $\mathcal{O}(D)$ is trivial. Conversely, a trivialization of $\mathcal{O}(D)$ changes the f_i by local units so that they glue to one global meromorphic function. Hence Cartier divisors modulo principal divisors are exactly holomorphic line bundles that admit a nonzero meromorphic section.

An indeterminacy calculation

The map

$$F : \mathbb{C}^2 \dashrightarrow \mathbb{P}^1, \qquad F(x, y) = [x : y]$$

is holomorphic away from the origin. On $\{x \neq 0\}$ its affine coordinate is y/x; on $\{y \neq 0\}$ it is x/y. No continuous extension exists at 0, since along the line $y = mx$ the limit is $[1 : m]$, depending on m. Blowing up the origin resolves the map: a point of the blow-up is $((x, y), [u : v])$ with $xv = yu$, and the extension is simply $((x, y), [u : v]) \mapsto [u : v]$. The exceptional curve records all limiting directions that were collapsed at the indeterminacy point.

3.3.4 The universal property of normalization in local algebra

Let X be reduced and irreducible, set $A = \mathcal{O}_{X,x}$, and denote by $\overline{A}$ the integral closure of A in its fraction field. Locally the normalization is the finite map associated with $A \subset \overline{A}$. Suppose a normal irreducible analytic space Y maps dominantly and holomorphically to X. At $y \in Y$ over x, pullback gives $A \to \mathcal{O}_{Y,y}$ and, by dominance, an embedding of the corresponding meromorphic function fields.

Every $b \in \overline{A}$ satisfies a monic equation

$$b^m + a_1 b^{m-1} + \cdots + a_m = 0, \qquad a_j \in A.$$

After pullback, b is a meromorphic germ on Y integral over $\mathcal{O}_{Y,y}$. Normality means precisely that every such integral meromorphic germ is holomorphic. Thus $A \to \mathcal{O}_{Y,y}$ extends to $\overline{A} \to \mathcal{O}_{Y,y}$. The extension is unique because the two maps agree in the meromorphic function field.

These local extensions agree on overlaps and define a holomorphic lift $Y \to \widetilde{X}$. On the dense regular locus the normalization is an isomorphism, so the lift is forced there; the identity principle gives global uniqueness. In the cusp example the integral element is $t = y/x$, and the lift records t as a genuine holomorphic coordinate. This proof isolates the roles of the hypotheses: reduced irreducibility supplies a function field, finiteness produces an analytic space, and normality of the source converts integral meromorphic functions into holomorphic ones.

3.3.5 Self-study workshop: distinguishing reduced, normal, Cartier, and smooth

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall local analytic rings, integral dependence, fraction fields, and the correspondence between locally principal divisors and line bundles. These adjectives answer different questions and should never be used as synonyms.

A diagnostic chain of examples. The space with local ring $\mathbb{C}\{z\}/(z^2)$ has one underlying point in its zero set but carries a nonzero nilpotent class z. It is not reduced. The cusp

$$C = \{y^2 = x^3\}$$

is reduced, but its local ring $A = \mathbb{C}\{t^2, t^3\}$ is not integrally closed: t belongs to its fraction field and satisfies $T^2 - t^2 = 0$ with coefficient $t^2 = x \in A$, yet $t \notin A$. Its normalization is $\mathbb{C}\{t\}$.

Normal does not mean smooth. The surface

$$X = \{xy = z^2\} \subset \mathbb{C}^3$$

is singular at the origin because its gradient vanishes there. It is nevertheless normal: it is a hypersurface, hence Cohen–Macaulay, and its singular locus has codimension two; Serre's criterion gives normality.

Finally, a Weil divisor records codimension-one subvarieties, while a Cartier divisor is locally one meromorphic equation modulo holomorphic units. On a smooth complex manifold every Weil divisor is Cartier, but on a normal singular space this may fail. The obstruction is local principality, not the existence of a set-theoretic equation. For a Cartier divisor (f_i), the quotients f_i/f_j are nowhere-vanishing holomorphic functions and therefore define a line bundle.

Checkpoint (four independent tests)

(1) Which example above is nonreduced?
(2) Why is the cusp's normalization map finite?
(3) Can a normal analytic space be singular?
(4) What local property upgrades a Weil divisor to a Cartier divisor?

Solution (checkpoint). The double point $\mathbb{C}\{z\}/(z^2)$ is nonreduced. For the cusp, $\mathbb{C}\{t\} = A + At$ is generated by two elements over A, hence the normalization map is finite. The quadric cone is normal and singular. A Weil divisor is Cartier exactly when its divisor ideal is locally generated by one non-zero-divisor, equivalently when it has local meromorphic equations whose ratios are units. ⋄

Exit criterion

Given a local analytic equation, you can test nilpotents and the Jacobian, exhibit an integral element for normalization, and explain why divisor class questions are about local principality.

Summary. Analytic spaces admit singularities and nilpotent directions. Normalization repairs integral closedness, and divisors record codimension-one data. A list of equations gives an upper bound for codimension; a regular sequence is what gives equality.

Exercise 3.44: A non-Cartier Weil divisor

In $X = \{z_1z_2 + z_3z_4 = 0\} \subset \mathbb{C}^4$, study the plane $A = \{z_1 = z_3 = 0\}$ and show that it is a Weil divisor but not Cartier at the origin.

Exercise 3.45: Indeterminacy

Study the zero set, pole set, and indeterminacy set of $f(z_1, z_2) = z_2/z_1$. Why can the origin be assigned neither 0 nor ∞?

Optional handout: Serre's criterion for normality

This handout is optional. No later argument in the core lectures depends on it.

For a Noetherian reduced ring, normality can be separated into codimension-one and depth conditions. Serre's criterion states

$$\text{normal} \quad \Longleftrightarrow \quad (R_1) + (S_2).$$

Here a Noetherian local ring $(R, \mathfrak{m})$ is *regular* if the minimal number of generators of $\mathfrak{m}$ equals $\dim R$. Condition (R_1) means that $R_\mathfrak{p}$ is regular for every prime $\mathfrak{p}$ of height at most one. A sequence $a_1, \dots, a_r \in \mathfrak{m}$ is *regular* if each a_j is a non-zero-divisor modulo $(a_1, \dots, a_{j-1})$; the depth of R is

the maximal length of such a sequence. Condition (S_2) means $\operatorname{depth} R_{\mathfrak{p}} \geq \min\{2, \dim R_{\mathfrak{p}}\}$ for all primes $\mathfrak{p}$.

Geometrically, (R_1) says that singularities occur in codimension at least two, while (S_2) prevents a function from being specified independently on pieces separated by a codimension-two defect. Together they explain why holomorphic functions on the regular locus of a normal space extend across codimension at least two when locally bounded.

Example 3.46: The node and the cusp

For a reduced complex curve, (S_2) is automatic and (R_1) requires regularity at every closed point. Hence a normal curve is smooth. Both the node $xy = 0$ and the cusp $y^2 = x^3$ are nonnormal, although for different reasons: the node has two branches, whereas the cusp is irreducible but misses the integral element t in its local ring.

A ring is *Cohen–Macaulay* if its depth equals its dimension. Hypersurfaces are Cohen–Macaulay and therefore satisfy (S_2). A reduced hypersurface is consequently normal precisely when its singular locus has codimension at least two inside the hypersurface.

Exercise 3.47: A normal singularity

Use Serre's criterion to show that the quadric cone $\{z_1 z_2 - z_3 z_4 = 0\} \subset \mathbb{C}^4$ is normal even though it is singular at the origin.

Exercise 3.48: Divisor line bundle

Given a Cartier divisor represented by nonzero meromorphic functions f_i on an open cover $\{U_i\}$, construct the holomorphic line bundle with transition functions f_i / f_j and its canonical meromorphic section. Check directly that a principal divisor gives the trivial line bundle.

3.4 Lecture 12: Hermitian and Kähler metrics; the Fubini–Study metric

Week 6 material • 90 minutes

Notation for this lecture

Throughout the notes, and in particular in this lecture, TX denotes the *real* tangent bundle of the complex manifold X, and $J = J_X : TX \to TX$ is its complex structure. We write

$$T_{\mathbb{C}}X := TX \otimes_{\mathbb{R}} \mathbb{C} = T^{1,0}X \oplus T^{0,1}X, \qquad v^{1,0} := \frac{1}{2}(v - \mathrm{i}Jv), \quad v^{0,1} := \frac{1}{2}(v + \mathrm{i}Jv).$$

Thus $T^{1,0}X$ and $T^{0,1}X$ are respectively the $+\mathrm{i}$ and $-\mathrm{i}$ eigensubbundles of the complex-linear extension of J, and $v^{0,1} = \overline{v^{1,0}}$ for $v \in TX$. Some references use TX for the holomorphic tangent bundle; we do not: that bundle is denoted $T^{1,0}X$ here.
A Riemannian metric g is a positive real inner product on TX; $g_{\mathbb{C}}$ denotes its complex-bilinear extension to $T_{\mathbb{C}}X$. A Hermitian metric h is a positive Hermitian inner product on $T^{1,0}X$, linear in the first argument and conjugate-linear in the second. Its coefficient convention is

$$h\left(\sum_j \xi^j \frac{\partial}{\partial z_j}, \sum_k \eta^k \frac{\partial}{\partial z_k}\right) = \sum_{j,k} h_{j\bar{k}} \xi^j \overline{\eta^k}.$$

The associated fundamental form is denoted ω, and $\mathrm{d}V_\omega = \omega^n/n!$. A real $(1,1)$-form is positive if $\omega(v, Jv) > 0$ for every nonzero $v \in TX$. The metric is Kähler when $\mathrm{d}\omega = 0$.
The symbol ∇ denotes the Levi–Civita connection of g and also its complex-linear extension to $T_{\mathbb{C}}X$. The equation $\nabla J = 0$ means $\nabla_U(JV) = J\nabla_U V$ for all real vector fields U, V. In coordinates, $h^{j\bar{k}}$ are the entries of the inverse matrix of $(h_{j\bar{k}})$, and $\partial_j := \partial/\partial z_j$ and $\partial_{\bar{j}} := \partial/\partial \bar{z}_j$ are the coordinate vector fields. Γ^m_{jk} are the connection coefficients defined by $\nabla_{\partial/\partial z_j}(\partial/\partial z_k) = \sum_m \Gamma^m_{jk} \partial/\partial z_m$ when the connection preserves type.
A local Kähler potential is a real function φ satisfying $\omega = \mathrm{d}\mathrm{d}^{\mathrm{c}}\varphi = (\mathrm{i}/2\pi)\partial\bar{\partial}\varphi$. ω_{FS} denotes the Fubini–Study form on $\mathbb{P}^n$, and $\Lambda \subset \mathbb{C}^n$ denotes a full lattice when complex tori occur. The symbol δ_{jk} is the Kronecker delta. For a matrix H, $H^* = \overline{H}^{\mathsf{T}}$. The standard Hermitian inner product and norm on $\mathbb{C}^n$ are $\langle \xi, w \rangle = \sum_j \xi_j \bar{w}_j$ and $|\xi|^2 = \langle \xi, \xi \rangle$. For a product $X_1 \times X_2$, pr_i is projection to the i-th factor. If F is smooth, $\mathrm{d}F_x$ is its real differential.

Prerequisite results used below

Poincaré lemma. On a star-shaped open subset of $\mathbb{R}^m$, every smooth closed k-form with $k \geq 1$ is exact.
Local Dolbeault lemma. If a smooth (p,q)-form α with $q \geq 1$ is $\bar{\partial}$-closed near a point, then after shrinking the neighborhood there is a smooth $(p, q-1)$-form β with $\alpha = \bar{\partial}\beta$.
Hermitian spectral theorem. Every Hermitian matrix is unitarily diagonalizable with real eigenvalues; it is positive definite exactly when all eigenvalues are positive, equivalently when $v^* A v > 0$ for every $v \neq 0$.

Why these ideas are needed

A complex atlas distinguishes holomorphic directions but says nothing about lengths, angles, volumes, or orthogonality. Analysis requires those notions, so we add a Hermitian metric compatible with J. Its fundamental form packages the metric in a way that interacts with

∂ and $\bar\partial$. The extra Kähler condition $\mathrm{d}\omega = 0$ may look restrictive, but it is precisely what connects local complex Hessians with symplectic geometry and global cohomology. It also has a fundamental Riemannian meaning: the Levi–Civita connection preserves the complex structure, $\nabla J = 0$. We therefore keep track of the real tangent bundle, its complexification, and the holomorphic tangent bundle separately before using local potentials. The Fubini–Study metric then shows that this structure is natural on projective space rather than an artifact of Euclidean coordinates.

Plan for the lecture. 0–20 min: the three equivalent forms of a Hermitian metric; 20–42: the Kähler condition and $\nabla J = 0$; 42–65: Kähler normal coordinates and the local expansion; 65–78: local potentials and Fubini–Study; 78–90: examples and the bridge to Hodge theory.

After the local analytic and sheaf-theoretic foundations, we now add geometry: lengths, angles, volume, and positivity. There are three standard descriptions of a Hermitian metric. They live on different bundles, and passing between them involves both the complexification of the real tangent bundle and a factor of 2. We begin by making these identifications explicit.

3.4.1 Which tangent bundle carries the metric?

Recall the convention fixed in Lecture 1: TX is the real tangent bundle. The complex-linear extension of J to $T_{\mathbb{C}}X = TX \otimes_{\mathbb{R}} \mathbb{C}$ has the splitting

$$T_{\mathbb{C}}X = T^{1,0}X \oplus T^{0,1}X, \qquad T^{1,0}X = \ker(J - \mathrm{i}), \quad T^{0,1}X = \ker(J + \mathrm{i}).$$

For a real tangent vector u the two projections are

$$u^{1,0} = \frac{1}{2}(u - \mathrm{i}Ju), \qquad u^{0,1} = \frac{1}{2}(u + \mathrm{i}Ju) = \overline{u^{1,0}}, \qquad u = u^{1,0} + u^{0,1}.$$

Thus a Riemannian metric is defined on TX, whereas a Hermitian inner product is defined on $T^{1,0}X$. Authors who write "a Hermitian metric on TX" often use TX to mean the holomorphic tangent bundle. In these notes that bundle is always written $T^{1,0}X$, in accordance with the convention of Lecture 1.

Definition 3.49: Hermitian metric: three equivalent descriptions

A Hermitian metric on a complex manifold X may be specified by any one of the following equivalent objects:

(1) a Riemannian metric g on the real bundle TX such that

$$g(Ju, Jv) = g(u, v) \qquad (u, v \in TX);$$

(2) a smoothly varying positive Hermitian inner product h on $T^{1,0}X$, linear in the first argument and conjugate-linear in the second;

(3) a real positive $(1,1)$-form ω on X.

The form ω is called the *fundamental form*. The metric is *Kähler* if $\mathrm{d}\omega = 0$.

Proposition 3.50: The conversion formulas

Let $g_{\mathbb{C}}$ be the complex-bilinear extension of g to $T_{\mathbb{C}}X$. The three objects in Definition 3.49 correspond by

$$h(\xi, \eta) = 2g_{\mathbb{C}}(\xi, \bar\eta), \qquad \xi, \eta \in T^{1,0}X, \tag{12.1}$$

$$\omega(u, v) = g(Ju, v), \qquad u, v \in TX, \tag{12.2}$$

$$g(u,v) = \omega(u, Jv), \qquad u, v \in TX. \tag{12.3}$$

In particular, for real u, v,

$$h(u^{1,0}, v^{1,0}) = g(u,v) - \mathrm{i}\omega(u,v). \tag{12.4}$$

In holomorphic coordinates,

$$h = \sum_{j,k} h_{j\bar{k}}\, \mathrm{d}z_j \otimes \mathrm{d}\bar{z}_k, \qquad H = (h_{j\bar{k}}) = H^* > 0,$$

and

$$\omega = \frac{\mathrm{i}}{2} \sum_{j,k} h_{j\bar{k}} \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k. \tag{12.5}$$

Proof. Suppose first that g is J-invariant. Substituting v by Jv gives

$$g(Ju, v) = -g(u, Jv), \tag{12.6}$$

so J is skew-adjoint for g. If $\xi \in T^{1,0}X$, then $\bar{\xi} \in T^{0,1}X$, and formula (12.1) is linear in ξ and conjugate-linear in η. Every $\xi \in T^{1,0}X$ is uniquely $u^{1,0}$ for the real vector $u = \xi + \bar{\xi}$. Using J-invariance and (12.6), we compute

$$\begin{aligned} 2g_{\mathbb{C}}(u^{1,0}, v^{0,1}) &= \frac{1}{2} g_{\mathbb{C}}(u - \mathrm{i}Ju, v + \mathrm{i}Jv) \\ &= \frac{1}{2}\{g(u,v) + \mathrm{i}g(u,Jv) - \mathrm{i}g(Ju,v) + g(Ju,Jv)\} \\ &= g(u,v) - \mathrm{i}g(Ju,v). \end{aligned}$$

This proves (12.4). In particular, $h(u^{1,0}, u^{1,0}) = g(u,u) > 0$ for $u \neq 0$, so h is positive.

Conversely, let h be given on $T^{1,0}X$ and define

$$g(u,v) := \operatorname{Re} h(u^{1,0}, v^{1,0}), \qquad \omega(u,v) := -\operatorname{Im} h(u^{1,0}, v^{1,0}). \tag{12.7}$$

Since $(Ju)^{1,0} = \mathrm{i}u^{1,0}$, sesquilinearity gives $g(Ju, Jv) = g(u,v)$. Hermitian symmetry shows that g is symmetric and ω is skew-symmetric, while positivity of h gives positivity of g. Moreover,

$$g(Ju, v) = \operatorname{Re}\Big(\mathrm{i}h(u^{1,0}, v^{1,0})\Big) = -\operatorname{Im} h(u^{1,0}, v^{1,0}) = \omega(u,v),$$

so the construction recovers (12.2).

It remains to start from a real positive $(1,1)$-form ω. For a real two-form, being of type $(1,1)$ is equivalent to

$$\omega(Ju, Jv) = \omega(u,v). \tag{12.8}$$

Replacing v by Jv in (12.8) gives $\omega(Ju, v) = -\omega(u, Jv)$. Therefore

$$g(v,u) = \omega(v, Ju) = -\omega(Ju, v) = \omega(u, Jv) = g(u,v),$$

and $g(u,u) = \omega(u, Ju) > 0$ for $u \neq 0$. A second use of (12.8) gives $g(Ju, Jv) = g(u,v)$. Thus (12.3) produces a J-invariant Riemannian metric, inverse to (12.2).

Finally, write $\partial_j = \partial/\partial z_j$. By definition $h_{j\bar{k}} = h(\partial_j, \partial_k)$ and $h_{j\bar{k}} = \overline{h_{k\bar{j}}}$. Expanding (12.7) in the real coordinate fields gives (12.5) as follows. For real u, v, put

$$S := \sum_{j,k} h_{j\bar{k}}\, \mathrm{d}z_j(u) \overline{\mathrm{d}z_k(v)} = h(u^{1,0}, v^{1,0}).$$

Hermitian symmetry shows that interchanging u, v gives $\bar{S}$, and therefore

$$\frac{\mathrm{i}}{2}\sum_{j,k} h_{j\bar{k}}(\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k)(u,v) = \frac{\mathrm{i}}{2}(S - \bar{S}) = -\operatorname{Im} S = \omega(u,v).$$

For example, when $n = 1$ and $h = \mathrm{d}z \otimes \mathrm{d}\bar{z}$, one obtains $g = \mathrm{d}x^2 + \mathrm{d}y^2$ and $\omega = \mathrm{d}x \wedge \mathrm{d}y = (\mathrm{i}/2)\mathrm{d}z \wedge \mathrm{d}\bar{z}$; this checks the factor $1/2$. □

Because ω is real of type $(1,1)$, the equation $\mathrm{d}\omega = 0$ is equivalent to $\partial\omega = 0$, and also to $\bar{\partial}\omega = 0$. Indeed, $\partial\omega$ and $\bar{\partial}\omega$ have the distinct types $(2,1)$ and $(1,2)$, and they are complex conjugates. The Riemannian volume form is

$$\mathrm{d}V_\omega = \frac{\omega^n}{n!}.$$

3.4.2 The Kähler condition and the equation $\nabla J = 0$

The definition $\mathrm{d}\omega = 0$ is expressed in the language of differential forms. The equivalent equation $\nabla J = 0$ explains its Riemannian meaning: the Levi–Civita connection transports complex directions complex-linearly.

Proposition 3.51: Coordinate Kähler equations

For a Hermitian metric $h = (h_{j\bar{k}})$, the following conditions are equivalent:

(1) $\mathrm{d}\omega = 0$;
(2) $\partial\omega = 0$;
(3) for all indices j, k, ℓ,

$$\frac{\partial h_{j\bar{k}}}{\partial z_\ell} = \frac{\partial h_{\ell\bar{k}}}{\partial z_j}. \tag{12.9}$$

Proof. Using (12.5),

$$\partial\omega = \frac{\mathrm{i}}{2}\sum_{\ell,j,k} \frac{\partial h_{j\bar{k}}}{\partial z_\ell}\mathrm{d}z_\ell \wedge \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k.$$

Fix $\ell < j$. The coefficient of $\mathrm{d}z_\ell \wedge \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$ is

$$\frac{\mathrm{i}}{2}\left(\frac{\partial h_{j\bar{k}}}{\partial z_\ell} - \frac{\partial h_{\ell\bar{k}}}{\partial z_j}\right).$$

Thus $\partial\omega = 0$ is equivalent to (12.9). The equivalence with $\mathrm{d}\omega = 0$ was explained above. □

Let ∇ be the Levi–Civita connection: it is the unique connection on TX such that

$$\nabla g = 0, \qquad \nabla_U V - \nabla_V U = [U, V]. \tag{12.10}$$

The second identity says that ∇ has zero torsion. We extend ∇ complex-linearly to $T_{\mathbb{C}}X$.

Theorem 3.52: Kähler metrics and parallel complex structures

For a Hermitian metric on a complex manifold, the following are equivalent:

(1) $\mathrm{d}\omega = 0$;
(2) $\nabla J = 0$, that is, $\nabla_U(JV) = J\nabla_U V$ for all real vector fields U, V;
(3) the complexified Levi–Civita connection preserves both subbundles $T^{1,0}X$ and $T^{0,1}X$.

When these conditions hold, in holomorphic coordinates

$$\nabla_{\partial_j}\partial_k = \sum_m \Gamma^m_{jk}\partial_m, \qquad \Gamma^m_{jk} = \sum_\ell h^{m\bar{\ell}}\frac{\partial h_{k\bar{\ell}}}{\partial z_j}, \tag{12.11}$$

$$\nabla_{\partial_{\bar{j}}}\partial_k = 0, \qquad \partial_{\bar{j}} := \frac{\partial}{\partial \bar{z}_j}. \tag{12.12}$$

Proof. First, $\nabla J = 0$ is equivalent to preservation of the eigensubbundles. Indeed, if $V \in T^{1,0}X$, then $JV = \mathrm{i}V$, so

$$J(\nabla_U V) = \nabla_U(JV) = \mathrm{i}\nabla_U V;$$

hence $\nabla_U V$ again has type $(1,0)$. The same argument treats type $(0,1)$. Conversely, if both eigensubbundles are preserved, decompose any complexified vector field into its two types; on each summand $\nabla J = 0$, hence it holds on real vector fields as well.

We now compare this condition with (12.9). Extend g complex-bilinearly and use the Koszul formula. Since holomorphic coordinate fields commute and $g_{\mathbb{C}}$ vanishes on two vectors of the same type, we obtain

$$\begin{aligned} 2g_{\mathbb{C}}(\nabla_{\partial_{\bar{a}}}\partial_j, \partial_k) &= \partial_j g_{\mathbb{C}}(\partial_{\bar{a}}, \partial_k) - \partial_k g_{\mathbb{C}}(\partial_{\bar{a}}, \partial_j) \\ &= \frac{1}{2}\left(\frac{\partial h_{k\bar{a}}}{\partial z_j} - \frac{\partial h_{j\bar{a}}}{\partial z_k}\right). \end{aligned} \tag{12.13}$$

Here we used $2g_{\mathbb{C}}(\partial_j, \partial_{\bar{k}}) = h_{j\bar{k}}$, which follows from formula (12.1). The pairing of $T^{0,1}X$ with $T^{1,0}X$ under $g_{\mathbb{C}}$ is nondegenerate. Therefore the left side of (12.13) vanishes for every k precisely when the $(0,1)$-component of $\nabla_{\partial_{\bar{a}}}\partial_j$ vanishes. Formula (12.13) shows that this is exactly the Kähler symmetry (12.9).

For completeness, no additional obstruction comes from $\nabla_{\partial_a}\partial_j$: pairing the Koszul formula with any ∂_k gives zero because every metric pairing involved has two $(1,0)$-arguments. Taking complex conjugates treats $T^{0,1}X$. Hence $\mathrm{d}\omega = 0$ is equivalent to preservation of type, and therefore to $\nabla J = 0$.

Assume now that these conditions hold. Applying the Koszul formula with the third argument $\partial_{\bar{\ell}}$ gives

$$2g_{\mathbb{C}}(\nabla_{\partial_j}\partial_k, \partial_{\bar{\ell}}) = \frac{1}{2}\left(\frac{\partial h_{k\bar{\ell}}}{\partial z_j} + \frac{\partial h_{j\bar{\ell}}}{\partial z_k}\right) = \frac{\partial h_{k\bar{\ell}}}{\partial z_j},$$

where the last equality is the Kähler symmetry. If $\nabla_{\partial_j}\partial_k = \sum_m \Gamma^m_{jk}\partial_m$, the left side is $\sum_m \Gamma^m_{jk} h_{m\bar{\ell}}$. Multiplication by the inverse matrix gives (12.11). Finally, pair $\nabla_{\partial_{\bar{j}}}\partial_k$ with both ∂_ℓ and $\partial_{\bar{\ell}}$. Formula (12.13) and its complex conjugate show that both pairings vanish; nondegeneracy gives (12.12). □

There is also a useful coordinate-free check of one implication. Since $\nabla g = 0$,

$$(\nabla_U \omega)(V, W) = g((\nabla_U J)V, W). \tag{12.14}$$

Thus $\nabla J = 0$ is equivalent to $\nabla\omega = 0$. For every torsion-free connection,

$$\mathrm{d}\omega(U, V, W) = (\nabla_U\omega)(V, W) + (\nabla_V\omega)(W, U) + (\nabla_W\omega)(U, V), \tag{12.15}$$

so $\nabla J = 0$ immediately implies $\mathrm{d}\omega = 0$. The coordinate calculation above proves the converse and uses essentially that J is the integrable complex structure coming from holomorphic coordinates.

Geometrically, $\nabla J = 0$ says that parallel transport commutes with J. Hence parallel transport is unitary, the holonomy group lies in $U(n)$, and the Levi–Civita connection induces a connection on $T^{1,0}X$ whose $(0,1)$ part is the usual $\bar{\partial}$-operator, as equation (12.12) shows.

3.4.3 Local potentials

Theorem 3.53: Local Kähler potentials

A real positive $(1,1)$-form ω is Kähler if and only if near every point there is a strictly psh real function φ such that

$$\omega = \mathrm{dd^c}\varphi = \frac{\mathrm{i}}{2\pi}\partial\bar\partial\varphi.$$

Proof. If φ is real, then $\mathrm{dd^c}\varphi$ is real and has type $(1,1)$. The identity $\mathrm{ddd^c}\varphi = 0$ gives closedness, while strict plurisubharmonicity says exactly that its coefficient matrix $(\varphi_{j\bar k})$ is positive definite. In fact, comparison with (12.5) gives $h_{j\bar k} = \pi^{-1}\varphi_{j\bar k}$. Conversely, on a coordinate ball the Poincaré lemma gives a real one-form η with $\omega = \mathrm{d}\eta$. The $(0,2)$ component of this equation says $\bar\partial\eta^{0,1} = 0$. By the local Dolbeault lemma from Lecture 4, $\eta^{0,1} = \bar\partial f$. Reality gives $\eta^{1,0} = \partial\bar f$, and hence

$$\omega = \partial\bar\partial f + \bar\partial\partial\bar f = \partial\bar\partial(f - \bar f) = 2\mathrm{i}\partial\bar\partial(\operatorname{Im} f).$$

Thus $\varphi = 4\pi \operatorname{Im} f$ is a local potential for the convention fixed in Lecture 2. Positivity of ω means precisely that the Hermitian matrix $(\varphi_{j\bar k})$ is positive definite. □

Two local potentials for the same form differ by a pluriharmonic function, locally the real part of a holomorphic function. Therefore local potentials need not glue even when their complex Hessians do; this failure encodes the global Kähler class.

3.4.4 Kähler normal coordinates and the second-order expansion

For a general Riemannian metric one can choose real normal coordinates, but those coordinates need not be holomorphic. The Kähler symmetry is exactly what makes it possible to remove the first derivatives by a *holomorphic* quadratic change of coordinates. The next result, including its refined expansion, follows the construction in [1, Chapter VI, Theorem 4.8].

Theorem 3.54: Kähler normal holomorphic coordinates

Let h be a Hermitian metric on X. It is Kähler if and only if, for every $p \in X$, there are holomorphic coordinates $z = (z_1, \dots, z_n)$ centered at p such that

$$h_{\ell\bar m}(z) = \delta_{\ell m} + O(|z|^2). \tag{12.16}$$

If h is Kähler, the coordinates can be chosen more precisely so that

$$h_{\ell\bar m}(z) = \delta_{\ell m} - \sum_{j,k} R_{j\bar k\ell\bar m}(p) z_j \bar z_k + O(|z|^3), \tag{12.17}$$

where, with the curvature convention

$$\mathcal{R}(U,V) = [\nabla_U, \nabla_V] - \nabla_{[U,V]}, \qquad R_{j\bar k\ell\bar m} = h\big(\mathcal{R}(\partial_j, \partial_{\bar k})\partial_\ell, \partial_m\big), \tag{12.18}$$

the displayed coefficients are the components of the curvature of $T^{1,0}X$. Coordinates satisfying (12.17) are called Kähler normal, or geodesic holomorphic, coordinates at p.

Proof. Sufficiency. If (12.16) holds at p, then every first derivative of every $h_{j\bar k}$ vanishes at p. Hence the coordinate Kähler equations (12.9) hold at p. If such coordinates exist at every point, then $\partial\omega$ vanishes at every point, so $\mathrm{d}\omega = 0$.

Necessity: removing the linear terms. Assume that h is Kähler. Start with arbitrary holomorphic coordinates x centered at p. A complex-linear change of the x-coordinates makes $h_{\ell\bar m}(0) = \delta_{\ell m}$. Put

$$A_{j\ell\bar m} := \frac{\partial h_{\ell\bar m}}{\partial x_j}(0).$$

Hermitian symmetry determines the antiholomorphic linear terms, and Taylor's formula gives

$$h_{\ell\bar{m}}(x) = \delta_{\ell m} + \sum_j \left(A_{j\ell\bar{m}} x_j + \overline{A_{jm\bar{\ell}}}\, \bar{x}_j \right) + O(|x|^2). \tag{12.19}$$

The Kähler equations at 0 say

$$A_{j\ell\bar{m}} = A_{\ell j\bar{m}}. \tag{12.20}$$

This symmetry is precisely the integrability condition needed to absorb the holomorphic linear term into the differential of a quadratic coordinate change. Define

$$z_m = x_m + \frac{1}{2} \sum_{j,\ell} A_{j\ell\bar{m}} x_j x_\ell. \tag{12.21}$$

The Jacobian of $x \mapsto z$ is the identity at 0, so the holomorphic inverse function theorem makes z a coordinate system near 0. By (12.20),

$$\mathrm{d}z_m = \mathrm{d}x_m + \sum_{j,\ell} A_{j\ell\bar{m}} x_j \, \mathrm{d}x_\ell. \tag{12.22}$$

Consequently,

$$\begin{aligned} \sum_m \mathrm{d}z_m \otimes \mathrm{d}\bar{z}_m = \sum_m \mathrm{d}x_m \otimes \mathrm{d}\bar{x}_m \\ + \sum_{j,\ell,m} A_{j\ell\bar{m}} x_j \mathrm{d}x_\ell \otimes \mathrm{d}\bar{x}_m \\ + \sum_{j,\ell,m} \overline{A_{jm\bar{\ell}}} \bar{x}_j \mathrm{d}x_\ell \otimes \mathrm{d}\bar{x}_m + O(|x|^2). \end{aligned}$$

Comparison with (12.19) yields

$$h = \sum_m \mathrm{d}z_m \otimes \mathrm{d}\bar{z}_m + O(|z|^2),$$

which is (12.16). Notice that the calculation exhibits the cancellation; it does not merely invoke the tensor transformation rule.

Removing the pure quadratic terms. Begin again with coordinates x that already satisfy $h_{\ell\bar{m}} = \delta_{\ell m} + O(|x|^2)$. Write the complete second-order Taylor expansion as

$$\begin{aligned} h_{\ell\bar{m}}(x) = \delta_{\ell m} + \sum_{j,k} B_{j\bar{k}\ell\bar{m}} x_j \bar{x}_k + \sum_{j,k} C_{jk\ell\bar{m}} x_j x_k \\ + \sum_{j,k} \overline{C_{jkm\bar{\ell}}} \bar{x}_j \bar{x}_k + O(|x|^3), \end{aligned} \tag{12.23}$$

where $C_{jk\ell\bar{m}} = C_{kj\ell\bar{m}}$. Differentiate the Kähler equation $\partial_j h_{\ell\bar{m}} = \partial_\ell h_{j\bar{m}}$ once more in a holomorphic direction and evaluate at 0. Comparing the coefficient of x_k in (12.23) shows

$$C_{jk\ell\bar{m}} = C_{\ell kj\bar{m}}.$$

Together with symmetry in j, k, this says that $C_{jk\ell\bar{m}}$ is invariant under every permutation of the three unbarred indices j, k, ℓ. Now set

$$z_m = x_m + \frac{1}{3} \sum_{j,k,\ell} C_{jk\ell\bar{m}} x_j x_k x_\ell. \tag{12.24}$$

Complete symmetry gives

$$\mathrm{d}z_m = \mathrm{d}x_m + \sum_{j,k,\ell} C_{jk\ell\bar{m}} x_j x_k \, \mathrm{d}x_\ell . \tag{12.25}$$

Expanding $\sum_m \mathrm{d}z_m \otimes \mathrm{d}\bar{z}_m$ as before shows that its pure holomorphic and antiholomorphic quadratic terms are exactly the two C-terms in (12.23). They are therefore absorbed by the coordinate change, whereas the mixed term cannot be removed in this way. In the new coordinates,

$$h_{\ell\bar{m}}(z) = \delta_{\ell m} + \sum_{j,k} B_{j\bar{k}\ell\bar{m}} z_j \bar{z}_k + O(|z|^3). \tag{12.26}$$

Identifying the mixed coefficient with curvature. Formula (12.11) gives

$$\Gamma^q_{j\ell} = \sum_s h^{q\bar{s}} \partial_j h_{\ell\bar{s}}.$$

In the coordinates of (12.26), $h(0) = I$ and $\partial h(0) = 0$, so $\Gamma^q_{j\ell}(0) = 0$. Equation (12.12) then gives

$$\begin{aligned} \mathcal{R}(\partial_j, \partial_{\bar{k}})\partial_\ell \big|_0 &= -\partial_{\bar{k}} \left(\sum_q \Gamma^q_{j\ell} \partial_q \right) \Big|_0 , \\ R_{j\bar{k}\ell\bar{m}}(p) &= -\frac{\partial^2 h_{\ell\bar{m}}}{\partial z_j \partial \bar{z}_k}(0) = -B_{j\bar{k}\ell\bar{m}}. \end{aligned} \tag{12.27}$$

Substitution into (12.26) proves (12.17). □

Example 3.55: Euclidean space and complex tori

On $\mathbb{C}^n$, $\varphi = \pi|z|^2$ gives $\omega_0 = (\mathrm{i}/2)\sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$. Translation invariance lets ω_0 descend to every complex torus $\mathbb{C}^n/\Lambda$.

Example 3.56: The Fubini–Study form

On the chart $U_0 \subset \mathbb{P}^n$, write $w_j = z_j/z_0$, set $\varphi(w) = \log(1 + |w|^2)$, and put

$$\omega_{\mathrm{FS}} = \mathrm{d}\mathrm{d}^{\mathrm{c}}\varphi = \frac{\mathrm{i}}{2\pi}\partial\bar{\partial}\varphi.$$

On an overlap, changing the homogeneous representative adds $\log|g|^2$ for a nowhere-zero holomorphic function g; its $\mathrm{d}\mathrm{d}^{\mathrm{c}}$ vanishes, so the local forms glue. Direct differentiation gives

$$\frac{\partial^2}{\partial w_j \partial \bar{w}_k} \log(1 + |w|^2) = \frac{(1 + |w|^2)\delta_{jk} - \bar{w}_j w_k}{(1 + |w|^2)^2}.$$

For $\xi \neq 0$, Cauchy–Schwarz yields

$$\sum_{j,k} \varphi_{j\bar{k}} \xi_j \bar{\xi}_k = \frac{(1 + |w|^2)|\xi|^2 - |\langle \xi, w \rangle|^2}{(1 + |w|^2)^2} \geq \frac{|\xi|^2}{(1 + |w|^2)^2} > 0.$$

Thus the form is globally defined and positive.

Proposition 3.57: Products and complex submanifolds

If (X_i, ω_i) are Kähler, then $\mathrm{pr}_1^*\,\omega_1 + \mathrm{pr}_2^*\,\omega_2$ is Kähler on $X_1 \times X_2$. If $\iota : Y \hookrightarrow X$ is a holomorphic immersion, then $\iota^*\omega$ is Kähler on Y.

Proof. Pullback preserves type and commutes with d, so all forms are closed and of type $(1,1)$. For $(v_1, v_2) \neq 0$, the product form evaluates on $((v_1, v_2), J(v_1, v_2))$ as the sum of two nonnegative terms, one positive. For an immersion, $\mathrm{d}\iota(v) \neq 0$ when $v \neq 0$, so $\iota^*\omega(v, Jv) = \omega(\mathrm{d}\iota(v), J\mathrm{d}\iota(v)) > 0$. □

Warning

A Hermitian metric need not be Kähler. Positivity is pointwise linear algebra, while $\mathrm{d}\omega = 0$ is a differential condition. Also, a global identity $\omega = \mathrm{d}\mathrm{d}^{\mathrm{c}}\varphi$ cannot hold for a Kähler form on a compact manifold, since Stokes would force $\int_X \omega^n = 0$.

3.4.5 Why the next step is Hodge theory

The form ω does more than measure complex tangent vectors. Its associated Riemannian metric g and orientation define the Hodge star, the formal adjoint d^*, and the elliptic operator

$$\Delta_{\mathrm{d}} = \mathrm{d}\mathrm{d}^* + \mathrm{d}^*\mathrm{d}.$$

On a compact oriented Riemannian manifold, Hodge theory says that every de Rham cohomology class has a unique Δ_{d}-harmonic representative. This is already a powerful bridge from topology to differential equations, but the Kähler condition makes the bridge substantially stronger. Since $\mathrm{d}\omega = 0$, the complex splitting $\mathrm{d} = \partial + \bar{\partial}$ is compatible with the metric in such a way that

$$\Delta_{\mathrm{d}} = 2\Delta_{\partial} = 2\Delta_{\bar{\partial}}.$$

Consequently harmonic forms split into their (p, q)-components, and topological cohomology acquires a decomposition by complex type. Lecture 13 first develops the Riemannian Hodge theorem without using complex geometry. Lecture 14 then identifies the additional identities supplied by the Kähler condition. Only after this analytic picture is complete will we return to sheaf cohomology and then introduce vector bundles systematically.

3.4.6 Pointwise Hermitian linear algebra

Let

$$\alpha = \frac{\mathrm{i}}{2}\sum_{j,k} a_{j\bar{k}}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$$

be a real $(1,1)$-form. Reality is equivalent to $A = (a_{j\bar{k}})$ being Hermitian, and positivity means $\xi^* A\xi > 0$ for every nonzero $\xi \in \mathbb{C}^n$. A unitary change of coordinates diagonalizes A at one point:

$$\alpha = \frac{\mathrm{i}}{2}\sum_j \lambda_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j.$$

Consequently,

$$\frac{\alpha^n}{n!} = \lambda_1 \cdots \lambda_n \left(\frac{\mathrm{i}}{2}\right)^n \mathrm{d}z_1 \wedge \mathrm{d}\bar{z}_1 \wedge \cdots \wedge \mathrm{d}z_n \wedge \mathrm{d}\bar{z}_n.$$

Thus positivity can be read either from the quadratic form or from positive eigenvalues; the determinant controls volume but does not by itself control every direction.

For a Hermitian metric h, a linear holomorphic coordinate change can arrange $h_{j\bar{k}} = \delta_{jk}$ at one point. Theorem 3.54 shows that the Kähler equations permit a quadratic holomorphic coordinate correction which also kills every first derivative. This extra cancellation is why curvature calculations and Laplacian identities are substantially cleaner in Kähler geometry than in general Hermitian geometry.

3.4.7 A detailed Fubini–Study calculation

Put $r^2 = |w|^2$ and $\Phi = \log(1 + r^2)$. Then

$$\Phi_{j\bar{k}} = \frac{(1 + r^2)\delta_{jk} - \bar{w}_j w_k}{(1 + r^2)^2}.$$

For $\xi \in \mathbb{C}^n$,

$$\sum_{j,k} \Phi_{j\bar{k}} \xi_j \bar{\xi}_k = \frac{(1 + r^2)|\xi|^2 - |\langle \xi, w \rangle|^2}{(1 + r^2)^2} \geq \frac{|\xi|^2}{(1 + r^2)^2} > 0,$$

where Cauchy–Schwarz was used in the inequality. This proves strict positivity without appealing to symmetry.

On the projective line the affine expression is

$$\omega_{\mathrm{FS}} = \frac{\mathrm{i}}{2\pi} \frac{\mathrm{d}w \wedge \mathrm{d}\bar{w}}{(1 + |w|^2)^2}.$$

Using $w = r\mathrm{e}^{\mathrm{i}\theta}$ and $\mathrm{i}\mathrm{d}w \wedge \mathrm{d}\bar{w} = 2r\mathrm{d}r \wedge \mathrm{d}\theta$ gives

$$\int_{\mathbb{P}^1} \omega_{\mathrm{FS}} = \frac{1}{\pi} \int_0^{2\pi} \int_0^{\infty} \frac{r}{(1 + r^2)^2} \, \mathrm{d}r \, \mathrm{d}\theta = 1.$$

The missing point at infinity has measure zero, while the change $w' = 1/w$ confirms smoothness there. This normalization is the one later used when the first Chern class of the hyperplane line bundle is introduced.

Remark (Local potentials and global classes)

On an overlap, two Kähler potentials differ by the real part of a holomorphic function. Their Hessians therefore glue even when the potentials do not. The failure of the potentials to glue is not a defect: it is precisely the cohomological information carried by the Kähler class.

3.4.8 Metric formulas in coordinates

Let $w = w(z)$ be a holomorphic change of coordinates. Since $\mathrm{d}z_j = \sum_a (\partial z_j / \partial w_a) \mathrm{d}w_a$, the Hermitian matrix transforms as

$$h'_{a\bar{b}} = \sum_{j,k} h_{j\bar{k}} \frac{\partial z_j}{\partial w_a} \overline{\frac{\partial z_k}{\partial w_b}}.$$

Thus $h' = J^{\mathsf{T}} h \, \overline{J}$, and positivity is coordinate independent: $\xi^{\mathsf{T}} h' \bar{\xi} = (J\xi)^{\mathsf{T}} h \overline{J\xi} > 0$.

For

$$\omega = \frac{\mathrm{i}}{2} \sum_{j,k} h_{j\bar{k}} \, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k,$$

direct differentiation gives

$$\partial\omega = \frac{\mathrm{i}}{2} \sum_{\ell,j,k} \frac{\partial h_{j\bar{k}}}{\partial z_\ell} \mathrm{d}z_\ell \wedge \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k.$$

Collecting the coefficient of $\mathrm{d}z_\ell \wedge \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$ shows that $\partial\omega = 0$ exactly when

$$\frac{\partial h_{j\bar{k}}}{\partial z_\ell} = \frac{\partial h_{\ell\bar{k}}}{\partial z_j} \quad \text{for all } j, k, \ell.$$

These are the coordinate Kähler equations. They are the integrability conditions for finding a local potential ϕ with $h_{j\bar{k}} = \pi^{-1} \partial^2 \phi / \partial z_j \partial \bar{z}_k$ under our $\mathrm{d}\mathrm{d}^{\mathrm{c}}$ convention.

The Fubini–Study Hessian

On the chart $U_0 \simeq \mathbb{C}^n$, put $r^2 = |z_1|^2 + \cdots + |z_n|^2$ and $\phi = \log(1 + r^2)$. First

$$\frac{\partial \phi}{\partial z_j} = \frac{\bar{z}_j}{1 + r^2}.$$

Differentiating once more yields

$$\phi_{j\bar{k}} = \frac{(1 + r^2)\delta_{jk} - \bar{z}_j z_k}{(1 + r^2)^2}.$$

For $\xi \in \mathbb{C}^n$,

$$\begin{aligned}\sum_{j,k} \phi_{j\bar{k}} \xi_j \bar{\xi}_k &= \frac{(1 + r^2)|\xi|^2 - |\langle \xi, z \rangle|^2}{(1 + r^2)^2} \\ &\geq \frac{(1 + r^2)|\xi|^2 - r^2|\xi|^2}{(1 + r^2)^2} = \frac{|\xi|^2}{(1 + r^2)^2} > 0.\end{aligned}$$

Thus $\omega_{\mathrm{FS}} = \mathrm{d}\mathrm{d}^{\mathrm{c}}\phi$ is positive. The inequality used is ordinary Cauchy–Schwarz, and it also gives the two eigenvalues: $(1 + r^2)^{-2}$ in the radial complex direction and $(1 + r^2)^{-1}$ on its orthogonal complement.

The potentials glue although the functions do not. On $U_0 \cap U_1$, with $w_0 = 1/z_1$ and $w_j = z_j/z_1$,

$$1 + \sum_{j=1}^{n} |z_j|^2 = |z_1|^2 \left(1 + |w_0|^2 + \sum_{j=2}^{n} |w_j|^2\right).$$

Therefore

$$\log(1 + |z|^2) = \log |z_1|^2 + \log(1 + |w|^2).$$

Since z_1 is holomorphic and nonvanishing on the overlap, $\mathrm{d}\mathrm{d}^{\mathrm{c}} \log |z_1|^2 = 0$ there. Hence the local $(1, 1)$-forms agree and define a global form on $\mathbb{P}^n$.

On $\mathbb{P}^1$ the calculation reduces to

$$\omega_{\mathrm{FS}} = \frac{\mathrm{i}}{2\pi} \frac{\mathrm{d}z \wedge \mathrm{d}\bar{z}}{(1 + |z|^2)^2} = \frac{1}{\pi} \frac{r\, \mathrm{d}r \wedge \mathrm{d}\theta}{(1 + r^2)^2}.$$

The missing point has measure zero, and

$$\int_{\mathbb{P}^1} \omega_{\mathrm{FS}} = \frac{1}{\pi} \int_0^{2\pi} \int_0^{\infty} \frac{r}{(1 + r^2)^2}\, \mathrm{d}r\, \mathrm{d}\theta = 2 \left[-\frac{1}{2(1 + r^2)} \right]_0^{\infty} = 1.$$

This normalization is the one later used when the first Chern class of the hyperplane line bundle is introduced.

Pullbacks, products, and submanifolds

If $F : X \to Y$ is holomorphic and ω_Y is Kähler, then $\mathrm{d}(F^*\omega_Y) = F^*(\mathrm{d}\omega_Y) = 0$. Positivity may become degenerate if F has a kernel. If F is an immersion, however,

$$(F^*\omega_Y)_x(v, Jv) = \omega_{Y, F(x)}(\mathrm{d}F_x v, J\mathrm{d}F_x v) > 0$$

for $v \neq 0$, so the pullback is Kähler. This proves the submanifold statement with the inclusion map.

For two Kähler manifolds, the product form is

$$\omega = \mathrm{pr}_1^*\, \omega_1 + \mathrm{pr}_2^*\, \omega_2.$$

At (x, y) and $(v, w) \neq 0$,

$$\omega((v,w), J(v,w)) = \omega_1(v, Jv) + \omega_2(w, Jw) > 0.$$

It is closed term by term. This coordinate-free calculation checks both defining conditions and explains why omitting either summand would make the form degenerate.

3.4.9 Local Kähler potentials with constants and reality shown

Let ω be a real closed $(1,1)$-form on a coordinate ball B. After shrinking B, the real Poincaré lemma gives a real one-form η with $\omega = \mathrm{d}\eta$. Decompose

$$\eta = \eta^{1,0} + \eta^{0,1}, \qquad \eta^{1,0} = \overline{\eta^{0,1}}.$$

The $(0,2)$ component of $\mathrm{d}\eta = \omega$ is $\bar\partial\eta^{0,1}$, hence it vanishes. The Dolbeault–Grothendieck lemma proved in Lecture 4 gives a smooth complex-valued function f such that $\eta^{0,1} = \bar\partial f$. Reality then gives $\eta^{1,0} = \partial\bar f$. Consequently

$$\begin{aligned}\omega &= \partial\eta^{0,1} + \bar\partial\eta^{1,0} \\ &= \partial\bar\partial f + \bar\partial\partial\bar f = \partial\bar\partial(f - \bar f).\end{aligned}$$

Set

$$\phi = \frac{2\pi}{\mathrm{i}}(f - \bar f) = 4\pi \operatorname{Im} f.$$

Then ϕ is real, and our convention $\mathrm{dd^c} = (\mathrm{i}/2\pi)\partial\bar\partial$ gives $\mathrm{dd^c}\phi = \omega$. If ω is positive, the Levi matrix of ϕ is positive, so ϕ is strictly psh.

If ϕ_1 and ϕ_2 are two local potentials, then $\partial\bar\partial(\phi_1 - \phi_2) = 0$. Put $u = \phi_1 - \phi_2$. Then $\bar\partial(\partial u) = 0$, so the coefficients of the $(1,0)$-form ∂u are holomorphic; moreover $\partial(\partial u) = 0$. The holomorphic Poincaré lemma on a ball gives a holomorphic function h with $\partial h = 2\partial u$. Hence $\partial(u - \operatorname{Re} h) = 0$. Since the expression in parentheses is real, it is constant, and the constant may be absorbed into h. Thus two real potentials differ locally by the real part of a holomorphic function. This is the precise ambiguity that permits local potentials to define one global Kähler form even though the functions themselves do not glue.

3.4.10 Self-study workshop: checking that a local Hermitian formula defines a global Kähler metric

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. You need Lecture 2's type decomposition, Hermitian matrices, and the rule that $\partial\bar\partial$ annihilates pluriharmonic functions. Recall that positivity is pointwise linear algebra, while closedness is a differential condition.

The three checks. In coordinates a real $(1,1)$-form has the shape

$$\omega = \frac{\mathrm{i}}{2}\sum_{j,k} g_{j\bar k}\,\mathrm{d}z_j \wedge \mathrm{d}\bar z_k.$$

It defines a Hermitian metric precisely when $G = (g_{j\bar k})$ is Hermitian positive definite. It is Kähler when additionally $\mathrm{d}\omega = 0$. If $\omega = \mathrm{dd^c}\varphi$, closedness follows from $\mathrm{d}^2 = 0$, but positivity still requires the complex Hessian of φ to be positive.

For $\mathbb{P}^1$, use $u = z_1/z_0$ on U_0 and $v = z_0/z_1 = 1/u$ on U_1. Put

$$\varphi_0 = \log(1 + |u|^2), \qquad \varphi_1 = \log(1 + |v|^2).$$

On the overlap,

$$\varphi_0 - \varphi_1 = \log |u|^2.$$

Since u is holomorphic and nonzero there, $\mathrm{dd^c} \log |u|^2 = 0$. Thus $\mathrm{dd^c}\varphi_0$ and $\mathrm{dd^c}\varphi_1$ glue. Direct differentiation gives

$$\frac{\partial^2 \varphi_0}{\partial u \, \partial \bar{u}} = \frac{1}{(1 + |u|^2)^2} > 0.$$

This proves both global well-definedness and positivity; citing the potential alone would prove neither.

The same logic works for a Kähler submanifold. If $\iota : Y \hookrightarrow X$ is a holomorphic immersion and ω is Kähler, then $\mathrm{d}(\iota^*\omega) = \iota^*(\mathrm{d}\omega) = 0$. For a nonzero real tangent vector $v \in T_y Y$, injectivity of $\mathrm{d}\iota_y$ and holomorphicity of ι give

$$(\iota^*\omega)(v, Jv) = \omega(\mathrm{d}\iota_y v, J\mathrm{d}\iota_y v) > 0.$$

Checkpoint (potential versus metric)

(1) Why may two local Kähler potentials differ without changing the metric?
(2) Show that $\varphi = |z_1|^2 + 2|z_2|^2$ gives a Kähler form on $\mathbb{C}^2$.
(3) What fails for $\varphi = |z_1|^2 - |z_2|^2$?

Solution (checkpoint). (1) Potentials differing by a pluriharmonic function have the same $\mathrm{dd^c}$. (2) Its complex Hessian is $\mathrm{diag}(1, 2)$, so $\mathrm{dd^c}\varphi$ is closed and positive definite. (3) The Hessian has eigenvalues $1, -1$; the form is closed and real but not positive, hence not Kähler. ◇

Exit criterion

For a metric given by local potentials, you can check overlap compatibility, differentiate the potential, diagonalize its Hermitian matrix, and distinguish closedness from positivity.

Summary. A Hermitian metric may be described equivalently on the real tangent bundle, on $T^{1,0}X$, or by a positive real $(1, 1)$-form. The Kähler condition is equivalent to $\nabla J = 0$ and produces holomorphic normal coordinates whose mixed quadratic term records curvature. Locally, Kähler forms are complex Hessians of strictly psh potentials; Euclidean space, tori, projective space, products, and submanifolds give the basic examples.

Exercise 3.58: Area of $\mathbb{P}^1$

Use the affine coordinate to compute $\int_{\mathbb{P}^1} \omega_{\mathrm{FS}}$. Convert to polar coordinates and verify smoothness at the point at infinity.

Exercise 3.59: Products

Prove directly that the product of two Kähler manifolds is Kähler for $\mathrm{pr}_1^* \omega_1 + \mathrm{pr}_2^* \omega_2$.

Optional handout: Resolution of singularities and weak factorization

This handout is optional. No later argument in the core lectures depends on it.

This optional handout preserves the birational material removed from the core schedule. A *modification* is a proper holomorphic map $\mu : \widetilde{X} \to X$ that is biholomorphic away from a nowhere-dense analytic subset. The blow-up of the origin in $\mathbb{C}^n$ is

$$\operatorname{Bl}_0 \mathbb{C}^n = \{(z, [\ell]) \in \mathbb{C}^n \times \mathbb{P}^{n-1} : z \in \ell\}.$$

In the chart $\ell_1 \neq 0$, it has coordinates $z_1 = t$, $z_j = t u_j$ for $j \geq 2$; the exceptional divisor is $E = \{t = 0\}$. The *strict transform* of a subvariety is the closure of its inverse image away from the center. These definitions are sufficient for the calculations below; the topic is enrichment rather than a prerequisite for the later analytic course.

Hironaka's resolution theorem says that, in characteristic zero, every reduced complex analytic space admits a proper modification from a smooth space, obtained locally by a finite sequence of blow-ups with smooth centers. The centers can be chosen inside the singular locus, so the modification is an isomorphism over the regular part.

For a divisor D, an embedded log resolution makes the total transform a simple normal-crossings divisor. Locally this means that, in suitable coordinates,

$$\mu^* D = \{z_1^{a_1} \cdots z_r^{a_r} = 0\}.$$

Analytic singularities then become monomial. This is why resolution is so effective for integrability, multiplier ideals, and change-of-variables formulas.

The weak factorization theorem goes in the opposite direction: a birational map between smooth complete algebraic varieties over a field of characteristic zero can be factored into blow-ups and blow-downs along smooth centers, away from the open set where the original map is an isomorphism. It says that blow-up is not merely one construction but a universal vocabulary for comparing smooth birational models.

Remark (Discrepancies)

Here K_X denotes the canonical divisor, locally represented by a meromorphic top-degree form. If E is a prime exceptional divisor, the number $a(E, X)$ below is called its discrepancy. For a modification $\mu : Y \to X$ of smooth n-folds,

$$K_Y = \mu^* K_X + \sum_E a(E, X) E.$$

The coefficients measure the Jacobian vanishing of μ. They govern the change-of-variables test for multiplier ideals and, in algebraic geometry, the classification of singularities as log canonical, klt, and so forth.

Exercise 3.60: Canonical divisor of a blow-up

Blow up a smooth codimension-r center. In local coordinates compute the Jacobian in one chart and show $K_{\widetilde{X}} = \pi^* K_X + (r-1)E$.

Exercise 3.61: Transition between blow-up charts

Write the transition map between the two charts of $\operatorname{Bl}_0 \mathbb{C}^2$ and identify the normal bundle of E as $\mathcal{O}_{\mathbb{P}^1}(-1)$.

Exercise 3.62: Three concurrent lines

Blow up three lines through the origin in $\mathbb{C}^2$ with distinct tangent directions. Prove that their strict transforms are disjoint and meet E transversely at three different points.

Chapter 4

Hodge Theory, Kähler Identities, and Cohomology

4.1 Lecture 13: Elliptic operators, harmonic forms, and Hodge theory

Week 7 material • 90 minutes

Notation for this lecture

M is a compact oriented Riemannian manifold without boundary, $\mathcal{A}^k(M)$ is the space of smooth k-forms, and $(\alpha,\beta)_{L^2} = \int_M \alpha \wedge *\bar{\beta}$ uses the Hodge star $*$, extended complex-linearly from real forms. For a differential operator P, P^* is its formal L^2 adjoint. The principal symbol $\sigma_P(x,\xi)$ is obtained here by retaining the highest-order terms and replacing $\partial/\partial x_j$ by the real covector component ξ_j; this convention suppresses the alternative universal factor i. The metric dual of ξ is $\xi^\sharp$, and $\iota_{\xi^\sharp}$ denotes contraction. The Hodge Laplacian is $\Delta_{\mathrm{d}} = \mathrm{d}\mathrm{d}^* + \mathrm{d}^*\mathrm{d}$ and $\mathcal{H}^k(M) = \ker(\Delta_{\mathrm{d}} : \mathcal{A}^k \to \mathcal{A}^k)$ is the space of harmonic k-forms. The Green operator G is the inverse of Δ_{d} on the orthogonal complement of $\mathcal{H}^k(M)$, and H denotes orthogonal projection onto harmonic forms. The space H^s_{loc} is the local Sobolev space of order s. De Rham cohomology is denoted $H^k_{\mathrm{dR}}(M,\mathbb{C})$, abbreviated to $H^k(M,\mathbb{C})$ when no other cohomology theory is in view, and $b_k = \dim_{\mathbb{C}} H^k_{\mathrm{dR}}(M,\mathbb{C})$ is the kth Betti number.

Prerequisite results used below

Elliptic regularity. If P is elliptic of order m with smooth coefficients and $Pu = f$ distributionally, then $f \in H^s_{\mathrm{loc}}$ implies $u \in H^{s+m}_{\mathrm{loc}}$. In particular, distributional solutions of $Pu = 0$ are smooth.

Spectral theorem for a self-adjoint elliptic operator. On a compact manifold, a nonnegative formally self-adjoint elliptic operator has finite-dimensional smooth kernel and a discrete sequence of real eigenvalues tending to infinity. Its eigenvectors form an orthonormal basis of L^2, and its inverse on the orthogonal complement of the kernel is the Green operator.

Why these ideas are needed

Cohomology is defined as a quotient of infinite-dimensional spaces, so it is not yet clear why its groups should be finite-dimensional or have canonical representatives. An elliptic Laplacian converts the quotient problem into a differential equation. Harmonic forms are simultaneously closed and orthogonal to exact forms; elliptic regularity makes weak solutions smooth, and the compact resolvent makes the harmonic space finite-dimensional. The Hodge decomposition is

introduced because it turns an abstract class into a unique geometric object that can be tested by metrics, curvature, and integration.

Plan for the lecture. 0–20 min: the L^2 inner product and Hodge star; 20–42: adjoints and Laplacians; 42–68: elliptic regularity and Hodge decomposition; 68–82: de Rham isomorphism; 82–90: the Green operator.

Lecture 12 attached a Riemannian metric and an orientation to every Hermitian manifold. We now temporarily forget the complex structure and ask a purely Riemannian question: can one choose a canonical differential form in each de Rham class? The answer is the Hodge theorem. Separating this analytic theorem from the Kähler identities is useful: compactness and ellipticity will do the work in this lecture, while the complex structure and the equation $\mathrm{d}\omega = 0$ enter only in Lecture 14.

Before introducing the Laplacian, recall what ellipticity tests. If a differential operator P of order ℓ has local highest-order part $\sum_{|I|=\ell} a_I(x)\partial_x^I$, its principal symbol is

$$\sigma_P(x,\xi) = \sum_{|I|=\ell} a_I(x)\xi^I, \qquad \xi \in T_x^*M.$$

Although the displayed expression uses coordinates, the symbol is intrinsic: it is the coefficient of t^ℓ in $\mathrm{e}^{-t\varphi}P(\mathrm{e}^{t\varphi}u)$ at x, where $\mathrm{d}\varphi_x = \xi$ and $u(x)$ is the vector on which the symbol acts. Thus P is elliptic when $\sigma_P(x,\xi)$ is invertible for every real nonzero covector ξ.

Let (M^m, g) be compact and oriented. The Hodge star is defined by

$$\alpha \wedge *\bar{\beta} = \langle \alpha, \beta \rangle_g \, \mathrm{d}V_g,$$

and on real k-forms satisfies $*^2 = (-1)^{k(m-k)}$. We extend $*$ complex-linearly; the conjugation in the displayed formula then makes the induced L^2 pairing Hermitian.

Definition 4.1: Formal adjoint and Hodge Laplacian

On r-forms the formal adjoint of d is

$$\mathrm{d}^* = (-1)^{m(r+1)+1} * \mathrm{d} * .$$

The Hodge Laplacian is

$$\Delta_{\mathrm{d}} = \mathrm{d}\mathrm{d}^* + \mathrm{d}^*\mathrm{d},$$

and $\mathcal{H}^k(M) = \ker(\Delta_{\mathrm{d}} : \mathcal{A}^k \to \mathcal{A}^k)$ is the space of harmonic k-forms.

The symbols of the two first-order operators are

$$\sigma_{\mathrm{d}}(x,\xi)\alpha = \xi \wedge \alpha, \qquad \sigma_{\mathrm{d}^*}(x,\xi)\alpha = \iota_{\xi^\sharp}\alpha.$$

The exterior-multiplication and contraction identity

$$\iota_{\xi^\sharp}(\xi \wedge \alpha) + \xi \wedge \iota_{\xi^\sharp}\alpha = |\xi|_g^2 \alpha$$

therefore gives $\sigma_{\Delta_{\mathrm{d}}}(x,\xi) = |\xi|_g^2 \,\mathrm{id}$. This local calculation is the reason elliptic theory applies to the de Rham complex.

Integration by parts gives

$$(\Delta_{\mathrm{d}}\alpha, \alpha) = \|\mathrm{d}\alpha\|_{L^2}^2 + \|\mathrm{d}^*\alpha\|_{L^2}^2 .$$

Thus on a compact manifold without boundary, α is harmonic exactly when $\mathrm{d}\alpha = \mathrm{d}^*\alpha = 0$.

Warning

On a compact boundaryless manifold, formal adjoints agree with Hilbert-space adjoints after closure. With a boundary or on an incomplete manifold, the operator domain contains boundary conditions. Lecture 27 cannot reuse the present integration-by-parts formula without checking domains.

Theorem 4.2: Hodge decomposition

On a compact oriented Riemannian manifold,

$$\mathcal{A}^k(M) = \mathcal{H}^k(M) \oplus \operatorname{Im} \mathrm{d} \oplus \operatorname{Im} \mathrm{d}^*$$

orthogonally in L^2. Every de Rham class has a unique harmonic representative, so

$$\mathcal{H}^k(M) \simeq H^k_{\mathrm{dR}}(M, \mathbb{C}).$$

Proof (analytic core). The symbol calculation above shows that Δ_{d} is elliptic. It is formally self-adjoint and nonnegative because

$$(\Delta_{\mathrm{d}}\alpha, \alpha)_{L^2} = \|\mathrm{d}\alpha\|^2_{L^2} + \|\mathrm{d}^*\alpha\|^2_{L^2}.$$

The spectral theorem for self-adjoint elliptic operators on a compact manifold gives an orthonormal basis of smooth eigenforms. The zero eigenspace $\mathcal{H}^k(M)$ is finite dimensional, and every other eigenvalue is positive. Let H be orthogonal projection onto $\mathcal{H}^k(M)$. On an eigenform with eigenvalue $\lambda > 0$ define $G_{\mathrm{d}}\alpha = \lambda^{-1}\alpha$, and put $G_{\mathrm{d}} = 0$ on $\mathcal{H}^k(M)$. Elliptic regularity shows that G_{d} maps smooth forms to smooth forms, and

$$\Delta_{\mathrm{d}} G_{\mathrm{d}} = G_{\mathrm{d}} \Delta_{\mathrm{d}} = I - H.$$

On smooth forms, $\mathrm{d}\Delta_{\mathrm{d}} = \Delta_{\mathrm{d}}\mathrm{d}$ and $\mathrm{d}^*\Delta_{\mathrm{d}} = \Delta_{\mathrm{d}}\mathrm{d}^*$; these identities follow by expanding both sides and using $\mathrm{d}^2 = (\mathrm{d}^*)^2 = 0$. Hence d and d^* preserve every eigenspace and commute with H and G_{d}. Consequently

$$\alpha = H\alpha + \Delta_{\mathrm{d}} G_{\mathrm{d}} \alpha = H\alpha + \mathrm{d}\mathrm{d}^* G_{\mathrm{d}} \alpha + \mathrm{d}^* \mathrm{d} G_{\mathrm{d}} \alpha.$$

The three summands are mutually orthogonal: harmonic forms are both closed and coclosed, while $(\mathrm{d}\beta, \mathrm{d}^*\gamma)_{L^2} = (\mathrm{d}^2\beta, \gamma)_{L^2} = 0$. This proves the asserted orthogonal decomposition.

If $\mathrm{d}\alpha = 0$, commutation gives $\mathrm{d} G_{\mathrm{d}} \alpha = 0$ and the coexact component vanishes:

$$\alpha = H\alpha + \mathrm{d}(\mathrm{d}^* G_{\mathrm{d}} \alpha).$$

Thus $H\alpha$ represents the same de Rham class. If a harmonic form is exact, $h = \mathrm{d}\gamma$, then $\|h\|^2 = (h, \mathrm{d}\gamma) = (\mathrm{d}^* h, \gamma) = 0$; this proves uniqueness. □

The analytic ingredients and the resulting decomposition are [1, Chapter VI, Corollary 2.4 and Theorems 3.16–3.17]. The proof above isolates their roles: compact elliptic theory constructs H and G_{d}, while the two commutation identities turn the spectral decomposition into a statement about de Rham cohomology.

Corollary 4.3: Poincaré duality from Hodge theory

The Hodge star maps $\mathcal{H}^k$ isomorphically to $\mathcal{H}^{m-k}$. Hence $b_k = b_{m-k}$ and the pairing $([\alpha], [\beta]) \mapsto \int_M \alpha \wedge \beta$ is nondegenerate.

Proof. The identities $*\Delta_{\mathrm{d}} = \Delta_{\mathrm{d}}*$ and $*^2 = (-1)^{k(m-k)}$ on k-forms show that $*$ restricts to an isomorphism $\mathcal{H}^k \simeq \mathcal{H}^{m-k}$. Hodge theory therefore gives $b_k = \dim \mathcal{H}^k = \dim \mathcal{H}^{m-k} = b_{m-k}$. If

$0 \neq [\alpha] \in H^k(M, \mathbb{C})$, take its nonzero harmonic representative α and put $\beta = *\overline{\alpha}$. Then β is harmonic and

$$\int_M \alpha \wedge \beta = \int_M \alpha \wedge *\overline{\alpha} = \|\alpha\|_{L^2}^2 > 0.$$

Thus no nonzero class lies in the left kernel of the pairing. Applying the same argument in degree $m - k$, or using equality of dimensions, eliminates the right kernel as well. □

Example 4.4: Flat tori

On $T^m = \mathbb{R}^m/\mathbb{Z}^m$ with its flat metric, harmonic forms are exactly constant-coefficient forms. Thus $H^k(T^m, \mathbb{C}) \simeq \Lambda^k \mathbb{C}^m$ and $b_k = \binom{m}{k}$.

Remark (Geometric interpretation)

A cohomology class contains infinitely many closed forms. Its harmonic representative is the unique member of least L^2 norm. Hodge theory replaces a topological quotient by the finite-dimensional kernel of an elliptic equation.

4.1.1 Why the Hodge Laplacian is elliptic

For a first-order operator P, the principal symbol $\sigma_P(x, \xi)$ keeps only the highest derivative and replaces $\partial/\partial x_j$ by multiplication by ξ_j. For the exterior derivative,

$$\sigma_{\mathrm{d}}(x, \xi)\alpha = \xi \wedge \alpha,$$

while the symbol of d^* is contraction by the metric dual $\xi^\sharp$. The identity

$$\iota_{\xi^\sharp}(\xi \wedge \alpha) + \xi \wedge \iota_{\xi^\sharp}\alpha = |\xi|^2 \alpha$$

therefore gives

$$\sigma_{\Delta_{\mathrm{d}}}(x, \xi) = |\xi|^2 \,\mathrm{id}\,.$$

It is invertible for $\xi \neq 0$, which is ellipticity.

Ellipticity yields the estimate

$$\|u\|_{H^{s+2}} \leq C_s\big(\|\Delta_{\mathrm{d}} u\|_{H^s} + \|u\|_{L^2}\big).$$

Thus a distributional solution of $\Delta_{\mathrm{d}} u = f$ gains two derivatives over f. Bootstrapping shows that harmonic distributions are smooth. Rellich compactness of $H^1 \hookrightarrow L^2$ on a compact manifold then forces the harmonic space to be finite-dimensional.

4.1.2 The Green operator as a chain homotopy

Let H be orthogonal projection onto harmonic forms and let G be the Green operator, zero on $\mathcal{H}^\bullet$ and inverse to Δ_{d} on its orthogonal complement. Then

$$I - H = \Delta_{\mathrm{d}} G = G \Delta_{\mathrm{d}}, \qquad [G, \mathrm{d}] = [G, \mathrm{d}^*] = 0.$$

For a closed form α this becomes

$$\alpha - H\alpha = \mathrm{d}\mathrm{d}^* G \alpha.$$

Therefore $K = \mathrm{d}^* G$ is a chain homotopy from the identity on the de Rham complex to harmonic projection. The abstract statement “cohomology equals harmonic forms” is encoded in this explicit operator identity.

For functions, harmonicity is especially rigid. If M is compact and connected,

$$0 = (\Delta f, f) = \|\mathrm{d}f\|^2$$

shows that harmonic functions are constant. For one-forms on a flat torus, constant coefficients are both closed and coclosed, producing the expected m-dimensional space. On a sphere, $H^1 = 0$, so every harmonic one-form vanishes although many eigen-one-forms with positive eigenvalue exist.

Remark (Spectral picture)

The compact resolvent of Δ_{d} gives a finite-dimensional zero eigenspace and positive eigenvalues $0 < \lambda_1 \le \lambda_2 \le \cdots \to \infty$; the zero eigenspace can be trivial in a given degree. Harmonic forms form that zero eigenspace, while the Green operator multiplies the λ_j-eigenspace by λ_j^{-1}. Hodge decomposition is therefore the zero-frequency part of a full elliptic spectral decomposition.

4.1.3 Formal adjoints and the principal symbol

For compactly supported forms, Stokes gives

$$0 = \int_M \mathrm{d}(\alpha \wedge *\bar{\beta}) = \int_M \mathrm{d}\alpha \wedge *\bar{\beta} + (-1)^{\deg \alpha} \int_M \alpha \wedge \mathrm{d}(*\bar{\beta}).$$

After applying $*^2 = (-1)^{k(m-k)}$ and collecting the degree signs, one obtains

$$\mathrm{d}^* = (-1)^{m(k+1)+1} * \mathrm{d} *$$

on k-forms. The precise exponent depends on the Hodge-star convention, but the integration identity $(\mathrm{d}\alpha, \beta) = (\alpha, \mathrm{d}^*\beta)$ is invariant.

The principal symbol of d at a covector ξ is exterior multiplication:

$$\sigma_1(\mathrm{d})(\xi)\alpha = \xi \wedge \alpha.$$

Its adjoint has symbol $\iota_{\xi^\sharp}$ in the same convention. Hence

$$\begin{aligned} \sigma_2(\Delta_{\mathrm{d}})(\xi) &= (\xi\wedge)\iota_{\xi^\sharp} + \iota_{\xi^\sharp}(\xi\wedge) \\ &= |\xi|^2 \,\mathrm{id}, \end{aligned}$$

because contraction and exterior multiplication satisfy the Clifford relation. This is invertible for $\xi \neq 0$, proving ellipticity.

The complete Hodge calculation on the circle

Write $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ with coordinate θ. A function has Fourier series $f = \sum_{k\in\mathbb{Z}} a_k e^{\mathrm{i}k\theta}$, and

$$\Delta f = -f'' = \sum_k k^2 a_k e^{\mathrm{i}k\theta}.$$

Thus harmonic functions are constants. A one-form is $g(\theta)\mathrm{d}\theta$; since $*\mathrm{d}\theta = 1$ and $*1 = \mathrm{d}\theta$,

$$\mathrm{d}^*(g\mathrm{d}\theta) = -g', \qquad \Delta(g\mathrm{d}\theta) = -g''\mathrm{d}\theta.$$

The harmonic one-forms are exactly $c\mathrm{d}\theta$.

For an arbitrary one-form, let

$$c = \frac{1}{2\pi}\int_0^{2\pi} g(\theta)\,\mathrm{d}\theta, \qquad h(\theta) = \int_0^{\theta} (g(t) - c)\,\mathrm{d}t.$$

The zero-average condition makes h periodic, and

$$g\mathrm{d}\theta = c\mathrm{d}\theta + \mathrm{d}h.$$

This is the Hodge decomposition in degree one, written without functional analysis.

The Green operator from the spectral theorem

On a compact manifold, self-adjoint elliptic theory gives an orthonormal basis $\{h_1, \ldots, h_b, \phi_1, \phi_2, \ldots\}$ with

$$\Delta h_a = 0, \qquad \Delta\phi_j = \lambda_j\phi_j, \qquad 0 < \lambda_1 \le \lambda_2 \le \cdots \to \infty.$$

Here $b = \dim\ker\Delta$ may be zero. For $\alpha = \sum_a c_a h_a + \sum_j a_j\phi_j$, define

$$H\alpha = \sum_a c_a h_a, \qquad G\alpha = \sum_{j\ge 1} \frac{a_j}{\lambda_j}\phi_j.$$

Then

$$\Delta G = G\Delta = I - H,$$

and elliptic estimates show that smooth α gives smooth $G\alpha$. Substituting $I - H = \mathrm{d}\mathrm{d}^*G + \mathrm{d}^*\mathrm{d}G$ produces the orthogonal decomposition explicitly.

The minimum-norm property

Let h be the harmonic representative of a de Rham class and let another representative be $h + \mathrm{d}\gamma$. Since $\mathrm{d}^*h = 0$,

$$(h, \mathrm{d}\gamma) = (\mathrm{d}^*h, \gamma) = 0.$$

Therefore

$$\|h + \mathrm{d}\gamma\|^2 = \|h\|^2 + \|\mathrm{d}\gamma\|^2 \ge \|h\|^2,$$

with equality only when $\mathrm{d}\gamma = 0$. Thus the harmonic representative is the unique minimum-norm element of its class.

On a flat torus, applying the Fourier calculation in every coordinate shows that all nonzero Fourier modes have positive Laplace eigenvalue $4\pi^2|k|^2$. Hence harmonic forms have only the zero mode and are exactly the constant-coefficient forms.

4.1.4 From the Green operator to a unique cohomology representative

The identity

$$I - H = \mathrm{d}\mathrm{d}^*G + \mathrm{d}^*\mathrm{d}G$$

becomes especially useful on closed forms. The operators d and Δ_{d} commute, so $\mathrm{d}G = G\mathrm{d}$ on the orthogonal complement of the harmonic forms. If $\mathrm{d}\alpha = 0$, then $\mathrm{d}G\alpha = 0$ and hence

$$\alpha = H\alpha + \mathrm{d}(\mathrm{d}^*G\alpha).$$

Thus $H\alpha$ is closed and differs from α by an exact form: it represents the same de Rham class. This proves existence of a harmonic representative, not just an orthogonal decomposition of arbitrary forms.

For uniqueness, suppose a harmonic form h is exact, say $h = \mathrm{d}\gamma$. Then

$$\|h\|^2 = (h, \mathrm{d}\gamma) = (\mathrm{d}^* h, \gamma) = 0,$$

so $h = 0$. Therefore two harmonic representatives of the same class coincide. We have proved that

$$\ker \Delta_{\mathrm{d}} \longrightarrow H^k_{\mathrm{dR}}(M, \mathbb{C}), \qquad h \longmapsto [h],$$

is an isomorphism. Ellipticity supplies finite-dimensionality and regularity; the two displayed calculations supply existence and uniqueness at the cohomological level.

4.1.5 Self-study workshop: from ellipticity to the Hodge decomposition

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall formal adjoints, principal symbols, Sobolev compactness on a compact manifold, and orthogonal projections in a Hilbert space. Self-adjointness alone is not enough; ellipticity supplies regularity and finite dimensionality.

The analytic chain. For the de Rham operator, the principal symbols are exterior multiplication and contraction:

$$\sigma_{\mathrm{d}}(\xi) = \xi\wedge, \qquad \sigma_{\mathrm{d}^*}(\xi) = \iota_{\xi^\sharp}.$$

The anticommutator identity

$$\iota_{\xi^\sharp}(\xi \wedge \alpha) + \xi \wedge \iota_{\xi^\sharp}\alpha = |\xi|^2 \alpha$$

gives

$$\sigma_{\Delta_{\mathrm{d}}}(\xi) = |\xi|^2 \,\mathrm{id}\,.$$

Thus the Hodge Laplacian is elliptic. On a compact manifold, elliptic estimates and Rellich compactness imply a discrete spectrum, smooth finite-dimensional kernel, and a Green operator G on the orthogonal complement of harmonic forms.

The identity

$$I = H + \Delta_{\mathrm{d}} G = H + \mathrm{d}\mathrm{d}^* G + \mathrm{d}^* \mathrm{d} G$$

is the Hodge decomposition. If α is closed, then $\mathrm{d}G\alpha = G\mathrm{d}\alpha = 0$, so

$$\alpha = H\alpha + \mathrm{d}(\mathrm{d}^* G\alpha).$$

Every de Rham class therefore has a harmonic representative. If a harmonic form is exact, say $\alpha = \mathrm{d}\beta$, then

$$\|\alpha\|^2 = (\alpha, \mathrm{d}\beta) = (\mathrm{d}^*\alpha, \beta) = 0,$$

so the representative is unique.

Checkpoint (identify each input)

(1) Which formula proves ellipticity of Δ_{d}?
(2) Why are harmonic forms smooth even if initially defined weakly?
(3) Why is compactness of the manifold relevant to finite-dimensionality?

Solution (checkpoint). The symbol identity $\sigma_{\Delta_{\mathrm{d}}}(\xi) = |\xi|^2 \,\mathrm{id}$ is invertible for $\xi \neq 0$. Elliptic regularity upgrades weak kernel elements to smooth forms. Compactness gives global elliptic estimates and compact Sobolev inclusion; on a noncompact manifold the kernel and continuous spectrum can be much larger. ◇

Exit criterion

You can compute the symbol of the Hodge Laplacian, explain the roles of ellipticity and compactness, derive the closed-form decomposition, and prove uniqueness of harmonic representatives.

Summary. Ellipticity supplies finite dimensionality, regularity, and closed range. Orthogonal decomposition then identifies de Rham cohomology with harmonic forms. Boundary conditions and completeness determine when integration by parts is legitimate.

Exercise 4.5: The circle

Find the harmonic zero- and one-forms on S^1 and verify the Hodge decomposition directly.

Exercise 4.6: Minimum norm

Prove that the harmonic representative of a closed form is the unique L^2 minimizer in its de Rham class.

Optional handout: Heat kernels and the index viewpoint

This handout is optional. No later argument in the core lectures depends on it.

For a nonnegative self-adjoint elliptic operator Δ on a compact manifold, the heat operator $\mathrm{e}^{-t\Delta}$ has a smooth kernel and trace

$$\operatorname{Tr}(\mathrm{e}^{-t\Delta}) = \sum_{j=0}^{\infty} \mathrm{e}^{-t\lambda_j}.$$

As $t \to \infty$, only the zero eigenvalue survives, so the heat operator converges to orthogonal projection onto harmonic forms. This gives a dynamic version of Hodge theory.

For the de Rham complex, take the supertrace over even and odd forms. Positive eigenvalues cancel in pairs under $\mathrm{d} + \mathrm{d}^*$, leaving

$$\operatorname{Str}(\mathrm{e}^{-t\Delta_{\mathrm{d}}}) = \sum_{k} (-1)^k \dim \mathcal{H}^k(M) = \chi(M).$$

The left side is independent of t. As $t \downarrow 0$, the heat kernel becomes local; its asymptotic expansion identifies the limiting density with the Euler form. This is the analytic core of the Gauss–Bonnet–Chern theorem.

More generally, for an elliptic complex D, the supertrace computes its Fredholm index. Applied to the Dolbeault operator it leads to Hirzebruch–Riemann–Roch. Applied to a spin Dirac operator it gives the Atiyah–Singer index theorem.

Remark (Why ellipticity is decisive)

Ellipticity simultaneously provides a discrete spectrum, smoothing of the heat operator, finite-dimensional harmonic spaces, and a local small-time expansion. Hodge theory and index theory are two uses of the same analytic structure.

Exercise 4.7: Heat equation on the circle

Write the Fourier-series heat kernel on S^1 and verify directly that it converges to the projection onto constants as $t \to \infty$.

4.2 Lecture 14: Kähler identities, the $\partial\bar\partial$-lemma, and Hard Lefschetz

Week 7 material • 90 minutes

Notation for this lecture

(X,ω) is a compact Kähler manifold of complex dimension n. The Lefschetz operator is $L(\alpha)=\omega\wedge\alpha$, its formal adjoint is Λ, and $H=[\Lambda,L]=\Lambda L-L\Lambda$ is the grading operator; on forms of total degree k, H acts by $n-k$. $[A,B]=AB-(-1)^{|A||B|}BA$ denotes the graded commutator for homogeneous operators, reducing to the ordinary commutator whenever one operator has even degree. In a holomorphic coordinate system we abbreviate $\partial_j=\partial/\partial z_j$ and $\partial_{\bar j}=\partial/\partial\bar z_j$, and write $\iota_j=\iota_{\partial_j}$ and $\iota_{\bar j}=\iota_{\partial_{\bar j}}$ for interior multiplication. The constant model used in the proof is $\omega_0=\mathrm{i}\sum_j \mathrm{d}z_j\wedge\mathrm{d}\bar z_j$; with the convention of Lecture 12 its associated Hermitian metric on $T^{1,0}$ is $h_0=2\sum_j \mathrm{d}z_j\otimes\mathrm{d}\bar z_j$. ∂^* and $\bar\partial^*$ are formal adjoints, while $\Delta'=\partial\partial^*+\partial^*\partial$ and $\Delta''=\bar\partial\bar\partial^*+\bar\partial^*\bar\partial$ are the two complex Laplacians. $\Delta_\partial=\Delta'$ and $\Delta_{\bar\partial}=\Delta''$, while $\Delta_{\mathrm{d}}=\mathrm{d}\mathrm{d}^*+\mathrm{d}^*\mathrm{d}$ is the Hodge Laplacian. $\mathcal{H}^{p,q}(X)=\ker\Delta''\cap\mathcal{A}^{p,q}(X)$ denotes harmonic forms of type (p,q). A form α is primitive if $\Lambda\alpha=0$. The primitive harmonic classes of total degree k are denoted $P^k(X,\mathbb{C})$; via Hodge theory they are the kernel of $\Lambda: H^k(X,\mathbb{C})\to H^{k-2}(X,\mathbb{C})$. The Green operator G is the inverse of the relevant Laplacian on the orthogonal complement of its harmonic kernel. $H^k_{\mathrm{dR}}(X,\mathbb{C})$ denotes complex de Rham cohomology and is abbreviated $H^k(X,\mathbb{C})$ in the Hard Lefschetz statements. We also write $\mathcal{H}^k_{\mathrm{d}}(X)=\ker(\Delta_{\mathrm{d}}:\mathcal{A}^k(X)\to\mathcal{A}^k(X))$ and $\mathcal{H}^*(X)=\bigoplus_k\mathcal{H}^k_{\mathrm{d}}(X)$. We write $b_k=\dim_{\mathbb{C}}H^k_{\mathrm{dR}}(X,\mathbb{C})$ and $h^{p,q}=\dim_{\mathbb{C}}H^{p,q}_{\bar\partial}(X)$.

Prerequisite results used below

Hodge decomposition. For the Hodge Laplacian on a compact oriented Riemannian manifold,

$$\mathcal{A}^k=\mathcal{H}^k\oplus\operatorname{Im}\mathrm{d}\oplus\operatorname{Im}\mathrm{d}^*,\qquad H^k_{\mathrm{dR}}(M,\mathbb{C})\simeq\mathcal{H}^k.$$

The three summands are mutually orthogonal and every harmonic form is smooth.

Lefschetz linear algebra. On a Hermitian vector space of complex dimension n, the operators $L=\omega\wedge$, $\Lambda=L^*$, and $H=[\Lambda,L]$ satisfy the $\mathfrak{sl}_2$ commutation relations. Every form is uniquely a sum of L^r times primitive forms. This pointwise statement is the algebraic input for Hard Lefschetz once the Kähler identities show that L preserves harmonic forms.

Why these ideas are needed

On a general complex manifold, de Rham and Dolbeault theory coexist but need not determine one another. The Kähler condition forces commutator identities relating ∂, $\bar\partial$, wedge by ω, contraction, and their adjoints. These identities imply equality of Laplacians and explain the exceptional rigidity of compact Kähler manifolds. The $\partial\bar\partial$-lemma packages the resulting exactness, while Hard Lefschetz turns multiplication by the Kähler class into cohomological isomorphisms. Both results are introduced as structural consequences of the same operator algebra, not as isolated miracles.

Plan for the lecture. 0–20 min: L, Λ, and Lefschetz linear algebra; 20–48: Kähler commutators; 48–65: three Laplacians; 65–78: the $\partial\bar\partial$-lemma; 78–90: Hard Lefschetz.

On a Kähler manifold (X^n,ω), let $L\alpha=\omega\wedge\alpha$, let Λ be its adjoint, and set $H=[\Lambda,L]$. Pointwise linear algebra gives an $\mathfrak{sl}_2$ representation.

4.2.1 The flat calculation of Akizuki–Nakano

The proof follows the operator calculation in [1, Chapter VI, §6.1]. We first work on an open set $\Omega \subset \mathbb{C}^n$ with

$$\omega_0 = \mathrm{i} \sum_{k=1}^{n} \mathrm{d}z_k \wedge \mathrm{d}\bar{z}_k.$$

This normalization makes the forms $\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$, with increasing multi-indices I, J, an orthonormal basis. If $u = \sum_{I,J} u_{I\bar{J}} \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$, then

$$\partial u = \sum_{I,J,k} \frac{\partial u_{I\bar{J}}}{\partial z_k} \mathrm{d}z_k \wedge \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J, \tag{14.1}$$

$$\bar{\partial} u = \sum_{I,J,k} \frac{\partial u_{I\bar{J}}}{\partial \bar{z}_k} \mathrm{d}\bar{z}_k \wedge \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J. \tag{14.2}$$

Lemma 4.8: Flat formal-adjoint formulas

For compactly supported forms on (Ω, ω_0),

$$\partial^* u = -\sum_k \iota_k \left(\frac{\partial u}{\partial \bar{z}_k} \right), \qquad \bar{\partial}^* u = -\sum_k \iota_{\bar{k}} \left(\frac{\partial u}{\partial z_k} \right). \tag{14.3}$$

Proof. Exterior multiplication by $\mathrm{d}z_k$ is pointwise adjoint to ι_k, and exterior multiplication by $\mathrm{d}\bar{z}_k$ is adjoint to $\iota_{\bar{k}}$. The volume form and coefficient metric are constant. Thus ordinary integration by parts moves $\partial/\partial\bar{z}_k$ to $-\partial/\partial z_k$, and moves $\partial/\partial z_k$ to $-\partial/\partial\bar{z}_k$. Apply this coefficient by coefficient in (14.1)–(14.2), and then move exterior multiplication across the pointwise inner product. This gives exactly (14.3). □

Lemma 4.9: Akizuki–Nakano identity in the constant model

If $L_0 u = \omega_0 \wedge u$, then

$$[\bar{\partial}^*, L_0] = \mathrm{i}\partial. \tag{14.4}$$

Proof. The degrees of L_0 and $\bar{\partial}^*$ are 2 and -1, so their graded commutator is the ordinary commutator. Since ω_0 has constant coefficients,

$$\begin{aligned} [\bar{\partial}^*, L_0]u &= -\sum_k \iota_{\bar{k}} \frac{\partial}{\partial z_k}(\omega_0 \wedge u) + \omega_0 \wedge \sum_k \iota_{\bar{k}} \frac{\partial u}{\partial z_k} \\ &= -\sum_k \iota_{\bar{k}} \left(\omega_0 \wedge \frac{\partial u}{\partial z_k} \right) + \omega_0 \wedge \sum_k \iota_{\bar{k}} \frac{\partial u}{\partial z_k}. \end{aligned}$$

For every vector V,

$$\iota_V(\alpha \wedge \beta) = (\iota_V \alpha) \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge \iota_V \beta. \tag{14.5}$$

Because $\deg \omega_0 = 2$, the terms in which the contraction acts on u cancel. Moreover,

$$\iota_{\bar{k}} \omega_0 = \mathrm{i} \sum_j \iota_{\bar{k}} (\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j) = -\mathrm{i}\, \mathrm{d}z_k, \tag{14.6}$$

where the minus sign comes from passing $\mathrm{d}z_j$. Hence

$$[\bar{\partial}^*, L_0]u = \mathrm{i} \sum_k \mathrm{d}z_k \wedge \frac{\partial u}{\partial z_k} = \mathrm{i}\partial u.$$

□

4.2.2 Global Kähler identities

Theorem 4.10: Kähler identities

With the standard adjoint convention,

$$[\Lambda, \partial] = \mathrm{i}\bar{\partial}^*, \qquad [\Lambda, \bar{\partial}] = -\mathrm{i}\partial^*, \qquad [L, \partial^*] = \mathrm{i}\bar{\partial}, \qquad [L, \bar{\partial}^*] = -\mathrm{i}\partial.$$

Proof. It is enough to globalize (14.4). Fix $x_0 \in X$. By Theorem 3.54, after a constant linear rescaling we may choose holomorphic coordinates centered at x_0 in which

$$\omega = \omega_0 + E, \qquad E = O(|z|^2), \qquad \partial E = O(|z|), \tag{14.7}$$

where $\omega_0 = \mathrm{i}\sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$. We explain why this second-order agreement is sufficient for the formal adjoint.

The pointwise inner product on forms and the volume density are smooth algebraic expressions in the metric and its inverse. In these coordinates their coefficient matrices equal the constant ones plus $O(|z|^2)$. Integrating by parts to compute the formal adjoint differentiates those matrices and the volume density once. Consequently $\bar{\partial}^*$ has the schematic but precise-order expansion

$$\bar{\partial}^* = \bar{\partial}_0^* + \sum_{|\alpha|=1} B_\alpha(z)\, \partial^\alpha + C(z), \tag{14.8}$$

where $B_\alpha(z) = O(|z|^2)$ and $C(z) = O(|z|)$. The matrices B_α, C act on the exterior-algebra indices. Both remainders must be recorded: differentiating a coefficient of the input form produces the first-order term, whereas differentiating the metric or density produces the zero-order term.

Now $L = L_0 + E\wedge$. The constant parts give $[\bar{\partial}_0^*, L_0] = \mathrm{i}\partial$. Every other contribution to $[\bar{\partial}^*, L]$ vanishes at x_0. Indeed, $E(x_0) = 0$ and $\partial E(x_0) = 0$, so $[\bar{\partial}_0^*, E\wedge]$ vanishes there; $B_\alpha(x_0) = 0$, including terms in which ∂^α lands on E; and $C(x_0) = 0$. Thus for every smooth form u,

$$[\bar{\partial}^*, L]u(x_0) = \mathrm{i}\partial u(x_0). \tag{14.9}$$

The point x_0 was arbitrary, so $[\bar{\partial}^*, L] = \mathrm{i}\partial$ globally.

Complex conjugation gives

$$[\partial^*, L] = -\mathrm{i}\bar{\partial}. \tag{14.10}$$

Taking formal adjoints of (14.9) and (14.10), and using $L^* = \Lambda$, gives

$$[\Lambda, \bar{\partial}] = -\mathrm{i}\partial^*, \qquad [\Lambda, \partial] = \mathrm{i}\bar{\partial}^*. \tag{14.11}$$

Reversing the order in (14.9)–(14.10) yields $[L, \bar{\partial}^*] = -\mathrm{i}\partial$ and $[L, \partial^*] = \mathrm{i}\bar{\partial}$.

□

Corollary 4.11: Equality and commutation of the Kähler Laplacians

On a Kähler manifold,

$$\Delta_\partial = \Delta_{\bar{\partial}} = \frac{1}{2}\Delta_{\mathrm{d}}. \tag{14.12}$$

Moreover Δ_{d} commutes with

$$*, \quad \partial, \quad \bar{\partial}, \quad \partial^*, \quad \bar{\partial}^*, \quad L, \quad \Lambda. \tag{14.13}$$

Proof. We keep the graded commutator throughout. Since $\partial, \bar{\partial}, \partial^*$ and $\bar{\partial}^*$ have odd degree,

$$\Delta_\partial = [\partial, \partial^*], \qquad \Delta_{\bar{\partial}} = [\bar{\partial}, \bar{\partial}^*]. \tag{14.14}$$

From $[\Lambda, \partial] = \mathrm{i}\bar{\partial}^*$ we have $\bar{\partial}^* = -\mathrm{i}[\Lambda, \partial]$, and hence

$$\Delta_{\bar{\partial}} = -\mathrm{i}[\bar{\partial}, [\Lambda, \partial]]. \tag{14.15}$$

Apply the graded Jacobi identity to $\bar{\partial}, \Lambda, \partial$. Their parities are odd, even, odd, and $[\partial, \bar{\partial}] = 0$, so Jacobi reduces to

$$-[\bar{\partial}, [\Lambda, \partial]] + [\partial, [\bar{\partial}, \Lambda]] = 0. \tag{14.16}$$

The Kähler identity $[\Lambda, \bar{\partial}] = -\mathrm{i}\partial^*$ gives $[\bar{\partial}, \Lambda] = \mathrm{i}\partial^*$. Substituting this and (14.16) into (14.15) yields

$$\Delta_{\bar{\partial}} = -\mathrm{i}[\partial, [\bar{\partial}, \Lambda]] = -\mathrm{i}[\partial, \mathrm{i}\partial^*] = [\partial, \partial^*] = \Delta_{\partial}. \tag{14.17}$$

Next we prove that the mixed terms vanish. Again using $\bar{\partial}^* = -\mathrm{i}[\Lambda, \partial]$,

$$[\partial, \bar{\partial}^*] = -\mathrm{i}[\partial, [\Lambda, \partial]]. \tag{14.18}$$

Jacobi applied to $\partial, \Lambda, \partial$ gives

$$-2[\partial, [\Lambda, \partial]] + [\Lambda, [\partial, \partial]] = 0.$$

But $[\partial, \partial] = 2\partial^2 = 0$, so (14.18) is zero. Taking adjoints gives $[\bar{\partial}, \partial^*] = 0$. Expanding $\mathrm{d} = \partial + \bar{\partial}$ and $\mathrm{d}^* = \partial^* + \bar{\partial}^*$ now gives

$$\begin{aligned} \Delta_{\mathrm{d}} &= [\partial + \bar{\partial}, \partial^* + \bar{\partial}^*] \\ &= \Delta_{\partial} + \Delta_{\bar{\partial}} + [\partial, \bar{\partial}^*] + [\bar{\partial}, \partial^*] = 2\Delta_{\partial} = 2\Delta_{\bar{\partial}}. \end{aligned}$$

The relations $[\partial, \Delta_{\partial}] = 0$ and $[\partial^*, \Delta_{\partial}] = 0$ follow directly from $\partial^2 = 0$ and the graded Jacobi identity; the barred statements are identical. Lecture 13 proved that the real Hodge Laplacian commutes with $*$. It remains to check L and Λ. Because $\partial\omega = 0$, one has $[\partial, L] = 0$. Using Jacobi and $[\partial^*, L] = -\mathrm{i}\bar{\partial}$,

$$[L, \Delta_{\partial}] = [L, [\partial, \partial^*]] = -[\partial, [\partial^*, L]] = \mathrm{i}[\partial, \bar{\partial}] = 0. \tag{14.19}$$

Equation (14.12) gives $[L, \Delta_{\mathrm{d}}] = 0$, and taking adjoints gives $[\Lambda, \Delta_{\mathrm{d}}] = 0$. □

The flat identity, its globalization by geodesic holomorphic coordinates, and the Jacobi computations above are the treatment of [1, Chapter VI, Lemma 6.3, Theorems 6.4 and 6.7, and Corollary 6.5], with the first-order remainder in (14.8) written explicitly. On a general Hermitian manifold one has only $\omega = \omega_0 + O(|z|)$; surviving first-order terms are torsion terms involving $\partial\omega$. This is exactly where the Kähler condition is used.

Theorem 4.12: Dolbeault Hodge theorem

Let X be a compact complex manifold equipped with a Hermitian metric. For every bidegree (p, q),

$$\mathcal{A}^{p,q}(X) = \mathcal{H}^{p,q}_{\bar{\partial}}(X) \oplus \operatorname{Im}\bar{\partial} \oplus \operatorname{Im}\bar{\partial}^*$$

orthogonally in L^2, where

$$\mathcal{H}^{p,q}_{\bar{\partial}}(X) = \ker(\Delta_{\bar{\partial}} : \mathcal{A}^{p,q}(X) \to \mathcal{A}^{p,q}(X)).$$

Every Dolbeault class has a unique harmonic representative; equivalently,

$$H^{p,q}_{\bar{\partial}}(X) \simeq \mathcal{H}^{p,q}_{\bar{\partial}}(X).$$

No Kähler hypothesis is needed for this theorem.

Proof. For a real covector ξ, write $\xi^{0,1} = \frac{1}{2}(\xi + \mathrm{i}\xi \circ J)$. The principal symbols of $\bar\partial$ and $\bar\partial^*$ are, up to the harmless symbol convention fixed in Lecture 13,

$$\sigma_{\bar\partial}(x,\xi)\alpha = \xi^{0,1} \wedge \alpha, \qquad \sigma_{\bar\partial^*}(x,\xi)\alpha = \iota_{(\xi^{0,1})^\sharp}\alpha.$$

Exterior multiplication and contraction satisfy the same anticommutator identity as in Lecture 13. Since $|\xi^{0,1}|^2 = \frac{1}{2}|\xi|^2$ for real ξ, it follows that

$$\sigma_{\Delta_{\bar\partial}}(x,\xi) = |\xi^{0,1}|^2 \,\mathrm{id} = \tfrac{1}{2}|\xi|^2 \,\mathrm{id}\,.$$

Thus $\Delta_{\bar\partial}$ is elliptic. It is formally self-adjoint and nonnegative, because

$$(\Delta_{\bar\partial}\alpha, \alpha)_{L^2} = \|\bar\partial\alpha\|^2_{L^2} + \|\bar\partial^*\alpha\|^2_{L^2}.$$

The spectral-theorem and Green-operator proof of the Hodge decomposition from Lecture 13 now applies verbatim with d replaced by $\bar\partial$. In particular,

$$\alpha = H_{\bar\partial}\alpha + \bar\partial\bar\partial^* G_{\bar\partial}\alpha + \bar\partial^*\bar\partial G_{\bar\partial}\alpha.$$

If $\bar\partial\alpha = 0$, then the last term vanishes and α differs from its harmonic projection by a $\bar\partial$-exact form. A harmonic $\bar\partial$-exact form is zero by integration by parts. This proves existence and uniqueness of harmonic representatives. □

Corollary 4.13: Kähler Hodge decomposition

If X is compact Kähler, then

$$H^k_{\mathrm{dR}}(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}_{\bar\partial}(X), \qquad \overline{H^{p,q}} = H^{q,p}.$$

Thus $b_k = \sum_{p+q=k} h^{p,q}$, and every odd Betti number is even.

Proof. The Kähler identities give $\Delta_{\mathrm{d}} = 2\Delta_{\bar\partial}$. Since $\Delta_{\bar\partial}$ preserves bidegree, the decomposition of a harmonic k-form into types is again harmonic:

$$\mathcal{H}^k_{\mathrm{d}}(X,\mathbb{C}) = \bigoplus_{p+q=k} \mathcal{H}^{p,q}_{\bar\partial}(X).$$

Real Hodge theory identifies the left side with de Rham cohomology, while Dolbeault Hodge theory identifies each summand with $H^{p,q}_{\bar\partial}(X)$. Complex conjugation commutes with the real Laplacian and interchanges types (p,q) and (q,p), proving $\overline{H^{p,q}} = H^{q,p}$. Taking dimensions gives the Betti-number formula. If k is odd, no pair (p,q) with $p+q=k$ has $p=q$; the summands occur in equal-dimensional conjugate pairs, so b_k is even. □

This is the compact Kähler Hodge decomposition of [1, Chapter VI, Theorem 8.5].

This rules out a Hopf surface as Kähler because $b_1 = 1$.

Theorem 4.14: The $\partial\bar\partial$-lemma

Let α be a d-closed pure-type form on a compact Kähler manifold. The following are equivalent:

$$\alpha \in \operatorname{Im}\mathrm{d}, \quad \alpha \in \operatorname{Im}\partial, \quad \alpha \in \operatorname{Im}\bar\partial, \quad \alpha \in \operatorname{Im}\partial\bar\partial.$$

In particular, a pure-type form $\alpha = \mathrm{d}\beta$ can be written $\alpha = \partial\bar\partial\gamma$.

Proof. If $\alpha = \partial\bar\partial\gamma$, then $\alpha = \mathrm{d}(\bar\partial\gamma) = \partial(\bar\partial\gamma) = \bar\partial(-\partial\gamma)$, so α is exact in each of the other three senses. We now prove the converse.

First suppose that the pure-type form α is d-exact. Its harmonic projection is zero. Let G_{d} be the Green operator for Δ_{d}. Since $\mathrm{d}\alpha = 0$, purity gives $\partial\alpha = \bar\partial\alpha = 0$. The Kähler identities and commutation of the Green operator with ∂ and $\bar\partial$ therefore give

$$\alpha = \Delta_{\mathrm{d}} G_{\mathrm{d}}\alpha = 2\Delta_{\bar\partial} G_{\mathrm{d}}\alpha = 2\bar\partial\bar\partial^{*} G_{\mathrm{d}}\alpha,$$

so α is $\bar\partial$-exact; the analogous formula with ∂ shows that it is ∂-exact.

It remains to show that a $\bar\partial$-closed, ∂-exact form is $\partial\bar\partial$-exact. Let $G_{\bar\partial}$ be the $\bar\partial$-Green operator. Exactness forces the harmonic projection of α to vanish, while $\bar\partial\alpha = 0$ gives

$$\alpha = \bar\partial\bar\partial^{*} G_{\bar\partial}\alpha.$$

Moreover $\partial\alpha = 0$ and $[G_{\bar\partial}, \partial] = 0$. Using $\bar\partial^{*} = -\mathrm{i}[\Lambda, \partial]$ we obtain

$$\bar\partial^{*} G_{\bar\partial}\alpha = \mathrm{i}\partial\Lambda G_{\bar\partial}\alpha, \qquad \alpha = -\mathrm{i}\partial\bar\partial\Lambda G_{\bar\partial}\alpha.$$

This is the desired representation. Interchanging ∂ and $\bar\partial$ proves the remaining case and completes all equivalences. □

This Green-operator form of the $\partial\bar\partial$-lemma is equivalent to [1, Chapter VI, Lemma 8.6]. Compactness supplies the Green operator, purity separates the two bidegrees in $\mathrm{d}\alpha = 0$, and the Kähler identities convert one adjoint into the missing differential.

Theorem 4.15: Hard Lefschetz

If X is compact Kähler of dimension n, then for $0 \le k \le n$,

$$L^{n-k} : H^{k}(X, \mathbb{C}) \xrightarrow{\sim} H^{2n-k}(X, \mathbb{C}).$$

Every cohomology class has a unique Lefschetz decomposition into powers of L times primitive classes.

Proof. Pointwise, L, Λ, and $H = [\Lambda, L]$ satisfy the $\mathfrak{sl}_2$ relations $[H, L] = -2L$ and $[H, \Lambda] = 2\Lambda$ in our convention, because H acts on total degree k as multiplication by $n - k$. A form of degree $k \le n$ is primitive if $\Lambda\alpha = 0$, equivalently $L^{n-k+1}\alpha = 0$. Finite-dimensional $\mathfrak{sl}_2$ representation theory gives, for $0 \le k \le n$, the orthogonal Lefschetz decomposition

$$\Lambda^{k} T_x^{*}X \otimes_{\mathbb{R}} \mathbb{C} = \bigoplus_{0 \le r \le \lfloor k/2 \rfloor} L^{r} P_x^{k-2r},$$

where, for $0 \le j \le n$,

$$P_x^{j} = \ker\Big(\Lambda : \Lambda^{j} T_x^{*}X \otimes_{\mathbb{R}} \mathbb{C} \longrightarrow \Lambda^{j-2} T_x^{*}X \otimes_{\mathbb{R}} \mathbb{C}\Big),$$

with the target understood as zero when $j < 2$, and shows that $L^{n-k} : \Lambda^{k} \to \Lambda^{2n-k}$ is an isomorphism on the corresponding irreducible strings.

The Kähler identities imply $[\Delta_{\mathrm{d}}, L] = [\Delta_{\mathrm{d}}, \Lambda] = 0$, so the finite-dimensional graded space $\mathcal{H}^{*}(X)$ of harmonic forms is stable under L, Λ, and H. It is therefore itself an $\mathfrak{sl}_2$-module. On every irreducible string generated by a primitive harmonic form of degree $j \le n$, the forms occur as

$$\alpha,\ L\alpha, \ldots, L^{n-j}\alpha.$$

The map L^{n-k} pairs the degree-k and degree-$(2n-k)$ positions in every string bijectively. Summing over the strings proves that $L^{n-k} : \mathcal{H}_{\mathrm{d}}^{k} \to \mathcal{H}_{\mathrm{d}}^{2n-k}$ is an isomorphism. The Hodge isomorphism then gives Hard Lefschetz on de Rham cohomology, and uniqueness of the irreducible-string decomposition gives the cohomological Lefschetz decomposition. □

The pointwise primitive decomposition and the cohomological conclusion are [1, Chapter VI, Corollary 5.17 and Theorem 8.17].

Warning

The $\partial\bar\partial$-lemma is not the local Dolbeault lemma. It depends on compactness and the Kähler identities. On a general compact complex manifold, a pure-type form that is d-exact and $\bar\partial$-closed need not be $\partial\bar\partial$-exact.

4.2.3 Primitive forms in low dimension

On a Kähler surface, every real $(1,1)$-form has a pointwise decomposition

$$\alpha = \alpha_0 + \frac{1}{2}(\Lambda\alpha)\omega, \qquad \Lambda\alpha_0 = 0.$$

The primitive part α_0 is orthogonal to ω. For a harmonic form the decomposition is global because L and Λ commute with the Laplacian. Hence

$$H^2(X,\mathbb{C}) = LH^0(X,\mathbb{C}) \oplus P^2(X,\mathbb{C}).$$

The Kähler class spans the first summand, and the primitive summand contains the trace-free degree-two information.

In complex dimension three, Hard Lefschetz includes the isomorphisms

$$L^3 : H^0 \xrightarrow{\sim} H^6, \qquad L^2 : H^1 \xrightarrow{\sim} H^5, \qquad L : H^2 \xrightarrow{\sim} H^4.$$

These statements imply both Poincaré duality and extra restrictions on the cup product. For example, multiplication by a Kähler class cannot annihilate a nonzero class in H^2.

4.2.4 Concrete consequences of the $\partial\bar\partial$-lemma

Let α and β be closed real $(1,1)$-forms on a compact Kähler manifold with the same de Rham class. Their difference is pure type and exact, so

$$\alpha - \beta = \mathrm{i}\partial\bar\partial\varphi = 2\pi\mathrm{d}\mathrm{d}^{\mathrm{c}}\varphi$$

for a real-valued smooth function φ, after adjusting the primitive by conjugation. Thus all Kähler forms in a fixed class are described by global Kähler potentials subject to the positivity inequality $\beta + \mathrm{i}\partial\bar\partial\varphi > 0$.

The lemma also identifies several cohomologies. A $\bar\partial$-closed (p,q)-form that is de Rham exact is $\partial\bar\partial$-exact, so the natural map from Bott–Chern classes to de Rham classes is injective in pure type. In particular, a holomorphic form on a compact Kähler manifold is closed: its $\bar\partial$ vanishes by holomorphicity, and the Kähler identity gives

$$\|\partial\alpha\|^2 \le (\Delta_\partial\alpha,\alpha) = (\Delta_{\bar\partial}\alpha,\alpha) = 0.$$

Combining Hodge symmetry with type decomposition gives

$$b_{2k+1} = 2\sum_{p<q,\ p+q=2k+1} h^{p,q},$$

so every odd Betti number is even. Hard Lefschetz gives further inequalities such as $b_k \le b_{k+2}$ for $k < n$, since multiplication by $[\omega]$ is injective in that range.

Warning

Hodge decomposition as a vector-space isomorphism is weaker than the $\partial\bar\partial$-lemma, and Poincaré duality alone is weaker than Hard Lefschetz. When using a topological obstruction to Kählerness, state exactly which consequence fails.

4.2.5 Lefschetz linear algebra in complex dimension two

At a point choose a unitary coframe θ_1, θ_2 with

$$\omega = \mathrm{i}\theta_1 \wedge \bar\theta_1 + \mathrm{i}\theta_2 \wedge \bar\theta_2.$$

A $(1,1)$-form has the decomposition

$$\alpha = \frac{\Lambda\alpha}{2}\,\omega + \alpha_0, \qquad \Lambda\alpha_0 = 0.$$

For example,

$$\mathrm{i}\theta_1 \wedge \bar\theta_1 - \mathrm{i}\theta_2 \wedge \bar\theta_2, \quad \theta_1 \wedge \bar\theta_2, \quad \theta_2 \wedge \bar\theta_1$$

are primitive. Direct wedge multiplication shows $\alpha_0 \wedge \omega = 0$, while $L^2(1) = \omega^2 \neq 0$. Thus the degree-two forms split into the one-dimensional Lefschetz direction $\mathbb{C}\omega$ and the primitive subspace.

The commutator can be checked on the vacuum 1:

$$[\Lambda, L]1 = \Lambda\omega = 2,$$

and on a top form: $[\Lambda, L]\omega^2 = -2\omega^2$. In general it acts on k-forms by $(n-k)$ with the corresponding convention, producing the finite $\mathfrak{sl}_2$ strings behind Hard Lefschetz.

Two global consequences with every integration step

If the Kähler form of a compact n-fold were exact, $\omega = \mathrm{d}\eta$, then

$$\omega^n = \mathrm{d}(\eta \wedge \omega^{n-1})$$

because $\mathrm{d}\omega = 0$. Stokes would give $\int_X \omega^n = 0$, contradicting positivity of the volume form. Therefore the Kähler class is nonzero.

For the first Betti number, Kähler Hodge decomposition gives

$$H^1(X, \mathbb{C}) = H^{1,0}(X) \oplus H^{0,1}(X).$$

Complex conjugation is an antilinear isomorphism between the two summands, so

$$b_1 = h^{1,0} + h^{0,1} = 2h^{1,0}.$$

This proves evenness and rules out every compact complex manifold with odd b_1 from being Kähler.

A concrete use of the $\partial\bar\partial$-lemma

Let ω_0 and ω_1 be cohomologous Kähler forms. Their difference $\alpha = \omega_1 - \omega_0$ is real, of type $(1,1)$, d-closed, and de Rham exact. The $\partial\bar\partial$-lemma gives a complex function f with $\alpha = \partial\bar\partial f$. Since $\overline{\partial\bar\partial f} = -\partial\bar\partial \bar f$ and α is real,

$$\partial\bar\partial\left(\frac{f - \bar f}{2}\right) = \alpha.$$

The function $\varphi = (2\pi/\mathrm{i})(f - \bar f)/2$ is real, and our normalization therefore gives

$$\omega_1 = \omega_0 + \mathrm{d}\mathrm{d}^{\mathrm{c}}\varphi.$$

The potential is unique modulo constants on a compact connected manifold: if $\mathrm{d}\mathrm{d}^{\mathrm{c}}\varphi = 0$, then φ is pluriharmonic and hence harmonic, so the maximum principle makes it constant.

Hard Lefschetz on projective space

The cohomology ring is

$$H^*(\mathbb{P}^n, \mathbb{C}) = \mathbb{C}[h]/(h^{n+1}), \qquad h = [\omega_{\mathrm{FS}}], \quad \deg h = 2.$$

In degree $2k$, the class h^k spans. The Lefschetz map is

$$L^{n-2k} : H^{2k}(\mathbb{P}^n) \longrightarrow H^{2n-2k}(\mathbb{P}^n), \qquad h^k \longmapsto h^{n-k},$$

which is an isomorphism between one-dimensional spaces. Odd cohomology vanishes. This elementary calculation is the model for the general harmonic-form theorem.

4.2.6 A dependency audit for the $\partial\bar\partial$-lemma

Let X be compact Kähler and let α be a pure-type form that is both d-closed and d-exact. Pure type and $\mathrm{d}\alpha = 0$ imply separately $\partial\alpha = 0$ and $\bar\partial\alpha = 0$. Exactness makes α orthogonal to all harmonic forms, so $H\alpha = 0$. If G is the Green operator, the Kähler equality $\Delta_{\mathrm{d}} = 2\Delta_\partial$ gives

$$\alpha = \Delta_{\mathrm{d}} G\alpha = 2\Delta_\partial G\alpha.$$

The operators G and ∂ commute. Since $\partial\alpha = 0$, we have $\partial G\alpha = 0$, and consequently

$$\alpha = 2\partial\partial^* G\alpha.$$

Now use the Kähler identity $[\Lambda, \bar\partial] = -\mathrm{i}\partial^*$, equivalently $\partial^* = \mathrm{i}[\Lambda, \bar\partial]$. Since G also commutes with $\bar\partial$ and $\bar\partial\alpha = 0$, one has $\bar\partial G\alpha = 0$. Hence

$$\begin{aligned}\partial^* G\alpha &= \mathrm{i}\big(\Lambda\bar\partial G\alpha - \bar\partial\Lambda G\alpha\big)\\ &= -\mathrm{i}\bar\partial\Lambda G\alpha,\end{aligned}$$

and therefore

$$\alpha = \partial\bar\partial\gamma, \qquad \gamma = -2\mathrm{i}\Lambda G\alpha.$$

This is the $\partial\bar\partial$-lemma. The proof uses compactness to obtain G, the Kähler metric to identify the Laplacians and supply the commutator identity, pure type to separate ∂ and $\bar\partial$, and exactness to remove the harmonic projection. Removing any one of these inputs invalidates a specific displayed step.

4.2.7 Self-study workshop: using the Kähler identities to manufacture a second primitive

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall the Lefschetz operator $L = \omega\wedge$ and its adjoint Λ, the Green operator, type decomposition, and commutation of the Kähler Laplacians. The $\partial\bar\partial$-lemma is a global theorem; the local Dolbeault lemma alone does not prove it.

A proof of one useful form of the lemma. Let α be a pure-type form on a compact Kähler manifold with $\bar\partial\alpha = 0$ and suppose $\alpha = \partial\beta$. Being ∂-exact, α is orthogonal to ∂-harmonic forms. Hodge theory for Δ_∂ gives

$$\alpha = \partial\partial^* G\alpha.$$

The Kähler identity

$$\partial^* = \mathrm{i}[\Lambda, \bar\partial]$$

and the commutation of G with $\bar\partial$ imply

$$\partial^* G\alpha = \mathrm{i}(\Lambda\bar\partial - \bar\partial\Lambda)G\alpha = -\mathrm{i}\bar\partial\Lambda G\alpha.$$

Therefore

$$\alpha = -\mathrm{i}\partial\bar\partial(\Lambda G\alpha).$$

This is the missing global primitive. The proof visibly uses compact Hodge theory, Kähler commutators, purity of type, and closedness.

The same commutator algebra yields $\Delta_{\mathrm{d}} = 2\Delta_\partial = 2\Delta_{\bar\partial}$. Consequently harmonic forms preserve bidegree, and de Rham cohomology decomposes into Dolbeault groups. The $\mathfrak{sl}_2$ relations among L, Λ, and the degree operator act on harmonic forms and give Hard Lefschetz:

$$L^{n-k} : H^k(X, \mathbb{C}) \xrightarrow{\sim} H^{2n-k}(X, \mathbb{C}).$$

Checkpoint (locate the global step)

(1) Why is a ∂-exact form orthogonal to ∂-harmonic forms?
(2) Where is $\bar\partial\alpha = 0$ used in the displayed proof?
(3) Which conclusion forces the odd Betti numbers of a compact Kähler manifold to be even?

Solution (checkpoint). If h is harmonic, then $\partial^* h = 0$, so $(\partial\beta, h) = (\beta, \partial^* h) = 0$. Closedness gives $\bar\partial G\alpha = G\bar\partial\alpha = 0$, eliminating the first commutator term. Hodge decomposition gives $H^m(X, \mathbb{C}) = \bigoplus_{p+q=m} H^{p,q}$ with complex conjugation pairing $H^{p,q}$ and $H^{q,p}$; when m is odd there are no diagonal terms $p = q$, so the dimension is even. ◇

Exit criterion

You can reconstruct the Green-operator proof of the $\partial\bar\partial$-lemma, name every Kähler identity used, and derive at least one topological consequence of Hodge decomposition or Hard Lefschetz.

Summary. The Kähler identities synchronize the de Rham, ∂, and $\bar\partial$ Laplacians. They produce Hodge type decomposition, the $\partial\bar\partial$-lemma, and Hard Lefschetz—testable topological consequences of being Kähler.

Exercise 4.16: Evenness of b_1

Using conjugation and harmonic one-forms directly, prove that b_1 is even on a compact Kähler manifold.

Exercise 4.17: A Kähler form is not exact

Prove that the Kähler form of a compact Kähler manifold cannot be de Rham exact.

Optional handout: Formality and Massey products

This handout is optional. No later argument in the core lectures depends on it.

A differential graded algebra $(A^\bullet, d)$ is formal if it is connected to its cohomology algebra $(H^\bullet(A), 0)$ by quasi-isomorphisms. The $\partial\bar\partial$-lemma implies that the de Rham algebra of a compact Kähler manifold is formal: use the subalgebra of ∂-closed forms and the maps to de Rham and $\bar\partial$ cohomology.

Formality is stronger than knowing the Betti numbers or the cup product. It implies that all higher Massey products vanish. A triple Massey product begins with classes a, b, c satisfying $ab = bc = 0$, chooses primitives $dx = ab$, $dy = bc$, and forms the secondary class represented by $ay \pm xc$. Its indeterminacy comes from the choices of primitives.

Theorem 4.18: Deligne–Griffiths–Morgan–Sullivan

Every compact Kähler manifold is formal over $\mathbb{R}$ and $\mathbb{C}$.

Proof. Work first over $\mathbb{R}$ and retain the course convention $\mathrm{d}^{\mathrm{c}} = (4\pi\mathrm{i})^{-1}(\partial - \bar{\partial})$. Let $K = \ker \mathrm{d}^{\mathrm{c}} \subset \mathcal{A}^{\bullet}(X)$. Because $\mathrm{dd}^{\mathrm{c}} + \mathrm{d}^{\mathrm{c}}\mathrm{d} = 0$, K is a differential graded subalgebra. The dd^{c}-lemma has two consequences.

First, the inclusion $(K, \mathrm{d}) \hookrightarrow (\mathcal{A}^{\bullet}, \mathrm{d})$ is a quasi-isomorphism. Indeed, every de Rham class has a harmonic representative, and a Kähler harmonic form is both d- and d^{c}-closed, proving surjectivity. If $\alpha \in K$ is d-closed and d-exact in $\mathcal{A}^{\bullet}$, then the dd^{c}-lemma gives $\alpha = \mathrm{dd}^{\mathrm{c}}\gamma = \mathrm{d}(\mathrm{d}^{\mathrm{c}}\gamma)$; its primitive $\mathrm{d}^{\mathrm{c}}\gamma$ lies in K, proving injectivity.

Second, let $H^{\bullet}_{\mathrm{d}^{\mathrm{c}}}(X) = \ker \mathrm{d}^{\mathrm{c}}/\mathrm{Im}\, \mathrm{d}^{\mathrm{c}}$ and send $\alpha \in K$ to its d^{c}-cohomology class. The induced map

$$(K, \mathrm{d}) \longrightarrow (H^{\bullet}_{\mathrm{d}^{\mathrm{c}}}(X), 0)$$

is a morphism of differential graded algebras: for $\mathrm{d}^{\mathrm{c}}\alpha = 0$, the form $\mathrm{d}\alpha$ is d-exact and d^{c}-closed, hence $\mathrm{d}\alpha = \mathrm{dd}^{\mathrm{c}}\gamma$ and is zero in d^{c}-cohomology. The same dd^{c}-lemma argument shows that this map induces an isomorphism on cohomology. We have therefore constructed a zigzag of DGA quasi-isomorphisms

$$(\mathcal{A}^{\bullet}(X), \mathrm{d}) \longleftarrow (K, \mathrm{d}) \longrightarrow (H^{\bullet}_{\mathrm{d}^{\mathrm{c}}}(X), 0).$$

The last algebra is naturally isomorphic to de Rham cohomology. This is formality over $\mathbb{R}$; tensoring with $\mathbb{C}$ proves the complex statement. □

This yields topological obstructions to Kähler structures that are invisible to Hodge number parity. Certain nilmanifolds have nonzero Massey products and hence cannot be Kähler, even when their odd Betti numbers happen to be even.

Remark (Limits of the obstruction)

Formality is necessary but not sufficient for a manifold to be Kähler. The Kähler cone, Hodge decomposition, and hard Lefschetz impose additional constraints.

Exercise 4.19: A formal algebra

Show that the de Rham algebra of a sphere is formal by constructing a quasi-isomorphism from a minimal algebra with one generator (and, in even dimension, one relation-killing generator) to differential forms.

4.3 Lecture 15: Dolbeault, Čech, and sheaf cohomology

Week 8 material • 90 minutes

Notation for this lecture

$\Gamma(X,\mathcal{F})$ is the space of global sections of a sheaf $\mathcal{F}$ and $H^q(X,\mathcal{F})$ its sheaf cohomology. A fine sheaf admits endomorphisms subordinate to every locally finite open cover and is acyclic for global sections. The sheaf Ω^p_X consists of holomorphic p-forms and $\mathcal{A}^{p,q}_X$ of smooth (p,q)-forms. Dolbeault cohomology is $H^{p,q}_{\bar\partial}(X) = \ker(\bar\partial : \mathcal{A}^{p,q} \to \mathcal{A}^{p,q+1})/\mathrm{Im}(\bar\partial : \mathcal{A}^{p,q-1} \to \mathcal{A}^{p,q})$. For a cover $\mathfrak{U} = \{U_i\}$, $\check{C}^q(\mathfrak{U},\mathcal{F})$ denotes Čech cochains; the shorter symbol $C^q(\mathfrak{U},\mathcal{F})$ has the same meaning in this lecture. The symbol δ is their coboundary; a Leray cover is one whose finite intersections are acyclic for $\mathcal{F}$. The Picard group is $\mathrm{Pic}(X) = H^1(X,\mathcal{O}_X^*)$, and $\mathrm{Pic}^0(X) = \ker(c_1 : \mathrm{Pic}(X) \to H^2(X,\mathbb{Z}))$ denotes its topologically trivial part.

Prerequisite results used below

Acyclic-resolution theorem. If $0 \to \mathcal{F} \to \mathcal{G}^0 \to \mathcal{G}^1 \to \cdots$ is an exact resolution and $H^q(X,\mathcal{G}^p) = 0$ for all $q > 0$ and $p \geq 0$, then $H^q(X,\mathcal{F})$ is naturally the degree-q cohomology of the complex of global sections $\Gamma(X,\mathcal{G}^\bullet)$.

Fine sheaves are acyclic. On a paracompact Hausdorff space, every fine sheaf $\mathcal{G}$ satisfies $H^q(X,\mathcal{G}) = 0$ for $q > 0$. Sheaves of smooth differential forms are fine because a smooth partition of unity acts by sheaf endomorphisms subordinate to the chosen cover.

Why these ideas are needed

Dolbeault cohomology begins with global differential forms, sheaf cohomology begins with derived global sections, and Čech cohomology begins with overlap data. These constructions look unrelated, yet they measure the same obstruction to gluing local holomorphic primitives. Comparing them gives a choice of computational language: solve differential equations when analysis is convenient, manipulate cocycles when a cover is explicit, and use exact sequences when the sheaf has a useful resolution. The comparison theorems justify moving between these models without changing the invariant being computed.

Plan for the lecture. 0–18 min: derived functors; 18–40: fine resolutions; 40–60: Dolbeault isomorphism; 60–78: the Čech–Dolbeault double complex; 78–90: long exact sequences and calculations.

Definition 4.20: Sheaf cohomology

The global-section functor $\Gamma(X,-)$ is left exact: a short exact sequence of sheaves

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

gives an exact sequence

$$0 \longrightarrow \Gamma(X,\mathcal{F}') \longrightarrow \Gamma(X,\mathcal{F}) \longrightarrow \Gamma(X,\mathcal{F}''),$$

but the last arrow need not be surjective. A sheaf $\mathcal{I}$ is *injective* if every morphism from a subsheaf extends across the containing sheaf. Choose an injective resolution

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I}^0 \xrightarrow{d_I} \mathcal{I}^1 \xrightarrow{d_I} \cdots .$$

The sheaf cohomology groups are

$$H^q(X,\mathcal{F}) = H^q\big(\Gamma(X,\mathcal{I}^\bullet), d_I\big).$$

The category of sheaves of abelian groups has enough injectives, and the comparison theorem for resolutions shows that the resulting groups do not depend on the chosen resolution. Thus $H^0(X,\mathcal{F}) = \Gamma(X,\mathcal{F})$, while the groups in positive degree measure the failure of global sections to preserve surjections.

Definition 4.21: Fine sheaf

A sheaf $\mathcal{F}$ is fine if, for every locally finite open cover $\{U_i\}$, there are endomorphisms $h_i : \mathcal{F} \to \mathcal{F}$ whose family is locally finite, whose supports lie in U_i, and whose locally finite sum is $\sum_i h_i = \mathrm{id}_{\mathcal{F}}$. Here the support of h_i is the complement of the largest open set on which the induced map on sections is zero. Modules over smooth functions are fine because an ordinary smooth partition of unity (ρ_i) acts by $h_i(s) = \rho_i s$. Fine sheaves are acyclic on paracompact Hausdorff spaces.

Proposition 4.22: Long exact sequence in sheaf cohomology

A short exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ induces a natural long exact sequence

$$0 \to H^0(X,\mathcal{F}') \to H^0(X,\mathcal{F}) \to H^0(X,\mathcal{F}'') \xrightarrow{\delta} H^1(X,\mathcal{F}') \to \cdots$$
$$\cdots \to H^q(X,\mathcal{F}'') \xrightarrow{\delta} H^{q+1}(X,\mathcal{F}') \to \cdots .$$

Proof. The horseshoe construction gives compatible injective resolutions whose complexes of global sections form a degreewise split short exact sequence

$$0 \longrightarrow \Gamma(X,\mathcal{I}'^{\bullet}) \longrightarrow \Gamma(X,\mathcal{I}^{\bullet}) \longrightarrow \Gamma(X,\mathcal{I}''^{\bullet}) \longrightarrow 0.$$

The usual connecting construction for complexes produces the displayed sequence. Concretely, represent a class in the quotient complex by a cocycle c, choose a cochain lift b, and observe that db maps to $dc = 0$; hence db comes from a cocycle a in the subcomplex. Set $\delta[c] = [a]$. Changing b or the representative c changes a by a coboundary, so this class is well defined.

For $q = 0$ the same mechanism has the familiar gluing interpretation. A global section of $\mathcal{F}''$ has local lifts s_i in $\mathcal{F}$; on overlaps the differences $s_i - s_j$ lie in $\mathcal{F}'$ and form a one-cocycle. Its class is $\delta(s)$, and it vanishes exactly when the local lifts can be modified to glue to one global lift. □

Proposition 4.23: Acyclic-resolution theorem

Suppose

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G}^0 \xrightarrow{d} \mathcal{G}^1 \xrightarrow{d} \cdots$$

is exact and $H^s(X,\mathcal{G}^q) = 0$ for every $s > 0$ and $q \geq 0$. Then there are natural isomorphisms

$$H^q(X,\mathcal{F}) \simeq H^q\big(\Gamma(X,\mathcal{G}^{\bullet}), d\big).$$

Proof. Put $\mathcal{Z}^q = \ker(d : \mathcal{G}^q \to \mathcal{G}^{q+1})$, with $\mathcal{Z}^0 \simeq \mathcal{F}$. Exactness gives

$$0 \longrightarrow \mathcal{Z}^q \longrightarrow \mathcal{G}^q \longrightarrow \mathcal{Z}^{q+1} \longrightarrow 0.$$

For $s > 0$, the long exact sequence and acyclicity of $\mathcal{G}^q$ give isomorphisms $H^s(X,\mathcal{Z}^{q+1}) \simeq H^{s+1}(X,\mathcal{Z}^q)$. At degree zero, the connecting map identifies, for $q \geq 1$,

$$\frac{\Gamma(X,\mathcal{Z}^q)}{d\Gamma(X,\mathcal{G}^{q-1})} \simeq H^1(X,\mathcal{Z}^{q-1}).$$

Iterating these identifications gives the asserted isomorphism in every degree; the connecting maps are natural, so the isomorphism is natural as well. □

The local Dolbeault lemma gives a fine resolution

$$0 \to \Omega_X^p \to \mathcal{A}_X^{p,0} \xrightarrow{\bar\partial} \mathcal{A}_X^{p,1} \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathcal{A}_X^{p,n} \to 0.$$

Theorem 4.24: Dolbeault isomorphism

For every complex manifold X and all $p, q \geq 0$,

$$H^q(X, \Omega_X^p) \simeq H^{p,q}_{\bar\partial}(X).$$

Proof. The inclusion $\Omega_X^p \hookrightarrow \mathcal{A}_X^{p,0}$ identifies a holomorphic p-form with a smooth $(p,0)$-form whose coefficients are annihilated by $\bar\partial$. The Dolbeault–Grothendieck lemma says that every $\bar\partial$-closed smooth (p,q)-form with $q > 0$ has a local $\bar\partial$-primitive. Hence

$$0 \longrightarrow \Omega_X^p \longrightarrow \mathcal{A}_X^{p,0} \xrightarrow{\bar\partial} \mathcal{A}_X^{p,1} \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathcal{A}_X^{p,n} \longrightarrow 0$$

is an exact sequence of sheaves. Each $\mathcal{A}_X^{p,q}$ is a module over the sheaf of smooth functions, so a smooth partition of unity acts on it; it is therefore fine and acyclic. The acyclic-resolution theorem identifies $H^q(X, \Omega_X^p)$ with the degree-q cohomology of the global complex $(\mathcal{A}^{p,\bullet}(X), \bar\partial)$, which is precisely $H^{p,q}_{\bar\partial}(X)$. □

This is the De Rham–Weil argument of [1, Chapter IV, Theorem 6.4 and formulas (6.13)–(6.15)]: local exactness supplies a resolution and partitions of unity supply acyclicity.

Theorem 4.25: Leray–Čech computation

If a cover $\mathfrak{U}$ is acyclic for $\mathcal{F}$, meaning every finite intersection has $H^q(-, \mathcal{F}) = 0$ for $q > 0$, then $\check H^q(\mathfrak{U}, \mathcal{F}) \simeq H^q(X, \mathcal{F})$. A Stein cover is acyclic for coherent sheaves.

Proof. Choose an injective resolution $0 \to \mathcal{F} \to \mathcal{I}^0 \to \mathcal{I}^1 \to \cdots$ and form the double complex

$$C^{r,s} = C^r(\mathfrak{U}, \mathcal{I}^s), \qquad D = \delta + (-1)^r d_{\mathcal{I}}.$$

Here $d_{\mathcal{I}} : \mathcal{I}^s \to \mathcal{I}^{s+1}$ is the differential of the chosen injective resolution; the sign ensures $D^2 = 0$ because the two directions commute before totalization. Compute the cohomology of its total complex in two ways. Each injective sheaf is flabby, hence its augmented Čech complex is exact. Taking horizontal cohomology first therefore leaves the complex of global sections $\Gamma(X, \mathcal{I}^\bullet)$, whose cohomology is $H^q(X, \mathcal{F})$.

Taking vertical cohomology first gives, on every intersection $U_{i_0 \dots i_r}$, the groups $H^s(U_{i_0 \dots i_r}, \mathcal{F})$. The acyclicity assumption kills every row with $s > 0$, while the row $s = 0$ is precisely the Čech complex $C^\bullet(\mathfrak{U}, \mathcal{F})$. Hence the same total cohomology is $\check H^q(\mathfrak{U}, \mathcal{F})$. The two identifications are natural and give the claimed isomorphism. If all finite intersections are Stein and $\mathcal{F}$ is coherent, Cartan's Theorem B supplies the required acyclicity. □

This is Leray's theorem in the form [1, Chapter IV, Theorem 5.17]. Notice that the hypothesis concerns every nonempty finite intersection, not only the individual members of the cover.

4.3.1 The double-complex dictionary

Consider $C^{r,s} = C^r(\mathfrak{U}, \mathcal{A}^{p,s})$ with horizontal differential $\delta : C^{r,s} \to C^{r+1,s}$ and vertical differential $(-1)^r \bar\partial : C^{r,s} \to C^{r,s+1}$. Because restriction commutes with $\bar\partial$, the unsigned operators commute;

the factor $(-1)^r$ therefore makes the total differential $D = \delta + (-1)^r\bar\partial$ satisfy $D^2 = 0$. Taking Čech cohomology first leaves only global smooth forms by fineness and produces the global Dolbeault complex. Taking Dolbeault cohomology first produces $C^r(\mathfrak{U}, \Omega_X^p)$. The two routes explain the Dolbeault isomorphism and give an explicit conversion between Čech and Dolbeault representatives.

Example 4.26: $H^1(\mathbb{P}^1, O(k))$

Cover $\mathbb{P}^1$ by $U_0, U_1 \simeq \mathbb{C}$, with overlap $\mathbb{C}^*$. A one-cocycle is a Laurent series. Subtracting the parts that extend to either chart leaves a finite block of negative powers. The result is

$$h^0(\mathcal{O}(k)) = \max\{k+1, 0\}, \qquad h^1(\mathcal{O}(k)) = \max\{-k-1, 0\}.$$

Warning

Čech cohomology for an arbitrary cover need not equal sheaf cohomology; refine and take a limit, or verify the Leray condition. Likewise, global sections do not generally preserve right exactness. Their failure is measured by H^1.

4.3.2 The explicit Čech–Dolbeault conversion

Let $\{U_i\}$ be a cover and let (g_{ij}) be a one-cocycle with values in $\mathcal{O}_X$, using the convention $g_{ij}+g_{jk} = g_{ik}$ and $g_{ji} = -g_{ij}$. Choose a smooth partition of unity (ρ_k) subordinate to the cover and put

$$b_i = \sum_k \rho_k g_{ik} \quad \text{on } U_i.$$

Each product is defined on $U_i \cap U_k$ and extends smoothly by zero across the complement of $\operatorname{supp}\rho_k$; local finiteness makes the sum smooth. The cocycle identity gives $b_i - b_j = g_{ij}$ on $U_i \cap U_j$. Therefore

$$\alpha|_{U_i} = \bar\partial b_i$$

defines one global scalar $(0,1)$-form: the local expressions agree because $\bar\partial g_{ij} = 0$. Moreover $\bar\partial\alpha = 0$. Changing the partition of unity or the cocycle representative changes α by a global $\bar\partial$-exact form.

Conversely, given a $\bar\partial$-closed $(0,1)$-form α, first refine to coordinate polydiscs, or assume the cover was chosen that way. The local Dolbeault–Grothendieck lemma gives primitives $\alpha = \bar\partial b_i$. Then $g_{ij} = b_i - b_j$ is holomorphic and satisfies the cocycle condition. The two constructions are inverse on cohomology. Higher degrees repeat this procedure along the rows and columns of the Čech–Dolbeault double complex.

4.3.3 The complete calculation for $O(k)$ on $\mathbb{P}^1$

Cover $\mathbb{P}^1$ by $U_0 \simeq \mathbb{C}_z$ and $U_\infty \simeq \mathbb{C}_w$, with $w = z^{-1}$. Choose frames so that $e_\infty = z^k e_0$. A one-cocycle is a holomorphic function on $\mathbb{C}^*$ times e_0, hence has a Laurent expansion

$$h(z) = \sum_{m\in\mathbb{Z}} a_m z^m.$$

A section on U_0 removes all terms with $m \geq 0$. A section $b(w)e_\infty = b(z^{-1})z^k e_0$ on U_∞ removes all terms with $m \leq k$. The surviving powers are therefore

$$z^{k+1}, z^{k+2}, \ldots, z^{-1} \quad \text{when } k \leq -2,$$

where the last exponent is negative. Consequently

$$h^1(\mathbb{P}^1, O(k)) = \begin{cases} 0, & k \geq -1, \\ -k-1, & k \leq -2. \end{cases}$$

Global sections are degree-k homogeneous polynomials when $k \geq 0$, so

$$h^0(\mathbb{P}^1, O(k)) = \begin{cases} k+1, & k \geq 0, \\ 0, & k < 0. \end{cases}$$

The difference $h^0 - h^1 = k + 1$ is Riemann–Roch in genus zero, and the symmetry $h^1(O(k)) = h^0(O(-k-2))$ anticipates Serre duality.

Remark (When Čech computes derived cohomology)

An arbitrary cover need not compute sheaf cohomology. It does when every finite intersection is acyclic for the sheaf, as in a Leray cover. For coherent sheaves on a Stein cover, Cartan B supplies this acyclicity. The cover is therefore part of the calculation, not harmless notation.

4.3.4 The contracting homotopy for a fine sheaf

Let $\{U_i\}$ be a locally finite cover with a subordinate partition of unity $\{\rho_i\}$. For a Čech q-cochain c, define

$$(hc)_{i_0 \dots i_{q-1}} = \sum_j \rho_j c_{j i_0 \dots i_{q-1}}.$$

Local finiteness makes the sum meaningful. A direct alternating-sum computation uses $\sum_j \rho_j = 1$ and gives

$$\delta h + h\delta = \mathrm{id}$$

in positive Čech degrees. Hence every Čech cocycle with values in a fine sheaf is a coboundary. On a paracompact Hausdorff space, sheaf cohomology is the direct limit of Čech cohomology over locally finite refinements. The contracting homotopy above exists on every such refinement, so every group in that direct system vanishes. Therefore a fine sheaf has no higher sheaf cohomology.

Explicit conversion between Čech and Dolbeault classes

Let (g_{ij}) be a holomorphic one-cocycle with $g_{ij} + g_{jk} + g_{ki} = 0$. Put

$$a_i = \sum_j \rho_j g_{ij}$$

on U_i, extending each term by zero where ρ_j vanishes. The cocycle identity gives

$$a_i - a_k = \sum_j \rho_j (g_{ij} - g_{kj}) = g_{ik}.$$

Therefore $\bar\partial a_i = \bar\partial a_k$ on overlaps, and the local forms glue to a global $(0,1)$-form

$$\alpha|_{U_i} = \bar\partial a_i.$$

It is $\bar\partial$-closed. Changing the partition or the representative cocycle changes α by a global $\bar\partial$-exact form, so $[\alpha]$ depends only on $[g]$.

Conversely, for a global $\bar\partial$-closed $(0,1)$-form α, refine the cover so that its members lie in coordinate polydiscs. The local Dolbeault–Grothendieck lemma then gives a_i with $\alpha = \bar\partial a_i$. Then

$$g_{ij} = a_i - a_j$$

is holomorphic and is a Čech cocycle. The two constructions are inverse on cohomology. This is the Dolbeault isomorphism in degree one with representatives, not merely an abstract identification.

The full $O(k)$ calculation on $\mathbb{P}^1$

Let $z = Z_1/Z_0$ on U_0 and $w = Z_0/Z_1 = z^{-1}$ on U_1. Choose frames $e_0 = Z_0^k$ and $e_1 = Z_1^k$, so

$$e_1 = z^k e_0.$$

A global section is a pair of entire functions $f_0(z), f_1(w)$ satisfying

$$f_0(z) = z^k f_1(1/z).$$

If $k \geq 0$, holomorphy at both 0 and ∞ forces f_0 to be a polynomial of degree at most k, giving the basis $1, z, \dots, z^k$. If $k < 0$, the same growth condition forces the section to vanish. Thus

$$h^0(\mathcal{O}(k)) = \max\{k+1, 0\}.$$

A one-cochain is a holomorphic function $h(z)$ on $\mathbb{C}^*$, expressed in the e_0 frame. Coboundaries have the form

$$h(z) = z^k f_1(1/z) - f_0(z),$$

with f_0, f_1 entire. In the Laurent series of h, the f_0 term removes every power z^m with $m \geq 0$, while $z^k f_1(1/z)$ removes every power with $m \leq k$. Only

$$z^{k+1}, z^{k+2}, \dots, z^{-1}$$

survive when $k \leq -2$. Therefore

$$H^1(\mathbb{P}^1, \mathcal{O}(k)) = \begin{cases} 0, & k \geq -1, \\ \operatorname{span}\{z^{k+1}, \dots, z^{-1}\}, & k \leq -2, \end{cases}$$

and $h^1 = \max\{-k-1, 0\}$.

The exponential sequence and $H^1(\mathbb{P}^1, \mathcal{O}) = H^2(\mathbb{P}^1, \mathcal{O}) = 0$ give

$$\operatorname{Pic}(\mathbb{P}^1) \simeq H^2(\mathbb{P}^1, \mathbb{Z}) \simeq \mathbb{Z}.$$

The generator is $\mathcal{O}(1)$, and the integer is the degree.

4.3.5 Why the Dolbeault complex computes sheaf cohomology

Fix p and let Ω_X^p denote the sheaf of holomorphic p-forms. The sequence of sheaves

$$0 \longrightarrow \Omega_X^p \longrightarrow \mathcal{A}_X^{p,0} \xrightarrow{\bar\partial} \mathcal{A}_X^{p,1} \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathcal{A}_X^{p,n} \longrightarrow 0$$

is exact. At degree zero, the kernel consists exactly of forms with holomorphic coefficients. In every positive antiholomorphic degree, exactness is precisely the Dolbeault–Grothendieck lemma, applied coefficientwise in a coordinate chart.

Each sheaf $\mathcal{A}_X^{p,q}$ is fine: if $\{\rho_i\}$ is a smooth partition of unity, multiplication by ρ_i is an endomorphism with support in the corresponding open set and the endomorphisms sum to the identity. The contracting-homotopy calculation above shows that fine sheaves have no higher cohomology. Thus this exact sequence is an acyclic resolution of Ω_X^p.

The abstract resolution theorem now says that sheaf cohomology is the cohomology of global sections of the resolution. Here the global-section complex is

$$0 \longrightarrow \mathcal{A}^{p,0}(X) \xrightarrow{\bar\partial} \mathcal{A}^{p,1}(X) \xrightarrow{\bar\partial} \cdots,$$

whose cohomology is the Dolbeault cohomology. Therefore

$$H^q(X, \Omega_X^p) \simeq H^{p,q}_{\bar\partial}(X).$$

The statement is global, but its proof has two distinct inputs: local exactness from Dolbeault–Grothendieck and global acyclicity from partitions of unity.

4.3.6 Self-study workshop: seeing the same cohomology class in Čech and Dolbeault form

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall Čech cocycles, partitions of unity, fine sheaves, the Dolbeault resolution, and the difference between local exactness and global exactness.

Convert a cocycle explicitly. Let (f_{ij}) be a Čech 1-cocycle of holomorphic functions for a locally finite cover (U_i), with $f_{ij} + f_{jk} = f_{ik}$, and choose a partition of unity (ρ_k). On U_i define

$$a_i = \sum_k \rho_k f_{ik}.$$

Here $\rho_k f_{ik}$ is initially defined on $U_i \cap U_k$ and is extended by zero: its support is contained in $\operatorname{supp}\rho_k \subset U_k$, so the extension is smooth across the boundary of U_k. Local finiteness makes the displayed sum finite near every point. On $U_i \cap U_j$, the cocycle identity gives

$$a_i - a_j = \sum_k \rho_k (f_{ik} - f_{jk}) = \sum_k \rho_k f_{ij} = f_{ij}.$$

The forms $\bar\partial a_i$ therefore agree on overlaps and glue to a global $\bar\partial$-closed $(0,1)$-form α. Changing the partition or representatives changes α by a global $\bar\partial$-exact form. This constructs the map

$$\check H^1(X, \mathcal{O}_X) \longrightarrow H^{0,1}_{\bar\partial}(X).$$

Conversely, if α is a global $\bar\partial$-closed $(0,1)$-form, first refine the cover to coordinate polydiscs. The local Dolbeault–Grothendieck lemma gives $\alpha = \bar\partial u_i$ on U_i. The differences $u_i - u_j$ are holomorphic and form a Čech cocycle. Changing local primitives changes it by a coboundary. The two constructions are inverse at the level of classes.

The abstract reason is that

$$0 \to \mathcal{O}_X \to \mathcal{A}^{0,0} \xrightarrow{\bar\partial} \mathcal{A}^{0,1} \to \cdots$$

is an exact resolution by fine, hence acyclic, sheaves. The coordinate construction above is the degree-one shadow of that theorem.

Checkpoint (local primitive, global obstruction)

(1) Why is a_i smooth rather than holomorphic?
(2) Why do the forms $\bar{\partial} a_i$ glue?
(3) If the resulting Dolbeault class vanishes, what does that say about (f_{ij})?

Solution (checkpoint). The partition functions ρ_k are smooth and usually not holomorphic. Since $a_i - a_j = f_{ij}$ and $\bar{\partial} f_{ij} = 0$, the local forms have zero difference. If $\alpha = \bar{\partial} u$ globally, then $a_i - u$ are holomorphic and their differences are f_{ij}, so the Čech cocycle is a coboundary. ⋄

Exit criterion

You can pass in both directions between a degree-one Čech cocycle and a Dolbeault class, verify all overlap signs, and explain the role of fine resolutions.

Summary. Dolbeault cohomology computes sheaf cohomology through a fine resolution; Čech theory makes gluing obstructions explicit on overlaps. The double complex shows that these are two directions through the same object.

Exercise 4.27: Exponential sequence and the Picard group

Derive $H^1(X, \mathcal{O}_X) \to \mathrm{Pic}(X) \xrightarrow{c_1} H^2(X, \mathbb{Z})$ from the exponential sequence and identify $\mathrm{Pic}^0(X)$.

Exercise 4.28: Explicit Čech–Dolbeault conversion

Given a one-cocycle $g_{ij} \in \mathcal{O}(U_{ij})$, use a partition of unity to construct a $(0,1)$-form representing its Dolbeault class.

Optional handout: Hypercohomology and two spectral sequences

This handout is optional. No later argument in the core lectures depends on it.

For a complex of sheaves $\mathcal{K}^\bullet$, its hypercohomology $\mathbb{H}^m(X, \mathcal{K}^\bullet)$ derives global sections and the internal differential at the same time. Resolve each $\mathcal{K}^q$ by injective or fine sheaves and take the cohomology of the resulting total complex.

Filtering the double complex in two directions gives two spectral sequences:

$$E_1^{p,q} = H^q(X, \mathcal{K}^p) \Longrightarrow \mathbb{H}^{p+q}(X, \mathcal{K}^\bullet),$$

and

$$'E_2^{p,q} = H^p(X, \mathcal{H}^q(\mathcal{K}^\bullet)) \Longrightarrow \mathbb{H}^{p+q}(X, \mathcal{K}^\bullet).$$

The first computes cohomology sheaf by sheaf; the second computes the cohomology sheaves first.

Example 4.29: The holomorphic de Rham complex

For $\mathcal{O}_X \xrightarrow{\partial} \Omega_X^1 \to \cdots \to \Omega_X^n$, the holomorphic Poincaré lemma says its cohomology sheaves are the constant sheaf $\mathbb{C}_X$ in degree zero and zero otherwise. Hence its hypercohomology is $H^*(X, \mathbb{C})$. The first spectral sequence is exactly the Frölicher spectral sequence $E_1^{p,q} = H^q(X, \Omega_X^p)$.

Hypercohomology is also the natural home for deformation complexes, variations of Hodge structure, and derived direct images. It records not only groups but the maps between them.

Exercise 4.30: A two-term complex

For $\mathcal{K}^\bullet = [\mathcal{F} \xrightarrow{u} \mathcal{G}]$ in degrees $0, 1$, write the first five terms of the long exact sequence relating $\mathbb{H}^*(\mathcal{K}^\bullet)$ to $H^*(\mathcal{F})$ and $H^*(\mathcal{G})$.

4.4 Lecture 16: Serre duality, the Frölicher spectral sequence, and curves

Week 8 material • 90 minutes

Notation for this lecture

X is a compact complex n-manifold, $K_X = \Omega_X^n$ its canonical line bundle, and Ω_X^p the sheaf of holomorphic p-forms. Scalar Serre duality pairs $H^q(X, \Omega_X^p)$ with $H^{n-q}(X, \Omega_X^{n-p})$. If L is a holomorphic line bundle in the sense of Lecture 11, then L^{-1} is its dual line bundle and $H^q(X, L)$ abbreviates the cohomology of its sheaf of holomorphic sections. We write $h^q(X, L) = \dim_{\mathbb{C}} H^q(X, L)$ and $\chi(X, L) = \sum_q (-1)^q h^q(X, L)$. For a compact connected Riemann surface, $g = h^1(X, \mathcal{O}_X)$ is the genus and $\deg L = \int_X c_1(L)$ is the degree of a line bundle. The Frölicher spectral sequence has first page $E_1^{p,q} = H^q(X, \Omega_X^p)$, differentials $d_r : E_r^{p,q} \to E_r^{p+r,q-r+1}$, and converges to $H_{\mathrm{dR}}^{p+q}(X, \mathbb{C})$. The Hodge numbers are $h^{p,q} = \dim_{\mathbb{C}} H^q(X, \Omega_X^p)$. The symbol $\mathrm{d}V$ denotes the volume form of a chosen Hermitian metric. Finally, $b_k = \dim_{\mathbb{C}} H_{\mathrm{dR}}^k(X, \mathbb{C})$ and $\chi_{\mathrm{top}}(X) = \sum_k (-1)^k b_k$.

Prerequisite results used below

Dolbeault theorem. For every p, the complex of sheaves of smooth (p, q)-forms is a fine resolution of Ω_X^p; hence $H^q(X, \Omega_X^p) \simeq H_{\bar\partial}^{p,q}(X)$.

Dolbeault Hodge theorem. After choosing a Hermitian metric on compact X, the Dolbeault Laplacian $\Delta_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$ is elliptic and self-adjoint. Every class in $H_{\bar\partial}^{p,q}(X)$ has a unique smooth harmonic representative, and these harmonic spaces are finite dimensional. No Kähler hypothesis is required for this statement.

Stokes pairing. If X is compact without boundary, α has type (p, q), and β has type $(n - p, n - q - 1)$, then $\int_X \bar\partial\alpha \wedge \beta = (-1)^{p+q+1} \int_X \alpha \wedge \bar\partial\beta$. Thus integration descends to Dolbeault cohomology and defines the pairing used in Serre duality.

Why these ideas are needed

Once cohomology classes have harmonic representatives, integration should reveal which groups are dual and how the different bidegrees fit together. Serre duality turns the integral pairing into a perfect pairing with coefficients in the canonical bundle. The Frölicher spectral sequence measures how far Dolbeault data is from already determining de Rham cohomology; its degeneration in the Kähler case explains Hodge-number symmetries. Curves provide a complete testing ground where these abstract mechanisms reduce to genus, degree, Riemann–Roch, and explicit dimensions.

Plan for the lecture. 0–22 min: the Serre pairing; 22–42: Riemann–Roch for curves; 42–62: Frölicher; 62–78: Kähler degeneration; 78–90: Hopf surfaces and non-Kähler diagnostics.

Theorem 4.31: Serre duality

Let X be a compact complex n-fold. For every p and q, the pairing

$$H^q(X, \Omega_X^p) \times H^{n-q}(X, \Omega_X^{n-p}) \longrightarrow \mathbb{C}, \qquad ([\alpha], [\beta]) \longmapsto \int_X \alpha \wedge \beta$$

is nondegenerate. Hence

$$H^q(X, \Omega_X^p)^* \simeq H^{n-q}(X, \Omega_X^{n-p}).$$

Proof. By the Dolbeault isomorphism from Lecture 15, represent the first class by a $\bar\partial$-closed form $\alpha \in \mathcal{A}^{p,q}(X)$ and the second by a $\bar\partial$-closed form $\beta \in \mathcal{A}^{n-p,n-q}(X)$. Their wedge product has type (n,n) and can be integrated. If α is replaced by $\alpha + \bar\partial\gamma$, the graded Leibniz rule gives

$$\bar\partial\gamma \wedge \beta = \bar\partial(\gamma \wedge \beta)$$

because the second term in the graded Leibniz rule contains $\bar\partial\beta = 0$. The integral of the right side vanishes by Stokes. If instead β is replaced by $\beta + \bar\partial\delta$, then $\bar\partial\alpha = 0$ and the graded Leibniz rule writes $\alpha \wedge \bar\partial\delta$, up to a degree sign, as $\bar\partial(\alpha \wedge \delta)$. Stokes again makes its integral zero. Thus the pairing is well defined in both variables.

Choose a Hermitian metric on X. The Dolbeault Hodge theorem identifies the two cohomology groups with spaces of harmonic forms. Complex conjugation followed by the Hodge star gives a conjugate-linear isometry

$$\# : \mathcal{A}^{p,q}(X) \longrightarrow \mathcal{A}^{n-p,n-q}(X), \qquad \#\alpha = *\overline{\alpha},$$

and the definition of the Hodge star gives

$$\alpha \wedge \#\alpha = |\alpha|^2 \, \mathrm{d}V.$$

We record the adjoint calculation because it is the analytic heart of the proof. If α has total degree $k = p + q$, then in real dimension $2n$ our conventions give $*^2 = (-1)^k$ on k-forms and $\bar\partial^* = - * \partial *$. Consequently

$$\# \, \bar\partial\alpha = (-1)^{k+1} \bar\partial^* \# \alpha, \qquad \# \, \bar\partial^*\alpha = (-1)^k \bar\partial \# \alpha.$$

For example,

$$\bar\partial^* \#\alpha = - * \partial * (*\overline{\alpha}) = (-1)^{k+1} * \partial\overline{\alpha} = (-1)^{k+1} \# \bar\partial\alpha,$$

and the second identity follows in the same way. When these formulas are applied to $\bar\partial\bar\partial^*\alpha$ and $\bar\partial^*\bar\partial\alpha$, the two degree signs cancel, giving

$$\# \Delta_{\bar\partial}\alpha = \Delta_{\bar\partial} \# \alpha.$$

Thus $\#$ maps harmonic forms isomorphically to harmonic forms of the complementary bidegree.

Let $0 \neq [\alpha] \in H^q(X, \Omega_X^p)$ and choose its nonzero harmonic representative. Then $\beta = \#\alpha$ is closed and

$$\int_X \alpha \wedge \beta = \|\alpha\|_{L^2}^2 > 0.$$

Thus the pairing has no left kernel. Starting with a nonzero harmonic form of type $(n-p, n-q)$ gives the same conclusion for the right kernel. Harmonic spaces on a compact manifold are finite dimensional, so the resulting injection into the dual is an isomorphism. Although a metric was used to prove nondegeneracy, the Stokes calculation shows that the pairing itself is canonical. □

This harmonic-form proof is [1, Chapter VI, Theorem 7.3]. No Kähler hypothesis is used: ellipticity of the Dolbeault Laplacian on a compact Hermitian manifold is enough.

Corollary 4.32: Serre duality for line bundles

Let L be a holomorphic line bundle on a compact complex n-fold. Then integration after the natural contraction $L \otimes L^{-1} \to \mathcal{O}_X$ gives a perfect pairing

$$H^q(X, L) \times H^{n-q}(X, K_X \otimes L^{-1}) \longrightarrow \mathbb{C}.$$

In particular, $H^q(X, L)^* \simeq H^{n-q}(X, K_X \otimes L^{-1})$.

Proof. Line bundles and their transition functions were introduced in Lecture 11. In a local holomorphic frame, $\bar\partial$ acts on the coefficient of an L-valued form, so the Dolbeault–Grothendieck lemma

proves exactness of the L-valued Dolbeault complex. Partitions of unity make its sheaves fine. Thus the proof of the Dolbeault isomorphism in Lecture 15 applies with coefficients in L.

A Hermitian metric on L is obtained by choosing local metrics and averaging them with a partition of unity. It identifies the conjugate fiber with the dual fiber, and hence combines with $*\overline{(\cdot)}$ to map harmonic L-valued $(0,q)$-forms to harmonic L^{-1}-valued $(n,n-q)$-forms. The pointwise identity after contracting the line-bundle factors is again $\alpha \wedge \#\alpha = |\alpha|^2 \mathrm{d}V$. The preceding nondegeneracy argument now applies word for word. Higher-rank bundles are postponed to Lecture 17. □

Theorem 4.33: Riemann–Roch for a compact curve

If X is a compact connected complex curve of genus g and L is a line bundle, then

$$h^0(L) - h^0(K_X \otimes L^{-1}) = \deg L + 1 - g.$$

In particular, $\deg K_X = 2g - 2$, and $H^0(X, L) = 0$ if $\deg L < 0$.

Proof (from Euler characteristic). For a divisor D, write $L(D) = L \otimes \mathcal{O}(D)$. For a point $p \in X$, the quotient in

$$0 \longrightarrow L \longrightarrow L(p) \longrightarrow L(p)|_p \longrightarrow 0$$

is a one-dimensional skyscraper sheaf. Additivity of Euler characteristic in the associated long exact sequence gives $\chi(L(p)) = \chi(L) + 1$. For an effective divisor $D = \sum_j m_j p_j$, iteration gives the exact sequence

$$0 \longrightarrow L \longrightarrow L(D) \longrightarrow L(D)|_D \longrightarrow 0,$$

where the last sheaf has length $\deg D = \sum_j m_j$, and hence $\chi(L(D)) = \chi(L) + \deg D$. Applying the same identity to $L(-D)$ and then twisting back by D handles negative coefficients as well.

We also justify the divisor description of an arbitrary line bundle without using Riemann–Roch. Fix $p \in X$. Iterating the point sequence gives $\chi(L(kp)) = \chi(L) + k$. Since the Dolbeault resolution of the line bundle $L(kp)$ on a curve has no $(0,q)$-terms for $q > 1$, this line bundle has no cohomology above degree one; hence $h^0(L(kp)) \geq \chi(L) + k > 0$ for k large. A nonzero section of $L(kp)$ is a meromorphic section of L with a pole of order at most k at p. Its divisor D identifies $L \simeq \mathcal{O}(D)$, and Poincaré–Lelong gives $\deg L = \deg D$.

Applying the Euler-characteristic identity to the positive and negative parts of D now gives

$$\chi(L) = \deg L + \chi(\mathcal{O}_X) = \deg L + 1 - g,$$

where connectedness gives $h^0(X, \mathcal{O}_X) = 1$ and the definition of genus gives $h^1(X, \mathcal{O}_X) = g$. Serre duality identifies $h^1(L) = h^0(K_X \otimes L^{-1})$, which proves the formula. Taking $L = K_X$ and using $h^0(K_X) = h^1(\mathcal{O}_X) = g$ gives $\deg K_X = 2g - 2$. Finally, a nonzero holomorphic section has an effective zero divisor of degree $\deg L$; hence no such section exists when $\deg L < 0$. □

This is the curve formula [1, Chapter VI, Theorem 10.8]; the argument above includes the Euler-characteristic bookkeeping and the existence of a meromorphic section rather than treating the divisor presentation of L as automatic.

Definition 4.34: Frölicher spectral sequence

Filter the total de Rham complex by holomorphic degree,

$$F^p \mathcal{A}^k(X) = \bigoplus_{a \geq p} \mathcal{A}^{a,k-a}(X).$$

The exterior derivative preserves this decreasing filtration because $\mathrm{d} = \partial + \bar{\partial}$. The resulting

spectral sequence is

$$E_1^{p,q} = H_{\bar\partial}^{p,q}(X) \Longrightarrow H_{\mathrm{dR}}^{p+q}(X,\mathbb{C}), \qquad d_1 = [\partial].$$

More precisely,

$$E_0^{p,q} = \mathcal{A}^{p,q}(X), \qquad d_0 = \bar\partial,$$

and the differential on the rth page has bidegree $(r, 1-r)$. Convergence means that de Rham cohomology has a decreasing filtration $F^p H_{\mathrm{dR}}^k$ for which

$$E_\infty^{p,q} \simeq F^p H_{\mathrm{dR}}^{p+q} / F^{p+1} H_{\mathrm{dR}}^{p+q}.$$

The first two obstruction steps make the pages concrete. If $\bar\partial\alpha = 0$, then

$$d_1[\alpha]_{\bar\partial} = [\partial\alpha]_{\bar\partial}.$$

If this class vanishes, choose a form β of type $(p+1, q-1)$ such that $\partial\alpha = -\bar\partial\beta$. The next component of $\mathrm{d}(\alpha+\beta)$ is $\partial\beta$, and its class in $E_2^{p+2,q-1}$ represents $d_2[\alpha]$. Thus later differentials record successive failures to complete a Dolbeault cocycle to a de Rham cocycle by adding terms of higher holomorphic degree.

For every compact complex manifold,

$$b_k = \sum_{p+q=k} \dim E_\infty^{p,q} \le \sum_{p+q=k} h^{p,q}.$$

Passing from one page to the next takes cohomology; a nonzero differential lowers the total dimensions on its source and target diagonals. Hence equality for every k is equivalent to degeneration at E_1. On a compact Kähler manifold, each Dolbeault class has a harmonic representative, and equality of the Laplacians makes that representative both ∂- and $\bar\partial$-closed. Every class is therefore a permanent cycle, while the harmonic type decomposition supplies the canonical direct sum decomposition of de Rham cohomology.

The construction and dimension inequality are [1, Chapter VI, formulas (11.1)–(11.2)]. Degeneration by itself identifies only the associated graded space of the filtered de Rham group; it does not in general choose a canonical splitting of that filtration. In the Kähler case the splitting comes from harmonic forms, not merely from the vanishing of the spectral sequence differentials.

Example 4.35: Diagnosing a Hopf surface

A Hopf surface has $b_1 = 1$. If it were Kähler, Hodge symmetry would give $b_1 = h^{1,0} + h^{0,1} = 2h^{1,0}$, a contradiction. Thus topology detects non-Kählerness. Conversely, E_1 degeneration alone does not always imply Kähler.

Warning

"Hodge decomposition" can mean the Riemannian harmonic decomposition, the type decomposition of cohomology, or E_1 degeneration of Frölicher. They synchronize naturally only in the compact Kähler case; always state which meaning is intended.

4.4.1 The Serre pairing in Dolbeault language

Represent $[\alpha] \in H_{\bar\partial}^{p,q}(X)$ and $[\beta] \in H_{\bar\partial}^{n-p,n-q}(X)$ by smooth closed forms. Wedge the form parts:

$$\langle [\alpha], [\beta] \rangle = \int_X \alpha \wedge \beta.$$

If α changes by $\bar\partial\gamma$, Stokes' theorem and $\bar\partial\beta = 0$ show that the integral is unchanged. Hodge theory identifies the dual of a harmonic (p,q)-form with a harmonic $(n-p,n-q)$-form by $\alpha \mapsto *\overline{\alpha}$, proving nondegeneracy.

For a line bundle L on a compact curve C, this becomes

$$H^1(C,L)^* \simeq H^0(C, K_C \otimes L^{-1}).$$

If $\deg L > 2g-2$, the bundle on the right has negative degree and no nonzero section, so $H^1(C,L) = 0$. Riemann–Roch then gives

$$h^0(C,L) = \deg L + 1 - g.$$

Taking $L = K_C$ gives $h^0(K_C) = g$, while taking $L = \mathcal{O}_C$ gives $h^1(\mathcal{O}_C) = g$.

4.4.2 What the Frölicher differentials measure

Filter the de Rham complex by holomorphic degree. The first page is

$$E_1^{p,q} = H^{p,q}_{\bar\partial}(X),$$

and d_1 is induced by ∂: if $\bar\partial\alpha = 0$, then $\bar\partial(\partial\alpha) = 0$ and $d_1[\alpha] = [\partial\alpha]$. A class killed by d_1 has a representative whose ∂ can be corrected by a $\bar\partial$-primitive. Later differentials record successive failures of these corrections.

On a compact Kähler manifold, every Dolbeault class has a harmonic representative and the Kähler identities make it both ∂- and $\bar\partial$-closed. Hence all differentials vanish at E_1, and

$$b_k = \sum_{p+q=k} h^{p,q}.$$

For a general compact complex manifold, the spectral sequence still converges to de Rham cohomology, but cancellations may occur. The alternating sum is unchanged:

$$\sum_{p,q} (-1)^{p+q} h^{p,q} = \sum_k (-1)^k b_k,$$

because every differential removes equal-dimensional pieces in adjacent total degrees.

Remark (Curves are automatically Kähler)

Every Hermitian form on a complex curve is a real two-form of top degree, hence is closed. Compact Riemann surfaces are therefore Kähler, their Frölicher sequence degenerates at E_1, and their Hodge diamond is determined by the genus.

4.4.3 Why the Serre pairing descends to cohomology

Let α be a (p,q)-form and β an $(n-p,n-q)$-form. Their wedge product is an (n,n)-form. If α is changed by $\bar\partial\gamma$ and $\bar\partial\beta = 0$, then

$$\bar\partial\gamma \wedge \beta = \bar\partial(\gamma \wedge \beta)$$

because the second term in the graded Leibniz rule contains $\bar\partial\beta = 0$. On compact X, Stokes gives

$$\int_X \bar\partial(\gamma \wedge \beta) = 0.$$

The same calculation in the second variable proves that the integral depends only on cohomology classes. Compactness is essential here; on a noncompact manifold a boundary-at-infinity term may remain.

Riemann–Roch on the sphere and an elliptic curve

For $X = \mathbb{P}^1$, $g = 0$ and $K_X = \mathcal{O}(-2)$. Riemann–Roch gives

$$h^0(\mathcal{O}(k)) - h^0(\mathcal{O}(-k-2)) = k+1.$$

Using the elementary fact that a negative-degree line bundle has no nonzero section recovers

$$h^0(\mathcal{O}(k)) = \max\{k+1, 0\},$$

and Serre duality gives the H^1 formula from Lecture 15.

On an elliptic curve E, $K_E \simeq \mathcal{O}_E$ and $g = 1$. For a line bundle L of positive degree d, the dual L^{-1} has negative degree, hence

$$h^1(L) = h^0(L^{-1}) = 0, \qquad h^0(L) = d.$$

For a degree-zero line bundle, a nonzero section has an effective divisor of degree zero and therefore no zeros; it trivializes the bundle. Consequently $h^0(L) = 1$ for $L \simeq \mathcal{O}_E$ and 0 for every nontrivial degree-zero L.

Computing the first Frölicher differential

Represent a class in $E_1^{p,q} = H_{\bar\partial}^{p,q}(X)$ by a form α with $\bar\partial\alpha = 0$. Since

$$\bar\partial(\partial\alpha) = -\partial(\bar\partial\alpha) = 0,$$

the form $\partial\alpha$ determines a Dolbeault class of type $(p+1, q)$. The first differential is

$$d_1[\alpha]_{\bar\partial} = [\partial\alpha]_{\bar\partial}.$$

If α is replaced by $\alpha + \bar\partial\gamma$, then

$$\partial(\alpha + \bar\partial\gamma) = \partial\alpha - \bar\partial(\partial\gamma),$$

so the resulting class is unchanged. Thus d_1 measures the first obstruction to choosing a representative that is closed under both ∂ and $\bar\partial$.

If X is compact Kähler, choose the harmonic representative of $[\alpha]$. Equality of the Laplacians makes it both ∂- and $\bar\partial$-closed, so $d_1 = 0$. The $\partial\bar\partial$-lemma eliminates all higher zigzag obstructions, giving $E_1 = E_\infty$.

Euler characteristic survives every differential

For any finite-dimensional spectral sequence, passing from a page to the next takes cohomology of a complex. The alternating sum of dimensions is unchanged by taking cohomology. Therefore

$$\sum_{p,q} (-1)^{p+q} \dim E_1^{p,q} = \sum_k (-1)^k \dim H^k_{\mathrm{dR}}(X, \mathbb{C}).$$

Substituting the first page gives

$$\sum_{p,q} (-1)^{p+q} h^{p,q} = \chi_{\mathrm{top}}(X),$$

whether or not the spectral sequence degenerates. Individual Betti numbers require control of differentials; the Euler characteristic does not.

4.4.4 Nondegeneracy of the Serre pairing via harmonic forms

Equip X with a Hermitian metric. The Hodge star and complex conjugation combine to give a conjugate-linear map

$$\mathcal{S} : \mathcal{A}^{p,q}(X) \longrightarrow \mathcal{A}^{n-p,n-q}(X), \qquad \mathcal{S}\alpha = *\overline{\alpha}.$$

By the defining identity for the Hodge star, pointwise

$$\alpha \wedge \mathcal{S}\alpha = |\alpha|^2 \, \mathrm{d}V.$$

Here $\mathrm{d}V$ denotes the Riemannian volume form; the displayed identity can be checked on the standard monomial basis in a unitary coordinate coframe.

If α has total degree k, the identities $*^2 = (-1)^k$ and $\bar\partial^* = - * \partial *$ give

$$\mathcal{S}\bar\partial\alpha = (-1)^{k+1}\bar\partial^*\mathcal{S}\alpha, \qquad \mathcal{S}\bar\partial^*\alpha = (-1)^k\bar\partial\mathcal{S}\alpha.$$

The signs cancel in the two terms of the Laplacian, so $\mathcal{S}\Delta_{\bar\partial} = \Delta_{\bar\partial}\mathcal{S}$ and $\mathcal{S}$ exchanges the kernels of the two relevant Dolbeault Laplacians. Therefore, if $0 \neq [\alpha] \in H^q(X, \Omega_X^p)$, choose its nonzero harmonic representative α and set $\beta = \mathcal{S}\alpha$. Then β is harmonic and represents a class in $H^{n-q}(X, \Omega_X^{n-p})$. Their Serre pairing is

$$\int_X \alpha \wedge \beta = \int_X |\alpha|^2 \, \mathrm{d}V > 0.$$

Thus no nonzero class in the first factor pairs trivially with every class in the second. Applying the same argument from the complementary bidegree proves nondegeneracy in the other factor. Finite-dimensionality then identifies each group with the dual of the other and completes the analytic proof of scalar Serre duality. The line-bundle form used for Riemann–Roch follows by the identical argument after contracting $L \otimes L^{-1}$.

4.4.5 Self-study workshop: Serre duality and curve calculations

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall Dolbeault representatives, the canonical line bundle, dual line bundles, Euler characteristics, and the first page $E_1^{p,q} = H^q(X, \Omega_X^p)$ of the Frölicher spectral sequence.

Check the Serre pairing by bidegree. Represent a class of $H^q(X, \Omega_X^p)$ by a (p, q)-form α and a class of $H^{n-q}(X, \Omega_X^{n-p})$ by an $(n-p, n-q)$-form β. Wedging gives an (n, n)-form:

$$([\alpha], [\beta]) \longmapsto \int_X \alpha \wedge \beta.$$

If α changes by $\bar\partial\gamma$, Stokes' theorem and $\bar\partial\beta = 0$ show that the integral is unchanged. Serre duality asserts that this pairing is perfect, not merely well defined.

On a compact Riemann surface of genus g, duality gives

$$h^1(L) = h^0(K_X \otimes L^{-1}).$$

Riemann–Roch then reads

$$h^0(L) - h^0(K_X \otimes L^{-1}) = \deg L + 1 - g.$$

If $\deg L > 2g-2$, the dual bundle has negative degree and no nonzero holomorphic section, so $h^1(L)=0$ and $h^0(L)=\deg L+1-g$. This is a complete dimension calculation from two structural theorems and one elementary fact about zeros.

The Frölicher differential $d_1 : H^q(X,\Omega_X^p) \to H^q(X,\Omega_X^{p+1})$ is induced by ∂. On a compact Kähler manifold the $\partial\bar\partial$-lemma forces all differentials to vanish and the sequence degenerates at E_1. On a general compact complex manifold this need not happen; the discrepancy measures the failure of Dolbeault groups to assemble directly into de Rham cohomology.

Checkpoint (a curve calculation)

Let X have genus g and let $\deg L = 2g$.
(1) Why is $H^1(X,L)=0$?
(2) Compute $h^0(X,L)$.
(3) What bidegree does the Serre-dual partner of a $(0,1)$ class have?

Solution (checkpoint). The bundle $K_X \otimes L^{-1}$ has degree $(2g-2)-2g=-2$, so it has no nonzero holomorphic sections; duality gives $H^1(X,L)=0$. Riemann–Roch gives $h^0(L)=2g+1-g=g+1$. Since $n=1$ and $q=1$, the dual partner has type $(1,0)$ with values in L^{-1}. ◇

Exit criterion

You can verify the Serre pairing by type and Stokes, combine it with Riemann–Roch for an explicit curve computation, and explain what degeneration of the Frölicher sequence means.

Summary. Serre duality pairs degrees q and $n-q$. Riemann–Roch converts a difference of analytic dimensions into degree and genus. The Frölicher spectral sequence measures how far Dolbeault data remain from a de Rham type decomposition.

Exercise 4.36: Line bundles of high degree

If $\deg L > 2g-2$, prove $H^1(X,L)=0$ and $h^0(L)=\deg L+1-g$.

Exercise 4.37: Euler characteristic

Prove $\chi_{\mathrm{top}}(X)=\sum_{p,q}(-1)^{p+q}h^{p,q}$ for every compact complex manifold, even when Frölicher does not degenerate at E_1.

Optional handout: Jacobians and the Abel–Jacobi map

This handout is optional. No later argument in the core lectures depends on it.

Let X be a compact Riemann surface of genus g. Its Jacobian is the complex torus

$$\mathrm{Jac}(X) = H^0(X,K_X)^*/H_1(X,\mathbb{Z}),$$

where a cycle γ acts on a holomorphic one-form by integration. Choosing a basis of holomorphic one-forms and a symplectic basis of $H_1(X,\mathbb{Z})$ gives the period-matrix description $\mathbb{C}^g/(\mathbb{Z}^g+\tau\mathbb{Z}^g)$ with $\operatorname{Im}\tau>0$.

Fix $p_0 \in X$. The Abel–Jacobi map sends

$$p \longmapsto \left(\omega \mapsto \int_{p_0}^{p}\omega\right) \mod H_1(X,\mathbb{Z}).$$

Changing the path changes the integral by a period, so the class is well defined. Extending additively gives a map from degree-zero divisors to the Jacobian. Here $\mathrm{Pic}^0(X)$ denotes holomorphic line bundles of degree zero modulo isomorphism, with tensor product as group law.

Theorem 4.38: Abel's theorem

A degree-zero divisor D is principal if and only if its Abel–Jacobi image is zero. Consequently $\mathrm{Pic}^0(X) \simeq \mathrm{Jac}(X)$.

Proof. Write $D = \sum_p n_p p$ with $\sum n_p = 0$. If $D = (f)$, the logarithmic differential df/f has residue n_p at p. Cut the surface along a symplectic basis $a_1, b_1, \dots, a_g, b_g$ to a polygon and apply the residue theorem to $(\int^z \omega)\, df/f$, where ω is holomorphic. The boundary terms are periods of ω multiplied by the integral multiples $2\pi\mathrm{i}$ coming from the change of $\log f$, while the residues give $2\pi\mathrm{i} \sum n_p \int_{p_0}^{p} \omega$. Thus the Abel–Jacobi value of D is a period, hence zero in the Jacobian.

Conversely, Riemann–Roch provides a differential of the third kind η with simple poles at the support of D and residues n_p: one may prescribe the principal parts because their total residue is zero, the only Serre-duality obstruction. Subtract a holomorphic differential so that all a-periods of η vanish. The Riemann bilinear relation then identifies its b-period vector, modulo $2\pi\mathrm{i}\mathbb{Z}^g$, with $2\pi\mathrm{i}$ times the Abel–Jacobi image of D. If that image is zero, a further integral-period normalization makes every period of η belong to $2\pi\mathrm{i}\mathbb{Z}$.

Fix a point outside the poles and define

$$f(z) = \exp\left(\int_{p_0}^{z} \eta \right).$$

Changing the path changes the integral by a multiple of $2\pi\mathrm{i}$, so f is single-valued. Near p, one has $\eta = n_p\, dz/z +$ a holomorphic form; hence $f = z^{n_p}$ times a nonvanishing holomorphic function. Therefore $(f) = D$, proving that D is principal.

Degree-zero divisors modulo principal divisors form $\mathrm{Pic}^0(X)$. The equivalence just proved identifies its kernel and image with those of the additive Abel–Jacobi map, giving $\mathrm{Pic}^0(X) \simeq \mathrm{Jac}(X)$. □

Example 4.39: Genus one

For an elliptic curve $X = \mathbb{C}/\Lambda$, the Abel–Jacobi map is an isomorphism $X \simeq \mathrm{Jac}(X)$ after choosing the origin. In higher genus it embeds X as a curve in a g-dimensional torus.

Exercise 4.40: Degree-zero line bundles

Show that the divisor $p - q$ determines the same point of the Jacobian as the difference of the Abel–Jacobi images of p and q.

Chapter 5

Hermitian Vector Bundles, Chern Curvature, and Positivity

5.1 Lecture 17: Vector bundles, connections, and Chern–Weil theory

Week 9 material • 90 minutes

Notation for this lecture

$E \to X$ is a complex vector bundle of rank r, $\mathcal{A}^k(E)$ the space of smooth E-valued k-forms, and $D : \mathcal{A}^k(E) \to \mathcal{A}^{k+1}(E)$ a connection. In a local frame $e = (e_1, \dots, e_r)$, a section is $s = e\sigma$ with a column of coefficients σ, and $Ds = e(\mathrm{d}\sigma + A\sigma)$ defines the matrix-valued connection one-form A. Under a frame change $e' = eg$, the gauge matrix is $g : U \to \mathrm{GL}(r, \mathbb{C})$. The curvature is $\Theta_D = D^2 = \mathrm{d}A + A \wedge A$; in the matrix product $A \wedge A$, matrix composition is combined with the exterior product. $D_{\mathrm{End}\,E}$ is the induced connection on endomorphisms, tr is matrix trace, and $c_k(E, D)$ and $\mathrm{ch}(E, D)$ denote the Chern–Weil forms, while $c_k(E)$ and $\mathrm{ch}(E)$ denote their de Rham classes. An invariant homogeneous polynomial of degree k is a symmetric k-linear map $P : \mathfrak{gl}(r, \mathbb{C})^k \to \mathbb{C}$ satisfying $P(g^{-1}B_1 g, \dots, g^{-1}B_k g) = P(B_1, \dots, B_k)$; when its arguments are endomorphism-valued forms, matrix multiplication is combined with the wedge product.

Prerequisite results used below

Existence of connections. Every smooth vector bundle over a paracompact manifold admits a connection. Indeed, local trivial connections can be combined with a smooth partition of unity because the Leibniz rule is preserved by an affine combination.

Covariant exterior derivative. A connection extends uniquely to $D : \mathcal{A}^k(E) \to \mathcal{A}^{k+1}(E)$ by $D(\alpha \otimes s) = \mathrm{d}\alpha \otimes s + (-1)^k \alpha \wedge Ds$. Its square is $\mathcal{A}^\bullet$-linear, hence is multiplication by an $\mathrm{End}\,E$-valued two-form Θ_D. The induced connection on $\mathrm{End}\,E$ satisfies the Bianchi identity $D_{\mathrm{End}\,E}\Theta_D = 0$.

Splitting principle (cohomological form). There is a flag bundle $p : F(E) \to X$ for which $p^* : H^*(X, \mathbb{Z}) \to H^*(F(E), \mathbb{Z})$ is injective and p^*E has a filtration whose successive quotients are line bundles. Consequently, an identity between characteristic classes may be checked after formally writing p^*E as a sum of line bundles. The classes $x_1, \dots, x_r$ of these line quotients are called the formal Chern roots; symmetric polynomials in the x_j descend uniquely to X.

Why these ideas are needed

Transition functions tell us how a vector bundle is glued, but an ordinary derivative of a section does not respect a change of frame. A connection corrects the derivative by a matrix-valued one-form so that differentiation becomes intrinsic. Curvature measures the failure of two covariant derivatives to commute and therefore records the obstruction to finding locally parallel frames. Chern–Weil theory then applies invariant polynomials to curvature and produces closed forms whose cohomology classes do not depend on the chosen connection. This is the bridge from local differential data to the topology of a bundle.

Plan for the lecture. 0–20 min: bundles and transition functions; 20–45: connections and local matrices; 45–65: curvature and Bianchi; 65–82: Chern–Weil theory; 82–90: preview of line bundles.

Lectures 13–16 treated scalar differential forms first, so the analytic and cohomological mechanisms are already visible without coefficient notation. We now introduce vector bundles from the beginning and explain how those mechanisms acquire coefficients. The order matters: a symbol such as $\bar\partial_E$ will be used only after the bundle E, its transition matrices, and the induced operator on local coefficients have been defined. Line bundles from Lecture 11 are the rank-one model; the new feature in higher rank is noncommutative matrix multiplication.

Definition 5.1: Complex vector bundle

A rank-r complex vector bundle $E \to X$ is locally $U_\alpha \times \mathbb{C}^r$. Writing a local frame as a row $e_\alpha = (e_{\alpha,1}, \dots, e_{\alpha,r})$, the frames are related by

$$e_\beta = e_\alpha g_{\alpha\beta}, \qquad g_{\alpha\beta} : U_\alpha \cap U_\beta \longrightarrow \mathrm{GL}(r,\mathbb{C}).$$

The transition matrices satisfy $g_{\alpha\alpha} = I$, $g_{\beta\alpha} = g_{\alpha\beta}^{-1}$, and $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps. Conversely, such a cocycle glues the products $U_\alpha \times \mathbb{C}^r$ to a bundle. The bundle is holomorphic when the transition matrices are holomorphic.

If $s = e_\alpha\sigma_\alpha = e_\beta\sigma_\beta$, then the coefficient columns satisfy $\sigma_\beta = g_{\alpha\beta}^{-1}\sigma_\alpha$. Thus the sheaf $\mathcal{O}(E)$ of holomorphic sections is a locally free $\mathcal{O}_X$-module. Conversely, local bases of a locally free sheaf reconstruct the bundle by these transition matrices. This is the bundle–sheaf dictionary from Lectures 8–10.

Definition 5.2: Connection

A connection on a smooth complex bundle is a $\mathbb{C}$-linear operator

$$D : \mathcal{A}^0(E) \longrightarrow \mathcal{A}^1(E), \qquad D(fs) = \mathrm{d}f \otimes s + fDs.$$

It extends uniquely to E-valued forms by the graded Leibniz rule.

Explicitly, if α is a scalar k-form and s a smooth section, then

$$D(\alpha \otimes s) = \mathrm{d}\alpha \otimes s + (-1)^k \alpha \wedge Ds.$$

The sign is forced by applying the Leibniz rule twice. It is also the sign needed for D^2 to be linear over smooth functions.

In a local frame $e = (e_1, \dots, e_r)$, write $s = e\sigma$. Then

$$Ds = e(\mathrm{d}\sigma + A\sigma), \qquad A \in \mathcal{A}^1(U, \mathrm{End}\,\mathbb{C}^r).$$

Under $e' = eg$,

$$A' = g^{-1}Ag + g^{-1}\mathrm{d}g.$$

The inhomogeneous term shows that A is not a tensor.

Definition 5.3: Curvature

The curvature is

$$\Theta_D = D^2 \in \mathcal{A}^2(X, \operatorname{End} E), \qquad \Theta = \mathrm{d}A + A \wedge A.$$

It transforms as $\Theta' = g^{-1}\Theta g$ and hence is tensorial.

Indeed, applying D twice to $e\sigma$ gives

$$D^2(e\sigma) = e\big(\mathrm{d}A + A \wedge A\big)\sigma;$$

the terms containing $\mathrm{d}\sigma$ cancel because A has degree one. Substituting $A' = g^{-1}Ag + g^{-1}\mathrm{d}g$ and $\mathrm{d}(g^{-1}) = -g^{-1}(\mathrm{d}g)g^{-1}$ gives the stated conjugation law. Thus A depends on the frame, whereas Θ_D is a global $\operatorname{End} E$-valued two-form.

Proposition 5.4: Bianchi identity

The induced connection satisfies $D_{\operatorname{End} E}\Theta = 0$. Consequently $\mathrm{d}\operatorname{tr}(\Theta^k) = 0$.

Proof. For an $\operatorname{End} E$-valued k-form Φ, the induced connection is

$$D_{\operatorname{End} E}\Phi = \mathrm{d}\Phi + A \wedge \Phi - (-1)^k \Phi \wedge A.$$

Since $\Theta = \mathrm{d}A + A \wedge A$ has degree two,

$$\begin{aligned} D_{\operatorname{End} E}\Theta &= \mathrm{d}(A \wedge A) + A \wedge (\mathrm{d}A + A \wedge A) - (\mathrm{d}A + A \wedge A) \wedge A \\ &= (\mathrm{d}A \wedge A - A \wedge \mathrm{d}A) + A \wedge \mathrm{d}A - \mathrm{d}A \wedge A = 0. \end{aligned}$$

The induced connection is a derivation, so $D_{\operatorname{End} E}(\Theta^k) = \sum_{\ell=0}^{k-1} \Theta^\ell (D\Theta)\Theta^{k-1-\ell} = 0$. Finally, trace removes the commutator with A:

$$\mathrm{d}\operatorname{tr}(\Theta^k) = \operatorname{tr}\big(D_{\operatorname{End} E}\Theta^k\big) = 0.$$

□

Theorem 5.5: Chern–Weil theory

The homogeneous components of

$$\det\left(I + \frac{\mathrm{i}}{2\pi}\Theta\right) = 1 + c_1(E, D) + \cdots + c_r(E, D)$$

are closed. Their de Rham classes are independent of D and lie in the image of integral cohomology. In particular,

$$c_1(E, D) = \frac{\mathrm{i}}{2\pi}\operatorname{tr}\Theta, \qquad \operatorname{ch}(E, D) = \operatorname{tr}\exp\left(\frac{\mathrm{i}}{2\pi}\Theta\right).$$

For example,

$$c_2(E, D) = \frac{1}{2}\left(\frac{\mathrm{i}}{2\pi}\right)^2 \left(\operatorname{tr}\Theta \wedge \operatorname{tr}\Theta - \operatorname{tr}(\Theta \wedge \Theta)\right).$$

Proof. Let P be an invariant homogeneous polynomial of degree k on $\mathfrak{gl}(r, \mathbb{C})$. The Bianchi identity and invariance of P give

$$\mathrm{d}P(\Theta, \dots, \Theta) = kP(D\Theta, \Theta, \dots, \Theta) = 0.$$

For a path $D_t = D_0 + tA$ of connections, where $A = D_1 - D_0 \in \mathcal{A}^1(\operatorname{End} E)$,

$$\Theta_t = \Theta_0 + tD_0A + t^2A \wedge A, \qquad \dot{\Theta}_t = D_tA.$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}P(\Theta_t, \dots, \Theta_t) = kP(D_tA, \Theta_t, \dots, \Theta_t) = k\mathrm{d}P(A, \Theta_t, \dots, \Theta_t),$$

where the final two expressions contain $k-1$ copies of Θ_t. Integrating from 0 to 1 shows that the forms for two connections differ by the exterior derivative of the Chern–Simons transgression form. Hence the de Rham class is connection independent. For a line bundle, the normalization $(\mathrm{i}/2\pi)\Theta$ represents the integral class obtained from its transition functions through the exponential sequence. After pulling E back to the flag bundle in the splitting principle, choose a smooth splitting of the filtration of p^*E and the direct-sum connection induced by connections on its line quotients. Connection independence allows us to compute with this particular connection. The determinant in the theorem then becomes $\prod_{j=1}^{r}(1+x_j)$, where each x_j is an integral first Chern class. Its homogeneous pieces are therefore integral; injectivity of pullback and naturality identify them with the integral Chern classes of E. □

The local connection, curvature, and first-Chern-class calculations above follow [1, Chapter V, §§1–9]; the proof separates the two logical parts of Chern–Weil theory: Bianchi gives closedness, while transgression gives connection independence.

Example 5.6: Flat bundles

If $\Theta = 0$, parallel frames exist locally and parallel transport around loops gives $\rho : \pi_1(X) \to \mathrm{GL}(r, \mathbb{C})$. Conversely, a representation defines $\widetilde{X} \times_\rho \mathbb{C}^r$. Real Chern classes of a flat bundle vanish, though integral Chern classes may be torsion.

Warning

D^2 vanishes on scalar functions but acts on bundle sections by curvature. Writing $D^2 = 0$ confuses a general connection with the exterior derivative. A connection matrix has a $g^{-1}\mathrm{d}g$ term under change of frame; its curvature does not. Tensoriality is the quickest consistency check.

5.1.1 Connections form an affine space

If D_0 and D_1 are two connections on E, then

$$(D_1 - D_0)(fs) = f(D_1 - D_0)s.$$

Their difference is therefore a tensor $A \in \mathcal{A}^1(X, \operatorname{End} E)$. Conversely, $D_0 + A$ is again a connection. Thus connections form an affine space modeled on endomorphism-valued one-forms. The curvature changes by

$$\Theta_{D_0+A} = \Theta_{D_0} + D_0A + A \wedge A.$$

The quadratic term records the noncommutativity of matrix multiplication; it vanishes for line bundles only because scalar one-forms wedge with themselves to zero.

Along a smooth path γ, parallel transport solves

$$D_{\dot\gamma}s = 0.$$

In a frame this is a linear ODE $\dot\sigma + A(\dot\gamma)\sigma = 0$. A flat connection has path-independent transport on simply connected open sets, producing parallel frames. Globally, transport around loops gives the holonomy representation of $\pi_1(X)$; flatness removes local curvature but not global monodromy.

5.1.2 Transgression in the simplest case

For line-bundle connections D_0 and $D_1 = D_0 + A$, the curvature relation simplifies to

$$\Theta_1 - \Theta_0 = \mathrm{d}A.$$

Hence

$$c_1(E, D_1) - c_1(E, D_0) = \mathrm{d}\left(\frac{\mathrm{i}}{2\pi}A\right).$$

This directly proves metric or connection independence of the first Chern class. For higher Chern forms, the path $D_t = D_0 + tA$ gives

$$P(\Theta_1) - P(\Theta_0) = \mathrm{d}\left(k\int_0^1 P(A, \Theta_t^{k-1})\,\mathrm{d}t\right)$$

for an invariant polynomial P of degree k. The expression in parentheses is the Chern–Simons transgression form.

The normalization $\mathrm{i}/(2\pi)$ is topological. For a line bundle on a closed oriented surface C,

$$\int_C \frac{\mathrm{i}}{2\pi}\Theta_D \in \mathbb{Z}.$$

Choosing a meromorphic section identifies this integer with its number of zeros minus poles. Chern–Weil theory and the divisor degree are therefore the smooth and analytic descriptions of the same class.

Remark (Gauge covariance)

The connection matrix changes inhomogeneously because it differentiates a moving frame; curvature changes by conjugation because it measures a tensorial failure of commutation. Traces of powers of curvature are invariant under conjugation, which is why they descend to global differential forms.

5.1.3 Local frames, sections, and the connection transformation law

Let $e_\alpha = (e_{\alpha,1}, \ldots, e_{\alpha,r})$ be a row of local frame vectors and suppose $e_\beta = e_\alpha g_{\alpha\beta}$. If a section is written $s = e_\alpha\sigma_\alpha = e_\beta\sigma_\beta$, then

$$\sigma_\alpha = g_{\alpha\beta}\sigma_\beta, \qquad \sigma_\beta = g_{\alpha\beta}^{-1}\sigma_\alpha.$$

On a triple overlap, equality of the two changes from γ to α gives $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$. This is precisely the cocycle condition needed for the quotient of $\bigsqcup U_\alpha \times \mathbb{C}^r$ to be a bundle.

Write

$$Ds = e_\alpha(\mathrm{d}\sigma_\alpha + A_\alpha\sigma_\alpha).$$

Using $e_\beta = e_\alpha g$ and $\sigma_\alpha = g\sigma_\beta$ gives

$$\begin{aligned} Ds &= e_\alpha\big(\mathrm{d}(g\sigma_\beta) + A_\alpha g\sigma_\beta\big) \\ &= e_\beta\big(\mathrm{d}\sigma_\beta + (g^{-1}A_\alpha g + g^{-1}\mathrm{d}g)\sigma_\beta\big). \end{aligned}$$

Therefore

$$A_\beta = g^{-1}A_\alpha g + g^{-1}\mathrm{d}g.$$

Squaring and using $\mathrm{d}(g^{-1}) = -g^{-1}(\mathrm{d}g)g^{-1}$ cancels every inhomogeneous term and yields

$$\Theta_\beta = g^{-1}\Theta_\alpha g.$$

The first formula proves that A is frame dependent; the second proves that curvature is an $\mathrm{End}\,E$-valued two-form.

Induced connections calculated on simple tensors

For $s \in \Gamma(E)$, $t \in \Gamma(F)$, and $\lambda \in \Gamma(E^*)$, define

$$D_{E\otimes F}(s \otimes t) = D_E s \otimes t + s \otimes D_F t,$$
$$(D_{E^*}\lambda)(s) = \mathrm{d}(\lambda(s)) - \lambda(D_E s).$$

Applying the connection once more and keeping the graded sign shows

$$\Theta_{E\otimes F} = \Theta_E \otimes \mathrm{id}_F + \mathrm{id}_E \otimes \Theta_F,$$
$$\Theta_{E^*} = -{}^{\mathsf{T}}\Theta_E.$$

For the determinant bundle, $\Theta_{\det E} = \operatorname{tr}\Theta_E$. Taking traces in the tensor-product and dual identities gives

$$c_1(E \otimes F) = \mathrm{rk}(F)c_1(E) + \mathrm{rk}(E)c_1(F), \qquad c_1(E^*) = -c_1(E).$$

A flat connection with nontrivial monodromy

On the trivial line bundle over $\mathbb{C}^*$ consider

$$D = \mathrm{d} + \lambda\frac{\mathrm{d}z}{z}, \qquad \lambda \in \mathbb{C}.$$

Since $\mathrm{d}(\mathrm{d}z/z) = 0$ and a scalar one-form wedges with itself to zero, $\Theta_D = 0$. A parallel section satisfies

$$\mathrm{d}s + \lambda\frac{\mathrm{d}z}{z}s = 0, \qquad s = z^{-\lambda}$$

on a chosen logarithm branch. Analytic continuation once around the origin multiplies s by $\mathrm{e}^{-2\pi\mathrm{i}\lambda}$. Thus zero curvature gives path-independent transport only on simply connected sets; globally, flatness leaves a representation of the fundamental group.

Transgression for the first two invariant polynomials

Let $D_t = D_0 + tA$. Its curvature is

$$\Theta_t = \Theta_0 + tD_0A + t^2A \wedge A, \qquad \dot\Theta_t = D_tA.$$

For the trace polynomial,

$$\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}\Theta_t = \operatorname{tr}(D_tA) = \mathrm{d}\operatorname{tr}A.$$

Hence

$$\operatorname{tr}\Theta_1 - \operatorname{tr}\Theta_0 = \mathrm{d}\operatorname{tr}A.$$

For the quadratic polynomial,

$$\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}(\Theta_t \wedge \Theta_t) = 2\operatorname{tr}(D_tA \wedge \Theta_t) = 2\mathrm{d}\operatorname{tr}(A \wedge \Theta_t),$$

where Bianchi removes $D_t\Theta_t$. Integration in t gives the explicit Chern–Simons primitive. Therefore the de Rham classes of the Chern–Weil forms do not depend on a connection, even though their local representatives do.

5.1.4 The Bianchi identity expanded in one frame

For an $\mathrm{End}\,E$-valued k-form Φ, define the covariant exterior derivative in a local frame by

$$D\Phi = \mathrm{d}\Phi + [A, \Phi], \qquad [A, \Phi] = A \wedge \Phi - (-1)^k \Phi \wedge A.$$

The bracket includes matrix multiplication as well as the wedge product. Since $\Theta = \mathrm{d}A + A \wedge A$ has degree two,

$$\begin{aligned} D\Theta &= \mathrm{d}(\mathrm{d}A + A \wedge A) + A \wedge \Theta - \Theta \wedge A \\ &= \mathrm{d}A \wedge A - A \wedge \mathrm{d}A + A \wedge \mathrm{d}A + A \wedge A \wedge A \\ &\qquad - \mathrm{d}A \wedge A - A \wedge A \wedge A = 0. \end{aligned}$$

This is the Bianchi identity. The cancellation shows why both the graded sign and the quadratic term in the curvature are necessary.

Taking the trace removes commutators, so

$$\mathrm{d}\,\mathrm{tr}(\Theta^k) = \mathrm{tr}(D\Theta^k) = k\,\mathrm{tr}((D\Theta) \wedge \Theta^{k-1}) = 0.$$

Thus the Chern–Weil forms are closed before one asks whether their cohomology classes depend on the connection. The transgression calculation above answers the latter question; Bianchi answers the former.

5.1.5 Self-study workshop: checking connections under a change of frame

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall vector-bundle transition functions, matrix-valued differential forms, the graded product rule, and conjugation of endomorphisms. Fix the column-vector convention for section coefficients before calculating.

The gauge transformation from first principles. Let $e = (e_1, \dots, e_r)$ be a local frame and write a section as $s = e\sigma$. A connection has the local form

$$D(e\sigma) = e(\mathrm{d}\sigma + A\sigma).$$

On an overlap choose $e' = eg$, so $\sigma' = g^{-1}\sigma$. Requiring the two formulas for Ds to agree gives

$$A' = g^{-1}Ag + g^{-1}\mathrm{d}g.$$

The inhomogeneous term is why a connection matrix is not a tensor. Applying D once more gives the curvature

$$\Theta = \mathrm{d}A + A \wedge A.$$

Substitution into the transformation law cancels all derivatives of g and yields

$$\Theta' = g^{-1}\Theta g.$$

Curvature is therefore an $\mathrm{End}\,E$-valued two-form.

The trace removes conjugation: $\mathrm{tr}\,\Theta' = \mathrm{tr}\,\Theta$. More generally invariant polynomials in Θ glue to global forms. The Bianchi identity $D\Theta = 0$ makes these forms closed. If D_t is a path of connections, differentiating an invariant polynomial produces an exact transgression term, so its de Rham class is independent of the chosen connection. This is the logical chain behind Chern–Weil theory:

$$\text{gauge covariance} \Longrightarrow \text{global form} \Longrightarrow \text{closed form} \Longrightarrow \text{topological class}.$$

Checkpoint (matrix order matters)

(1) Why is $g^{-1}\mathrm{d}g$ present in A' but absent in Θ'?
(2) For a line bundle, what becomes of $A \wedge A$?
(3) If $\Theta = 0$, do all integral Chern classes necessarily vanish?

Solution (checkpoint). The derivative term is forced by differentiating the changed frame; in curvature its contributions cancel against those from $A' \wedge A'$. In rank one the scalar one-form satisfies $A \wedge A = 0$. Flatness makes all real Chern–Weil classes vanish, but integral Chern classes may remain torsion because de Rham cohomology does not detect torsion. ◇

Exit criterion

Starting from $e' = eg$, you can derive the connection and curvature transformation laws in the correct matrix order and explain why invariant polynomials define connection-independent cohomology classes.

Summary. A connection differentiates sections and curvature measures the failure of second derivatives to commute. Chern–Weil theory converts invariant polynomials in curvature into topological characteristic classes. Every positivity and vanishing theorem below is built on this curvature operator.

Exercise 5.7: Direct sum, tensor product, and dual

Write the induced connections and curvatures on $E \oplus F$, $E \otimes F$, and E^*.

Exercise 5.8: Transition functions of a line bundle

Starting from $g_{\alpha\beta} \in \mathcal{O}^*(U_{\alpha\beta})$, construct the corresponding class in $H^2(X, \mathbb{Z})$ via the exponential sequence and compare it with c_1.

Optional handout: Holonomy, flat bundles, and monodromy

This handout is optional. No later argument in the core lectures depends on it.

A connection D transports vectors along a piecewise smooth path γ by solving $D_{\dot\gamma} s = 0$. Parallel transport depends on the path. Transport around loops based at x generates the holonomy group $\mathrm{Hol}_x(D) \subset \mathrm{GL}(E_x)$.

Curvature is the infinitesimal holonomy. For a small coordinate parallelogram spanned by v, w, parallel transport around its boundary is

$$\mathrm{id} - \Theta_D(v, w)\ \mathrm{Area} + O(\mathrm{diam}^3).$$

The Ambrose–Singer theorem makes this precise: the Lie algebra of the holonomy group is generated by parallel translates of curvature endomorphisms.

If D is flat, transport depends only on the homotopy class of a path and produces a representation $\rho : \pi_1(X, x) \to \mathrm{GL}(E_x)$. Conversely, $E_\rho = \widetilde{X} \times_\rho \mathbb{C}^r$ carries a canonical flat connection. Thus flat bundles are linear representations of the fundamental group expressed geometrically.

Remark (Unitary local systems)

A flat bundle admits a parallel Hermitian metric exactly when the monodromy is conjugate to a unitary representation. Such bundles have a particularly clean Hodge theory. The nonunitary

case leads to harmonic metrics and nonabelian Hodge theory.

Chern–Weil forms vanish for a flat connection, so real Chern classes vanish. Integral classes can still be torsion because de Rham cohomology does not see torsion.

Exercise 5.9: Flat line bundles on a circle

Classify flat complex line bundles with connection over S^1 by their monodromy $\lambda \in \mathbb{C}^*$. Which of them admit a parallel Hermitian metric?

5.2 Lecture 18: Chern connections, line bundles, and the first Chern class

Week 9 material • 90 minutes

Notation for this lecture

$(E, \bar\partial_E, h)$ is a holomorphic vector bundle with Hermitian metric h, always taken conjugate-linear in the first argument and linear in the second. The Chern connection is $D = D' + D''$, where $D'' = \bar\partial_E$ and D' has type $(1,0)$. In a holomorphic row frame $e = (e_1, \dots, e_r)$, a section is $e\sigma$ for a coefficient column σ, the metric matrix is $H = (h(e_\alpha, e_\beta))$, and $h(e\sigma, e\tau) = \sigma^* H \tau$. The matrix $A = H^{-1}\partial H$ is the connection matrix in the convention of this text, and $\Theta(E,h) = \bar\partial A$ is its curvature. For form-valued arguments, h is extended by combining the Hermitian pairing with the wedge product:

$$h(\alpha \otimes s, \beta \otimes t) = \bar\alpha \wedge \beta\, h(s,t).$$

This is the meaning of $\mathrm{d}h(s,t) = h(Ds,t) + h(s,Dt)$. For a line bundle frame e we write $|e|_h^2 = \mathrm{e}^{-\varphi}$. The first Chern form is $c_1(E,h) = (\mathrm{i}/2\pi)\,\mathrm{tr}\,\Theta(E,h)$. For a meromorphic section s, $\mathrm{div}(s)$ is its divisor and $[\,\mathrm{div}(s)\,]$ the associated integration current.

Prerequisite results used below

Type decomposition for connections. On a complex manifold every complex connection decomposes uniquely as $D = D' + D''$, with D' raising the first bidegree and D'' raising the second. A holomorphic structure is equivalently an operator $\bar\partial_E : \mathcal{A}^{0,0}(E) \to \mathcal{A}^{0,1}(E)$ satisfying the Leibniz rule and $\bar\partial_E^2 = 0$.

Metric-compatibility identity. A connection D preserves h precisely when $\mathrm{d}h(s,t) = h(Ds,t) + h(s,Dt)$ for all smooth sections s, t. In a holomorphic frame this linear identity uniquely determines the $(1,0)$ part once $D'' = \bar\partial_E$ is fixed; this is the key input in the Chern connection theorem.

Why these ideas are needed

A holomorphic bundle already has a $\bar\partial$ operator, while a Hermitian metric gives orthogonality and adjoints. Rather than choose an arbitrary connection, we seek one compatible with both structures. The Chern connection is uniquely forced by these requirements, so its curvature is the canonical differential representative of the bundle's complex geometry. For a line bundle the matrix formulas collapse to derivatives of a single weight, making signs and normalization completely explicit. The first Chern class then turns this local curvature computation into an integral topological invariant.

Plan for the lecture. 0–20 min: metric compatibility and type; 20–45: local formula for the Chern connection; 45–65: line-bundle curvature; 65–82: bundle-valued Poincaré–Lelong; 82–90: $\mathcal{O}(k)$.

Definition 5.10: Hermitian holomorphic vector bundle

A Hermitian holomorphic vector bundle is a holomorphic bundle E with a smooth Hermitian metric h. Throughout, h is conjugate-linear in its first argument and linear in its second. In a holomorphic row frame $e = (e_1, \dots, e_r)$ it is represented by the positive Hermitian matrix $H = (h(e_j, e_k))$, so that $h(e\sigma, e\tau) = \sigma^* H \tau$ for coefficient columns σ, τ.

Theorem 5.11: Chern connection

There is a unique connection $D = D' + \bar\partial_E$ whose $(0,1)$ part is the holomorphic structure and which satisfies

$$\mathrm{d}h(s,t) = h(Ds,t) + h(s,Dt).$$

In a holomorphic frame,

$$D' = \partial + H^{-1}\partial H, \qquad \Theta(E,h) = \bar\partial(H^{-1}\partial H).$$

Its curvature has type $(1,1)$.

Proof. In a holomorphic frame write a section as a column vector and put $D' = \partial + A$. Metric compatibility for arbitrary vectors s, t says

$$\partial(s^*Ht) = (\bar\partial s)^*Ht + s^*H(\partial t + At).$$

Expanding the left side gives

$$(\bar\partial s)^*Ht + s^*(\partial H)t + s^*H\partial t.$$

Comparison of the terms containing no derivative of s or t gives $HA = \partial H$. Hence $A = H^{-1}\partial H$, proving uniqueness. Conversely, define $D = \bar\partial + \partial + H^{-1}\partial H$. Substitution into the displayed identity and its conjugate verifies metric compatibility, so the connection exists.

Its $(0,2)$ curvature is $\bar\partial_E^2 = 0$. For $A = H^{-1}\partial H$,

$$\partial A + A \wedge A = \partial(H^{-1}\partial H) + (H^{-1}\partial H) \wedge (H^{-1}\partial H) = 0,$$

because $\partial H^{-1} = -H^{-1}(\partial H)H^{-1}$. Thus the $(2,0)$ component also vanishes, and the remaining curvature is $\Theta = \bar\partial(H^{-1}\partial H)$ of type $(1,1)$. □

This computation is the Chern-connection formula in [1, Chapter V, §12]. It also shows why both hypotheses are needed: the holomorphic structure fixes D'', and metric compatibility then determines D'.

Example 5.12: Line-bundle curvature

In a local holomorphic frame e, write $|e|_h^2 = \mathrm{e}^{-\varphi}$. Then

$$D'e = -\partial\varphi \otimes e, \qquad \Theta(L,h) = \partial\bar\partial\varphi, \qquad c_1(L,h) = \frac{\mathrm{i}}{2\pi}\Theta = \mathrm{dd^c}\varphi.$$

If $e' = ge$, then $\varphi' = \varphi - \log|g|^2$, so $\mathrm{dd^c}\varphi$ is globally defined.

Warning

Some authors use $|e|^2 = \mathrm{e}^{+\varphi}$ or define curvature with the opposite sign. Never memorize "curvature is $\pm\partial\bar\partial\varphi$" apart from both the metric exponent and curvature convention. Here $|e|^2 = \mathrm{e}^{-\varphi}$.

Theorem 5.13: Bundle-valued Poincaré–Lelong

If s is a nonzero meromorphic section of (L,h), then

$$\mathrm{dd^c}\log|s|_h^2 = [\mathrm{div}(s)] - c_1(L,h).$$

In cohomology, $[\mathrm{div}(s)] = c_1(L) \in H^2(X,\mathbb{Z})$.

Proof. Choose a local holomorphic frame e with $|e|_h^2 = \mathrm{e}^{-\varphi}$ and write $s = fe$. Then

$$\log |s|_h^2 = \log |f|^2 - \varphi.$$

The scalar Poincaré–Lelong formula gives $\mathrm{dd^c} \log |f|^2 = [\mathrm{div}(f)]$, while $\mathrm{dd^c}\varphi = c_1(L,h)$. Their difference is the asserted identity. Under a frame change, the two scalar terms change by opposite copies of $\log |g|^2$, so the local identities glue. Taking de Rham classes kills the exact left-hand side and gives $[\mathrm{div}(s)] = c_1(L)$. □

The function $\log |s|_h^2$ is locally the difference of the logarithm of a holomorphic function and the logarithm of another holomorphic function, plus a smooth weight; it is therefore locally integrable. Hence the left side is a well-defined current even across the zero and pole sets. This is the line-bundle form of [1, Chapter V, §13].

Example 5.14: $\mathcal{O}(-1)$ and $\mathcal{O}(1)$

The fiber of $\mathcal{O}_{\mathbb{P}^n}(-1)$ over $[z]$ is the line $\mathbb{C}z \subset \mathbb{C}^{n+1}$, and $\mathcal{O}(1) = \mathcal{O}(-1)^*$. The standard dual metric satisfies

$$c_1(\mathcal{O}(1), h_{\mathrm{FS}}) = \omega_{\mathrm{FS}}, \qquad c_1(\mathcal{O}(k)) = k[\omega_{\mathrm{FS}}].$$

For $k \geq 0$, $H^0(\mathbb{P}^n, \mathcal{O}(k))$ is the space of homogeneous polynomials of degree k.

Proposition 5.15: Changing the metric

If $h' = h\mathrm{e}^{-\psi}$, then

$$c_1(L,h') = c_1(L,h) + \mathrm{dd^c}\psi.$$

Thus curvature forms of one line bundle vary inside a fixed de Rham class. A Kähler potential equation is a search for a suitable metric.

Proof. In a holomorphic frame write $|e|_h^2 = \mathrm{e}^{-\varphi}$. Then $|e|_{h'}^2 = \mathrm{e}^{-(\varphi+\psi)}$, so the line-bundle curvature formula gives

$$c_1(L,h') = \mathrm{dd^c}(\varphi + \psi) = c_1(L,h) + \mathrm{dd^c}\psi.$$

Since the difference is exact, both metrics represent the same de Rham class. □

5.2.1 Normal holomorphic frames and curvature at a point

Given a Hermitian holomorphic bundle (E,h) and $x \in X$, a constant linear change first makes $H(x) = I$. A holomorphic change of frame of the form $g(z) = I + \sum_j g_j z_j$ can then arrange $\partial H(x) = 0$. In such a normal frame the Chern connection matrix vanishes at x, and

$$\Theta(E,h)_x = \bar\partial (H^{-1}\partial H)_x$$

depends only on the mixed second derivatives of the metric. Curvature is therefore the second-order obstruction to finding a local holomorphic orthonormal frame.

For a line bundle with $|e|^2 = \mathrm{e}^{-\varphi}$, multiplying e by a holomorphic unit changes φ by $-\log |g|^2$. One can kill the constant and holomorphic linear terms of φ at a point, leaving

$$\varphi(z) = \sum_{j,k} \varphi_{j\bar k}(0) z_j \bar z_k + \mathrm{Re}\, q_2(z) + O(|z|^3),$$

where q_2 is a holomorphic quadratic polynomial. Only the mixed Hermitian quadratic term contributes to $\partial\bar\partial\varphi$; a further holomorphic frame change can also remove $\mathrm{Re}\, q_2$ if desired.

5.2.2 The metric on $\mathcal{O}(1)$ in coordinates

On $U_0 \subset \mathbb{P}^n$, let e_0 be the dual of the tautological vector $(1, w_1, \dots, w_n)$. Its squared norm in $\mathcal{O}(1)$ is

$$|e_0|^2 = \frac{1}{1+|w|^2} = \mathrm{e}^{-\varphi}, \qquad \varphi = \log(1+|w|^2).$$

Hence

$$c_1(\mathcal{O}(1), h_{\mathrm{FS}}) = \mathrm{dd}^{\mathrm{c}} \log(1+|w|^2) = \omega_{\mathrm{FS}}.$$

On $U_0 \cap U_j$, the frame changes by the holomorphic unit w_j, and the potential changes by $-\log|w_j|^2$. Its dd^{c} vanishes on the overlap, so the curvature forms glue.

If P is a homogeneous polynomial of degree k, it defines a section of $\mathcal{O}(k)$. Bundle-valued Poincaré–Lelong gives

$$[P=0] = k\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}} \log |P|^2_{h_{\mathrm{FS}}^k}.$$

Integrating against $\omega_{\mathrm{FS}}^{n-1}$ shows that a degree-k hypersurface has cohomology class $k[\omega_{\mathrm{FS}}]$. The exact potential term disappears by Stokes, while its logarithmic singularity remembers the divisor.

Warning

A topologically trivial line bundle can carry a nonflat metric, but its curvature is exact. Positivity of an exact curvature form is impossible on a compact manifold: integrating its top power would be both positive and zero. Topology constrains which metric curvature signs can occur globally.

5.2.3 The Chern connection in scalar and diagonal examples

For a line bundle frame e with $|e|_h^2 = \mathrm{e}^{-\varphi}$, write $D'e = Ae$. Metric compatibility gives

$$\partial |e|_h^2 = h(D'e, e) = A\mathrm{e}^{-\varphi},$$

so $A = -\partial\varphi$. Therefore

$$\Theta = \bar\partial A = -\bar\partial\partial\varphi = \partial\bar\partial\varphi, \qquad \frac{\mathrm{i}}{2\pi}\Theta = \mathrm{dd}^{\mathrm{c}}\varphi.$$

For the trivial bundle on $\mathbb{C}^n$ with $|1|^2 = \mathrm{e}^{-|z|^2}$, this gives

$$c_1(L,h) = \mathrm{dd}^{\mathrm{c}}|z|^2 = \frac{\mathrm{i}}{2\pi}\sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar z_j > 0.$$

The de Rham class is nevertheless zero because the form is globally exact. Curvature of a metric is differential-geometric data; its cohomology class is the topological invariant.

For a rank-two bundle with a diagonal metric

$$H = \begin{pmatrix} \mathrm{e}^{-\varphi_1} & 0 \\ 0 & \mathrm{e}^{-\varphi_2} \end{pmatrix},$$

one has

$$H^{-1}\partial H = \begin{pmatrix} -\partial\varphi_1 & 0 \\ 0 & -\partial\varphi_2 \end{pmatrix}, \qquad \Theta = \begin{pmatrix} \partial\bar\partial\varphi_1 & 0 \\ 0 & \partial\bar\partial\varphi_2 \end{pmatrix}.$$

Thus an orthogonal holomorphic direct sum has block-diagonal Chern curvature.

The metric on $\mathcal{O}(-1)$ and $\mathcal{O}(1)$

On $U_0 \subset \mathbb{P}^n$ with $w_j = z_j/z_0$, the tautological frame of $\mathcal{O}(-1)$ is

$$e_0(w) = (1, w_1, \dots, w_n) \in \mathbb{C}^{n+1}.$$

The metric induced from the Euclidean norm is

$$|e_0|^2 = 1 + |w|^2 = \mathrm{e}^{-\varphi_-}, \qquad \varphi_- = -\log(1 + |w|^2).$$

Hence

$$c_1(\mathcal{O}(-1), h_-) = -\mathrm{dd}^{\mathrm{c}} \log(1 + |w|^2) = -\omega_{\mathrm{FS}}.$$

The dual frame e_0^* of $\mathcal{O}(1)$ has squared norm $(1 + |w|^2)^{-1} = \mathrm{e}^{-\varphi_+}$ with $\varphi_+ = \log(1 + |w|^2)$, so

$$c_1(\mathcal{O}(1), h_+) = \omega_{\mathrm{FS}}.$$

On $U_0 \cap U_1$, the tautological frames satisfy $e_1 = w_1^{-1} e_0$. Therefore

$$|e_1|^2 = |w_1|^{-2} |e_0|^2, \qquad \varphi_{-,1} = \varphi_{-,0} + \log |w_1|^2.$$

The added term is pluriharmonic on the overlap, so the curvature forms agree. This is the metric version of the transition-function calculation.

Chern class from logarithms of transition functions

For a line bundle with transition functions g_{ij}, choose local logarithms $g_{ij} = \mathrm{e}^{2\pi \mathrm{i} f_{ij}}$. On triple overlaps,

$$n_{ijk} = f_{ij} + f_{jk} + f_{ki} \in \mathbb{Z}.$$

The integers form a Čech two-cocycle representing $c_1(L)$. If a Hermitian metric has weights φ_i, the compatibility

$$\varphi_j = \varphi_i - \log |g_{ij}|^2$$

implies that $\mathrm{dd}^{\mathrm{c}} \varphi_i$ glue. A partition-of-unity comparison between the logarithmic cocycle and these local weights shows that the de Rham class of $\mathrm{dd}^{\mathrm{c}} \varphi_i$ is the image of (n_{ijk}) in $H^2(X, \mathbb{R})$. Thus the Čech and curvature constructions yield the same first Chern class.

A divisor calculation on $\mathbb{P}^1$

The section of $\mathcal{O}(k)$ represented by the homogeneous polynomial Z_1^k vanishes to order k at $[1 : 0]$ and nowhere else. Bundle-valued Poincaré–Lelong gives

$$[k[1 : 0]] = c_1(\mathcal{O}(k), h) + \mathrm{dd}^{\mathrm{c}} \log |Z_1^k|_h^2.$$

Integrating over $\mathbb{P}^1$ kills the exact term, and the normalization $\int_{\mathbb{P}^1} \omega_{\mathrm{FS}} = 1$ yields

$$k = \int_{\mathbb{P}^1} c_1(\mathcal{O}(k)) = k \int_{\mathbb{P}^1} \omega_{\mathrm{FS}}.$$

This checks simultaneously the divisor degree, curvature normalization, and tensor power formula.

5.2.4 Checking the Chern connection under a holomorphic frame change

Let e be a holomorphic row frame, let $H = (h(e_j, e_k))$, and write the Chern connection matrix as

$$A = H^{-1}\partial H.$$

If $e' = eg$ with g holomorphic, then the metric matrix in the new frame is $H' = g^*Hg$. Since the entries of g^* are antiholomorphic, $\partial g^* = 0$, and therefore

$$\begin{aligned}(H')^{-1}\partial H' &= g^{-1}H^{-1}(g^*)^{-1}\partial(g^*Hg)\\ &= g^{-1}H^{-1}(\partial H)g + g^{-1}\partial g\\ &= g^{-1}Ag + g^{-1}\partial g.\end{aligned}$$

This is exactly the transformation law of a connection. It proves that the local formula $H^{-1}\partial H$ defines one global connection rather than unrelated matrices in different frames.

Because A has type $(1,0)$ and the $(0,1)$ part of the Chern connection is $\bar\partial$, its curvature can be written

$$\Theta = \bar\partial A$$

in a holomorphic frame; the $(2,0)$ term $\partial A + A \wedge A$ vanishes by differentiating $H^{-1}H = I$. Applying the connection transformation law and using $\bar\partial g = 0$ gives

$$\Theta' = g^{-1}\Theta g.$$

Hence $\operatorname{tr}\Theta$ is frame independent, while for a line bundle the scalar curvature itself is frame independent. This calculation is the bridge from the metric matrix to the global first Chern form.

5.2.5 Self-study workshop: deriving the Chern connection from the metric

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall holomorphic frames, Hermitian matrices, type decomposition of a connection, and Lecture 12's local potentials. The Chern connection is characterized simultaneously by metric compatibility and by $D'' = \bar\partial$.

Solve the compatibility equation. In a holomorphic frame let the metric matrix be $H = (h(e_\alpha, e_\beta))$. Write the $(1,0)$ connection matrix as A. Metric compatibility and the condition $D'' = \bar\partial$ give

$$\partial H = HA, \qquad A = H^{-1}\partial H.$$

The curvature has type $(1,1)$:

$$\Theta = \bar\partial(H^{-1}\partial H).$$

For a line bundle with $|e|_h^2 = \mathrm{e}^{-\varphi}$,

$$A = -\partial\varphi, \qquad \Theta = \partial\bar\partial\varphi, \qquad c_1(L,h) = \frac{\mathrm{i}}{2\pi}\Theta = \mathrm{dd^c}\varphi.$$

Thus positivity of a line bundle metric is exactly strict plurisubharmonicity of its local weight.

For $\mathcal{O}(1) \to \mathbb{P}^n$, the standard metric has affine weight

$$\varphi(z) = \log(1 + |z|^2).$$

Its curvature is the Fubini–Study form. On overlaps the weights change by $\log|g_{\alpha\beta}|^2$ for a nonvanishing holomorphic transition function, whose dd^{c} vanishes. Hence the curvature glues even though no global potential exists. Integrating over a projective line gives

$$\int_{\mathbb{P}^1} c_1(\mathcal{O}(1)) = 1,$$

which fixes the sign and normalization.

Checkpoint (a sign ledger)

(1) If the metric is written $|e|^2 = \mathrm{e}^{-\varphi}$, what is the connection one-form?
(2) Why does $\partial\bar\partial\varphi$ have the sign shown above?
(3) What changes for the dual metric on L^*?

Solution (checkpoint). The form is $A = H^{-1}\partial H = -\partial\varphi$. Since $\bar\partial\partial = -\partial\bar\partial$, one gets $\bar\partial(-\partial\varphi) = \partial\bar\partial\varphi$. The dual metric has weight $-\varphi$, so its curvature and first Chern form are the negatives of those of L. ◇

Exit criterion

You can derive the Chern connection rather than quote it, convert a metric weight into curvature, check its frame invariance, and normalize $c_1(\mathcal{O}(1))$ on a projective line.

Summary. The holomorphic structure and Hermitian metric uniquely determine the Chern connection. For line bundles, curvature is the complex Hessian of the weight. The first Chern class unifies divisors, curvature, and integral topology.

Exercise 5.16: Curvature of $\mathcal{O}(1)$

On $U_0 \subset \mathbb{P}^n$, use the frame $(1, w)$ of $\mathcal{O}(-1)$, compute the dual metric, and recover $c_1(\mathcal{O}(1)) = \omega_{\mathrm{FS}}$.

Exercise 5.17: A curved metric on the trivial line bundle

On $\mathbb{C}^n \times \mathbb{C}$, set $|1|_h^2 = \mathrm{e}^{-|z|^2}$. Compute the curvature and explain how a topologically trivial line bundle can have a nonflat metric.

Optional handout: The Chern character and Riemann–Roch

This handout is optional. No later argument in the core lectures depends on it.

The total Chern class is multiplicative in short exact sequences, while the Chern character is additive:

$$c(E) = \det\left(I + \frac{\mathrm{i}}{2\pi}\Theta_E\right), \qquad \mathrm{ch}(E) = \mathrm{tr}\exp\left(\frac{\mathrm{i}}{2\pi}\Theta_E\right).$$

Under the splitting principle, if $x_1, \dots, x_r$ are formal Chern roots, then

$$c(E) = \prod_j (1 + x_j), \qquad \mathrm{ch}(E) = \sum_j \mathrm{e}^{x_j}.$$

Hence

$$\mathrm{ch}(E) = r + c_1(E) + \tfrac{1}{2}(c_1(E)^2 - 2c_2(E)) + \cdots.$$

The Hirzebruch–Riemann–Roch theorem expresses the holomorphic Euler characteristic as

$$\chi(X,E)=\sum_q (-1)^q h^q(X,E)=\int_X \operatorname{ch}(E)\operatorname{td}(T^{1,0}X),$$

where the Todd class has expansion $\operatorname{td}(T^{1,0}X)=1+\frac{1}{2}c_1(T^{1,0}X)+\frac{1}{12}(c_1(T^{1,0}X)^2+c_2(T^{1,0}X))+\cdots$.

Example 5.18: A line bundle on a surface

For a line bundle L on a smooth compact complex surface,

$$\chi(L)=\chi(\mathcal{O}_X)+\frac{1}{2}c_1(L)\cdot(c_1(L)-c_1(K_X)).$$

This is the formula used in the capstone lecture. If L is sufficiently positive, Kodaira vanishing turns the Euler characteristic into the actual number of sections.

The Grothendieck–Riemann–Roch theorem extends this to a proper holomorphic map $f:X\to Y$ by comparing $\operatorname{ch}(f_!E)$ with the fiber integral of $\operatorname{ch}(E)\operatorname{td}(T_f)$.

Exercise 5.19: Projective space

Use $c_1(T_{\mathbb{P}^1})=2[\omega_{\mathrm{FS}}]$ to recover $\chi(\mathbb{P}^1,\mathcal{O}(k))=k+1$ from Riemann–Roch.

5.3 Lecture 19: Griffiths and Nakano positivity; curvature formulas

Week 10 material • 90 minutes

Notation for this lecture

$E \to X$ is a Hermitian holomorphic bundle of rank r over an n-fold. At a chosen point, in holomorphic coordinates and a frame orthonormal at that point, our coefficient convention is

$$\mathrm{i}\Theta(E,h) = \mathrm{i}\sum_{j,k=1}^{n}\sum_{\alpha,\beta=1}^{r} R_{j\bar{k}\alpha\bar{\beta}}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k \otimes e^*_\alpha \otimes e_\beta.$$

The Hermitian array $R_{j\bar{k}\alpha\bar{\beta}}$ is called the curvature tensor in these coordinates. Its Griffiths quadratic form is

$$Q_{\mathrm{Gr}}(\xi, v) = \sum R_{j\bar{k}\alpha\bar{\beta}}\xi_j\bar{\xi}_k v_\alpha \bar{v}_\beta,$$

and its Nakano quadratic form is $Q_{\mathrm{Nak}}(u) = \sum R_{j\bar{k}\alpha\bar{\beta}}u_{j\alpha}\bar{u}_{k\beta}$. Griffiths positivity tests the quadratic form only on decomposable tensors $\xi \otimes v$ with $\xi \in T^{1,0}_x X$ and $v \in E_x$. Nakano positivity tests it on every tensor $u = (u_{j\alpha}) \in T^{1,0}_x X \otimes E_x$. E^*, $E \otimes F$, and $\det E = \bigwedge^r E$ denote the dual, tensor product, and determinant bundles. The second fundamental form of a holomorphic subbundle is denoted β, $T^{1,0}X$ is the holomorphic tangent bundle, and $\mathrm{Ric}(\omega)$ is the Ricci form of a Kähler metric.

Prerequisite results used below

Tensor-product curvature rule. For bundles with connections, $\Theta(E \otimes F) = \Theta(E) \otimes I_F + I_E \otimes \Theta(F)$, while $\Theta(E^*) = -\Theta(E)^{\mathsf{T}}$ and $\Theta(\det E) = \operatorname{tr}\Theta(E)$; here I_E and I_F are the identity endomorphisms of the indicated bundles.

Orthogonal subbundle formula. If $S \subset E$ is a holomorphic subbundle and β its second fundamental form for the orthogonal splitting $E = S \oplus S^\perp$, then

$$\langle \mathrm{i}\Theta(S)(\xi,\bar{\xi})s, s\rangle = \langle \mathrm{i}\Theta(E)(\xi,\bar{\xi})s, s\rangle - \|\beta_\xi s\|^2.$$

The negative square explains why Griffiths negativity descends to subbundles and positivity descends to quotient bundles. All adjoints and projections are taken with the stated Hermitian metrics.

Why these ideas are needed

For line bundles, positive curvature has only one natural meaning. In higher rank the curvature tensor has both tangent and bundle indices, and different applications test different combinations of them. Griffiths positivity tests decomposable tensors and is natural for projectivized bundles; Nakano positivity tests the full tensor product and is the condition that enters the Bochner estimate for bundle-valued forms. Introducing both notions before using them prevents a false transfer of line-bundle intuition. The subbundle, quotient, dual, and tensor-product formulas show which positivity properties survive basic geometric operations.

Plan for the lecture. 0–22 min: two notions of vector-bundle positivity; 22–42: line bundles and duals; 42–65: subbundle and quotient curvature; 65–80: tensor operations and Demailly–Skoda; 80–90: sign audit.

Locally write

$$\mathrm{i}\Theta(E,h) = \mathrm{i}\sum_{j,k,\alpha,\beta} R_{j\bar{k}\alpha\bar{\beta}}\, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k \otimes e_\alpha^* \otimes e_\beta .$$

Metric compatibility implies the Hermitian symmetry $R_{j\bar{k}\alpha\bar{\beta}} = \overline{R_{k\bar{j}\beta\bar{\alpha}}}$, so the expressions below are real.

Definition 5.20: Griffiths and Nakano positivity

E is Griffiths semipositive if, for all $\xi \in T^{1,0}X$ and $v \in E$,

$$\sum_{j,k,\alpha,\beta} R_{j\bar{k}\alpha\bar{\beta}}\xi_j\bar{\xi}_k v_\alpha \bar{v}_\beta \geq 0.$$

It is Nakano semipositive if, for every $u = (u_{j\alpha}) \in T^{1,0}X \otimes E$,

$$\sum_{j,k,\alpha,\beta} R_{j\bar{k}\alpha\bar{\beta}}u_{j\alpha}\bar{u}_{k\beta} \geq 0.$$

Strict positivity means strict inequality on nonzero vectors.

These two curvature tests, including the distinction between decomposable and arbitrary tensors, follow the conventions of [1, Chapter VII, §6].

Nakano positivity is generally stronger than Griffiths positivity; for rank one both are ordinary positivity of the curvature form. Duality exchanges Griffiths positivity and negativity. Nakano duality involves an index transposition and is more subtle.

Theorem 5.21: Curvature of a subbundle and quotient

For a Hermitian holomorphic sequence $0 \to S \to E \to Q \to 0$, use the smooth orthogonal splitting $E \simeq S \oplus Q$ and let $\beta \in \mathcal{A}^{1,0}(\mathrm{Hom}(S,Q))$ be the second fundamental form. Then

$$D_E = \begin{pmatrix} D_S & -\beta^* \\ \beta & D_Q \end{pmatrix}, \qquad \Theta_E = \begin{pmatrix} \Theta_S - \beta^* \wedge \beta & -D'\beta^* \\ D''\beta & \Theta_Q - \beta \wedge \beta^* \end{pmatrix}.$$

Here D' and D'' are the type components of the induced connection on the relevant Hom bundle. Equivalently,

$$\left\langle \mathrm{i}\Theta_S(\xi,\bar{\xi})s, s\right\rangle = \left\langle \mathrm{i}\Theta_E(\xi,\bar{\xi})s, s\right\rangle - \left\|\beta_\xi s\right\|^2,$$

and, if $q \in Q_x$ is identified with its orthogonal lift in E_x,

$$\left\langle \mathrm{i}\Theta_Q(\xi,\bar{\xi})q, q\right\rangle = \left\langle \mathrm{i}\Theta_E(\xi,\bar{\xi})q, q\right\rangle + \left\|\beta^*_{\bar{\xi}} q\right\|^2.$$

Hence quotients of Griffiths-semipositive bundles remain semipositive, and subbundles of Griffiths-seminegative bundles remain seminegative.

Proof. The orthogonal splitting is smooth but generally not holomorphic. Metric compatibility forces the off-diagonal blocks of D_E to be adjoints with opposite signs, giving

$$D_E = \begin{pmatrix} D_S & -\beta^* \\ \beta & D_Q \end{pmatrix}.$$

Concretely, $\beta = \mathrm{pr}_Q \circ D'_E|_S$, while metric compatibility forces the upper-right entry to be $-\beta^*$. Square

this matrix using the graded rule: when a connection passes a one-form, a minus sign appears. The diagonal blocks are $\Theta_S - \beta^* \wedge \beta$ and $\Theta_Q - \beta \wedge \beta^*$, and the off-diagonal blocks are $-D'\beta^*$ and $D''\beta$. Evaluate the upper-left block on $(\xi, \bar{\xi})$ and a section $s \in S$; the term $\mathrm{i}\beta^* \wedge \beta$ contributes $-\left\|\beta_\xi s\right\|^2$ after solving for Θ_S. This proves the stated subbundle identity. Solving the lower-right block for Θ_Q gives the quotient identity with the opposite, positive square term. □

This block computation is the exact-sequence curvature formula of [1, Chapter V, §14]. The two displayed quadratic identities are often safer to use than the wedge-matrix formula because they make the signs visible.

Theorem 5.22: Demailly–Skoda positivity improvement

If E is Griffiths semipositive, then $E \otimes \det E$ is Nakano semipositive.

Proof. At a point choose holomorphic coordinates and a unitary bundle frame, and use the coefficient convention in the notation box. The curvature of $E \otimes \det E$ is

$$\widetilde{R}_{j\bar{k}\alpha\bar{\beta}} = R_{j\bar{k}\alpha\bar{\beta}} + \delta_{\alpha\beta} \sum_{\gamma} R_{j\bar{k}\gamma\bar{\gamma}}.$$

Fix an arbitrary tensor $u = (u_{j\alpha}) \in T^{1,0}X \otimes E$. For $v \in \mathbb{C}^r$ put $\xi_j(v) = \sum_\alpha u_{j\alpha}\bar{v}_\alpha$. Griffiths semipositivity gives

$$R_{j\bar{k}\alpha\bar{\beta}}\xi_j(v)\overline{\xi_k(v)}v_\alpha\bar{v}_\beta \geq 0.$$

Average with respect to probability measure on the unit sphere in $\mathbb{C}^r$. Unitary invariance (or a direct Gaussian calculation followed by radial normalization) gives the fourth-moment identity

$$\int \bar{v}_\gamma v_\delta v_\alpha \bar{v}_\beta = \frac{\delta_{\gamma\delta}\delta_{\alpha\beta} + \delta_{\gamma\alpha}\delta_{\delta\beta}}{r(r+1)}.$$

After substitution, the averaged inequality becomes

$$\frac{1}{r(r+1)} \sum_{j,k,\alpha,\beta} \left(R_{j\bar{k}\alpha\bar{\beta}} + \delta_{\alpha\beta} \sum_{\gamma} R_{j\bar{k}\gamma\bar{\gamma}} \right) u_{j\alpha}\bar{u}_{k\beta} \geq 0.$$

This is exactly Nakano semipositivity of $E \otimes \det E$. □

This is [1, Chapter VII, Theorem 8.1]. The same averaging argument gives strict Nakano positivity when the Griffiths inequality is strict: for $u \neq 0$, the averaged decomposable test cannot vanish identically. The theorem therefore converts the geometrically natural Griffiths condition into the form used by Nakano-type Bochner identities.

Example 5.23: Curvature of the tangent bundle

On a Kähler manifold, the Chern curvature of $T^{1,0}X$ is the complex part of the Riemann tensor. Its Griffiths quadratic form gives holomorphic bisectional curvature; equal directions give holomorphic sectional curvature. We use the differential- geometric Ricci form

$$\mathrm{Ric}(\omega) = -\mathrm{i}\partial\bar{\partial}\log\det(g_{j\bar{k}}) = -2\pi\mathrm{dd}^{\mathrm{c}}\log\det(g_{j\bar{k}}), \qquad [\mathrm{Ric}(\omega)] = 2\pi c_1(X).$$

Thus $(2\pi)^{-1}\,\mathrm{Ric}(\omega)$, rather than $\mathrm{Ric}(\omega)$ itself, is the Chern–Weil representative of $c_1(X)$ in our dd^{c} normalization.

Warning

Vector-bundle positivity does not mean that every matrix entry is a positive $(1,1)$-form. State whether the curvature is tested only on decomposable tensors $\xi \otimes v$ (Griffiths) or on arbitrary tensors in $T^{1,0}X \otimes E$ (Nakano).

5.3.1 How to read the four curvature indices

At a point choose unitary frames of $T^{1,0}X$ and E. The tensor $R_{j\bar{k}\alpha\bar{\beta}}$ defines a Hermitian form on $T^{1,0}X \otimes E$. Griffiths positivity tests only rank-one matrices

$$u_{j\alpha} = \xi_j v_\alpha,$$

whereas Nakano positivity tests every matrix $(u_{j\alpha})$. If either $\dim X = 1$ or $\operatorname{rank} E = 1$, every such matrix has rank at most one and the two tests agree. In higher dimensions and ranks, arbitrary tensors are sums of decomposable tensors, but positivity on each summand does not control cross terms; this is the gap between the notions.

For a line bundle the curvature tensor has no bundle indices. On an (n,q)-form, the commutator $[\mathrm{i}\Theta(L), \Lambda]$ sums q curvature eigenvalues. For a vector bundle, those eigenvalues are coupled to the bundle indices, and Nakano positivity is the condition naturally seen by this commutator. This explains the choice of positivity in Lecture 25 rather than merely naming one notion as "stronger."

5.3.2 Curvature under the standard bundle operations

The induced connections give

$$\begin{aligned}\Theta(E \otimes F) &= \Theta(E) \otimes \mathrm{id}_F + \mathrm{id}_E \otimes \Theta(F),\\ \Theta(E^*) &= -{}^{\mathsf{T}}\Theta(E),\\ \Theta(\det E) &= \operatorname{tr}\Theta(E).\end{aligned}$$

Thus Griffiths positivity is preserved by tensor products of positive bundles, and dualization reverses its sign. The determinant records the trace over bundle directions; its curvature is the first Chern form.

The tautological sequence on projective space,

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow \underline{\mathbb{C}}^{n+1} \longrightarrow Q \longrightarrow 0,$$

starts with a flat ambient bundle. The subbundle formula makes $\mathcal{O}(-1)$ Griffiths negative by the square of its second fundamental form, while the quotient formula makes Q Griffiths positive. Dualizing recovers positivity of $\mathcal{O}(1)$. This gives a geometric curvature proof of Fubini–Study positivity.

Tracing the curvature of $T^{1,0}X$ in its bundle indices gives the Ricci form:

$$\operatorname{Ric}(\omega) = \mathrm{i}\operatorname{tr}\Theta(T^{1,0}X).$$

Tracing once more against the Kähler metric gives scalar curvature. Holomorphic sectional, bisectional, Ricci, and scalar curvature are therefore successive tests or traces of the same Chern curvature tensor, with progressively less information.

Remark (A reliable decision rule)

For geometric maps and quotients, Griffiths positivity is often the natural language. For L^2 estimates on bundle-valued (n,q)-forms, compute the Nakano curvature operator. If only Griffiths positivity is available, a theorem such as Demailly–Skoda is the bridge; omitting that bridge is

a genuine gap.

5.3.3 Reading the curvature tensor as two quadratic forms

At one point identify $T^{1,0}_x X \simeq \mathbb{C}^n$ and $E_x \simeq \mathbb{C}^r$. Curvature defines a Hermitian form Q on $\mathbb{C}^n \otimes \mathbb{C}^r$ by

$$Q(u) = \sum_{j,k,\alpha,\beta} R_{j\bar{k}\alpha\bar{\beta}} u_{j\alpha} \bar{u}_{k\beta}.$$

Nakano semipositivity says $Q(u) \geq 0$ for every matrix $u = (u_{j\alpha})$. Griffiths semipositivity tests only rank-one matrices $u_{j\alpha} = \xi_j v_\alpha$. This proves directly that Nakano positivity implies Griffiths positivity. When $r = 1$ or $n = 1$, every matrix has rank at most one, so the two notions coincide.

For a line bundle L and a vector bundle E,

$$R^{E\otimes L}_{j\bar{k}\alpha\bar{\beta}} = R^{E}_{j\bar{k}\alpha\bar{\beta}} + R^{L}_{j\bar{k}} \delta_{\alpha\beta}.$$

If $\mathrm{i}\Theta_L \geq \varepsilon\omega$, then on an arbitrary tensor u the second term contributes at least $\varepsilon \sum_{j,\alpha} |u_{j\alpha}|^2$. This is why tensoring by a sufficiently positive line bundle can dominate a bounded negative part of vector-bundle curvature.

The tautological subbundle as a curvature test

The tautological line bundle $S = \mathcal{O}_{\mathbb{P}^n}(-1)$ sits inside the flat bundle $\mathbb{P}^n \times \mathbb{C}^{n+1}$. The ambient curvature is zero. The subbundle formula gives

$$\langle \mathrm{i}\Theta_S(\xi, \bar{\xi}) s, s \rangle = -\|\beta_\xi s\|^2 \leq 0.$$

Thus S is Griffiths negative. Its dual $S^* = \mathcal{O}(1)$ is positive. In the affine chart, the second fundamental form differentiates the line $\mathbb{C}(1, w)$ in directions orthogonal to itself. Its squared norm is exactly the Fubini–Study Hessian

$$\frac{(1 + |w|^2)|\xi|^2 - |\langle \xi, w \rangle|^2}{(1 + |w|^2)^2}.$$

This recovers the coordinate formula for ω_{FS} from the abstract subbundle identity.

Curvature under dual, tensor, and determinant

Metric compatibility with the evaluation pairing gives

$$\Theta_{E^*} = -{}^{\mathsf{T}}\Theta_E.$$

Consequently, if E is Griffiths positive and $0 \neq \lambda \in E^*_x$, choose the metric-dual vector v; then

$$\langle \mathrm{i}\Theta_{E^*}(\xi, \bar{\xi})\lambda, \lambda \rangle = -\langle \mathrm{i}\Theta_E(\xi, \bar{\xi}) v, v \rangle < 0.$$

Thus duality exchanges Griffiths positivity and negativity.

For the determinant bundle,

$$\Theta_{\det E} = \mathrm{tr}_E\, \Theta_E,$$

so Griffiths positivity of E implies positivity of $\det E$: sum the Griffiths quadratic form over an orthonormal basis of E_x. Combining this trace term with the original curvature is exactly the tensor appearing in the Demailly–Skoda theorem.

Subbundle and quotient signs in a rank-two model

Let $E = X \times \mathbb{C}^2$ be flat and let S be a holomorphic line subbundle locally spanned by

$$s(z) = \frac{(1, f(z))}{\sqrt{1 + |f(z)|^2}}.$$

The normalization is smooth rather than holomorphic, which is allowed for the orthogonal splitting. Projecting ∂s to $S^\perp$ gives

$$\|\beta_{\partial/\partial z} s\|^2 = \frac{|f'(z)|^2}{(1 + |f(z)|^2)^2}.$$

Since E is flat, the subbundle curvature is the negative of this quantity, whereas the quotient line has the positive of it. This explicit square verifies the signs in the block-curvature formula.

5.3.4 Squaring the block connection without hiding the signs

Let $S \subset E$ be a holomorphic subbundle and use the orthogonal smooth splitting $E = S \oplus Q$. The Chern connection has block form

$$D_E = \begin{pmatrix} D_S & -\beta^* \\ \beta & D_Q \end{pmatrix},$$

where the $(1,0)$-form $\beta \in \mathcal{A}^{1,0}(\mathrm{Hom}(S, Q))$ is the second fundamental form and β^* is its Hermitian adjoint of type $(0,1)$. Squaring this matrix with graded wedge multiplication gives

$$\Theta_E = \begin{pmatrix} \Theta_S - \beta^* \wedge \beta & -D'\beta^* \\ D''\beta & \Theta_Q - \beta \wedge \beta^* \end{pmatrix}.$$

The displayed wedge signs can be deceptive, so evaluate on a tangent vector $\xi \in T^{1,0}_x X$. For $s \in S_x$ one obtains

$$\langle \mathrm{i}\Theta_S(\xi, \bar{\xi}) s, s \rangle = \langle \mathrm{i}\Theta_E(\xi, \bar{\xi}) s, s \rangle - \|\beta_\xi s\|^2.$$

For $q \in Q_x$, using its orthogonal lift to E_x gives

$$\langle \mathrm{i}\Theta_Q(\xi, \bar{\xi}) q, q \rangle = \langle \mathrm{i}\Theta_E(\xi, \bar{\xi}) q, q \rangle + \|\beta^*_{\bar{\xi}} q\|^2.$$

Thus a subbundle loses curvature and a quotient gains curvature. When E is flat, these formulas reduce exactly to the negative and positive squares computed in the rank-two example; that example is therefore a sign check for the general identity.

5.3.5 Self-study workshop: keeping Griffiths and Nakano positivity separate

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall the Chern curvature tensor, tensor products of Hermitian vector spaces, and positive Hermitian forms. The distinction in this lecture is entirely about which tensors are used to test the same four-index curvature array.

Write both quadratic forms. At a point choose unitary coordinates and a unitary frame, and write

$$\mathrm{i}\Theta(E) = \sum_{j,k,\alpha,\beta} c_{j\bar{k}\alpha\bar{\beta}}\, \mathrm{i}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k \otimes e^*_\alpha \otimes e_\beta.$$

Griffiths positivity tests only decomposable tensors:

$$\sum c_{j\bar{k}\alpha\bar{\beta}}\xi_j\bar{\xi}_k v_\alpha \bar{v}_\beta > 0 \quad \text{for } \xi \neq 0,\ v \neq 0.$$

Nakano positivity tests every matrix $u = (u_{j\alpha}) \in T^{1,0}X \otimes E$:

$$\sum c_{j\bar{k}\alpha\bar{\beta}} u_{j\alpha}\overline{u_{k\beta}} > 0.$$

Since decomposable tensors are a subset of all tensors, Nakano positivity implies Griffiths positivity. The converse is false in general. For a line bundle every tensor is decomposable in the fiber index, so the notions coincide.

Curvature formulas must also track subbundle signs. If $S \subset E$ is a holomorphic subbundle with orthogonal second fundamental form B, then schematically

$$\Theta_S = \mathrm{pr}_S\, \Theta_E|_S - B^* \wedge B.$$

The negative square means a flat ambient bundle can have negatively curved subbundles; dualizing gives positive quotient bundles. This is the local mechanism behind positivity of the tautological quotient on projective space.

Checkpoint (test space before sign)

(1) Which positivity notion is naturally used in the Bochner estimate for vector-bundle-valued (n, q)-forms?
(2) Why do Griffiths and Nakano positivity agree in rank one?
(3) What happens to Griffiths curvature under dualizing?

Solution (checkpoint). The standard vector-bundle BKN estimate requires Nakano positivity of the relevant curvature operator. In rank one there is no independent fiber matrix index, so every test reduces to a tangent vector times a scalar. The dual curvature is the negative transpose, so Griffiths positivity becomes Griffiths negativity. ◇

Exit criterion

You can write both curvature quadratic forms with indices, state the implication in the correct direction, and determine the curvature sign of a subbundle, quotient, or dual in a model case.

Summary. Griffiths positivity tests decomposable tensors, while Nakano positivity tests all tensors and is the form usually needed by Bochner vanishing. The second fundamental form measures exactly how much curvature a subbundle loses, or a quotient gains, relative to its ambient bundle.

Exercise 5.24: The quotient line bundle

Use the quotient curvature formula to reprove positivity of the Fubini–Study metric on $\mathcal{O}_{\mathbb{P}^n}(1)$.

Exercise 5.25: Tensor-product curvature

Prove $\Theta(E \otimes F) = \Theta(E) \otimes \mathrm{id}_F + \mathrm{id}_E \otimes \Theta(F)$ and discuss preservation of Griffiths positivity.

Optional handout: Positivity notions and the Griffiths conjecture

This handout is optional. No later argument in the core lectures depends on it.

Vector bundles carry several inequivalent notions of positivity. We use the convention that $\mathbb{P}(E^*)$ parametrizes lines in E^*, so $\mathcal{O}_{\mathbb{P}(E^*)}(1)$ is the corresponding quotient of π^*E. Besides Griffiths and Nakano positivity of curvature, a holomorphic bundle E is algebraically ample when this line bundle is ample. Griffiths positivity implies ampleness, because its curvature induces positive curvature in the tautological quotient directions.

The converse is the Griffiths conjecture: every ample vector bundle should admit a Griffiths-positive Hermitian metric. It is known in important special cases but remains open in general. This contrasts with line bundles, where Kodaira's theorem proves the equivalence.

Proposition 5.26: Curvature of the tautological quotient

Let $\pi : \mathbb{P}(E^*) \to X$, where $\mathbb{P}(E^*)$ parametrizes lines $\ell \subset E^*_x$. Then $\mathcal{O}(-1)$ is the tautological subline of π^*E^* and $\mathcal{O}(1) = \mathcal{O}(-1)^*$; equivalently, evaluation makes $\mathcal{O}(1)$ a quotient of π^*E. A Griffiths-positive metric on E induces a metric whose curvature is positive in horizontal directions and whose vertical restriction is the Fubini–Study form. Thus $\mathcal{O}_{\mathbb{P}(E^*)}(1)$ is positive.

Proof. At a point $[\ell] \in \mathbb{P}(E^*)$, choose a unit covector $\xi \in \ell$, put $v = \xi^\sharp \in E_x$ using the metric, and split a tangent vector τ into horizontal and vertical parts using the Chern connection. Choose a unitary frame for E whose connection matrix vanishes at the base point. Differentiating the logarithm of the quotient norm gives the quotient-curvature formula

$$\mathrm{i}\Theta_{\mathcal{O}(1)}(\tau,\bar\tau) = \frac{\langle \mathrm{i}\Theta_E(\tau_h,\bar\tau_h)v, v\rangle}{|v|^2} + \omega_{\mathrm{FS}}(\tau_v,\bar\tau_v).$$

The first term is positive for every nonzero horizontal vector by Griffiths positivity. The second is the squared norm of the second fundamental form of the quotient; on each projective fiber it is exactly the Fubini–Study metric and is positive for every nonzero vertical vector. Since the connection splitting has no mixed term at the chosen point, their sum is positive for every nonzero τ. The computation is frame invariant, so it proves positivity globally. □

Other useful notions include dual Nakano positivity, RC-positivity, and positivity of direct images. They are designed for different Bochner identities; there is no universal strongest notion that is best for every application.

Warning (Projectivization convention)

Some authors let $\mathbb{P}(E)$ parametrize lines in E, others one-dimensional quotients. The symbols $\mathcal{O}(1)$ and $\mathcal{O}(-1)$ switch roles. State the convention before making a positivity assertion.

5.4 Lecture 20: Positive line bundles, Kodaira embedding, and projectivity

Week 10 material • 90 minutes

Notation for this lecture

X is a compact complex manifold of dimension n, $L \to X$ is a holomorphic line bundle, $L^k = L^{\otimes k}$, and $\mathcal{O}(L^k)$ is its sheaf of holomorphic sections, so $H^0(X, L^k) = \Gamma(X, \mathcal{O}(L^k))$ is its space of global holomorphic sections. The fiber at x is L^k_x, whereas $\mathcal{O}(L^k)_x$ is the stalk of holomorphic section germs. A Hermitian metric is positive if its Chern form is positive; L is ample if some power gives a projective embedding. For sections $s_0, \dots, s_N$ without a common zero, $\Phi_{|L^k|} : X \to \mathbb{P}^N$, $x \mapsto [s_0(x) : \cdots : s_N(x)]$, is the associated Kodaira map. $\mathcal{I}_{\{x\}} \subset \mathcal{O}_X$ is the ideal sheaf of the reduced point and $\mathfrak{m}_x = (\mathcal{I}_{\{x\}})_x \subset \mathcal{O}_{X,x}$ is the maximal ideal of the local ring. The first-jet space is

$$J^1_x(L^k) = \mathcal{O}(L^k)_x / \mathfrak{m}_x^2 \mathcal{O}(L^k)_x.$$

ev_x denotes evaluation at x, and $\mathcal{O}_{\mathbb{P}^N}(1)$ is the hyperplane line bundle. We write $h^q(X, L^k) = \dim_{\mathbb{C}} H^q(X, L^k)$, $\chi(X, L^k) = \sum_q (-1)^q h^q(X, L^k)$, $K_X = \bigwedge^n (T^{1,0}X)^*$ for the canonical bundle, and $\mathrm{td}(T^{1,0}X)$ for the Todd class of the holomorphic tangent bundle.

Prerequisite results used below

Finite-dimensionality. For a holomorphic line bundle on compact X, the vector space $H^0(X, L^k)$ is finite dimensional. This follows, for example, by applying elliptic theory to the $\bar\partial$ operator, whose kernel on smooth sections is exactly $H^0(X, L^k)$. Hence the complete linear system has a finite basis.

Embedding criterion. Let X be compact and let $s_0, \dots, s_N$ be global sections of a line bundle with no common zero. If the associated map $\Phi : X \to \mathbb{P}^N$ separates distinct points and its differential is injective at every point, then Φ is a holomorphic embedding. Indeed, it is an injective immersion, and compactness makes it a homeomorphism onto its image.

Jet interpretation. Surjectivity of $H^0(X, L^k) \to J^1_x(L^k)$ means that global sections prescribe an arbitrary value and first derivative at x. Surjectivity onto $L^k_x \oplus L^k_y$ prescribes independent values at distinct points x, y. These two statements give immersion and point separation, respectively.

Analytic input: the high-power $\bar\partial$ estimate. If (L, h) is positive on compact X, then $\omega = c_1(L, h)$ is a Kähler form. Use ω for the volume form and for the metrics on forms. There are constants $k_0, C > 0$ such that for $k \geq k_0$ and every $\bar\partial$-closed $g \in \mathcal{A}^{0,1}(X, L^k)$ there is $u \in \mathcal{A}^{0,0}(X, L^k)$ with

$$\bar\partial u = g, \qquad \|u\|^2_{L^2} \leq \frac{C}{k} \|g\|^2_{L^2}.$$

The same conclusion holds with the logarithmic weights used below: one first regularizes the logarithm, observes that the positive term $k\, c_1(L, h)$ dominates all fixed negative curvature terms, obtains estimates independent of the regularization, and passes to a weak limit. This is the precise analytic input from the Hörmander–Kodaira theory developed in Lectures 25–27. The constants may depend on the fixed metric, coordinate charts, and cutoffs, but not on k or on the regularization parameter.

Why these ideas are needed

Curvature positivity is differential and local, whereas projective embeddings require many global holomorphic sections. Kodaira's theorem explains why these apparently different conditions meet: positive curvature yields analytic estimates, the estimates remove cohomological obstructions, and the resulting sections separate points and tangent vectors. The distinction between base-point freeness, point separation, and first-jet separation is essential because each addresses a different failure of the map to projective space. This lecture makes the course's model local-to-global mechanism visible in one theorem.

Plan for the lecture. 0–18 min: analytic positivity and ampleness; 18–40: separating points and tangent vectors; 40–62: the Kodaira map; 62–78: analytic inputs in the embedding proof; 78–90: positive classes on a blow-up.

Definition 5.27: Positive and ample line bundles

A holomorphic line bundle $L \to X$ is *positive* if it has a smooth Hermitian metric with $c_1(L,h) > 0$. On a compact complex manifold, L is *ample* if the complete linear system of some power L^k defines a holomorphic embedding into projective space.

Proposition 5.28: Jet criterion for a projective embedding

Let $V \subset H^0(X, L^k)$ be a finite-dimensional space of sections on a compact complex manifold. Suppose:

(1) for every $x \in X$, the evaluation map $V \to L^k_x$ is surjective;
(2) for every pair $x \neq y$, the map $V \to L^k_x \oplus L^k_y$ is surjective;
(3) for every $x \in X$, the map $V \to J^1_x(L^k)$ is surjective.

Then any basis $s_0, \dots, s_N$ of V defines a holomorphic embedding

$$\Phi_V : X \longrightarrow \mathbb{P}^N, \qquad x \longmapsto [s_0(x) : \cdots : s_N(x)].$$

Proof. The first condition says that the sections have no common zero, so Φ_V is defined everywhere. If $x \neq y$, the second condition gives a section vanishing at x and nonzero at y; hence the two evaluation functionals on V are not proportional and $\Phi_V(x) \neq \Phi_V(y)$.

Fix $0 \neq \xi \in T^{1,0}_x X$. Choose $s_0 \in V$ with $s_0(x) \neq 0$ and use the target chart where the s_0-coordinate is nonzero. By the third condition there is $s \in V$ with $s(x) = 0$ and $\mathrm{d}s_x(\xi) \neq 0$, where the derivative is computed in any local holomorphic frame of L^k. This derivative is frame-independent at x: the extra term produced by differentiating a change of frame is multiplied by $s(x) = 0$. Since $s(x) = 0$,

$$\mathrm{d}\left(\frac{s}{s_0}\right)_x (\xi) = \frac{\mathrm{d}s_x(\xi)}{s_0(x)} \neq 0.$$

Thus $\mathrm{d}\Phi_{V,x}$ is injective. The map is an injective immersion, and compactness of X makes it a homeomorphism onto its image. Therefore it is a holomorphic embedding. □

The jet space in the third condition fits into a canonical exact sequence

$$0 \longrightarrow L^k_x \otimes T^{*1,0}_x X \longrightarrow J^1_x(L^k) \longrightarrow L^k_x \longrightarrow 0.$$

A local frame identifies it with a direct sum, but that splitting depends on the frame. The exact sequence, rather than a chosen splitting, is intrinsic.

Theorem 5.29: Kodaira embedding theorem

A line bundle on a compact complex manifold is positive if and only if it is ample. Consequently, a compact Kähler manifold is projective if and only if it has an integral Kähler class.

Here an *integral Kähler class* means a Kähler class in the image of the coefficient map $H^2(X,\mathbb{Z}) \to H^2(X,\mathbb{R})$.

Proof. Assume first that L has a metric of positive curvature and set $\omega = c_1(L,h)$; this is a Kähler form on X. At $x \in X$, choose holomorphic coordinates and a frame e such that the metric weight has the expansion $\varphi(z) = \sum \lambda_j |z_j|^2 + O(|z|^3)$ with every $\lambda_j > 0$. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n)$, put $z^\alpha = z_1^{\alpha_1} \cdots z_n^{\alpha_n}$ and $|\alpha| = \sum_j \alpha_j$. For $|\alpha| \le 1$, multiply $z^\alpha e^k$ by a cutoff χ which equals one on a smaller concentric ball and is supported in a ball of radius $(\log k)/\sqrt{k}$. Thus $\bar\partial\chi$ is supported in the surrounding annulus, where the factor $\mathrm{e}^{-k\varphi}$ is smaller than every fixed power of k.

Apply the high-power Hörmander estimate stated in the prerequisite box to solve

$$\bar\partial u_{k,\alpha} = \bar\partial(\chi z^\alpha e^k).$$

The curvature contribution growing with k is $k\Theta(L)$; it dominates both the fixed curvature terms in the estimate and the bounded negative part introduced by the cutoff weight. Since the $\bar\partial$ error is supported where the Gaussian factor is smaller than every power of k, the estimate gives $\|u_{k,\alpha}\| = O(k^{-M})$ for every prescribed $M > 0$; the implicit constant may depend on M and on the fixed coordinate and cutoff data, but not on k. Insert the singular factor $(n+1)\log|z|^2$ in the weight. Finite weighted norm then forces the value and first jet of $u_{k,\alpha}$ at x to vanish; equivalently, one may use the local mean-value and Cauchy inequalities to deduce the same jet estimate from the L^2 bound. Thus

$$s_{k,\alpha} = \chi z^\alpha e^k - u_{k,\alpha}$$

is global holomorphic and realizes the prescribed zero- or first-order jet at x.

For two distinct points, choose disjoint coordinate balls and repeat the construction with a weight singular at both points. The resulting sections separate their values. The zero-jet sections remove base points, the two-point construction separates points, and the first-jet sections make the Kodaira map an immersion. These properties are open in the points and tangent directions. Compactness of X, of $X \times X$ away from a small diagonal neighborhood, and of the unit tangent sphere bundle shows that one sufficiently large k works everywhere. An injective immersion from compact X to projective space is an embedding; hence L is ample.

The jet-surjectivity conclusion is [1, Chapter VII, Theorem 13.1 and Corollary 13.3]. Demailly obtains it by blowing up the prescribed points and applying vanishing; the argument above uses the equivalent singular-weight mechanism directly on X. The radial integrability calculation which forces the correction term to have zero first jet is given below.

Conversely, suppose $L^k = \Phi^* O(1)$ for a holomorphic embedding Φ. Pull back the Fubini–Study metric. Its curvature is positive because $\mathrm{d}\Phi$ is injective. Define a metric on L by taking the positive kth root of the metric on L^k in every local frame. Its curvature is $k^{-1}\Phi^*\omega_{\mathrm{FS}} > 0$, so L is positive.

Finally, let $[\omega] \in H^2(X,\mathbb{Z})$ be a Kähler class. Under the map $H^2(X,\mathbb{Z}) \to H^2(X,\mathcal{O}_X)$ its image is represented by the $(0,2)$ component of $[\omega]$, hence is zero because ω has type $(1,1)$. Exactness of the exponential sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_X \xrightarrow{f \mapsto \mathrm{e}^{2\pi \mathrm{i} f}} \mathcal{O}_X^* \longrightarrow 1$$

therefore gives a holomorphic line bundle L with $c_1(L) = [\omega]$. Choose any Hermitian metric h_0 on L and put $\alpha = c_1(L,h_0)$. The real $(1,1)$-form $\omega - \alpha$ is exact, so the $\partial\bar\partial$-lemma on the compact Kähler manifold gives $\omega - \alpha = \mathrm{d}\mathrm{d}^{\mathrm{c}}\psi$ for a real smooth function ψ. By the metric-change formula, $h = h_0 \mathrm{e}^{-\psi}$ has $c_1(L,h) = \omega > 0$, and the first part embeds X. Conversely, a projective embedding pulls the integral Kähler class $c_1(\mathcal{O}(1)) = [\omega_{\mathrm{FS}}]$ back to X. □

The final paragraph is Kodaira's projectivity criterion [1, Chapter VII, §14]. The type-$(1,1)$ hypothesis is essential: it is exactly what kills the image of the integral class in $H^2(X, \mathcal{O}_X)$.

Warning

"Many sections" is not enough. Base-point freeness only defines a map, separation of points gives injectivity, and separation of first jets gives an immersion. Finally, use compactness of X and Hausdorffness of projective space to conclude that an injective immersion is an embedding.

Corollary 5.30: Asymptotic section growth

If L is positive and $\dim_{\mathbb{C}} X = n$, then

$$h^0(X, L^k) = \frac{k^n}{n!} \int_X c_1(L)^n + O(k^{n-1}).$$

This follows from Hirzebruch–Riemann–Roch plus higher-cohomology vanishing, or from Bergman-kernel asymptotics.

Proof. Hirzebruch–Riemann–Roch gives

$$\chi(X, L^k) = \int_X \mathrm{e}^{kc_1(L)} \operatorname{td}(T^{1,0}X) = \frac{k^n}{n!} \int_X c_1(L)^n + O(k^{n-1}).$$

For $k \gg 1$, the line bundle $L^k \otimes K_X^{-1}$ is positive: the term $k\Theta(L)$ dominates the fixed curvature of K_X^{-1}. Using Kodaira vanishing, which is proved in Lecture 26, and applying it to $L^k = K_X \otimes (L^k \otimes K_X^{-1})$ gives $H^q(X, L^k) = 0$ for $q > 0$. Hence $h^0(X, L^k) = \chi(X, L^k)$, proving the formula. Thus the present corollary records the numerical consequence of that later analytic input. □

Example 5.31: Kähler classes on a blow-up

Let $\pi : \widetilde{\mathbb{P}^2} \to \mathbb{P}^2$ be the blow-up of one point, let E be the exceptional divisor, and put $H = \pi^* c_1(\mathcal{O}_{\mathbb{P}^2}(1))$. The class $H - \varepsilon[E]$ is Kähler for $0 < \varepsilon \ll 1$. If $\varepsilon = a/b \in \mathbb{Q}$, then $bH - a[E]$ is an integral positive class corresponding to $\pi^*\mathcal{O}(b) \otimes \mathcal{O}(-aE)$.

Remark (Geometric interpretation)

Positive curvature localizes a peak section near one point. Solving $\bar\partial$ corrects the nonholomorphic error created by the cutoff. Kodaira's theorem is therefore the principle "positive curvature creates global coordinates."

5.4.1 Jet separation as a cohomology calculation

Let $Z \subset X$ be a zero-dimensional subscheme. The exact sequence

$$0 \longrightarrow \mathcal{I}_Z\mathcal{O}(L^k) \longrightarrow \mathcal{O}(L^k) \longrightarrow \mathcal{O}(L^k)/\mathcal{I}_Z\mathcal{O}(L^k) \longrightarrow 0$$

shows that prescribed data on Z lift to a global section whenever

$$H^1(X, \mathcal{I}_Z\mathcal{O}(L^k)) = 0.$$

Take $Z = \{x\}$ to remove a base point, $Z = \{x, y\}$ to prescribe values at two points, and Z defined by $\mathcal{I}_{\{x\}}^2$ to prescribe a value and first derivative at x. Kodaira's analytic construction proves these surjections directly by solving $\bar\partial$; Serre vanishing proves them algebraically once ampleness is known.

If a basis of sections gives $\Phi_k : X \to \mathbb{P}^N$, then

$$L^k \simeq \Phi_k^* O(1).$$

The point-separation condition makes Φ_k injective, and surjectivity onto the first-jet spaces $J_x^1(L^k)$ makes $\mathrm{d}\Phi_k$ injective. The bundle identity also proves the converse direction of Kodaira. Pull the positive Fubini–Study metric back through the embedding; its curvature stays positive because $\mathrm{d}\Phi_k$ is injective. This gives a positive metric on L^k, and dividing its local weights by k gives a positive metric on L.

5.4.2 Numerical shadows of positivity

If L is positive and $V \subset X$ is an irreducible compact subvariety of dimension $p > 0$, then on the regular locus

$$\int_V c_1(L,h)^p > 0.$$

When V is singular this is interpreted through $[V]$ and Wirtinger's theorem. Thus analytic positivity implies positive intersection numbers on every subvariety. For projective manifolds, the Nakai–Moishezon criterion gives a converse using all such intersection numbers.

The asymptotic formula

$$h^0(X, L^k) = \frac{k^n}{n!} \int_X c_1(L)^n + O(k^{n-1})$$

has a local interpretation. A positive metric provides approximately one independent peak section per quantum cell of volume k^{-n}; integrating the local density yields the leading coefficient. Riemann–Roch packages the same count cohomologically.

On $\mathrm{Bl}_p \mathbb{P}^2$, let H be the pullback of the hyperplane class and let $[E]$ be the class of the exceptional divisor. A section of the line bundle with first Chern class $aH - b[E]$ is a degree-a polynomial whose derivatives of order $< b$ vanish at p. The number of imposed linear conditions is

$$1 + 2 + \cdots + b = \frac{b(b+1)}{2}.$$

This concrete linear-system description makes the exceptional coefficient a jet condition rather than an abstract intersection number.

Warning

An integral $(1,1)$ class determines a holomorphic line bundle only after using the exponential sequence and the type condition. A positive real class need not be rational, so a compact Kähler manifold need not be projective. Kodaira's criterion requires an integral Kähler class, not merely a Kähler form.

5.4.3 Base points and jets as exact sequences

For $x \in X$, let $\mathcal{I}_{\{x\}} \subset \mathcal{O}_X$ be the ideal sheaf of the reduced point and let $i_x : \{x\} \hookrightarrow X$ be the inclusion. Evaluation at x is encoded by

$$0 \longrightarrow \mathcal{I}_{\{x\}} O(L^k) \longrightarrow O(L^k) \longrightarrow (i_x)_* L_x^k \longrightarrow 0.$$

The resulting cohomology segment is

$$H^0(X, L^k) \longrightarrow L_x^k \longrightarrow H^1(X, \mathcal{I}_{\{x\}} O(L^k)).$$

If the last group vanishes, evaluation is surjective, so some section is nonzero at x and x is not a base point.

For first jets use

$$0 \longrightarrow \mathcal{I}_{\{x\}}^2 \mathcal{O}(L^k) \longrightarrow \mathcal{O}(L^k) \longrightarrow \mathcal{O}(L^k)/\mathcal{I}_{\{x\}}^2 \mathcal{O}(L^k) \longrightarrow 0.$$

Its stalk at x is the first-jet space $J_x^1(L^k) = \mathcal{O}(L^k)_x/\mathfrak{m}_x^2 \mathcal{O}(L^k)_x$. There is a canonical exact sequence

$$0 \longrightarrow L_x^k \otimes T_x^{*1,0} X \longrightarrow J_x^1(L^k) \longrightarrow L_x^k \longrightarrow 0.$$

After choosing a local frame the sequence splits and a jet becomes a value together with a first derivative, but the splitting is not canonical. Surjectivity of global sections onto $J_x^1(L^k)$ realizes arbitrary first-jet data. For two points $x \neq y$, the ideal sheaves are comaximal and

$$\mathcal{O}_X/(\mathcal{I}_{\{x\}}\mathcal{I}_{\{y\}}) \simeq (i_x)_*\mathbb{C} \oplus (i_y)_*\mathbb{C};$$

the corresponding quotient of $\mathcal{O}(L^k)$ records independent values in $L_x^k \oplus L_y^k$. These are the precise sheaf-theoretic conditions behind base-point freeness, immersion, and point separation.

The Veronese embedding checked in coordinates

The complete linear system of $\mathcal{O}_{\mathbb{P}^n}(k)$ consists of all degree-k monomials Z^I. The associated map is

$$\nu_k : \mathbb{P}^n \to \mathbb{P}^N, \qquad [Z_0 : \cdots : Z_n] \longmapsto [Z^I]_{|I|=k}, \quad N = \binom{n+k}{k} - 1.$$

It is well defined because scaling Z by λ scales every coordinate by the same factor λ^k. On U_0, divide by Z_0^k; among the affine target coordinates are

$$1, \quad z_1, \ldots, z_n, \quad z_1^2, z_1 z_2, \ldots.$$

The first-degree entries show that the map is injective on this chart and that its Jacobian contains the identity matrix. Since every point lies in such a chart, ν_k is an injective immersion; compactness makes it an embedding.

The hyperplane bundle pulls back as

$$\nu_k^* \mathcal{O}_{\mathbb{P}^N}(1) = \mathcal{O}_{\mathbb{P}^n}(k),$$

and hence $\nu_k^*[\omega_{\mathrm{FS}}] = k[\omega_{\mathrm{FS}}]$. This is both a transition- function calculation and a Chern-class check.

Why one exponent works globally

The peak-section argument is stronger than the assertion that each point has some possibly unrelated exponent. Fix a finite collection of coordinate balls with slightly smaller concentric balls still covering X. On this finite collection, the smallest curvature eigenvalue has a common positive lower bound; the coordinate, metric, cutoff, and elliptic-estimate constants also have common bounds. Consequently there is one integer k_0 such that, for every $k \geq k_0$, the same construction generates first jets at every point of every smaller ball. No multiplication by an unproved globally nonvanishing section is needed.

For pairs in a fixed neighborhood of the diagonal, the uniform first-jet estimate and the holomorphic inverse-function theorem give local separation. The complement of that neighborhood in $X \times X$ is compact. The two-point peak construction is locally uniform there, so finitely many product neighborhoods and the maximum of their thresholds give point separation for every k above one further integer. Taking the larger threshold proves that one complete linear system $|L^k|$ is simultaneously base-point free, immersive, and point separating.

A numerical calculation on a blow-up

On $\widetilde{\mathbb{P}^2}$ let H be the pullback of the class of a line, let E be the exceptional curve, and write $[E]$ for its divisor class. For $c_1(L) = aH - b[E]$,

$$L \cdot E = b, \qquad L \cdot (H - E) = a - b, \qquad L^2 = a^2 - b^2.$$

Therefore a Kähler class must satisfy $b > 0$ and $a - b > 0$. Conversely, when $a > b > 0$, a small multiple of a curvature form positive along E can be combined with the pullback of Fubini–Study to produce a Kähler representative. If a, b are integers, sections correspond to degree-a homogeneous polynomials vanishing to order at least b at the blown-up point. The intersection inequalities are the numerical shadow of having enough such sections.

5.4.4 Why the singular weight forces the prescribed jet

We spell out the local step in the Kodaira construction. Fix $x \in X$, choose coordinates $z = (z_1, \dots, z_n)$ centered at x, and trivialize L. Let $p(z)$ be a holomorphic polynomial of degree at most m, representing the jet we want. With a cutoff χ equal to one near x, the smooth section χp is not holomorphic, but

$$g = \bar\partial(\chi p)$$

is supported away from x and satisfies $\bar\partial g = 0$.

Choose a weight whose local singular part is

$$\psi = (n + m) \log |z|^2.$$

After smoothing and adding the weight of L^k, positivity of L makes the total curvature positive when k is sufficiently large. Hörmander's theorem then gives a solution u of $\bar\partial u = g$ with

$$\int_X |u|^2 \mathrm{e}^{-\psi} < \infty.$$

Near x the right side g vanishes, so elliptic regularity makes u holomorphic there. If the first nonzero homogeneous term of u has degree ℓ, then its radial contribution to the integral is comparable to

$$\int_0^{\varepsilon} r^{2\ell}\, r^{-2(n+m)}\, r^{2n-1}\, \mathrm{d}r = \int_0^{\varepsilon} r^{2\ell-2m-1}\, \mathrm{d}r.$$

This is finite exactly when $\ell > m$. Hence all Taylor coefficients of u through degree m vanish. The corrected section

$$s = \chi p - u$$

is global and holomorphic, and it has the same m-jet as p at x. Taking $m = 0$ produces a nonvanishing section; $m = 1$ gives tangent-vector separation. The singular weight is therefore not a technical decoration: its precise exponent converts an L^2 estimate into a jet-vanishing statement.

5.4.5 Self-study workshop: reducing an embedding theorem to two jet-surjectivity statements

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall base points, first jets, ideal sheaves of points, positive line-bundle metrics, and the compactness principle that turns local choices into finitely many global sections.

What an embedding must prove. A map

$$\Phi = [s_0 : \cdots : s_N] : X \longrightarrow \mathbb{P}^N$$

defined by sections of L^m is an embedding if it is everywhere defined, separates points, and separates tangent vectors. These conditions are encoded by coherent quotients. At a point x,

$$0 \longrightarrow \mathcal{I}_{\{x\}}\mathcal{O}(L^m) \longrightarrow \mathcal{O}(L^m) \longrightarrow (i_x)_* L_x^m \longrightarrow 0$$

tests base-point freeness, where $i_x : \{x\} \hookrightarrow X$. For distinct x, y, use the quotient by $\mathcal{I}_{\{x\}}\mathcal{I}_{\{y\}}$ to prescribe independent values. To control first derivatives at x, use

$$J_x^1(L^m) = \mathcal{O}(L^m)_x / \mathfrak{m}_x^2 \mathcal{O}(L^m)_x.$$

Surjectivity onto this jet space says that the sections realize every value and cotangent direction, hence $\mathrm{d}\Phi_x$ is injective.

Positive curvature supplies the analytic vanishing needed for these restriction maps. One chooses singular weights representing the point or jet conditions, solves the resulting $\bar{\partial}$ error with an L^2 estimate, and obtains global holomorphic sections. Compactness then reduces the point and tangent separation conditions to finitely many neighborhoods. Taking a common sufficiently large power produces one finite-dimensional linear system.

The model is $\mathcal{O}(1)$ on $\mathbb{P}^n$. Its sections $Z_0, \ldots, Z_n$ have no common zero, their ratios give the affine coordinates, and their derivatives span every cotangent space. The associated map is the identity projective embedding.

Checkpoint (three logically different conditions)

(1) Which quotient records first-order jets at x?
(2) Why does point separation alone not imply that Φ is an immersion?
(3) Where is compactness used after local sections have been constructed?

Solution (checkpoint). The quotient stalk is $J_x^1(L^m) = \mathcal{O}(L^m)_x / \mathfrak{m}_x^2 \mathcal{O}(L^m)_x$. Point separation controls only values at distinct points; a map can still have a zero differential. Compactness extracts finitely many sections and allows one exponent to work simultaneously over all points and tangent directions. ◇

Exit criterion

You can decompose Kodaira embedding into base-point freeness, point separation, and tangent separation, express each by an ideal-sheaf sequence, and identify the analytic and compactness inputs.

Summary. On compact manifolds, differential-geometric positivity of a line bundle and algebraic ampleness are equivalent. The proof must control values and first jets simultaneously by L^2 methods; merely writing the projective map does not establish an embedding.

Exercise 5.32: The Veronese embedding

Write the Kodaira map for $\mathcal{O}_{\mathbb{P}^n}(k)$ and prove that it is the degree-k Veronese embedding.

Exercise 5.33: Integral Kähler classes

Explain why an integral class $[\omega] \in H^2(X, \mathbb{Z})$ is the first Chern class of a line bundle. Identify the relevant part of the exponential long exact sequence.

Optional handout: Peak sections and Bergman kernel asymptotics

This handout is optional. No later argument in the core lectures depends on it.

Let (L, h) be positive over a compact Kähler n-fold, with curvature form ω. Choose an L^2-orthonormal basis $s_1^{(k)}, \dots, s_{N_k}^{(k)}$ of $H^0(X, L^k)$. The Bergman density is

$$\rho_k(x) = \sum_{j=1}^{N_k} |s_j^{(k)}(x)|_{h^k}^2 .$$

It is basis independent and equals the squared norm of the evaluation functional.

The Tian–Catlin–Zelditch expansion states, in a common normalization,

$$\rho_k(x) \sim k^n + a_1(x) k^{n-1} + a_2(x) k^{n-2} + \cdots ,$$

with a_1 a constant multiple of the scalar curvature of ω. Differentiated expansions imply

$$\frac{1}{k} \Phi_k^* \omega_{\mathrm{FS}} \longrightarrow \omega$$

smoothly, where Φ_k is the Kodaira map.

The local model is Bargmann–Fock space on $\mathbb{C}^n$ with Gaussian weight $\mathrm{e}^{-k|z|^2}$. A peak section looks like a constant or prescribed polynomial times this Gaussian in normal coordinates. A cutoff creates an exponentially small $\bar\partial$-error; Hörmander's estimate corrects it. Rescaling by $z \mapsto z/\sqrt{k}$ converts the local calculation to the universal Gaussian model.

Remark (Balanced metrics)

If ρ_k is constant, the metric is called k-balanced. Balanced metrics are finite- dimensional moment-map zeros and approximate constant-scalar-curvature Kähler metrics under suitable stability hypotheses.

Exercise 5.34: Bergman density of $\mathbb{P}^1$

Use rotational symmetry and $\int_{\mathbb{P}^1} \rho_k \omega_{\mathrm{FS}} = h^0(\mathcal{O}(k))$ to determine ρ_k for $\mathcal{O}(k)$ with the Fubini–Study metric.

Chapter 6

Currents, Positivity, Monge–Ampère Operators, and Lelong Numbers

6.1 Lecture 21: Currents, the Poincaré–Lelong formula, and positivity

Week 11 material • 90 minutes

Notation for this lecture

X is a complex n-manifold and $\Lambda^{p,q}T^*_xX$ is the space of complex covectors of bidegree (p,q) at x. A decomposable $(p,0)$-form has the form $\eta_1 \wedge \cdots \wedge \eta_p$. Strongly positive, positive, and weakly positive refer to the three cones defined in this lecture; in bidegree $(1,1)$ they coincide. The notation $T \geq 0$ means that a current T is positive when paired with every compactly supported strongly positive test form of complementary bidegree. $\omega_0 = (\mathrm{i}/2)\sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$ is the standard Euclidean Kähler form, V_{Reg} is the smooth locus of an analytic set V, $V_{\mathrm{Sing}} := V \setminus V_{\mathrm{Reg}}$ is its singular locus, $[V]$ is its integration current, and $\|T\|_K$ is the mass of T on a compact set K relative to a specified Hermitian metric. $\mathcal{A}^{p,q}_c(X)$ denotes compactly supported smooth (p,q)-forms, $\langle T, \eta\rangle$ is the action of a current on a test form, $T_\nu \rightharpoonup T$ means weak convergence on every test form, δ_a is the Dirac mass at a, and $\mathrm{div}(f)$ is the zero-pole divisor of a meromorphic function.

On a real m-manifold, a current of degree r acts on test forms of degree $m-r$. On a complex n-manifold, a current of bidegree (r,s) acts on test forms of bidegree $(n-r,n-s)$. A current of *bidimension* (p,p) therefore has bidegree $(n-p,n-p)$ and acts on test forms of bidegree (p,p). If T has bidimension (p,p), its trace measure relative to a Hermitian form ω is $\sigma_T = T\wedge\omega^p/p!$.

Prerequisite results used below

Hermitian spectral theorem. Every Hermitian form has an orthonormal eigenbasis and real eigenvalues. It is positive semidefinite exactly when its quadratic form is nonnegative on every vector.

Dual-cone principle. If C is a closed convex cone in a finite-dimensional real vector space V, its dual is $C^\vee = \{\lambda \in V^* : \lambda(v) \geq 0 \text{ for all } v \in C\}$. A current is positive precisely because its value on the cone of strongly positive complementary test forms is nonnegative; this is the distributional version of membership in a dual cone.

Topology on test forms. When all test forms are supported in a fixed compact subset of one coordinate chart, convergence in $\mathcal{A}^{p,q}_c(X)$ means uniform convergence of every coordinate coefficient and all its derivatives. Continuity of a current is therefore a quantitative local condition. The first definition below records the corresponding estimate and defines the order of

a current.

Why these ideas are needed

Smooth forms cannot represent a divisor or a logarithmic pole, so the first task is to extend differential forms to currents while preserving Stokes' formula. The Poincaré–Lelong formula then identifies the current of zeros and poles with the dd^c of a singular potential. For a smooth $(1,1)$-form, positivity can then be checked by a Hermitian matrix. Currents may have measure coefficients, so there is no pointwise matrix to diagonalize. Positivity must instead be defined by pairing with complementary test forms. The choice of strongly positive tests is forced by duality and by the requirement that integration over a complex subvariety be positive. Establishing the linear-algebra cones first makes later wedge products, weak limits, and curvature inequalities well-typed and prevents several inequivalent notions of positivity from being silently confused.

Plan for the lecture. 0–18 min: test forms and currents; 18–34: differentiation and weak convergence; 34–50: integration currents and Poincaré–Lelong; 50–68: positivity cones; 68–80: positive currents and mass; 80–90: Wirtinger's theorem.

Smooth forms cannot record divisors or concentration of curvature. A current is the coordinate-free version of a distribution-valued differential form.

Definition 6.1: Currents

Let M be an oriented real m-manifold. A current of degree q is a continuous linear functional on compactly supported smooth $(m-q)$-forms. If X has complex dimension n, a current of bidegree (p,q) acts on compactly supported forms of type $(n-p,n-q)$.
Concretely, continuity means the following. In every coordinate chart and for every compact set K in that chart, there are an integer $N \geq 0$ and a constant C_K such that

$$|\langle T,\eta\rangle| \leq C_K \max_{I,\,|\alpha|\leq N} \sup_K |\partial^\alpha \eta_I| \qquad \text{when } \operatorname{supp}\eta \subset K,$$

where the η_I are the coordinate coefficients of η. The least possible N locally is the *order* of T. Order zero means that only the C^0-norms of the coefficients are needed.

A locally integrable q-form α defines $T_\alpha(\eta) = \int_M \alpha \wedge \eta$. Weak convergence $T_\nu \rightharpoonup T$ means convergence of these pairings for every test form. Thus a locally integrable function is a degree-zero current, the Dirac mass at a point of a real m-manifold has degree m, and integration over an oriented real k-submanifold has degree $m-k$. On a complex n-fold, δ_a has bidegree (n,n), whereas a complex k-fold defines a current of bidegree $(n-k,n-k)$.

Definition 6.2: Differential of a current

For a current T of total degree q, define

$$\langle \mathrm{d}T,\eta\rangle = (-1)^{q+1}\langle T,\mathrm{d}\eta\rangle.$$

The operators ∂ and $\bar\partial$ are defined by taking the corresponding type components. These conventions make the derivative of a smooth form viewed as a current agree with its classical derivative.

If $V \subset X$ is an analytic set of pure dimension k, its integration current is

$$[V](\eta) = \int_{V_{\mathrm{Reg}}} \eta, \qquad \eta \in \mathcal{A}_c^{k,k}(X).$$

It has bidegree $(n-k, n-k)$ and is positive. If V is smooth and has no boundary, closedness is exactly Stokes' theorem. For singular V, Stokes applies on V_{Reg} but one must still control the ends approaching V_{Sing}. Local volume bounds and the Skoda–El Mir extension theorem, proved in Lecture 22, show that no boundary current is created there; hence $[V]$ is closed.

Theorem 6.3: Poincaré–Lelong formula

For a nonzero meromorphic function f on X,

$$\mathrm{dd^c} \log |f|^2 = [\mathrm{div}(f)]$$

as currents of bidegree $(1,1)$. In particular, $\mathrm{dd^c} \log |z|^2 = \delta_0$ on $\mathbb{C}$.

Proof. If $f = g/h$ locally, with g and h holomorphic and not identically zero, then $\log |f|^2 := \log |g|^2 - \log |h|^2$ is locally integrable. Changing the quotient representation adds and subtracts the same locally integrable logarithm, so the associated current is intrinsic. It is therefore enough to prove the formula for a holomorphic function and then subtract the formula for h.

On $\mathbb{C}^*$, the real-coordinate formula for $\mathrm{d^c}$ gives $\mathrm{d^c} \log |z|^2 = (2\pi)^{-1} \mathrm{d}\theta$. Pair with a compactly supported test function and apply Stokes on a punctured disc. The inner boundary contributes

$$\lim_{\varepsilon \downarrow 0} \frac{1}{2\pi} \int_{|z|=\varepsilon} \chi \, \mathrm{d}\theta = \chi(0),$$

while all terms containing derivatives of χ tend to zero. Thus $\mathrm{dd^c} \log |z|^2 = \delta_0$.

Near a smooth point of a divisor, write $f = z_1^m g$ with g holomorphic and nonvanishing. A local holomorphic logarithm of g gives $\mathrm{dd^c} \log |g|^2 = 0$, and the one-variable calculation in the normal direction gives $m[z_1 = 0]$. After summing the irreducible components and applying the same argument to $1/f$ at poles, the difference between the two sides is a closed order-zero $(1,1)$-current supported on the singular locus of the divisor, which has codimension at least two. Here is the needed special case of the dimension principle. On a smooth stratum written as $\{z_1 = \cdots = z_r = 0\}$ with $r \geq 2$, multiplication by each z_j and $\bar{z}_j$ annihilates an order-zero current supported on the stratum. Differentiating $z_j S = \bar{z}_j S = 0$ and using $\mathrm{d}S = 0$ gives $\mathrm{d}z_j \wedge S = \mathrm{d}\bar{z}_j \wedge S = 0$ for every $j \leq r$. A current of bidegree $(1,1)$ cannot contain all these distinct normal factors when $r \geq 2$, so it vanishes on the stratum. Induction over the singular strata eliminates the remaining support. Hence no residual term occurs and the formula holds everywhere. □

In intermediate bidegrees, "positive" is not a unique notion. Let V have complex dimension n, oriented so that $(\mathrm{i}\mathrm{d}z_1 \wedge \mathrm{d}\bar{z}_1) \wedge \cdots \wedge (\mathrm{i}\mathrm{d}z_n \wedge \mathrm{d}\bar{z}_n)$ is positive.

Definition 6.4: Three positivity cones for (p,p)-forms

A real (p,p)-form α is:

(1) *strongly positive* if it is a nonnegative combination of forms $\mathrm{i}^{p^2} \eta_1 \wedge \cdots \wedge \eta_p \wedge \bar{\eta}_1 \wedge \cdots \wedge \bar{\eta}_p$ with $\eta_j \in V^{*1,0}$;

(2) *positive* if it is a nonnegative combination of $\mathrm{i}^{p^2} \xi \wedge \bar{\xi}$, $\xi \in \Lambda^{p,0} V^*$;

(3) *weakly positive* if $\alpha \wedge \beta$ is a nonnegative volume form for every strongly positive $(n-p, n-p)$-form β.

The cones satisfy

$$\mathrm{SP}^p \subset \mathrm{P}^p \subset \mathrm{WP}^p,$$

with equality for $p = 0, 1, n-1, n$. For $\alpha = \mathrm{i}\sum a_{j\bar{k}}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$, positivity is equivalent to semipositivity of the Hermitian matrix $(a_{j\bar{k}})$.

Proposition 6.5: Operations preserving positivity

The wedge of strongly positive forms is strongly positive. The wedge of a positive form with a strongly positive form is positive. Complementary weakly and strongly positive forms wedge to a nonnegative volume form. Holomorphic linear pullback preserves all three cones.

Proof. A strongly positive form is a nonnegative sum of decomposable generators. Wedging two such generators concatenates their $(1,0)$ factors and, after the sign i^{p^2} is combined with i^{q^2}, gives the generator in degree $p+q$. Hence the first assertion follows by bilinearity. Write a positive form as a sum of $\mathrm{i}^{p^2}\xi \wedge \bar{\xi}$ and a strongly positive form as a sum of decomposable terms; their wedge again has the form $\mathrm{i}^{(p+q)^2}\eta \wedge \bar{\eta}$.

The definition of weak positivity is exactly nonnegativity after wedging with every complementary strongly positive generator, which proves the third assertion. Finally, a holomorphic linear pullback takes a $(1,0)$ covector to a $(1,0)$ covector and commutes with wedge and conjugation. It therefore preserves the generators of the strong and ordinary cones. For the weak cone, test after pullback against a complementary decomposable form and extend a basis from the image of the map; the test reduces to the original weak-positivity inequality. Thus all three cones are preserved. □

Warning

The wedge of two weakly positive forms need not be weakly positive in intermediate degree. Before multiplying, verify that one factor is strongly positive or reduce the question to a diagonal matrix calculation.

Definition 6.6: Positive current

A (p,p)-current T is positive if $\langle T, \eta\rangle \geq 0$ for every compactly supported strongly positive $(n-p, n-p)$ test form η. Thus the positive-current cone is dual to the cone of strongly positive test forms.

The coefficients of a positive current are complex measures, and their total variations on compact sets are controlled by a trace measure. Thus positivity improves a general distribution by one full regularity level.

Let ω be Hermitian and let T have bidimension (p,p). Define

$$\sigma_T = \frac{1}{p!}T \wedge \omega^p, \qquad \|T\|_K = \sigma_T(K).$$

Different Hermitian metrics give equivalent masses on relatively compact sets.

Theorem 6.7: Wirtinger's theorem

If $Z \subset X$ is analytic of pure dimension p, then

$$\mathrm{Vol}_\omega(Z \cap K) = \frac{1}{p!}\int_{Z_{\mathrm{Reg}}\cap K} \omega^p = \|[Z]\|_K\,.$$

In Euclidean space, a complex p-plane has the orientation calibrated by the Kähler form among all real $2p$-planes.

Proof. Work at one point in a unitary real basis $e_1, Je_1, \ldots, e_n, Je_n$, so $\omega = \sum_j e_j^* \wedge (Je_j)^*$. On an oriented orthonormal real $2p$-frame $v_1, \ldots, v_{2p}$, the value of $\omega^p/p!$ is the Pfaffian of the skew matrix

$(\omega(v_a, v_b))$. Its absolute value is at most one; equality holds exactly when the plane is J-invariant with its complex orientation. This is Wirtinger's linear algebra inequality. Applying it to $T_x Z_{\mathrm{Reg}}$ gives equality pointwise on a complex submanifold. Integrating over $Z_{\mathrm{Reg}} \cap K$ gives the mass formula; the singular set has zero $2p$-dimensional measure. □

Example 6.8: Three sources of positive currents

(1) A psh function u gives the closed positive current $\mathrm{dd^c}u$.
(2) An analytic set Z gives the integration current $[Z]$.
(3) The semipositive curvature of a Hermitian line bundle is a smooth positive $(1,1)$-form.

They represent potential theory, algebraic cycles, and differential geometry.

6.1.1 Duality of cones and coefficient measures

The definition of a positive current uses strongly positive test forms because the two cones are dual. At a point, a weakly positive (p,p)-form α satisfies

$$\alpha \wedge \mathrm{i}^{(n-p)^2} \eta_1 \wedge \cdots \wedge \eta_{n-p} \wedge \bar{\eta}_1 \wedge \cdots \wedge \bar{\eta}_{n-p} \geq 0$$

for every collection of $(1,0)$ covectors η_j. Replacing the η_j by linear combinations and polarizing controls every coefficient of α by its diagonal coefficients. The same argument for a current shows that its diagonal coefficients are positive measures and its off-diagonal coefficients are complex measures satisfying local Cauchy–Schwarz bounds.

Concretely, for a positive $(1,1)$-current $T = \mathrm{i}\sum T_{j\bar{k}} \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$, positivity of the 2×2 principal blocks gives

$$|T_{j\bar{k}}|(K)^2 \leq T_{j\bar{j}}(K') T_{k\bar{k}}(K')$$

after inserting a cutoff equal to one on $K \Subset K'$. Thus the trace measure $T \wedge \omega^{n-1}$ controls all coefficients. This is the local estimate behind weak compactness of positive currents.

6.1.2 Calibration and two geometric consequences

Wirtinger's inequality says that $\omega^p/p!$ has comass one: on every oriented real $2p$-plane P,

$$\left.\frac{\omega^p}{p!}\right|_P \leq \mathrm{d}V_P,$$

with equality precisely for a complex p-plane with its complex orientation. Therefore complex submanifolds minimize volume in their real homology class. If Z' is a compact real cycle homologous to a compact complex p-fold Z, then

$$\mathrm{Vol}(Z) = \int_Z \frac{\omega^p}{p!} = \int_{Z'} \frac{\omega^p}{p!} \leq \mathrm{Vol}(Z').$$

Closedness of ω is what makes the middle integral homological.

There is also a useful nonexistence test. If a Kähler form on X is exact, $\omega = \mathrm{d}\gamma$, then X contains no compact positive-dimensional complex subvariety. Indeed, for a compact p-dimensional analytic set Z,

$$0 < \int_Z \frac{\omega^p}{p!} = \frac{1}{p!} \int_Z \mathrm{d}(\gamma \wedge \omega^{p-1}) = 0,$$

a contradiction. In particular, $\mathbb{C}^n$ and Stein manifolds equipped with exact Kähler forms cannot contain compact complex curves.

Remark (Three levels of positivity)

For $(1,1)$-forms and for complementary degree $(n-1,n-1)$, the strong, ordinary, and weak cones coincide. The distinction appears only in intermediate degrees and only when $n \geq 4$. Consequently most curvature calculations with $(1,1)$-forms may use matrix positivity, while products of higher-degree currents require the full cone dictionary.

6.1.3 Positive forms reduced to matrices

For a real $(1,1)$-form write

$$\alpha = \mathrm{i} \sum_{j,k=1}^{n} a_{j\bar{k}}\, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k, \qquad A = (a_{j\bar{k}}) = A^*.$$

Evaluating on $v, \bar{v}$ gives

$$-\mathrm{i}\alpha(v,\bar{v}) = \sum_{j,k} a_{j\bar{k}} v_j \bar{v}_k.$$

Hence positivity of α is exactly semipositivity of A. A unitary coordinate change diagonalizes A, so at one point

$$\alpha = \mathrm{i} \sum_{j=1}^{n} \lambda_j\, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j, \qquad \lambda_j \geq 0.$$

This proves that for bidegree $(1,1)$ the strong, ordinary, and weak cones coincide.

If $\beta = \mathrm{i}\sum b_{j\bar{k}} \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$ is another positive form, unitary coordinates for α show directly that $\alpha \wedge \beta$ is strongly positive when one factor is positive of degree one. For a positive definite ω, the trace is characterized by

$$\alpha \wedge \frac{\omega^{n-1}}{(n-1)!} = (\mathrm{tr}_\omega\, \alpha)\frac{\omega^n}{n!}.$$

In unitary coordinates this is simply $\mathrm{tr}_\omega\, \alpha = \lambda_1 + \cdots + \lambda_n$.

For a smooth real function u,

$$(\mathrm{dd}^{\mathrm{c}} u)^n = \frac{n!}{\pi^n} \det(u_{j\bar{k}})\, \mathrm{d}V_{\mathbb{R}^{2n}},$$

with the present convention. The determinant appears because only terms containing each $\mathrm{d}z_j$ and $\mathrm{d}\bar{z}_k$ once survive in the wedge expansion; their signed sum is the Leibniz formula for $\det H_u$.

Positivity of integration currents in local coordinates

Let $V^p \subset X^n$ be a complex submanifold and choose coordinates with $V = \{z_{p+1} = \cdots = z_n = 0\}$. A compactly supported strongly positive test form of bidegree (p,p) is a sum of terms

$$\eta = (\mathrm{i})^{p^2}\, \theta_1 \wedge \cdots \wedge \theta_p \wedge \bar{\theta}_1 \wedge \cdots \wedge \bar{\theta}_p$$

with nonnegative coefficients. Pulling back to V replaces each θ_j by its tangential part. The resulting top-degree form is

$$|\det B|^2\, (\mathrm{i})^{p^2} \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_p \wedge \mathrm{d}\bar{z}_1 \wedge \cdots \wedge \mathrm{d}\bar{z}_p \geq 0.$$

Therefore $[V](\eta) = \int_V \eta \geq 0$. Positivity is independent of singularities: for an analytic set, integrate over the regular locus; its singular locus has zero $2p$-dimensional measure.

Why the coefficients of a positive current are measures

Let T have bidegree (q,q). Fix a compact K in a coordinate ball and the Euclidean Kähler form ω_0. Every coefficient test form can be written as a bounded linear combination of strongly positive forms after adding a sufficiently large multiple of ω_0^{n-q}. Positivity then gives

$$|\langle T,\eta\rangle| \le C_K \|\eta\|_{C^0}\langle T,\chi\omega_0^{n-q}\rangle,$$

where $\chi = 1$ on K. Thus T has order zero. The Riesz representation theorem makes each coordinate coefficient a complex Radon measure, and the positive trace measure $T\wedge\omega_0^{n-q}/(n-q)!$ controls all their total variations.

For $V=\{w=f(z)\}\subset\mathbb{C}^2$, the graph parametrization $\iota(z)=(z,f(z))$ gives

$$\iota^*\left(\frac{\mathrm{i}}{2}(\mathrm{d}z\wedge\mathrm{d}\bar z+\mathrm{d}w\wedge\mathrm{d}\bar w)\right)=\frac{\mathrm{i}}{2}(1+|f'(z)|^2)\mathrm{d}z\wedge\mathrm{d}\bar z.$$

Hence the mass of $[V]$ over a set is the ordinary area integral with density $1+|f'|^2$. This is the coordinate form of Wirtinger's calibration formula.

6.1.4 Poincaré–Lelong with the normalization derived

The constant in the Poincaré–Lelong formula is easiest to fix in one variable. Regularize the singular function by

$$u_\varepsilon(z)=\log(|z|^2+\varepsilon^2).$$

A direct calculation, using $\mathrm{d}\mathrm{d}^{\mathrm{c}}=(\mathrm{i}/2\pi)\partial\bar\partial$ and $\mathrm{i}\,\mathrm{d}z\wedge\mathrm{d}\bar z=2\,\mathrm{d}A$, gives

$$\mathrm{d}\mathrm{d}^{\mathrm{c}}u_\varepsilon=\frac{1}{\pi}\frac{\varepsilon^2}{(|z|^2+\varepsilon^2)^2}\,\mathrm{d}A.$$

This is a positive smooth measure, and polar coordinates show that its total mass is

$$\frac{1}{\pi}\int_0^{2\pi}\int_0^{\infty}\frac{\varepsilon^2 r}{(r^2+\varepsilon^2)^2}\,\mathrm{d}r\,\mathrm{d}\theta=1.$$

Its mass outside every fixed disc tends to zero, so $\mathrm{d}\mathrm{d}^{\mathrm{c}}u_\varepsilon\rightharpoonup\delta_0$. Hence

$$\mathrm{d}\mathrm{d}^{\mathrm{c}}\log|z|^2=\delta_0.$$

If a one-variable holomorphic function has $f(z)=z^m g(z)$ with $g(0)\neq 0$, then

$$\log|f|^2=m\log|z|^2+\log|g|^2.$$

The last term is pluriharmonic because g has a local holomorphic logarithm. Therefore $\mathrm{d}\mathrm{d}^{\mathrm{c}}\log|f|^2=m\delta_0$. In several variables, at a smooth point of a divisor choose coordinates in which $f=z_1^m g$ with g nonvanishing. Applying the one-variable identity in the normal coordinate and Fubini's theorem against test forms gives $m[z_1=0]$. The singular locus of the divisor has real codimension at least four and contributes no additional divisorial current. Summing over the irreducible components proves

$$\boxed{\mathrm{d}\mathrm{d}^{\mathrm{c}}\log|f|^2=[\mathrm{div}(f)]}.$$

Thus multiplicities and the factor 2π follow from the calculation rather than from an unstated convention.

6.1.5 Self-study workshop: testing positivity on the correct cone

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall Hermitian matrices, decomposable $(p,0)$-forms, duality between forms and currents, and the orientation supplied by a complex structure. In bidegree $(1,1)$ all standard positivity notions coincide; in middle bidegrees they need not.

Begin with the matrix case. A real $(1,1)$-form can be written

$$\alpha = \mathrm{i}\sum_{j,k} a_{j\bar{k}}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k.$$

After a unitary change of coordinates, $A = (a_{j\bar{k}})$ is diagonal:

$$\alpha = \mathrm{i}\sum_{j} \lambda_j\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j.$$

The condition $\alpha(v,\bar{v}) \geq 0$ for every v is equivalent to $\lambda_j \geq 0$ for all j. Wedge products then give

$$\alpha^p = p!\,\mathrm{i}^{p^2}\sum_{|I|=p}\left(\prod_{j\in I}\lambda_j\right)\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_I,$$

so powers of a positive $(1,1)$-form are strongly positive.

A current T of bidegree (p,p) is positive if $T(\eta) \geq 0$ for every compactly supported strongly positive test form of complementary bidegree. The complementary degree is essential: on an n-fold, T acts on $(n-p,n-p)$ forms. For an analytic set V of codimension p,

$$[V](\eta) = \int_{V_{\mathrm{Reg}}} \eta \geq 0$$

because restriction of a strongly positive form to the complex tangent space has the complex orientation and nonnegative density.

Checkpoint (degree and duality)

(1) On a threefold, which bidegree of test form pairs with a positive $(2,2)$-current?
(2) If α has eigenvalues $1,2,0$, is it positive? Strictly positive?
(3) Why is the wedge of two positive $(1,1)$-forms positive?

Solution (checkpoint). The complementary bidegree is $(1,1)$. The form is positive semidefinite but not strictly positive because of the zero eigenvalue. Diagonal approximation or the decomposable-form definition shows that a wedge of positive $(1,1)$-forms is strongly positive; continuity handles semidefinite limits. ◇

Exit criterion

You can diagonalize a $(1,1)$-form, test positivity through eigenvalues, identify the correct dual test cone for a current, and verify positivity of an integration current.

6.1.6 Weak convergence and order of a current

The topology on currents is weak: $T_\nu \rightharpoonup T$ means $\langle T_\nu, \eta\rangle \to \langle T, \eta\rangle$ for every compactly supported smooth test form η. Differentiation is continuous in this topology because

$$\langle \mathrm{d}T_\nu, \eta\rangle = (-1)^{\deg T_\nu + 1}\langle T_\nu, \mathrm{d}\eta\rangle.$$

This is the key advantage over pointwise convergence: singular limits commute with linear differential operators automatically.

For example, let ρ_ε be a nonnegative approximate identity on $\mathbb{R}^m$. Then $\rho_\varepsilon \mathrm{d}V \rightharpoonup \delta_0$. No subsequence converges as smooth forms, but the total mass remains visible. As another example, if smooth oriented submanifolds V_ν converge with controlled volume, their integration currents may have a weak limit even when their tangent spaces oscillate.

A current has order zero when it extends continuously to compactly supported continuous test forms. Measures and positive currents have order zero. Derivatives of measures need not: δ_0' detects the derivative of a test function. This distinction explains why multiplication by a smooth function is always allowed, whereas multiplication by a nonsmooth function requires additional hypotheses.

6.1.7 The one-variable Poincaré–Lelong constant

The normalization can be checked without memorization. On $\mathbb{C}^*$,

$$\mathrm{d}^{\mathrm{c}} \log |z|^2 = \frac{1}{2\pi}\mathrm{d}\theta.$$

Let χ be a test function supported in a disc and delete D_ε. Stokes' theorem, with the inner circle carrying the boundary orientation of the punctured disc, gives

$$\begin{aligned}\langle \mathrm{dd}^{\mathrm{c}} \log |z|^2, \chi\rangle &= \lim_{\varepsilon \downarrow 0} \int_{|z|>\varepsilon} \log |z|^2 \, \mathrm{dd}^{\mathrm{c}}\chi \\ &= \lim_{\varepsilon \downarrow 0} \frac{1}{2\pi}\int_{|z|=\varepsilon} \chi \, \mathrm{d}\theta = \chi(0).\end{aligned}$$

Terms containing derivatives of χ tend to zero because $\varepsilon|\log \varepsilon| \to 0$. Hence the coefficient of δ_0 is exactly one.

If $f = z_1^m g$ with g nonvanishing, then

$$\log |f|^2 = m \log |z_1|^2 + \log |g|^2.$$

The last term is pluriharmonic, and slicing in the z_1 direction yields $\mathrm{dd}^{\mathrm{c}} \log |f|^2 = m[z_1 = 0]$. This local factorization also shows why multiplicities, not merely zero sets, occur in the divisor.

Warning

Weak convergence is compatible with linear operations, not nonlinear ones. From $T_\nu \rightharpoonup T$ and $S_\nu \rightharpoonup S$ one cannot infer $T_\nu \wedge S_\nu \rightharpoonup T \wedge S$; the product on the right may not even exist. Lectures 23 and 31 introduce two different frameworks that recover selected products under extra control.

6.1.8 Currents calculated from the definition

On an oriented real m-manifold, a degree-k current acts on compactly supported $(m-k)$-forms. A locally integrable k-form α defines

$$T_\alpha(\eta) = \int_X \alpha \wedge \eta.$$

For a smooth $(m-k-1)$-form η, the definition of the differential is chosen so that $T_{\mathrm{d}\alpha} = \mathrm{d}T_\alpha$. Indeed, compact support and Stokes give

$$0 = \int_X \mathrm{d}(\alpha \wedge \eta) = \int_X \mathrm{d}\alpha \wedge \eta + (-1)^k \int_X \alpha \wedge \mathrm{d}\eta.$$

Thus

$$(\mathrm{d}T_\alpha)(\eta) = (-1)^{k+1} T_\alpha(\mathrm{d}\eta) = \int_X \mathrm{d}\alpha \wedge \eta.$$

This is the sign test that fixes the convention for all later current identities.

For the Dirac mass at $0 \in \mathbb{R}$, $\delta_0(\varphi) = \varphi(0)$ and

$$\delta_0'(\varphi) = -\varphi'(0).$$

If H is the Heaviside function, then

$$\langle H', \varphi \rangle = -\int_0^\infty \varphi'(x)\,\mathrm{d}x = \varphi(0),$$

so $H' = \delta_0$. The boundary term is exactly what survives after integration by parts; this one-dimensional model is the prototype for currents of integration.

Let $V^p \subset X^n$ be an oriented smooth submanifold without boundary. Its integration current has bidimension (p,p) and bidegree $(n-p, n-p)$:

$$[V](\eta) = \int_V \eta, \qquad \eta \in \mathcal{A}_c^{p,p}(X).$$

By Stokes,

$$\mathrm{d}[V](\eta) = \pm[V](\mathrm{d}\eta) = \pm \int_V \mathrm{d}\eta = 0.$$

If V has boundary, the same computation gives $\mathrm{d}[V] = [\partial V]$ with the orientation sign dictated by the convention. Thus closedness of an analytic-cycle current is a geometric Stokes statement.

The constant in Poincaré–Lelong

On $\mathbb{C}^*$ write $z = re^{\mathrm{i}\theta}$. Since $\log|z|^2 = 2\log r$, the real-coordinate formula from Lecture 2 gives

$$\mathrm{d}^{\mathrm{c}} \log|z|^2 = \frac{1}{2\pi}\,\mathrm{d}\theta.$$

It is closed on $\mathbb{C}^*$, but it has nonzero period:

$$\int_{|z|=\varepsilon} \mathrm{d}^{\mathrm{c}} \log|z|^2 = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\theta = 1.$$

Let χ be a compactly supported test function. Apply Stokes on $\{|z| > \varepsilon\}$ and then let $\varepsilon \downarrow 0$. Since $\mathrm{dd}^{\mathrm{c}} \log|z|^2 = 0$ away from 0, only the inner boundary remains, and its orientation gives

$$\langle \mathrm{dd}^{\mathrm{c}} \log|z|^2, \chi \rangle = \lim_{\varepsilon \to 0} \int_{|z|=\varepsilon} \chi\, \mathrm{d}^{\mathrm{c}} \log|z|^2 = \chi(0).$$

Therefore

$$\boxed{\mathrm{dd}^{\mathrm{c}} \log|z|^2 = \delta_0.}$$

If $f(z) = z^m g(z)$ with $g(0) \neq 0$, then

$$\log|f|^2 = m \log|z|^2 + \log|g|^2.$$

A local holomorphic logarithm of g shows $\mathrm{dd}^{\mathrm{c}} \log|g|^2 = 0$, hence $\mathrm{dd}^{\mathrm{c}} \log|f|^2 = m\delta_0$. Factoring at every zero and pole proves the one-dimensional divisor formula.

From local equations to a global divisor current

Let D be a divisor with local defining functions f_α. On an overlap, $f_\alpha = g_{\alpha\beta} f_\beta$ with $g_{\alpha\beta}$ holomorphic and nowhere zero. Then

$$\log|f_\alpha|^2 - \log|f_\beta|^2 = \log|g_{\alpha\beta}|^2, \qquad \mathrm{dd^c}\log|g_{\alpha\beta}|^2 = 0.$$

Thus the local currents $\mathrm{dd^c}\log|f_\alpha|^2$ glue. At a smooth point choose coordinates with $f_\alpha = z_1^m$; the one-variable computation in the normal direction and Fubini's theorem give $m[z_1 = 0]$. The singular locus has smaller dimension and does not change the current. This proves the local-to-global content of

$$\mathrm{dd^c}\log|f|^2 = [\mathrm{div}(f)].$$

For a section $s = f_\alpha e_\alpha$ of a Hermitian line bundle with $|e_\alpha|_h^2 = e^{-\varphi_\alpha}$,

$$\log|s|_h^2 = \log|f_\alpha|^2 - \varphi_\alpha.$$

Applying $\mathrm{dd^c}$ yields

$$\mathrm{dd^c}\log|s|_h^2 = [\mathrm{div}(s)] - \mathrm{dd^c}\varphi_\alpha.$$

The last term is the local curvature representative of $c_1(L,h)$, so rearranging gives the bundle-valued Poincaré–Lelong formula and explains why the two apparently local terms combine to a global identity.

6.1.9 Self-study workshop: using currents without suppressing test forms or constants

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall integration of compactly supported top-degree forms, distributional derivatives, Stokes' theorem, and the normalization $\mathrm{dd^c}\log|z|^2 = \delta_0$. Every current identity should be tested by pairing both sides with a smooth compactly supported form of complementary degree.

From functions to divisors. A locally integrable function u defines a degree-zero current $T_u(\eta) = \int u\eta$. Its derivative is defined by duality, with the sign chosen so that Stokes' theorem remains true. For a hypersurface $V = \{f = 0\}$, the integration current acts by

$$[V](\eta) = \int_{V_{\mathrm{Reg}}} \eta.$$

The singular set has smaller real dimension and does not affect the integral.

The local model contains the multiplicity. If $f(z) = z^m g(z)$ in one variable, with $g(0) \neq 0$, then

$$\log|f|^2 = m\log|z|^2 + \log|g|^2.$$

The last term is pluriharmonic near the origin because a nonvanishing holomorphic function has a local holomorphic logarithm. Consequently

$$\mathrm{dd^c}\log|f|^2 = m\delta_0.$$

In several variables the same factorization at a smooth generic point of each irreducible component gives $\mathrm{dd^c}\log|f|^2 = [\mathrm{div}(f)]$. This explains why the coefficient is the vanishing order rather than merely 1.

Metric Poincaré–Lelong is obtained by comparing local frames. If a line bundle section is $s = f_\alpha e_\alpha$ and $|e_\alpha|_h^2 = e^{-\varphi_\alpha}$, then

$$\log|s|_h^2 = \log|f_\alpha|^2 - \varphi_\alpha, \qquad \mathrm{dd^c}\log|s|_h^2 = [Z_s] - c_1(L,h).$$

Both terms on the right are global even though the two logarithms on the left were written in a frame.

Checkpoint (test the normalization)

Let $f(z) = z^3 e^z$ on a disc about 0.
(1) Compute $\mathrm{dd^c} \log |f|^2$ as a current.
(2) Explain why $\mathrm{dd^c} \log |e^z|^2 = 0$ by an actual calculation.
(3) State the bidegree of the current obtained.

Solution (checkpoint). We have $\log |f|^2 = 3 \log |z|^2 + z + \bar{z}$. Since $\partial\bar{\partial}(z + \bar{z}) = 0$, the answer is $3\delta_0$. It is a current of bidegree $(1, 1)$, acting on compactly supported functions when the ambient complex dimension is one. ◇

Exit criterion

You can translate a formal current equality into an equality on test forms, recover the multiplicity in Poincaré–Lelong, and keep the course's 2π normalization consistent.

Summary. Currents extend differential forms so that weak limits, singular potentials, and analytic cycles can be differentiated. Poincaré–Lelong converts zeros and poles into a complex Hessian. Positive currents are dual to strongly positive tests, have measure coefficients, and possess a natural mass measured geometrically by Wirtinger's theorem.

Exercise 6.9: The Heaviside distribution

Compute the distributional derivative of $H = 1_{[0,\infty)}$ on $\mathbb{R}$ and compare it with the differential convention for currents.

Exercise 6.10: Divisor of a rational function

Compute $\mathrm{dd^c} \log |z_1^2 z_2/(z_1 - z_2)|^2$ and state precisely in what sense the identity holds.

Exercise 6.11: Matrix criterion

Prove that the three positivity notions for a $(1, 1)$-form all agree with positive semidefiniteness of its coefficient matrix.

Exercise 6.12: Products of analytic sets

For analytic $Z_i \subset X_i$, prove $[Z_1 \times Z_2] = [Z_1] \times [Z_2]$ and compute the mass for a product metric.

Optional handout: Calibrations and complex submanifolds

This handout is optional. No later argument in the core lectures depends on it.

A closed p-form ϕ on a Riemannian manifold is a calibration if its restriction to every oriented unit p-plane is at most the Euclidean volume form. A submanifold is calibrated if equality holds on each tangent plane. Stokes' theorem then shows that it minimizes volume in its homology class.

On a Kähler manifold (X^n, ω), the form $\omega^p/p!$ calibrates complex p-dimensional submanifolds. Pointwise this is Wirtinger's inequality. If Z is complex and Z' is homologous to Z, then

$$\mathrm{Vol}(Z) = \int_Z \frac{\omega^p}{p!} = \int_{Z'} \frac{\omega^p}{p!} \le \mathrm{Vol}(Z').$$

Thus complex submanifolds are automatically minimal, although generally not uniquely volume minimizing.

Example 6.13: Projective degree

For a p-dimensional projective variety $Z \subset \mathbb{P}^N$,

$$\deg Z = \int_Z \omega_{\mathrm{FS}}^p.$$

The algebraic degree is therefore a calibrated volume. Compactness theorems for analytic cycles can be viewed as compactness under a uniform calibrated-volume bound.

Positive currents generalize calibrated submanifolds: the trace mass $T \wedge \omega^p/p!$ retains the same volume control even when the geometric object is singular or appears only as a weak limit.

Exercise 6.14: Minimality

Use the first variation of volume, or the calibration argument, to show that a complex curve in a Kähler surface has zero mean curvature on its smooth locus.

Optional handout: When can distributions be multiplied?

This handout is optional. No later argument in the core lectures depends on it.

The obstruction to multiplying currents is microlocal. A distribution can be singular at a point only in certain cotangent directions; these pairs form its wavefront set $\mathrm{WF}(u) \subset T^*M \setminus 0$. Smoothness near x is equivalent to the absence of all covectors over x from $\mathrm{WF}(u)$.

Theorem 6.15: Hörmander's product criterion

The product uv is canonically defined if there is no (x, ξ) such that $(x, \xi) \in \mathrm{WF}(u)$ and $(x, -\xi) \in \mathrm{WF}(v)$.

Proof. We use the pullback theorem for distributions. If $F : N \to M$ is smooth and

$$N_F = \{(F(y), \xi) : {}^t dF_y \xi = 0\}$$

does not meet $\mathrm{WF}(w)$, then F^*w is defined and depends continuously on w in the corresponding wavefront topology. Locally, write a regularized distribution as a Fourier integral. In the conic region where ${}^t dF_y \xi \neq 0$, repeated integration by parts in y gives arbitrary decay in ξ; in the complementary region the assumed wavefront cutoff makes the Fourier transform rapidly decreasing. The regularizations therefore converge independently of the cutoff. This proves the pullback theorem and also shows

$$\mathrm{WF}(F^*w) \subset \{(y, {}^t dF_y \xi) : (F(y), \xi) \in \mathrm{WF}(w)\}.$$

Now $u \otimes v$ is a well-defined distribution on $M \times M$, and

$$\mathrm{WF}(u \otimes v) \subset (\mathrm{WF}(u) \cup 0) \times (\mathrm{WF}(v) \cup 0) \setminus 0.$$

For the diagonal $\Delta(x) = (x, x)$, the conormal bundle is $N^*\Delta = \{(x, x; \xi, -\xi) : \xi \neq 0\}$. The hypothesis says exactly that this bundle misses $\mathrm{WF}(u \otimes v)$. Hence $uv := \Delta^*(u \otimes v)$ is canonically defined. For smooth functions this pullback is ordinary pointwise multiplication, so density and continuity show that the definition is the required extension of the usual product. □

Example 6.16: Transverse submanifolds

If smooth oriented submanifolds $A, B \subset M$ meet transversely, their integration currents have conormal wavefront sets, the criterion holds, and $[A] \wedge [B] = [A \cap B]$ with the intersection orientation. If they fail to be transverse, the product may require an excess-intersection construction and is not a naive distributional product.

This point of view complements Bedford–Taylor theory. Wavefront transversality is a condition on singular directions; Bedford–Taylor instead exploits positivity and local boundedness to define nonlinear products even when ordinary microlocal transversality is not available.

Exercise 6.17: Delta functions

Explain why the criterion permits $f\delta_0$ for smooth f but does not define δ_0^2. Compare with an arbitrary sequence of mollifiers.

6.2 Lecture 22: Closed positive currents, local potentials, and compactness

Week 11 material • 90 minutes

Notation for this lecture

T denotes a closed positive current; in the local-potential discussion it is real of bidegree $(1,1)$. On a coordinate ball, a local potential is a psh function u satisfying $T = \mathrm{dd^c}u$. When T has bidimension (p,p), it acts on (p,p) test forms and its trace measure is $\sigma_T = T \wedge \omega^p/p!$ for the chosen Hermitian form ω. A quasi-psh function is locally the sum of a psh and a smooth function. A pluriharmonic function is locally the real part of a holomorphic function. V_{Reg} and $V_{\mathrm{Sing}} := V \setminus V_{\mathrm{Reg}}$ denote the smooth and singular loci of an analytic set V. $T_j \rightharpoonup T$ means weak convergence against test forms, and $\|T\|_K := \sigma_T(K)$ is its mass on K. The support $\operatorname{supp} T$ is the complement of the largest open set on which T vanishes. An inequality $T \geq \gamma$ means that $T - \gamma$ is positive, where γ is a fixed smooth real $(1,1)$-form. For local regularization, ρ_ε is a mollifier and $u_\varepsilon = u * \rho_\varepsilon$. If $E \subset X$ is closed and T is defined on $X \setminus E$, its *trivial extension* $\widetilde{T}$ is obtained by extending every coefficient measure by zero across E. For a current already defined on X, $1_E T$ denotes restriction of its coefficient measures to the Borel set E. A closed set E is *locally complete pluripolar* if near every point there is a psh function ψ with $E = \{\psi = -\infty\}$.

Prerequisite results used below

Poincaré lemma for currents. On a ball, every closed current of positive degree is the differential of a current of one lower degree. The statement follows by dualizing the usual homotopy operator and is valid for currents of each fixed type after using the local Dolbeault lemma.
Compactness of normalized psh functions. Let u_j be psh on a domain $\Omega \subset \mathbb{C}^n$ and locally uniformly bounded above. Then either $u_j \to -\infty$ locally uniformly along a subsequence, or a subsequence converges in L^1_{loc} to a psh function. A normalization such as $\sup_{B'} u_j = 0$ rules out the first alternative.

Why these ideas are needed

Positive currents are useful only if they can be described locally and survive limits. Closedness provides local potentials, reducing a current to $\mathrm{dd^c}u$ for a plurisubharmonic function u; additive pluriharmonic ambiguity is the expected gauge freedom. Weak compactness is the mechanism that extracts limits from bounded families, while positivity supplies the mass estimates needed for that compactness. These ideas explain how singular Hermitian metrics can be approximated without losing curvature inequalities and prepare the limiting arguments in the L^2 chapters.

Plan for the lecture. 0–20 min: local potentials; 20–42: weak convergence and mass bounds; 42–60: regularization; 60–78: support and dimension principles; 78–90: closure and examples.

Theorem 6.18: Local psh potentials

Every closed positive $(1,1)$-current is locally $T = \mathrm{dd^c}u$ for a psh function u. On a connected small neighborhood, u is unique up to a pluriharmonic distribution.

Proof. The distributional Poincaré lemma gives $T = \mathrm{d}S$. Type decomposition and the distributional

Dolbeault lemma give $S^{0,1} = \bar\partial v$. Reality gives $S^{1,0} = \partial \bar v$, and therefore

$$T = \partial\bar\partial(v - \bar v).$$

Set $u = (2\pi/\mathrm{i})(v - \bar v) = 4\pi \operatorname{Im} v$. Since $\mathrm{dd^c} = (\mathrm{i}/2\pi)\partial\bar\partial$, this gives $T = \mathrm{dd^c} u$ with the constant visible. Positivity makes the complex Hessian of the real distribution u semipositive. Regularizing on a smaller ball and passing the submean inequalities to the limit produces its unique psh upper-semicontinuous representative. □

Theorem 6.19: Weak compactness of positive currents

Let T_j be positive currents of fixed bidegree. If $\sup_j \|T_j\|_K < \infty$ for every compact K, then a subsequence converges weakly to a positive current T. If every T_j is closed, so is T.

Proof. The trace measures control the total variations of all coordinate coefficients. Apply Banach–Alaoglu on each member of a countable compact exhaustion and take a diagonal subsequence. Positivity is preserved on passing to limits of nonnegative tests, and differentiation is continuous in the distribution topology. □

For a locally uniformly bounded-above sequence of psh functions, normalize $\sup_K u_j = 0$. Hartogs' lemma and L^1_{loc} compactness give a subsequence that either tends to $-\infty$ or converges in L^1_{loc} to a psh function. In the second case $\mathrm{dd^c} u_j \rightharpoonup \mathrm{dd^c} u$.

Extending an order-zero current by zero is measure-theoretically harmless, but its differential can acquire a boundary term on the omitted set. The next theorem says that positivity, finite mass, and a pluripolar defining function rule out precisely that defect.

Theorem 6.20: Skoda–El Mir extension theorem

Let $E \subset X$ be closed and locally complete pluripolar, and let T be a closed positive current on $X \setminus E$. Suppose that the trivial extension $\widetilde T$ has locally finite mass near E. Then $\widetilde T$ is a closed positive current on X.
Consequently, if S is a closed positive current on X, then both $1_{X\setminus E} S$ and $1_E S$ are closed positive currents.

Proof (cutoff argument). Positivity and local finite mass make the extension by zero an order-zero positive current, so only closedness needs proof. Work in a coordinate ball, choose a psh function $\psi \le -1$ with $E = \{\psi = -\infty\}$, and let χ be a smooth increasing function that is zero on $(-\infty, -2]$ and one on $[-1, \infty)$. The cutoffs $\chi_\nu = \chi(\psi/\nu)$ vanish near E and increase pointwise to one on $X \setminus E$.

For a fixed test form η, closedness of T away from E gives

$$\langle \mathrm{d}(\chi_\nu T), \eta\rangle = \langle \mathrm{d}\chi_\nu \wedge T, \eta\rangle.$$

The derivative is supported in the shell $A_\nu = \{-2\nu < \psi < -\nu\}$. The Cauchy–Schwarz inequality for positive currents, applied after writing the derivative in terms of the bounded psh weight $\exp(\psi/\nu)$, gives

$$|\langle \mathrm{d}\chi_\nu \wedge T, \eta\rangle|^2 \le C_\eta\, \sigma_T(A_\nu),$$

where C_η is independent of ν and σ_T is a local trace measure. For $\nu = 2^j$ the shells are disjoint. Local finiteness of σ_T therefore implies $\sigma_T(A_{2^j}) \to 0$. Since $\chi_{2^j} T \rightharpoonup \widetilde T$, passage to the limit gives $\mathrm{d}\widetilde T = 0$.

For the consequence, apply the theorem to $S|_{X\setminus E}$. Its trivial extension is $1_{X\setminus E} S$ and has mass bounded by that of S. Hence it is closed and positive; subtracting it from S gives the closed positive current $1_E S$. □

Corollary 6.21: Integration over an analytic set is closed

If $V \subset X$ is a pure-dimensional analytic set, then $[V]$ is a closed positive current.

Proof. On V_{Reg} this is Stokes' theorem. The singular locus is locally complete pluripolar. Weierstrass parametrization gives locally finite volume of V_{Reg} near V_{Sing}, so $[V_{\mathrm{Reg}}]$ has locally finite mass. The Skoda–El Mir theorem extends it across V_{Sing} as a closed positive current, which is precisely $[V]$. □

Theorem 6.22: Support and dimension principles

Let T be a closed positive current.

(1) If T has bidimension (p,p) and $\operatorname{supp} T$ lies in an analytic set of dimension $< p$, then $T = 0$.

(2) If A is irreducible of pure dimension p and $\operatorname{supp} T \subset A$, then $T = c[A]$ for some $c \geq 0$.

Proof. Work first on the regular stratum and choose coordinates in which $A = \{z_{p+1} = \cdots = z_n = 0\}$. Positive currents have order zero, so $z_j T = 0$ for $j > p$, differentiation gives $\mathrm{d}z_j \wedge T = \mathrm{d}(z_j T) - z_j \mathrm{d}T = 0$. Positivity and polarization then force every coefficient containing a normal differential to vanish. If $\dim A < p$, no tangential coefficient of the required bidimension remains and $T = 0$. If $\dim A = p$, the only possible term is $\mu[A]$ for a positive measure density μ on A. The equation $\mathrm{d}T = 0$ says that every distributional derivative of μ along A is zero, so μ is constant on each connected piece of A_{Reg}.

The difference between T and these constant multiples is supported on A_{Sing}, whose dimension is smaller. Induction on dimension eliminates that residual current. If A is irreducible, its regular locus is connected, so only one constant occurs. This proves both assertions. □

The first statement forbids an area-type mass from being compressed into too small an analytic set; the second is the local rigidity behind Siu decomposition.

Proposition 6.23: A safe regularization statement

On a coordinate domain, convolution with a nonnegative radial mollifier turns a closed positive current into smooth closed positive forms on smaller domains, converging weakly to the original current. On a manifold, patched convolution introduces small errors in closedness or positivity; exponential maps, connections, or Demailly's regularization control them.

Proof. Write $T = \sum T_{I\bar{J}}\, \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ and let ρ_ε be a nonnegative radial approximate identity. On the set whose distance from the boundary exceeds ε, define

$$T_\varepsilon = T * \rho_\varepsilon = \sum (T_{I\bar{J}} * \rho_\varepsilon)\, \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J .$$

Convolution commutes with constant-coefficient differentiation, so $\mathrm{d}T_\varepsilon = (\mathrm{d}T) * \rho_\varepsilon = 0$. For every strongly positive test form η at z, translation identifies $\langle T_\varepsilon(z), \eta \rangle$ with the average, using a nonnegative kernel, of the corresponding nonnegative pairings for T; hence $T_\varepsilon \geq 0$. Approximate-identity convergence gives $T_\varepsilon \rightharpoonup T$.

On a manifold, coordinate convolution must be multiplied by partitions of unity. Differentiating the partition functions produces commutator terms, and coordinate changes do not preserve the constant kernel. These are precisely the small errors mentioned in the statement; a connection or exponential-map construction estimates them by $O(\varepsilon)$ on compact sets. □

Warning

Partitions of unity do not commute with d. Patching local convolutions usually destroys both closedness and positivity. A global regularization theorem is valuable precisely because it quantifies these errors.

6.2.1 Compactness through normalized potentials

For $(1,1)$-currents one can see weak compactness directly at the level of potentials. On a ball B, write $T_j = \mathrm{dd}^{\mathrm{c}} u_j$ and normalize $\sup_{B'} u_j = 0$ on a smaller concentric ball B'. A mass bound for T_j controls the L^1 norm of u_j on still smaller balls by the submean inequality and the Riesz representation. The compactness theorem for psh functions gives either $u_j \to -\infty$ locally uniformly or a subsequence converging in L^1_{loc}. The normalization excludes the first alternative, and continuity of dd^{c} on distributions yields

$$T_j = \mathrm{dd}^{\mathrm{c}} u_j \rightharpoonup \mathrm{dd}^{\mathrm{c}} u.$$

This proof also records the local singularity data, which a coefficientwise Banach–Alaoglu argument does not.

Potentials are unique only modulo pluriharmonic functions. Normalizing by a supremum, an average, or a value at a nonsingular point chooses a representative. On overlaps, one must control the pluriharmonic differences before taking a global limit; compactness of the currents alone does not select global potentials.

6.2.2 Extension across a pluripolar set

An important complement to the support theorem is the Skoda–El Mir extension principle. Let $E \subset X$ be a closed complete pluripolar set and let T be a closed positive current on $X \setminus E$. If T has locally finite mass near E, define its trivial extension by

$$\langle \widetilde{T}, \eta \rangle = \int_{X \setminus E} T \wedge \eta.$$

Then $\widetilde{T}$ is again closed and positive on X.

The issue is closedness: differentiating a cutoff that vanishes near E produces a boundary error. Write locally $E \subset \{\psi = -\infty\}$ for a psh ψ and use convex cutoffs $\chi_\nu(\psi)$ that rise slowly from zero to one. Positivity bounds the error terms containing $\mathrm{dd}^{\mathrm{c}} \chi_\nu(\psi)$ by the mass of T in shrinking sublevel shells. Finite mass makes these errors tend to zero, proving $\mathrm{d}\widetilde{T} = 0$.

For an analytic set A, apply this principle to $[A_{\mathrm{Reg}}]$. The singular set has smaller dimension and is complete pluripolar locally; local volume bounds provide the finite-mass hypothesis. One recovers that $[A]$ is a closed positive current without integrating across the singular locus directly.

Warning

Finite mass near the exceptional set is indispensable. Positivity alone does not prevent infinite concentration, and the trivial extension of a closed current can acquire a boundary term. Always distinguish extension as an order-zero current from extension as a *closed* current.

6.2.3 Constructing a local psh potential

Let T be a real closed $(1,1)$-current on a ball $B \subset \mathbb{C}^n$. The Poincaré lemma for currents first gives a real one-current A with $T = \mathrm{d}A$. Decompose $A = A^{1,0} + A^{0,1}$. Since T has no $(2,0)$ or $(0,2)$ component,

$$\partial A^{1,0} = 0, \qquad \bar{\partial} A^{0,1} = 0.$$

After shrinking the ball, the Dolbeault lemma gives $A^{0,1} = \bar{\partial} v$; reality gives $A^{1,0} = \partial \bar{v}$. Thus

$$T = \partial \bar{\partial} v + \bar{\partial} \partial \bar{v} = \partial \bar{\partial} (v - \bar{v}).$$

Multiplying by the normalization constant produces a real distribution u such that $T = \mathrm{dd^c} u$. If $T \geq 0$, then every complex-line restriction of u has positive distributional Laplacian. Its upper-semicontinuous representative is therefore psh.

Two potentials differ by a pluriharmonic function. Indeed, $\mathrm{dd^c}(u - v) = 0$; locally $\partial(u - v)$ is a holomorphic closed one-form, hence the differential of a holomorphic function g, and $u - v = \Re g$+constant.

Weak compactness with a mass bound

Let T_j be positive currents of fixed bidegree on Ω and assume

$$\sup_j \int_K T_j \wedge \omega_0^p < \infty$$

for every $K \Subset \Omega$. The coefficient estimate from Lecture 21 bounds every coordinate measure on K. Banach–Alaoglu gives weak-* compactness of each coefficient measure. Choose a countable exhaustion by compact coordinate balls and diagonalize to obtain a subsequence converging on all compactly supported test forms.

The limit T is positive because $\langle T, \eta \rangle = \lim_j \langle T_j, \eta \rangle \geq 0$ for every strongly positive test form. If $\mathrm{d}T_j = 0$, then

$$\langle \mathrm{d}T, \eta \rangle = \pm \langle T, \mathrm{d}\eta \rangle = \pm \lim_j \langle T_j, \mathrm{d}\eta \rangle = 0.$$

Thus closedness also passes to the limit. The mass hypothesis is indispensable: $j\delta_0$ has no weakly convergent subsequence of finite currents.

Regularization in one coordinate ball

Extend a current T slightly beyond a relatively compact ball and convolve its coefficient measures with a standard kernel ρ_ε. For every test form η,

$$\langle T * \rho_\varepsilon, \eta \rangle = \langle T, \check{\rho}_\varepsilon * \eta \rangle.$$

Convolution commutes with d, ∂, and $\bar{\partial}$, so closedness is preserved. If T is positive, translations preserve positivity and the convolution is an average of positive translates, hence remains positive. Moreover $T * \rho_\varepsilon \to T$ weakly.

On a manifold, coordinate convolution does not agree on overlaps. A partition of unity introduces small negative errors in the derivatives; global regularization therefore generally produces

$$T_\varepsilon \geq -\varepsilon\omega$$

rather than exact positivity. Tracking this error is the key point in Demailly's regularization theorem.

A support-principle calculation

Let T be a closed current of bidegree (q, q) supported on a smooth complex submanifold Y of codimension r. In coordinates $Y = \{z_1 = \cdots = z_r = 0\}$, multiplication by each z_j annihilates an order-zero current supported on Y. Closedness then removes normal differential components. If $q < r$, there are not enough normal factors to support a nonzero current, so $T = 0$. If $q = r$ and T is positive, only the transverse top component remains and $T = c[Y]$ with $c \geq 0$ locally. Closedness forces c to be constant along each connected component. This is the local model behind the dimension and support principles.

6.2.4 The local potential calculation with the constant retained

Return to a real closed $(1,1)$-current T on a ball. The preceding construction gave a distribution v with

$$T = \partial\bar\partial(v - \bar v).$$

Define

$$u = \frac{2\pi}{\mathrm{i}}(v - \bar v) = 4\pi \operatorname{Im} v.$$

Then u is real and, with the convention declared in Lecture 1,

$$\mathrm{d}\mathrm{d}^{\mathrm{c}} u = \frac{\mathrm{i}}{2\pi}\partial\bar\partial\left(\frac{2\pi}{\mathrm{i}}(v - \bar v)\right) = T.$$

This makes the normalization completely explicit.

If T is positive, regularize u in a smaller coordinate ball. Since

$$\mathrm{d}\mathrm{d}^{\mathrm{c}} u_\varepsilon = T * \rho_\varepsilon \geq 0,$$

every u_ε is smooth psh. The regularized upper limit gives an upper-semicontinuous representative of u, and the linewise submean inequalities pass to it. It is therefore psh and still satisfies $\mathrm{d}\mathrm{d}^{\mathrm{c}} u = T$ distributionally. This avoids the unjustified operation of restricting an arbitrary current directly to a complex line.

If u_1 and u_2 are two real potentials, then $h = u_1 - u_2$ satisfies $\partial\bar\partial h = 0$. The distributional Dolbeault–Grothendieck lemma and elliptic regularity imply that h is smooth and locally the real part of a holomorphic function, up to a constant. Hence local potentials are unique modulo pluriharmonic functions, exactly as in the smooth Kähler case.

6.2.5 Self-study workshop: constructing potentials and taking weak limits safely

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall the Poincaré lemma for currents, positivity of $\mathrm{d}\mathrm{d}^{\mathrm{c}} u$ for psh u, weak convergence against test forms, and compactness of bounded sets of measures on compact subsets.

Local potential construction. Let T be a real closed $(1,1)$-current on a coordinate ball. The current Poincaré lemma first writes $T = \mathrm{d}S$. Splitting S by type and using $\partial T = \bar\partial T = 0$, one solves a local $\bar\partial$ equation for the unwanted type components and obtains

$$T = \mathrm{d}\mathrm{d}^{\mathrm{c}} u$$

for a real distribution u. If $T \geq 0$, elliptic regularization and the mean inequality show that u has a psh representative. The potential is not unique: u may be changed by a pluriharmonic function. A normalization such as $\int_B u \,\mathrm{d}V = 0$ is therefore required before compactness of potentials can be claimed.

Now let $T_\nu \geq 0$ be closed and suppose their masses are uniformly bounded on every compact subset. In coordinates, positivity controls every coefficient measure by the trace measure

$$\sigma_{T_\nu} = T_\nu \wedge \omega_0^{n-1}.$$

Weak compactness of measures gives convergent subsequences of all coefficients. Positivity passes to the limit because evaluation on a fixed strongly positive test form is a nonnegative real number. Closedness passes because

$$(\mathrm{d}T_\nu)(\eta) = \pm T_\nu(\mathrm{d}\eta) = 0$$

and the right side is stable under weak convergence.

Checkpoint (what compactness really needs)

(1) Why does adding an arbitrary constant destroy compactness of a family of psh potentials but not change its currents?
(2) Which scalar measure controls the coefficient measures of a positive $(1,1)$-current?
(3) Why is a weak limit of positive closed currents again closed?

Solution (checkpoint). Constants disappear under dd^c, so unnormalized potentials can drift to $+\infty$ or $-\infty$. The trace measure $T \wedge \omega_0^{n-1}$ controls the coefficients by positivity. Closedness follows by moving d to each fixed test form and passing to the limit. ⋄

Exit criterion

You can state the ambiguity of a local psh potential, choose a normalization, and give the coefficient-by-coefficient compactness argument for positive closed currents.

Summary. A closed positive $(1,1)$-current is locally the complex Hessian of a psh potential. Mass bounds give weak compactness, while support theorems give geometric rigidity. Together these tools control singular metrics and degenerate limits.

Exercise 6.24: Normalization and compactness

On the unit ball compare the normalized limits of $u_j = j^{-1}\log|z_1|$ and $v_j = j\log|z_1|$.

Exercise 6.25: Currents supported at a point

Classify closed positive (p,p)-currents in $\mathbb{C}^n$ supported at the origin. Which bidegrees can be nonzero?

Optional handout: Compactness of cycles and Bishop's theorem

This handout is optional. No later argument in the core lectures depends on it.

Weak compactness of positive currents has a geometric refinement. Bishop's compactness theorem states, in a local form, that a sequence of pure p-dimensional analytic sets with locally uniformly bounded volume has a subsequence converging to an analytic cycle. The limit is stronger than an arbitrary positive current: it has integer multiplicities.

The proof combines three ingredients. Wirtinger's theorem converts volume into the mass of integration currents. Weak compactness produces a closed positive limit. Generic linear projections and the area formula control the number of sheets away from a small branch set. Weierstrass preparation then shows that the limiting support is analytic and that sheet numbers give integer multiplicities.

Example 6.26: Graphs of holomorphic maps

Let $f_j : U \to V$ be holomorphic maps whose graphs have locally bounded volume in $U \times V$. A subsequence of the graphs converges to an analytic cycle. If the limit contains exactly one sheet over generic points of U and has no vertical component, it is the graph of a meromorphic map. This is a basic compactness mechanism in moduli problems.

Warning (Current limits need not remain smooth)

Smooth submanifolds can converge to a reducible or singular cycle, and multiplicity can appear. For example, the divisors $\{z^2 = \varepsilon_j\}$ converge as currents to $2[\{z = 0\}]$ as $\varepsilon_j \to 0$.

Exercise 6.27: Multiplicity in a limit

Test the currents $[z^m = \varepsilon]$ against a compactly supported $(n-1, n-1)$-form and verify convergence to $m[z = 0]$.

6.3 Lecture 23: The Bedford–Taylor Monge–Ampère operator

Week 12 material • 90 minutes

Notation for this lecture

$\Omega \subset \mathbb{C}^n$ is a domain, $u_1, \dots, u_q$ are locally bounded psh functions, and T is a closed positive current of bidimension (p,p), where $q \le p$. Thus $\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T$ has bidimension $(p-q, p-q)$. The Bedford–Taylor product is defined recursively by $\mathrm{dd^c}u_1 \wedge T = \mathrm{dd^c}(u_1 T)$ and iteration; it is not an a priori product of distributions. The Monge–Ampère measure is $\mathrm{MA}(u) = (\mathrm{dd^c}u)^n$. The notation $K \Subset \Omega$ means that K is compact and contained in Ω, and $\|u\|_{L^\infty(K)}$ is the essential supremum on K. The symbol $\mathrm{d}\lambda$ denotes Euclidean Lebesgue measure on $\mathbb{C}^n \simeq \mathbb{R}^{2n}$. Weak convergence of the resulting measures is denoted $\rightharpoonup$. CLN abbreviates the Chern–Levine–Nirenberg local mass estimate. For a compact $K \subset \Omega$, the relative Bedford–Taylor capacity is

$$\mathrm{Cap}(K,\Omega) := \sup\left\{ \int_K (\mathrm{dd^c}v)^n : v \in \mathrm{PSH}(\Omega),\ 0 \le v \le 1 \right\}.$$

Prerequisite results used below

Integration by parts for currents. If S is a current and ϕ is a compactly supported smooth form of complementary degree, then $\langle \mathrm{d}S, \phi\rangle = (-1)^{\deg S+1}\langle S, \mathrm{d}\phi\rangle$. The analogous formulas for ∂, $\bar\partial$, and $\mathrm{dd^c}$ determine all signs in the Bedford–Taylor recursion.

Monotone convergence of measures. If positive Radon measures μ_j increase to μ on Borel sets, then $\int f\,\mathrm{d}\mu_j \to \int f\,\mathrm{d}\mu$ for every nonnegative measurable f. For decreasing locally bounded psh approximants, Bedford–Taylor continuity is a deeper analogue and is stated and proved in this lecture.

Why these ideas are needed

The complex Monge–Ampère expression $(\mathrm{dd^c}u)^n$ is central to nonlinear complex geometry, but ordinary distribution theory does not permit arbitrary products of distributional coefficients. The Bedford–Taylor construction succeeds for locally bounded plurisubharmonic potentials by defining wedge products recursively through integration by parts. Local boundedness is the hypothesis that makes the product unambiguous, and monotone continuity is what makes approximation useful. The point of the construction is to retain both nonlinearity and positivity after smoothness has been lost.

Plan for the lecture. 0–20 min: why currents cannot simply be multiplied; 20–45: recursive definition; 45–63: the CLN estimate; 63–78: monotone continuity; 78–90: comparison and capacity.

For $u \in C^2 \cap \mathrm{PSH}(\Omega)$,

$$(\mathrm{dd^c}u)^n = \frac{n!}{\pi^n}\det(u_{j\bar k})\,\mathrm{d}\lambda$$

under our normalization and Euclidean volume convention. If u is merely locally bounded, its Hessian coefficients are distributions and the determinant cannot be multiplied term by term.

Definition 6.28: Bedford–Taylor product

Let $u_1, \dots, u_q \in \mathrm{PSH}(\Omega) \cap L^\infty_{\mathrm{loc}}$ and let T be closed positive. Define recursively

$$\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T := \mathrm{dd^c}\big(u_1\,\mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T\big).$$

Local boundedness makes multiplication of the measure coefficients by u_1 meaningful.

Proposition 6.29: The ordered product is positive and closed

For the displayed order of the factors, the recursive product is a closed positive current and commutes with restriction to open subsets.

Proof. Induct on q. Put $R = \mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T$, already defined by induction. Because u_1 is locally bounded, $u_1 R$ is a current of order zero, so $\mathrm{dd^c}(u_1 R)$ is defined and is closed. Choose smooth psh functions $u_{1,\nu} \downarrow u_1$ on a smaller coordinate set. Since R is fixed and has measure coefficients, bounded convergence gives $u_{1,\nu} R \rightharpoonup u_1 R$. Consequently

$$\mathrm{dd^c}u_{1,\nu} \wedge R = \mathrm{dd^c}(u_{1,\nu} R) \rightharpoonup \mathrm{dd^c}(u_1 R).$$

Each current on the left is closed and positive, so the limit is closed and positive. This establishes the recursive step without using a theorem about simultaneous limits. The construction uses only multiplication of a measure current by a bounded Borel function and distributional $\mathrm{dd^c}$. Both operations commute with restriction, so the ordered products agree on overlaps. Symmetry is not formal and will follow only after the monotone-continuity theorem below. □

The preceding argument varies one potential while the remaining current is fixed. To pass several singular factors to a limit, one first needs a mass bound uniform in all regularization parameters. This is the role of the next estimate.

Theorem 6.30: Chern–Levine–Nirenberg estimate

Let $L, K \subset \Omega$ be compact with $L \subset K^\circ$. If the u_j are bounded on K and T is closed positive, then

$$\left\| \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T \right\|_L \leq C_{L,K} \prod_{j=1}^{q} \|u_j\|_{L^\infty(K)} \, \|T\|_K .$$

Proof. Write the bidimension of T as (p, p). It is enough to treat one factor and iterate. Subtract $c = \sup_K u_1$. The new function has the same $\mathrm{dd^c}$ and satisfies $-2\|u_1\|_{L^\infty(K)} \leq u_1 - c \leq 0$. To simplify notation, denote it again by u_1. Choose $\chi \in C_c^\infty(K^\circ)$ with $0 \leq \chi \leq 1$ and $\chi = 1$ near L. The recursive definition and closedness of T give

$$0 \leq \int_L \mathrm{dd^c}u_1 \wedge T \wedge \omega^{p-1} \leq \int_K \chi \, \mathrm{dd^c}u_1 \wedge T \wedge \omega^{p-1} = \int_K u_1 \, \mathrm{dd^c}\chi \wedge T \wedge \omega^{p-1}.$$

On the compact set K, choose C so that $-C\omega \leq \mathrm{dd^c}\chi \leq C\omega$. Taking the absolute value of the last integral and using positivity bounds it by $2C\|u_1\|_{L^\infty(K)}\|T\|_K$. For several factors, choose nested compact sets $L = K_0 \Subset K_1 \Subset \cdots \Subset K_q = K$ and repeat the one-factor estimate, moving one $\mathrm{dd^c}$ at a time onto a cutoff. Multiplying the constants gives the stated bound. No smooth approximation or monotone-continuity theorem is used here: integration by parts is precisely the recursive definition of the product. Thus the CLN estimate is available before, and can legitimately be used in, the limiting argument that follows. □

With uniform local mass now established, simultaneous decreasing approximation can be treated without multiplying two independently varying measures.

Theorem 6.31: Bedford–Taylor monotone continuity

If $u_j^{(k)} \downarrow u_j$ are locally bounded psh functions, then

$$\mathrm{dd^c}u_1^{(k)} \wedge \cdots \wedge \mathrm{dd^c}u_q^{(k)} \wedge T \rightharpoonup \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T.$$

The corresponding statement with one undifferentiated potential in front also holds.

Proof. We prove first the statement with the undifferentiated factor; applying $\mathrm{dd^c}$ then gives the displayed convergence. For one factor, $u_1^{(k)}T \rightharpoonup u_1 T$ by bounded convergence with respect to the fixed coefficient measures of T. Suppose the assertion is known for $q-1$ factors and put

$$S_k = \mathrm{dd^c}u_2^{(k)} \wedge \cdots \wedge \mathrm{dd^c}u_q^{(k)} \wedge T, \qquad S = \mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T.$$

The induction hypothesis gives $S_k \rightharpoonup S$, and the CLN estimate gives relative compactness of $u_1^{(k)}S_k$ on smaller coordinate sets.

We use an elementary measure lemma. If $f_k \downarrow f$ are upper-semicontinuous functions bounded on compact sets, $\mu_k \rightharpoonup \mu$ are positive measures, and $f_k \le 0$, then every weak limit ϑ of $f_k\mu_k$ satisfies $\vartheta \le f\mu$. Indeed, fix k_0 and decrease continuous functions g_ℓ to f_{k_0}. For $k \ge k_0$, $f_k\mu_k \le g_\ell\mu_k$; first let $k \to \infty$, then $\ell \to \infty$, and finally $k_0 \to \infty$. Applying this lemma after wedging with every smooth strongly positive test form shows that any cluster limit Θ of $u_1^{(k)}S_k$ satisfies

$$\Theta \le u_1 S. \tag{15.1}$$

It remains to prove the reverse inequality. Since the question is local, fix a compact set L in a smoothly bounded strictly pseudoconvex domain $\Omega = \{\psi < 0\}$, where ψ is smooth, strictly psh, and $\mathrm{d}\psi \neq 0$ on $\partial\Omega$. Adding constants, assume $-M \le u_j^{(k)}, u_j \le -1$. For a small $\delta > 0$ with $L \subset \{\psi < -\delta\}$, set $A = M/\delta$ and replace every function by $\max\{A\psi, u_j^{(k)}\}$ or $\max\{A\psi, u_j\}$. This does not change it on L and makes all functions equal to $A\psi$ on one fixed boundary collar. Put $\beta = \mathrm{dd^c}\psi > 0$ and let (m,m) be the bidimension of S_k.

Choose smooth psh regularizations $u_j^{(k,\varepsilon)}$ from above which retain the common boundary value; one first regularizes in the interior and then glues to $A\psi$ on the collar with a regularized maximum. Repeated integration by parts has no boundary term and, using $u_j \le u_j^{(k,\varepsilon)}$ at each replacement, gives

$$\begin{aligned}\int_\Omega u_1 S \wedge \beta^m &\le \int_\Omega u_1^{(k,\varepsilon_1)}\mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T \wedge \beta^m \\ &\le \int_\Omega u_1^{(k,\varepsilon_1)}\mathrm{dd^c}u_2^{(k,\varepsilon_2)} \wedge \cdots \wedge \mathrm{dd^c}u_q^{(k,\varepsilon_q)} \wedge T \wedge \beta^m.\end{aligned}$$

For example, the first integration step replaces $u_1^{(k,\varepsilon_1)}\mathrm{dd^c}u_2$ by $u_2\mathrm{dd^c}u_1^{(k,\varepsilon_1)}$; the next inequality replaces u_2 by its larger regularization. Cycling through the factors produces the last line. Let $\varepsilon_q, \ldots, \varepsilon_1$ tend to zero in that order. The one-factor convergence already proved at the start of the induction yields

$$\int_\Omega u_1 S \wedge \beta^m \le \int_\Omega u_1^{(k)} S_k \wedge \beta^m.$$

After taking the limit along the chosen subsequence, (15.1) and this trace-mass inequality force $\Theta = u_1 S$: a positive current whose trace against the strictly positive form β^m has zero mass is zero. Every cluster limit is therefore the same, so $u_1^{(k)}S_k \rightharpoonup u_1 S$. Applying $\mathrm{dd^c}$ proves the theorem. □

Corollary 6.32: Symmetry of Bedford–Taylor products

The recursive product is symmetric in $u_1, \dots, u_q$ and hence does not depend on the order chosen in its definition.

Proof. On a smaller coordinate domain choose simultaneous smooth psh regularizations $u_j^{(k)} \downarrow u_j$. Their classical wedge product is symmetric. Monotone continuity identifies its limit with the recursively defined product in every chosen order. Adjacent transpositions therefore leave the limit unchanged. □

Remark (Polarization)

Mixed Monge–Ampère terms can be recovered from diagonal ones by polarization. In degree two,

$$2\,\mathrm{dd^c}v \wedge \mathrm{dd^c}w = (\mathrm{dd^c}(v+w))^2 - (\mathrm{dd^c}v)^2 - (\mathrm{dd^c}w)^2.$$

Expanding the right-hand side is the safest way to keep the factor 2.

Theorem 6.33: Local comparison principle

If $\Omega \Subset \mathbb{C}^n$ is a bounded domain, $u, v \in \mathrm{PSH}(\Omega) \cap L^\infty(\Omega)$, and $\liminf_{z\to\partial\Omega}(u-v) \geq 0$, then

$$\int_{\{u<v\}} (\mathrm{dd^c}v)^n \leq \int_{\{u<v\}} (\mathrm{dd^c}u)^n.$$

Proof. First suppose u, v are smooth and $u \geq v$ near the boundary. For a regular value $\varepsilon > 0$, set $E_\varepsilon = \{u + \varepsilon < v\} \Subset \Omega$ and

$$S = \sum_{j=0}^{n-1} (\mathrm{dd^c}v)^j \wedge (\mathrm{dd^c}u)^{n-1-j}.$$

This is a closed positive $(n-1, n-1)$-form, and the algebraic identity

$$(\mathrm{dd^c}v)^n - (\mathrm{dd^c}u)^n = \mathrm{dd^c}(v-u) \wedge S$$

gives, by Stokes,

$$\int_{E_\varepsilon} \left((\mathrm{dd^c}v)^n - (\mathrm{dd^c}u)^n\right) = \int_{\partial E_\varepsilon} \mathrm{d^c}(v-u) \wedge S.$$

On the boundary, $h = v - u - \varepsilon$ vanishes and is positive on the interior side. Its outward normal derivative is therefore nonpositive. Contracting the strongly positive form S with the complex tangent directions shows that the boundary integrand is this derivative times a nonnegative density. Hence the right-hand side is nonpositive and

$$\int_{E_\varepsilon} (\mathrm{dd^c}v)^n \leq \int_{E_\varepsilon} (\mathrm{dd^c}u)^n. \tag{15.2}$$

Sard's theorem approximates an arbitrary ε by regular values.

If u and v are continuous and psh, their decreasing smooth regularizations converge uniformly on compact sets by Dini's theorem. The boundary inequality for a fixed E_ε is therefore preserved after changing ε by an arbitrarily small amount, and monotone continuity passes (15.2) to the limit. A bounded psh function is quasi-continuous for Bedford–Taylor capacity: for every $\delta > 0$ it is continuous outside an open set G_δ with $\mathrm{Cap}(G_\delta, \Omega) < \delta$. Moreover, if $|w| \leq M$, then scaling $(w+M)/(2M)$ in the definition of capacity gives

$$\int_{G_\delta} (\mathrm{dd^c}w)^n \leq (2M)^n \,\mathrm{Cap}(G_\delta, \Omega).$$

Apply the continuous argument on $\Omega \setminus G_\delta$, absorb the omitted mass with this estimate, and let first $\delta \downarrow 0$ and then $\varepsilon \downarrow 0$. The sets E_ε increase to $\{u < v\}$, so monotone convergence of measures proves the assertion. □

The capacity used in the proof was declared in the notation ledger. It is tailored to the Monge–Ampère operator because its defining test measures are precisely $(\mathrm{dd^c} u)^n$ with normalized bounded psh potentials. A locally pluripolar set has capacity zero; conversely, a Borel set of relative capacity zero is pluripolar. Thus capacity is the correct replacement for measure when bounded psh functions are made continuous outside a negligible exceptional set.

Warning

Local boundedness is a hypothesis, not a technical convenience. In complex dimension $n \geq 2$, for $u = \log|z|$ a classical product $(\mathrm{dd^c} u)^n$ at the pole can depend on the approximation. Unbounded potentials require Cegrell classes or non-pluripolar products.

6.3.1 The determinant hidden in the wedge product

In complex dimension two, write at a point

$$\mathrm{dd^c} u = \frac{\mathrm{i}}{2\pi}\left(u_{1\bar{1}}\mathrm{d}z_1 \wedge \mathrm{d}\bar{z}_1 + u_{1\bar{2}}\mathrm{d}z_1 \wedge \mathrm{d}\bar{z}_2 + u_{2\bar{1}}\mathrm{d}z_2 \wedge \mathrm{d}\bar{z}_1 + u_{2\bar{2}}\mathrm{d}z_2 \wedge \mathrm{d}\bar{z}_2\right).$$

Only complementary indices survive after wedging, and reordering gives

$$(\mathrm{dd^c} u)^2 = \frac{2}{\pi^2}\left(u_{1\bar{1}}u_{2\bar{2}} - |u_{1\bar{2}}|^2\right)\mathrm{d}\lambda.$$

The coefficient is nonnegative exactly when the Levi matrix is semipositive. In dimension n, antisymmetrization produces $n!\det(u_{j\bar{k}})$. Bedford–Taylor's definition preserves this geometric determinant while avoiding products of distributional coefficients.

If $u(z) = \sum_j \lambda_j |z_j|^2$, then

$$(\mathrm{dd^c} u)^n = \frac{n!}{\pi^n}(\lambda_1 \cdots \lambda_n)\mathrm{d}\lambda.$$

This model is useful for checking constants, homogeneity, and the effect of a degenerate Levi direction.

6.3.2 Locality, capacity, and uniqueness

The Bedford–Taylor product is local in the plurifine topology. One basic consequence is the maximum formula: on the open set $\{u > v\}$,

$$(\mathrm{dd^c}\max\{u, v\})^n = (\mathrm{dd^c} u)^n,$$

while interchanging u and v gives $(\mathrm{dd^c}\max\{u, v\})^n = (\mathrm{dd^c} v)^n$ on $\{v > u\}$. Possible additional mass is concentrated on the contact set $\{u = v\}$. In one variable, $\max(\log|z|, 0)$ illustrates this phenomenon: it is harmonic on both sides of the unit circle and its Laplacian is carried by the contact circle.

Capacity is tailored to make monotone limits stable. If a sequence of uniformly bounded psh functions converges in Bedford–Taylor capacity, then outside a set of arbitrarily small capacity the convergence is uniform enough to pass products against the uniformly controlled Monge–Ampère measures. This is the nonlinear replacement for ordinary dominated convergence.

The comparison principle gives uniqueness for the Dirichlet problem. Suppose $u, v \in \mathrm{PSH}(\Omega) \cap C(\overline{\Omega})$ have the same boundary values and

$$(\mathrm{dd^c} u)^n = (\mathrm{dd^c} v)^n.$$

Apply comparison to u and $v + \varepsilon\rho$, where $\rho < 0$ is a strictly psh defining function. The strict extra mass rules out $u < v$ in the interior. Reversing the roles and letting $\varepsilon \downarrow 0$ gives $u = v$.

Remark (Global mass)

On a compact Kähler manifold, bounded ω-psh potentials satisfy

$$\int_X (\omega + \mathrm{dd^c} u)^n = \int_X \omega^n.$$

Approximate by smooth potentials and use Stokes term by term. The equality can fail for more singular products unless the chosen definition retains full Monge–Ampère mass.

6.3.3 The Bedford–Taylor recursion with the integration step shown

Let $u_1, \dots, u_k$ be locally bounded psh functions. Starting from a closed positive current T, define recursively

$$\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_k \wedge T := \mathrm{dd^c}\left(u_1\, \mathrm{dd^c} u_2 \wedge \cdots \wedge \mathrm{dd^c} u_k \wedge T\right).$$

The product $u_1 S$ is meaningful because S has measure coefficients and u_1 is locally bounded. For a compactly supported test form η,

$$\langle \mathrm{dd^c}(u_1 S), \eta \rangle = \langle u_1 S, \mathrm{dd^c} \eta \rangle.$$

This formula is the definition; no product of distributional second derivatives is taken.

The first limiting step involves only one varying function and one fixed current. Indeed, if $u_\nu \downarrow u$ are smooth psh functions, then $u_\nu S \to uS$ by bounded convergence with respect to the fixed coefficient measures of S. Hence

$$\mathrm{dd^c} u_\nu \wedge S = \mathrm{dd^c}(u_\nu S) \rightharpoonup \mathrm{dd^c}(uS).$$

This proves positivity of one recursive step without multiplying two varying measures.

Symmetry is a separate integration-by-parts lemma. For bounded psh u, v, a closed positive current S, and a compactly supported smooth test form η, regularize u and v on nested coordinate balls. In the smooth case $\mathrm{dd^c} u_\nu \wedge \mathrm{dd^c} v_\mu \wedge S$ is unchanged when the two factors are interchanged. Move derivatives onto η before taking either limit. The ordered CLN estimate proved below bounds every cutoff term uniformly, while bounded convergence applies whenever only one regularization parameter varies. First let $\nu \to \infty$, then $\mu \to \infty$; reversing the order gives the same current. Thus

$$\mathrm{dd^c} u \wedge \mathrm{dd^c} v \wedge S = \mathrm{dd^c} v \wedge \mathrm{dd^c} u \wedge S.$$

Adjacent transpositions give symmetry for any number of factors. Notice the logical order: the recursive product and its ordered mass bound come first; symmetry is then proved, rather than assumed as formal commutativity.

The smooth determinant and its normalization

For $u \in C^2$ psh on $\mathbb{C}^n$,

$$\mathrm{dd^c} u = \frac{\mathrm{i}}{2\pi} \sum_{j,k} u_{j\bar{k}}\, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k.$$

Diagonalize H_u at a point with eigenvalues $\lambda_1, \dots, \lambda_n$. Then

$$(\mathrm{dd^c} u)^n = n! \left(\prod_j \lambda_j \right) \bigwedge_{j=1}^{n} \frac{\mathrm{i}}{2\pi} \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j = \frac{n!}{\pi^n} \det(H_u)\, \mathrm{d}V.$$

For $u = |z|^2$, this gives $(\mathrm{dd^c}|z|^2)^n = n!\pi^{-n}\mathrm{d}V$. Integrating over the Euclidean unit ball, whose volume is $\pi^n/n!$, gives total mass one.

The Monge–Ampère mass of a radial maximum

In one variable let

$$u_R(z) = \max\{\log|z|^2, \log R^2\}.$$

The function is constant for $|z| < R$ and harmonic for $|z| > R$. Its outward radial derivative jumps from 0 to $2/R$ at the circle. Equivalently, $\mathrm{d^c} \log|z|^2 = (2\pi)^{-1}\mathrm{d}\theta$ outside. Stokes therefore gives

$$\int_{\mathbb{C}} \chi\, \mathrm{dd^c} u_R = \frac{1}{2\pi} \int_0^{2\pi} \chi(Re^{\mathrm{i}\theta})\, \mathrm{d}\theta.$$

Thus $\mathrm{dd^c} u_R$ is normalized arclength measure on $|z| = R$, with total mass one. The mass sits precisely on the contact set where the two psh branches meet.

The comparison principle in one dimension

Let u, v be bounded subharmonic functions on a disc with $\liminf_{z\to\partial D}(u - v) \geq 0$. For $\varepsilon > 0$ set $E_\varepsilon = \{u < v - \varepsilon\} \Subset D$. Approximate the maximum $w = \max\{u, v - \varepsilon\}$ from above by smooth subharmonic functions. Green's formula on E_ε gives

$$\int_{E_\varepsilon} \mathrm{dd^c} v \leq \int_{E_\varepsilon} \mathrm{dd^c} u.$$

Indeed, on its boundary the two functions agree and the outward normal derivative of the upper envelope cannot jump downward. Letting $\varepsilon \downarrow 0$ and using monotone convergence yields

$$\int_{\{u<v\}} \mathrm{dd^c} v \leq \int_{\{u<v\}} \mathrm{dd^c} u.$$

Bedford–Taylor's higher-dimensional proof repeats this maximum construction with mixed products and uses the recursive integration-by-parts definition at every step.

6.3.4 The Chern–Levine–Nirenberg induction

The local mass estimate is what makes the limiting arguments above legitimate. Choose $K = K_0 \Subset K_1 \Subset \cdots \Subset K_k \Subset \Omega$ and cutoffs χ_j such that $\chi_j = 1$ on K_{j-1} and $\operatorname{supp}\chi_j \subset K_j$. Let T be closed positive and let $u_1, \dots, u_k$ be locally bounded psh. Adding constants does not change $\mathrm{dd^c} u_j$, so on K_k we may suppose $0 \leq u_j \leq 2\|u_j\|_{L^\infty(K_k)}$.

For $k = 1$, positivity and integration by parts give

$$\begin{aligned} \int_{K_0} \mathrm{dd^c} u_1 \wedge T \wedge \omega^{m-1} &\leq \int \chi_1 \mathrm{dd^c} u_1 \wedge T \wedge \omega^{m-1} \\ &= \int u_1 \mathrm{dd^c} \chi_1 \wedge T \wedge \omega^{m-1}. \end{aligned}$$

The fixed cutoff satisfies $-C_1\omega \le \mathrm{dd^c}\chi_1 \le C_1\omega$. Consequently the last expression is bounded by

$$2C_1\|u_1\|_{L^\infty(K_1)}\int_{K_1} T\wedge\omega^m.$$

For the induction step, regard $S = \mathrm{dd^c}u_2\wedge\cdots\wedge\mathrm{dd^c}u_k\wedge T$ as the closed positive current and repeat the calculation with u_1 and χ_1. Apply the induction hypothesis on the larger compact set to obtain

$$\int_K \mathrm{dd^c}u_1\wedge\cdots\wedge\mathrm{dd^c}u_k\wedge T\wedge\omega^{m-k} \le C_{K,K_k}\prod_{j=1}^{k}\|u_j\|_{L^\infty(K_k)}\int_{K_k} T\wedge\omega^m.$$

For nonsmooth u_j, the same calculation is already legitimate: by definition, pairing $\mathrm{dd^c}(u_jS)$ with a cutoff means pairing u_jS with the cutoff's $\mathrm{dd^c}$. Thus no monotone-continuity theorem is needed to prove the estimate. This logical order matters: the CLN bound is subsequently used to prove monotone continuity, so invoking that theorem here would be circular.

6.3.5 Self-study workshop: defining Monge–Ampère products by recursion

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall locally bounded psh functions, positive currents, weak convergence, and integration by parts for currents. Products of arbitrary distributions are not defined, so every wedge product here needs a construction.

The recursive definition and its estimate. Let $u_1,\dots,u_q$ be locally bounded psh functions and let T be a closed positive current. Assuming

$$S = \mathrm{dd^c}u_2\wedge\cdots\wedge\mathrm{dd^c}u_q\wedge T$$

has already been defined, set

$$\mathrm{dd^c}u_1\wedge S := \mathrm{dd^c}(u_1S).$$

Local boundedness makes u_1S a current of order zero. Closedness of S ensures that the definition agrees with the classical wedge product when the potentials are smooth. For a test form η,

$$\langle\mathrm{dd^c}(u_1S),\eta\rangle = \langle u_1S,\mathrm{dd^c}\eta\rangle,$$

up to the fixed degree convention. This formula supplies the local mass estimates needed for continuity.

If $u_j^{(\nu)}\downarrow u_j$, Bedford–Taylor monotone convergence states that the corresponding products converge weakly. The order of recursion does not matter; the proof regularizes, uses symmetry in the smooth case, and passes to the limit with Chern–Levine–Nirenberg bounds. None of these statements follows from formal commutativity of symbols.

For smooth u, the top power is a determinant:

$$(\mathrm{dd^c}u)^n = \frac{n!}{\pi^n}\det(u_{j\bar k})\,\mathrm{d}\lambda,$$

where $\mathrm{d}\lambda$ is Euclidean Lebesgue measure. Positivity of the Hessian guarantees that this is a positive measure.

Checkpoint (why boundedness appears)

(1) Where is local boundedness used in the recursive definition?
(2) Why can one not simply multiply the coefficient distributions of two currents?
(3) What convergence direction is built into the basic Bedford–Taylor theorem?

Solution (checkpoint). It makes the product $u_1 S$ well defined because S has measure coefficients. Products of distributions have no canonical definition and can depend on regularization. The basic theorem uses decreasing sequences of locally bounded psh functions; more general convergence requires capacity or energy hypotheses. ⋄

Exit criterion

You can write the recursion on a test form, identify the role of boundedness and closedness, and explain why monotone convergence is a theorem rather than a formal limit operation.

Summary. Bedford–Taylor defines Monge–Ampère products by recursive integration by parts, not multiplication of distributions. CLN gives compactness, decreasing continuity legitimizes approximation, and comparison gives uniqueness.

Exercise 6.34: Monge–Ampère of a maximum

Compute $\mathrm{dd}^{\mathrm{c}} \max(\log|z|, 0)$ on $\mathbb{C}$ and locate its mass.

Exercise 6.35: Comparison in one dimension

Reduce the local comparison principle for $n = 1$ to Green's formula and complete the proof.

Optional handout: Finite-energy classes in pluripotential theory

This handout is optional. No later argument in the core lectures depends on it.

Bedford–Taylor theory treats locally bounded psh functions. On a compact Kähler manifold, many natural solutions are unbounded, so one uses non-pluripolar products and energy classes. For ω-psh u, define truncations $u_j = \max(u, -j)$. The non-pluripolar Monge–Ampère measure is

$$\mathrm{MA}(u) = \lim_{j\to\infty} 1_{\{u>-j\}}(\omega + \mathrm{dd}^{\mathrm{c}} u_j)^n.$$

It never charges the pole set $\{u = -\infty\}$.

The full-mass class $\mathcal{E}(X, \omega)$ consists of those u for which $\int_X \mathrm{MA}(u) = \int_X \omega^n$. The finite-energy class $\mathcal{E}^1$ additionally requires

$$\int_X |u|\ \mathrm{MA}(u) < \infty$$

after a normalization such as $\sup_X u = 0$. The Aubin–Mabuchi energy extends to $\mathcal{E}^1$ and makes the complex Monge–Ampère equation amenable to variational methods.

Remark (Variational principle)

To solve $\mathrm{MA}(u) = \mu$, one may maximize $E(u) - \int_X u\,\mathrm{d}\mu$ on $\mathcal{E}^1$. Concavity of E, compactness of normalized sublevel sets, and a finite-energy condition on μ give existence; the comparison principle gives uniqueness up to constants.

This framework contains singular Kähler–Einstein metrics and geodesics in the space of Kähler potentials. It is genuinely beyond the locally bounded theory but preserves its two central principles: monotone approximation and comparison.

Exercise 6.36: Bounded potentials

Show that every bounded ω-psh function belongs to $\mathcal{E}^1(X, \omega)$.

6.4 Lecture 24: Lelong numbers, Jensen's formula, and Siu's theorem

Week 12 material • 90 minutes

Notation for this lecture

T is a closed positive current of bidimension (p,p) near a point a, $\beta_a = \mathrm{dd^c}|z-a|^2$, and $\alpha_a = \mathrm{dd^c}\log|z-a|^2$ on $X\setminus\{a\}$. Products involving α_a at a are defined by replacing the logarithm with $\max\{\log|z-a|^2, \log\varepsilon^2\}$ and then letting $\varepsilon\downarrow 0$. The Euclidean ball of center a and radius r is denoted $B(a,r)$, and $\Theta_T(a,r) = r^{-2p}\int_{B(a,r)} T\wedge\beta_a^p$ is its normalized mass at scale r. Its Lelong number is $\nu(T,a) = \lim_{r\downarrow 0}\Theta_T(a,r)$, with the normalization stated in the definition below. For a psh function u we write $\nu(u,a) = \nu(\mathrm{dd^c}u,a)$ and $E_c(u) := E_c(\mathrm{dd^c}u)$, and $E_c(T) = \{x : \nu(T,x)\geq c\}$ is the upper level set. In the Siu decomposition $T = R + \sum_j \lambda_j[A_j]$, the A_j are irreducible p-dimensional analytic sets, λ_j are their generic Lelong numbers, and R is the residual positive current. For a $(1,1)$-current, $p = n-1$ and the A_j are prime divisors, often denoted D_j. For a Borel set A, $1_A T$ means restriction of the coefficient measures of the order-zero current T to A; this operation does not preserve closedness for an arbitrary set. Its use for analytic sets is justified by the Skoda–El Mir theorem in Lecture 22.

Prerequisite results used below

Poincaré–Lelong formula. For a nonzero meromorphic function f, $\mathrm{dd^c}\log|f|^2 = [\mathrm{div}(f)]$ with the normalization of these notes. In particular, if $f(z) = z^m g(z)$ and $g(0)\neq 0$, then $\nu(\mathrm{dd^c}\log|f|^2, 0) = m$.

Jensen formula. If f is holomorphic on a neighborhood of $\overline{D(0,R)}$, $f(0)\neq 0$, and a_k are its zeros counted with multiplicity, then

$$\log|f(0)| = \frac{1}{2\pi}\int_0^{2\pi}\log|f(Re^{i\theta})|\,d\theta - \sum_{|a_k|<R}\log\frac{R}{|a_k|}.$$

Applied to local potentials, this identity gives the monotonicity of normalized masses used to define Lelong numbers.

Why these ideas are needed

A positive current can be singular at a point without carrying an obvious atom, so we need an invariant numerical measure of its concentration. The Lelong number is a scale-normalized density designed to recover the vanishing order of a holomorphic function through Poincaré–Lelong. Jensen's formula explains the link between radial growth and concentrated mass. Siu's theorem supplies the unexpected geometric payoff: the loci where the Lelong number exceeds a fixed threshold are analytic. Thus a quantitative invariant of a weak object produces rigid complex subvarieties.

Plan for the lecture. 0–22 min: the mass definition of Lelong number; 22–42: psh slopes and Jensen–Lelong; 42–60: comparison; 60–78: Siu analyticity; 78–90: decomposition and midterm review.

The exponent $2p$ in the definition below is forced by the real scaling of a complex p-plane. What requires proof is that the corresponding normalized mass is monotone, hence has a limit.

Proposition 6.37: Lelong–Jensen monotonicity formula

Let T be a closed positive current of bidimension (p,p) in a coordinate ball centered at a. If $0 < r_1 < r_2$ and $B(a,r_2)$ lies in the coordinate ball, then

$$\Theta_T(a,r_2) - \Theta_T(a,r_1) = \int_{\{r_1<|z-a|<r_2\}} T \wedge \alpha_a^p \geq 0. \tag{16.1}$$

In particular, $r \mapsto \Theta_T(a,r)$ is nondecreasing.

Proof. Put $\varphi = \log|z-a|^2$ and, for $0 < \varepsilon < r$, set $\varphi_\varepsilon = \max\{\varphi, \log\varepsilon^2\}$. The Bedford–Taylor product

$$N_T(a,r) := \int_{B(a,r)} T \wedge (\mathrm{dd^c}\varphi_\varepsilon)^p$$

is independent of ε: two truncations agree near $\partial B(a,r)$, and repeated integration by parts moves every $\mathrm{dd^c}$ onto the other factors, which are closed. The change-of-weight formula with the convex increasing function $\chi(t) = \mathrm{e}^t$ gives

$$\int_{B(a,r)} T \wedge \left(\mathrm{dd^c}(\mathrm{e}^{\varphi})\right)^p = \mathrm{e}^{p\log r^2} N_T(a,r) = r^{2p} N_T(a,r).$$

Since $\mathrm{e}^{\varphi} = |z-a|^2$, the left side is $\int_{B(a,r)} T \wedge \beta_a^p$. Hence $N_T(a,r) = \Theta_T(a,r)$. On the annulus $r_1 < |z-a| < r_2$, the truncations equal φ, so subtraction gives (16.1). The form $\alpha_a = \mathrm{dd^c}\varphi$ is semipositive away from a; positivity of T makes the last integral nonnegative. □

Definition 6.38: Lelong number

The Lelong number of T at a is

$$\nu(T,a) = \lim_{r\downarrow 0} \frac{1}{r^{2p}} \int_{B(a,r)} T \wedge \beta_a^p,$$

where the limit exists by the preceding monotonicity formula. This normalization satisfies $\nu([\mathbb{C}^p],0) = 1$.

Theorem 6.39: Slope characterization for psh functions

For psh u,

$$\nu(u,a) := \nu(\mathrm{dd^c}u,a) = \liminf_{z\to a} \frac{u(z)}{\log|z-a|^2} = \lim_{r\downarrow 0} \frac{\sup_{|z-a|=r} u(z)}{\log r^2}.$$

Proof. Translate a to the origin. Let $A(r)$ be the spherical mean of u and $M(r) = \sup_{|z|=r} u(z)$. Write $\mu = \mathrm{dd^c}u$ and $\beta_0 = \mathrm{dd^c}|z|^2$. Jensen's formula, first for smooth u and then by convolution, gives

$$A(r) - A(r_0) = 2\int_{r_0}^{r} \left(\frac{1}{t^{2n-2}} \int_{B(0,t)} \mu \wedge \beta_0^{n-1} \right) \frac{\mathrm{d}t}{t}$$

with the constants corresponding to our convention $\mathrm{dd^c}\log|z|^2 = \delta_0$. The normalized mass in the integrand is nondecreasing and tends to $\nu(u,0)$. Dividing by $\log r^2$ and applying the elementary Cesàro lemma therefore gives $A(r)/\log r^2 \to \nu(u,0)$.

The submean inequality on concentric balls and convexity of radial maxima give the radial comparison estimate

$$A(r) \leq M(r) \leq A(2r) + O(1)$$

after u is normalized above on one fixed ball. The additive term disappears after division by $\log r^2$, and replacing r by $2r$ does not change the limit. Hence $M(r)/\log r^2 \to \nu(u,0)$. Finally, for $|z| = r$ one has $u(z) \le M(r)$; because $\log r^2 < 0$, division reverses the inequality. This gives the lower bound for the pointwise $\liminf$, while choosing on each sphere a point where the supremum is attained gives the reverse bound. All three displayed quantities therefore equal $\nu(u,0)$. □

Example 6.40: Analytic singularities

If $u = c\log\sum_{j=1}^{N}|f_j|^2 + O(1)$, then $\nu(u,a) = c\min_j \operatorname{ord}_a(f_j)$ in the $\log|z|^2$ convention. In particular, $\nu(\log|f|^2, a) = \operatorname{ord}_a(f)$.

Theorem 6.41: Comparison of singularities

If $u \le v + O(1)$ near a, then $\nu(u,a) \ge \nu(v,a)$: the more negative singularity has the larger Lelong number. Mixed Monge–Ampère comparison generalizes this relation to mass inequalities.

Proof. Choose C with $u \le v + C$ near a. On the sphere of radius r, $\sup u \le \sup v + C$. Since $\log r^2 < 0$, division reverses the inequality:

$$\frac{\sup u}{\log r^2} \ge \frac{\sup v}{\log r^2} + \frac{C}{\log r^2}.$$

Letting $r \downarrow 0$ and using the slope characterization makes the last term vanish and yields $\nu(u,a) \ge \nu(v,a)$. □

Warning

Because $\log r^2 < 0$, dividing an inequality by $\log r^2$ reverses its direction. Also, Lelong number zero does not imply local boundedness; for example $u = -\sqrt{-\log|z|}$ is still singular.

Proposition 6.42: Closedness and size of Lelong upper-level sets

For every $c > 0$, the set $E_c(T) = \{x : \nu(T,x) \ge c\}$ is closed. If T has bidimension (p,p), then $E_c(T)$ has locally finite $2p$-dimensional Hausdorff measure $\mathcal{H}^{2p}$.

Proof. Fix a_0 with $\nu(T,a_0) < c$. Monotonicity supplies a radius $s > 0$ with $\Theta_T(a_0,s) < c$. Choose $r < s$ so close to s that $(s/r)^{2p}\Theta_T(a_0,s) < c$. If $|a - a_0| < s - r$, then $B(a,r) \subset B(a_0,s)$; since $\beta_a = \beta_{a_0}$ in Euclidean coordinates,

$$\nu(T,a) \le \Theta_T(a,r) \le \left(\frac{s}{r}\right)^{2p}\Theta_T(a_0,s) < c.$$

Thus the complement of $E_c(T)$ is open.

For the measure bound, fix a compact set K in a coordinate ball and choose a maximal finite family $a_1,\dots,a_N \in E_c(T)\cap K$ whose mutual distances are at least 2ε. The balls $B(a_j,\varepsilon)$ are disjoint, whereas the balls $B(a_j,2\varepsilon)$ cover $E_c(T)\cap K$. Monotonicity gives

$$\int_{B(a_j,\varepsilon)} T\wedge\beta_0^p \ge c\varepsilon^{2p}.$$

If K_ε is the ε-neighborhood of K, summing the disjoint-ball inequalities yields $Nc\varepsilon^{2p} \le (T\wedge\beta_0^p)(K_\varepsilon)$. The covering by 2ε-balls therefore has uniformly bounded $2p$-dimensional Hausdorff content. Letting $\varepsilon \downarrow 0$ proves local finiteness of $\mathcal{H}^{2p}$. □

Theorem 6.43: Siu's analyticity theorem

For a closed positive current T and $c > 0$, the upper level set

$$E_c(T) = \{x \in X : \nu(T, x) \geq c\}$$

is analytic; if T has bidimension (p, p), then $\dim E_c(T) \leq p$. Thus $x \mapsto \nu(T, x)$ has a strong form of upper semicontinuity in the analytic Zariski topology.

Proof (proof with the analytic inputs stated). We first treat a $(1,1)$-current locally written $T = \mathrm{dd}^c u$. Work on a relatively compact pseudoconvex ball Ω and then restrict to a smaller ball $\Omega' \Subset \Omega$. For $m \geq 1$, let $\mathcal{H}_m$ be the Hilbert space of holomorphic functions on Ω with $\int_\Omega |f|^2 e^{-2mu}\, \mathrm{d}\lambda < \infty$, choose an orthonormal basis $(g_{m,j})$, and set on Ω'

$$u_m = \frac{1}{2m} \log \sum_j |g_{m,j}|^2.$$

The analytic input is the Bergman approximation lemma. It follows from the submean inequality and a weighted $\bar\partial$ estimate: for $B(z, r) \Subset \Omega$,

$$u(z) - \frac{C}{m} \leq u_m(z) \leq \sup_{B(z,r)} u + \frac{1}{m} \log \frac{C}{r^n},$$

and consequently

$$\nu(u, z) - \frac{n}{m} \leq \nu(u_m, z) \leq \nu(u, z). \tag{16.2}$$

The upper bound comes directly from the submean inequality. For the lower bound, solve $\bar\partial$ for a cutoff equal to one near z with weight $2mu + 2n \log|\zeta - z|$. The solution vanishes at z, and subtracting it from the cutoff gives a holomorphic competitor with prescribed value and controlled weighted norm. The extremal characterization of the Bergman kernel yields the lower bound. The required weighted estimate is proved in Lecture 27; equation (16.2) is the exact interface needed here, so no later conclusion is being used implicitly.

The function u_m has analytic singularities: locally its singularity is defined by the coherent ideal generated by the $g_{m,j}$; coherence makes this ideal locally finitely generated even though the displayed basis can be infinite. More explicitly, $\nu(u_m, z) \geq a$ means that every local generator has vanishing order at least $\lceil 2ma \rceil$. This is the vanishing of finitely many jets and therefore defines an analytic set. The two inequalities give the exact identity

$$E_c(u) = \bigcap_{m \geq m_0} E_{c - n/m}(u_m).$$

Here m_0 is chosen so that $c - n/m_0 > 0$. For $N \geq m_0$, let $A_N = \bigcap_{m_0 \leq m \leq N} E_{c-n/m}(u_m)$. The A_N are analytic and form a decreasing sequence. The ascending chain of their ideal sheaves stabilizes locally by Noetherianity, so $\bigcap_N A_N$ is analytic. This proves the theorem for $(1,1)$ currents.

For a positive current of arbitrary bidimension, the second input is Lelong's projection lemma. In the form used here it states that, after localization and a generic linear projection to a $(p+1)$-dimensional space, suitable pushforwards of a bidimension (p,p) current are positive $(1,1)$-currents; finitely many such projections detect the inequality $\nu(T, x) \geq c$. It also gives the Hausdorff-mass bound needed in the reduction. A complete construction of the projection family and the associated parameter-space potentials is given in [1, Chapter III, §8]; the precise interface used here is the finite detection of the Lelong threshold by $(1,1)$ pushforwards. Applying the $(1,1)$ result to these pushforwards produces local analytic equations for $E_c(T)$, and intrinsicness of the Lelong number makes

the local sets agree on overlaps. This projection lemma is the geometric reduction in the general-current proof; the Bergman approximation above is the analytic core after the reduction. Finally, the preceding Hausdorff-measure proposition and analyticity imply $\dim E_c(T) \le p$. □

Theorem 6.44: Siu decomposition

If T is a closed positive current of bidimension (p,p), then uniquely

$$T = \sum_j \lambda_j [A_j] + R,$$

where the A_j are irreducible p-dimensional analytic sets and $\lambda_j > 0$ are the generic Lelong numbers along them. For every $c > 0$, $E_c(R)$ has dimension $< p$.

Proof. For each rational $c > 0$, Siu's theorem makes $E_c(T)$ analytic. List all its p-dimensional irreducible components as c varies; local finiteness of the components and second countability of X make this a countable family (A_j). For an irreducible analytic set A, put

$$\lambda_A := \inf_{x \in A} \nu(T,x).$$

For every rational $c > \lambda_A$, the set $A \cap E_c(T)$ is a proper analytic subset of A. Hence $\nu(T,x) = \lambda_A$ outside a countable union of proper analytic subsets of A; this is the precise meaning of the generic Lelong number. Write $\lambda_j = \lambda_{A_j}$.

We use the component-extraction lemma. If A is an irreducible p-dimensional analytic set and λ is the generic Lelong number of T along A, then

$$1_A T = \lambda [A], \qquad T - \lambda [A] \ge 0. \tag{16.3}$$

Here $1_A T$ means restriction of the measure coefficients of the order-zero current T to A. Apply Skoda–El Mir to $T|_{X\setminus A}$: its extension $1_{X\setminus A}T$ is closed and positive, so $1_A T = T - 1_{X\setminus A}T$ is closed; it is positive directly because it is the restriction of the coefficient measures of T to A. The support theorem gives

$$1_A T = \mu [A]$$

for some $\mu \ge 0$.

We compare μ with $\lambda = \inf_A \nu(T,\cdot)$. Since $T \ge 1_A T = \mu[A]$ and $\nu([A],x) = 1$ at a general smooth point of A, one has $\lambda \ge \mu$. For the reverse inequality, work near a smooth point in coordinates where $A = \mathbb{C}^p \times \{0\}$ and put $\beta = \mathrm{dd^c}|z|^2$. The density bound $\nu(T,a) \ge \lambda$ gives, for every sufficiently small r and every $a \in A$,

$$\int_{B(a,r)} T \wedge \beta^p \ge \lambda r^{2p}.$$

Multiply by a nonnegative compactly supported continuous function $f(a)$ on A and integrate in a. Fubini's theorem introduces

$$g_r(z) = \int_{A \cap B(z,r)} f(a)\, \mathrm{d}V_A(a),$$

where $\mathrm{d}V_A$ is Euclidean volume on A. If $v_p = \pi^p/p!$ is the volume of the unit ball in $\mathbb{C}^p$, then $g_r/(v_p r^{2p})$ converges to f on A and to zero off A, with a uniform bound. Since $\beta^p|_A = v_p^{-1}\mathrm{d}V_A$, dominated convergence therefore yields

$$\int_A f\, 1_A T \wedge \beta^p \ge \lambda \int_A f\, [A] \wedge \beta^p.$$

Using $1_A T = \mu[A]$ gives $\mu \ge \lambda$. Thus $\mu = \lambda$, and (16.3) follows because $T - 1_A T = 1_{X\setminus A}T$ is positive.

Apply (16.3) successively. For every N,

$$R_N := T - \sum_{j=1}^{N} \lambda_j [A_j] \geq 0.$$

At the N-th step, a general point of A_N lies on none of $A_1, \dots, A_{N-1}$, so subtracting the earlier components has not changed the generic Lelong number along A_N. The component-extraction lemma therefore applies to the successive residual current. The sequence (R_N) decreases in the positive-current order, and its masses on each compact set are bounded by the mass of T. Weak compactness therefore gives a closed positive cluster limit. The limit is unique: pairing R_N with any strongly positive test form gives a decreasing sequence, and arbitrary test forms are linear combinations of strongly positive ones. Equivalently, the increasing partial sums are locally weakly convergent. Thus

$$R = T - \sum_{j} \lambda_j [A_j] \geq 0$$

is closed.

If $E_c(R)$ had a p-dimensional component, its generic Lelong number would have been one of the λ_j and would already have been removed. Therefore $\dim E_c(R) < p$. Finally, the support principle says that the coefficient of $[A_j]$ in any positive decomposition is its generic Lelong number along A_j. Hence both the coefficients and the residual current are unique. □

6.4.1 Monotonicity and density

For a closed positive current T of bidimension (p, p) in a Euclidean ball, set

$$\Theta_T(a, r) = \frac{1}{r^{2p}} \int_{B(a,r)} T \wedge (\mathrm{dd}^{\mathrm{c}} |z - a|^2)^p.$$

The Jensen–Lelong formula shows that $r \mapsto \Theta_T(a, r)$ is nondecreasing. The Lelong number is its limit at zero, so it is an infinitesimal density rather than an arbitrary singularity index. For a smooth positive form of positive bidegree the numerator is $O(r^{2n})$ and the Lelong number vanishes; the exceptional bidimension $p = n$ case is a closed nonnegative function, hence a constant, whose density equals that constant. For the integration current of a smooth p-fold the numerator is exactly $r^{2p} + o(r^{2p})$, giving density one.

If an analytic set A has multiplicity m at a, then

$$\nu([A], a) = m.$$

One way to see this is to project generically to a p-plane. A nearby regular value has m preimages counted with multiplicity, and the normalized volume ratio tends to that degree. Thus the analytic multiplicity and the metric density are the same integer.

6.4.2 What the number sees and what it misses

For an ideal $\mathfrak{a} = (f_1, \dots, f_N)$ and $u = c \log \sum |f_j|^2 + O(1)$, the Lelong number is

$$\nu(u, a) = c \, \operatorname{ord}_a(\mathfrak{a}), \qquad \operatorname{ord}_a(\mathfrak{a}) = \min_j \operatorname{ord}_a(f_j).$$

It records only the least vanishing order. Different ideals may therefore have the same Lelong number while having very different zero schemes or integral closures.

The example

$$u(z) = -\sqrt{-\log |z|}$$

near the origin is unbounded but satisfies $u(z)/\log|z|^2 \to 0$; hence its Lelong number is zero. Lelong numbers detect logarithmic poles, not every psh singularity. Multiplier ideals and non-pluripolar products will retain finer integrability and mass information.

For a positive closed $(1,1)$-current, Siu decomposition takes the especially concrete form

$$T = \sum_D \nu(T, D)[D] + R,$$

where D runs over prime divisors and R has zero generic Lelong number along every divisor. The residual current may still have positive Lelong numbers on subsets of codimension at least two. Thus "residual" means no divisorial component, not smooth or nonsingular.

Remark (Blowing up a point)

Pulling a psh singularity to the blow-up separates radial order from directional behavior. The generic Lelong number along the exceptional divisor equals the Lelong number at the center, while exceptional points can carry larger directional singularities. This is a first glimpse of how modifications resolve analytic singularities.

6.4.3 Checking the normalization on a linear space

Let $L = \mathbb{C}^p \times \{0\} \subset \mathbb{C}^n$ and $\beta = \mathrm{dd^c}|z|^2$. Restriction to L gives

$$\beta|_L = \frac{1}{\pi}\sum_{j=1}^{p} \mathrm{d}x_j \wedge \mathrm{d}y_j, \qquad \beta^p|_L = \frac{p!}{\pi^p}\,\mathrm{d}V_{\mathbb{R}^{2p}}.$$

Since $\mathrm{Vol}(B_r^{2p}) = \pi^p r^{2p}/p!$,

$$\frac{1}{r^{2p}}\int_{B_r} [L] \wedge \beta^p = 1.$$

This verifies the normalization $\nu([L], 0) = 1$ directly rather than by declaration.

For an analytic curve C through 0, a generic linear projection to one coordinate disc has degree equal to the multiplicity $m_0(C)$. Away from its branch values, the curve consists of $m_0(C)$ graphs, each asymptotic to a complex line. The area formula and the preceding normalization give

$$\nu([C], 0) = m_0(C).$$

For the cusp $(x, y) = (t^2, t^3)$, a generic projection has degree two, hence $\nu([C], 0) = 2$.

Analytic singularities and vanishing order

Let $u = c\log\sum_{j=1}^N |f_j|^2 + O(1)$ and put $m = \min_j \mathrm{ord}_0 f_j$. Taylor estimates give constants $C_1, C_2 > 0$ and, after choosing a direction on which one initial form does not vanish,

$$\sum_j |f_j(z)|^2 \le C_1|z|^{2m}, \qquad \sup_{|z|=r}\sum_j |f_j(z)|^2 \ge C_2 r^{2m}.$$

Taking logarithms shows

$$\sup_{|z|=r} u(z) = cm\log r^2 + O(1).$$

Dividing by $\log r^2$ and using the slope characterization yields $\nu(u, 0) = cm$.

The converse implication is false: zero Lelong number does not mean boundedness. For $0 < |z| < e^{-1}$ let

$$u(z) = -\sqrt{-\log|z|^2}.$$

As a convex increasing function of $\log|z|^2$ near $-\infty$, it is subharmonic and tends to $-\infty$. Yet

$$\frac{u(z)}{\log|z|^2} = \frac{1}{\sqrt{-\log|z|^2}} \longrightarrow 0,$$

so $\nu(u,0) = 0$.

A Siu decomposition one can read off

On $\mathbb{C}^2$ take

$$u = \log|z_1^2 z_2^3|^2 + |z|^2.$$

Poincaré–Lelong and smoothness of the last term give

$$\mathrm{dd^c} u = 2[z_1 = 0] + 3[z_2 = 0] + \mathrm{dd^c}|z|^2.$$

At a generic point of $\{z_1 = 0\}$ the Lelong number is 2; at a generic point of $\{z_2 = 0\}$ it is 3; at the origin it is 5. The Siu decomposition extracts the two divisorial currents with coefficients 2 and 3, leaving the smooth residual current. Accordingly,

$$E_c(\mathrm{dd^c} u) = \begin{cases} \{z_1 z_2 = 0\}, & 0 < c \le 2, \\ \{z_2 = 0\}, & 2 < c \le 3, \\ \{0\}, & 3 < c \le 5, \\ \varnothing, & c > 5. \end{cases}$$

Every upper-level set is analytic, and the jump at the intersection illustrates why the residual condition is stated in terms of dimension rather than pointwise zero Lelong number.

Monotonicity gives existence of the density

For a closed positive current T of bidimension (p,p) centered at the origin, the Jensen–Lelong formula shows that

$$\theta_T(r) = r^{-2p} \int_{B_r} T \wedge \beta^p$$

is nondecreasing in r. Hence $\lim_{r \downarrow 0} \theta_T(r)$ exists as the infimum of the finite-radius densities. Closedness removes interior derivative terms in Stokes' formula, while positivity makes the remaining annular integral nonnegative. Both hypotheses are essential: without positivity the density can oscillate, and without closedness there is an uncontrolled error term.

6.4.4 Where monotonicity comes from in the Jensen–Lelong argument

Let T be a closed positive current of bidimension (p,p) near the origin and put

$$\varphi(z) = \log|z|^2, \qquad \varphi_\varepsilon = \max\{\varphi, \log\varepsilon^2\}, \qquad \beta = \mathrm{dd^c}|z|^2.$$

For $0 < \varepsilon < r$, define

$$N_T(r) = \int_{|z|<r} T \wedge (\mathrm{dd^c}\varphi_\varepsilon)^p.$$

Two choices of ε agree near $|z| = r$. Repeated integration by parts and closedness of T show that their integrals are equal, so $N_T(r)$ is well defined.

The exact change-of-weight calculation uses $\chi(t) = \mathrm{e}^t$. Since $\chi \circ \varphi = |z|^2$ and $\chi'(\log r^2) = r^2$, integration by parts on the sublevel set $\{\varphi < \log r^2\}$ gives

$$\int_{|z|<r} T \wedge \beta^p = r^{2p} N_T(r).$$

There is no unspecified dimensional constant: the normalization $\beta = \mathrm{dd^c}|z|^2$ fixes it. Consequently

$$\Theta_T(0,r) = N_T(r), \qquad \Theta_T(0,r_2) - \Theta_T(0,r_1) = \int_{r_1<|z|<r_2} T \wedge (\mathrm{dd^c} \log |z|^2)^p \geq 0.$$

Positivity gives the last inequality, while closedness is what permits the integration by parts. This proves both monotonicity and existence of the density limit without appealing to a hidden constant.

6.4.5 Self-study workshop: extracting multiplicity from density

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall positive closed currents, Poincaré–Lelong, monotonicity of normalized masses, and orders of vanishing of holomorphic tuples.

The scale-invariant quantity. For a positive closed current T of bidimension (p,p) near 0, the Lelong number is the limit of a normalized mass

$$\nu(T,0) = \lim_{r \downarrow 0} \frac{1}{r^{2p}} \int_{B(0,r)} T \wedge (\mathrm{dd^c}|z|^2)^p,$$

with the dimensional constant determined by the convention ledger. The exponent $2p$ is forced by real scaling on a complex p-plane. The Jensen–Lelong formula proves monotonicity, so the limit exists.

For a psh function with analytic singularities,

$$u = \log\bigl(|f_1|^2 + \cdots + |f_N|^2\bigr) + O(1),$$

the Lelong number at the origin is the minimum vanishing order of the tuple. For

$$u(z_1,z_2) = \log(|z_1|^4 + |z_2|^6)$$

the generic leading order is two, hence $\nu(u,0) = 2$. Along the special line $z_1 = 0$ the restriction has order three, but the ambient Lelong number records the least radial growth, not the largest order on a special direction.

For an analytic cycle, the density equals algebraic multiplicity. Thus Lelong numbers translate analytic concentration into local intersection data. Siu's theorem adds a global rigidity statement: for $c > 0$, the upper level set $\{x : \nu(T,x) \geq c\}$ is analytic. Upper semicontinuity here is in the analytic Zariski sense, which is much stronger than ordinary topological semicontinuity.

Checkpoint (calculate before quoting Siu)

(1) What is $\nu(\mathrm{dd^c} \log |z^m|^2, 0)$ in one variable?
(2) Why does adding a bounded function not change a Lelong number?
(3) For the analytic singularity above, why does the order-three axis not change the ambient value two?

Solution (checkpoint). Poincaré–Lelong gives $m\delta_0$, whose density is m. A bounded summand is negligible after division by $\log|z|$ and has no point mass under the density limit. The Lelong number measures the uniform leading radial rate; generic directions with $z_1 \neq 0$ already realize order two, while extra vanishing on one direction only makes the singularity stronger there. ◇

Exit criterion

You can recover the scaling exponent in the density, compute Lelong numbers for divisors and analytic singularities, and state precisely what Siu's analyticity theorem adds.

Summary. The Lelong number is the normalized infinitesimal density of a positive current and the logarithmic slope of a psh singularity. Siu's theorem makes its threshold sets analytic, and Siu decomposition extracts all pure-dimensional analytic mass.

Exercise 6.45: Lelong number of an integration current

Prove $\nu([A], x) = 1$ at a smooth point. Compute the Lelong number at the origin for $A = \{y^2 = x^3\}$.

Exercise 6.46: A singularity with zero Lelong number

Verify that $u(z) = -(-\log|z|)^\alpha$, $0 < \alpha < 1$, is psh on a small punctured disc with its natural extension, and compute its Lelong number at the origin.

Optional handout: Tangent cones, multiplicity, and Lelong numbers

This handout is optional. No later argument in the core lectures depends on it.

Let $A \subset \mathbb{C}^n$ be a pure p-dimensional analytic set through the origin. Rescale A by $z \mapsto z/r$. Along a sequence $r_j \downarrow 0$, the integration currents have locally bounded mass and converge to a positive conic current. The resulting analytic cycle is the tangent cone $C_0 A$.

The Lelong number $\nu([A], 0)$ equals the algebraic multiplicity of A at the origin. One way to see this is to choose a generic projection $\pi : \mathbb{C}^n \to \mathbb{C}^p$. Near the origin, A is a branched cover of a ball in $\mathbb{C}^p$; the number of sheets, counted with multiplicity, is constant off the branch locus and equals both the Hilbert–Samuel multiplicity and the normalized density of $[A]$.

Example 6.47: The cusp

For $A = \{y^2 = x^3\}$, the lowest homogeneous term is y^2. Hence the tangent cone is the x-axis with multiplicity two, and $\nu([A], 0) = 2$. The set-theoretic tangent line alone would miss this multiplicity.

For a general positive current, tangent currents need not be unique. The Lelong number is nevertheless unique because it records only the normalized radial mass. This is one reason Lelong numbers are robust invariants even when the full tangent geometry is not.

Exercise 6.48: A plane branch

For the curve parametrized by $t \mapsto (t^m, t^k)$ with $m < k$ and $\gcd(m, k) = 1$, determine its tangent cone and its Lelong number at the origin.

Chapter 7

Bochner Identities, Vanishing Theorems, and Hörmander L^2 Methods

7.1 Lecture 25: The Bochner–Kodaira–Nakano identity

Week 13 material • 90 minutes

Notation for this lecture

(X, ω) is a Kähler manifold and (E, h) a Hermitian holomorphic bundle with Chern connection $D = D' + D''$ and $D'' = \bar\partial_E$. $\Delta' = D'D'^* + D'^*D'$ and $\Delta'' = D''D''^* + D''^*D''$ act on E-valued forms. $L = \omega\wedge$ and $\Lambda = \Lambda_\omega = L^*$ are the Lefschetz operators, $\Theta(E,h) = D^2$ is curvature, and $B = [\mathrm{i}\Theta(E,h), \Lambda]$ is the curvature endomorphism on forms. For homogeneous operators A, C, brackets denote the graded commutator $[A, C] = AC - (-1)^{|A||C|}CA$, where $|A|, |C| \in \mathbb{Z}/2\mathbb{Z}$ are their parities. The Kähler volume form is $\mathrm{d}V_\omega = \omega^n/n!$, and $\|\cdot\|$ and $\langle\cdot,\cdot\rangle$ are the pointwise or integrated Hermitian norm and pairing indicated by context. On a domain with boundary, $\operatorname{Dom}\bar\partial^*$ includes the $\bar\partial$-Neumann boundary condition. BKN abbreviates the Bochner–Kodaira–Nakano identity.

Prerequisite results used below

Kähler commutator identities. With the conventions of Lecture 14,

$$[\Lambda, D''] = -\mathrm{i}(D')^*, \qquad [\Lambda, D'] = \mathrm{i}(D'')^*,$$

together with their adjoints. These identities hold on smooth compactly supported bundle-valued forms and extend to the relevant operator domains.

Adjoint norm identity. For any densely defined closed operator A on a Hilbert space and any $u \in \operatorname{Dom}(A^*A) \cap \operatorname{Dom}(AA^*)$, $\langle (AA^* + A^*A)u, u\rangle = \|Au\|^2 + \|A^*u\|^2$. On the larger form domain $\operatorname{Dom} A \cap \operatorname{Dom} A^*$, the right-hand side is the closed quadratic form associated with $AA^* + A^*A$. Combining this with the curvature commutator yields the integrated BKN estimate after boundary or cutoff terms are accounted for.

Why these ideas are needed

To solve a $\bar\partial$ equation or prove that a cohomology class vanishes, we need an estimate that converts curvature positivity into control of derivatives. The Bochner–Kodaira–Nakano identity is that conversion mechanism. It compares the two natural bundle-valued Laplacians and

isolates curvature in the zero-order commutator $[\mathrm{i}\Theta(E,h),\Lambda]$. Pairing the identity with a form turns operator algebra into an energy formula. Boundary terms, completeness, and bidegree are emphasized because they determine whether a pointwise positive tensor actually yields a usable global coercive estimate.

Plan for the lecture. 0–18 min: bundle-valued forms and adjoints; 18–42: the BKN identity; 42–62: the curvature commutator; 62–78: the basic estimate; 78–90: boundary, completeness, and torsion.

Let (X,ω) be Kähler and (E,h) a Hermitian holomorphic bundle with Chern connection $D = D' + D''$, $D'' = \bar\partial_E$. Define

$$\Delta' = D'D'^{*} + D'^{*}D', \qquad \Delta'' = D''D''^{*} + D''^{*}D''.$$

Theorem 7.1: Bochner–Kodaira–Nakano identity

On compactly supported smooth E-valued forms,

$$\boxed{\Delta'' = \Delta' + [\mathrm{i}\Theta(E,h),\Lambda_\omega]}.$$

If X is compact without boundary, or complete and justified by controlled cutoffs, then

$$\|D''u\|^2 + \|D''^{*}u\|^2 = \|D'u\|^2 + \|D'^{*}u\|^2 + \int_X \langle [\mathrm{i}\Theta(E),\Lambda]u,u\rangle\,\mathrm{d}V_\omega.$$

Proof. The bundle-valued Kähler identities are $D''^{*} = -\mathrm{i}[\Lambda, D']$ and $D'^{*} = \mathrm{i}[\Lambda, D'']$. The operators D', D'' and their adjoints are odd, whereas Λ is even. Hence the Laplacians are graded commutators, and the graded Jacobi identity gives

$$\begin{aligned}\Delta'' &= [D'', D''^{*}] = -\mathrm{i}[D'', [\Lambda, D']]\\ &= -\mathrm{i}\big([[D'',\Lambda], D'] + [\Lambda, [D'', D']]\big).\end{aligned}$$

Now $[D'',\Lambda] = \mathrm{i}D'^{*}$ and $[D'', D'] = D''D' + D'D'' = D^2 = \Theta(E,h)$; the pure-type curvature components vanish because $D'^2 = D''^2 = 0$. Since $\Theta(E,h)$ and Λ are even,

$$\begin{aligned}\Delta'' &= [D'^{*}, D'] - \mathrm{i}[\Lambda, \Theta(E,h)]\\ &= \Delta' + [\mathrm{i}\Theta(E,h),\Lambda].\end{aligned}$$

Pairing with a compactly supported form and integrating gives the norm identity because $(D'D'^{*}u,u) = \|D'^{*}u\|^2$, $(D'^{*}D'u,u) = \|D'u\|^2$, and the two D'' terms give $\|D''^{*}u\|^2$ and $\|D''u\|^2$. On a compact manifold there is no boundary. On a complete manifold the cutoff argument in Lecture 28 extends the identity to the graph domain. On a non-Kähler Hermitian manifold, the Kähler commutators acquire torsion operators such as $\tau = [\Lambda, \partial\omega]$ and their adjoints. □

The compact-support identity just proved is the Kähler specialization of the Hermitian Bochner–Kodaira–Nakano formula. In the non-Kähler case the torsion commutators cannot be suppressed. A coordinate-free derivation, together with the Hermitian correction terms, is given in [1, Chapter VII, Theorems 1.1–1.2 and Corollary 1.3].

Definition 7.2: Curvature operator

On a fixed bidegree, set

$$B_{E,\omega} = [\mathrm{i}\Theta(E,h),\Lambda_\omega].$$

This is a fiberwise self-adjoint Hermitian operator. Its positivity must be checked on the *bide-*

gree being estimated.

Example 7.3: Diagonalization for a line bundle on (n, q)-forms

At a point choose a unitary frame with $\mathrm{i}\Theta(L) = \mathrm{i}\sum_j \lambda_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$. For the coefficient u_J of an (n, q)-form, the eigenvalue of B is $\sum_{j\in J} \lambda_j$. Hence $\mathrm{i}\Theta(L) \geq c\omega$ implies $B \geq qc\,\mathrm{id}$ on (n, q)-forms.

The same sum does not apply to arbitrary (p, q)-forms. Even for positive line-bundle curvature, B can be negative in low total degree.

Corollary 7.4: Basic L^2 estimate

If $B \geq 0$ on E-valued (n, q)-forms, then for $u \in C_c^\infty(X, \Lambda^{n,q}T^*X \otimes E)$,

$$\|D''u\|^2 + \|D''^*u\|^2 \geq \int_X \langle Bu, u\rangle \, \mathrm{d}V_\omega.$$

If X is compact without boundary or complete, the same estimate holds for every $u \in \mathrm{Dom}(D'') \cap \mathrm{Dom}(D''^*)$ of bidegree (n, q). The omitted derivative terms are nonnegative; they are not asserted to vanish.

Proof. For a compactly supported smooth form, pair the BKN identity with u:

$$\|D''u\|^2 + \|D''^*u\|^2 = \|D'u\|^2 + \|D'^*u\|^2 + (Bu, u).$$

Discarding the two nonnegative derivative terms gives the assertion. On a compact manifold, smooth forms are a core for the closed operators. On a complete manifold, apply the identity to controlled cutoffs $\chi_\nu u$; the commutator errors tend to zero in the graph norm. Approximation then proves the estimate on the stated domain. □

This is the fundamental inequality used throughout the subsequent vanishing and existence arguments; compare [1, Chapter VII, formula (2.1)]. Notice the order of implication: the pointwise identity is proved first on compactly supported smooth forms, and only then is it extended to a closed-operator domain by compactness or completeness.

Warning

Keep three levels separate: Griffiths or Nakano positivity of the curvature tensor, positivity of the commutator B on one bidegree, and an integrated coercive lower bound. Only the last directly gives closed range and a solution operator.

7.1.1 Boundary and completeness

On a smoothly bounded domain, the domain of D''^* carries the $\bar{\partial}$-Neumann boundary condition, and the Morrey–Kohn identity has an additional boundary Levi term. Pseudoconvexity makes this term nonnegative; it does not make it disappear. On a complete boundaryless manifold, cutoffs $\chi_\nu \to 1$ with $|\mathrm{d}\chi_\nu| \to 0$ extend compact-support identities to the graph domain of the closed operators.

7.1.2 Computing the curvature commutator on a basis

Suppose L is a line bundle and at one point choose unitary coordinates with

$$\mathrm{i}\Theta(L) = \mathrm{i}\sum_{j=1}^{n} \lambda_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j.$$

Let $u_{I\bar{J}}\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ be a basis (p,q)-form. Wedge by the jth curvature term and contraction by Λ count whether j occurs in I or J. The result is

$$[\mathrm{i}\Theta(L), \Lambda](\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J) = \left(\sum_{j\in I} \lambda_j + \sum_{j\in J} \lambda_j - \sum_{j=1}^{n} \lambda_j\right) \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J.$$

For $p = n$, the first and last sums cancel, leaving $\sum_{j\in J} \lambda_j$. This is why positive line-bundle curvature gives a positive operator on (n,q)-forms for $q \geq 1$. For $p = q = 0$, the eigenvalue is $-\sum_j \lambda_j$, so the same curvature is negative on functions. The bidegree cannot be suppressed.

For a vector bundle, diagonalizing the tangent indices does not generally diagonalize the bundle indices. On an E-valued (n,q)-form the quadratic expression of $[\mathrm{i}\Theta(E), \Lambda]$ is the Nakano curvature form applied to the coefficient array. This is the precise bridge from tensor positivity to the analytic energy estimate.

7.1.3 Three settings for integration by parts

On a compact manifold without boundary, smooth forms lie in the domains of both D'' and D''^* and Stokes creates no boundary term. On a complete noncompact manifold, one first applies the compact-support identity to $\chi_\nu u$ and uses cutoffs with $|\mathrm{d}\chi_\nu| \to 0$. On a domain with boundary, the Hilbert adjoint has the $\bar{\partial}$-Neumann boundary condition. For a scalar $(0,1)$-form $v = \sum v_j \mathrm{d}\bar{z}_j$ on $\Omega = \{\rho < 0\}$ it reads, up to normalization,

$$\sum_j v_j \rho_j = 0 \quad \text{on } \partial\Omega.$$

The coefficients are therefore complex tangential at the boundary.

The Morrey–Kohn formula contains

$$\int_{\partial\Omega} \sum_{j,k} \rho_{j\bar{k}} v_j \bar{v}_k \, ddS.$$

Pseudoconvexity makes this term nonnegative on tangential vectors. It does not make the term zero; compact support away from the boundary is what removes it. These three settings lead to the same basic estimate by different domain arguments.

Remark (Coercivity versus semipositivity)

If $B \geq c\,\mathrm{id}$ with $c > 0$, then the basic estimate controls the full L^2 norm and gives closed range. If only $B \geq 0$, it controls no component in $\ker B$. The identity remains true, but a vanishing or existence conclusion requires an additional source of coercivity.

7.1.4 The curvature commutator on a basis

At a point choose unitary coordinates and suppose a line bundle has

$$\mathrm{i}\Theta(L) = \mathrm{i}\sum_{j=1}^{n} \lambda_j \, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j.$$

Let e_j denote exterior multiplication by $\mathrm{d}z_j$, let $\bar{e}_j$ denote exterior multiplication by $\mathrm{d}\bar{z}_j$, and let e_j^*, $\bar{e}_j^*$ be the adjoint contractions. Then

$$L_\omega = \mathrm{i}\sum_j e_j\bar{e}_j, \qquad \Lambda_\omega = -\mathrm{i}\sum_j \bar{e}_j^* e_j^*.$$

On a basis form $\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$, the elementary anticommutation relations give

$$[\mathrm{i}\Theta(L), \Lambda] = \sum_j \lambda_j (e_j e_j^* + \bar{e}_j \bar{e}_j^* - 1).$$

Its eigenvalue is therefore

$$\sum_{j\in I} \lambda_j + \sum_{j\in J} \lambda_j - \sum_{j=1}^{n} \lambda_j.$$

When $I = \{1, \dots, n\}$, this reduces to $\sum_{j\in J} \lambda_j$. Hence on (n, q)-forms the smallest eigenvalue is the sum of the q smallest curvature eigenvalues. In particular, $\mathrm{i}\Theta(L) \geq c\omega$ implies $B \geq qc\,\mathrm{id}$ on this degree.

This computation also explains why positivity cannot be transferred blindly to an arbitrary bidegree. On functions $(p, q) = (0, 0)$, the eigenvalue is $-\sum_j \lambda_j$, which is negative for positive curvature.

From the operator identity to the energy identity

For a compactly supported smooth E-valued form u, take the L^2 inner product of

$$\Delta'' u = \Delta' u + Bu$$

with u. By the definition of the formal adjoints,

$$(\Delta'' u, u) = \|D'' u\|^2 + \|D''^* u\|^2,$$
$$(\Delta' u, u) = \|D' u\|^2 + \|D'^* u\|^2.$$

Therefore

$$\|D'' u\|^2 + \|D''^* u\|^2 = \|D' u\|^2 + \|D'^* u\|^2 + (Bu, u).$$

The estimate comes from discarding the first two terms on the right after explicitly recognizing them as squared norms. They are generally not zero.

The boundary term and its domain

Let $\Omega = \{\rho < 0\} \Subset \mathbb{C}^n$ have smooth boundary and let $v = \sum_j v_j \mathrm{d}\bar{z}_j$. Membership in the Hilbert-space domain of $\bar{\partial}^*$ imposes the $\bar{\partial}$-Neumann condition

$$\sum_j v_j \rho_j = 0 \qquad \text{on } \partial\Omega.$$

Thus the coefficient vector of v is complex tangential. Integration by parts gives the boundary contribution

$$\int_{\partial\Omega} \sum_{j,k} \rho_{j\bar{k}} v_j \bar{v}_k \frac{\mathrm{e}^{-\varphi}}{|\partial\rho|}\, \mathrm{d}\sigma.$$

Pseudoconvexity makes this integral nonnegative on the tangential subspace. It does not make the term vanish. If v is compactly supported in the interior, the term is absent because no boundary is reached.

On a complete manifold without boundary, choose cutoffs χ_ν with $|\mathrm{d}\chi_\nu| \to 0$ and apply the compact-support identity to $\chi_\nu u$. The commutators

$$[D'', \chi_\nu]u = \bar{\partial}\chi_\nu \wedge u, \qquad [D''^*, \chi_\nu]u = -\iota_{(\partial\chi_\nu)^\sharp} u$$

go to zero in L^2. This is the step that extends the identity to the graph domain.

7.1.5 How the BKN operator identity is assembled

Let $D = D' + D''$ be the Chern connection. On a Kähler manifold the commutator identities are

$$[\Lambda, D'] = \mathrm{i}D''^{*}, \qquad [\Lambda, D''] = -\mathrm{i}D'^{*}.$$

They are checked at one point in holomorphic normal coordinates and a unitary bundle frame. Since both sides are tensorial first-order operators, that calculation gives the global identities.

Insert these expressions for the adjoints into $\Delta'' = D''D''^{*} + D''^{*}D''$ and expand the graded commutators. Terms involving $D'D'$ and $D''D''$ vanish in the required types. The remaining mixed terms group as

$$[D'D'' + D''D', \Lambda].$$

The curvature identity for the Chern connection is $D'D'' + D''D' = \Theta(E)$. After multiplying by the conventional factor i, this gives

$$\boxed{\Delta'' = \Delta' + [\mathrm{i}\Theta(E), \Lambda]}.$$

No positivity has been used yet: this is an operator identity. Positivity enters only after taking the inner product with u, recognizing the two terms from Δ' as squared norms, and estimating the zeroth-order curvature commutator. Keeping these stages separate prevents the common mistake of treating the BKN formula itself as an inequality.

7.1.6 Self-study workshop: reading every term in the Bochner–Kodaira–Nakano identity

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall the Chern connection, formal and Hilbert-space adjoints, the Kähler commutators, and positivity operators on bundle-valued forms. Also decide at the outset whether the manifold is compact without boundary, complete with cutoffs, or a domain with a boundary condition.

From the operator identity to an estimate. For a Hermitian holomorphic bundle (E, h) on a Kähler manifold, the identity is

$$\Delta'' = \Delta' + [\mathrm{i}\Theta(E, h), \Lambda].$$

Pairing an E-valued form u with this equality gives

$$\|\bar\partial u\|^2 + \|\bar\partial^{*} u\|^2 = \|D'u\|^2 + \|D'^{*}u\|^2 + \big([\mathrm{i}\Theta(E, h), \Lambda]u, u\big),$$

under the compact-support or domain hypotheses that justify integration by parts. The connection-derivative terms are nonnegative; they are discarded for an inequality, not declared zero.

The curvature operator must be evaluated on the actual bidegree. For a line bundle, choose coordinates with

$$\mathrm{i}\Theta(L) = \mathrm{i}\sum_{j=1}^{n} \lambda_j \, \mathrm{d}z_j \wedge \mathrm{d}\bar z_j.$$

On an (n, q) basis vector $\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n \wedge \mathrm{d}\bar z_J$, the commutator has eigenvalue

$$\sum_{j \in J} \lambda_j.$$

Thus positive curvature gives a strictly positive operator for $q \geq 1$. For a vector bundle the fiber indices couple to the form indices, which is why Nakano rather than merely Griffiths positivity appears.

On a smoothly bounded domain, membership in $\operatorname{Dom}\bar\partial^{*}$ imposes the $\bar\partial$-Neumann condition. The integration formula contains a boundary Levi term; pseudoconvexity makes it nonnegative on the tangential coefficients. On a complete manifold, cutoff functions replace the boundary argument.

Checkpoint (do not erase positive terms)

(1) Why may the D' terms be omitted from a lower bound?
(2) Why is the boundary term not said to vanish on a pseudoconvex domain?
(3) On $(n, 2)$-forms for a positive line bundle, which curvature eigenvalues are added?

Solution (checkpoint). They are squared norms and hence nonnegative. Pseudoconvexity supplies a nonnegative Levi form on tangential components; it does not make that form zero. On the component indexed by $J = \{j_1, j_2\}$, the eigenvalue is $\lambda_{j_1} + \lambda_{j_2}$. ◇

Exit criterion

You can state the identity with its domain mechanism, compute the curvature commutator on a basis, and identify exactly which nonnegative derivative or boundary terms are discarded in obtaining an estimate.

Summary. The BKN identity links the $\bar\partial$-Laplacian to covariant derivatives and a curvature commutator. Geometric positivity becomes functional-analytic coercivity only after bidegree, boundary, and completeness have all been audited.

Exercise 7.5: Eigenvalues of the curvature operator

In the diagonal frame above, compute the eigenvalues of B on the standard basis of (n, q)-forms using wedge and contraction.

Exercise 7.6: The non-Kähler warning

Find the torsion terms in the Hermitian BKN formula and explain why they vanish when $\partial\omega = 0$.

Optional handout: Weitzenböck formulas and the classical Bochner method

This handout is optional. No later argument in the core lectures depends on it.

The Bochner–Kodaira–Nakano identity is one member of a broad family of Weitzenböck formulas. They compare a geometrically defined Laplacian with a rough Laplacian plus curvature:

$$\Delta = \nabla^*\nabla + \mathcal{R}.$$

The first term is nonnegative after integration. The sign of the zeroth-order curvature operator $\mathcal{R}$ therefore controls harmonic sections.

For a real one-form α on a Riemannian manifold,

$$\Delta_{\mathrm{d}}\alpha = \nabla^*\nabla\alpha + \mathrm{Ric}^\sharp(\alpha).$$

If M is compact and $\mathrm{Ric} > 0$, a harmonic one-form satisfies

$$0 = (\Delta\alpha, \alpha) = \|\nabla\alpha\|^2 + \int_M \mathrm{Ric}(\alpha^\sharp, \alpha^\sharp),$$

so $\alpha = 0$ and $b_1(M) = 0$. If $\mathrm{Ric} \geq 0$, every harmonic one-form is parallel everywhere, and its dual vector lies in the Ricci nullspace at every point.

Example 7.7: Holomorphic tensors

For a holomorphic section s of a Hermitian bundle, a Bochner formula has the shape

$$\tfrac{1}{2}\Delta|s|^2 = |Ds|^2 - \langle \mathcal{R}s, s\rangle$$

with sign depending on the bundle and Laplacian convention. At a maximum of $|s|^2$, curvature of the appropriate sign forces s to vanish or become parallel. This yields rigidity results without computing cohomology.

Warning (Pointwise versus integrated arguments)

The maximum-principle version uses a pointwise scalar identity. The Hodge-theoretic version uses integration and an operator on forms. Their curvature signs often look opposite because the bundles and Laplacian conventions differ.

Exercise 7.8: Bochner on functions

For this exercise only, put $\Delta_{\mathrm{geom}} = \operatorname{div}\nabla$; on functions this is $-\Delta_{\mathrm{d}}$ with the nonnegative Hodge-Laplacian convention of Lecture 13. Derive

$$\frac{1}{2}\Delta_{\mathrm{geom}}|\mathrm{d}f|^2 = |\operatorname{Hess} f|^2 + \langle \mathrm{d}f, \mathrm{d}\Delta_{\mathrm{geom}} f\rangle + \operatorname{Ric}(\nabla f, \nabla f)$$

and apply it to a harmonic function on a compact manifold.

7.2 Lecture 26: Kodaira–Akizuki–Nakano and related vanishing theorems

Week 13 material • 90 minutes

Notation for this lecture

X is a compact Kähler n-fold, $K_X = \Omega_X^n$ its canonical bundle, and L or E a positive holomorphic bundle with its chosen Hermitian metric. Bundle-valued Dolbeault cohomology is $H^q(X, \Omega_X^p \otimes E)$, represented by harmonic (p,q)-forms. The curvature operator is $B = [\mathrm{i}\Theta(E), \Lambda]$. KAN means Kodaira–Akizuki–Nakano; its basic conclusion is vanishing in the range $p + q > n$ for a positive line bundle. Kodaira vanishing is the special statement $H^q(X, K_X \otimes L) = 0$ for $q > 0$. E^* and $\det E$ are the dual and determinant bundles, and Serre duality identifies the dual of $H^q(X, E)$ with $H^{n-q}(X, K_X \otimes E^*)$. In unitary coordinates and a unitary frame (e_α), write the Hermitian coefficients of $\mathrm{i}\Theta(E)$ as $\Theta_{j\bar{k}\alpha\bar{\beta}}$. The bundle is *Nakano positive* if

$$\sum_{j,k,\alpha,\beta} \Theta_{j\bar{k}\alpha\bar{\beta}} \xi^{j\alpha} \overline{\xi^{k\beta}} > 0$$

for every nonzero coefficient array $\xi = (\xi^{j\alpha}) \in T^{1,0}X \otimes E$. Griffiths positivity tests only decomposable arrays $\xi^{j\alpha} = v^j s^\alpha$; this is a weaker condition. The notation $B(x) > 0$ means that the Hermitian endomorphism B on the fixed bidegree is positive definite at the point x.

Prerequisite results used below

Dolbeault–Hodge theorem. On a compact Hermitian manifold, every Dolbeault cohomology class with coefficients in a Hermitian holomorphic bundle has a unique Δ''-harmonic representative.

Coercivity implies vanishing. If the BKN identity gives $\|\bar\partial u\|^2 + \|\bar\partial^* u\|^2 \geq c\|u\|^2$ with $c > 0$ on a fixed bidegree, then a harmonic form u in that bidegree is zero. By the Dolbeault–Hodge theorem, the corresponding cohomology group vanishes. For a positive line bundle with unit curvature eigenvalues, the curvature operator on (p,q)-forms is multiplication by $p + q - n$.

Unique continuation for harmonic forms. The Dolbeault Laplacian Δ'' is a second-order elliptic operator. On a connected manifold, a smooth solution of $\Delta'' u = 0$ which vanishes on a nonempty open set vanishes everywhere. This is used only in the pointwise strictness refinement below.

Why these ideas are needed

Positive curvature should impose global restrictions, but the route from a pointwise inequality to the disappearance of a cohomology group is not automatic. Hodge theory chooses a harmonic representative, and the BKN identity shows that such a form has zero energy. If the curvature commutator is uniformly positive on the relevant bidegree, zero energy forces the representative to vanish. Kodaira, Nakano, and KAN vanishing are variations of this one mechanism. Tracking the bidegree and the exact curvature eigenvalue explains both the vanishing range and its limitations.

Plan for the lecture. 0–20 min: harmonic representatives and the vanishing mechanism; 20–42: KAN; 42–60: Kodaira and Nakano vanishing; 60–76: partial positivity; 76–90: examples and limitations.

The universal template is simple: represent a cohomology class by a harmonic u; BKN gives

$0 \geq \int \langle Bu, u\rangle$; strict positivity of B forces $u = 0$.

Theorem 7.9: Kodaira–Akizuki–Nakano vanishing

Let X be a compact Kähler n-fold and L a positive line bundle. Then

$$H^q(X, \Omega_X^p \otimes L) = 0, \qquad p + q > n.$$

Proof. Choose a positive metric on L and use its curvature form $\omega = \mathrm{i}\Theta(L)$ as the Kähler metric. At a fixed point, take an ω-unitary coframe $\mathrm{d}z_1, \ldots, \mathrm{d}z_n$. All curvature eigenvalues are then equal to 1, so the basis calculation from Lecture 25 gives

$$[\mathrm{i}\Theta(L), \Lambda](\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J) = \big(|I| + |J| - n\big)\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J.$$

Let u be the $\bar{\partial}$-harmonic representative of a class in $H^q(X, \Omega_X^p \otimes L)$; the Dolbeault–Hodge theorem gives such a unique u, and $|I| = p$, $|J| = q$. Pairing BKN with u yields

$$0 = \|D'u\|^2 + \|D'^{*}u\|^2 + (p + q - n)\|u\|^2.$$

When $p + q > n$, every term on the right is nonnegative and the last coefficient is positive. Hence $u = 0$, so the original cohomology class is zero. □

This curvature-operator proof is the argument in [1, Chapter VII, Theorem 3.3]. Choosing $\omega = \mathrm{i}\Theta(L)$ is not merely a normalization: it turns every curvature eigenvalue into 1 and makes the numerical range $p + q > n$ visible without an additional comparison constant.

Corollary 7.10: Kodaira vanishing

If L is positive, then

$$H^q(X, K_X \otimes L) = 0 \quad (q \geq 1).$$

By Serre duality, $H^q(X, L^{-1}) = 0$ for $q < n$.

Proof. Put $p = n$ in Kodaira–Akizuki–Nakano vanishing. Then $p + q > n$ exactly when $q > 0$, and $\Omega_X^n = K_X$. Serre duality gives

$$H^q(X, L^{-1})^* \simeq H^{n-q}(X, K_X \otimes L),$$

whose right side vanishes when $q < n$. □

Theorem 7.11: Nakano vanishing

If a Hermitian holomorphic vector bundle E is Nakano positive, then

$$H^q(X, K_X \otimes E) = 0 \qquad (q \geq 1).$$

Proof. Represent a class by its harmonic E-valued (n, q)-form u. We first spell out the pointwise linear algebra. At a point $x \in X$, choose an ω-unitary coframe $\mathrm{d}z_1, \ldots, \mathrm{d}z_n$ and an h-unitary frame $e_1, \ldots, e_r$ of E_x, where $r = \operatorname{rank} E$; the corresponding dual frame is denoted by $e_1^*, \ldots, e_r^*$. Write

$$\mathrm{i}\Theta(E, h)_x = \mathrm{i}\sum_{j,k=1}^{n}\sum_{\alpha,\beta=1}^{r} c_{j\bar{k}\alpha\bar{\beta}}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k \otimes e_\alpha^* \otimes e_\beta.$$

For

$$u = \sum_{|J|=q,\alpha} u_{J,\alpha}\,\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n \wedge \mathrm{d}\bar{z}_J \otimes e_\alpha,$$

extend the coefficients alternately to ordered q-tuples and denote them by $u_{jS,\alpha}$ when S has length $q-1$. Wedge–contraction calculus gives

$$\langle [\mathrm{i}\Theta(E,h),\Lambda]u,u\rangle_x = \sum_{|S|=q-1}\sum_{j,k,\alpha,\beta} c_{j\bar{k}\alpha\bar{\beta}}u_{jS,\alpha}\overline{u_{kS,\beta}}.$$

For each fixed S, the inner sum is the Nakano curvature quadratic form applied to the tensor $(u_{jS,\alpha})_{j,\alpha} \in T^{1,0}_x X \otimes E_x$. Thus Nakano positivity implies that $B = [\mathrm{i}\Theta(E),\Lambda]$ is positive definite on every (n,q)-space with $q \geq 1$. (For $q = 1$ the two quadratic forms agree directly; for $q > 1$ the displayed expression is a sum of such forms, so "precisely" would be too strong.) Compactness of the unit sphere bundle supplies a uniform $c > 0$ with $B \geq c\,\mathrm{id}$. The integrated BKN identity therefore gives

$$0 = \|D''u\|^2 + \|D''^{*}u\|^2 \geq (Bu,u) \geq c\|u\|^2.$$

Thus $u = 0$ and the cohomology class vanishes. □

The coefficient identity above is formula (7.1), and the resulting vanishing is the Nakano case of Theorem 7.3 and Corollary 7.5 in [1, Chapter VII, §7].

The bidegree (n,q) is chosen so Nakano positivity matches $B = [\mathrm{i}\Theta(E),\Lambda]$. Griffiths positivity alone is generally insufficient; Demailly–Skoda first upgrades $E \otimes \det E$ to Nakano positivity.

Example 7.12: Basic vanishing on projective space

$\mathcal{O}(k)$ is positive for $k > 0$, so $H^q(\mathbb{P}^n, K_{\mathbb{P}^n} \otimes \mathcal{O}(k)) = 0$ for $q > 0$. Since $K_{\mathbb{P}^n} = \mathcal{O}(-n-1)$, this produces many vanishings for $\mathcal{O}(m)$; Serre duality recovers the edge cases of Bott's formula.

Theorem 7.13: Spectral vanishing criterion

Let L be a Hermitian line bundle on a compact Kähler manifold. At each point choose a unitary coframe in which $\mathrm{i}\Theta(L) = \mathrm{i}\sum_j \lambda_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$. If, for some $c > 0$, every pair of multi-indices $|I| = p$, $|J| = q$ satisfies

$$\sum_{j\in I}\lambda_j + \sum_{j\in J}\lambda_j - \sum_{j=1}^{n}\lambda_j \geq c,$$

then $H^q(X,\Omega^p_X \otimes L) = 0$. In particular, on (n,q)-forms it is enough that the sum of the q smallest curvature eigenvalues be uniformly positive.

Proof. Wedge and contraction on the basis form $\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ show that the displayed number is exactly the eigenvalue of $B = [\mathrm{i}\Theta(L),\Lambda]$. Thus $B \geq c\,\mathrm{id}$ on (p,q)-forms. A harmonic representative u satisfies the BKN identity, hence $0 \geq c\,\|u\|^2$, so $u = 0$. □

Corollary 7.14: Semipositivity with strictness at one point

Let X be connected and let (E,h) be a Hermitian holomorphic bundle on the compact Kähler manifold (X,ω). Fix a bidegree (p,q) and let $B = [\mathrm{i}\Theta(E,h),\Lambda_\omega]$ act on E-valued (p,q)-forms. If $B \geq 0$ everywhere and $B(x_0) > 0$ at one point $x_0 \in X$, then

$$H^q(X,\Omega^p_X \otimes E) = 0.$$

Proof. Let u be the harmonic representative of a cohomology class. The integrated BKN identity gives

$$0 = \|D'u\|^2 + \|D'^{*}u\|^2 + \int_X \langle Bu,u\rangle\,\mathrm{d}V_\omega.$$

Every term is nonnegative. Since $B(x_0)$ is positive definite and B is continuous, there are a neighborhood U of x_0 and a constant $c > 0$ such that $B \geq c\,\mathrm{id}$ on U. The integral identity therefore gives

$\int_U |u|^2 \mathrm{d}V_\omega = 0$, so smoothness of the harmonic form implies $u = 0$ on U. Unique continuation for the elliptic equation $\Delta'' u = 0$ gives $u = 0$ on the connected manifold X. □

This is the direct BKN form of the "semipositive everywhere, strictly positive at one point" principle; compare [1, Chapter VII, Corollaries 2.3 and 2.5]. The final unique continuation step is essential: positivity at one point first gives vanishing only on an open neighborhood.

Remark (Partial positivity)

Knowing only the number of positive eigenvalues does not by itself give the displayed uniform inequality: negative eigenvalues may dominate. More flexible partial-positivity theorems use high tensor powers, a metric adapted to the positive directions, or q-convexity. The spectral criterion above is the exact statement supplied directly by BKN.

Warning (Dual curvature)

Dual curvature is $\Theta(E^*) = -{}^{\mathrm{t}}\Theta(E)$. Griffiths positivity therefore dualizes to Griffiths negativity. Nakano duality also transposes the tangent and bundle indices and cannot be inferred by deleting the minus sign.

Warning

If curvature is only semipositive, BKN says merely that harmonic representatives lie in $\ker B$, not that they vanish. One needs a uniform positive lower bound, strict positivity somewhere plus unique continuation, or another geometric argument.

7.2.1 The vanishing region in the Hodge diamond

For a positive line bundle L, Kodaira–Akizuki–Nakano gives

$$H^{p,q}(X, L) = 0 \qquad \text{when } p + q > n.$$

Serre duality converts this to the negative-bundle statement

$$H^{p,q}(X, L^{-1}) = 0 \qquad \text{when } p + q < n.$$

Thus positive curvature kills the upper-right triangle of the bundle-valued Hodge diamond, while negative curvature kills the lower-left triangle. The middle diagonal $p + q = n$ is not controlled by this argument because the curvature commutator has zero eigenvalue there.

On $\mathbb{P}^n$, take $L = \mathcal{O}(k)$ with $k > 0$ and $K_{\mathbb{P}^n} = \mathcal{O}(-n-1)$. Kodaira vanishing gives

$$H^q(\mathbb{P}^n, \mathcal{O}(k-n-1)) = 0 \qquad (q > 0).$$

Equivalently, $H^q(\mathbb{P}^n, \mathcal{O}(m)) = 0$ for $q > 0$ once $m > -n-1$. Serre duality recovers the top-degree groups for very negative m. Bott's formula supplies the remaining exact boundary cases; curvature already explains the large vanishing regions.

7.2.2 From vanishing to generation and embeddings

Vanishing becomes geometric when inserted into an exact sequence. For a point x,

$$0 \to L^k \otimes \mathcal{I}_x \to L^k \to L^k_x \to 0$$

shows that L^k is generated at x if $H^1(X, L^k \otimes \mathcal{I}_x) = 0$. Replacing $\mathcal{I}_x$ by $\mathcal{I}_x^2$ controls first jets. The ideal sheaves are not line bundles, so Kodaira vanishing does not apply to them directly. One resolves

the ideals by vector bundles, uses Serre vanishing for high powers, or constructs the desired jets analytically by weighted $\bar{\partial}$ estimates.

This distinction is important: a cohomology vanishing theorem for smooth bundles is not automatically a vanishing theorem for every coherent sheaf. The passage to coherent ideals requires a resolution or a singular metric whose multiplier ideal is the desired sheaf.

Partial positivity works by the same spectral logic. On (n, q)-forms, if the sum of the q smallest curvature eigenvalues is uniformly positive, the commutator is coercive even when some individual eigenvalues are negative. Merely knowing that a fixed number of eigenvalues are positive is insufficient unless the negative ones are quantitatively controlled.

Remark (One proof template)

Choose the cohomology realization, select a harmonic representative, compute the curvature operator on its exact bidegree, and prove a uniform positive lower bound. Every vanishing theorem in this lecture differs only in one of these four entries.

7.2.3 KAN vanishing with the eigenvalue calculation exposed

Choose the metric on a positive line bundle L so that $\mathrm{i}\Theta(L) = \omega$ at the level of the chosen normalization. In a unitary frame, all curvature eigenvalues are 1. The calculation from Lecture 25 gives on a (p, q) basis form

$$B = [\mathrm{i}\Theta(L), \Lambda] = (p + q - n)I.$$

Let u be the harmonic representative of a class in $H^q(X, \Omega_X^p \otimes L)$. Then $D''u = D''^*u = 0$, so BKN gives

$$0 = \|D'u\|^2 + \|D'^*u\|^2 + (p + q - n)\|u\|^2.$$

When $p + q > n$, all three terms are nonnegative and the last has a strictly positive coefficient. Hence $u = 0$. This proves the exact vanishing range.

At the boundary $p+q = n$, curvature contributes zero. The identity then says only $D'u = D'^*u = 0$; it does not force u to vanish. For example, $H^0(\mathbb{P}^1, K_{\mathbb{P}^1} \otimes \mathcal{O}(2)) = H^0(\mathbb{P}^1, \mathcal{O}) = \mathbb{C}$ lies exactly on this boundary.

Kodaira vanishing checked on $\mathbb{P}^1$

On $\mathbb{P}^1$, $K = \mathcal{O}(-2)$ and a positive line bundle is $L = \mathcal{O}(k)$ with $k > 0$. Kodaira vanishing asserts

$$H^1(\mathbb{P}^1, K \otimes L) = H^1(\mathbb{P}^1, \mathcal{O}(k - 2)) = 0.$$

The explicit Čech calculation gives $h^1(\mathcal{O}(m)) = \max\{-m - 1, 0\}$. Substituting $m = k - 2$ gives zero exactly for $k \geq 1$, confirming the theorem including the smallest positive bundle.

By Serre duality,

$$H^0(\mathbb{P}^1, L^{-1})^* simeq H^1(\mathbb{P}^1, K \otimes L),$$

so the same vanishing says that a negative-degree line bundle has no nonzero global section.

Nakano vanishing and compact uniformity

Suppose E is Nakano positive. On E-valued (n, q)-forms, $q \geq 1$, its curvature commutator is pointwise positive definite. The unit sphere bundle in this finite-rank vector bundle is compact because X is compact. The smallest eigenvalue therefore has a positive global minimum $c > 0$, giving

$$(Bu, u) \geq c\|u\|^2.$$

This compactness step upgrades pointwise strict positivity to the uniform coercivity actually used in the integrated proof.

If E is only Griffiths positive, this conclusion is not available because the curvature is tested only on decomposable tensors. Demailly–Skoda replaces E by $E \otimes \det E$, which is Nakano positive, and then the argument applies to

$$H^q(X, K_X \otimes E \otimes \det E), \qquad q > 0.$$

Partial positivity as an ordered-eigenvalue test

On (n, q)-forms with line-bundle curvature eigenvalues $\lambda_1 \le \cdots \le \lambda_n$, the least eigenvalue of B is

$$\lambda_1 + \cdots + \lambda_q.$$

Thus having at least $n-q+1$ positive eigenvalues is not sufficient if the remaining negative eigenvalues have large magnitude. The direct BKN criterion is the uniform inequality

$$\lambda_1 + \cdots + \lambda_q \ge c > 0.$$

Once this holds, the harmonic-representative proof is identical to KAN. This calculation separates a precise spectral hypothesis from the looser slogan "sufficiently many positive directions."

7.2.4 Deriving the dual vanishing range

Let L be positive. KAN gives

$$H^q(X, \Omega_X^p \otimes L) = 0 \qquad \text{when } p + q > n.$$

For L^{-1}, use Serre duality rather than changing a sign in the Bochner formula. The contraction pairing gives

$$(\Omega_X^p)^* \otimes K_X \simeq \Omega_X^{n-p}.$$

Therefore

$$\begin{aligned} H^q(X, \Omega_X^p \otimes L^{-1})^* &\simeq H^{n-q}\left(X, (\Omega_X^p \otimes L^{-1})^* \otimes K_X\right) \\ &\simeq H^{n-q}(X, \Omega_X^{n-p} \otimes L). \end{aligned}$$

KAN annihilates the last group when

$$(n-p) + (n-q) > n, \qquad \text{equivalently} \qquad p + q < n.$$

Hence

$$H^q(X, \Omega_X^p \otimes L^{-1}) = 0 \quad \text{for } p + q < n.$$

The strict inequality reverses because both bidegrees are replaced by their complements. At $p+q = n$ neither KAN nor its dual forces vanishing; examples on projective space show that this is a genuine boundary rather than a weakness of the proof.

7.2.5 Self-study workshop: turning curvature positivity into cohomology vanishing

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall harmonic representatives, Dolbeault cohomology with bundle coefficients, the curvature eigenvalue calculation from Lecture 25, and Serre duality. A vanishing theorem is an estimate applied to a harmonic representative.

Kodaira vanishing in four lines with all inputs named. Let L be a positive line bundle on a compact Kähler n-fold and let $q \geq 1$. A class in $H^q(X, K_X \otimes L)$ has an L-valued harmonic (n,q) representative u. The BKN identity and positivity give, for some $c > 0$,

$$0 = \|\bar\partial u\|^2 + \|\bar\partial^* u\|^2 \geq \big([\mathrm{i}\Theta(L), \Lambda]u, u\big) \geq c\|u\|^2.$$

Compactness is what turns pointwise positive definiteness into one uniform constant c. Hence $u = 0$ and

$$H^q(X, K_X \otimes L) = 0 \qquad (q \geq 1).$$

Serre duality identifies $H^q(X, K_X \otimes L)^*$ with $H^{n-q}(X, L^{-1})$. Setting $p = n - q$ therefore gives the equivalent negative-bundle statement $H^p(X, L^{-1}) = 0$ for $p < n$.

The Kodaira–Akizuki–Nakano theorem keeps general bidegree and concludes

$$H^q(X, \Omega_X^p \otimes L) = 0 \qquad \text{when } p + q > n.$$

The inequality $p + q > n$ is not a dimension slogan; it is exactly the range where the curvature commutator is positive on (p,q)-forms. For vector bundles, the analogous argument needs positivity of the operator on the relevant tensor space, usually Nakano positivity.

Checkpoint (audit a vanishing claim)

(1) Where is compactness used in the proof above?
(2) Does positivity of L imply $H^0(X, K_X \otimes L) = 0$?
(3) Why can Griffiths positivity not simply replace Nakano positivity in the vector-bundle proof?

Solution (checkpoint). Compactness gives Hodge representatives and a uniform positive lower bound for the curvature operator. The theorem makes no vanishing claim in degree zero; positive bundles often have many sections. Griffiths positivity tests only decomposable tensors, while the BKN quadratic form tests arbitrary coupled tangent-fiber coefficients. ◇

Exit criterion

You can derive Kodaira vanishing from a harmonic representative, translate it by Serre duality, and recover the bidegree range and positivity notion in KAN and vector-bundle variants.

Summary. KAN, Kodaira, and Nakano vanishing come from one energy identity; they differ only in bundle, bidegree, and positivity condition. "Compute B first, then claim vanishing" is safer than memorizing ranges.

Exercise 7.15: Check on $\mathbb{P}^1$

Use the explicit cohomology calculation in Lecture 15 to verify Kodaira vanishing on $\mathbb{P}^1$.

Exercise 7.16: A basic Fano consequence

If $-K_X$ is positive, prove $H^q(X, \mathcal{O}_X) = 0$ for $q > 0$ and deduce $\chi(\mathcal{O}_X) = 1$.

Optional handout: Lefschetz hyperplane theorems from positivity

This handout is optional. No later argument in the core lectures depends on it.

Let X be a smooth projective n-fold and $Y \subset X$ a smooth ample divisor. The topological Lefschetz hyperplane theorem states that

$$H^k(X, \mathbb{Z}) \longrightarrow H^k(Y, \mathbb{Z})$$

is an isomorphism for $k < n - 1$ and injective for $k = n - 1$. There is also a weak homotopy version for π_k.

The complement $X \setminus Y$ is affine and carries a strictly psh exhaustion coming from $-\log |s_Y|^2$. Morse theory for such an exhaustion has handles of real index at most n. Consequently X is obtained from a neighborhood of Y by attaching cells of dimension at least n, which yields the cohomological range.

There is an algebraic companion. From

$$0 \to \Omega^p_X(-Y) \to \Omega^p_X \to \Omega^p_X|_Y \to 0$$

and the conormal sequence on Y, Kodaira–Akizuki–Nakano vanishing kills the outer cohomology groups in a range. The resulting long exact sequences prove the Hodge pieces of the restriction theorem.

Example 7.17: Hypersurfaces in projective space

A smooth hypersurface $Y \subset \mathbb{P}^n$ has the same cohomology as $\mathbb{P}^n$ below its middle dimension. All new topology is concentrated in the primitive middle cohomology. For curves in $\mathbb{P}^2$, the middle degree is already 1, where the genus appears.

Exercise 7.18: Fundamental group

Assuming the homotopy form of Lefschetz, show that a smooth hypersurface of complex dimension at least two in $\mathbb{P}^n$ is simply connected.

7.3 Lecture 27: Hörmander's weighted L^2 estimate

Week 14 material • 90 minutes

Notation for this lecture

$\Omega \subset \mathbb{C}^n$ is a pseudoconvex domain, $\varphi \in C^2(\Omega)$ a strictly psh weight, and $\mathrm{d}\lambda$ Euclidean Lebesgue measure. The weighted inner product and norm are $(u,v)_\varphi = \int_\Omega \langle u, v\rangle \mathrm{e}^{-\varphi}\mathrm{d}\lambda$ and $\|u\|_\varphi^2 = (u,u)_\varphi$. $L^2_{0,q}(\Omega, \mathrm{e}^{-\varphi})$ denotes the Hilbert space of measurable $(0,q)$-forms with finite weighted norm; all unbounded $\bar\partial$ operators below are their maximal distributional realizations. $H_\varphi = (\varphi_{j\bar k})$ is the positive-definite Levi matrix and H_φ^{-1} its ordinary matrix inverse. A semipositive Levi matrix needs a range condition and a generalized inverse; that variant is not being used in the main theorem. The closed operators $T = \bar\partial : L^2_{0,0} \to L^2_{0,1}$ and $S = \bar\partial : L^2_{0,1} \to L^2_{0,2}$ satisfy $ST = 0$; T^*, S^* are their Hilbert-space adjoints, $\bar\partial_\varphi^* = T^*$ on $(0,1)$-forms, and $\mathrm{Dom}\, T^*$ is its operator domain. The source form is $f = \sum_j f_j \mathrm{d}\bar z_j$ with $\bar\partial f = 0$, and $A_f = \int_\Omega f^* H_\varphi^{-1} f\, \mathrm{e}^{-\varphi}\mathrm{d}\lambda$ is its inverse-Hessian energy.

Prerequisite results used below

Hilbert-complex existence lemma. Let $T : H_0 \to H_1$ and $S : H_1 \to H_2$ be densely defined closed operators with $ST = 0$. If a vector $f \in \ker S$ satisfies

$$|(f,v)_{H_1}|^2 \le C\big(\|T^* v\|_{H_0}^2 + \|Sv\|_{H_2}^2\big)$$

for every $v \in \mathrm{Dom}\, T^* \cap \mathrm{Dom}\, S$, then there is $u \in H_0$ with $Tu = f$ and $\|u\|_{H_0}^2 \le C$. The minimum-norm solution is obtained by orthogonal projection to $(\ker T)^\perp$.

Morrey–Kohn–Hörmander estimate. On a smooth bounded pseudoconvex domain and for a C^2 psh weight φ, every smooth $(0,1)$-form v in the $\bar\partial$-Neumann domain satisfies

$$\|\bar\partial v\|_\varphi^2 + \|\bar\partial_\varphi^* v\|_\varphi^2 \ge \int_\Omega v^* H_\varphi v\, \mathrm{e}^{-\varphi}\mathrm{d}\lambda.$$

Why these ideas are needed

Vanishing on a compact manifold says that a $\bar\partial$ equation is solvable, but it does not quantify the solution and does not address noncompact domains. Hörmander's method introduces a weight so that curvature can be strengthened where control is needed. The basic estimate bounds the solution by the inverse Levi matrix of the weight, retaining directional information that a scalar lower bound would discard. A Hilbert-space duality argument then converts an a priori estimate for test forms into an actual solution. This architecture will be reused for singular metrics and extension problems.

Plan for the lecture. 0–18 min: theorem and convention ledger; 18–42: Morrey–Kohn estimate; 42–65: curvature Cauchy–Schwarz; 65–80: Hilbert-complex duality; 80–90: exhaustion and passage to the limit.

7.3.1 Chosen version and hypothesis ledger

Fix a pseudoconvex domain $\Omega \subset \mathbb{C}^n$, Lebesgue measure $\mathrm{d}\lambda$, weight $\mathrm{e}^{-\varphi}$, and a strictly psh $\varphi \in C^2(\Omega)$. Let

$$f = \sum_{j=1}^{n} f_j \mathrm{d}\bar{z}_j, \qquad \bar{\partial} f = 0$$

in distributions. Write $H_\varphi = (\varphi_{j\bar{k}})$ and assume

$$A_f := \int_\Omega f^* H_\varphi^{-1} f \, \mathrm{e}^{-\varphi} \mathrm{d}\lambda < \infty.$$

Theorem 7.19: Hörmander's scalar $(0,1)$ estimate

Under these assumptions, there is $u \in L^2(\Omega, \mathrm{e}^{-\varphi})$ such that

$$\bar{\partial} u = f \quad \text{in distributions,} \qquad \boxed{\int_\Omega |u|^2 \mathrm{e}^{-\varphi} \mathrm{d}\lambda \le A_f}.$$

The choice $u \perp \ker \bar{\partial}$ is the unique minimum-norm solution.

Proof (including the operator-domain and existence steps).

The basic estimate and operator domains. For a compactly supported smooth $(0,1)$-form $v = \sum v_j \mathrm{d}\bar{z}_j$, the formal weighted adjoint is

$$\bar{\partial}_\varphi^* v = -\sum_j \mathrm{e}^{\varphi} \frac{\partial}{\partial z_j} (\mathrm{e}^{-\varphi} v_j) = -\sum_j \left(\frac{\partial v_j}{\partial z_j} - \varphi_j v_j \right).$$

On a domain with boundary one must use the closed Hilbert adjoint and its $\bar{\partial}$-Neumann boundary condition.

The Morrey–Kohn–Hörmander identity is

$$\|\bar{\partial} v\|_\varphi^2 + \|\bar{\partial}_\varphi^* v\|_\varphi^2 = \sum_{j,k} \int_\Omega \left| \frac{\partial v_j}{\partial \bar{z}_k} \right|^2 \mathrm{e}^{-\varphi} \mathrm{d}\lambda + \int_\Omega v^* H_\varphi v \, \mathrm{e}^{-\varphi} \mathrm{d}\lambda + \text{boundary Levi term}.$$

Pseudoconvexity makes the boundary term nonnegative on complex tangential components. Hence, for $v \in \mathrm{Dom}(\bar{\partial}) \cap \mathrm{Dom}(\bar{\partial}_\varphi^*)$,

$$Q(v) := \|\bar{\partial} v\|_\varphi^2 + \|\bar{\partial}_\varphi^* v\|_\varphi^2 \ge \int_\Omega v^* H_\varphi v \, \mathrm{e}^{-\varphi} \mathrm{d}\lambda.$$

The identity is first proved for smooth forms satisfying the normal boundary condition. Friedrichs approximation in the graph norm then extends the inequality to the displayed Hilbert-space domains; this density step is what permits nonsmooth L^2 data later in the proof.

Curvature Cauchy–Schwarz and Hilbert-complex duality. For every such v,

$$|(f, v)_\varphi|^2 = \left| \int_\Omega \left\langle H_\varphi^{-1/2} f, H_\varphi^{1/2} v \right\rangle \mathrm{e}^{-\varphi} \mathrm{d}\lambda \right|^2 \le A_f \, Q(v).$$

Set $T = \bar{\partial} : L^2_{0,0} \to L^2_{0,1}$ and $S = \bar{\partial} : L^2_{0,1} \to L^2_{0,2}$. Then $ST = 0$ and $Sf = 0$. Let P be the orthogonal projection of $L^2_{0,1}$ onto the closed subspace $\ker S$. For $v \in \mathrm{Dom}\, T^*$, write $v_0 = Pv$ and

$v_1 = v - v_0$. Since $\operatorname{im} T \subset \ker S$ and $v_1 \perp \ker S$, the definition of the adjoint gives $v_1 \in \operatorname{Dom} T^*$ and $T^* v_1 = 0$. Thus $v_0 \in \operatorname{Dom} T^* \cap \ker S$. Here, by definition, $\ker S = \{w \in \operatorname{Dom} S : Sw = 0\}$; hence v_0 also belongs to $\operatorname{Dom} S$, so the basic estimate really is applicable to v_0. We have

$$(f, v)_\varphi = (f, v_0)_\varphi, \qquad T^* v = T^* v_0, \qquad S v_0 = 0.$$

Applying the preceding estimate to v_0 gives $|(f, v)_\varphi|^2 \le A_f \, \|T^* v\|_\varphi^2$. Consequently

$$\ell(T^* v) := (f, v)_\varphi$$

is a well-defined bounded functional of norm at most $A_f^{1/2}$ on $\operatorname{im} T^*$. Hahn–Banach extends ℓ to $L^2_{0,0}$, and the Riesz theorem gives $u \in L^2_{0,0}$ with $(u, T^* v)_\varphi = (f, v)_\varphi$ for every $v \in \operatorname{Dom} T^*$. By the definition of the Hilbert adjoint, $u \in \operatorname{Dom} T$, $Tu = f$, and $\|u\|_\varphi^2 \le A_f$.

This functional-analytic step is the concrete form of the closed Hilbert-complex existence theorem in [1, Chapter VIII, Theorem 1.2]. In particular, the projection P is not a formal Hodge decomposition: it is merely the orthogonal projection onto the closed kernel of the closed operator S.

Warning

A differential identity does not itself produce a solution. The Hahn–Banach/Riesz duality step is the logical core of existence, and $\bar\partial f = 0$ is needed to eliminate the $S^* g$ component. The estimate alone does not provide boundary smoothness.

Exhaustion of a general pseudoconvex domain. The geometric input is worth making explicit. By the smooth exhaustion theorem proved earlier, a pseudoconvex domain admits a smooth strictly psh exhaustion $\rho : \Omega \to \mathbb{R}$. Choose regular values $c_1 < c_2 < \cdots \to +\infty$ and set

$$\Omega_\nu = \{z \in \Omega : \rho(z) < c_\nu\}.$$

Then $\Omega_\nu \Subset \Omega_{\nu+1}$, their union is Ω, and $\partial\Omega_\nu = \{\rho = c_\nu\}$ is smooth by the regular-value theorem. On a complex-tangential vector ξ, the boundary Levi form is $(\mathrm{i}\partial\bar\partial\rho)(\xi, \bar\xi) > 0$, so every Ω_ν is strictly pseudoconvex. Solve on each Ω_ν with the restricted distributionally closed form $f|_{\Omega_\nu}$. The preceding graph-domain argument applies directly to this L^2 datum and gives the same global bound A_f. For every fixed m, the restrictions $u_\nu|_{\Omega_m}$, $\nu \ge m$, are bounded in the single Hilbert space $L^2(\Omega_m, \mathrm{e}^{-\varphi})$; extract a weak subsequence and then diagonalize in m. Compactly supported tests pass the equation to the limit, and weak lower semicontinuity on Ω_m, followed by monotone convergence in m, preserves the global estimate. This explains why pseudoconvexity, rather than smooth boundary, is the final hypothesis. The orthogonal projection of any solution onto $(\ker \bar\partial)^\perp$ remains a solution and can only decrease the norm. It is the unique minimum-norm solution. □

The bundle-valued and weakly pseudoconvex versions of this argument, including the inverse-curvature estimate and the exhaustion passage, appear in [1, Chapter VIII, Theorems 6.1, 6.5, and 6.9]. In Theorem 6.9 the bidegree restriction $q \ge 1$ is indispensable: otherwise the requested solution would have antiholomorphic degree -1.

If $H_\varphi \ge c \, \mathrm{id}_{\mathbb{C}^n}$, then $\|u\|_\varphi^2 \le c^{-1} \, \|f\|_\varphi^2$. Replacing the weight by $\mathrm{e}^{-2\varphi}$ doubles the Hessian and changes the constant accordingly.

7.3.2 Deriving the weighted adjoint

Let u be a compactly supported smooth function and $v = \sum_j v_j \mathrm{d}\bar{z}_j$. Integration by parts in the weighted inner product gives

$$\begin{aligned}(\bar{\partial}u, v)_\varphi &= \sum_j \int_\Omega \frac{\partial u}{\partial \bar{z}_j} \bar{v}_j \mathrm{e}^{-\varphi} \mathrm{d}\lambda \\ &= \int_\Omega u \, \overline{-\sum_j \left(\frac{\partial v_j}{\partial z_j} - \varphi_j v_j \right)} \mathrm{e}^{-\varphi} \mathrm{d}\lambda.\end{aligned}$$

Hence

$$\bar{\partial}^*_\varphi v = -\sum_j \left(\frac{\partial v_j}{\partial z_j} - \varphi_j v_j \right).$$

The zero-order term comes from differentiating the actual exponent $\mathrm{e}^{-\varphi}$. Replacing it by $\mathrm{e}^{-2\varphi}$ replaces φ_j by $2\varphi_j$ and doubles the Levi contribution in the basic estimate.

7.3.3 Why the inverse Hessian is the sharp datum

The inequality

$$|(f, v)_\varphi|^2 \le \left(\int_\Omega f^* H_\varphi^{-1} f \, \mathrm{e}^{-\varphi} \mathrm{d}\lambda \right) \left(\int_\Omega v^* H_\varphi v \, \mathrm{e}^{-\varphi} \mathrm{d}\lambda \right)$$

is pointwise Cauchy–Schwarz in the metric H_φ. It adapts to anisotropic curvature. If

$$H_\varphi = \operatorname{diag}(\lambda_1, \dots, \lambda_n),$$

then the energy is

$$\sum_j \frac{|f_j|^2}{\lambda_j};$$

components in strongly curved directions are cheaper to solve than components in weakly curved directions. Replacing this by $c^{-1}|f|^2$ after using $H_\varphi \ge c \, \mathrm{id}$ loses that information.

For the Gaussian weight $\varphi = t|z|^2$, one has $H_\varphi = t \, \mathrm{id}$ and obtains the normalization test

$$\|u\|^2_\varphi \le t^{-1} \|f\|^2_\varphi.$$

If instead $\varphi = |z_1|^2$ on $\mathbb{C}^n$, the Hessian has a kernel. A source with a $\mathrm{d}\bar{z}_2$ component has infinite inverse-Hessian energy. Adding $\varepsilon|z|^2$ makes the matrix invertible, but the corresponding energy grows like ε^{-1}; no uniform limit follows.

7.3.4 What the theorem does and does not return

The Hilbert-complex argument produces a weak solution in the maximal domain of $\bar{\partial}$. Testing against compactly supported smooth forms gives $\bar{\partial}u = f$ in distributions. If f and φ are smooth, interior elliptic regularity supplies a smooth solution locally, but smoothness up to the boundary is a separate $\bar{\partial}$-Neumann regularity problem.

Among all solutions, the one orthogonal to the weighted Bergman space $A^2(\Omega, \mathrm{e}^{-\varphi}) = \ker \bar{\partial}$ has minimum norm. Indeed, every other solution is $u + h$ with h holomorphic, and orthogonality gives

$$\|u + h\|^2_\varphi = \|u\|^2_\varphi + \|h\|^2_\varphi.$$

Warning

The compatibility condition $\bar\partial f = 0$, membership in the inverse-curvature energy space, and the boundary or exhaustion mechanism are independent hypotheses. None can be inferred merely from the formal equation or from strict plurisubharmonicity of the weight.

7.3.5 The scalar estimate with every logical step separated

Use the closed maximal operators

$$T = \bar\partial : L^2_{0,0}(\Omega, \mathrm{e}^{-\varphi}) \to L^2_{0,1}(\Omega, \mathrm{e}^{-\varphi}), \qquad S = \bar\partial : L^2_{0,1} \to L^2_{0,2}.$$

They are densely defined and closed, and $ST = 0$. A test form belongs to $\operatorname{Dom} T^* \cap \operatorname{Dom} S$; on a smooth boundary its normal component obeys the $\bar\partial$-Neumann condition. The Morrey–Kohn–Hörmander identity then gives

$$Q(v) := \|T^* v\|^2_\varphi + \|Sv\|^2_\varphi \geq \int_\Omega v^* H_\varphi v \, \mathrm{e}^{-\varphi} \mathrm{d}\lambda.$$

For a $\bar\partial$-closed f with finite inverse-Hessian energy, fiberwise Cauchy–Schwarz gives

$$\begin{aligned} |(f, v)_\varphi|^2 &= \left| \int_\Omega (H_\varphi^{-1/2} f)^* (H_\varphi^{1/2} v) \mathrm{e}^{-\varphi} \mathrm{d}\lambda \right|^2 \\ &\leq A_f \int_\Omega v^* H_\varphi v \, \mathrm{e}^{-\varphi} \mathrm{d}\lambda \leq A_f \, Q(v). \end{aligned}$$

We now remove the Sv term before applying Hahn–Banach. Let P be the orthogonal projection of $L^2_{0,1}$ onto $\ker S$. Given $v \in \operatorname{Dom} T^*$, put $v_0 = Pv$ and $v_1 = v - v_0$. Since $\operatorname{im} T \subset \ker S$ and $v_1 \perp \ker S$, for every $w \in \operatorname{Dom} T$ one has

$$(Tw, v_1)_\varphi = 0.$$

This is exactly the adjoint-domain criterion: $v_1 \in \operatorname{Dom} T^*$ and $T^* v_1 = 0$. Hence $v_0 \in \operatorname{Dom} T^* \cap \ker S$ and

$$(f, v)_\varphi = (f, v_0)_\varphi, \qquad T^* v = T^* v_0.$$

Applying the estimate to v_0 gives

$$|(f, v)_\varphi|^2 \leq A_f \, \|T^* v\|^2_\varphi \qquad (v \in \operatorname{Dom} T^*).$$

Therefore $\ell(T^* v) = (f, v)_\varphi$ is well defined on $\operatorname{im} T^*$ and has norm at most $A_f^{1/2}$. Hahn–Banach extends it to $L^2_{0,0}$, and Riesz gives $u \in L^2_{0,0}$ such that

$$(u, T^* v)_\varphi = (f, v)_\varphi \qquad (v \in \operatorname{Dom} T^*).$$

By the definition of the adjoint, $u \in \operatorname{Dom} T$ and

$$Tu = f, \qquad \|u\|^2_\varphi \leq A_f \, .$$

Finally replace u by its orthogonal projection onto $(\ker T)^\perp$ to obtain the unique minimum-norm solution.

An equality case for the Gaussian weight

On $\mathbb{C}^n$ take $\varphi = t|z|^2$, $t > 0$, and $f = \mathrm{d}\bar{z}_1$. Then $H_\varphi = t\,\mathrm{id}$ and f is $\bar\partial$-closed. The function $u = \bar{z}_1$ satisfies $\bar\partial u = f$. Rotational symmetry makes u orthogonal to every entire holomorphic function in the Gaussian space, so it is the canonical solution. Moreover,

$$\int_{\mathbb{C}^n} |\bar{z}_1|^2 \mathrm{e}^{-t|z|^2}\mathrm{d}\lambda = \frac{1}{t}\int_{\mathbb{C}^n} \mathrm{e}^{-t|z|^2}\mathrm{d}\lambda,$$

while

$$A_f = \int f^*(t\,\mathrm{id})^{-1} f\,\mathrm{e}^{-t|z|^2}\mathrm{d}\lambda = \frac{1}{t}\int \mathrm{e}^{-t|z|^2}\mathrm{d}\lambda.$$

Thus equality holds and the factor t^{-1} is sharp. If the actual weight is $\mathrm{e}^{-2t|z|^2}$, the Hessian is $2t\,\mathrm{id}$ and the sharp factor becomes $(2t)^{-1}$.

The exhaustion limit written against a test form

Choose smooth strictly pseudoconvex $\Omega_1 \Subset \Omega_2 \Subset \cdots \nearrow \Omega$ and solve $\bar\partial u_\nu = f$ on Ω_ν with

$$\|u_\nu\|^2_{L^2(\Omega_m,\mathrm{e}^{-\varphi})} \le A_f \qquad (\nu \ge m).$$

For each fixed m, weak compactness gives a subsequence converging on Ω_m; a diagonal subsequence gives one u on all of Ω. If $\eta \in C_c^\infty(\Omega)$, then $\operatorname{supp}\eta \subset \Omega_m$ for some m, and

$$(u, \bar\partial^*\eta) = \lim_\nu (u_\nu, \bar\partial^*\eta) = \lim_\nu (f,\eta) = (f,\eta).$$

Therefore $\bar\partial u = f$ distributionally. Weak lower semicontinuity on every Ω_m and then monotone convergence in m preserve the global estimate.

7.3.6 From a weak solution to the regularity actually needed

The functional-analytic argument initially produces $u \in L^2_{\mathrm{loc}}$ with $\bar\partial u = f$ only distributionally. When u is a function and f is smooth, apply the formal adjoint locally:

$$\Delta_{\bar\partial} u = \bar\partial^*\bar\partial u = \bar\partial^* f.$$

On functions, the Dolbeault Laplacian is a fixed positive multiple of the scalar Laplacian. The right-hand side is smooth, so interior elliptic regularity gives $u \in H^s_{\mathrm{loc}}$ for every s, and Sobolev embedding yields $u \in C^\infty$. The original equation then holds pointwise.

For higher-degree forms, choose the minimum-norm solution. It lies orthogonally to $\ker\bar\partial$ in its own degree. Since the image of $\bar\partial$ from the preceding degree is contained in this kernel, u is orthogonal to that image. The definition of the Hilbert adjoint therefore gives the canonical gauge condition $\bar\partial^* u = 0$ in the weak sense. The system again becomes

$$\Delta_{\bar\partial} u = \bar\partial^* f,$$

and the same interior elliptic estimate applies. If f lies only in a Sobolev space, the estimate gives the corresponding gain of derivatives. Thus existence, the L^2 bound, and smooth regularity are three distinct steps; the last is supplied by ellipticity after the Hilbert-space argument is complete.

7.3.7 Self-study workshop: auditing a weighted $\bar\partial$ existence proof

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Before starting, record the domain, boundary or exhaustion mechanism, source and target bidegrees, weight exponent, Hilbert adjoint, curvature matrix, and desired regularity. Here the convention is the norm with $\mathrm{e}^{-\varphi}$ on a pseudoconvex domain $\Omega \subset \mathbb{C}^n$.

The complete logical chain. The data are a $(0,1)$-form f satisfying $\bar\partial f = 0$ weakly and

$$A_f = \int_\Omega \langle H_\varphi^{-1} f, f\rangle e^{-\varphi}\, d\lambda < \infty, \qquad H_\varphi = (\varphi_{j\bar k}) > 0.$$

The inverse is legitimate only because the Hessian is positive definite. For $v \in \operatorname{Dom}\bar\partial \cap \operatorname{Dom}\bar\partial_\varphi^*$, the Morrey–Kohn–Hörmander estimate is

$$Q(v) := \|\bar\partial v\|_\varphi^2 + \|\bar\partial_\varphi^* v\|_\varphi^2 \geq \int_\Omega \langle H_\varphi v, v\rangle e^{-\varphi}\, d\lambda.$$

The derivative term has been discarded as nonnegative; the boundary Levi term is nonnegative by pseudoconvexity and the adjoint boundary condition.

Fiberwise and then integral Cauchy–Schwarz gives

$$|(f,v)_\varphi|^2 \leq A_f\, Q(v).$$

Let P be the orthogonal projection onto $\ker\bar\partial$ in degree $(0,1)$. For $v \in \operatorname{Dom}\bar\partial_\varphi^*$, the component $v - Pv$ is orthogonal to $\operatorname{im}\bar\partial$ and hence has adjoint equal to zero. Because $f \in \ker\bar\partial$, applying the estimate to Pv yields

$$|(f,v)_\varphi|^2 \leq A_f \|\bar\partial_\varphi^* v\|_\varphi^2.$$

This defines a bounded functional $\ell(\bar\partial_\varphi^* v) = (f,v)_\varphi$ on the range of the adjoint. Hahn–Banach and Riesz produce u, and the definition of the Hilbert adjoint gives

$$\bar\partial u = f, \qquad \|u\|_\varphi^2 \leq A_f\,.$$

Exhaustion of a nonsmooth pseudoconvex domain requires uniform estimates, diagonal weak convergence, distributional passage of the equation, and weak lower semicontinuity. Smoothness of u is a separate regularity question.

Checkpoint (find the first invalid step)

(1) A proof says that psh φ makes H_φ^{-1} exist. What is wrong?
(2) A proof changes $e^{-\varphi}$ to $e^{-2\varphi}$ but keeps the same Hessian. What must change?
(3) A proof obtains the basic inequality and immediately asserts existence. What step is missing?

Solution (checkpoint). Psh gives only semipositivity, so the Hessian may have a kernel; one needs strict positivity or a justified regularization or range condition. The full exponent is 2φ, so its Hessian contribution and the sharp constant are doubled. The bounded graph functional, Hahn–Banach/Riesz representation, and use of $\bar\partial f = 0$ are the missing Hilbert-complex step. ◇

Exit criterion

You can reproduce the theorem with every hypothesis, explain the curvature Cauchy–Schwarz inequality, carry out the duality step, and reject proofs that hide a degenerate inverse, boundary term, or nonuniform limit.

Summary. Hörmander's method has three indispensable components: the Morrey–Kohn basic estimate, Cauchy–Schwarz with the inverse curvature, and Hilbert-complex duality. Closed operators, domains, exhaustion, and weak limits are part of the proof, not optional details.

Exercise 7.20: Gaussian weight

For $\varphi = t|z|^2$, derive $\|u\|_t^2 \le t^{-1}\,\|f\|_t^2$. Check the constant when the weight is $\mathrm{e}^{-2t|z|^2}$.

Exercise 7.21: The compatibility condition

For $n \ge 2$, explain why $\bar\partial f = 0$ is necessary for $\bar\partial u = f$ and give an explicit $(0,1)$-form violating it.

Optional handout: Canonical solutions and the Bergman projection

This handout is optional. No later argument in the core lectures depends on it.

Let $T = \bar\partial$ on weighted L^2 functions. If $f \in \operatorname{im} T$, all solutions of $Tu = f$ form the affine space $u_0 + \ker T$. There is a unique solution $u_{\min} \perp \ker T$, characterized by minimum norm. If the $\bar\partial$-Neumann operator N_1 exists, then

$$u_{\min} = \bar\partial_\varphi^* N_1 f.$$

The weighted Bergman space $A^2(\Omega, \mathrm{e}^{-\varphi}) = \ker \bar\partial \subset L^2$ is closed. Let P_φ be the orthogonal projection onto it. For a smooth function g in the operator domain,

$$P_\varphi g = g - \bar\partial_\varphi^* N_1 \bar\partial g.$$

Thus solving $\bar\partial$ and projecting onto holomorphic functions are the same analytic operation viewed from complementary subspaces.

If $K_\varphi(z,w)$ is the integral kernel of P_φ, then

$$K_\varphi(z,z) = \sup\{|f(z)|^2 : f \in A^2, \|f\|_\varphi \le 1\}.$$

Evaluation from a point and the minimum-norm extension problem are dual. This is the link between Hörmander estimates, Ohsawa–Takegoshi extension, and positivity of Bergman kernels.

Remark (Regularity)

The L^2 estimate proves existence and norm control. Smoothness up to the boundary is a separate issue governed by regularity of N_q. On weakly pseudoconvex domains, the Bergman projection can fail to preserve high Sobolev regularity.

Exercise 7.22: The unit disc

Compute the unweighted Bergman kernel of the unit disc from the orthonormal basis $\sqrt{(k+1)/\pi}\, z^k$ and verify the extremal characterization on the diagonal.

7.4 Lecture 28: Complete Kähler manifolds, singular weights, and multiplier ideals

Week 14 material • 90 minutes

Notation for this lecture

(X,ω) is a complete Kähler manifold, (E,h) a Hermitian holomorphic bundle, $K_X=\Omega_X^n$ the canonical bundle, and $\mathrm{d}V_\omega=\omega^n/n!$ the Kähler volume form. A possibly singular line-bundle metric is written $h=\mathrm{e}^{-\varphi}$ in a local frame. A *quasi-psh* (qpsh) weight is locally the sum of a psh function and a smooth function; equivalently, $\mathrm{i}\partial\bar\partial\varphi\geq -C\omega$ locally for some $C\geq 0$. Its multiplier ideal is $\mathcal{I}(\varphi)_x=\{f\in\mathcal{O}_{X,x}:|f|^2\mathrm{e}^{-\varphi}$ is locally integrable near $x\}$. χ_ν denotes a compactly supported cutoff tending to one with $|\mathrm{d}\chi_\nu|_\omega\to 0$, and weak convergence in weighted L^2 is denoted $\rightharpoonup$. The inequality $B=[\mathrm{i}\Theta(E,h),\Lambda_\omega]\geq c\,\mathrm{id}$ is an operator inequality on the stated bidegree. When B is positive definite, B^{-1} denotes its fiberwise inverse obtained from the spectral decomposition. Nadel vanishing refers to cohomology with the factor $\mathcal{I}(\varphi)$.

Prerequisite results used below

Hopf–Rinow theorem. For a connected Riemannian manifold, metric completeness is equivalent to compactness of closed bounded sets. Consequently a complete manifold admits compactly supported cutoffs $\chi_\nu\to 1$ whose gradients tend to zero uniformly.

Weak compactness in Hilbert space. Every bounded sequence in a Hilbert space has a weakly convergent subsequence, and the norm is weakly lower semicontinuous: $u_j\rightharpoonup u$ implies $\|u\|\leq\liminf_j\|u_j\|$. These two facts allow estimates proved for compactly supported smooth forms and smooth weights to pass first to a complete exhaustion and then to a singular-weight limit.

Why these ideas are needed

The most useful weights in complex geometry are often singular, and the spaces on which estimates are applied are often noncompact. Completeness supplies cutoff functions whose derivatives tend to zero, allowing integration by parts without an artificial boundary. Regularization replaces a singular weight by smooth ones, while uniform estimates and weak compactness permit passage to the limit. Multiplier ideals record exactly which holomorphic germs remain square-integrable against the singular metric. These concepts are introduced together because each controls one stage of the same approximation-and-limit argument.

Plan for the lecture. 0–20 min: the complete Kähler bundle-valued theorem; 20–42: inverse curvature operator; 42–62: cutoffs and weak limits; 62–78: regularizing singular weights; 78–90: multiplier ideals and Nadel vanishing.

Theorem 7.23: Bundle-valued L^2 estimate on a complete Kähler manifold

Let (X,ω) be complete Kähler, (E,h) Hermitian holomorphic, and $q\geq 1$. On E-valued (n,q)-forms let $B=[\mathrm{i}\Theta(E,h),\Lambda_\omega]$. Suppose B is positive definite almost everywhere. If f is $\bar\partial$-closed, belongs to $L^2_{n,q}(X,E;\omega,h)$, and satisfies

$$A_f=\int_X\left\langle B^{-1}f,f\right\rangle_{\omega,h}\mathrm{d}V_\omega<\infty,$$

then there is an E-valued $(n,q-1)$-form u with

$$\bar\partial u=f,\qquad \|u\|^2_{L^2}\leq A_f\,.$$

Proof. Choose cutoffs $0 \le \chi_\nu \le 1$ with compact support, $\chi_\nu \to 1$, and $|\mathrm{d}\chi_\nu| \le C/\nu$. Apply BKN to $\chi_\nu v$. Since $D''(\chi_\nu v) = \chi_\nu D'' v + \bar\partial\chi_\nu \wedge v$ and the analogous commutator formula holds for D''^*, the error terms are bounded by $C\nu^{-1}\,\|v\|$ times the graph norm. Letting $\nu \to \infty$ gives, first for smooth graph-domain forms and then by closure,

$$Q(v) := \|D''v\|^2 + \|D''^* v\|^2 \ge \int_X \langle Bv, v\rangle\, \mathrm{d}V_\omega.$$

Fiberwise spectral decomposition defines the positive operator B^{-1} almost everywhere. Curvature Cauchy–Schwarz gives

$$|(f, v)|^2 \le \left(\int_X \langle B^{-1} f, f\rangle\right)\left(\int_X \langle Bv, v\rangle\right) \le A_f\, Q(v).$$

Apply the Hilbert-complex lemma to $T = D'' : L^2_{n,q-1}(E) \to L^2_{n,q}(E)$ and $S = D'' : L^2_{n,q}(E) \to L^2_{n,q+1}(E)$. The condition $D''f = 0$ is exactly $f \in \ker S$. Hahn–Banach and Riesz therefore produce $u \in \operatorname{Dom} T$ with $Tu = f$ and $\|u\|^2 \le A_f$. Taking $u \perp \ker T$ gives the minimum-norm solution. □

The explicit L^2 hypothesis on f is needed for the global Hilbert-complex proof: finite inverse-curvature energy alone need not imply $f \in L^2$ when the eigenvalues of B are unbounded. On a weakly pseudoconvex manifold one can instead assume $f \in L^2_{\mathrm{loc}}$ and exhaust by relatively compact sublevel sets; this is the more general form of [1, Chapter VIII, Theorems 6.1 and 6.5].

Adding a scalar weight $\mathrm{e}^{-\varphi}$ replaces h by $h\mathrm{e}^{-\varphi}$ and adds

$$\mathrm{i}\partial\bar\partial\varphi \otimes \mathrm{id}_E$$

to the curvature term.

Warning

If B is only semipositive, B^{-1} is undefined on its kernel. One must prove f lies in the range and use the Moore–Penrose inverse, add $\varepsilon\,\mathrm{id}$ with an ε-independent energy bound, or find another coercive term. Writing $(B + \varepsilon\,\mathrm{id})^{-1}$ and merely saying "let $\varepsilon \to 0$" is incomplete.

When a singular weight is regularized, it is smooth only away from an analytic set. The original metric ω on that complement need not be complete, so the preceding theorem cannot yet be applied. The following construction supplies the missing complete metric and explains why logarithmic singularities are useful twice: first in the metric, and later in multiplier ideals.

Lemma 7.24: A complete metric on the complement of an analytic set

Let (X, ω) be a compact Kähler manifold and let $Z \subsetneq X$ be an analytic subset. Then $X \setminus Z$ carries a complete Kähler metric $\widehat{\omega} \ge \omega$.

Proof. Choose a qpsh function ψ with analytic singularities precisely along Z, smooth on $X \setminus Z$, and

$$\mathrm{i}\partial\bar\partial\psi \ge -C\omega.$$

Such a function is obtained locally from $\log \sum_\alpha |g_\alpha|^2$, where the g_α generate the ideal of Z, and then glued after adding smooth local corrections. Subtract a constant so that $\psi < -\mathrm{e}$ and put $\chi(t) = -\log(-t)$ for $t < 0$. Since

$$\chi'(t) = -\frac{1}{t} > 0, \qquad \chi''(t) = \frac{1}{t^2} > 0,$$

the chain rule gives on $X \setminus Z$

$$\mathrm{i}\partial\bar\partial\chi(\psi) = -\frac{1}{\psi}\,\mathrm{i}\partial\bar\partial\psi + \frac{\mathrm{i}\partial\psi \wedge \bar\partial\psi}{\psi^2}.$$

The coefficient $-1/\psi$ is at most e^{-1}. Hence, for a fixed sufficiently large A, the form

$$\widehat{\omega} = A\omega + \mathrm{i}\partial\bar{\partial}[-\log(-\psi)]$$

is Kähler and may be arranged to dominate ω.

It remains to check completeness. The last positive term in the Hessian is

$$\mathrm{i}\partial \log(-\psi) \wedge \bar{\partial} \log(-\psi).$$

If a curve in $X \setminus Z$ approaches Z, then $\psi \to -\infty$ and $\log(-\psi) \to +\infty$. Its $\widehat{\omega}$-length therefore dominates, up to a fixed factor, the total variation of $\log(-\psi)$ and is infinite. Since X is compact, every curve leaving all compact subsets of $X \setminus Z$ approaches Z. The metric is complete. □

The construction is modeled on [1, Chapter VIII, Lemma 7.2]: a convex function of a logarithmic pole provides completeness transverse to the analytic set, while an exhaustion term provides completeness at any noncompact outer boundary. In the present compact setting there is no outer boundary to treat.

The next elementary removal statement is what permits us to solve on the smooth locus of a singular weight and then return to the original manifold.

Lemma 7.25: L^2 removal across an analytic set

Let Z be a proper analytic subset of a coordinate ball $U \subset \mathbb{C}^n$. Suppose u and f have locally square-integrable coefficients on U and $\bar{\partial} u = f$ on $U \setminus Z$ in the sense of distributions. Then $\bar{\partial} u = f$ on all of U.

Proof. On every compact subset $K \Subset U$ there are Lipschitz cutoffs $0 \le \gamma_\nu \le 1$ which vanish on a neighborhood of $Z \cap K$, tend to one almost everywhere, and satisfy

$$\|\mathrm{d}\gamma_\nu\|_{L^2(K)} \longrightarrow 0.$$

Near a smooth point of a hypersurface this is the familiar logarithmic cutoff in the two real normal variables: interpolate between radii ϵ^2 and ϵ as a function of $\log r$. Its Dirichlet energy is $O(|\log \epsilon|^{-1})$. A stratification of Z, followed by a finite covering of K, gives the stated cutoffs for a general analytic set.

Let ψ be a compactly supported smooth test form. The equation on $U \setminus Z$, tested against $\gamma_\nu \psi$, contains one additional term involving $u \wedge \bar{\partial}\gamma_\nu$. Its absolute value is at most

$$C_\psi \, \|u\|_{L^2(K)} \, \|\mathrm{d}\gamma_\nu\|_{L^2(K)},$$

which tends to zero. The remaining terms converge by dominated convergence, since $u, f \in L^2_{\mathrm{loc}}$. Passing to the limit gives the distributional equation on U. □

Definition 7.26: Multiplier ideal

For a qpsh weight φ, define

$$\mathcal{I}(\varphi)_x = \{f \in \mathcal{O}_{X,x} : |f|^2 \mathrm{e}^{-\varphi} \in L^1_{\mathrm{loc}}\}.$$

This is a coherent ideal sheaf. If $\varphi = c \log \sum |g_j|^2 + O(1)$, it measures the vanishing order required to compensate the analytic singularity.

7.4.1 A ledger for singular-weight limits

For a nonsmooth psh weight φ:

(1) choose smooth approximations φ_ν on relatively compact sets and record their monotonicity and curvature loss;
(2) add a small strict term if needed, while maintaining a uniform bound for $\langle B_\nu^{-1} f, f\rangle$;
(3) solve the regularized problems and extract weakly in one fixed weighted space;
(4) pass $\bar\partial u_\nu = f_\nu$ against compactly supported test forms;
(5) preserve the estimate by weak lower semicontinuity or Fatou's lemma.

The regularization theorem used below is more precise than ordinary convolution. We use the decreasing equisingular form of Demailly regularization: for a qpsh weight on a compact Kähler manifold, there are weights φ_ν with analytic singularities, smooth on $X \setminus Z_\nu$, such that $\varphi_\nu \downarrow \varphi$ almost everywhere and in L^1, with the required multiplier-ideal control and curvature loss $\varepsilon_\nu \downarrow 0$. This is a stronger input than the nonmonotone working version used for current approximation in Lecture 31. Thus, if $\mathrm{i}\Theta(L, \mathrm{e}^{-\varphi}) \geq \varepsilon\omega$, the approximants may be chosen so that

$$\mathrm{i}\Theta(L, \mathrm{e}^{-\varphi_\nu}) \geq (\varepsilon - \varepsilon_\nu)\omega.$$

The analytic set Z_ν retains the singularities; its complement admits a complete Kähler metric by Lemma 7.24, which is precisely where the complete estimate applies.

Theorem 7.27: Nadel vanishing

Let X be compact Kähler and let L carry a singular Hermitian metric $h = \mathrm{e}^{-\varphi}$ such that, as currents,

$$\mathrm{i}\Theta(L, h) \geq \varepsilon\omega$$

for some $\varepsilon > 0$. Then

$$H^q(X, K_X \otimes L \otimes \mathcal{I}(\varphi)) = 0, \qquad q \geq 1.$$

Proof (using the singular L^2 resolution). Let $\mathcal{L}^{n,q}(L, h)$ be the sheaf of measurable L-valued (n, q)-forms whose coefficients and distributional $\bar\partial$ are locally square integrable with weight $\mathrm{e}^{-\varphi}$. Its degree-zero kernel is $K_X \otimes L \otimes \mathcal{I}(\varphi)$: in a local frame, a holomorphic coefficient f belongs to the kernel exactly when $|f|^2 \mathrm{e}^{-\varphi}$ is integrable. Smooth partitions of unity act on the $\mathcal{L}^{n,q}$, so these sheaves are fine.

The analytic input is the singular Hörmander lemma: if $\mathrm{i}\Theta(L, h) \geq \varepsilon\omega$, every $\bar\partial$-closed, locally weighted-L^2 L-valued (n, q)-form, $q \geq 1$, is locally $\bar\partial$ of a weighted-L^2 form. On compactly supported data the solution satisfies the same inverse-curvature estimate as in the smooth theorem, with a harmless loss such as $2/(q\varepsilon)$.

Here are the limit steps in that lemma. Equisingular regularization gives weights φ_ν that are smooth off analytic sets Z_ν, decrease to φ, and satisfy, after discarding finitely many indices, $\mathrm{i}\Theta(L, \mathrm{e}^{-\varphi_\nu}) \geq (\varepsilon/2)\omega$. For each ν, choose a complete Kähler metric $\widehat{\omega}_\nu$ on $X \setminus Z_\nu$ by Lemma 7.24 and put $\omega_{\nu,\delta} = \omega + \delta\widehat{\omega}_\nu$ for $\delta > 0$. This metric is complete. Apply the inverse-curvature estimate rather than asserting a lower bound relative to the larger metric $\omega_{\nu,\delta}$. At a point choose an ω-unitary coframe in which

$$\omega_{\nu,\delta} = \mathrm{i}\sum_{j=1}^{n} a_j \, \mathrm{d}z_j \wedge \mathrm{d}\bar z_j, \qquad a_j \geq 1.$$

For a coefficient f_J of an (n, q)-form, the inverse-curvature density supplied by the lower bound $(\varepsilon/2)\omega$ is at most

$$\frac{2|f_J|^2}{\varepsilon\big(\prod_{j\in J} a_j\big)\big(\sum_{j\in J} a_j^{-1}\big)}\,\mathrm{d}V_\omega \leq \frac{2}{q\varepsilon}|f_J|^2\,\mathrm{d}V_\omega.$$

Indeed, $\big(\prod_{j\in J} a_j\big)\big(\sum_{j\in J} a_j^{-1}\big) = \sum_{j\in J}\prod_{k\in J\setminus\{j\}} a_k \geq q$. Summing over J gives a bound independent of δ and then of ν.

We now separate the two limits. Fix ν and let $\delta \downarrow 0$. Weak compactness on relatively compact subsets of $X \setminus Z_\nu$, followed by a diagonal argument, gives a solution u_ν there with the same bound. A qpsh weight is locally bounded above, so weighted local L^2 bounds imply ordinary local L^2 bounds for both u_ν and the datum. Hence Lemma 7.25 extends the equation distributionally across Z_ν.

Next let $\nu \to \infty$. Write $h_\nu = \mathrm{e}^{-\varphi_\nu}$. Since $\varphi_\nu \downarrow \varphi$, the densities h_ν increase to $h = \mathrm{e}^{-\varphi}$. For each fixed m and every $\nu \geq m$,

$$\|u_\nu\|_{L^2(\omega,h_m)} \leq \|u_\nu\|_{L^2(\omega,h_\nu)} \leq C\|f\|_{L^2(\omega,h)}.$$

Extract weakly first in the fixed Hilbert space $L^2(\omega, h_m)$ and diagonalize in m. Since $h_m \leq h_{m+1}$, the inclusions $L^2(\omega, h_{m+1}) \hookrightarrow L^2(\omega, h_m)$ are continuous; uniqueness of weak limits makes the diagonal limits compatible and hence defines one form u. Testing against compactly supported smooth forms gives $\bar\partial u = f$. For every m, weak lower semicontinuity gives the bound with h_m; monotone convergence $h_m \uparrow h$ then gives the desired bound with the original singular metric. This proves the singular Hörmander lemma and hence local exactness.

Consequently

$$0 \to K_X \otimes L \otimes \mathcal{I}(\varphi) \to \mathcal{L}^{n,0}(L,h) \xrightarrow{\bar\partial} \mathcal{L}^{n,1}(L,h) \to \cdots$$

is a fine resolution. The same regularization argument on compact X solves every closed global section in positive degree. Thus the global-section complex is exact in degrees $q \geq 1$. It computes sheaf cohomology, and the asserted groups vanish. □

7.4.2 Completeness as a cutoff theorem

Fix a base point on a complete Riemannian manifold and let $r(x)$ be the distance to it. Choose $\chi \in C_c^\infty(\mathbb{R})$ with $\chi = 1$ on $(-\infty, 1]$ and set $\chi_\nu(x) = \chi(r(x)/\nu)$. Before smoothing, this is Lipschitz and satisfies $|\mathrm{d}\chi_\nu| \leq C/\nu$ almost everywhere. Completeness makes closed bounded balls compact, so the cutoffs have compact support. To smooth them, cover the compact transition annulus by finitely many coordinate balls, convolve in each chart, and patch with a partition of unity. Choosing the convolution radii sufficiently small gives $\widetilde{\chi}_\nu \in C_c^\infty(X)$ which is still 1 on the inner ball, vanishes outside a slightly enlarged outer ball, and satisfies $|\mathrm{d}\widetilde{\chi}_\nu| \leq C'/\nu$ with C' independent of ν.

For a graph-domain form v,

$$D''(\widetilde{\chi}_\nu v) = \widetilde{\chi}_\nu D'' v + \bar\partial\widetilde{\chi}_\nu \wedge v,$$

and the adjoint has the analogous contraction error. The error norms are bounded by $C\nu^{-1}\|v\|$, so compact-support identities pass to the closed-operator domain. Completeness is used exactly here; it is not a curvature assumption.

7.4.3 Multiplier ideals as radial integrability thresholds

On $\mathbb{C}^n$, take $\varphi = c\log|z|^2 + O(1)$ with $c > 0$. If a holomorphic germ f vanishes to order m, then its leading size is $|f| \asymp r^m$ in generic directions, and

$$|f|^2\mathrm{e}^{-\varphi}\mathrm{d}\lambda \asymp r^{2m-2c+2n-1}\mathrm{d}r\,\mathrm{d}S.$$

This is integrable at zero exactly when $m > c - n$. Hence

$$\mathcal{I}(c\log|z|^2)_0 = \mathfrak{m}_0^{\max\{0,\lfloor c\rfloor - n+1\}}.$$

For $n = 1$, this reduces to $(z^{\lfloor c\rfloor})$. The strict inequality at the borderline exponent explains the jump when c crosses an integer.

For a general analytic singularity $\varphi = c \log \sum |g_j|^2 + O(1)$, a log resolution turns the problem into a collection of such monomial integrability tests along exceptional divisors. The Jacobian of the resolution contributes the discrepancy term; this is why multiplier ideals reflect both vanishing orders and the geometry of the modification.

7.4.4 A safe singular-limit template

All approximations must be compared in fixed Hilbert spaces. If φ_ν are regularized weights, record whether $\mathrm{e}^{-\varphi_\nu}$ increases or decreases, the curvature loss in $\mathrm{i}\partial\bar\partial\varphi_\nu$, and a bound for

$$\int \langle B_\nu^{-1} f_\nu, f_\nu \rangle \mathrm{e}^{-\varphi_\nu}.$$

Only a parameter-independent bound permits weak compactness. If $\varphi_\nu \downarrow \varphi$, then $h_\nu = \mathrm{e}^{-\varphi_\nu} \uparrow h = \mathrm{e}^{-\varphi}$. For each fixed m, the tail of the solution sequence is bounded in $L^2(h_m)$. A diagonal weak limit, lower semicontinuity in every h_m, and monotone convergence $h_m \uparrow h$ recover the norm for the singular metric. Passing the equation against compactly supported tests, crossing analytic pole sets, smoothness, and minimum-norm character are separate verification steps.

Remark (Why Nadel vanishing is a sheaf theorem)

The weighted L^2 complex resolves $K_X \otimes L \otimes \mathcal{I}(\varphi)$ because degree-zero holomorphic coefficients are exactly those integrable against the singular metric. The multiplier ideal is not an extra decoration on the conclusion; it is the kernel of the analytic resolution.

7.4.5 How completeness supplies admissible cutoffs

Fix a base point and let $r(x)$ be its Riemannian distance. Completeness makes closed metric balls compact by Hopf–Rinow. Smooth r outside a harmless neighborhood of the cut locus and choose $\chi \in C_c^\infty(\mathbb{R})$ with $\chi = 1$ on $(-\infty, 1]$ and $\chi = 0$ on $[2, \infty)$. Set

$$\chi_\nu(x) = \chi(r(x)/\nu).$$

Then $\chi_\nu \to 1$, the supports are compact, and

$$|\mathrm{d}\chi_\nu| \le \frac{C}{\nu}.$$

For a graph-domain form v,

$$D''(\chi_\nu v) = \chi_\nu D'' v + \bar\partial\chi_\nu \wedge v,$$
$$D''^*(\chi_\nu v) = \chi_\nu D''^* v - \iota_{(\partial\chi_\nu)^\sharp} v.$$

The error norms are at most $C\nu^{-1}\|v\|$. Applying BKN to $\chi_\nu v$ and sending $\nu \to \infty$ proves the basic estimate on the closed operator domain. Completeness is used exactly to obtain this compactly supported approximation.

Why an L^2 equation crosses the singular set

When a regularized weight is smooth only on $X \setminus Z$, solving there is not yet the desired statement. The missing step is Lemma 7.25. Its logarithmic cutoffs are tailored to the real codimension-two normal direction of a divisor: their gradients tend to zero in L^2, so the cutoff error pairs with an L^2 solution and disappears. Higher-codimension strata are no worse. Thus local L^2 control of both sides, rather than a pointwise extension theorem, is what carries the distributional $\bar\partial$ equation across the analytic set.

Multiplier ideals from a radial integral

On $\mathbb{C}^n$ let $\varphi = c\log|z|^2$ and let a holomorphic germ f vanish to order m. On a set of angular directions of positive measure, $|f(z)|$ is comparable to r^m. Therefore local integrability is decided by

$$\int_0^{\varepsilon} r^{2m} r^{-2c} r^{2n-1}\,\mathrm{d}r < \infty.$$

This holds exactly when

$$2m - 2c + 2n > 0, \qquad \text{or equivalently} \qquad m > c - n.$$

Hence

$$\boxed{\mathcal{I}(c\log|z|^2)_0 = \mathfrak{m}_0^{\max\{0,\lfloor c-n\rfloor+1\}}.}$$

For example, in one variable $c = 1$ requires one zero, while $0 < c < 1$ requires none. The strict inequality explains the jump when c crosses an integer threshold.

A safe singular-weight limiting argument

Suppose φ is psh and choose decreasing regularizations φ_ν with analytic singularities such that

$$\mathrm{i}\partial\bar\partial\varphi_\nu \geq -\varepsilon_\nu\omega, \qquad \varepsilon_\nu \downarrow 0.$$

Add a fixed strictly psh term if necessary, but compute the actual curvature operator B_ν of the full exponent. The indispensable hypothesis is a uniform bound

$$\sup_\nu \int \langle B_\nu^{-1} f_\nu, f_\nu\rangle \mathrm{e}^{-\varphi_\nu}\mathrm{d}V < \infty.$$

On the smooth locus $X \setminus Z_\nu$, first solve with the complete metrics $\omega + \delta\widehat{\omega}_\nu$ supplied by Lemma 7.24. Let $\delta \downarrow 0$ with ν fixed and use Lemma 7.25 to extend the distributional equation across Z_ν.

Only then let $\nu \to \infty$. Put $h_\nu = \mathrm{e}^{-\varphi_\nu}$. The decrease $\varphi_\nu \downarrow \varphi$ gives $h_\nu \uparrow h = \mathrm{e}^{-\varphi}$. A uniform h_ν-bound controls u_ν in every fixed earlier space $L^2(\omega, h_m)$ once $\nu \geq m$. Diagonal weak compactness, weak lower semicontinuity in each h_m, and monotone convergence $h_m \uparrow h$ preserve the final estimate. If the inverse energy instead blows up like ε_ν^{-1}, no conclusion follows from this argument.

There is also a useful pointwise check when the complete auxiliary metric is $\omega' = \mathrm{i}\sum a_j \mathrm{d}z_j\wedge\mathrm{d}\bar z_j$ in an ω-unitary coframe, with every $a_j \geq 1$. On an (n,q) coefficient indexed by J, the volume and form-norm factors combine with the inverse curvature to give the denominator

$$\left(\prod_{j\in J} a_j\right)\left(\sum_{j\in J} a_j^{-1}\right) = \sum_{j\in J}\prod_{k\in J\setminus\{j\}} a_k \geq q.$$

This is the concrete reason the inverse-curvature energy remains uniformly bounded as the auxiliary complete metric degenerates back to ω.

Why Nadel vanishing is a sheaf statement

Locally, a holomorphic L-valued $(n,0)$-form belongs to the weighted L^2 kernel exactly when its scalar coefficient lies in $\mathcal{I}(\varphi)$. Thus the complex of weighted forms begins with

$$0 \to K_X \otimes L \otimes \mathcal{I}(\varphi) \to \mathcal{L}^{n,0}_{(2)}(L, \mathrm{e}^{-\varphi}) \xrightarrow{\bar\partial} \mathcal{L}^{n,1}_{(2)} \to \cdots.$$

The singular Hörmander estimate proves local exactness in positive degree. Smooth partitions of unity preserve local weighted L^2 integrability, so the later sheaves are fine. Taking global sections therefore computes the sheaf cohomology.

On compact X, the curvature lower bound $i\Theta(L, e^{-\varphi}) \geq \varepsilon\omega$ makes the global $\bar{\partial}$ equation solvable in every degree $q \geq 1$. Hence the cohomology of the global-section complex vanishes, yielding

$$H^q(X, K_X \otimes L \otimes \mathcal{I}(\varphi)) = 0.$$

Both ingredients are needed: the multiplier ideal identifies the correct degree-zero kernel, and the L^2 estimate makes the fine resolution exact.

7.4.6 Why the multiplier ideal is coherent: a proof ledger

Fix a relatively compact pseudoconvex coordinate ball U and define

$$\mathcal{H}_U = \left\{ f \in \mathcal{O}(U) : \int_U |f|^2 e^{-\varphi}\, dV < \infty \right\}.$$

Because a psh function is locally bounded above, the weighted norm controls the ordinary L^2 norm on smaller balls. The submean inequality and Cauchy estimates therefore make evaluation of every fixed jet a continuous functional on $\mathcal{H}_U$.

Let $\mathcal{J}$ be the ideal sheaf generated by the members of $\mathcal{H}_U$. The ring $\mathcal{O}_{U,x}$ is Noetherian, so at a fixed x finitely many functions $f_1, \ldots, f_N \in \mathcal{H}_U$ generate the stalk $\mathcal{J}_x$. The nontrivial point is to compare this ideal with all locally square-integrable germs.

Take $F \in \mathcal{I}(\varphi)_x$, multiply a representative by a cutoff, and solve its $\bar{\partial}$-error on U with the weight φ plus a logarithmic pole at x. Increasing the pole order produces, for every k, a function $F_k \in \mathcal{H}_U$ such that

$$F - F_k \in \mathfrak{m}_x^k \mathcal{I}(\varphi)_x.$$

It follows that

$$\mathcal{I}(\varphi)_x \subseteq \mathcal{J}_x + \mathfrak{m}_x^k \mathcal{I}(\varphi)_x \quad \text{for every } k.$$

Conversely, $\mathcal{J}_x \subseteq \mathcal{I}(\varphi)_x$ by construction and $\mathfrak{m}_x^k \mathcal{I}(\varphi)_x \subseteq \mathcal{I}(\varphi)_x$, so the sum on the right is contained in $\mathcal{I}(\varphi)_x$. Since $\mathfrak{m}_x^k \mathcal{I}(\varphi)_x \subseteq \mathfrak{m}_x^k \mathcal{O}_{U,x}$, the class of every $F \in \mathcal{I}(\varphi)_x$ lies in $\bigcap_k \mathfrak{m}_x^k(\mathcal{O}_{U,x}/\mathcal{J}_x)$. The Krull intersection theorem for the Noetherian module $\mathcal{O}_{U,x}/\mathcal{J}_x$ makes this intersection zero. Hence $\mathcal{I}(\varphi)_x = \mathcal{J}_x$.

The finite functions $f_1, \ldots, f_N$ define a coherent image sheaf $\operatorname{im}(\mathcal{O}_U^N \to \mathcal{O}_U)$. Repeating the cutoff-and-division estimate on a smaller ball gives equality with $\mathcal{I}(\varphi)$ at nearby points. Thus the multiplier ideal is coherent. The proof uses four inputs in order: weighted L^2 approximation, jet forcing by logarithmic poles, Noetherian local algebra, and Oka coherence for a finite presentation.

7.4.7 Self-study workshop: passing from smooth complete estimates to singular metrics

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall complete Kähler cutoffs, bundle-valued (n, q) estimates, weak compactness in Hilbert space, monotone comparison of weights, and local integrability of singular powers.

Two limiting mechanisms that must not be conflated. Completeness provides cutoffs $\chi_\nu \to 1$ with compact support and controlled gradient. Apply the compact-support BKN identity to $\chi_\nu u$. Commutators such as $[\bar\partial, \chi_\nu]u = \bar\partial\chi_\nu \wedge u$ tend to zero because $|\mathrm{d}\chi_\nu|$ is uniformly small on the exhausting regions. This extends integration by parts to the maximal graph domain.

For a singular psh weight φ, choose smooth approximants φ_ν on the complements of their analytic pole sets and add a small strict term if necessary. One must record:

$$\text{curvature loss}, \quad \sup_\nu A_{f,\nu} < \infty, \quad u_\nu \rightharpoonup u, \quad \bar\partial u = f, \quad \|u\|_\varphi^2 \le \liminf_\nu \|u_\nu\|_{\varphi_\nu}^2.$$

The direction of approximation determines whether Fatou's lemma or comparison of weighted norms applies. For the decreasing choice $\varphi_\nu \downarrow \varphi$, the densities $h_\nu = \mathrm{e}^{-\varphi_\nu}$ increase to $h = \mathrm{e}^{-\varphi}$. Thus, for fixed m and $\nu \ge m$, an h_ν-bound also gives an h_m-bound. One extracts weakly in each fixed $L^2(h_m)$, diagonalizes, and finally uses monotone convergence in m. Merely writing $(B + \varepsilon\,\mathrm{id})^{-1}$ and sending $\varepsilon \to 0$ is invalid if the energy blows up on $\ker B$. If the approximants are smooth only off analytic sets, one must also use logarithmic cutoffs to prove that the limiting L^2 equation extends distributionally across those sets.

Multiplier ideals record the local integrability required by the singular metric. In one variable, for $\varphi = c\log|z|^2$ and $f = z^m$,

$$|f|^2\mathrm{e}^{-\varphi}\,\mathrm{d}A \asymp r^{2m-2c+1}\,\mathrm{d}r\,\mathrm{d}\theta.$$

The integral converges exactly when $2m - 2c + 1 > -1$, or $m > c - 1$. Thus $\mathcal{I}(\varphi)_0$ consists of germs whose order satisfies this inequality.

Checkpoint (uniformity before convergence)

(1) Which property of completeness is used in the BKN proof?
(2) Why is weak convergence alone insufficient when the weights vary?
(3) Compute the smallest integer $m \ge 0$ with $z^m \in \mathcal{I}(2.4\log|z|^2)_0$.

Solution (checkpoint). Completeness supplies compactly supported cutoffs whose gradient errors tend to zero. One must compare the varying weighted norms and use the correct monotonicity before lower semicontinuity applies. The condition is $m > 1.4$, so the smallest integer is $m = 2$. ◇

Exit criterion

You can separate the cutoff argument from the singular-weight approximation, list all uniform estimates needed for a weak limit, and compute a multiplier ideal by a radial integral.

Summary. Completeness replaces compactness in justifying integration by parts, and the inverse curvature operator retains the sharp estimate. Singular weights require uniform estimates and weak limits; multiplier ideals encode the integrability threshold.

Exercise 7.28: A one-dimensional multiplier ideal

Compute $\mathcal{I}(c\log|z|^2)_0$. Remember that the area element is $r\mathrm{d}r\mathrm{d}\theta$.

Exercise 7.29: Complete cutoffs

Prove that a complete Riemannian manifold admits $\chi_\nu \in C_c^\infty$, $0 \le \chi_\nu \le 1$, $\chi_\nu \to 1$, and $|\mathrm{d}\chi_\nu| \le C/\nu$ (first use the distance function, then smooth it).

Optional handout: The strong openness theorem

This handout is optional. No later argument in the core lectures depends on it.

Multiplier ideals satisfy an elementary monotonicity: if $\psi \le \varphi + O(1)$, then $\mathcal{I}(\psi) \subset \mathcal{I}(\varphi)$. A much deeper property is strong openness.

Theorem 7.30: Strong openness

For every psh function φ,

$$\mathcal{I}(\varphi) = \bigcup_{\varepsilon>0} \mathcal{I}((1+\varepsilon)\varphi).$$

Equivalently, if $|f|^2 \mathrm{e}^{-\varphi}$ is locally integrable, then it remains integrable after making the singularity slightly stronger.

Proof (one-dimensional proof and the higher-dimensional quoted input). In one complex dimension there is a complete elementary proof. Write $f(z) = z^m g(z)$ with $g(0) \neq 0$. The Riesz decomposition and Jensen's formula show that, for a subharmonic φ,

$$|f|^2 e^{-\varphi} \in L^1_{\mathrm{loc}}(0) \quad \Longleftrightarrow \quad \nu(\varphi, 0) < m + 1. \tag{28.1}$$

For the nontrivial direction, compare circular means of φ with $\nu(\varphi,0)\log r^2$ and integrate in polar coordinates; equality at the threshold gives the divergent integral $\int_0 r^{-1}\,dr$. If (28.1) holds, choose $p > 1$ so close to 1 that $p\nu(\varphi,0) < m+1$. Applying (28.1) to $p\varphi$ gives $f \in \mathcal{I}(p\varphi)_0$, proving strong openness on a Riemann surface.

In higher dimension, integrability is not determined by one Lelong number, and a proof by restriction to a single curve is invalid. Guan and Zhou prove the theorem by an induction on dimension built around a movable L^2 extension estimate. The key quantitative input controls extensions from varying level hypersurfaces by the mass of $|f|^2 e^{-\varphi}$ on those hypersurfaces. Fubini selects levels on which that mass decays; Ohsawa–Takegoshi extends the restrictions with uniform control; coherence and Noetherian stabilization convert failure of openness into a fixed holomorphic obstruction. The uniform extensions violate that obstruction, closing the induction. This paragraph identifies the quoted higher-dimensional mechanism rather than reproducing the long quantitative induction; importantly, it does not use the false claim that arbitrary holomorphic interpolants must have a universal $|a|^{-2}$ lower bound. □

Example 7.31: Monomial weights

For $\varphi = \sum_j c_j \log|z_j|^2$ and $f = z_1^{m_1}\cdots z_n^{m_n}$, Fubini's theorem gives integrability exactly when $m_j > c_j - 1$ for every j. These strict inequalities visibly persist after replacing every c_j by $(1+\varepsilon)c_j$ for sufficiently small $\varepsilon > 0$.

Strong openness implies that jumping numbers of multiplier ideals are genuine thresholds and underlies precise semicontinuity statements in families.

Exercise 7.32: Radial singularity

Determine for which integers m the germ z^m belongs to $\mathcal{I}(c\log|z|^2)$ and verify strong openness directly in this example.

Chapter 8

Extension, the Calabi–Yau Equation, and Modern Positivity

8.1 Lecture 29: Introduction to the Ohsawa–Takegoshi extension method

Week 15 material • 90 minutes

Notation for this lecture

$\Omega \subset \mathbb{C}^n$ is a pseudoconvex domain and $Y = \Omega \cap \{z_n = 0\}$ the model hypersurface; $z' = (z_1, \dots, z_{n-1})$. f is holomorphic on Y, F is a holomorphic extension to Ω, φ is a psh weight, and $\mathrm{d}\lambda_k$ is Lebesgue measure in complex dimension k. Its Levi matrix is $H_\varphi = (\varphi_{j\bar{k}})$. In the invariant formulation, $Y = \{\sigma = 0\} \subset X$ is the smooth zero set of a transverse section $\sigma \in H^0(X, E)$ of a rank-r Hermitian vector bundle E. Along Y, the differential $D\sigma : T^{1,0}X|_Y \to E|_Y$ vanishes on $T^{1,0}Y$ and induces an isomorphism $N_{Y/X} \to E|_Y$. Its determinant $\Lambda^r(D\sigma) \in \det N_{Y/X}^* \otimes \det E|_Y$ is the normal Jacobian used in adjunction. This restriction is independent of the connection chosen to form $D\sigma$. F_ε^0 is a cutoff extension, $g_\varepsilon = \bar\partial F_\varepsilon^0$ its error, and u_ε the correction solving $\bar\partial u_\varepsilon = g_\varepsilon$. OT and BKN abbreviate Ohsawa–Takegoshi and Bochner–Kodaira–Nakano. The Bergman space is $A^2(U) = \mathcal{O}(U) \cap L^2(U)$ with the measure or weight stated in context.

Prerequisite results used below

Hörmander weighted existence theorem. If $\Omega \subset \mathbb{C}^n$ is pseudoconvex, $\varphi \in C^2(\Omega)$ is strictly psh, and $f = \sum f_j \mathrm{d}\bar z_j$ is $\bar\partial$-closed with

$$\int_\Omega f^* H_\varphi^{-1} f \, \mathrm{e}^{-\varphi} \mathrm{d}\lambda < \infty,$$

then there is u with $\bar\partial u = f$ and $\int_\Omega |u|^2 \mathrm{e}^{-\varphi} \mathrm{d}\lambda$ bounded by the displayed inverse-Hessian integral.

Adjunction for a transverse zero set. If $Y = \{\sigma = 0\}$ is the smooth zero set of a transverse section of a rank-r bundle E, then $K_Y \simeq (K_X \otimes \det E)|_Y$. Locally, division by $\Lambda^r(D\sigma)$ identifies an ambient top form with determinant-bundle value along Y with its adjunction trace. For $r = 1$, this is division by $\mathrm{d}\sigma$ in a local trivialization.

Why these ideas are needed

Solving $\bar\partial u = f$ manufactures holomorphic objects, but many geometric problems also prescribe their values on a hypersurface or submanifold. Pure sheaf theory gives extension on a Stein space without controlling its size. Ohsawa–Takegoshi adds an L^2 estimate whose constant is independent of the chosen datum. The singular normal weight forces the correction term to have zero trace, while a twisted BKN estimate pays for the cutoff error. Quantitative extension is what makes the theorem stable enough for Bergman kernels, singular metrics, and algebraic applications.

Plan for the lecture. 0–18 min: why extension from a submanifold is difficult; 18–38: the hyperplane theorem; 38–62: singular weights and twisted estimates; 62–78: zero sets of bundle sections; 78–90: applications and scaling checks.

Cartan B already extends holomorphic functions from a closed submanifold of a Stein manifold. Ohsawa–Takegoshi adds quantitative L^2 control, which is indispensable for singular metrics and algebraic geometry.

Theorem 8.1: Ohsawa–Takegoshi, hyperplane version

Let $\Omega \subset \mathbb{C}^n$ be pseudoconvex with $|z_n| < 1$, let $Y = \Omega \cap \{z_n = 0\}$, and let φ be psh. If $f \in O(Y)$ and

$$\int_Y |f|^2 \mathrm{e}^{-\varphi} \mathrm{d}\lambda_Y < \infty,$$

then there is $F \in O(\Omega)$ with $F|_Y = f$ and

$$\int_\Omega |F|^2 \mathrm{e}^{-\varphi} \mathrm{d}\lambda \le C_{\mathrm{OT}} \int_Y |f|^2 \mathrm{e}^{-\varphi} \mathrm{d}\lambda_Y .$$

After coordinate normalization, C_{OT} is independent of f and φ; refined versions achieve the optimal constant.

Here $\mathrm{d}\lambda_Y$ is the Euclidean measure in the coordinates $z' = (z_1, \dots, z_{n-1})$ on Y, and $\varphi|_Y$ is the upper-semicontinuous restriction. The finiteness assumption is part of the statement: if the restriction of φ is identically $-\infty$ on a component, nonzero data there do not have finite weighted norm. This hypersurface form is the model case of [15]; the bundle-valued L^2 method and the complete-metric step are developed in [1, Chapter VIII, §7].

Proof (the twisted L^2 argument). We prove a slightly stronger weighted estimate with a universal, not necessarily optimal, constant. Exhaust Ω by smoothly bounded strictly pseudoconvex domains and regularize φ from above. Uniform estimates and a diagonal weak limit reduce the proof to the case in which the domain and weight are smooth. Rescale the normal coordinate so that $s = z_n$ satisfies $|s| < \mathrm{e}^{-1}$.

The twisted basic estimate is the following consequence of the BKN identity. For positive smooth functions η, λ and compactly supported $(n,1)$-forms v, let $\|\cdot\|_\varphi$ be the L^2 norm with density $\mathrm{e}^{-\varphi}$, let $\bar\partial^*_\varphi$ be the Hilbert adjoint of $\bar\partial$ for this norm, and let Λ be contraction with the Euclidean Kähler form. Then

$$\|\sqrt{\eta+\lambda}\,\bar\partial^*_\varphi v\|^2_\varphi + \|\sqrt{\eta}\,\bar\partial v\|^2_\varphi \ge \big\langle [\eta \mathrm{i}\partial\bar\partial\varphi - \mathrm{i}\partial\bar\partial\eta - \mathrm{i}\lambda^{-1}\partial\eta\wedge\bar\partial\eta, \Lambda] v, v\big\rangle. \tag{29.1}$$

It follows by applying BKN to $\sqrt{\eta}, v$ and estimating the cross term with $2|ab| \le \lambda|a|^2 + \lambda^{-1}|b|^2$. Notice that no positivity of $\mathrm{i}\partial\bar\partial\eta$ is being assumed.

Set

$$\chi_0(t) = t - \log(1-t), \quad \sigma_\varepsilon = \log(|s|^2+\varepsilon^2), \quad \eta_\varepsilon = \varepsilon - \chi_0(\sigma_\varepsilon), \quad \lambda_\varepsilon = \frac{\chi_0'(\sigma_\varepsilon)^2}{\chi_0''(\sigma_\varepsilon)}. \tag{29.2}$$

Here $1 \le \chi_0' \le 2$ and $\chi_0'' = (1-t)^{-2} > 0$. Apply (29.1) with the additional singular metric factor $|s|^{-2}$. Direct differentiation gives

$$\mathrm{i}\partial\bar\partial\sigma_\varepsilon = \frac{\varepsilon^2 \mathrm{i}\,\mathrm{d}s\wedge\mathrm{d}\bar s}{(|s|^2+\varepsilon^2)^2}.$$

Indeed,

$$\partial\sigma_\varepsilon = \frac{\bar s\,\mathrm{d}s}{|s|^2+\varepsilon^2}, \qquad \partial\eta_\varepsilon = -\chi_0'(\sigma_\varepsilon)\partial\sigma_\varepsilon,$$

and

$$-\mathrm{i}\partial\bar\partial\eta_\varepsilon = \chi_0'(\sigma_\varepsilon)\mathrm{i}\partial\bar\partial\sigma_\varepsilon + \chi_0''(\sigma_\varepsilon)\mathrm{i}\partial\sigma_\varepsilon\wedge\bar\partial\sigma_\varepsilon.$$

By the definition of λ_ε, $\lambda_\varepsilon^{-1}\partial\eta_\varepsilon\wedge\bar\partial\eta_\varepsilon = \chi_0''(\sigma_\varepsilon)\partial\sigma_\varepsilon\wedge\bar\partial\sigma_\varepsilon$. The last term therefore cancels the negative square in (29.1). The part left by the twist is exactly

$$\chi_0'(\sigma_\varepsilon)\frac{\varepsilon^2\mathrm{i}\,\mathrm{d}s\wedge\mathrm{d}\bar s}{(|s|^2+\varepsilon^2)^2} \ge 0.$$

The psh curvature of φ contributes another nonnegative term. This normal approximate delta mass is what pays for the derivative of the cutoff.

On each coordinate neighborhood of Y, extend f constantly in the normal coordinate and glue these local extensions with a smooth partition of unity. All local extensions agree on Y and the derivatives of the partition sum to zero, so the resulting smooth extension $\widetilde f$ satisfies $\bar\partial\widetilde f|_Y = 0$, hence $\bar\partial\widetilde f = O(|s|)$. Choose a cutoff θ supported in $[0,1)$ and equal to 1 on $[0,1/2]$. For

$$F_\varepsilon^0 = \theta(|s|^2/\varepsilon^2)\widetilde f, \qquad g_\varepsilon = \bar\partial F_\varepsilon^0,$$

the main term of g_ε is a scalar multiple of $\bar\partial\eta_\varepsilon\,\widetilde f$, supported where $|s| \asymp \varepsilon$; the term containing $\bar\partial\widetilde f$ is $O(\varepsilon)$. Let B_ε denote the nonnegative commutator on $(n,1)$-forms supplied by the right-hand side of (29.1), and interpret B_ε^{-1} on its positive range. Curvature Cauchy–Schwarz in that operator inequality, followed by integration in the normal disc, yields

$$\int_\Omega \langle B_\varepsilon^{-1} g_\varepsilon, g_\varepsilon\rangle |s|^{-2}\mathrm{e}^{-\varphi}\mathrm{d}\lambda \le C\int_Y |f|^2\mathrm{e}^{-\varphi}\mathrm{d}\lambda_Y. \tag{29.3}$$

Let us record the scale cancellation hidden in this line. On the annulus $|s| \asymp \varepsilon$,

$$|\bar\partial\theta(|s|^2/\varepsilon^2)| \lesssim \varepsilon^{-1}, \qquad \frac{\varepsilon^2}{(|s|^2+\varepsilon^2)^2} asymp \varepsilon^{-2}.$$

Thus the cutoff error has squared size $O(\varepsilon^{-2})$, the inverse of the normal curvature contributes $O(\varepsilon^2)$, and the normal annulus has area $O(\varepsilon^2)$. The extra metric factor $|s|^{-2}$ contributes $O(\varepsilon^{-2})$, so the four powers cancel. The term containing $\bar\partial\widetilde f = O(|s|)$ is smaller. A partition of unity in the tangential variables and the submean inequality for the psh weight turn this slice computation into (29.3). This is the quantitative reason the bound is independent of ε; throughout this proof C denotes a positive constant independent of the regularization and exhaustion parameters. The Hilbert-space existence lemma used here says that the basic estimate bounds the linear functional $v \mapsto (g_\varepsilon, v)_\varphi$ in the graph norm of $\bar\partial_\varphi^*$; Riesz representation then produces a solution with the same bound. It therefore solves $\bar\partial u_\varepsilon = g_\varepsilon$ and gives

$$\int_\Omega \frac{|u_\varepsilon|^2\mathrm{e}^{-\varphi}}{|s|^2(\eta_\varepsilon+\lambda_\varepsilon)}\mathrm{d}\lambda \le C\int_Y |f|^2\mathrm{e}^{-\varphi}\mathrm{d}\lambda_Y. \tag{29.4}$$

On $\Omega\setminus Y$, the difference $F_\varepsilon = F_\varepsilon^0 - u_\varepsilon$ is holomorphic. Estimate (29.4) makes u_ε, and hence F_ε, locally L^2 across Y; the L^2 removable-singularity theorem extends F_ε holomorphically across Y. Thus $u_\varepsilon = F_\varepsilon^0 - F_\varepsilon$ is smooth near Y. For each fixed $\varepsilon > 0$, the factor $\eta_\varepsilon + \lambda_\varepsilon$ is bounded above on

a sufficiently small neighborhood of Y. Fubini's theorem and the nonintegrability of $|s|^{-2}$ therefore force $u_\varepsilon|_Y = 0$. Consequently $F_\varepsilon|_Y = f$.

From (29.2), $\eta_\varepsilon + \lambda_\varepsilon \le C(-\log(|s|^2+\varepsilon^2))^2$. Combining (29.4) with the elementary cutoff estimate and then taking weak limits gives

$$\int_\Omega \frac{|F|^2 \mathrm{e}^{-\varphi}}{|s|^2(-\log|s|)^2}\mathrm{d}\lambda \le C\int_Y |f|^2\mathrm{e}^{-\varphi}\mathrm{d}\lambda_Y. \tag{29.5}$$

Because $|s|^2(-\log|s|)^2$ is bounded above on $0<|s|<\mathrm{e}^{-1}$, (29.5) implies the ordinary estimate in the theorem after changing the universal constant. Montel convergence preserves the restriction to Y, while Fatou's lemma preserves the bound. Finally pass through the regularized weights and the exhaustion; all constants are independent of both parameters. □

8.1.1 Why ordinary Hörmander is not enough

Start with a smooth extension $\widetilde{f}$ and try to solve $\bar\partial u = \bar\partial(\chi\widetilde{f})$ so that $F = \chi\widetilde{f} - u$. To retain $F|_Y = f$, one must force $u|_Y = 0$. A singular weight containing $\log|z_n|^2$ makes finite weighted norm force divisibility by z_n. The resulting curvature degenerates, so one uses a *twisted* Bochner identity: multiply the basic estimate by an auxiliary function η and complete squares between errors from $\partial\eta$ and $\partial\bar\partial\eta$.

Theorem 8.2: Zero set of a transverse bundle section

Let X be weakly pseudoconvex and Kähler, let (E,h_E) have rank r, and let $\sigma\in H^0(X,E)$ be transverse to the zero section, so $Y=\sigma^{-1}(0)$ is smooth of codimension r. Let (L,h_L) be a Hermitian holomorphic line bundle and give $A=\det E\otimes L$ the product metric. Write $\{\,\cdot,\cdot\,\}$ for the h_E-pairing of E-valued forms and $|\sigma|$ for the corresponding norm. Suppose

$$\mathrm{i}\Theta(A) + r\mathrm{i}\partial\bar\partial\log|\sigma|^2 \ge 0.$$

Assume also that a continuous function $\alpha\ge 1$ satisfies

$$\mathrm{i}\Theta(A) + r\mathrm{i}\partial\bar\partial\log|\sigma|^2 \ge \alpha^{-1}\frac{\{\mathrm{i}\Theta(E)\sigma,\sigma\}}{|\sigma|^2}, \qquad |\sigma|\le \mathrm{e}^{-\alpha}.$$

Then every $f\in H^0(Y,K_Y\otimes L|_Y)$ with finite L^2 norm extends to $F\in H^0(X,K_X\otimes\det E\otimes L)$, in the adjunction sense $F|_Y/\Lambda^r(D\sigma)=f$, and

$$\int_X \frac{|F|^2}{|\sigma|^{2r}(-\log|\sigma|)^2} \le C_r\int_Y |f|^2.$$

The pointwise norms and volume forms are induced by ω, h_E, h_L. The constant C_r depends only on r.

Proof (twisted curvature computation and limiting argument). Put $\sigma_\varepsilon=\log(|\sigma|^2+\varepsilon^2)$ and use the functions $\eta_\varepsilon,\lambda_\varepsilon$ from (29.2), together with the singular factor $|\sigma|^{-2r}$ in the metric of A. Let D' be the $(1,0)$-part of the Chern connection of (E,h_E). The Chern-connection calculation is

$$\mathrm{i}\partial\bar\partial\sigma_\varepsilon = \frac{\mathrm{i}\{D'\sigma,D'\sigma\}}{|\sigma|^2+\varepsilon^2} - \frac{\mathrm{i}\{D'\sigma,\sigma\}\wedge\{\sigma,D'\sigma\}}{(|\sigma|^2+\varepsilon^2)^2} - \frac{\{\mathrm{i}\Theta(E)\sigma,\sigma\}}{|\sigma|^2+\varepsilon^2}.$$

The fiberwise Lagrange inequality

$$\mathrm{i}\{D'\sigma,\sigma\}\wedge\{\sigma,D'\sigma\} \le |\sigma|^2\mathrm{i}\{D'\sigma,D'\sigma\}$$

and the two stated curvature hypotheses give exactly the positive operator required by the twisted estimate (29.1). This is why the curvature of E, the factor r, and the scaling condition on σ occur in the theorem.

Choose a smooth ambient representative of the adjunction data whose $\bar\partial$ vanishes to first order on Y, cut it off in $\{|\sigma| < \varepsilon\}$, and solve its $\bar\partial$ error on $X \setminus Y$. The inverse-curvature estimate and integration in the r normal variables give a bound independent of ε; the normal integral is the numerical constant C_r. The correction is integrable against $|\sigma|^{-2r}$, so its trace on Y is zero. The bounds $\eta_\varepsilon + \lambda_\varepsilon \le C(-\log(|\sigma|^2 + \varepsilon^2))^2$ then give the displayed weighted estimate. Weak compactness supplies F, and adjunction identifies its trace with f. The minimum-norm solution is global, so no separate patching of local holomorphic extensions is needed. □

The theorem is written so that every bundle in the adjunction map is visible. In particular, f is intrinsic data in $K_Y \otimes L|_Y$, whereas F takes values in $K_X \otimes \det E \otimes L$. Omitting $\det E$ would make the quotient by $\Lambda^r(D\sigma)$ coordinate dependent. The curvature calculation is the higher-codimension analogue of the logarithmic-weight calculation in [1, Chapter VIII, Theorem 7.1].

The proof uses both $D''\sigma = 0$ and the pointwise Cauchy–Schwarz inequality

$$\mathrm{i}\{D'\sigma, \sigma\} \wedge \{\sigma, D'\sigma\} \le |\sigma|^2\, \mathrm{i}\{D'\sigma, D'\sigma\}.$$

The factor $|\sigma|^2$ is forced by scaling $\sigma \mapsto \lambda\sigma$.

Corollary 8.3: Three consequences of extension

(1) For hypersurfaces and smooth complete intersections satisfying the stated curvature hypotheses, restriction is surjective with control of the minimum L^2 norm.
(2) Extension from a point gives lower bounds for weighted Bergman kernels.
(3) Together with additional extension and positivity theorems, OT is a principal input in results on psh variation of Bergman kernels and deformation invariance of pluricanonical sections.

Proof. For (1), represent the prescribed section on the submanifold and apply the relevant OT theorem; the constructed extension lies in the affine space of all extensions and orthogonal projection to the complement of the kernel of restriction gives the unique minimum-norm representative. For (2), take $Y = \{x\}$ and normalize the value to one. The reciprocal of the minimum squared norm is the weighted Bergman kernel $K_\varphi(x)$, so the extension estimate gives its lower bound. Statement (3) records uses of OT, not formal consequences of the version proved here: psh variation additionally uses Berndtsson's positivity theorem for direct images, and deformation invariance uses an iterated extension argument together with the relevant family and singular-metric hypotheses. □

Warning

Ohsawa–Takegoshi is not obtained by simply inserting a singular weight into Lecture 27. One must design a twisted weight, keep constants uniform during regularization, and prove that the limiting correction vanishes on the submanifold. Extending functions, top forms, or adjoint-bundle sections also changes Jacobian factors and weights.

8.1.2 Why the singular norm forces the correct trace

In one normal variable s, suppose a holomorphic function g satisfies

$$\int_{|s|<\varepsilon} \frac{|g(s)|^2}{|s|^2}\, \mathrm{d}\lambda(s) < \infty.$$

Write $g(s) = s^m a(s)$ with $a(0) \neq 0$. The radial integral behaves like

$$\int_0^{\varepsilon} r^{2m-2} r \, \mathrm{d}r,$$

which is finite exactly when $m \geq 1$. Thus $g(0) = 0$. With parameters along a hypersurface, Fubini's theorem gives the same conclusion almost everywhere and holomorphicity gives it everywhere.

In codimension r, the normal volume element is comparable to $\rho^{2r-1}\mathrm{d}\rho$, while the singular factor is $|\sigma|^{-2r}$. A correction with nonzero trace would again produce the divergent integral $\int \rho^{-1}\mathrm{d}\rho$. This is the analytic reason the L^2 correction vanishes on Y. The logarithmic square in the final estimate is just strong enough to make the extended section itself integrable after the correction has acquired the required vanishing.

8.1.3 Point extension and the Bergman kernel

For a weighted domain, the Bergman kernel on the diagonal has the extremal description

$$K_\varphi(x) = \sup_{0 \neq F \in A^2(\Omega, \mathrm{e}^{-\varphi})} \frac{|F(x)|^2 \mathrm{e}^{-\varphi(x)}}{\|F\|_\varphi^2} = \frac{\mathrm{e}^{-\varphi(x)}}{\inf\{\|F\|_\varphi^2 : F(x) = 1\}}.$$

An extension theorem from the point x gives a competitor with controlled norm and therefore a lower bound for $K_\varphi(x)$. Conversely, the reproducing kernel gives the unique minimum-norm function with prescribed value.

On the unit disc with unweighted Lebesgue area, rotational symmetry shows that the constant function 1 is orthogonal to every holomorphic function vanishing at the origin. It is therefore the minimum-norm extension of the value 1, with norm squared π. Hence $K_\Delta(0) = 1/\pi$. Under a dilation $z \mapsto Rz$, the minimum norm scales by R^2 and the kernel by R^{-2}, illustrating the normal-volume scaling in Ohsawa–Takegoshi.

If the defining section is rescaled, $\sigma \mapsto a\sigma$, the adjunction trace and the denominator $|\sigma|^{2r}$ both change. Checking that these powers cancel is a fast way to detect a missing determinant or Jacobian factor in a bundle-valued extension statement.

Warning

Qualitative extension follows from Cartan B on a Stein ambient space, but it neither selects the minimum-norm extension nor controls behavior as the weight or submanifold varies. Those quantitative features are the content of Ohsawa–Takegoshi.

8.1.4 A product-domain extension computed exactly

Let $\Omega = \Delta^{n-1} \times \Delta_R$ with unweighted Lebesgue measure and $Y = \{z_n = 0\}$. Given $f \in A^2(\Delta^{n-1})$, the constant-in-the-normal-direction extension

$$F(z', z_n) = f(z')$$

satisfies

$$\int_\Omega |F|^2 \mathrm{d}\lambda = \pi R^2 \int_Y |f|^2 \mathrm{d}\lambda_Y.$$

It is the minimum-norm extension. Indeed, every other extension has a power series

$$G(z', z_n) = f(z') + \sum_{k \geq 1} a_k(z') z_n^k.$$

The monomials in z_n are orthogonal in the disc, so Fubini gives

$$\|G\|^2 = \pi R^2 \|f\|_Y^2 + \sum_{k\geq 1} \frac{\pi R^{2k+2}}{k+1} \|a_k\|_Y^2 \geq \pi R^2 \|f\|_Y^2.$$

This model explains the normal-volume factor and gives a scaling check for every general extension theorem.

The cutoff error near a hypersurface

Let $s = z_n$ and choose a smooth ambient extension $\widetilde{f}$ with $\bar\partial \widetilde{f} = O(|s|)$. For a cutoff χ set

$$F_\varepsilon^0 = \chi(|s|^2/\varepsilon^2)\widetilde{f}.$$

Then

$$\bar\partial F_\varepsilon^0 = \chi'(|s|^2/\varepsilon^2)\frac{s}{\varepsilon^2}\widetilde{f}\,\mathrm{d}\bar s + \chi(|s|^2/\varepsilon^2)\bar\partial\widetilde{f}.$$

The first term has size $O(\varepsilon^{-1})$ but is supported in an annulus of normal area $O(\varepsilon^2)$; the second has size $O(\varepsilon)$. An ordinary unweighted estimate would control the correction but would not force its trace to vanish. The singular factor $|s|^{-2}$ changes the problem: finite norm then implies divisibility by s.

If a holomorphic correction has expansion $u(s) = a_0 + a_1 s + \cdots$, then

$$\int_{|s|<\delta} \frac{|u(s)|^2}{|s|^2}\mathrm{d}\lambda(s) < \infty$$

forces $a_0 = 0$, because a nonzero constant term produces $2\pi|a_0|^2 \int_0^\delta r^{-1}\mathrm{d}r = \infty$. This is the exact analytic mechanism that preserves the prescribed trace.

Local adjunction in coordinates

Suppose $Y = \{z_{n-r+1} = \cdots = z_n = 0\}$ and the defining bundle section is

$$\sigma = (z_{n-r+1}, \ldots, z_n).$$

Then

$$\Lambda^r(\mathrm{d}\sigma) = \mathrm{d}z_{n-r+1} \wedge \cdots \wedge \mathrm{d}z_n.$$

If

$$F = f(z')\,\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n,$$

its adjunction trace is

$$\frac{F|_Y}{\Lambda^r(\mathrm{d}\sigma)} = f(z',0)\,\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_{n-r},$$

a section of K_Y. Under a change of defining equations $\widetilde\sigma = A\sigma$, the determinant $\det A$ multiplies $\Lambda^r(\mathrm{d}\sigma)$, while the factor $\det E$ in the ambient bundle changes by the same amount. This proves that the adjunction quotient is coordinate independent.

Point extension and the Bergman extremal problem

For a weighted Bergman space, evaluation at x is a bounded functional whenever a mean-value estimate holds. Riesz gives a vector K_x with $F(x) = (F, K_x)_\varphi$. The minimum-norm function satisfying $F(x) = 1$ is

$$\frac{K_x}{K_\varphi(x,x)}, \qquad \min_{F(x)=1} \|F\|_\varphi^2 = \frac{1}{K_\varphi(x,x)}.$$

An Ohsawa–Takegoshi extension from the point produces an admissible F of controlled norm, hence a lower bound on $K_\varphi(x,x)$. This is a quantitative conclusion that Cartan's qualitative extension theorem cannot supply.

Passing from regularized extensions to the final one

The preceding estimates are first proved with smooth data, whereas the weight in an extension theorem is usually only plurisubharmonic. We now record the limiting argument, since three different compactness principles enter and should not be conflated.

Choose relatively compact pseudoconvex domains $\Omega_1 \Subset \Omega_2 \Subset \cdots$ exhausting Ω. On Ω_ν, regularize the given weight φ from above by smooth strictly plurisubharmonic weights φ_ν, arranged so that $\varphi_\nu \searrow \varphi$ on every fixed compact subset. The smooth extension theorem gives holomorphic functions $F_\nu \in \mathcal{O}(\Omega_\nu)$ with

$$F_\nu|_{Y\cap\Omega_\nu} = f, \qquad \int_{\Omega_\nu} |F_\nu|^2 \mathrm{e}^{-\varphi_\nu}\mathrm{d}\lambda \le C \int_Y |f|^2 \mathrm{e}^{-\varphi}\mathrm{d}\lambda_Y, \tag{29.6}$$

where the constant C is independent of ν.

Fix $K \Subset \Omega$, and choose a slightly larger compact set $K \Subset K' \Subset \Omega$. For all large ν, the functions φ_ν are bounded above on K' by one constant $M_{K'}$. Consequently (29.6) bounds the unweighted L^2-norm of F_ν on K':

$$\int_{K'} |F_\nu|^2 \mathrm{d}\lambda \le \mathrm{e}^{M_{K'}} \int_{K'} |F_\nu|^2 \mathrm{e}^{-\varphi_\nu}\mathrm{d}\lambda.$$

The submean inequality on coordinate balls whose doubles lie in K' then gives a uniform supremum bound on K. Cauchy's inequalities give bounds for all derivatives on a still smaller compact set. Thus Montel's theorem and a diagonal subsequence produce a holomorphic limit $F \in \mathcal{O}(\Omega)$, with convergence uniform on compact subsets. Since the restriction of every F_ν to Y is f, the same compact convergence shows $F|_Y = f$.

It remains to retain the estimate. On a fixed Ω_j, pointwise convergence and $\varphi_\nu \to \varphi$ give

$$|F_\nu|^2 \mathrm{e}^{-\varphi_\nu} \longrightarrow |F|^2 \mathrm{e}^{-\varphi}.$$

Fatou's lemma, followed by monotone convergence as $j \to \infty$, yields

$$\int_\Omega |F|^2 \mathrm{e}^{-\varphi}\mathrm{d}\lambda \le C \int_Y |f|^2 \mathrm{e}^{-\varphi}\mathrm{d}\lambda_Y.$$

The logical division is worth remembering: Montel produces a holomorphic limit, compact convergence preserves the trace, and Fatou preserves the norm inequality.

8.1.5 Self-study workshop: locating the Hörmander estimate inside Ohsawa–Takegoshi

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall adjunction, singular bundle metrics, cutoff errors, twisted BKN estimates, and weak compactness. The bare scalar Hörmander theorem does not by itself give an extension with the correct trace and uniform constant.

The extension architecture. Let $Y = \{\sigma = 0\} \subset X$ be a smooth hypersurface and let f be holomorphic data on Y. First use adjunction to interpret the target correctly:

$$K_Y \simeq (K_X \otimes \mathcal{O}(Y))|_Y.$$

Choose a smooth ambient representative $\widetilde{f}$ with the prescribed trace and multiply it by a cutoff depending on $|\sigma|/\varepsilon$. The resulting form F^0_ε has correct trace but is not holomorphic:

$$g_\varepsilon = \bar\partial F^0_\varepsilon$$

is supported in the thin annulus where $|\sigma| \asymp \varepsilon$.

The singular weight is engineered so that solving $\bar\partial u_\varepsilon = g_\varepsilon$ costs exactly the normal factor needed to cancel the cutoff derivative. A twisted BKN inequality, not an untwisted invocation of Hörmander, produces a bound independent of ε. Then

$$F_\varepsilon = F^0_\varepsilon - u_\varepsilon$$

is holomorphic. The singular norm forces the minimum solution u_ε to have zero trace on Y, so $F_\varepsilon|_Y = f$. Weak compactness and lower semicontinuity give a global limit with the extension estimate.

In the product model $X = Y \times \Delta$, the constant extension $F(y, w) = f(y)$ lets one check every factor directly. Integrating in w isolates the normal numerical constant and shows what the general theorem is trying to preserve.

Checkpoint (why the construction is twisted)

(1) Why not begin with a globally holomorphic extension?
(2) What must be uniform as $\varepsilon \downarrow 0$?
(3) How does the singular norm protect the trace after subtracting u_ε?

Solution (checkpoint). A global holomorphic extension is precisely the desired conclusion; only a smooth ambient representative is initially available. The L^2 bound for the correction and the curvature lower bound must be independent of ε. The singular weight makes any correction with nonzero trace have infinite or prohibitively large normal energy, so the minimum solution vanishes along Y in the required adjunction sense. ◇

Exit criterion

You can identify the smooth extension, cutoff error, singular weight, twisted estimate, trace argument, and weak limit as six distinct steps, and you can explain why ordinary Hörmander solvability is only one ingredient.

Summary. OT upgrades $\bar\partial$ solvability to quantitative extension with a precise normal singularity. Holomorphicity, curvature compensation, uniform estimates, and limiting integrability are all essential.

Exercise 8.4: Point extension and the Bergman kernel

Take Y to be one point and explain how the OT estimate gives a lower bound for the weighted Bergman kernel.

Exercise 8.5: Scaling audit

For a trivial line bundle with $\sigma(z) = az$, compare the powers of a on both sides of the corrected inequality.

Optional handout: The twisted estimate behind extension

This handout is optional. No later argument in the core lectures depends on it.

Here is the algebraic heart of the Ohsawa–Takegoshi method. Let $\eta > 0$ and let $\lambda > 0$ be auxiliary functions. Applying the Bochner identity to $\sqrt{\eta}, v$ and completing the square yields a twisted estimate of the form

$$\left\|\sqrt{\eta}\,\bar\partial v\right\|^2 + \left\|\sqrt{\eta+\lambda}\,\bar\partial^* v\right\|^2 \geq \left([\eta\,\mathrm{i}\Theta - \mathrm{i}\partial\bar\partial\eta - \lambda^{-1}\mathrm{i}\partial\eta\wedge\bar\partial\eta, \Lambda]v, v\right).$$

The exact bundle curvature is inserted in Θ. The term with λ^{-1} is the price of Cauchy–Schwarz when differentiating $\sqrt{\eta}$.

For a hypersurface defined by $w = 0$, regularize $\sigma_\varepsilon = \log(|w|^2+\varepsilon^2)$. Choose $\chi_0(t) = t-\log(1-t)$ and set

$$\eta_\varepsilon = \varepsilon - \chi_0(\sigma_\varepsilon), \qquad \lambda_\varepsilon = \frac{\chi_0'(\sigma_\varepsilon)^2}{\chi_0''(\sigma_\varepsilon)}.$$

Then

$$-\mathrm{i}\partial\bar\partial\eta - \lambda^{-1}\mathrm{i}\partial\eta\wedge\bar\partial\eta$$

contains a positive multiple of $\varepsilon^2\mathrm{i}dw\wedge d\bar w/(|w|^2+\varepsilon^2)^2$. This approximate delta mass controls the normal trace on Y.

Start with a cutoff extension $\chi(|w|^2/\varepsilon^2)f$. Its $\bar\partial$ is supported in an ε-tube around Y. The twisted estimate and Hilbert-space duality give a correction u_ε with a uniform singular weighted bound. For each fixed ε, the factor $|w|^{-2}$ is nonintegrable while $\eta_\varepsilon+\lambda_\varepsilon$ is finite on $w = 0$; smoothness of the correction therefore forces its trace on Y to vanish. Extracting a weak limit gives the extension and the logarithmically weighted estimate, which in turn implies the ordinary L^2 bound on a normalized normal disc.

Remark (Invariance of plurigenera)

Applied fiberwise to adjoint bundles, extension is a central analytic input in the deformation invariance of plurigenera. The complete theorem also requires the appropriate singular metrics and an iterated extension argument. In its familiar smooth form the family is a smooth projective morphism with connected fibers; singular-family versions require additional hypotheses. Invariance is therefore not a formal consequence of the hyperplane statement.

Exercise 8.6: Approximate delta mass

Show that $\varepsilon^2\mathrm{i}dw\wedge d\bar w/(|w|^2+\varepsilon^2)^2$ has uniformly bounded mass on discs and converges, after normalization, to δ_0.

8.2 Lecture 30: The Calabi–Yau theorem and the continuity method

Week 15 material • 90 minutes

Notation for this lecture

(X, ω) is a compact Kähler manifold of complex dimension n. For a real function φ, $\omega_\varphi = \omega + \mathrm{i}\partial\bar\partial\varphi$, and the condition $\omega_\varphi > 0$ means that this is a Kähler form. $F \in C^\infty(X, \mathbb{R})$ is the prescribed log-density, $V = \int_X \omega^n$, $t \in [0, 1]$ is the continuity parameter, and c_t is the constant normalizing total volume. $\Delta_\omega f = \mathrm{tr}_\omega(\mathrm{i}\partial\bar\partial f)$ is the geometric-sign complex Laplacian; Δ_{d} is the nonnegative Hodge Laplacian. The symbol $\mathrm{tr}_\omega \eta$ is the trace of a $(1, 1)$-form η with respect to ω. $C^{k,\alpha}$ denotes the usual Hölder space and $\mathrm{Ric}(\omega)$ the Ricci form; $c_1(X) = c_1(T^{1,0}X)$ is the first Chern class of the holomorphic tangent bundle.

Prerequisite results used below

Banach-space implicit function theorem. Let $\Phi : U \subset B_1 \to B_2$ be continuously differentiable between Banach spaces. If $D\Phi(x_0) : B_1 \to B_2$ is an isomorphism, then Φ is a C^1 diffeomorphism between neighborhoods of x_0 and $\Phi(x_0)$.

Elliptic Schauder estimate. For a uniformly elliptic second-order linear operator L with $C^{k,\alpha}$ coefficients and a fixed normalization eliminating its kernel,

$$\|u\|_{C^{k+2,\alpha}} \le C\big(\|Lu\|_{C^{k,\alpha}} + \|u\|_{C^0}\big).$$

Together with compact embedding of Hölder spaces, this turns uniform a priori bounds into closedness of the continuity path.

Why these ideas are needed

A Kähler class fixes total volume but contains many metrics. Prescribing the volume form, equivalently the Ricci curvature up to cohomology, leads to a nonlinear complex Monge–Ampère equation. The continuity method connects an obvious solution to the desired one: the implicit-function theorem gives openness, while a priori estimates prevent solutions from degenerating and give closedness. Separating the C^0, second-order, and higher-order estimates reveals what each controls. The Calabi–Yau theorem is therefore both a geometric existence result and a model strategy for nonlinear elliptic equations.

Plan for the lecture. 0–18 min: the Calabi conjecture and Monge–Ampère; 18–35: continuity method; 35–58: C^0 estimate; 58–75: second-order estimate; 75–85: higher regularity; 85–90: geometric consequences.

Let (X^n, ω) be compact Kähler. To avoid ambiguity in 2π factors, write

$$\omega_\varphi = \omega + \mathrm{i}\partial\bar\partial\varphi = \omega + 2\pi\mathrm{dd}^{\mathrm{c}}\varphi.$$

In this lecture

$$\Delta_\omega f = \mathrm{tr}_\omega(\mathrm{i}\partial\bar\partial f)$$

is the scalar complex Laplacian, with $\Delta_\omega f \le 0$ at a local maximum. It is therefore a geometric-sign operator; with the Riemannian and form conventions of Lecture 13, the nonnegative Hodge Laplacian on functions is $\Delta_{\mathrm{d}} = -2\Delta_\omega$. We keep the subscripts to prevent these two signs from being confused.

Theorem 8.7: Calabi–Yau theorem

Let $F \in C^\infty(X, \mathbb{R})$ satisfy

$$\int_X e^F \omega^n = \int_X \omega^n.$$

There is a unique smooth real function φ with $\sup_X \varphi = 0$ such that

$$\omega_\varphi > 0, \qquad \omega_\varphi^n = e^F \omega^n.$$

The integral condition is necessary: expanding $\omega_\varphi^n - \omega^n$ as $i\partial\bar\partial\varphi$ times a closed sum and applying Stokes gives $\int_X \omega_\varphi^n = \int_X \omega^n$. The normalization $\sup_X \varphi = 0$ removes the only ambiguity, since adding a constant does not change ω_φ.

Proof (continuity method and a priori estimates).

Uniqueness. Let $u = \varphi - \psi$. The algebraic identity

$$\omega_\varphi^n - \omega_\psi^n = i\partial\bar\partial u \wedge \sum_{j=0}^{n-1} \omega_\varphi^j \wedge \omega_\psi^{n-1-j}$$

and Stokes' theorem give

$$0 = \int_X u(\omega_\varphi^n - \omega_\psi^n) = -\sum_{j=0}^{n-1} \int_X i\partial u \wedge \bar\partial u \wedge \omega_\varphi^j \wedge \omega_\psi^{n-1-j}.$$

Every term on the right is nonpositive, so $\partial u = 0$ and u is constant. The normalization $\sup_X \varphi = \sup_X \psi = 0$ gives $u = 0$.

The continuity path. Consider

$$\omega_{\varphi_t}^n = e^{tF + c_t} \omega^n, \qquad t \in [0, 1],$$

where c_t equalizes total volume. At $t = 0$, $\varphi = 0$. The linearization is

$$D\left[\log \frac{\omega_\varphi^n}{\omega^n}\right](\dot\varphi) = \Delta_{\omega_\varphi} \dot\varphi.$$

The Laplacian is invertible modulo constants, so the implicit-function theorem proves openness.

More precisely,

$$c_t = -\log\left(\frac{1}{V}\int_X e^{tF} \omega^n\right), \qquad V = \int_X \omega^n.$$

Fix the additive constant in φ by $\int_X \varphi\, \omega^n = 0$ and write $C_0^{k+2,\alpha}(X)$ for the functions in $C^{k+2,\alpha}(X)$ satisfying this zero-mean condition. Treat the normalizing constant as an unknown. The linearization in $(\dot\varphi, \dot c)$ is

$$(\dot\varphi, \dot c) \longmapsto \Delta_{\omega_{\varphi_t}} \dot\varphi - \dot c.$$

This maps $C_0^{k+2,\alpha}(X) \oplus \mathbb{R}$ isomorphically onto $C^{k,\alpha}(X)$: integration against $\omega_{\varphi_t}^n$ determines $\dot c$, self-adjoint elliptic theory solves the resulting zero-mean equation, and the fixed ω^n-mean determines the additive constant of $\dot\varphi$. The Banach-space implicit-function theorem therefore makes the solution set open. This formulation avoids the incorrect assertion that $\Delta_{\omega_{\varphi_t}}$ preserves zero mean with respect to the fixed volume form ω^n.

A priori estimates for closedness.

(1) C^0 **estimate.** Normalize $\sup_X \varphi = 0$ and put $u = -\varphi \geq 0$. Use integration by parts in

$$\int_X u^p (\omega_\varphi^n - \omega^n)$$

together with

$$\omega_\varphi^n - \omega^n = \mathrm{i}\partial\bar\partial\varphi \wedge \sum_{j=0}^{n-1} \omega_\varphi^j \wedge \omega^{n-1-j}.$$

If $S = \sum_{j=0}^{n-1} \omega_\varphi^j \wedge \omega^{n-1-j}$, then $\mathrm{d}S = 0$ and Stokes' theorem gives the exact energy identity

$$\int_X u^p (\omega_\varphi^n - \omega^n) = p \int_X u^{p-1} \mathrm{i}\partial u \wedge \bar\partial u \wedge S. \tag{30.1}$$

The term of S equal to ω^{n-1} controls the background gradient. Since $\omega_\varphi^n = \mathrm{e}^{tF+c_t}\omega^n$ and c_t is uniformly bounded, the left-hand side is at most $C\int_X u^p\omega^n$. Using $|\mathrm{d}u^{(p+1)/2}|^2 = \frac{(p+1)^2}{4} u^{p-1}|\mathrm{d}u|^2$ therefore yields

$$\left\|\mathrm{d}u^{(p+1)/2}\right\|_{L^2}^2 \leq C\frac{(p+1)^2}{p} \int_X u^p\omega^n, \qquad p \geq 1. \tag{30.2}$$

Here is the Green-kernel step behind the starting L^1 bound. Let $G_\omega(x,y)$ be a Green kernel for Δ_ω; adding a constant in the y-variable does not change the representation formula, so normalize it by $G_\omega \leq 0$. If x_0 is a maximum point of φ, then $\varphi(x_0) = 0$, while $\omega_\varphi > 0$ gives $\Delta_\omega\varphi \geq -n$. Therefore

$$-\frac{1}{V}\int_X \varphi\,\omega^n = \int_X G_\omega(x_0,y)\Delta_\omega\varphi(y)\,\omega^n(y) \leq n\int_X [-G_\omega(x_0,y)]\,\omega^n(y) \leq C.$$

Compactness makes the last bound uniform in x_0. Thus $\int_X u\,\omega^n \leq C$. Apply the Sobolev inequality to $u^{(p+1)/2}$ and use Poincaré's inequality after subtracting its average. If $\beta = n/(n-1)$ for $n > 1$ and if $\beta > 1$ is fixed when $n = 1$, this gives

$$\|u\|_{L^{(p+1)\beta}}^{p+1} \leq C\frac{(p+1)^2}{p}\|u\|_{L^p}^p + C\|u\|_{L^{(p+1)/2}}^{p+1}. \tag{30.3}$$

Both exponents on the right are smaller than the new exponent on the left. Hölder interpolation and iteration of $p+1 \mapsto \beta(p+1)$, starting from the L^1 bound, give $\|\varphi\|_{L^\infty} \leq C$. Evaluating the equation only at a maximum controls neither the infimum nor the oscillation.

(2) **Second-order estimate.** Put $Q = \log \mathrm{tr}_\omega\,\omega_\varphi - A\varphi$. The Chern–Lu calculation and a lower bound for the bisectional curvature of ω yield

$$\Delta_{\omega_\varphi} \log \mathrm{tr}_\omega\,\omega_\varphi \geq -C_1\,\mathrm{tr}_{\omega_\varphi}\,\omega - C_2.$$

Since $\Delta_{\omega_\varphi}\varphi = n - \mathrm{tr}_{\omega_\varphi}\,\omega$, choosing $A > C_1$ and applying the maximum principle bounds $\mathrm{tr}_{\omega_\varphi}\,\omega$ at a maximum point x_0 of Q. If $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of g_φ relative to g, the algebraic inequality

$$\sum_j \lambda_j \leq C_n \left(\sum_j \lambda_j^{-1}\right)^{n-1} \prod_j \lambda_j$$

and the determinant equation bound $\mathrm{tr}_\omega\,\omega_\varphi$ at x_0. Since $Q(x) \leq Q(x_0)$, the C^0 estimate then transfers this forward trace bound to every $x \in X$. Finally the determinant is bounded below and the sum of the eigenvalues is bounded above, so each λ_j is bounded above and below uniformly.

(3) **Higher-order estimates.** Uniform eigenvalue bounds make the complex Monge–Ampère equation uniformly elliptic and concave in the complex Hessian. The Evans–Krylov theorem gives a $C^{2,\alpha}$ estimate from the C^{α} norm of the right-hand side. Differentiate the equation; each derivative satisfies a linear uniformly elliptic equation with controlled coefficients. Schauder estimates then give uniform $C^{k,\alpha}$ bounds for every k. Calabi's third-order estimate is an alternative route to the first Hölder control.

The estimates are uniform in t. If t_j tends to a boundary point of the solution set, compactness in $C^{k,\alpha}$ gives a smooth limiting solution. Thus the solution set is closed and, since $[0,1]$ is connected, it is all of $[0,1]$. □

The existence theorem and the original a priori-estimate strategy are due to Yau [14]. The presentation above uses the modern Evans–Krylov route after the two-sided metric estimate; Yau's third-order estimate followed by Schauder theory is an alternative at that stage.

Warning

A determinant bound alone does not bound every eigenvalue: control of a product does not control the largest and smallest factors. The second-order trace estimate and the determinant equation must be used together.

Corollary 8.8: Prescribing Ricci curvature

With the convention fixed in Lecture 19,

$$\mathrm{Ric}(\omega_\varphi) = \mathrm{Ric}(\omega) - \mathrm{i}\partial\bar\partial \log \frac{\omega_\varphi^n}{\omega^n},$$

every real closed $(1,1)$-form representing $2\pi c_1(X)$ is the Ricci form of a unique Kähler metric in each Kähler class. The corresponding potential is unique up to an additive constant. If $c_1(X) = 0$, each Kähler class contains a unique Ricci-flat metric.

Proof. Let η be a real closed $(1,1)$-form representing $2\pi c_1(X)$. Then $\mathrm{Ric}(\omega) - \eta$ is cohomologically trivial. The $\partial\bar\partial$-lemma gives a real smooth function G with

$$\mathrm{Ric}(\omega) - \eta = \mathrm{i}\partial\bar\partial G.$$

Add a constant to G so that $\int_X \mathrm{e}^G \omega^n = \int_X \omega^n$, and apply the Calabi–Yau theorem to obtain $\omega_\varphi^n = \mathrm{e}^G \omega^n$. The transformation formula then gives

$$\mathrm{Ric}(\omega_\varphi) = \mathrm{Ric}(\omega) - \mathrm{i}\partial\bar\partial G = \eta.$$

If two metrics in the same Kähler class have Ricci form η, the logarithm of the ratio of their volume forms is pluriharmonic and hence constant. Equal total volumes make the constant zero, and uniqueness in the Calabi–Yau theorem makes the metrics equal. Taking $\eta = 0$ proves the final assertion when $c_1(X) = 0$. □

Remark (Geometric interpretation)

Locally the equation prescribes the determinant of a complex Hessian; globally it redistributes volume inside a fixed Kähler class. The continuity method splits existence into linear invertibility and nonlinear a priori estimates.

8.2.1 Complex dimension one as a linear model

On a compact Riemann surface, every $(1,1)$-form is a function times ω. The Monge–Ampère equation becomes

$$\omega + \mathrm{i}\partial\bar\partial\varphi = \mathrm{e}^F\omega, \qquad \mathrm{i}\partial\bar\partial\varphi = (\mathrm{e}^F - 1)\omega.$$

This is a Poisson equation. Its right-hand side has integral zero exactly when $\int_X \mathrm{e}^F\omega = \int_X \omega$, and Hodge theory then gives a solution unique modulo constants. The determinant nonlinearity begins only in complex dimension at least two.

At a point in higher dimension, let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of ω_φ relative to ω. The equation controls

$$\lambda_1 \cdots \lambda_n = \mathrm{e}^F,$$

while the second-order estimates control $\sum \lambda_j$ and $\sum \lambda_j^{-1}$. A product bound alone allows one eigenvalue to approach infinity while another approaches zero. Trace and determinant together give uniform ellipticity.

8.2.2 The a priori estimate ladder

The closedness argument uses each estimate to unlock the next:

$$\begin{gathered} C^0 \text{ control of } \varphi \\ \Downarrow \\ \text{trace bounds for } \omega_\varphi \text{ relative to } \omega \\ \Downarrow \\ \text{uniform ellipticity and Evans–Krylov } C^{2,\alpha} \\ \Downarrow \\ \text{Schauder estimates in every order.} \end{gathered}$$

The arrows are logical dependencies. For example, the maximum-principle quantity $\log \mathrm{tr}_\omega\, \omega_\varphi - A\varphi$ cannot give a global trace bound until the oscillation of φ is known.

The Ricci transformation formula follows directly from determinants:

$$\begin{aligned} \mathrm{Ric}(\omega_\varphi) - \mathrm{Ric}(\omega) &= -\mathrm{i}\partial\bar\partial \log\det(g + \varphi_{j\bar k}) + \mathrm{i}\partial\bar\partial \log\det g \\ &= -\mathrm{i}\partial\bar\partial \log \frac{\omega_\varphi^n}{\omega^n}. \end{aligned}$$

Thus prescribing the volume ratio prescribes the change in Ricci curvature. The $\partial\bar\partial$-lemma supplies the desired logarithmic ratio from a cohomological Ricci target, and Calabi–Yau solves the resulting nonlinear equation.

Remark (Continuity method checklist)

Specify the Banach spaces and normalization, prove the linearization is invertible, derive estimates independent of the path parameter, take a convergent subsequence, and verify that positivity survives. “Open and closed” is the last line of the argument, not the proof itself.

8.2.3 The volume constraint from Stokes

For any smooth φ with $\omega_\varphi = \omega + \mathrm{i}\partial\bar\partial\varphi > 0$, expand

$$\omega_\varphi^n - \omega^n = \mathrm{i}\partial\bar\partial\varphi \wedge \sum_{j=0}^{n-1} \omega_\varphi^j \wedge \omega^{n-1-j}.$$

Every form in the sum is closed. Hence the right side is an exterior derivative, and compactness plus Stokes gives

$$\int_X \omega_\varphi^n = \int_X \omega^n.$$

Therefore the condition $\int_X \mathrm{e}^F \omega^n = \int_X \omega^n$ is necessary. It is also the only integral obstruction in the Calabi–Yau theorem.

Linearization of the determinant

Let $G(t) = (g_{j\bar{k}} + t\psi_{j\bar{k}})$. Jacobi's formula gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \det G(t) = \det G(t)\ \mathrm{tr}\big(G(t)^{-1} G'(t)\big).$$

At $t = 0$,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 \log \frac{(\omega + \mathrm{i}\partial\bar{\partial}(\varphi + t\psi))^n}{\omega^n} = g_\varphi^{j\bar{k}} \psi_{j\bar{k}} = \Delta_{\omega_\varphi} \psi.$$

Constants form the kernel, but the phrase "restrict to zero-mean functions" needs a measure: Δ_{ω_φ} is self-adjoint for ω_φ^n, not for the fixed measure ω^n. Keep the convenient normalization $\int_X \psi\, \omega^n = 0$ and include the volume-normalizing constant as an unknown. The actual linear map is

$$C_0^{k+2,\alpha}(X) \oplus \mathbb{R} \longrightarrow C^{k,\alpha}(X), \qquad (\psi, a) \longmapsto \Delta_{\omega_\varphi} \psi - a, \tag{30.4}$$

where $C_0^{k+2,\alpha}(X)$ denotes functions having zero ω^n-mean. Given h, integration against ω_φ^n first determines

$$a = -\frac{\int_X h\, \omega_\varphi^n}{\int_X \omega_\varphi^n}.$$

The function $h + a$ then has zero ω_φ^n-mean, so the Fredholm alternative solves $\Delta_{\omega_\varphi} \psi = h + a$ modulo constants; the fixed ω^n-mean selects one solution. Thus (30.4), rather than the Laplacian between two fixed zero-mean spaces, is the isomorphism used for openness.

Complex dimension one is a Poisson equation

On a compact Riemann surface every real $(1,1)$-form is a function times ω. Write

$$\mathrm{i}\partial\bar{\partial}\varphi = (\Delta_\omega \varphi)\omega.$$

Then the Monge–Ampère equation is simply

$$\Delta_\omega \varphi = \mathrm{e}^F - 1.$$

The right side has integral zero by the volume condition. Hodge theory solves this equation uniquely after fixing the mean of φ. Positivity follows from the equation itself: $\omega_\varphi = \mathrm{e}^F \omega > 0$. There is no separate nonlinear second-order estimate; the determinant is a single eigenvalue.

Why trace and determinant are both needed

At a point diagonalize ω_φ relative to ω, with eigenvalues $\lambda_j > 0$. The equation bounds the product

$$\lambda_1 \cdots \lambda_n = \mathrm{e}^F.$$

The second-order estimate bounds $\sum_j \lambda_j = \operatorname{tr}_\omega \omega_\varphi$. If the product is bounded below and the sum is bounded above, every λ_j is bounded above and

$$\lambda_j \geq \frac{\prod_k \lambda_k}{(\sum_k \lambda_k)^{n-1}}$$

gives a uniform lower bound. This yields uniform ellipticity. Without the trace estimate, the tuple $(R, R^{-1}, 1, \ldots, 1)$ keeps determinant one while degenerating.

Uniqueness as an energy calculation

If $\omega_\varphi^n = \omega_\psi^n$ and $u = \varphi - \psi$, then

$$0 = \int_X u(\omega_\varphi^n - \omega_\psi^n) = -\sum_{j=0}^{n-1} \int_X \mathrm{i}\partial u \wedge \bar\partial u \wedge \omega_\varphi^j \wedge \omega_\psi^{n-1-j}.$$

Every measure on the right is nonnegative, so every integral vanishes. In particular $\partial u = 0$ and u is constant. The chosen normalization removes that constant. This proof uses only positivity, Stokes, and the algebraic difference-of-powers identity.

Closedness of the continuity set as a sequence argument

Let t_j belong to the solvable set and suppose $t_j \to t_\infty$. Choose the normalized solutions φ_j of

$$(\omega + \mathrm{i}\partial\bar\partial\varphi_j)^n = \mathrm{e}^{t_j F + c_{t_j}} \omega^n, \qquad \int_X \varphi_j \omega^n = 0, \tag{30.5}$$

where c_t is the constant making the two sides have equal total volume. To prove closedness, it is not enough to say that a subsequence converges: the topology of the convergence must be strong enough to pass the determinant equation.

The C^0-estimate first gives $\|\varphi_j\|_{L^\infty} \leq C_0$. The Laplacian estimate then gives

$$\operatorname{tr}_\omega \omega_{\varphi_j} \leq C_2.$$

Because the determinant in (30.5) is bounded above and below, the eigenvalue argument from the preceding subsection gives constants independent of j such that

$$C^{-1}\omega \leq \omega_{\varphi_j} \leq C\omega. \tag{30.6}$$

Thus the complex Monge–Ampère operator is uniformly elliptic along the sequence. The Evans–Krylov estimate (or the corresponding complex $C^{2,\alpha}$ estimate) now gives a uniform $C^{2,\alpha}$-bound. After this point the equation is a uniformly elliptic equation with controlled coefficients, so Schauder estimates, differentiated successively, give

$$\|\varphi_j\|_{C^k(X)} \leq C_k \qquad \text{for every } k.$$

Arzelà–Ascoli and a diagonal subsequence therefore give $\varphi_j \to \varphi_\infty$ in C^∞. We may pass directly to the wedge power in (30.5), obtaining

$$(\omega + \mathrm{i}\partial\bar\partial\varphi_\infty)^n = \mathrm{e}^{t_\infty F + c_{t_\infty}} \omega^n.$$

The normalization also passes to the limit. Hence t_∞ is solvable. The proof makes the role of each estimate explicit: C^0 controls the potential, the trace plus determinant give uniform ellipticity, $C^{2,\alpha}$ supplies compactness strong enough for the nonlinear equation, and Schauder theory supplies smoothness.

8.2.4 Self-study workshop: organizing the continuity method around a priori estimates

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall Kähler potentials, the determinant of a Hermitian matrix, elliptic linearization, the implicit function theorem in Banach spaces, and compactness from uniform Hölder estimates.

The continuity method as a dependency chain. For

$$(\omega + 2\pi \mathrm{dd}^{\mathrm{c}}\varphi)^n = \mathrm{e}^F \omega^n, \qquad \omega + 2\pi \mathrm{dd}^{\mathrm{c}}\varphi > 0,$$

integration and Stokes force the volume constraint $\int_X \mathrm{e}^F \omega^n = \int_X \omega^n$. Normalize $\sup_X \varphi = 0$ to remove the additive constant.

Embed the equation in a path

$$(\omega + 2\pi \mathrm{dd}^{\mathrm{c}}\varphi_t)^n = \mathrm{e}^{tF + c_t} \omega^n, \qquad t \in [0, 1],$$

where c_t enforces equal total volume. The solution set is nonempty at $t = 0$. Differentiating the logarithm of the determinant gives

$$\left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{s=0} \log \frac{(\omega_\varphi + s\mathrm{i}\partial\bar{\partial}\psi)^n}{\omega^n} = \Delta_{\omega_\varphi}\psi.$$

Constants form the kernel. Since the natural mean for this Laplacian uses ω_φ^n, one either works with that varying mean or adjoins the scalar normalization c_t to the linearized map. The resulting normalized operator is an isomorphism, so the implicit function theorem gives openness.

Closedness is the hard part. The estimates occur in a strict order:

$$C^0 \Longrightarrow \Delta\varphi \text{ bound} \Longrightarrow C^{2,\alpha} \Longrightarrow C^{k,\alpha}.$$

The second-order estimate uses both a trace bound and the determinant equation; a determinant alone does not control individual eigenvalues. Compactness then extracts a limit that remains positive and solves the equation.

Uniqueness follows from an energy argument. If φ_0, φ_1 solve the same equation, integrate their difference against the difference of the Monge–Ampère measures to obtain a sum of nonnegative gradient terms; all vanish, so the potentials differ by a constant.

Checkpoint (open is not closed)

(1) Why is the linearized operator considered on mean-zero functions?
(2) Why does a determinant bound not control the largest eigenvalue?
(3) Which estimate prevents the limiting Kähler form from degenerating?

Solution (checkpoint). Constants lie in the kernel of the Laplacian, and normalization removes them. A product of positive eigenvalues can stay fixed while one tends to infinity and another to zero. The second-order estimate, combined with the determinant equation, gives two-sided eigenvalue control and preserves positivity in the limit. ◇

Exit criterion

You can derive the volume condition and linearization, distinguish openness from closedness, state the estimate ladder in the correct order, and prove uniqueness up to constants.

Summary. The logical skeleton is: continuity path, openness by the Laplacian, uniform C^0 and trace estimates, and fully nonlinear regularity. This is the summit of smooth complex Monge–Ampère theory.

Exercise 8.9: Linearization

Compute $\frac{\mathrm{d}}{\mathrm{d}t}\big|_0 \log\det(g_{j\bar{k}} + t\psi_{j\bar{k}})$ carefully.

Exercise 8.10: Complex dimension one

Reduce the equation on a compact Riemann surface to a Poisson equation and explain why the genuinely higher-dimensional difficulties disappear.

Optional handout: Moment maps, Mabuchi energy, and K-stability

This handout is optional. No later argument in the core lectures depends on it.

The Calabi–Yau theorem solves a prescribed volume-form equation in a fixed Kähler class. A harder problem asks for a canonical metric, such as a constant-scalar-curvature Kähler metric. The scalar curvature appears as an infinite-dimensional moment map for the action of Hamiltonian symplectomorphisms on compatible complex structures.

On the space of Kähler potentials $\mathcal{H} = \{\varphi : \omega_\varphi = \omega + \mathrm{i}\partial\bar{\partial}\varphi > 0\}$, the Mabuchi functional is characterized by

$$\frac{\mathrm{d}}{\mathrm{d}t} M(\varphi_t) = -\int_X \dot{\varphi}_t (S(\omega_{\varphi_t}) - \overline{S}) \frac{\omega_{\varphi_t}^n}{n!}.$$

Its critical points are constant-scalar-curvature metrics. Along suitable geodesics in $\mathcal{H}$, the functional is convex.

Algebraic one-parameter degenerations of (X, L) are test configurations. Their Donaldson–Futaki invariants are slopes at infinity of the corresponding energy. The Yau–Tian–Donaldson principle predicts that existence of a canonical metric is equivalent to an algebraic stability condition. This is a theorem for Fano Kähler–Einstein metrics and in several other important settings.

Remark (Relation to Monge–Ampère)

Geodesics in $\mathcal{H}$ satisfy a homogeneous complex Monge–Ampère equation on $X\times$ an annulus. They are generally only weak, so finite-energy pluripotential theory becomes part of the metric geometry of $\mathcal{H}$.

Exercise 8.11: First variation of volume

For a smooth path φ_t, prove $\frac{\mathrm{d}}{\mathrm{d}t}\int_X \omega_{\varphi_t}^n = 0$ by differentiating and applying Stokes' theorem.

8.3 Lecture 31: Nef, big, and psef cones; regularization; non-pluripolar products

Week 16 material • 90 minutes

Notation for this lecture

X is a compact Kähler manifold of complex dimension n, ω a reference Kähler form, and $\alpha \in H^{1,1}(X,\mathbb{R})$ a real cohomology class represented smoothly by θ. The set $\mathrm{PSH}(X,\theta) = \{u : \theta + \mathrm{dd^c}u \geq 0\}$ consists of θ-psh potentials, and $V_\theta = \big(\sup\{u \in \mathrm{PSH}(X,\theta) : u \leq 0\}\big)^*$ has minimal singularities, where the star denotes upper-semicontinuous regularization. Psef means pseudoeffective. A function has analytic singularities if locally it equals $c\log\sum_j |f_j|^2 + v$ with $c > 0$, holomorphic f_j, and smooth v. $\nu(u,x)$ is its Lelong number, 1_A denotes restriction by the indicator of A, angle brackets denote the non-pluripolar product, and $\mathrm{Vol}(\alpha)$ is the non-pluripolar volume defined below. An *almost-positive* closed real $(1,1)$-current is one bounded below by a smooth real $(1,1)$-form. Weak convergence of currents is denoted by $\rightharpoonup$ and means convergence against compactly supported smooth test forms. On a one-point blow-up, H and E are the line and exceptional classes.

Prerequisite results used below

Bergman extremal principle. If $\mathcal{H}$ is a Hilbert space of holomorphic functions with bounded point evaluation at x and (f_j) is an orthonormal basis, then $K(x,x) = \sum_j |f_j(x)|^2$ and $K(x,x) = \sup_{\|f\|\leq 1} |f(x)|^2$.

Bedford–Taylor monotone continuity. If $u_{j,\nu}$ are locally bounded psh functions decreasing to u_j, then

$$\mathrm{dd^c}u_{1,\nu} \wedge \cdots \wedge \mathrm{dd^c}u_{p,\nu} \rightharpoonup \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_p.$$

This theorem applies on the bounded truncation loci used in the non-pluripolar construction; it does not define an unrestricted product of arbitrary singular currents.

Why these ideas are needed

Strictly positive smooth forms are not preserved under many limits and birational operations. Nef, pseudoeffective, and big classes separate three weaker forms of positivity: approximation by almost-positive forms, representation by positive currents, and domination of a Kähler form by a current. Potentials with minimal singularities choose a least-singular representative, while regularization makes it accessible to smooth arguments. Non-pluripolar products are needed because ordinary wedge products can charge the very pole sets where singular potentials are undefined. These tools let intersection theory survive controlled singularities.

Plan for the lecture. 0–20 min: four positivity cones; 20–42: potentials with minimal singularities; 42–62: Demailly regularization; 62–78: non-pluripolar products; 78–90: volume and the blow-up example.

Let X be compact Kähler and $\alpha \in H^{1,1}(X,\mathbb{R})$.

Definition 8.12: Kähler, nef, pseudoeffective, and big classes

(1) α is Kähler if it contains a smooth positive definite representative.
(2) α is nef if, for every $\varepsilon > 0$, it has a smooth representative $\theta_\varepsilon \geq -\varepsilon\omega$.
(3) α is pseudoeffective (psef) if it contains a closed positive $(1,1)$-current.

(4) α is big if it contains a Kähler current $T \geq \varepsilon\omega$.

Theorem 8.13: Interiors of the positivity cones

On a compact Kähler manifold, the Kähler cone is the interior of the nef cone, and the big cone is the interior of the psef cone. Kähler implies nef and big implies psef, but nef and psef are not interchangeable.

Proof. Fix the reference Kähler class $[\omega]$ and any norm on the finite-dimensional space $H^{1,1}(X,\mathbb{R})$. Choose smooth closed representatives $\eta_1,\dots,\eta_N$ of a cohomology basis. Then a class $\sum_j t_j[\eta_j]$ with small norm has the smooth representative $\sum_j t_j\eta_j$ with small C^0-norm. A Kähler form remains positive after adding such a form, so every Kähler class is an interior point of the nef cone. Conversely, if α is interior to the nef cone, then $\alpha - \delta[\omega]$ is nef for some $\delta > 0$. By nefness it has, for every $0 < \varepsilon < \delta$, a smooth representative bounded below by $-\varepsilon\omega$. Adding $\delta\omega$ gives a smooth representative of α bounded below by $(\delta-\varepsilon)\omega$, so α is Kähler.

If α contains a Kähler current $T \geq \delta\omega$, adding any sufficiently small smooth closed form preserves positivity; hence α is interior to the psef cone. Conversely, if α is interior to the psef cone, then $\alpha - \delta[\omega]$ is psef for some $\delta > 0$. Choose a closed positive current S in that class. Then $S + \delta\omega \in \alpha$ and $S + \delta\omega \geq \delta\omega$, so α is big. □

For a psef class and a smooth representative $\theta \in \alpha$, define

$$V_\theta = \left(\sup\{u \in \mathrm{PSH}(X,\theta) : u \leq 0\}\right)^*.$$

The star denotes upper-semicontinuous regularization. The envelope has minimal singularities: every $u \in \mathrm{PSH}(X,\theta)$ satisfies $u \leq V_\theta + C$.

Theorem 8.14: Demailly regularization, working version

Let $T = \theta + \mathrm{dd}^{\mathrm{c}}u$ be a closed almost-positive $(1,1)$-current and let γ be a smooth closed real $(1,1)$-form such that $T \geq \gamma$. There are u_m with analytic singularities such that

$$T_m = \theta + \mathrm{dd}^{\mathrm{c}}u_m \geq \gamma - \varepsilon_m\omega, \qquad \varepsilon_m \downarrow 0,$$

and $u_m \to u$ in L^1, $T_m \rightharpoonup T$. With the normalization below, their Lelong numbers satisfy

$$\nu(u,x) - \frac{n}{m} \leq \nu(u_m,x) \leq \nu(u,x), \qquad x \in X.$$

Monotone or equisingular refinements require additional construction.

Proof (with the regularization gluing lemma as the global input). We describe the two parts of the construction and indicate precisely where the global curvature bound enters. On a coordinate ball $B \Subset X$, after adding a smooth local potential of $\theta - \gamma$, reduce to a psh function v. Let

$$\mathcal{H}_{m,B} = \left\{ f \in \mathcal{O}(B) : \int_B |f|^2 \mathrm{e}^{-2mv}\, \mathrm{d}\lambda < \infty \right\}$$

and choose an orthonormal basis $(f_{m,j})$. The Bergman approximation

$$v_m = \frac{1}{2m}\log\sum_j |f_{m,j}|^2$$

has analytic singularities. The submean inequality gives, on a smaller ball,

$$v_m(z) \leq \sup_{|\zeta - z| < r} v(\zeta) + \frac{C + 2n|\log r|}{2m}. \tag{31.1}$$

Conversely, apply the point-extension form of Ohsawa–Takegoshi from Lecture 29 to construct a holomorphic function of weighted norm at most $Ce^{-2mv(z)}$ and value 1 at z. The extremal characterization of the Bergman kernel then gives

$$v(z) - \frac{C}{m} \le v_m(z). \tag{31.2}$$

Equations (31.1)–(31.2) imply $v_m \to v$ in L^1_{loc}. Applying the same bounds after restriction to complex lines and using the slope description of Lelong numbers yields

$$\nu(v, z) - \frac{n}{m} \le \nu(v_m, z) \le \nu(v, z). \tag{31.3}$$

The passage from these local ideals to a global current is subtler than an ordinary partition of unity: multiplying potentials by cutoffs destroys plurisubharmonicity. We therefore isolate the precise global input, usually called the regularization gluing lemma. Choose finitely many nested coordinate balls $U_j' \Subset U_j'' \Subset U_j$ of radius comparable to δ. After restoring smooth local potentials for θ and γ and making the required pluriharmonic transition corrections, the local Bergman weights can be shifted by constants so that on every overlap

$$|v_{j,m} - v_{k,m}| \le C\left(\varepsilon(\delta)\delta^2 + \frac{|\log \delta|}{m}\right), \tag{31.4}$$

where $\varepsilon(\delta) \to 0$ measures the oscillation of the smooth coefficients. The constants and the three nested radii are chosen so that, near the boundary of a chart, that chart's weight is strictly dominated by one from a neighboring chart.

Now replace the maximum of the corrected local weights by a smooth convex regularized maximum. Such a function is nondecreasing in each argument and is translation invariant. Its complex Hessian is a convex combination of the input Hessians plus a nonnegative quadratic "variance" term. Consequently a common lower curvature bound is preserved. The only losses come from the smooth local corrections and have size

$$C\left(\varepsilon(\delta) + \frac{1}{m\delta^2}\right)\omega. \tag{31.5}$$

The strict domination near chart boundaries makes the regularized maximum equal to an already-defined neighboring branch there, so the branches glue rather than merely agree up to a bounded function. Because all Bergman weights have the same coefficient $1/(2m)$ and are locally logarithms of finitely generated ideals, the gluing lemma also preserves analytic singularities. This last assertion is not a consequence of an arbitrary smooth maximum; it is part of the compatible-ideal construction.

Choose $\delta = \delta_m \downarrow 0$ slowly enough that $m\delta_m^2 \to \infty$. Both terms in (31.5) then tend to zero, and the resulting global functions satisfy

$$\theta + \mathrm{dd^c} u_m \ge \gamma - \varepsilon_m \omega.$$

The local L^1 estimates give global L^1 convergence; continuity of $\mathrm{dd^c}$ on distributions gives $T_m \rightharpoonup T$, and (31.3) gives the asserted control of Lelong numbers. The full compatible-ideal gluing argument is Demailly's regularization theorem [18]; equations (31.4)–(31.5) record exactly the two estimates needed from it. Monotone or equisingular refinements require a further diagonal construction, which is why those properties are not claimed here. □

Definition 8.15: Non-pluripolar product

Let $1 \le p \le n$ and let $T_j = \theta_j + \mathrm{dd^c} u_j$, $1 \le j \le p$, be closed positive $(1,1)$-currents. On a coordinate ball U, choose smooth functions ρ_j with $\theta_j = \mathrm{dd^c}\rho_j$ and put $\psi_j = \rho_j + u_j$, so

$T_j = \mathrm{dd^c}\psi_j$ on U. For $k \geq 1$, define

$$U_k = \bigcap_{j=1}^{p} \{\psi_j > -k\}, \qquad \mu_k = 1_{U_k} \bigwedge_{j=1}^{p} \mathrm{dd^c} \max(\psi_j, -k).$$

The wedge in μ_k is the Bedford–Taylor product of the locally bounded psh truncations. Bedford–Taylor locality gives $\mu_{k+1} = \mu_k$ on U_k. The increasing limit, extended by zero across the union of the pole sets, is denoted

$$\langle T_1 \wedge \cdots \wedge T_p \rangle .$$

On a relatively compact overlap, two choices of local potentials differ by a bounded pluriharmonic function. Their level-k finite loci therefore contain one another after replacing k by $k + C$; locality and compatibility show that the two limits agree. Hence these local currents glue. The resulting current is closed, positive, and assigns no mass to pluripolar sets.

Two assertions in this definition are theorems rather than notation. In the present compact Kähler setting, the positive currents μ_k have locally uniformly bounded mass, so their increasing limit has locally finite coefficients. Second, although multiplication by 1_{U_k} creates an apparent boundary, the compatible limit is closed. The proof uses the plurifine locality and monotone continuity of Bedford–Taylor products; see [19, §1]. Without this finite-mass/closedness theorem, the displayed limit would define at most a formal positive set function.

The local potentials are essential. The expression $\max(u_j, -k)$ need not be θ_j-psh because a constant need not be θ_j-psh. When a big class and a minimal reference potential V_{θ_j} are fixed, the equivalent relative truncations $\max(u_j, V_{\theta_j} - k)$ are often convenient; the local definition above makes clear which ordinary psh functions enter Bedford–Taylor theory.

Warning

The non-pluripolar product is not the classical product of currents. For example, $\mathrm{dd^c} \log |f|^2 = [D]$ contains divisor mass, while the non-pluripolar bracket discards mass on the pole set. The angle brackets change the definition; they are not typographical decoration.

Definition 8.16: Volume of a pseudoeffective class

$$\mathrm{Vol}(\alpha) = \int_X \langle (\theta + \mathrm{dd^c} V_\theta)^n \rangle.$$

It is independent of θ. The class is big exactly when $\mathrm{Vol}(\alpha) > 0$, and if α is nef then $\mathrm{Vol}(\alpha) = \int_X \alpha^n$.

Independence of the smooth representative follows by translating potentials when θ changes by a global $\mathrm{dd^c}$-term. Independence from the chosen current with minimal singularities, positivity of the volume exactly on the big cone, and the nef intersection formula are substantive results; the non-pluripolar formulation is proved in [19], building on the numerical theory in [13]. In particular, "big if and only if positive volume" is not a formal consequence of the definition.

Example 8.17: Cones on the blown-up plane

On $X = \mathrm{Bl}_p \mathbb{P}^2$, write a class as $\alpha = aH - bE$. The effective curve cone is generated by E and

$H - E$. Consequently

positivity of α	inequalities
psef	$a \geq 0,\ b \leq a,$
big	$a > 0,\ b < a,$
nef	$a \geq b \geq 0,$
Kähler	$a > b > 0.$

For $b \geq 0$ in the big cone, the class is nef and $\mathrm{Vol}(\alpha) = a^2 - b^2$. For $b < 0$, its surface Zariski decomposition is $\alpha = aH + (-b)E = P + N$ with nef part $P = aH$, effective negative part $N = (-b)E$, and $P \cdot N = 0$. Hence $\mathrm{Vol}(\alpha) = a^2$. In particular $H + E$ is big with volume 1 although $(H + E)^2 = 0$.

8.3.1 The cones on the one-point blow-up

Let $X = \mathrm{Bl}_p \mathbb{P}^2$. The effective cone is generated by E and $H - E$: every plane curve of degree d and multiplicity m at p satisfies $d \geq m \geq 0$, while E supplies the exceptional ray. In the coordinates $\alpha = aH - bE$ this gives

$$\alpha \text{ is psef} \quad \Longleftrightarrow \quad a \geq 0, \quad a - b \geq 0.$$

Its interior is the big cone: $a > 0$ and $a > b$.

For nefness, test against the two curve classes E and $H - E$:

$$\alpha \cdot E = b, \qquad \alpha \cdot (H - E) = a - b.$$

Hence α is nef exactly when $a \geq b \geq 0$, and it is Kähler exactly when $a > b > 0$. This exhibits two different boundaries. The class E is psef but not nef because $E^2 = -1$; the class H is nef and big but not Kähler because $H \cdot E = 0$.

The class $H + E$ is big although $(H + E)^2 = 0$. Its divisorial Zariski decomposition is $H + E = H + E$, with nef positive part H and negative part E; its volume is $H^2 = 1$. This example shows why self-intersection alone does not test bigness for a non-nef class.

8.3.2 What the non-pluripolar product discards

Locally let $T = [D] = \mathrm{dd^c} \log |f|^2$. Outside D, the current vanishes. The non-pluripolar construction truncates the potential, computes on the finite locus, and then extends by zero across the pole set. Consequently

$$\langle [D] \rangle = 0$$

in the non-pluripolar sense, even though the ordinary current $[D]$ is nonzero. For $T = \theta + [D]$ with a smooth semipositive part, the non-pluripolar product retains the products built on the smooth part away from D but assigns no mass to D itself.

This loss is deliberate. Classical self-intersections of a divisor depend on global intersection theory and may be negative, whereas a non-pluripolar Monge–Ampère measure is designed to remain positive and stable under truncation. The angle brackets therefore encode a different operation, not an approximation to every classical intersection product.

Potentials with minimal singularities are the least singular representatives of a psef class. In a nef class they can be taken with arbitrarily small negative smooth error; in a big non-nef class their singularities record the fixed negative locus. The volume measures the movable positive part rather than divisorial mass forced into every representative.

Warning

The identity $\mathrm{Vol}(\alpha) = \int_X \alpha^n$ is valid for nef classes. For a general big class, the ordinary top intersection can be smaller, zero, or otherwise fail to represent the positive movable mass; use the non-pluripolar product of a minimally singular current.

8.3.3 The four cones compared by explicit representatives

If α is Kähler, its positive smooth representative also proves that it is nef, big, and psef. If α is big, the Kähler current is in particular a positive current, so it is psef. The reverse implications fail on a blow-up:

$$[E] \text{ is psef but not nef because } E^2 = -1,$$

while H is nef and big but not Kähler because $H \cdot E = 0$.

Write a class on $X = \mathrm{Bl}_p \mathbb{P}^2$ as $aH - bE$. The effective cone is generated by E and $H - E$. Indeed, a nonexceptional plane curve of degree d and multiplicity m at p has $d \geq m \geq 0$ and class

$$dH - mE = d(H - E) + (d - m)E,$$

which lies in the same cone. Solving

$$aH - bE = xE + y(H - E)$$

gives $y = a$ and $x = a - b$. Therefore

$$aH - bE \text{ is psef} \quad \Longleftrightarrow \quad a \geq 0, a - b \geq 0.$$

Nefness is dual to nonnegative intersection with the two generating curve rays:

$$(aH - bE) \cdot E = b, \qquad (aH - bE) \cdot (H - E) = a - b.$$

Hence the nef cone is $a \geq b \geq 0$, and its interior, the Kähler cone, is $a > b > 0$.

The truncation calculation for a divisor

Locally let $u = \log |z_1|^2$, so $\mathrm{dd^c} u = [z_1 = 0]$. Define

$$u_k = \max\{u, -k\}.$$

The Bedford–Taylor measure $\mathrm{dd^c} u_k$ is supported on the contact hypersurface $|z_1| = \mathrm{e}^{-k/2}$. The non-pluripolar construction restricts it to $\{u > -k\} = \{|z_1| > \mathrm{e}^{-k/2}\}$, where $u_k = u$ and $\mathrm{dd^c} u = 0$. Thus

$$1_{\{u > -k\}} \mathrm{dd^c} u_k = 0$$

for every k, and the limit is

$$\langle [z_1 = 0] \rangle = 0.$$

The ordinary current retains its divisor mass; the non-pluripolar bracket deliberately removes mass carried by the pole set.

For several currents, choose local psh potentials $T_j = \mathrm{dd^c} \psi_j$. If $k < \ell$, then on the common set $\cap_j \{\psi_j > -k\}$ both truncations $\max(\psi_j, -k)$ and $\max(\psi_j, -\ell)$ equal ψ_j. Bedford–Taylor locality therefore makes the level-k and level-ℓ products agree there. This compatibility allows the increasing union of finite loci to define one current.

Volume and Zariski decomposition on the blow-up

For a nef class $aH - bE$,

$$\mathrm{Vol}(aH - bE) = (aH - bE)^2 = a^2 - b^2.$$

For the big non-nef class $H + E$, the divisorial Zariski decomposition is

$$H + E = P + N, \qquad P = H, \quad N = E.$$

Here P is nef, N has negative-definite intersection matrix, and $P \cdot N = 0$. The volume sees only the movable positive part:

$$\mathrm{Vol}(H + E) = P^2 = 1,$$

although $(H + E)^2 = 0$. This example proves concretely that top self-intersection is not the volume formula outside the nef cone.

Minimal singularities in the same class

If $u, v \in \mathrm{PSH}(X, \theta)$ and $u \le v + C$, then u is at least as singular as v. The envelope

$$V_\theta = \sup\{w \in \mathrm{PSH}(X, \theta) : w \le 0\}^*$$

dominates every normalized competitor and hence has minimal singularities. If θ is Kähler, the zero function is a competitor and $V_\theta = 0$. For a big class with a fixed divisorial component, every positive current has a logarithmic pole along that component, and the envelope records the smallest unavoidable coefficient.

Why a Bergman approximant has analytic singularities

Let $B \Subset \Omega \subset \mathbb{C}^n$ be a coordinate ball and let $\varphi \in \mathrm{PSH}(\Omega)$. For a positive integer m, define the Hilbert space

$$\mathcal{H}_m = \left\{ f \in \mathcal{O}(\Omega) : \int_\Omega |f|^2 \mathrm{e}^{-2m\varphi} \mathrm{d}\lambda < \infty \right\}$$

and choose an orthonormal basis $(f_{m,j})_{j \ge 1}$. Its diagonal Bergman kernel and associated approximant are

$$B_m(z) = \sum_{j \ge 1} |f_{m,j}(z)|^2, \qquad \varphi_m(z) = \frac{1}{2m} \log B_m(z). \tag{31.6}$$

The series converges locally uniformly after taking the logarithm in the usual upper-semicontinuous sense, because point evaluation is bounded on every smaller ball by the submean inequality.

The functions in $\mathcal{H}_m$ generate a coherent ideal sheaf. After shrinking around a fixed point, choose finitely many local generators $g_1, \dots, g_N$. Local division estimates compare the full sum in (31.6) with the finite sum of squares; hence

$$\varphi_m = \frac{1}{2m} \log\bigl(|g_1|^2 + \cdots + |g_N|^2\bigr) + O(1) \tag{31.7}$$

locally. A function admitting a presentation of the form (31.7) is said to have *analytic singularities*. Notice that the finite list of generators is only local; no finite-dimensionality of the global Hilbert space is being asserted.

There are two complementary estimates behind the approximation. Cauchy's submean inequality gives, for $0 < r < \mathrm{dist}(z, \partial\Omega)$,

$$\varphi_m(z) \le \sup_{|\zeta - z| < r} \varphi(\zeta) + \frac{1}{2m} \log \frac{C}{r^{2n}}. \tag{31.8}$$

Conversely, a point-extension estimate constructs a holomorphic function with prescribed value at z and controlled weighted norm, giving

$$\varphi_m(z) \geq \varphi(z) - \frac{C}{m}$$

on a smaller ball, after harmless changes of the constant dictated by the chosen regularization. Thus $\varphi_m \to \varphi$ pointwise and in L^1_{loc}, while each φ_m has algebraically controlled poles. This is the bridge from an arbitrary plurisubharmonic weight to multiplier ideals, Lelong numbers, and divisorial calculations.

8.3.4 Self-study workshop: comparing positivity cones on one blow-up

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Recall positive currents, Kähler classes, intersection numbers on surfaces, divisorial Zariski decomposition, and truncation of unbounded psh potentials.

A cone picture with coordinates. Let $X = \mathrm{Bl}_p \mathbb{P}^2$, with H the pullback of a line and E the exceptional curve. The intersection table is

$$H^2 = 1, \qquad H \cdot E = 0, \qquad E^2 = -1.$$

For $\alpha = aH - bE$,

$$\alpha \cdot E = b, \qquad \alpha \cdot (H - E) = a - b.$$

Since the Mori cone is generated by E and $H - E$, the class is nef exactly when

$$b \geq 0, \qquad a - b \geq 0.$$

It is Kähler when both inequalities are strict. The effective cone is generated by E and $H - E$; writing

$$aH - bE = a(H - E) + (a - b)E$$

shows that α is psef exactly when $a \geq 0$ and $a - b \geq 0$. Its interior is the big cone. This example makes the inclusions

$$\text{Kähler} \subset \text{nef} \subset \text{psef}$$

visible and shows that nefness and bigness ask different questions.

For an unbounded local psh potential u, the non-pluripolar product is built from $u_k = \max\{u, -k\}$ on the sets $\{u > -k\}$ and extended by zero across the discarded singular set before taking a limit. On a compact manifold one first writes each current locally as $\mathrm{dd}^c u$; directly taking $\max(u, -k)$ for a global θ-psh potential need not preserve θ-plurisubharmonicity. If $u = \log |s_D|^2 + O(1)$ has a divisorial pole, the construction does not assign the self-product an artificial mass on D. Hence the non-pluripolar product is not the same as every formal current wedge.

Checkpoint (plot the inequalities)

(1) Is H nef? Is it Kähler on the blow-up?
(2) Is E psef? Is it nef?
(3) Why can the non-pluripolar product differ from an ordinary intersection number?

Solution (checkpoint). H is nef but not Kähler because $H \cdot E = 0$. The curve class E is effective, hence psef, but $E^2 = -1$, so it is not nef. Non-pluripolar products deliberately discard mass concentrated on pluripolar sets, while cohomological intersections still record divisorial self-intersection. ◇

Exit criterion

You can derive the nef and psef inequalities from generators, locate boundary classes, and explain the truncation principle and the loss of divisorial mass in non-pluripolar products.

Summary. Kähler and nef describe smooth positivity, while big and psef describe positivity by currents. Demailly regularization replaces arbitrary positive currents by analytic-singularity models, and non-pluripolar products safely ignore unstable mass on pole sets.

Exercise 8.18: Nef test on a surface

Use the curve classes H, E, and $H - E$ to test the nef and Kähler conditions for $aH - bE$.

Exercise 8.19: Compatibility of truncations

If $k < \ell$, prove that the two Bedford–Taylor products agree on $\cap_j \{\psi_j > -k\}$ for local psh potentials $T_j = \mathrm{dd}^c \psi_j$.

Optional handout: Divisorial Zariski decomposition

This handout is optional. No later argument in the core lectures depends on it.

On a projective surface, a pseudoeffective divisor class α admits a Zariski decomposition

$$\alpha = P + N,$$

where P is nef, $N = \sum a_i C_i$ is effective with negative-definite intersection matrix, and $P \cdot C_i = 0$. The positive part carries the movable positivity; the negative part records fixed divisorial components.

In higher dimension, Boucksom's divisorial decomposition uses generic Lelong numbers:

$$N(\alpha) = \sum_D \nu(\alpha, D)[D], \qquad Z(\alpha) = \alpha - \{N(\alpha)\}.$$

The sum runs over prime divisors and is locally finite in the appropriate sense. The class $Z(\alpha)$ is modified nef: it has no unavoidable divisorial singularities, though it need not be nef.

Example 8.20: The exceptional class

On $X = \mathrm{Bl}_p \mathbb{P}^2$, the class E is pseudoeffective and its entire Zariski decomposition is negative: $P = 0$, $N = E$. For a big class $aH - bE$ with $a > b > 0$, the class is already nef, so $N = 0$.

The volume ignores the negative divisorial part on a surface: $\mathrm{Vol}(\alpha) = P^2$. More generally, volume measures the asymptotic movable sections of a big line bundle and equals the total non-pluripolar mass of a current with minimal singularities.

Exercise 8.21: A two-curve decomposition

Let C_1, C_2 have negative-definite intersection matrix. Given a class α, solve the linear equations $(\alpha - a_1 C_1 - a_2 C_2) \cdot C_j = 0$ and determine when $a_1, a_2 \geq 0$.

8.4 Lecture 32: A capstone example, a map of the subject, and next directions

Week 16 material • 90 minutes

Notation for this lecture

$X = \mathrm{Bl}_p \mathbb{P}^2$ is the blow-up of the projective plane at p, $\pi : X \to \mathbb{P}^2$ its blow-down map, $H = \pi^*[\text{line}]$, and E the exceptional curve. The dot is the intersection product and K_X the canonical class. The Hodge number is $h^{p,q} = \dim_{\mathbb{C}} H^q(X, \Omega_X^p)$. For integers a, b, $L = aH - bE$ also denotes the line bundle $\pi^*\mathcal{O}_{\mathbb{P}^2}(a) \otimes \mathcal{O}_X(-bE)$ when no confusion can arise. $h^q(X, L) = \dim_{\mathbb{C}} H^q(X, L)$ and $\chi(L) = \sum_q (-1)^q h^q(X, L)$ is its holomorphic Euler characteristic. s_E is the canonical section of $\mathcal{O}_X(E)$, $[E]$ the integration current, and $c_1(\mathcal{O}(E), h)$ the first Chern form of a chosen metric h.

Prerequisite results used below

Surface Riemann–Roch. For a line bundle L on a smooth compact complex surface,

$$\chi(X, L) = \chi(X, \mathcal{O}_X) + \frac{1}{2} L \cdot (L - K_X).$$

Kodaira vanishing. If A is a positive line bundle on a compact Kähler n-fold, then $H^q(X, K_X \otimes A) = 0$ for every $q > 0$.

Adjunction for a smooth curve. For a smooth curve C in a smooth surface, $K_C \simeq (K_X \otimes \mathcal{O}_X(C))|_C$ and $2g(C) - 2 = C \cdot (C + K_X)$, where $g(C)$ is the genus. These three statements are the external inputs used in the capstone intersection and dimension calculations.

Why these ideas are needed

The course has introduced several languages, and their value is clearest when a single problem forces them to interact. The capstone calculation begins with a geometric positivity question, translates it into curvature and a $\bar\partial$ estimate, interprets the resulting solution through cohomology, and checks the answer on explicit intersection numbers. The final map of the subject is meant as a decision tool: it identifies which local model, positivity notion, compactness principle, or cohomology theorem should be tried next. The goal is not another list of results but a reusable workflow for unfamiliar problems.

Plan for the lecture. 0–15 min: review of the toolchain; 15–48: the calculation on $\mathrm{Bl}_p \mathbb{P}^2$; 48–68: curvature and vanishing; 68–80: psh functions and positive currents; 80–90: research directions and closure.

Let $X = \mathrm{Bl}_p \mathbb{P}^2$, with $\pi : X \to \mathbb{P}^2$, $H = \pi^*[\text{line}]$, and exceptional curve E.

8.4.1 Topology and intersection theory

$$H^2 = 1, \qquad H \cdot E = 0, \qquad E^2 = -1, \qquad K_X = \pi^* K_{\mathbb{P}^2} + E = -3H + E.$$

These formulas can all be read in local coordinates. Blow up the affine origin by the charts

$$(x, t) \longmapsto (x, xt), \qquad (s, y) \longmapsto (sy, y), \qquad s = t^{-1}.$$

The exceptional curve is $x = 0$ in the first chart and $y = 0$ in the second. Since $y = tx$ on the overlap, its normal bundle is $\mathcal{O}_{\mathbb{P}^1}(-1)$; hence $E^2 = -1$. A general line avoids p, so its pullback is disjoint

from E, giving $H \cdot E = 0$, and two general lines still meet once, giving $H^2 = 1$. Finally

$$\pi^*(\mathrm{d}x \wedge \mathrm{d}y) = \mathrm{d}x \wedge \mathrm{d}(xt) = x\,\mathrm{d}x \wedge \mathrm{d}t.$$

The factor x says that the Jacobian divisor is E, which proves $K_X = \pi^* K_{\mathbb{P}^2} + E$ rather than merely quoting the blow-up formula.

The blow-up replaces one point by a copy of $\mathbb{P}^1$, so its topological Euler characteristic is $3 - 1 + 2 = 4$. Holomorphic 1- and 2-forms on X restrict to $\mathbb{P}^2 \setminus \{p\}$. In a coordinate ball around p, write such a form in the frame $\mathrm{d}z_I$; Hartogs extends each holomorphic coefficient across the missing point. Uniqueness on overlaps makes the extended coefficients a global form on $\mathbb{P}^2$. Since $\mathbb{P}^2$ has no holomorphic forms in degrees 1 or 2, this proves $h^{1,0} = h^{2,0} = 0$. Hodge symmetry gives $b_1 = b_3 = 0$. Moreover X is connected and is a compact oriented real four-manifold, so $b_0 = b_4 = 1$. The identity

$$4 = \chi_{\mathrm{top}}(X) = b_0 - b_1 + b_2 - b_3 + b_4$$

therefore gives $b_2 = 2$. Since the intersection matrix of H, E is $\mathrm{diag}(1, -1)$, the two classes are independent and hence form a basis of $H^2(X, \mathbb{R})$. Finally, the blow-up is projective and therefore Kähler, so Hodge decomposition yields $b_2 = h^{2,0} + h^{1,1} + h^{0,2}$. Thus

$$h^{0,0} = h^{2,2} = 1, \qquad h^{1,1} = 2, \qquad h^{1,0} = h^{0,1} = h^{2,0} = h^{0,2} = 0.$$

8.4.2 Kähler classes, line bundles, and projective embeddings

$\alpha = aH - bE$ is Kähler exactly when $a > b > 0$. For integers $a > b > 0$, $L = \pi^* \mathcal{O}_{\mathbb{P}^2}(a) \otimes \mathcal{O}_X(-bE)$ is positive. Its sections are homogeneous polynomials of degree a vanishing to order at least b at p. High powers embed the blow-up by Kodaira.

Example 8.22: A Riemann–Roch and vanishing calculation

Let $a > b > 0$ be integers. Then $A = L \otimes K_X^{-1}$ has class $(a+3)H - (b+1)E$ and is positive because $a + 3 > b + 1 > 0$. Thus $L = K_X \otimes A$, and Kodaira vanishing gives $H^q(X, L) = 0$ for $q > 0$. Surface Riemann–Roch says

$$\chi(L) = \chi(\mathcal{O}_X) + \frac{1}{2} L \cdot (L - K_X),$$

so

$$h^0(L) = 1 + \frac{1}{2}\left(a^2 - b^2 + 3a - b\right).$$

Indeed, the monomials in two affine variables of total degree below b number $1 + 2 + \cdots + b = b(b+1)/2$. Therefore the same answer is

$$\binom{a+2}{2} - \binom{b+1}{2},$$

the number of degree-a plane polynomials after imposing vanishing of order at least b at p. This independently checks both the intersection calculation and the cohomological vanishing result.

8.4.3 Positive currents and singular potentials

$[E]$ is a closed positive current in class E, but $E^2 = -1$, so a psef class need not be nef. For the canonical section s_E of $\mathcal{O}(E)$,

$$\mathrm{dd^c} \log |s_E|_h^2 = [E] - c_1(\mathcal{O}(E), h).$$

Thus the exceptional curve is simultaneously a divisor, the analytic part of a positive current, and the logarithmic pole set of a qpsh potential. Siu decomposition is completely explicit here.

Summary.

Input	Typical output
psh / strict pseudoconvexity	Stein geometry, $\bar\partial$ solutions, holomorphic functions, extension
closed positive currents / Lelong numbers	analytic threshold sets, divisorial components, singular metrics
Chern curvature	characteristic classes, bundle positivity, Bochner coercivity
Kähler identities	Hodge decomposition, $\partial\bar\partial$-lemma, Hard Lefschetz
L^2 estimates	vanishing, peak sections, Kodaira embedding, OT extension
Monge–Ampère	prescribed volume and Ricci-flat metrics in Kähler classes

8.4.4 Four routes for further study

(1) **Kähler–Einstein metrics and stability:** from Calabi–Yau to the Fano case and K-stability;
(2) **complex algebraic geometry:** from ample, nef, and big classes to the minimal model program and nonvanishing;
(3) **pluripotential theory:** from Bedford–Taylor to finite-energy classes, geodesics, and Mabuchi geometry;
(4) **Hodge theory:** from pure structures to variations, degenerations, and mixed Hodge structures.

Warning (final audit)

For any formula, ask: What is the bidegree? What are the $\mathrm{d^c}$ and curvature normalizations? Which positivity cone or tensor test is intended? Is integration by parts justified by boundary conditions or completeness? How is a singular product defined? Why do local solutions glue globally? These six questions catch most hidden errors.

8.4.5 A complete strict-transform calculation

Let $C \subset \mathbb{P}^2$ be a degree-d curve with multiplicity m at the blown-up point, and let $\widetilde{C} = dH - mE$. Then

$$\widetilde{C} \cdot E = m, \qquad \widetilde{C}^2 = d^2 - m^2.$$

The first equality counts tangent directions with multiplicity on the exceptional curve. Since $K_X = -3H + E$, adjunction gives

$$\begin{aligned} 2p_a(\widetilde{C}) - 2 &= \widetilde{C} \cdot (\widetilde{C} + K_X) \\ &= d(d-3) + m(1-m), \end{aligned}$$

and hence

$$p_a(\widetilde{C}) = \frac{(d-1)(d-2)}{2} - \frac{m(m-1)}{2}.$$

Blowing up subtracts the genus contribution of an ordinary multiplicity-m point. If further infinitely near singularities remain, their multiplicities are subtracted on subsequent blow-ups.

For $L = aH - bE$, surface Riemann–Roch gives

$$\chi(L) = 1 + \frac{1}{2}(a^2 - b^2 + 3a - b).$$

The same number arises from plane polynomials:

$$\binom{a+2}{2} - \binom{b+1}{2} = 1 + \frac{1}{2}(a^2 - b^2 + 3a - b).$$

The equality aligns intersection theory with the number of jets forced to vanish at the center. When higher cohomology vanishes, it is the actual dimension of the linear system.

8.4.6 A problem-solving map for the whole course

When faced with a new complex-geometric problem, the following sequence usually locates the correct tool.

(1) Identify the local object: a holomorphic function, psh potential, bundle metric, analytic ideal, or differential form.
(2) Record its invariant: divisor, current, curvature tensor, cohomology class, or Lelong number.
(3) Determine the available positivity: Levi, Kähler, Griffiths, Nakano, nef, big, or positivity of a degree-specific commutator.
(4) Choose the globalization mechanism: exact sequence, Cartan theorem, Hodge representative, L^2 solution, compactness of currents, or continuity method.
(5) Audit the limit: topology of convergence, uniform estimate, singular set, normalization, and definition of every nonlinear product.

The exceptional divisor runs through this entire list. Locally it is the vanishing of a coordinate; as a Cartier divisor it defines $\mathcal{O}(E)$; its canonical section produces $[E]$ by Poincaré–Lelong; its normal curvature is negative; its class spans a new $H^{1,1}$ direction; and its negative self-intersection separates psef from nef. One object is being viewed through several compatible languages.

Remark (A final convention ledger)

This course uses $\mathrm{d}^{\mathrm{c}} = (4\pi\mathrm{i})^{-1}(\partial - \bar\partial)$, $\mathrm{dd}^{\mathrm{c}} = (\mathrm{i}/2\pi)\partial\bar\partial$, and $|e|_h^2 = \mathrm{e}^{-\varphi}$ for a line-bundle weight. Before importing a formula from another reference, test it on $\log|z|^2$, $\mathcal{O}(1)$ on $\mathbb{P}^1$, and the Gaussian weight. These three models detect nearly every sign or 2π discrepancy.

8.4.7 An explicit projective embedding of the blow-up

Take $p = [1:0:0] \in \mathbb{P}^2$ and

$$L = 2H - E.$$

Sections of L are conics through p. A basis is

$$Z_0Z_1, \quad Z_0Z_2, \quad Z_1^2, \quad Z_1Z_2, \quad Z_2^2,$$

so $h^0(L) = 5$. The induced rational map on $\mathbb{P}^2$ is undefined only at p and becomes a morphism on the blow-up.

On the affine chart $Z_0 = 1$, put $x = Z_1$, $y = Z_2$. In the blow-up chart $y = xv$, pull back the five sections:

$$(x, xv, x^2, x^2v, x^2v^2).$$

Because L subtracts one copy of E, divide by the common exceptional factor x. The lifted map is

$$(x, v) \longmapsto [1 : v : x : xv : xv^2].$$

On $E = \{x = 0\}$ this becomes $[1 : v : 0 : 0 : 0]$, so distinct tangent directions remain distinct. In the affine target chart where the first coordinate is nonzero, the first two coordinates v, x already have Jacobian

$$\frac{\partial(v,x)}{\partial(x,v)} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which has full rank. The second blow-up chart treats the missing vertical direction. Away from E, the map agrees with the conic system and separates points. Hence $2H - E$ embeds the blow-up as the cubic scroll in $\mathbb{P}^4$.

Riemann–Roch checked against monomial counting

For $L = aH - bE$ with $a \geq b \geq 0$, a degree-a polynomial vanishes to order at least b at $p = [1 : 0 : 0]$ exactly when every monomial $Z_0^{a-i-j} Z_1^i Z_2^j$ has $i + j \geq b$. The excluded monomials number

$$1 + 2 + \cdots + b = \frac{b(b+1)}{2}.$$

Therefore

$$h^0(L) = \binom{a+2}{2} - \binom{b+1}{2} = 1 + \frac{1}{2}(a^2 - b^2 + 3a - b)$$

whenever the imposed conditions are independent. For $a = 2, b = 1$, this gives five, matching the displayed basis. Surface Riemann–Roch gives the same number once higher cohomology vanishes.

Chern numbers and the blow-up formula

Since $K_X = -3H + E$,

$$c_1(X)^2 = (-K_X)^2 = (3H - E)^2 = 9 - 1 = 8.$$

The topological Euler characteristic is

$$c_2(X) = \chi_{\mathrm{top}}(X) = \chi_{\mathrm{top}}(\mathbb{P}^2) - 1 + \chi_{\mathrm{top}}(\mathbb{P}^1) = 3 - 1 + 2 = 4.$$

Noether's formula then checks

$$\chi(\mathcal{O}_X) = \frac{c_1^2 + c_2}{12} = \frac{8 + 4}{12} = 1,$$

consistent with $h^{0,0} = 1$ and $h^{0,1} = h^{0,2} = 0$.

One curve viewed in every language

Let $C \subset \mathbb{P}^2$ be a degree-d curve with multiplicity m at p. In local coordinates its lowest homogeneous part of degree m determines the intersection points of the strict transform with E. Globally,

$$[\widetilde{C}] = dH - mE, \qquad \widetilde{C} \cdot E = m, \qquad \widetilde{C}^2 = d^2 - m^2.$$

Its canonical section gives

$$\mathrm{dd^c} \log |s_{\widetilde{C}}|_h^2 = [\widetilde{C}] - c_1(\mathcal{O}(\widetilde{C}), h).$$

If $d > m > 0$, the class has positive intersection with both E and $H - E$ and lies in the Kähler cone. Kodaira then converts a positive metric into projective sections; Poincaré–Lelong converts any such section back into the divisor current. This closes the course's cycle from coordinates to currents, curvature, cohomology, and embeddings.

The blow-up charts used in the capstone calculations

The blow-up of the origin in $\mathbb{C}^2$ is the incidence variety

$$\widetilde{\mathbb{C}^2} = \big\{((x,y),[u:v]) \in \mathbb{C}^2 \times \mathbb{P}^1 : xv = yu\big\},$$

with projection $\pi((x,y),[u:v]) = (x,y)$. On the chart $u \neq 0$, put $t = v/u$. The incidence equation becomes

$$y = xt, \qquad (x,t) \longmapsto (x,xt),$$

and the exceptional curve is $E = \{x = 0\}$. On the chart $v \neq 0$, put $s = u/v$; then

$$x = sy, \qquad (s,y) \longmapsto (sy,y), \qquad E = \{y = 0\}.$$

On the overlap, $s = t^{-1}$ and $y = xt$. These transition formulas show directly that $E \simeq \mathbb{P}^1$, with t and s its two affine coordinates. The normal coordinate changes from x to $y = xt$, so the normal bundle has transition function t; with the standard convention this is $\mathcal{O}_{\mathbb{P}^1}(-1)$, and hence $E^2 = -1$.

Now let a plane curve be given near the origin by

$$f(x,y) = f_m(x,y) + f_{m+1}(x,y) + \cdots,$$

where f_k is homogeneous of degree k and $f_m \neq 0$. In the first chart,

$$f(x,xt) = x^m\big(f_m(1,t) + x f_{m+1}(1,t) + \cdots\big). \tag{32.1}$$

The factor x^m defines the exceptional component of the total transform. After dividing it out, the remaining factor defines the strict transform. Its intersection with E is given by $f_m(1,t) = 0$, so the points of intersection, counted with multiplicity and including the second chart, are precisely the tangent directions of the original curve. Formula (32.1) therefore proves both

$$\pi^*[C] = [\widetilde{C}] + m[E] \qquad \text{and} \qquad \widetilde{C} \cdot E = m.$$

For a degree-d projective curve this becomes $[\widetilde{C}] = dH - mE$. The same two charts thus justify the divisor class, intersection numbers, and local section formulas used throughout the capstone; none of those identities is merely formal notation.

8.4.8 Self-study workshop: solving one problem in all four languages

This workshop is part of the core notes. Work through it with paper and pencil before attempting the end-of-lecture exercises.

Prerequisite ledger. Bring together strict transforms, line bundles and sections, intersection theory, positive currents, Riemann–Roch, and projective embeddings. This workshop is a diagnostic for the whole course.

The class $dH - mE$ from four viewpoints. Let $\pi : X = \mathrm{Bl}_p \mathbb{P}^2 \to \mathbb{P}^2$ and let $L = dH - mE$ with $d \geq m \geq 0$.

Local analytic language. A section is represented by a homogeneous polynomial of degree d vanishing to order at least m at p. In blow-up coordinates $y = xv$, the pullback has a common factor x^m; division by this factor gives the local strict-transform section.

Sheaf language. The space of sections is

$$H^0(X,L) \simeq H^0(\mathbb{P}^2, \mathcal{I}_p^m(d)).$$

The quotient $\mathcal{O}_{\mathbb{P}^2}/\mathfrak{m}_p^m$ records the jet conditions removed from all degree-d polynomials.

Intersection language. One has

$$L^2 = d^2 - m^2, \qquad L \cdot E = m, \qquad L \cdot (H - E) = d - m.$$

Thus L is nef for $d \geq m \geq 0$ and ample for $d > m > 0$.

Current and metric language. A divisor $C \in |L|$ gives a positive current $[C]$ in $c_1(L)$. A smooth positive metric on L gives a Kähler representative of the same class when L is ample. Poincaré–Lelong compares them through the logarithm of the section norm.

For $d = 2, m = 1$, conics through p form a five-dimensional space. The resulting map embeds X as the cubic scroll in $\mathbb{P}^4$: locally the five monomials become $(1, v, x, xv, xv^2)$ after removing the exceptional factor, which separates both the point on E and its normal direction.

Checkpoint (capstone calculation)

(1) How many independent conditions does vanishing to order at least m impose on a plane polynomial?
(2) For $L = 3H - 2E$, compute L^2, $L \cdot E$, and $L \cdot (H - E)$.
(3) Which theorem relates a divisor current and the curvature of a metric on its line bundle?

Solution (checkpoint). The forbidden monomials in two local variables have total degree below m, so their number is $1 + 2 + \cdots + m = m(m+1)/2$. For $3H - 2E$, the three numbers are $9 - 4 = 5$, 2, and 1; the class is ample. The metric Poincaré–Lelong formula writes $\mathrm{dd^c} \log |s|_h^2 = [\mathrm{div}(s)] - c_1(L, h)$. ◇

Exit criterion

Given $dH - mE$, you can translate it into jets, monomial counts, strict transforms, intersection numbers, a divisor current, and a projective map, explaining which theorem connects each pair of languages.

Summary. $\mathrm{Bl}_p \mathbb{P}^2$ exhibits blow-up, divisors, Chern classes, the Kähler cone, Hodge numbers, vanishing, and positive currents in one model. The course goal is not to memorize 32 theorems but to break a problem along the chain: local analysis, curvature, cohomology, global geometry.

Exercise 8.23: Capstone calculation

Let $C \subset \mathbb{P}^2$ have degree d and multiplicity m at p. Compute the class of its strict transform, its intersection with E, its self-intersection, and its arithmetic genus; check the result with adjunction.

Exercise 8.24: Final project outline

Choose one of the four routes and write a five-page outline specifying a main theorem, three prerequisite tools, one normalization trap, and one explicit example.

Optional handout: Eight possible graduate projects

This handout is optional. No later argument in the core lectures depends on it.

Each project below can be developed from a concrete calculation to a modern theorem. None is needed for the core notes.

(1) **Toric Kähler geometry.** Compute the Fubini–Study moment polytope, Legendre-transform a toric potential, and state Abreu's scalar-curvature equation.

(2) **Higgs bundles.** Study flat rank-two bundles on a curve, formulate Hitchin's equations, and explain the nonabelian Hodge correspondence.
(3) **Variation of Hodge structure.** Derive Griffiths transversality for a family of curves and compute the period map in one example.
(4) **Singular Kähler–Einstein metrics.** Solve a one-dimensional conical model and explain how non-pluripolar products define the global equation.
(5) **Bergman kernels.** Compute kernels for the disc, ball, and projective space, then study positivity under variation of the weight.
(6) **Multiplier ideals.** Resolve a plane-curve singularity, compute its multiplier ideals, and identify its jumping numbers.
(7) **Birational surfaces.** Determine the nef, effective, and movable cones of a blow-up of $\mathbb{P}^2$ at several special points.
(8) **Complex dynamics.** Pull back positive currents under a rational map, construct a Green current, and relate its Lelong numbers to exceptional orbits.

A useful project structure. Begin with one explicit model and a complete calculation. State one general theorem with all hypotheses. Identify the analytic engine (for example L^2 estimates or Monge–Ampère comparison), the cohomological engine, and the compactness mechanism. End with one question whose answer is not formal from the notes.

Remark (What counts as understanding)

A successful project should translate the same object into at least three of the course's languages. For example, a divisor can be treated as an ideal sheaf, a line bundle with section, a cohomology class, and a positive current.

Chapter A

Background Lemmas, Assignments, and Teaching Notes

A.1 Notation and normalization at a glance

Notation	Meaning and convention
$\partial, \bar\partial$	$\mathrm{d} = \partial + \bar\partial$ and $\partial/\partial z_j = \frac{1}{2}(\partial_{x_j} - \mathrm{i}\partial_{y_j})$
$\mathrm{d}^{\mathrm{c}}, \mathrm{dd}^{\mathrm{c}}$	$\mathrm{d}^{\mathrm{c}} = (4\pi\mathrm{i})^{-1}(\partial - \bar\partial)$ and $\mathrm{dd}^{\mathrm{c}} = (\mathrm{i}/2\pi)\partial\bar\partial$
ω_φ	In the Calabi–Yau unit, $\omega + \mathrm{i}\partial\bar\partial\varphi = \omega + 2\pi\mathrm{dd}^{\mathrm{c}}\varphi$
$[Z]$	Integration current over Z_{Reg} with the orientation induced by the complex structure
$\nu(T, x)$	Lelong number normalized by $\nu([\mathbb{C}^p], 0) = 1$
$\Theta(E, h)$	Chern curvature $\bar\partial(H^{-1}\partial H)$; for $\lvert e\rvert^2 = \mathrm{e}^{-\varphi}$, $c_1(L, h) = (\mathrm{i}/2\pi)\Theta = \mathrm{dd}^{\mathrm{c}}\varphi$
L, Λ	$L = \omega \wedge -$ and $\Lambda = L^*$
Δ''	$D''D''^* + D''^*D''$; in the Kähler case $\Delta_{\mathrm{d}} = 2\Delta' = 2\Delta''$
$\mathrm{PSH}(X, \theta)$	θ-psh functions: $\theta + \mathrm{dd}^{\mathrm{c}}u \geq 0$
$\mathcal{I}(\varphi)$	Multiplier ideal of germs f with $\lvert f\rvert^2\mathrm{e}^{-\varphi} \in L^1_{\mathrm{loc}}$
$\langle T_1 \wedge \cdots \wedge T_p \rangle$	Non-pluripolar product; it assigns no mass to pluripolar sets

Warning

If one switches to the common convention $\mathrm{d}^{\mathrm{c}} = \frac{\mathrm{i}}{2}(\bar\partial - \partial)$, every dd^{c} formula changes by a factor involving π. Do not replace the symbol while retaining the constants in Poincaré–Lelong, Chern forms, or Monge–Ampère measures.

A.2 Four prerequisite patches

A.2.1 Index rules in Hermitian linear algebra

Let $e_1, \dots, e_n$ be a unitary $(1, 0)$ coframe. Wedge and contraction satisfy

$$[\bar e_j \wedge, \iota_{\bar e_k^\sharp}]_+ = \delta_{jk}, \qquad [e_j \wedge, \iota_{e_k^\sharp}]_+ = \delta_{jk}.$$

Every Kähler identity and curvature-commutator calculation reduces to these anticommutation rules. Compute first on one basis vector and extend linearly; manipulating a large index sum at once is the easiest way to lose a sign.

A.2.2 Distributional convergence and compactness of positive measures

$T_j \rightharpoonup T$ means $\langle T_j, \eta\rangle \to \langle T, \eta\rangle$ for every compactly supported smooth test form η. Differentiation is continuous: $\mathrm{d}T_j \rightharpoonup \mathrm{d}T$. A family of positive measures with locally uniform mass bounds is relatively compact by Riesz representation and Banach–Alaoglu. Coefficients of a positive current are controlled by its trace measure, so the same principle applies coefficientwise.

A.2.3 Short exact sequences and their long exact sequences

From $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ one obtains

$$0 \to H^0(\mathcal{F}') \to H^0(\mathcal{F}) \to H^0(\mathcal{F}'') \xrightarrow{\delta} H^1(\mathcal{F}') \to \cdots .$$

To construct δ in Čech language, lift a section s'' locally to s_i. The differences $s_i - s_j$ lie in $\mathcal{F}'$ and form a one-cocycle. Its class vanishes exactly when the lifts can be adjusted to glue. Whenever a restriction map is claimed surjective, identify the H^1 group that has been killed.

A.2.4 The Hilbert-complex existence lemma

Let $T : H_0 \to H_1$ and $S : H_1 \to H_2$ be closed densely defined operators with $ST = 0$. Suppose $f \in \ker S$ and

$$|(f, v)|^2 \le C\big(\|T^* v\|^2 + \|Sv\|^2\big)$$

for $v \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$. Then there is $u \in \operatorname{Dom} T$ with $Tu = f$ and $\|u\|^2 \le C$.

Indeed, view $v \mapsto (f, v)$ as a bounded functional on $\{(T^*v, Sv)\} \subset H_0 \oplus H_2$. Hahn–Banach and Riesz represent its extension by (u, g), giving $f = Tu + S^*g$ weakly. Because $f \in \ker S$ and $\overline{\operatorname{Im} S^*} = (\ker S)^\perp$, one has $S^*g = 0$. Taking $u \perp \ker T$ gives the minimum-norm solution.

A.3 Detailed background results used in the core

A.3.1 Partitions of unity and fine sheaves

Lemma A.1: Smooth partitions of unity

Let M be a Hausdorff second-countable smooth manifold and $\{U_i\}_{i\in I}$ an open cover. There is a locally finite refinement $\{V_j\}_{j\in J}$ and smooth functions $\rho_j \ge 0$ such that $\operatorname{supp}\rho_j \subset V_j$ and $\sum_j \rho_j = 1$.

Proof. Second countability and local Euclidean compactness make M paracompact, so the cover has a locally finite refinement by precompact coordinate balls W_j with $\overline{W_j} \subset V_j \subset U_{i(j)}$. Choose a smooth bump function χ_j supported in V_j and positive on $\overline{W_j}$. The locally finite sum $\chi = \sum_j \chi_j$ is smooth and strictly positive. Set $\rho_j = \chi_j/\chi$. Local finiteness justifies both the sum and all derivatives. □

If $\mathcal{F}$ is a sheaf of $\mathcal{A}_X$-modules, multiplication by ρ_j gives endomorphisms supported in V_j whose sum is the identity. Thus $\mathcal{F}$ is fine. The next theorem explains why fine resolutions compute cohomology.

Theorem A.2: Acyclic resolution theorem

Suppose

$$0 \to \mathcal{F} \to \mathcal{A}^0 \xrightarrow{d} \mathcal{A}^1 \xrightarrow{d} \cdots$$

is an exact complex of sheaves and each $\mathcal{A}^q$ is acyclic for global sections. Then

$$H^q(X, \mathcal{F}) \simeq H^q\big(\Gamma(X, \mathcal{A}^\bullet), d\big).$$

Proof. Let $\mathcal{Z}^q = \ker(\mathcal{A}^q \to \mathcal{A}^{q+1})$. Exactness gives short exact sequences

$$0 \to \mathcal{Z}^q \to \mathcal{A}^q \to \mathcal{Z}^{q+1} \to 0, \qquad \mathcal{Z}^0 \simeq \mathcal{F}.$$

The long exact sequence and $H^r(X, \mathcal{A}^q) = 0$ for $r > 0$ give dimension-shifting isomorphisms $H^r(X, \mathcal{Z}^{q+1}) \simeq H^{r+1}(X, \mathcal{Z}^q)$ for $r \geq 1$. In degree zero, the same long exact sequence identifies

$$\frac{\Gamma(X, \mathcal{Z}^{q+1})}{d\Gamma(X, \mathcal{A}^q)} \simeq H^1(X, \mathcal{Z}^q).$$

Iterating from $\mathcal{Z}^0 = \mathcal{F}$ gives the claimed isomorphism in every degree. This proof also shows naturality with respect to morphisms of resolutions. □

A.3.2 Closed operators and adjoints

Let $T : \operatorname{Dom} T \subset H_0 \to H_1$ be densely defined. Its graph norm is $\|u\|_T^2 = \|u\|_{H_0}^2 + \|Tu\|_{H_1}^2$. The operator is closed exactly when $\operatorname{Dom} T$ is complete in this norm. Its adjoint is defined by

$$\operatorname{Dom} T^* = \{v \in H_1 : u \mapsto (Tu, v) \text{ is bounded in the } H_0 \text{ norm}\}.$$

For such v, Riesz gives the unique T^*v satisfying $(Tu, v) = (u, T^*v)$.

Lemma A.3: Closed-range criterion

For a closed densely defined operator T, the range of T is closed if and only if there is $C > 0$ such that

$$\|u\| \leq C\,\|Tu\| \quad \text{for all } u \in \operatorname{Dom} T \cap (\ker T)^\perp.$$

In this case $\operatorname{im} T = (\ker T^*)^\perp$.

Proof. Restrict T to $\operatorname{Dom} T \cap (\ker T)^\perp$. It is a closed bijection onto $\operatorname{im} T$. If the range is closed, the closed graph theorem makes the inverse bounded, giving the estimate. Conversely, if Tu_j is Cauchy, the estimate makes the components of u_j orthogonal to the kernel Cauchy; closedness of T then puts the limit in the range. The identity $(\operatorname{im} T)^\perp = \ker T^*$ follows directly from the definition of the adjoint, and closedness removes the closure on the range. □

This is the functional-analytic step behind statements such as “coercivity gives a solution.” Boundary conditions are encoded in $\operatorname{Dom} T^*$; they are not visible in the formal differential expression alone.

A.3.3 The elliptic package

Theorem A.4: Elliptic regularity and the Fredholm alternative

Let $P : C^\infty(E) \to C^\infty(F)$ be an elliptic differential operator of order m on a compact manifold without boundary. For every real s there is C_s such that

$$\|u\|_{H^{s+m}} \le C_s\big(\|Pu\|_{H^s} + \|u\|_{H^s}\big).$$

The extension $P : H^{s+m}(E) \to H^s(F)$ is Fredholm. If Pu is smooth and u is a distribution, then u is smooth.

Proof.

Construction of a parametrix. The principal symbol $\sigma_m(P)(x,\xi)$ is invertible for $\xi \neq 0$. In local coordinates, invert it for large $|\xi|$ and correct lower-order errors recursively to construct a pseudodifferential parametrix Q of order $-m$ such that $QP = I - K_1$ and $PQ = I - K_2$, where K_1, K_2 are smoothing. Mapping properties of pseudodifferential operators give

$$\|u\|_{H^{s+m}} \le C\big(\|Pu\|_{H^s} + \|K_1 u\|_{H^{s+m}}\big) \le C_s\big(\|Pu\|_{H^s} + \|u\|_{H^s}\big),$$

because a smoothing operator maps H^s to every higher Sobolev space.

Rellich compactness makes K_1 and K_2 compact between the relevant Sobolev spaces. The two parametrix identities say that P is invertible modulo compact operators; Atkinson's theorem therefore makes $P : H^{s+m}(E) \to H^s(F)$ Fredholm. Equivalently, its kernel and cokernel are finite dimensional and its range is closed. If Pu is smooth and u is a distribution, choose s with $u \in H^s$. The identity $u = QPu + K_1 u$ improves u by m derivatives; iteration puts u in every Sobolev space, and Sobolev embedding makes it smooth.

For a self-adjoint nonnegative elliptic operator such as a Hodge Laplacian, the theorem implies finite-dimensional kernel and closed range. Spectral theory gives an orthonormal basis of smooth eigenvectors with eigenvalues tending to infinity. On the orthogonal complement of the kernel, the Green operator is the bounded inverse. This supplies the analytic input used in Lectures 13–16. □

A.3.4 Monotone limits of positive objects

Two limiting principles recur throughout the notes.

Lemma A.5: Weak lower semicontinuity

If $u_j \rightharpoonup u$ weakly in a Hilbert space, then $\|u\| \le \liminf_j \|u_j\|$. If T is closed and $Tu_j = f$ while u_j converges weakly and is uniformly bounded in the graph norm, then $Tu = f$.

Proof. The norm is the supremum of $|(u, v)|$ over unit vectors v, so weak convergence gives the first assertion. A bounded graph-norm sequence has weakly convergent subsequences in both H_0 and H_1. The graph of a closed linear operator is a closed linear subspace and hence weakly closed, proving the second assertion. □

For positive measures, local mass bounds give weak-* compactness. For psh functions, normalization prevents convergence to $-\infty$ and gives L^1_{loc} compactness. For L^2 solutions, weak lower semicontinuity preserves the estimate. These are three forms of the same compactness strategy: obtain a coercive local bound, extract weakly, pass the linear equation against tests, and recover nonlinear information from positivity or monotonicity.

A.4 Proof of Oka's coherence theorem

This appendix supplies the technical proof of Theorem 3.5. It is separated from the classroom text because the main conceptual use of the theorem is simple—coherent analytic sheaves are stable under finite kernels and cokernels—whereas the proof requires a careful induction with Weierstrass division. We use the notation $z' = (z_1, \dots, z_{n-1})$ and $w = z_n$.

A.4.1 The algebraic criterion

Lemma A.6: The row-kernel criterion for coherence

Let $\mathcal{A}$ be a sheaf of rings. Suppose that, locally, the kernel of every morphism

$$\mathcal{A}^{\oplus m} \longrightarrow \mathcal{A}$$

is of finite type. Then the kernel of every morphism $\mathcal{A}^{\oplus m} \to \mathcal{A}^{\oplus q}$ is locally of finite type. Consequently $\mathcal{A}$ is coherent, and kernels, images, and cokernels of morphisms between locally finitely presented $\mathcal{A}$-modules are locally finitely presented.

Proof. The assertion about a finite matrix follows by induction on the number q of rows. It is the hypothesis when $q = 1$. Write a matrix with q rows as (A', ℓ), where A' has $q - 1$ rows and ℓ is the last row. By induction, $K' = \ker A'$ is locally generated by finitely many sections. After shrinking, choose a surjection $p : \mathcal{A}^{\oplus s} \to K'$. Then

$$p\big(\ker(\ell \circ p)\big) = \ker(A', \ell).$$

Indeed, the inclusion from left to right is immediate. Conversely, a section in the right-hand kernel belongs to K', so it has a local preimage under p; that preimage lies in $\ker(\ell \circ p)$. The row-kernel hypothesis makes this latter kernel finitely generated, and its image is therefore finitely generated.

Now choose finite presentations of two modules and lift a morphism between them to their finite free presentations. The kernel is obtained from kernels of the finite matrices just treated, and the cokernel is visibly a quotient of a finite free module by a finitely generated submodule. Repeating the construction gives finite presentations. The image is the quotient by the kernel. This is the usual coherent-ring criterion, written in a form that applies on a common neighborhood. □

Thus it suffices to prove that the relations among any finite row of holomorphic functions are locally finitely generated.

A.4.2 Reduction of a relation to a finite remainder

Fix the origin in a polydisc in $\mathbb{C}^n$, and let $f = (f_1, \dots, f_m)$ be a row of holomorphic functions. Its relation sheaf is

$$\mathcal{R}(f) = \left\{ a = (a_1, \dots, a_m) : \sum_{j=1}^{m} a_j f_j = 0 \right\}. \tag{A.1}$$

We prove by induction on n that $\mathcal{R}(f)$ is of finite type near the origin. For $n = 0$ this is finite-dimensional linear algebra. If every germ $(f_j)_0$ is zero, all the functions vanish after shrinking and $\mathcal{R}(f) = \mathcal{O}^{\oplus m}$. Otherwise, reorder the row so that $(f_m)_0 \neq 0$.

By a generic complex-linear change of coordinates, f_m is regular of some order d in w. Weierstrass preparation gives

$$f_m = uP, \qquad P(z', w) = w^d + c_1(z')w^{d-1} + \cdots + c_d(z'), \tag{A.2}$$

where u is a holomorphic unit and $c_j(0) = 0$. Multiplying the last component of a relation by the unit u identifies the relation sheaf of the original row with that of $(f_1, \dots, f_{m-1}, P)$. We may therefore assume $f_m = P$.

Let $a = (a_1, \dots, a_m)$ satisfy (A.1). For $j < m$, Weierstrass division gives unique holomorphic functions q_j and unique remainders r_j such that

$$a_j = Pq_j + r_j, \qquad r_j(z', w) = \sum_{\nu=0}^{d-1} b_{j\nu}(z')w^\nu. \tag{A.3}$$

Let $e_1, \dots, e_m$ be the standard frame. For every $j < m$ the vector

$$E_j := Pe_j - f_j e_m \tag{A.4}$$

is an explicit relation. Subtracting $q_j E_j$ from a replaces its j-th component by r_j. Hence every relation is a sum of the elementary relations E_j and a reduced relation whose first $m-1$ components have degree less than d in w.

For a reduced relation, the last component exists precisely when

$$P \text{ divides } \sum_{j=1}^{m-1} r_j f_j. \tag{A.5}$$

Divide the right-hand side by P. Its remainder is

$$c_0(z') + c_1(z')w + \cdots + c_{d-1}(z')w^{d-1}. \tag{A.6}$$

The division operator by the fixed monic polynomial P is $\mathcal{O}_{\mathbb{C}^{n-1}}$-linear. Therefore the coefficient vector $(c_0, \dots, c_{d-1})$ is obtained from the finite vector

$$b = (b_{10}, \dots, b_{1,d-1}, \dots, b_{m-1,0}, \dots, b_{m-1,d-1})$$

by a finite matrix $B(z')$ of holomorphic functions in z'. Condition (A.5) is exactly

$$B(z')b(z') = 0. \tag{A.7}$$

By the induction hypothesis in dimension $n-1$ and Lemma A.6, the kernel of B is generated, on a smaller z'-polydisc, by finitely many vectors $b^{(1)}, \dots, b^{(N)}$. Convert each $b^{(k)}$ into polynomials $r_j^{(k)}(z', w)$ of degree less than d. Since its remainder vanishes, the sum $\sum_{j<m} r_j^{(k)} f_j$ is divisible by P; let

$$h_k = \frac{1}{P} \sum_{j=1}^{m-1} r_j^{(k)} f_j.$$

Then

$$R_k = (r_1^{(k)}, \dots, r_{m-1}^{(k)}, -h_k) \tag{A.8}$$

is a holomorphic relation. Equations (A.3)–(A.7) show that every relation is an $\mathcal{O}$-linear combination of the finitely many E_j and R_k.

A.4.3 Why the generators work on a neighborhood

The preceding calculation must be a sheaf argument, not merely a calculation in the single local ring $\mathcal{O}_{\mathbb{C}^n,0}$. Choose radii so that P has no zero on the boundary circle in the w-variable. The Cauchy-integral formulas in Weierstrass division then produce the quotient and remainder on one fixed smaller

product polydisc. Their coefficients satisfy uniform Cauchy bounds there. The lower-dimensional induction supplies the generators of $\ker B$ on one fixed smaller z'-polydisc. After taking the product with a smaller w-disc, all functions in (A.3)–(A.8) are therefore holomorphic on the same neighborhood.

We have proved that the kernel of every finite row $\mathcal{O}^{\oplus m} \to \mathcal{O}$ is locally of finite type. Lemma A.6 now shows that $\mathcal{O}_X$ is coherent and that finite kernels, images, and cokernels of locally finitely presented analytic modules remain locally finitely presented. This completes the proof of Oka's coherence theorem.

A.5 Correspondence with Demailly's book

Lectures	Topic	Main source or extension
1–4	complex manifolds and foundational several complex variables	Ch. I §§1–3, with the several-variable Cauchy theory, normal-family arguments, Hartogs phenomena, and elementary $\bar\partial$ method written out in detail
5–7	subharmonic/psh, pseudoconvex, and the Levi problem	Ch. I §§4–7
8–12	basic sheaf theory, Stein and Cartan A/B, coherence, analytic spaces, Hermitian and Kähler metrics	Ch. II §§1–9 and Ch. VI §§4–5; the sheaf foundations are expanded for readers with no prior exposure, and blow-ups from Ch. II §10 are retained in an optional handout
13–16	positive currents, Monge–Ampère, Lelong, Siu	Ch. III §§1–8; capacity and later pluripotential theory added
17–20	bundles, Chern curvature, positivity, embedding	Ch. V and Ch. VII §§6–14
21–24	Hodge, Kähler identities, cohomology, duality	selections from Ch. IV and Ch. VI
25–28	BKN, vanishing, Hörmander, multiplier ideals	Ch. VII §§1–9 and Ch. VIII §§1–6; Nadel added
29	Ohsawa–Takegoshi	Ch. VIII §7 plus the structure of twisted estimates
30	Calabi–Yau	added; connected to Ch. III Monge–Ampère theory
31	positivity cones, regularization, non-pluripolar products	modern extension of Ch. III/VII/VIII
32	capstone on the blown-up plane	synthesis of Ch. II/V/VI/VII

A.6 Optional: q-convexity and Andreotti–Grauert finiteness

Definition A.7: q-convexity

A smooth function on an n-fold is strongly q-convex if its Levi form has at least $n-q+1$ positive eigenvalues. A space is q-convex if it has such an exhaustion outside a compact set, and q-complete if the exhaustion is strongly q-convex everywhere. Stein/strictly psh corresponds to $q = 1$. Usually $1 \le q \le n$; if one formally allows $q = n + 1$, the eigenvalue condition is vacuous, whereas $q = 0$ would demand the impossible total of $n + 1$ positive eigenvalues.

Theorem A.8: Andreotti–Grauert, course version

If X is a q-convex complex space and $\mathcal{F}$ coherent, then $H^k(X, \mathcal{F})$ is finite dimensional for $k \ge q$. If X is q-complete, then $H^k(X, \mathcal{F}) = 0$ for $k \ge q$.

Proof (using the Andreotti–Grauert bump lemma). We isolate the analytic input, the Andreotti–Grauert bump lemma. A *strongly q-convex bump pair* is $A \Subset B$ for which $B \setminus A$ lies in one coordinate ball and the new boundary is cut out by a function whose Levi form has at least $n - q + 1$ eigenvalues bounded below by a positive constant. The bump lemma says that, for a coherent sheaf $\mathcal{F}$, the restriction map in degrees $k \ge q$ has closed image and finite-dimensional kernel and cokernel; if the bump can be pushed through a region with no compact exceptional set, the cokernel is zero. For a sequence of bumps escaping every compact set, the restriction of a degree-k class, $k \ge q$, to a sufficiently far tail can be corrected to zero. On a merely q-convex space, a finite-dimensional obstruction may remain on the exceptional compact set; this is exactly the distinction between finiteness and vanishing.

Its proof resolves $\mathcal{F}$ locally by finite free sheaves and combines a specially chosen Leray cover with weighted $\bar\partial$ estimates on the bump. The linear-algebra lemma for a Levi form with $n - q + 1$ positive eigenvalues supplies coercivity in cohomological degree at least q; a compact restriction operator accounts for the finite-dimensional error. Successive corrections through the free resolution give closed range and the stated finite obstruction for $\mathcal{F}$. This analytic bump lemma is the substantial part of the Andreotti–Grauert theorem; it is not a consequence of counting alternating Čech indices.

Now perturb the exhaustion slightly, without losing its strict Levi margin, so that its critical points are isolated. Between regular levels, compactness of a level band allows it to be crossed by finitely many strongly q-convex bumps. Near a critical point, real Morse coordinates together with a local Levi-positive quadratic correction produce the same finite bump chain. Thus, after a level containing the exceptional compact set, the restriction-correction maps in degrees $k \ge q$ have only finite-dimensional obstructions. All possible obstructions are already represented on one fixed compact sublevel; coherence and the finite covering used above make that space finite dimensional. Passing to the exhaustion proves $\dim H^k(X, \mathcal{F}) < \infty$ for a q-convex space.

If X is q-complete there is no exceptional compact set. Given a cohomology class, represent it by a cocycle on some sublevel. Repeatedly apply the surjective restriction-correction part of the bump lemma while pushing the support of the corrected cocycle to larger levels. The uniform estimates make the corrections a convergent series on compact subsets, and the limit is a global coboundary. Hence $H^k(X, \mathcal{F}) = 0$ for every $k \ge q$. □

A.7 Biweekly assignments and selected solutions

Assignment 1 (Weeks 1–2)

Required: 1.14, 1.25, 1.37, and 1.46.

Solution (1.46). For $k < 0$, the Laurent coefficient $a_k(z_1)$ is holomorphic on Δ. The solid arm $\Delta_r \times \Delta$ shows that it vanishes first on the open set Δ_r. The identity theorem in the z_1 variable then gives $a_k \equiv 0$ on Δ. Hence no negative Laurent term survives and the Taylor series in z_2 fills the hole. ◇

Assignment 2 (Weeks 3–4)

Required: 2.14, 2.26, 2.42, and 3.19.

Solution (2.26). The theorem on finite upper envelopes proves that $\max(u, v)$ is psh. For a counterexample, take $u(z) = \operatorname{Re} z$ and $v(z) = -\operatorname{Re} z$. Both are pluriharmonic, but $\min(u, v) = -|\operatorname{Re} z|$ has a downward corner along the imaginary axis; its distributional Laplacian contains a negative measure. ◇

Assignment 3 (Weeks 5–6)

Required: 2.57, 3.32, 3.45, and 3.58.

Solution (3.58). In the affine coordinate $z = re^{i\theta}$, $\omega_{\mathrm{FS}} = \pi^{-1} r(1+r^2)^{-2}\mathrm{d}r \wedge \mathrm{d}\theta$. Thus its integral is $2\int_0^\infty r(1+r^2)^{-2}\mathrm{d}r = 1$. The coordinate $1/z$ verifies smoothness at infinity. ◇

Assignment 4 (Weeks 7–8)

Required: 6.11, 6.24, 6.34, and 6.45.

Solution (6.34). $u = \max(\log|z|, 0)$ is harmonic on either side of $|z| = 1$. The distributional Laplacian comes only from the jump in normal derivative at $r = 1$, so $\mathrm{dd}^{\mathrm{c}} u$ is one half of normalized Haar measure on the unit circle. Its total mass is $1/2$; using $\log|z|^2$ instead would give mass 1 in our normalization. ◇

Assignment 5 (Weeks 9–10)

Required: 5.7, 5.16, 5.24, and 5.32.

Solution (5.32). A basis of $H^0(\mathbb{P}^n, \mathcal{O}(k))$ consists of all z^I, $|I| = k$. The map $[z] \mapsto [z^I]_{|I|=k}$ has no base points and separates homogeneous points. On $z_0 \neq 0$, the coordinate monomials contain $1, w_j$ after a common factor, so the differential has full rank. Hence this is the Veronese embedding. ◇

Assignment 6 (Weeks 11–12)

Required: 4.6, 4.17, 4.28, and 4.36.

Solution (4.28). Choose a subordinate partition ρ_k and set $a_i = \sum_k \rho_k g_{ik}$. Then $a_i - a_j = g_{ij}$. The forms $\bar\partial a_i$ agree on overlaps and glue to a global $\bar\partial$-closed $(0,1)$-form. Changing the partition changes it only by a $\bar\partial$-exact form. ◇

Assignment 7 (Weeks 13–14)

Required: 7.5, 7.15, 7.20, and 7.28.

Solution (7.28). If $f = z^m g$ with $g(0) \neq 0$, then $|f|^2 e^{-\varphi} \sim r^{2m-2c}$. Multiplication by the area element $r\mathrm{d}r\mathrm{d}\theta$ is integrable exactly when $2m - 2c + 1 > -1$, or $m > c - 1$. Hence $\mathcal{I}(c\log|z|^2) = (z^{\lfloor c \rfloor})$; for integral c, the strict inequality still gives $m \geq c$. ◇

Assignment 8 (Weeks 15–16)

Required: 8.5, 8.9, 8.18, and 8.23.

Solution (8.23). The strict transform has class $\widetilde{C} = dH - mE$. Thus $\widetilde{C} \cdot E = m$ and $\widetilde{C}^2 = d^2 - m^2$. Adjunction gives

$$2p_a(\widetilde{C}) - 2 = \widetilde{C} \cdot (\widetilde{C} + K_X) = d^2 - m^2 - 3d + m,$$

so $p_a = (d-1)(d-2)/2 - m(m-1)/2$. ⋄

A.8 Suggestions for instructors

- Begin each lecture with a ten-minute unit test: $|z|^2$, $\log|z|^2$, $O(1)$, or a flat torus.
- Every two weeks, ask a student to diagnose one Warning: identify the first invalid step, then repair it.
- A Week 8 take-home problem can audit a short proof missing upper semicontinuity, a boundary condition, or the correct curvature degree.
- Keep the Hörmander hypothesis ledger visible throughout Weeks 13–14.
- If sheaf theory is new to the class, shorten the spectral-sequence material in Lecture 16 and add Čech computations to Lectures 9 and 15.
- For an algebraic-geometry audience, treat Lecture 30 as optional and expand ample/nef/big cones and intersection theory in Lectures 20 and 31.

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Complete Solutions to Exercises

This chapter contains solutions to every exercise in the lectures and in the optional handouts. Each heading links back to the original exercise. The solutions use the normalizations fixed at the beginning of the book; in particular, $\mathrm{d}^{\mathrm{c}} = (4\pi\mathrm{i})^{-1}(\partial - \bar\partial)$ and $\mathrm{dd}^{\mathrm{c}} = (\mathrm{i}/2\pi)\partial\bar\partial$.

Lecture 1

Exercise 1.14: The two charts of the projective line

Solution. Set

$$U_0 = \{[z_0 : z_1] : z_0 \neq 0\}, \qquad U_1 = \{[z_0 : z_1] : z_1 \neq 0\}.$$

The coordinates $w = z_1/z_0$ on U_0 and $\zeta = z_0/z_1$ on U_1 identify both charts with $\mathbb{C}$. On the overlap, $w \neq 0$, and $\zeta = w^{-1}$; this is a holomorphic transition map with holomorphic inverse.

Identify the unit sphere $S^2 \subset \mathbb{R}^3$ with the one-point compactification of $\mathbb{C}$ by inverse stereographic projection

$$w \longmapsto \frac{1}{1+|w|^2}\big(2\Re w, 2\Im w, |w|^2 - 1\big), \qquad \infty \longmapsto (0,0,1).$$

The coordinate near the north pole is $\zeta = 1/w$, exactly the second projective chart. The complex coordinate changes are therefore the same, so this identification is a biholomorphism from $\mathbb{P}^1$ to the Riemann sphere. ◇

Exercise 1.15: Tangent spaces

Solution. Write $F = (F_1, \dots, F_r)$ and suppose $\mathrm{d}F_x$ has rank r. A tangent vector $v \in T_x^{1,0}X$ is the velocity at zero of a holomorphic curve $\gamma : (\mathbb{C}, 0) \to (X, x)$. Differentiating $F \circ \gamma = 0$ gives $\mathrm{d}F_x(v) = 0$, so $T_x^{1,0}X \subseteq \ker \mathrm{d}F_x$.

Conversely, the holomorphic implicit-function theorem gives local coordinates $(z', z'') \in \mathbb{C}^{n-r} \times \mathbb{C}^r$, centered at x, in which X is $z'' = 0$. In these coordinates $\ker \mathrm{d}F_x$ is spanned by $\partial/\partial z'_1, \dots, \partial/\partial z'_{n-r}$, and these vectors are tangent to the coordinate curves in X. Hence

$$T_x^{1,0}X = \ker(\mathrm{d}F_x : \mathbb{C}^n \longrightarrow \mathbb{C}^r).$$

◇

Exercise 1.16: Cauchy–Riemann equations as complex linearity

Solution. Use real coordinates $z_j = x_j + \mathrm{i}y_j$ and $w_a = s_a + \mathrm{i}t_a$, and write $h^a = u^a + \mathrm{i}v^a$. On ∂_{x_j},

$$\mathrm{d}h\, J_0(\partial_{x_j}) = \sum_a \left(u^a_{y_j}\partial_{s_a} + v^a_{y_j}\partial_{t_a} \right),$$

whereas

$$J_0\,\mathrm{d}h(\partial_{x_j}) = \sum_a \left(-v^a_{x_j}\partial_{s_a} + u^a_{x_j}\partial_{t_a}\right).$$

Equality for every a, j is equivalent to $u^a_{x_j} = v^a_{y_j}$ and $u^a_{y_j} = -v^a_{x_j}$, the Cauchy–Riemann equations. On $\partial_{y_j} = J_0\partial_{x_j}$ the same conclusion follows from $J_0^2 = -\mathrm{id}$. Thus the real differential commutes with J_0 exactly when all components of h are holomorphic. ◇

Exercise 1.17: $\bar\partial$ detects antiholomorphic variation

Solution. The Wirtinger rules treat z and $\bar z$ as independent variables. Therefore

f	∂f	$\bar\partial f$
z^2	$2z\,\mathrm{d}z$	0
$\bar z$	0	$\mathrm{d}\bar z$
$\lvert z\rvert^2$	$\bar z\,\mathrm{d}z$	$z\,\mathrm{d}\bar z$.

Intrinsically, ∂f and $\bar\partial f$ are the projections of $\mathrm{d}f$ to the $+\mathrm{i}$ and $-\mathrm{i}$ eigensubbundles of J^*, respectively. Since a holomorphic coordinate change has complex-linear differential, it preserves these eigenspaces. The three computations therefore express invariant facts, not accidents of the coordinate z. ◇

Exercise 1.19: A nonintegrable model

Solution. Using $[A, fB] = A(f)B + f[A, B]$, we obtain

$$[L_1, L_2] = \left[\frac{\partial}{\partial\bar z_1}, \frac{\partial}{\partial\bar z_2} + \bar z_1\frac{\partial}{\partial z_1}\right] = \frac{\partial}{\partial z_1}.$$

This vector is not in the complex span of L_1, L_2: if $aL_1 + bL_2 = \partial_{z_1}$, comparison of the $\partial_{\bar z_1}$ and $\partial_{\bar z_2}$ components forces $a = b = 0$, a contradiction. Thus $T^{0,1}$ is not closed under Lie brackets. By the Newlander–Nirenberg integrability criterion the almost complex structure is not integrable. ◇

Lecture 2

Exercise 1.25: Coefficients on a bidisc

Solution. If $f = \sum_\alpha c_\alpha z^\alpha$, the polydisc coefficient estimate gives

$$\lvert c_{(4,2)}\rvert \le 12\,2^{-4}3^{-2} = \frac{1}{12}.$$

Since $\partial^4_{z_1}\partial^2_{z_2} f(0) = 4!2!c_{(4,2)}$, it follows that

$$\lvert\partial^4_{z_1}\partial^2_{z_2} f(0)\rvert \le 4.$$

The factorial appears in the derivative but not in the Taylor coefficient. ◇

Exercise 1.26: A power series domain

Solution. The series factors into two geometric series:

$$\sum_{j,k\ge0}\frac{z_1^j z_2^k}{2^{j+k}} = \left(\sum_{j\ge0}(z_1/2)^j\right)\left(\sum_{k\ge0}(z_2/2)^k\right) = \frac{1}{(1-z_1/2)(1-z_2/2)}.$$

Absolute convergence holds exactly on the polydisc $\lvert z_1\rvert < 2$, $\lvert z_2\rvert < 2$. On a compact subset, choose $\rho_j < 2$ with $\lvert z_j\rvert \le \rho_j$ there. The series and each differentiated series are dominated by products of convergent geometric series with polynomial factors in j, k, so termwise differentiation is valid. ◇

Exercise 1.27: Sign check

Solution. Locally write α as a sum of terms $a\,\eta$, where η is a wedge of $p+q$ coordinate one-forms. Since $\bar\partial$ is the component of the exterior derivative of bidegree $(0,1)$, it is a graded derivation of total degree one. Consequently

$$\bar\partial(a\eta\wedge\beta)=\bar\partial(a\eta)\wedge\beta+(-1)^{p+q}a\eta\wedge\bar\partial\beta.$$

Summing the local terms proves

$$\bar\partial(\alpha\wedge\beta)=\bar\partial\alpha\wedge\beta+(-1)^{p+q}\alpha\wedge\bar\partial\beta.$$

The sign involves the total degree $p+q$, not merely the antiholomorphic degree q. ◇

Exercise 1.28: Reality of $\mathrm{d^c}$

Solution. For $z=x+\mathrm{i}y$,

$$\partial u=\frac{1}{2}(u_x-\mathrm{i}u_y)(\mathrm{d}x+\mathrm{i}\mathrm{d}y),\qquad \bar\partial u=\frac{1}{2}(u_x+\mathrm{i}u_y)(\mathrm{d}x-\mathrm{i}\mathrm{d}y).$$

Hence $\partial u-\bar\partial u=\mathrm{i}(u_x\mathrm{d}y-u_y\mathrm{d}x)$, and therefore

$$\mathrm{d^c}u=\frac{1}{4\pi}(u_x\mathrm{d}y-u_y\mathrm{d}x).$$

When u is real, both coefficients are real, so $\mathrm{d^c}u$ is a real one-form. Differentiation also gives $\mathrm{dd^c}u=(4\pi)^{-1}(u_{xx}+u_{yy})\mathrm{d}x\wedge\mathrm{d}y$. ◇

Lecture 3

Exercise 1.36: A quantitative derivative bound

Solution. Fix z with $|z_1|,|z_2|\le 1/2$. For every $0<\rho<1/2$, the closed polydisc $P(z;(\rho,\rho))$ lies in Δ^2. Cauchy's estimate gives

$$|\partial_{z_1}^2\partial_{z_2}f(z)|\le 2!\rho^{-3}.$$

Letting $\rho\uparrow 1/2$ yields the bound $2!(1/2)^{-3}=16$. ◇

Exercise 1.37: An implicit curve

Solution. The derivative of $F(z_1,z_2)=z_1^2+z_2^2-1$ with respect to z_1 equals 2 at $(1,0)$, so the implicit theorem gives the branch

$$g(z_2)=\sqrt{1-z_2^2},\qquad g(0)=1.$$

Differentiating $g^2+z_2^2=1$ gives $g'(0)=0$ and $g''(0)=-1$. Analytic continuation around either branch point 1 or -1 changes the sign of the square root. Hence there is no single-valued holomorphic branch on all of $\mathbb{C}\setminus\{\pm1\}$. ◇

Exercise 1.38: Normality with a moving pole

Solution. For $|a| \le 1/2$ and $z \in \Delta^n$,

$$|1 - az_1| \ge 1 - |a||z_1| > \frac{1}{2}, \qquad |f_a(z)| < 2.$$

The family is uniformly bounded, hence normal by Montel. More directly, any sequence a_ν has a convergent subsequence because the closed parameter disc is compact. If $a_\nu \to a$, then

$$f_{a_\nu} - f_a = \frac{(a_\nu - a)z_1}{(1 - a_\nu z_1)(1 - az_1)}$$

converges uniformly on Δ^n (and therefore on compact subsets) to zero. ◇

Lecture 4

Exercise 1.46: Filling a Hartogs hole

Solution. For every $k < 0$, the coefficient $a_k(z_1)$ is holomorphic on Δ, because it is a Cauchy integral over a fixed circle in the annular arm. When $|z_1| < r$, the solid arm supplies a function holomorphic for all $|z_2| < 1$, so its Laurent series has no negative terms. Thus $a_k = 0$ first on the open disc Δ_r. The identity principle in the z_1 variable gives $a_k \equiv 0$ on Δ. ◇

Exercise 1.47: A solvable and an unsolvable form

Solution. Direct calculation gives

$$\bar\partial\alpha = \mathrm{d}\bar z_2 \wedge \mathrm{d}\bar z_1 + \mathrm{d}\bar z_1 \wedge \mathrm{d}\bar z_2 = 0,$$

and $\alpha = \bar\partial(\bar z_1 \bar z_2)$. All its solutions are $u = \bar z_1 \bar z_2 + h$, where h is holomorphic. On the other hand,

$$\bar\partial\gamma = \mathrm{d}\bar z_2 \wedge \mathrm{d}\bar z_1 \neq 0,$$

so γ cannot equal $\bar\partial u$ because $\bar\partial^2 u = 0$. ◇

Exercise 1.48: Cauchy–Pompeiu check

Solution. Let $D = \{|\zeta| < R\}$ and $|z| < R$. On $|\zeta| = R$ we have $\bar\zeta = R^2/\zeta$. Therefore the boundary term is

$$\frac{1}{2\pi\mathrm{i}} \int_{|\zeta|=R} \frac{R^2}{\zeta(\zeta - z)}\,\mathrm{d}\zeta = 0,$$

because the residues at 0 and z cancel (the case $z = 0$ follows directly). Since $\partial\bar\zeta/\partial\bar\zeta = 1$, formula (4.1) becomes

$$\bar z = \frac{1}{2\pi\mathrm{i}} \int_D \frac{1}{\zeta - z}\,\mathrm{d}\zeta \wedge \mathrm{d}\bar\zeta,$$

which is the required recovery of $f(z) = \bar z$ from the interior defect term. ◇

Lecture 5

Exercise 2.14: Convexity of circular means

Solution. Put $v(w) = u(e^w)$ on a vertical strip in the $w = t + \mathrm{i}\theta$ plane. A holomorphic pullback of a subharmonic function is subharmonic, so v is subharmonic and 2π-periodic in θ. Its circular mean is

$$m(t) = M_u(0, e^t) = \frac{1}{2\pi}\int_0^{2\pi} v(t + \mathrm{i}\theta)\,\mathrm{d}\theta.$$

For smooth u, periodicity and integration by parts give

$$m''(t) = \frac{1}{2\pi}\int_0^{2\pi} v_{tt}\,\mathrm{d}\theta = \frac{1}{2\pi}\int_0^{2\pi} (v_{tt} + v_{\theta\theta})\,\mathrm{d}\theta \geq 0.$$

For general subharmonic u, regularize on slightly smaller annuli. The smooth means are convex and converge in L^1_{loc}, hence their distributional second derivatives converge to the nonnegative distribution m''. Therefore m is convex. Since $t = \log r$, this is the desired convexity in $\log r$. ◇

Exercise 2.15: Decreasing limits

Solution. Let $u_j \downarrow u$. If $u \equiv -\infty$ on a connected component, there is nothing to prove there. Otherwise fix a closed disc $\overline{D(a,r)}$ in the component. A decreasing limit of upper-semicontinuous functions is upper semicontinuous. Since u_1 is upper semicontinuous, choose a finite constant C with $u_1 \leq C$ on the boundary circle. Set

$$q_j(\theta) = C - u_j(a + r\mathrm{e}^{\mathrm{i}\theta}) \geq 0.$$

The sequence q_j increases pointwise to $q(\theta) = C - u(a + r\mathrm{e}^{\mathrm{i}\theta})$. If $u(a) > -\infty$, the submean inequality for u_j, rewritten in terms of q_j, is

$$\frac{1}{2\pi}\int_0^{2\pi} q_j(\theta)\,\mathrm{d}\theta \leq C - u_j(a).$$

Monotone convergence and $u_j(a) \downarrow u(a)$ give

$$\frac{1}{2\pi}\int_0^{2\pi} q(\theta)\,\mathrm{d}\theta \leq C - u(a) < \infty.$$

Subtracting from C yields the submean inequality for u. If $u(a) = -\infty$, that inequality is automatic. Thus every admissible disc satisfies the submean inequality, and u is subharmonic on the component. ◇

Exercise 2.17: Green function of the unit disc

Solution. If $|z| = 1$, then

$$|z - \zeta|^2 = (z - \zeta)(\bar{z} - \bar{\zeta}) = (1 - \bar{\zeta}z)(1 - \zeta\bar{z}) = |1 - \bar{\zeta}z|^2,$$

so $G_{\mathbb{D}}(z, \zeta) = 0$ on $\partial\mathbb{D}$. In the z variable, $1 - \bar{\zeta}z$ is holomorphic and has no zero in $\mathbb{D}$. Thus $\log|1 - \bar{\zeta}z|^2$ is harmonic there. Poincaré–Lelong gives

$$\mathrm{dd}^{\mathrm{c}}_z G_{\mathbb{D}}(z, \zeta) = \mathrm{dd}^{\mathrm{c}}_z \log|z - \zeta|^2 - \mathrm{dd}^{\mathrm{c}}_z \log|1 - \bar{\zeta}z|^2 = \delta_\zeta.$$

Hence this is the Green function with the normalization used in the notes. ◇

Lecture 6

Exercise 2.26: Maximum versus minimum

Solution. Both $u_1 = \log|z_1|$ and $u_2 = \log|z_2|$ are psh. The maximum of finitely many psh functions is upper semicontinuous and restricts to the maximum of subharmonic functions on every complex line; the latter is subharmonic. Therefore $\max(u_1, u_2)$ is psh.

For the failure of minima, take the psh functions 0 and $\log|z|$ on $\mathbb{C}$. Their minimum is

$$v(z) = \begin{cases} \log|z|, & |z| \le 1, \\ 0, & |z| \ge 1. \end{cases}$$

The radial derivative drops from 1 on the inner side of the unit circle to 0 on the outer side. Consequently Δv has a negative measure on $|z| = 1$. Equivalently, the mean-value inequality fails at points of that circle. Thus v is not subharmonic and hence not psh. ◇

Exercise 2.27: A pole on projective space

Solution. On $U_0 = \{z_0 \neq 0\}$, put $w = z_1/z_0$. Then

$$u = \log|w|^2 - \log(1 + |w|^2).$$

Poincaré–Lelong and the definition of the Fubini–Study form yield

$$\mathrm{dd^c} u = \delta_{w=0} - \omega_{\mathrm{FS}}, \qquad \omega_{\mathrm{FS}} + \mathrm{dd^c} u = [w = 0].$$

On $U_1 = \{z_1 \neq 0\}$, with $\zeta = z_0/z_1$,

$$u = -\log(1 + |\zeta|^2),$$

which is smooth and satisfies $\mathrm{dd^c} u = -\omega_{\mathrm{FS}}$. The divisor $\{z_1 = 0\}$ does not meet this chart, so the same current identity holds there with zero right-hand side. The two calculations agree on the overlap and prove globally $\omega_{\mathrm{FS}} + \mathrm{dd^c} u = [z_1 = 0]$. ◇

Exercise 2.29: Envelope in the disc

Solution. The answer is the continuous radial function

$$h^*_{K,\mathbb{D}}(z) = \begin{cases} -1, & |z| \le r, \\ \dfrac{\log|z|}{-\log r}, & r < |z| < 1. \end{cases}$$

The second branch is harmonic on the annulus, equals -1 at $|z| = r$, and tends to 0 at $|z| = 1$. The outward radial derivative jumps upward at $|z| = r$, so the piecewise function is subharmonic. It is therefore an admissible competitor.

Conversely, let v be admissible. On the inner boundary of the annulus it is at most -1, and throughout the disc it is at most 0, hence its boundary limsup at $|z| = 1$ is at most 0. Subtracting the displayed harmonic branch and applying the maximum principle directly to v on $r < |z| < 1$ gives the required pointwise upper bound. Hence the displayed function is the upper-semicontinuous regularization of the envelope. ◇

Lecture 7

Exercise 2.42: Ball and polydisc

Solution. On the unit ball $B \subset \mathbb{C}^n$,

$$\Phi_B(z) = -\log(1 - |z|^2)$$

is smooth, tends to $+\infty$ at the boundary, and has Levi matrix

$$\frac{\delta_{jk}}{1 - |z|^2} + \frac{\bar{z}_j z_k}{(1 - |z|^2)^2},$$

which is positive definite. Thus it is a strictly psh exhaustion.

On the polydisc $\mathbb{D}^n$, one convenient smooth choice is

$$\Phi_P(z) = -\sum_{j=1}^{n} \log(1 - |z_j|^2).$$

Its Levi matrix is diagonal with strictly positive entries $(1 - |z_j|^2)^{-2}$, and it is exhaustive because at least one term tends to $+\infty$ at the boundary. A more geometric but nonsmooth candidate is $\max_j\{-\log(1 - |z_j|^2)\}$. Replacing the maximum by the regularized maximum M_ε from Lecture 6 gives a smooth psh function within a bounded error; adding a small multiple of $|z|^2$ makes strict positivity manifest if necessary. ◇

Exercise 2.43: Changing a defining function

Solution. Let $\widetilde{\rho} = a\rho$, where $a > 0$ near the boundary. The product rule gives

$$\partial\bar{\partial}(a\rho) = a\,\partial\bar{\partial}\rho + \rho\,\partial\bar{\partial}a + \partial a \wedge \bar{\partial}\rho + \partial\rho \wedge \bar{\partial}a.$$

At a boundary point x, $\rho(x) = 0$. If $\xi \in T^{1,0}_x(\partial\Omega)$, then $\partial\rho(\xi) = 0$, and its conjugate also vanishes. Evaluating the preceding identity on $(\xi, \bar{\xi})$ therefore eliminates the last three terms and gives

$$\mathcal{L}_{a\rho,x}(\xi) = a(x)\mathcal{L}_{\rho,x}(\xi).$$

Thus the sign of the boundary Levi form is independent of a positive change of defining function. ◇

Lecture 8

Exercise 3.18: Open Riemann surfaces

Solution. The Behnke–Stein theorem states that every connected noncompact Riemann surface is a Stein manifold. One formulation of its proof constructs a strictly subharmonic exhaustion by successively enlarging Runge relatively compact domains; the Levi problem then identifies the resulting surface as Stein.

The boundary Levi form is tested on the complex tangent space $T^{1,0}(\partial\Omega)$, which has complex dimension $n - 1$. For $n = 1$ this space is zero, so Levi semipositivity has no nonzero direction on which it could fail. Thus every smoothly bounded domain in a Riemann surface is pseudoconvex in the boundary Levi sense. The theorem is global: it supplies the required holomorphic separation and exhaustion for the whole open surface. ◇

Exercise 3.19: Closed complex submanifolds

Solution. Let $\Phi : X \to \mathbb{R}$ be a smooth strictly psh exhaustion. Since Y is a complex submanifold, the Levi form of $\Phi|_Y$ is the restriction of the positive definite Levi form of Φ, hence is positive definite. Since Y is closed,

$$\{y \in Y : \Phi(y) \le c\} = Y \cap \{x \in X : \Phi(x) \le c\}$$

is closed in a compact set and therefore compact. Thus $\Phi|_Y$ is a smooth strictly psh exhaustion. The solution of the Levi problem says that a complex manifold admitting such an exhaustion is Stein, so Y is Stein. ◇

Exercise 3.21: Polynomial convexity of the ball

Solution. Let $K = \overline{B}$ and let $a \notin K$, so $|a| > 1$. The holomorphic linear polynomial

$$p_a(z) = \sum_{j=1}^{n} z_j \bar{a}_j$$

satisfies $p_a(a) = |a|^2$, whereas Cauchy–Schwarz gives $\sup_{z \in K} |p_a(z)| \le |a|$. Since $|a|^2 > |a|$, the point a is not in the polynomial hull of K. Every point outside K can be separated in this way, so the polynomial hull is exactly K; hence the closed unit ball is polynomially convex. ◇

Lecture 9

Exercise 2.60: Three levels of one function

Solution. On the disc $D(0, r)$ the section is the restriction

$$f|_{D(0,r)} : D(0, r) \longrightarrow \mathbb{C}, \qquad z \longmapsto z^3 + 2z.$$

As a section, it records the value of the function at every point of the chosen disc. Passing to the germ at 0 forgets its values away from an unspecified neighborhood of 0 and retains only its equivalence class in $\mathcal{O}_{\mathbb{C},0} = \mathbb{C}\{z\}$:

$$f_0 = z^3 + 2z.$$

In this particular example the restriction to any disc determines the original entire function by the identity theorem, but that recovery uses analytic continuation; it is not part of the definition of a germ.

The maximal ideal is $\mathfrak{m}_0 = (z)$. Reduction modulo its successive powers gives

$$[f_0]_{\mathfrak{m}_0} = 0, \qquad [f_0]_{\mathfrak{m}_0^2} = 2z, \qquad [f_0]_{\mathfrak{m}_0^4} = 2z + z^3.$$

The quotient by $\mathfrak{m}_0$ keeps only the value $f(0)$; the quotient by $\mathfrak{m}_0^2$ keeps the value and the first derivative, equivalently the first jet; and the quotient by $\mathfrak{m}_0^4$ keeps the Taylor polynomial through degree three. For the present cubic the last quotient loses no Taylor coefficient, whereas for a general germ it discards every term of order at least four. ◇

Exercise 2.57: The exponential sequence

Solution. The map $\mathbb{Z} \to \mathcal{O}_X$ sends an integer to the corresponding locally constant holomorphic function and is injective. If $\exp(2\pi i f) = 1$, then the holomorphic function f takes values in the discrete set $\mathbb{Z}$, so it is locally constant and integer-valued. Thus the kernel at $\mathcal{O}_X$ is precisely $\mathbb{Z}$.

It remains to prove surjectivity onto $\mathcal{O}_X^*$ as a sheaf. If g is nonvanishing near x, shrink to a simply connected coordinate ball on which g has no zero. Then g'/g has a holomorphic primitive in one variable, and the analogous closed holomorphic one-form dg/g has a primitive in several variables. Hence there is a holomorphic logarithm h with $e^h = g$. Taking $f = h/(2\pi\mathrm{i})$ gives $g = e^{2\pi\mathrm{i}f}$. Exactness is stalkwise, so this proves exactness of the sheaf sequence. ◇

Exercise 2.58: Nakayama at a stalk

Solution. Choose classes $\bar{s}_1, \dots, \bar{s}_r$ forming a basis of $\mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x$, and lift them to germs $s_i \in \mathcal{F}_x$. Nakayama's lemma says that the s_i generate the finite $\mathcal{O}_{X,x}$-module $\mathcal{F}_x$. Represent the germs by sections on a common neighborhood U and form

$$\mathcal{O}_U^{\oplus r} \longrightarrow \mathcal{F}|_U, \qquad (a_i) \longmapsto \sum_i a_i s_i.$$

Its cokernel $\mathcal{G}$ is coherent. Since $\mathcal{G}_x = 0$, the support of $\mathcal{G}$, a closed analytic subset of U, does not contain x. After shrinking U, the cokernel vanishes. Thus the lifted sections generate $\mathcal{F}$ throughout a neighborhood of x. ◇

Exercise 3.48: Divisor line bundle

Solution. Let D be represented on U_i by nonzero meromorphic functions f_i, with $g_{ij} = f_i/f_j \in \mathcal{O}^*(U_i \cap U_j)$. The identities $g_{ij}g_{jk} = g_{ik}$ glue the trivial line bundles. To fix conventions, choose frames e_i satisfying

$$e_j = g_{ij}e_i = \frac{f_i}{f_j}e_i.$$

Then $f_i e_i = f_j e_j$ on overlaps, so these local expressions define a canonical meromorphic section s_D. Its local coefficient is f_i, hence $\operatorname{div}(s_D) = D$.

If $D = \operatorname{div}(f)$ is principal, take $f_i = f|_{U_i}$. Then every $g_{ij} = 1$, so the gluing is the trivial line bundle, and s_D is the global meromorphic function f times its constant frame. Changing the local equations by units changes frames but not the isomorphism class. ◇

Lecture 10

Exercise 3.32: Node and cusp

Solution. The node $N = \{xy = 0\}$ is the union of the two coordinate axes, so it has two irreducible components. Since $d(xy)_0 = 0$, its Zariski tangent space is all of $\mathbb{C}^2$. Its normalization is the disjoint union $\mathbb{C} \sqcup \mathbb{C}$, with maps $t \mapsto (t, 0)$ and $s \mapsto (0, s)$; the two preimages of the origin record the two branches.

The cusp $C = \{y^2 = x^3\}$ is irreducible: its local equation has the primitive parametrization $\nu(t) = (t^2, t^3)$, and the corresponding subring $\mathbb{C}\{t^2, t^3\} \subset \mathbb{C}\{t\}$ is a domain. Again the differential of the defining equation vanishes at the origin, so $T_0C = \mathbb{C}^2$. Its normalization is the single copy $\mathbb{C} \to C$ just displayed. It is bijective near zero but not an isomorphism because $t = y/x$ is integral over $\mathbb{C}\{t^2, t^3\}$ and does not belong to that ring. ◇

Exercise 3.33: A division calculation

Solution. Regard the expressions as polynomials in y with holomorphic coefficients in x. Since

$$(y^2 - x)(y^2 + x) = y^4 - x^2,$$

we have

$$y^4 + x = (y^2 - x)(y^2 + x) + (x^2 + x).$$

Thus the quotient is $Q = y^2 + x$ and the remainder is $R = x^2 + x$. The remainder has degree zero in y, which is strictly less than the degree two of the Weierstrass polynomial P; uniqueness follows from the division theorem. ◇

Exercise 3.35: A monomial intersection

Solution. Every convergent power series is congruent modulo (x^a, y^b) to a unique linear combination of the monomials

$$x^i y^j, \qquad 0 \le i < a, \quad 0 \le j < b.$$

They span because higher powers lie in the ideal, and they are independent because no nonzero combination of them belongs to the monomial ideal. Hence

$$\dim_{\mathbb{C}} \mathbb{C}\{x, y\}/(x^a, y^b) = ab.$$

The curves $x^a = 0$ and $y^b = 0$ meet only at the origin, and the length of this local quotient is their local intersection multiplicity. Thus $i_0(x^a, y^b) = ab$, including the scheme-theoretic multiplicities of both axes. ◇

Lecture 11

Exercise 3.44: A non-Cartier Weil divisor

Solution. The hypersurface X has dimension three, while $A = \{z_1 = z_3 = 0\}$ is a two-dimensional irreducible plane contained in X. Therefore A is a prime Weil divisor.

At the origin let

$$R = \mathbb{C}\{z_1, z_2, z_3, z_4\}/(z_1 z_2 + z_3 z_4), \qquad I = (z_1, z_3) \subset R.$$

If A were Cartier, I would be principal in the local ring R. But the classes of z_1, z_3 are linearly independent in $I/\mathfrak{m}I$: the defining relation and every element of $\mathfrak{m}I$ have order at least two, so they impose no relation on these linear terms. Hence $\dim_{\mathbb{C}} I/\mathfrak{m}I = 2$. Nakayama's lemma says that I needs at least two generators and is not principal. Thus A is not Cartier at the origin. ◇

Exercise 3.45: Indeterminacy

Solution. As a meromorphic map to $\mathbb{P}^1$, the function is represented by $[z_2 : z_1]$. Its zero set is $\{z_2 = 0, z_1 \neq 0\}$, its pole set is $\{z_1 = 0, z_2 \neq 0\}$, and its indeterminacy set is the common zero set of the two homogeneous coordinates, namely $\{(0, 0)\}$.

The origin cannot be filled in continuously, much less holomorphically. Along $z_2 = 0$ the value is 0; along $z_1 = 0$ it is ∞; and along $z_2 = az_1$ it is the arbitrary value a. Thus neither 0, nor ∞, nor any finite value is compatible with all approaches. ◇

Exercise 3.47: A normal singularity

Solution. Let $F = z_1 z_2 - z_3 z_4$. Its gradient is

$$(z_2, z_1, -z_4, -z_3),$$

so the hypersurface is singular precisely at the origin. The cone has complex dimension three, so its singular locus has codimension three. It is therefore regular in codimension one, which is Serre's condition (R_1).

Every hypersurface local ring is Cohen–Macaulay, hence satisfies (S_k) for all k up to its dimension, in particular (S_2). The polynomial F is reduced, and Serre's criterion $(R_1) + (S_2)$ now proves that the cone is normal. It is nevertheless singular at the origin because $dF_0 = 0$, so its Zariski tangent space has dimension four instead of three. ◇

Lecture 12

Exercise 3.58: Area of $\mathbb{P}^1$

Solution. On the affine chart $w = z_1/z_0$,

$$\omega_{\mathrm{FS}} = \frac{\mathrm{i}}{2\pi}\frac{\mathrm{d}w \wedge \mathrm{d}\bar{w}}{(1+|w|^2)^2}.$$

Since $\mathrm{i}\mathrm{d}w \wedge \mathrm{d}\bar{w} = 2r\mathrm{d}r \wedge \mathrm{d}\theta$,

$$\int_{\mathbb{P}^1} \omega_{\mathrm{FS}} = \frac{1}{\pi}\int_0^{2\pi}\int_0^{\infty} \frac{r}{(1+r^2)^2}\,\mathrm{d}r\,\mathrm{d}\theta = 1.$$

In the coordinate $\zeta = 1/w$, the form becomes $\frac{\mathrm{i}}{2\pi}(1+|\zeta|^2)^{-2}\mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta}$, so it is smooth at infinity and no mass is omitted. ◇

Exercise 3.59: Products

Solution. Put $\omega = \mathrm{pr}_1^*\,\omega_1 + \mathrm{pr}_2^*\,\omega_2$. It is real of type $(1,1)$ and $\mathrm{d}\omega = 0$. For $(v_1, v_2) \neq 0$,

$$\omega((v_1,v_2), J(v_1,v_2)) = \omega_1(v_1, J_1 v_1) + \omega_2(v_2, J_2 v_2) > 0,$$

because at least one component is nonzero. Hence ω is Kähler. ◇

Exercise 3.61: Transition between blow-up charts

Solution. In the first chart write $(x,y) = (u, ut)$; in the second write $(x,y) = (sv, v)$. On the overlap $u, v \neq 0$,

$$v = ut, \qquad s = t^{-1}, \qquad u = sv.$$

The exceptional divisor is $u = 0$ in the first chart and $v = 0$ in the second, with coordinate transition $s = 1/t$, so $E \simeq \mathbb{P}^1$.

The normal coordinates satisfy $v = tu$. Thus the transition of the normal line in the two standard charts is multiplication by t, the transition function of the tautological line bundle $\mathcal{O}_{\mathbb{P}^1}(-1)$ in these frame conventions. Equivalently, the conormal bundle I_E/I_E^2 has transition t^{-1} and is $\mathcal{O}(1)$. Therefore $N_{E/\mathrm{Bl}_0\,\mathbb{C}^2} \simeq \mathcal{O}_{\mathbb{P}^1}(-1)$. ◇

Exercise 3.62: Three concurrent lines

Solution. After a linear change of coordinates, assume none of the three lines is vertical and write them as $y = m_i x$ with distinct m_i. In the blow-up chart $(x,y) = (u, ut)$, the total transform has equation $u(t - m_i) = 0$; its strict transform is $t = m_i$. These three curves are disjoint because the constants m_i are distinct. Each meets $E = \{u = 0\}$ at $(0, m_i)$, giving three different points. The local equations $u = 0$ and $t - m_i = 0$ have independent differentials, so the intersection is transverse. A vertical line is treated in the other chart and corresponds to the point at infinity of E. ◇

Exercise 3.60: Canonical divisor of a blow-up

Solution. Choose coordinates in which the smooth center is $Z = \{z_1 = \cdots = z_r = 0\}$. In the blow-up chart corresponding to z_1, write

$$z_1 = u, \qquad z_j = uv_j \ (2 \le j \le r), \qquad z_k = z_k \ (k > r).$$

Then

$$\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n = u^{r-1}\mathrm{d}u \wedge \mathrm{d}v_2 \wedge \cdots \wedge \mathrm{d}v_r \wedge \mathrm{d}z_{r+1} \wedge \cdots \wedge \mathrm{d}z_n.$$

The Jacobian therefore vanishes to order $r-1$ along $E = \{u = 0\}$. The same calculation in every chart glues, and the transformation rule for canonical forms gives

$$K_{\widetilde{X}} = \pi^* K_X + (r-1)E.$$

No other divisor occurs because the blow-up is an isomorphism away from E. ◇

Lecture 13

Exercise 4.5: The circle

Solution. Use the coordinate $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ and the standard metric. On functions, $\Delta f = -f''$, so periodic harmonic functions are constants. On one-forms, $\Delta(a(\theta)\mathrm{d}\theta) = -a''(\theta)\mathrm{d}\theta$, so harmonic one-forms are the constant multiples of $\mathrm{d}\theta$.

The decomposition is explicit. For a function f, write $f = \bar{f} + (f - \bar{f})$. The zero-mean part is coexact because one can choose a periodic b with $-b' = f - \bar{f}$, giving $f - \bar{f} = d^*(b\mathrm{d}\theta)$. For a one-form $a\mathrm{d}\theta$, let $\bar{a} = (2\pi)^{-1}\int_0^{2\pi} a\, d\theta$. The zero-mean function $a - \bar{a}$ has a periodic primitive F, and

$$a\mathrm{d}\theta = \bar{a}\mathrm{d}\theta + dF.$$

These summands are orthogonal, verifying Hodge decomposition in degrees zero and one. ◇

Exercise 4.6: Minimum norm

Solution. Let h be the harmonic representative of a class and let α be any other closed representative. Then $\alpha = h + d\beta$. Since $d^*h = 0$,

$$(h, d\beta) = (d^*h, \beta) = 0.$$

Therefore

$$\|\alpha\|_{L^2}^2 = \|h\|_{L^2}^2 + \|d\beta\|_{L^2}^2 \ge \|h\|_{L^2}^2.$$

Equality forces $d\beta = 0$, hence $\alpha = h$. Thus the harmonic representative is the unique minimizer, not merely a critical point. ◇

Exercise 4.7: Heat equation on the circle

Solution. For $\Delta = -\partial_\theta^2$, the orthonormal eigenfunctions are $(2\pi)^{-1/2}e^{ik\theta}$ with eigenvalues k^2. Hence

$$K_t(\theta, \phi) = \frac{1}{2\pi}\sum_{k\in\mathbb{Z}} e^{-k^2 t}e^{ik(\theta-\phi)}.$$

Termwise differentiation is valid for $t > 0$ and verifies $(\partial_t + \Delta_\theta)K_t = 0$. For $f(\phi) = \sum_k \hat{f}_k e^{ik\phi}$,

$$(e^{-t\Delta}f)(\theta) = \sum_{k\in\mathbb{Z}} e^{-k^2 t}\hat{f}_k e^{ik\theta}.$$

As $t \to \infty$, every term with $k \neq 0$ tends to zero, while the $k = 0$ term is $\hat{f}_0 = (2\pi)^{-1} \int f$. Thus the heat operator converges in L^2, and smoothly for smooth f, to the orthogonal projection onto the constant functions. ◇

Lecture 14

Exercise 4.16: Evenness of b_1

Solution. On a compact Kähler manifold, $\Delta_{\mathrm{d}} = 2\Delta_{\bar\partial}$, and the Laplacian preserves bidegree. Hence every complex harmonic one-form splits uniquely as $\alpha = \alpha^{1,0} + \alpha^{0,1}$, with both components harmonic. Therefore

$$\mathcal{H}^1(X, \mathbb{C}) = \mathcal{H}^{1,0}(X) \oplus \mathcal{H}^{0,1}(X).$$

Complex conjugation is an antilinear isomorphism $\mathcal{H}^{1,0} \to \mathcal{H}^{0,1}$, so the two spaces have equal complex dimension. By the Hodge theorem,

$$b_1 = \dim_{\mathbb{C}} \mathcal{H}^1(X, \mathbb{C}) = h^{1,0} + h^{0,1} = 2h^{1,0},$$

which is even. ◇

Exercise 4.17: A Kähler form is not exact

Solution. Suppose $\omega = d\eta$ on a compact Kähler n-fold. Since $d\omega = 0$,

$$\omega^n = d(\eta \wedge \omega^{n-1}).$$

Stokes' theorem would imply $\int_X \omega^n = 0$. But $\omega^n/n!$ is the positive Riemannian volume form, so its integral is strictly positive. This contradiction proves that $[\omega] \neq 0$ in de Rham cohomology. ◇

Exercise 4.19: A formal algebra

Solution. Let Ω be a volume form on S^n normalized to have integral one. If n is odd, the minimal algebra is $A = (\Lambda(x_n), d = 0)$; graded commutativity already gives $x^2 = 0$. The map $x \mapsto \Omega$ induces the isomorphism in degrees $0, n$, so it is a quasi-isomorphism $A \to \mathcal{A}^*(S^n)$.

If n is even, take

$$A = (\Lambda(x_n, y_{2n-1}), d), \qquad dx = 0, \quad dy = x^2.$$

The extra generator kills the unwanted square of x. Define $x \mapsto \Omega$ and $y \mapsto 0$. This respects the differential because $\Omega \wedge \Omega = 0$ on the n-manifold. The cohomology of A is one dimensional in degrees $0, n$ and zero otherwise, so the map is again a quasi-isomorphism. Composing with the projection to cohomology proves formality of the de Rham algebra of every sphere. ◇

Lecture 15

Exercise 4.27: Exponential sequence and the Picard group

Solution. The long exact sequence associated with $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^* \to 1$ contains

$$H^1(X, \mathbb{Z}) \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow H^1(X, \mathcal{O}_X^*) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \longrightarrow H^2(X, \mathcal{O}_X).$$

Since $H^1(X, \mathcal{O}_X^*) = \mathrm{Pic}(X)$, this is the requested segment. The kernel of c_1 consists of topologically trivial holomorphic line bundles and is the image of $H^1(X, \mathcal{O}_X)$. Therefore

$$\mathrm{Pic}^0(X) = \ker c_1 \simeq H^1(X, \mathcal{O}_X)/\mathrm{im}\, H^1(X, \mathbb{Z}),$$

with the quotient understood by the exact image of the integral group. On a compact Kähler manifold this quotient is a complex torus. ◇

Exercise 4.28: Explicit Čech–Dolbeault conversion

Solution. Let $\{\rho_k\}$ be a smooth partition of unity subordinate to the cover and use the additive cocycle convention $g_{jk} - g_{ik} + g_{ij} = 0$. On U_i, define the smooth function

$$b_i = \sum_k \rho_k g_{ik},$$

where a summand is used only on the overlap on which it is defined. On $U_i \cap U_j$, the cocycle identity gives

$$b_i - b_j = \sum_k \rho_k (g_{ik} - g_{jk}) = \sum_k \rho_k g_{ij} = g_{ij}.$$

Because g_{ij} is holomorphic, $\bar\partial b_i = \bar\partial b_j$ on overlaps. Thus the local forms glue to a global $(0,1)$-form α, defined by $\alpha|_{U_i} = \bar\partial b_i$. It is $\bar\partial$-closed. Changing the partition or replacing g by a Čech coboundary changes α by a global $\bar\partial$-exact form. Hence $[\alpha]_{\bar\partial}$ is exactly the Dolbeault class corresponding to $[g]$. ◇

Exercise 4.30: A two-term complex

Solution. The complex $\mathcal{K}^\bullet = [\mathcal{F} \xrightarrow{u} \mathcal{G}]$ fits into a distinguished triangle whose associated long exact sequence begins

$$0 \longrightarrow \mathbb{H}^0(X, \mathcal{K}^\bullet) \longrightarrow H^0(X, \mathcal{F}) \xrightarrow{H^0(u)} H^0(X, \mathcal{G}) \longrightarrow \mathbb{H}^1(X, \mathcal{K}^\bullet) \longrightarrow H^1(X, \mathcal{F})$$

$$\xrightarrow{H^1(u)} H^1(X, \mathcal{G}) \longrightarrow \mathbb{H}^2(X, \mathcal{K}^\bullet) \longrightarrow H^2(X, \mathcal{F}) \longrightarrow \cdots.$$

In particular, $\mathbb{H}^0 = \ker(H^0(\mathcal{F}) \to H^0(\mathcal{G}))$, while $\mathbb{H}^1$ is an extension of the cokernel in degree zero by the kernel in degree one. This displays both the sections and the obstruction created by the map u. ◇

Lecture 16

Exercise 4.36: Line bundles of high degree

Solution. Serre duality gives

$$H^1(X, L)^* \simeq H^0(X, K_X \otimes L^{-1}).$$

The latter bundle has degree $2g - 2 - \deg L < 0$. A nonzero holomorphic section would have an effective zero divisor of that negative degree, which is impossible. Hence both $H^0(K_X \otimes L^{-1})$ and $H^1(X, L)$ vanish. Riemann–Roch now yields

$$h^0(L) = h^0(L) - h^1(L) = \deg L + 1 - g.$$

◇

Exercise 4.37: Euler characteristic

Solution. For any finite-dimensional cochain complex, the alternating sum of the dimensions of the cochain spaces equals the alternating sum of the dimensions of its cohomology. Apply this on each page of the Frölicher spectral sequence, with total degree $p+q$. Passing from E_r to $E_{r+1} = H(E_r, d_r)$ does not change the total alternating sum. Therefore

$$\sum_{p,q} (-1)^{p+q} \dim E_1^{p,q} = \sum_{p,q} (-1)^{p+q} \dim E_\infty^{p,q}.$$

The left side is $\sum_{p,q} (-1)^{p+q} h^{p,q}$. The E_∞-terms are the graded pieces of a filtration of $H^{p+q}_{\mathrm{dR}}(X, \mathbb{C})$, so the right side is $\sum_k (-1)^k b_k = \chi_{\mathrm{top}}(X)$. Degeneration at E_1 is not needed because Euler characteristic is unchanged by every differential. ◇

Exercise 4.40: Degree-zero line bundles

Solution. Fix the base point p_0. For a holomorphic one-form ω, the Abel–Jacobi image of $p - q$ is

$$\int_q^p \omega \mod \text{periods}.$$

Choose paths from p_0 to p and q. Concatenating the first with the reverse of the second gives a path from q to p, so

$$\int_q^p \omega = \int_{p_0}^p \omega - \int_{p_0}^q \omega \mod H_1(X, \mathbb{Z}).$$

Changing any path adds a period and hence does not change the Jacobian point. Thus the class of the divisor $p - q$ is exactly $AJ(p) - AJ(q)$. ◇

Lecture 17

Exercise 5.7: Direct sum, tensor product, and dual

Solution. For sections s of E, t of F, and λ of E^*, define

$$D_{E\oplus F}(s,t) = (D_E s, D_F t),$$

$$D_{E\otimes F}(s \otimes t) = D_E s \otimes t + s \otimes D_F t,$$

and characterize the dual connection by

$$\mathrm{d}(\lambda(s)) = (D_{E^*}\lambda)(s) + \lambda(D_E s).$$

Squaring and using the graded Leibniz rule, the two mixed tensor-product terms cancel. Thus

$$\Theta_{E\oplus F} = \Theta_E \oplus \Theta_F, \qquad \Theta_{E\otimes F} = \Theta_E \otimes \mathrm{id}_F + id_E \otimes \Theta_F.$$

For the dual, applying D^2 to the scalar pairing gives

$$(\Theta_{E^*}\lambda)(s) = -\lambda(\Theta_E s).$$

In a frame this is $\Theta_{E^*} = -{}^t\Theta_E$. ◇

Exercise 5.8: Transition functions of a line bundle

Solution. Refine the cover so that each $g_{\alpha\beta}$ has a holomorphic logarithm and write

$$g_{\alpha\beta} = \exp(2\pi \mathrm{i} f_{\alpha\beta}).$$

On a triple overlap, the multiplicative cocycle identity implies

$$n_{\alpha\beta\gamma} = f_{\alpha\beta} + f_{\beta\gamma} + f_{\gamma\alpha} \in \mathbb{Z}.$$

The integers form a Čech two-cocycle. Changing the logarithms changes n by an integral coboundary, so $[n] \in H^2(X, \mathbb{Z})$ is well defined. It is exactly the connecting homomorphism

$$H^1(X, \mathcal{O}_X^*) \longrightarrow H^2(X, \mathbb{Z})$$

in the exponential sequence.

This connecting class is the topological first Chern class $c_1(L)$. To compare with Chern–Weil theory, choose a Hermitian metric and local connection forms. Their differences on overlaps are $g_{\alpha\beta}^{-1} dg_{\alpha\beta}$; the normalized curvature $(\mathrm{i}/2\pi)\Theta$ is global, and the Čech–de Rham comparison sends its de Rham class to the same integer cocycle n. Thus both constructions give $c_1(L)$. ⋄

Exercise 5.9: Flat line bundles on a circle

Solution. Since $\pi_1(S^1) = \mathbb{Z}$, a rank-one complex representation is determined by the image $\lambda \in \mathbb{C}^*$ of 1. The associated flat line bundle is

$$(\mathbb{R} \times \mathbb{C})/((t+1, v) \sim (t, \lambda v)).$$

Parallel transport once around the circle multiplies a fiber by λ, and a flat bundle with connection is recovered from this monodromy. Since $\mathbb{C}^*$ is abelian, conjugacy does not further identify different λ's.

A parallel Hermitian norm must satisfy $|\lambda v| = |v|$, so necessarily $|\lambda| = 1$. Conversely, when $|\lambda| = 1$, the standard norm on $\mathbb{R} \times \mathbb{C}$ is invariant under the gluing and descends to a parallel Hermitian metric. Thus precisely the unitary monodromies admit such a metric. ⋄

Lecture 18

Exercise 5.16: Curvature of $\mathcal{O}(1)$

Solution. On U_0, write a line as $[1 : w_1 : \cdots : w_n]$. The tautological frame $e = (1, w)$ of $\mathcal{O}(-1)$ has squared norm

$$|e|^2 = 1 + |w|^2.$$

The dual frame e^* of $\mathcal{O}(1)$ therefore has

$$|e^*|^2 = (1 + |w|^2)^{-1} = e^{-\varphi}, \qquad \varphi = \log(1 + |w|^2).$$

The line-bundle curvature formula gives

$$c_1(\mathcal{O}(1), h_{\mathrm{FS}}) = \mathrm{d}\mathrm{d}^{\mathrm{c}}\varphi = \frac{\mathrm{i}}{2\pi} \partial\bar{\partial} \log(1 + |w|^2) = \omega_{\mathrm{FS}}.$$

The potentials differ by logarithms of transition functions on overlaps, so their $\mathrm{d}\mathrm{d}^{\mathrm{c}}$'s glue globally. ⋄

Exercise 5.17: A curved metric on the trivial line bundle

Solution. Here $|1|_h^2 = e^{-\varphi}$ with $\varphi = |z|^2$. Hence

$$\Theta(L,h) = \partial\bar\partial|z|^2 = \sum_{j=1}^{n} \mathrm{d}z_j \wedge \mathrm{d}\bar z_j, \qquad c_1(L,h) = \mathrm{dd^c}|z|^2.$$

This curvature is nonzero and positive. Nevertheless the underlying bundle has the global nowhere-vanishing frame 1, so it is topologically and holomorphically trivial. There is no contradiction: topology fixes only the cohomology class of the curvature. The displayed Chern form is exact, so its de Rham class is zero even though the chosen connection is not flat. ◇

Exercise 5.19: Projective space

Solution. Put $\omega = [\omega_{\mathrm{FS}}]$, so $\int_{\mathbb{P}^1} \omega = 1$. In complex dimension one,

$$\mathrm{ch}(\mathcal{O}(k)) = e^{k\omega} = 1 + k\omega, \qquad \mathrm{td}(T_{\mathbb{P}^1}) = 1 + \frac{1}{2}c_1(T_{\mathbb{P}^1}) = 1 + \omega.$$

Riemann–Roch takes the degree-two component of the product:

$$\chi(\mathbb{P}^1, \mathcal{O}(k)) = \int_{\mathbb{P}^1} (1 + k\omega)(1 + \omega) = \int_{\mathbb{P}^1} (k+1)\omega = k + 1.$$

The formula holds for every integer k, including the range in which H^1 rather than H^0 contributes. ◇

Lecture 19

Exercise 5.24: The quotient line bundle

Solution. At $[\ell] \in \mathbb{P}^n$, restriction of linear functionals gives a quotient

$$(\mathbb{C}^{n+1})^* \times \mathbb{P}^n \longrightarrow \ell^* = \mathcal{O}(1)_{[\ell]}.$$

Equip the trivial bundle with its flat Hermitian metric and $\mathcal{O}(1)$ with the quotient metric. The ambient curvature is zero. The quotient curvature formula therefore says that, for a tangent vector ξ and a quotient vector q,

$$\langle \mathrm{i}\Theta_{\mathcal{O}(1)}(\xi,\bar\xi)q, q\rangle = \|\beta_\xi^* q\|^2 \geq 0.$$

If $\xi \neq 0$ and $q \neq 0$, variation of the line ℓ makes the second fundamental form nonzero, so the inequality is strict. In the affine frame this quotient metric is $(1+|w|^2)^{-1}$; its curvature is exactly ω_{FS}. Thus the quotient formula recovers positivity of $\mathcal{O}(1)$. ◇

Exercise 5.25: Tensor-product curvature

Solution. The tensor-product connection is $D(s \otimes t) = D_E s \otimes t + s \otimes D_F t$. Applying D again, the two mixed terms occur with opposite signs because $D_E s$ has degree one. They cancel, leaving

$$\Theta(E \otimes F) = \Theta(E) \otimes \mathrm{id}_F + id_E \otimes \Theta(F).$$

For a fixed tangent vector ξ, Griffiths semipositivity says that the Hermitian endomorphisms $A_E = \mathrm{i}\Theta_E(\xi,\bar\xi)$ and $A_F = \mathrm{i}\Theta_F(\xi,\bar\xi)$ are positive semidefinite. On $E \otimes F$, the curvature endomorphism is the Kronecker sum $A_E \otimes I + I \otimes A_F$, whose eigenvalues are all sums $\lambda_i(A_E) + \lambda_j(A_F) \geq 0$. Hence tensor products preserve Griffiths semipositivity. If one factor is strictly positive and the other semipositive, the tensor product is strictly Griffiths positive. ◇

Lecture 20

Exercise 5.32: The Veronese embedding

Solution. A basis of $H^0(\mathbb{P}^n, O(k))$ is formed by the monomials $z^\alpha = z_0^{\alpha_0} \cdots z_n^{\alpha_n}$ with $|\alpha| = k$. Thus the Kodaira map is

$$\nu_k([z_0 : \cdots : z_n]) = [z^\alpha]_{|\alpha|=k} \in \mathbb{P}^{\binom{n+k}{k}-1}.$$

Scaling z by λ scales every coordinate by λ^k, so the map is well defined. On the chart $z_i \neq 0$, divide all target coordinates by z_i^k. Among the resulting coordinate ratios are $z_i^{k-1} z_j / z_i^k = z_j / z_i$ for every $j \neq i$. Hence the original affine coordinates can be recovered from the image. This proves injectivity locally and shows that the differential has full rank. The projective map is therefore an injective immersion; compactness of $\mathbb{P}^n$ makes it an embedding. It is precisely the degree-k Veronese embedding. ◇

Exercise 5.33: Integral Kähler classes

Solution. The relevant part of the exponential long exact sequence is

$$H^1(X, O_X^*) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \longrightarrow H^2(X, O_X).$$

The first group is $\mathrm{Pic}(X)$. Exactness says that an integral class is the first Chern class of a holomorphic line bundle exactly when its image in $H^2(X, O_X)$ vanishes.

On a compact Kähler manifold, the map to $H^2(X, O_X) \simeq H^{0,2}(X)$ takes the $(0,2)$-component of a de Rham class. A Kähler class is of type $(1,1)$, so this component is zero. Therefore an integral Kähler class lies in the image of c_1 and equals $c_1(L)$ for some holomorphic line bundle L. If h_0 is any smooth metric on L, the $\partial\bar\partial$-lemma writes $\omega - c_1(L, h_0) = \mathrm{d}\mathrm{d}^{\mathrm{c}}\psi$. The changed metric $h_0 e^{-\psi}$ then has curvature $c_1(L, h_0 e^{-\psi}) = \omega > 0$, so L is positive. ◇

Exercise 5.34: Bergman density of $\mathbb{P}^1$

Solution. The Fubini–Study metric and volume form are invariant under the transitive action of $\mathrm{PU}(2)$. The induced action on $H^0(\mathbb{P}^1, O(k))$ is unitary, so the basis-independent Bergman density ρ_k is invariant and therefore constant. Since $\int_{\mathbb{P}^1} \omega_{\mathrm{FS}} = 1$ and $h^0(\mathbb{P}^1, O(k)) = k + 1$,

$$\rho_k \int_{\mathbb{P}^1} \omega_{\mathrm{FS}} = \int_{\mathbb{P}^1} \rho_k \omega_{\mathrm{FS}} = k + 1.$$

Thus $\rho_k \equiv k + 1$ for the normalizations in the handout. ◇

Lecture 21

Exercise 6.9: The Heaviside distribution

Solution. For $\varphi \in C_c^\infty(\mathbb{R})$,

$$\langle H', \varphi \rangle = -\int_0^\infty \varphi'(x)\,\mathrm{d}x = \varphi(0).$$

Thus $H' = \delta_0$. A distribution is a degree-zero current in real dimension one, so this is exactly the sign in the current differential convention. ◇

Exercise 6.10: Divisor of a rational function

Solution. For $f = z_1^2 z_2/(z_1 - z_2)$,

$$\operatorname{div}(f) = 2\{z_1 = 0\} + \{z_2 = 0\} - \{z_1 - z_2 = 0\}.$$

Therefore

$$\mathrm{dd^c} \log \left| \frac{z_1^2 z_2}{z_1 - z_2} \right|^2 = 2[z_1 = 0] + [z_2 = 0] - [z_1 - z_2 = 0].$$

This is an identity of closed real $(1,1)$-currents on $\mathbb{C}^2$, with the logarithm locally integrable and $\mathrm{dd^c}$ interpreted distributionally. ◇

Exercise 6.17: Delta functions

Solution. A smooth function has empty wavefront set, so $\langle f\delta_0, \varphi\rangle = f(0)\varphi(0)$ is well defined. In contrast, $\mathrm{WF}(\delta_0)$ contains $(0,\xi)$ and $(0,-\xi)$ for every $\xi \neq 0$, so the product criterion fails for δ_0^2. If $\rho_\varepsilon(x) = \varepsilon^{-m}\rho(x/\varepsilon)$, then

$$\int \rho_\varepsilon^2 \varphi = \varepsilon^{-m} \int \rho(y)^2 \varphi(\varepsilon y)\, \mathrm{d}y,$$

which generally diverges like $\varepsilon^{-m}\varphi(0)\int \rho^2$. Hence no canonical distributional square is obtained by smoothing. ◇

Exercise 6.11: Matrix criterion

Solution. Write $\alpha = \mathrm{i}\sum_{j,k} a_{j\bar{k}}\, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$, where $A = (a_{j\bar{k}})$ is Hermitian. For $v \in \mathbb{C}^n$, restriction to the complex line spanned by v gives

$$\alpha(v, \bar{v}) = \sum_{j,k} a_{j\bar{k}} v_j \bar{v}_k = v^* A v.$$

Thus weak positivity is equivalent to $v^*Av \geq 0$ for every v, namely $A \geq 0$.

If $A \geq 0$, the spectral theorem gives $A = U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n) U$, with $\lambda_j \geq 0$. For the transformed $(1,0)$-forms η_j,

$$\alpha = \mathrm{i} \sum_j \lambda_j \eta_j \wedge \bar{\eta}_j,$$

which is strongly positive and hence positive. The general inclusions $\mathrm{SP}^1 \subset \mathrm{P}^1 \subset \mathrm{WP}^1$, together with the first paragraph, prove that all three notions coincide exactly with positive semidefiniteness of A. ◇

Exercise 6.12: Products of analytic sets

Solution. On the regular locus of $Z_1 \times Z_2$, Fubini's theorem gives, for every compactly supported test form Φ,

$$\langle [Z_1] \times [Z_2], \Phi \rangle = \int_{(Z_1)_{\mathrm{Reg}}} \int_{(Z_2)_{\mathrm{Reg}}} \Phi = \int_{(Z_1 \times Z_2)_{\mathrm{Reg}}} \Phi.$$

The singular subsets have measure zero, so this is $[Z_1 \times Z_2]$.

Let $\dim Z_i = p_i$, use the product metric $\omega = \pi_1^*\omega_1 + \pi_2^*\omega_2$, and take product compact sets $K_1 \times K_2$. In the expansion of $\omega^{p_1+p_2}$, only the term with degrees p_1, p_2 survives on the product. Hence

$$\frac{1}{(p_1+p_2)!} \int_{(Z_1 \cap K_1)\times(Z_2 \cap K_2)} \omega^{p_1+p_2} = \left(\frac{1}{p_1!} \int_{Z_1 \cap K_1} \omega_1^{p_1} \right) \left(\frac{1}{p_2!} \int_{Z_2 \cap K_2} \omega_2^{p_2} \right).$$

Thus the product mass is the product of the two masses. ◇

Exercise 6.14: Minimality

Solution. On a Kähler surface, ω is closed and has comass one. Wirtinger's theorem says that its restriction to every oriented real two-plane is at most the area form, with equality exactly on complex lines. A smooth complex curve C is therefore calibrated by ω.

If C_t is a compactly supported variation of a relatively compact part of the smooth locus, then $C_t - C$ is a boundary. Stokes' theorem gives $\int_{C_t} \omega = \int_C \omega$, while calibration gives $\mathrm{Area}(C_t) \geq \int_{C_t} \omega = \mathrm{Area}(C)$. Hence the first variation at $t = 0$ vanishes. The first-variation formula is

$$\left.\frac{d}{dt}\right|_0 \mathrm{Area}(C_t) = -\int_C \langle H, V\rangle \, \mathrm{d}A$$

for arbitrary compactly supported normal variation field V. Therefore $H = 0$ on the smooth locus. ⋄

Lecture 22

Exercise 6.24: Normalization and compactness

Solution. Both sequences have supremum zero on the unit ball in the limiting boundary sense. For $u_j = j^{-1} \log |z_1|$, local integrability of $\log |z_1|$ gives

$$\|u_j\|_{L^1(K)} = j^{-1} \, \|\log |z_1|\|_{L^1(K)} \longrightarrow 0$$

on every compact K. Thus $u_j \to 0$ in L^1_{loc}, and $\mathrm{dd}^{\mathrm{c}} u_j = (2j)^{-1}[z_1 = 0] \rightharpoonup 0$.

For $v_j = j \log |z_1|$, every point with $|z_1| < 1$ satisfies $v_j(z) \to -\infty$; on $z_1 = 0$ the value is already $-\infty$. Thus $v_j \to -\infty$ locally pointwise, and its local L^1-norm diverges. Also $\mathrm{dd}^{\mathrm{c}} v_j = (j/2)[z_1 = 0]$, whose mass is unbounded. The comparison shows why a supremum normalization alone does not give compactness without a fixed interior normalization or an exclusion of the $-\infty$ alternative. ⋄

Exercise 6.25: Currents supported at a point

Solution. Let T have bidegree (p, p), so its bidimension is $(n - p, n - p)$. If $p < n$, its bidimension is positive, while its support has complex dimension zero. The support theorem then gives $T = 0$.

If $p = n$, a positive (n, n)-current is a positive measure. A positive measure supported at the origin is $c\delta_0$ for a unique $c \geq 0$, and every top-degree current is automatically closed. Therefore the only nonzero possibilities are

$$p = n, \qquad T = c\delta_0 \quad (c > 0).$$

This also explains why one cannot place a positive closed area-type current at a single point in lower bidegree. ⋄

Exercise 6.27: Multiplicity in a limit

Solution. Choose one coordinate z transverse to the hypersurfaces. For $\varepsilon \neq 0$, the equation $z^m = \varepsilon$ has m distinct roots $a_1(\varepsilon), \ldots, a_m(\varepsilon)$, all tending to zero. Thus

$$[z^m = \varepsilon] = \sum_{j=1}^{m} [z = a_j(\varepsilon)].$$

If η is a compactly supported $(n - 1, n - 1)$-test form, continuity under translation gives

$$\langle [z = a_j(\varepsilon)], \eta \rangle = \int_{z = a_j(\varepsilon)} \eta \longrightarrow \int_{z=0} \eta.$$

Summing the m limits yields $\langle [z^m = \varepsilon], \eta\rangle \to m\langle [z = 0], \eta\rangle$. Hence the currents converge to $m[z = 0]$; the multiplicity survives even though the supports converge set-theoretically to one hyperplane. $\diamond$

Lecture 23

Exercise 6.34: Monge–Ampère of a maximum

Solution. Let $u(z) = \max(\log|z|, 0)$. It is harmonic on $|z| < 1$ and on $|z| > 1$. For a radial function, $\Delta u = r^{-1}(ru')'$. Here ru' jumps from 0 to 1 at $r = 1$, so distributionally

$$\Delta u = \frac{1}{r}\delta_{r=1}.$$

Since $\mathrm{d}\mathrm{d}^{\mathrm{c}} u = (4\pi)^{-1}\Delta u\, \mathrm{d}x \wedge \mathrm{d}y$,

$$\mathrm{d}\mathrm{d}^{\mathrm{c}} \max(\log|z|, 0) = \frac{\mathrm{d}\theta}{4\pi} \quad \text{on } |z| = 1.$$

All the mass is uniformly distributed on the unit circle, and the total mass is $1/2$. Had the exercise used $\log|z|^2$, the total mass would be 1, in accordance with Poincaré–Lelong. $\diamond$

Exercise 6.35: Comparison in one dimension

Solution. First assume u, v are smooth and $u \geq v$ near the boundary. For a regular value $\varepsilon > 0$, set $E_\varepsilon = \{u + \varepsilon < v\} \Subset \Omega$ and $w = v - u - \varepsilon$. Then $w > 0$ in E_ε, $w = 0$ on its boundary, and its outward normal derivative is nonpositive. Green's formula gives

$$\int_{E_\varepsilon} \mathrm{d}\mathrm{d}^{\mathrm{c}}(v - u) = \frac{1}{4\pi}\int_{\partial E_\varepsilon} \frac{\partial w}{\partial n}\,\mathrm{d}s \leq 0.$$

Therefore $\int_{E_\varepsilon} \mathrm{d}\mathrm{d}^{\mathrm{c}} v \leq \int_{E_\varepsilon} \mathrm{d}\mathrm{d}^{\mathrm{c}} u$. Approximate arbitrary levels by regular values and let $\varepsilon \downarrow 0$. For bounded subharmonic u, v, use decreasing smooth regularizations on relatively compact subdomains; weak convergence of their Riesz measures and monotone continuity then give

$$\int_{\{u<v\}} \mathrm{d}\mathrm{d}^{\mathrm{c}} v \leq \int_{\{u<v\}} \mathrm{d}\mathrm{d}^{\mathrm{c}} u.$$

This is precisely the local comparison principle for $n = 1$. $\diamond$

Exercise 6.36: Bounded potentials

Solution. Normalize u by $\sup_X u = 0$ and suppose $-M \leq u \leq 0$. For every integer $j > M$, the truncation $u_j = \max(u, -j)$ equals u. Thus the non-pluripolar product agrees with the ordinary Bedford–Taylor measure $(\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u)^n$. Approximation by smooth bounded potentials and Stokes' theorem give

$$\int_X (\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u)^n = \int_X \omega^n,$$

so $u \in \mathcal{E}(X, \omega)$. Finally,

$$\int_X |u|\,\mathrm{MA}(u) \leq M\int_X \omega^n < \infty.$$

Hence $u \in \mathcal{E}^1(X, \omega)$. $\diamond$

Lecture 24

Exercise 6.45: Lelong number of an integration current

Solution. At a smooth point $x \in A$ of complex dimension p, choose holomorphic coordinates in which A is tangent to $\mathbb{C}^p \times \{0\}$. After rescaling by r^{-1}, the submanifold converges in C^1 on compact sets to that plane. Wirtinger's formula then shows that its normalized mass ratio converges to the mass ratio of the plane, which is one by the normalization of the Lelong number. Hence $\nu([A], x) = 1$.

For the cusp $y^2 = x^3$, the normalization is $t \mapsto (t^2, t^3)$. A generic projection to the x-axis has degree two near the origin, so the multiplicity is two. Equivalently, the lowest homogeneous term of the equation is y^2, and the tangent cone is the x-axis counted twice. Therefore $\nu([A], 0) = 2$. ⋄

Exercise 6.46: A singularity with zero Lelong number

Solution. Write $t = \log r < 0$ and $\chi(t) = -(-t)^\alpha$. Then

$$\chi'(t) = \alpha(-t)^{\alpha-1} > 0, \qquad \chi''(t) = \alpha(1-\alpha)(-t)^{\alpha-2} > 0.$$

A radial function $\chi(\log|z|)$ is subharmonic on the punctured disc exactly when χ is convex; thus u is subharmonic there. Since $u \to -\infty$ at zero and is not identically $-\infty$, defining $u(0) = -\infty$ gives its natural subharmonic extension.

Let $s = -\log r \to \infty$. The logarithmic slope is

$$\frac{u(r)}{\log r^2} = \frac{-s^\alpha}{-2s} = \frac{1}{2}s^{\alpha-1} \longrightarrow 0.$$

Therefore $\nu(u, 0) = 0$, even though u is unbounded at the origin. ⋄

Exercise 6.48: A plane branch

Solution. Under the rescaling appropriate to the lowest order, the parametrization (t^m, t^k), with $m < k$, approaches the x-axis. A generic projection to the x-axis is $t \mapsto t^m$, an m-sheeted map away from the origin. Therefore the tangent cycle is

$$C_0 A = m[\{y = 0\}],$$

not merely the reduced x-axis. The Lelong number equals this analytic multiplicity, so $\nu([A], 0) = m$. The coprimality assumption ensures that the parametrization is primitive and the branch is irreducible rather than a multiple parametrization of a simpler branch. ⋄

Lecture 25

Exercise 7.5: Eigenvalues of the curvature operator

Solution. Let

$$e_J = \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n \wedge \mathrm{d}\bar{z}_J, \qquad |J| = q.$$

Write $L_j = \mathrm{i}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j \wedge$ and let Λ_j be the corresponding contraction. On a form already containing every $\mathrm{d}z_j$, the commutator $[L_j, \Lambda_j]$ contributes one exactly when $\mathrm{d}\bar{z}_j$ is also present, and contributes zero otherwise. Consequently

$$[\mathrm{i}\Theta(L), \Lambda]e_J = \sum_{j=1}^{n} \lambda_j [L_j, \Lambda_j] e_J = \left(\sum_{j \in J} \lambda_j\right) e_J.$$

Thus the standard (n, q)-basis diagonalizes B, with eigenvalues the sums of the q curvature eigenvalues indexed by J. In particular, if every $\lambda_j \geq c$, then $B \geq qcI$. ◇

Exercise 7.6: The non-Kähler warning

Solution. On a Hermitian manifold define the zeroth-order torsion operator

$$\tau = [\Lambda, \partial\omega \wedge \cdot].$$

The Kähler commutators acquire τ and τ^*; for example, $[\Lambda, D''] = -\mathrm{i}(D'^* + \tau^*)$, with the conjugate identity for $[\Lambda, D']$. A useful complete form of the Hermitian identity is

$$\Delta'' = \Delta'_\tau + [\mathrm{i}\Theta(E), \Lambda] + T_\omega, \qquad \Delta'_\tau = [D' + \tau, (D' + \tau)^*],$$

where

$$T_\omega = [\Lambda, [\Lambda, \tfrac{\mathrm{i}}{2}\partial\bar\partial\omega \wedge \cdot]] - [\partial\omega \wedge \cdot, (\partial\omega \wedge \cdot)^*]$$

is a zeroth-order operator. Equivalently, expanding Δ'_τ displays the first-order commutators involving D', D'', and τ^*.

If $\partial\omega = 0$, reality gives $\bar\partial\omega = 0$, so $\partial\bar\partial\omega = 0$, $\tau = 0$, and $T_\omega = 0$. The formula then reduces exactly to the Kähler BKN identity $\Delta'' = \Delta' + [\mathrm{i}\Theta(E), \Lambda]$. ◇

Exercise 7.8: Bochner on functions

Solution. Write $\Delta_{\mathrm{geom}} = \operatorname{div}\nabla = -\Delta_{\mathrm{d}}$ on functions, as specified in the exercise. At a point choose a geodesic orthonormal frame e_i. Differentiating $|df|^2 = \sum_i (e_i f)^2$ twice gives the square $|\operatorname{Hess} f|^2$ and the term $\langle df, d\Delta_{\mathrm{geom}} f\rangle$. Commuting the remaining third covariant derivatives uses

$$\nabla_i\nabla_j\nabla_i f - \nabla_j\nabla_i\nabla_i f = R_{iji}{}^k \nabla_k f,$$

whose contraction is $\operatorname{Ric}(\nabla f, \nabla f)$. Thus

$$\frac{1}{2}\Delta_{\mathrm{geom}}|df|^2 = |\operatorname{Hess} f|^2 + \langle df, d\Delta_{\mathrm{geom}} f\rangle + \operatorname{Ric}(\nabla f, \nabla f).$$

If f is harmonic on a compact connected manifold, the maximum principle already gives that f is constant; equivalently, $0 = -\int f\Delta_{\mathrm{geom}} f = \int |df|^2$. Under the additional assumption $\operatorname{Ric} \geq 0$, integrating the displayed Bochner identity also forces $\operatorname{Hess} f = 0$ and the Ricci term to vanish, hence again $df = 0$. ◇

Lecture 26

Exercise 7.15: Check on $\mathbb{P}^1$

Solution. Every positive line bundle on $\mathbb{P}^1$ is $L = \mathcal{O}(k)$ with $k > 0$, and $K_{\mathbb{P}^1} = \mathcal{O}(-2)$. Kodaira vanishing predicts

$$H^1(\mathbb{P}^1, K_{\mathbb{P}^1} \otimes L) = H^1(\mathbb{P}^1, \mathcal{O}(k-2)) = 0.$$

Lecture 15 computed $h^1(\mathcal{O}(m)) = \max\{-m-1, 0\}$. Substituting $m = k-2$ gives $\max\{1-k, 0\} = 0$ for every $k \geq 1$, verifying the theorem, including the borderline case $k = 1$. ◇

Exercise 7.16: A basic Fano consequence

Solution. Apply Kodaira vanishing with the positive line bundle $L = -K_X = K_X^{-1}$. Since $K_X \otimes L \simeq \mathcal{O}_X$, it gives

$$H^q(X, \mathcal{O}_X) = 0 \qquad (q > 0).$$

On a compact connected complex manifold every holomorphic function is constant, so $h^0(X, \mathcal{O}_X) = 1$. Therefore

$$\chi(\mathcal{O}_X) = \sum_q (-1)^q h^q(X, \mathcal{O}_X) = 1.$$

◇

Exercise 7.18: Fundamental group

Solution. Let $Y \subset \mathbb{P}^n$ be a smooth hypersurface of complex dimension $m = n - 1 \geq 2$. The homotopy Lefschetz theorem says that the inclusion $Y \hookrightarrow \mathbb{P}^n$ induces an isomorphism on π_j for $j < m$ and a surjection for $j = m$. Since $1 < m$,

$$\pi_1(Y) \simeq \pi_1(\mathbb{P}^n) = 0.$$

Thus Y is simply connected. The condition $m \geq 2$ is essential: smooth plane curves can have positive genus and nontrivial fundamental group. ◇

Lecture 27

Exercise 7.20: Gaussian weight

Solution. For $\varphi = t|z|^2$,

$$H_\varphi = (\varphi_{j\bar{k}}) = tI, \qquad H_\varphi^{-1} = t^{-1} I.$$

Hörmander's estimate therefore gives

$$\|u\|_t^2 = \int |u|^2 e^{-t|z|^2}\, d\lambda \leq \frac{1}{t} \int |f|^2 e^{-t|z|^2}\, d\lambda = t^{-1} \|f\|_t^2.$$

If the weight is $e^{-2t|z|^2}$, the actual exponent is $\widetilde{\varphi} = 2t|z|^2$, so $H_{\widetilde{\varphi}} = 2tI$. The sharp coefficient furnished by the same argument is therefore $(2t)^{-1}$, not t^{-1}. ◇

Exercise 7.21: The compatibility condition

Solution. If $\bar{\partial} u = f$ in distributions, then

$$\bar{\partial} f = \bar{\partial}^2 u = 0.$$

Thus $\bar{\partial}$-closedness is necessary independently of estimates or boundary conditions. In $\mathbb{C}^n$ with $n \geq 2$, take

$$f = \bar{z}_1 \, \mathrm{d}\bar{z}_2.$$

Then

$$\bar{\partial} f = \mathrm{d}\bar{z}_1 \wedge \mathrm{d}\bar{z}_2 \neq 0.$$

Consequently no distribution, and hence no L^2 function, can solve $\bar{\partial} u = f$. ◇

Exercise 7.22: The unit disc

Solution. Summing the rank-one kernels of the stated orthonormal basis gives

$$K_{\mathbb{D}}(z,w) = \frac{1}{\pi}\sum_{k=0}^{\infty}(k+1)(z\bar{w})^k = \frac{1}{\pi(1-z\bar{w})^2}.$$

In particular, $K_{\mathbb{D}}(z,z) = \left[\pi(1-|z|^2)^2\right]^{-1}$.

For every $f \in A^2(\mathbb{D})$, the reproducing identity and Cauchy–Schwarz give

$$|f(z)| = |\langle f, K_z\rangle| \le \|f\|\,\|K_z\| = \|f\|\sqrt{K(z,z)}.$$

Equality for unit norm is attained by $f = K_z/\sqrt{K(z,z)}$. Therefore

$$K(z,z) = \sup\{|f(z)|^2 : f \in A^2(\mathbb{D}),\ \|f\| \le 1\},$$

which verifies the extremal characterization. ◇

Lecture 28

Exercise 7.28: A one-dimensional multiplier ideal

Solution. Let a nonzero germ f have order m, so $|f(z)|^2 \asymp r^{2m}$. Since $e^{-c\log|z|^2} = r^{-2c}$, local integrability is equivalent to

$$\int_0^{\varepsilon} r^{2m-2c} r\,dr < \infty.$$

This holds exactly when $2m - 2c + 1 > -1$, or $m > c - 1$. For $c \ge 0$ and integral m, this is $m \ge \lfloor c \rfloor$. Hence

$$\mathcal{I}(c\log|z|^2)_0 = (z^{\lfloor c\rfloor}).$$

The equality includes the zero germ. ◇

Exercise 7.29: Complete cutoffs

Solution. Fix $o \in M$ and let $r(x) = d(o,x)$. Completeness and Hopf–Rinow imply that closed r-balls are compact, so r is proper; it is also one-Lipschitz. Choose $\eta \in C^\infty(\mathbb{R})$ with $0 \le \eta \le 1$, $\eta = 1$ on $(-\infty, 1]$, $\eta = 0$ on $[2,\infty)$, and $|\eta'| \le C$. The Lipschitz functions $\eta(r/\nu)$ already have compact support and satisfy the desired gradient bound almost everywhere, but are not smooth on the cut locus.

We recall why the required smoothing preserves the gradient bound. Cover M by relatively compact normal-coordinate balls, mollify r in each chart, and choose the local smoothing errors before patching by a locally finite partition of unity. The errors may be chosen so small that the terms obtained by differentiating the partition functions have total norm at most 1 at every point. Local convolution does not increase the Lipschitz constant by more than an arbitrarily small prescribed error. The patched function may therefore be chosen smooth and proper, with $|\rho - r| \le 1$ and $|\mathrm{d}\rho| \le 2$. Then

$$\chi_\nu(x) = \eta(\rho(x)/\nu)$$

is smooth and compactly supported, tends pointwise to one, and satisfies

$$|\mathrm{d}\chi_\nu| \le \frac{2\|\eta'\|_\infty}{\nu}.$$

After shifting the index, it also equals one on any prescribed compact set for all sufficiently large ν. ◇

Exercise 7.32: Radial singularity

Solution. The computation in the preceding exercise gives

$$z^m \in \mathcal{I}(c \log |z|^2) \quad \Longleftrightarrow \quad m > c - 1 \quad \Longleftrightarrow \quad m \geq \lfloor c \rfloor.$$

If this strict inequality holds and $c > 0$, choose

$$0 < \varepsilon < \frac{m + 1 - c}{c}.$$

Then $m > (1+\varepsilon)c-1$, so $z^m \in \mathcal{I}((1+\varepsilon)c \log |z|^2)$. When $c = 0$, every holomorphic germ is integrable because the weight is 1. The reverse inclusion follows from monotonicity because increasing c makes the singular weight stronger. Therefore

$$\mathcal{I}(c \log |z|^2) = \bigcup_{\varepsilon>0} \mathcal{I}((1 + \varepsilon)c \log |z|^2),$$

verifying strong openness directly in this radial case. ◇

Lecture 29

Exercise 8.4: Point extension and the Bergman kernel

Solution. Fix $p \in \Omega$ and a local frame for the weight. Apply Ohsawa–Takegoshi to the value $f(p) = 1$ on $Y = \{p\}$. It gives a holomorphic F with $F(p) = 1$ and

$$\|F\|_\varphi^2 \leq C_{\mathrm{OT}} e^{-\varphi(p)}.$$

The weighted Bergman kernel has the extremal characterization

$$K_\varphi(p, p) = \sup_{G \neq 0} \frac{|G(p)|^2}{\|G\|_\varphi^2}.$$

Using F as a competitor yields

$$K_\varphi(p, p) \geq \frac{1}{C_{\mathrm{OT}} e^{-\varphi(p)}} = C_{\mathrm{OT}}^{-1} e^{\varphi(p)}.$$

Equivalently, the intrinsic weighted density satisfies $K_\varphi(p, p)e^{-\varphi(p)} \geq C_{\mathrm{OT}}^{-1}$. Thus quantitative point extension is dual to a quantitative lower bound for evaluation. ◇

Exercise 8.5: Scaling audit

Solution. For the trivial line bundle with its flat metric, let $\sigma(z) = az$. Then $D'\sigma = a\,\mathrm{d}z$, while $|\sigma|^2 = |a|^2|z|^2$. Therefore

$$\mathrm{i}\{D'\sigma, \sigma\} \wedge \{\sigma, D'\sigma\} = |a|^4|z|^2\,\mathrm{i}\,\mathrm{d}z \wedge \mathrm{d}\bar{z}$$

up to the harmless pairing convention. On the right-hand side of the corrected inequality,

$$|\sigma|^2\,\mathrm{i}\{D'\sigma, D'\sigma\} = (|a|^2|z|^2)(|a|^2\mathrm{i}\,\mathrm{d}z \wedge \mathrm{d}\bar{z}),$$

which also scales as $|a|^4$. The factor $|\sigma|^2$ is therefore forced. If it were omitted, the right side would scale only as $|a|^2$, and the alleged inequality could not be invariant under replacing a by a large constant. ◇

Exercise 8.6: Approximate delta mass

Solution. On the disc $|w| < R$, use $w = re^{i\theta}$ and $i\,dw \wedge d\bar{w} = 2r\,dr\,d\theta$. The mass is

$$\int_{|w|<R} \frac{\varepsilon^2 i\,dw \wedge d\bar{w}}{(|w|^2+\varepsilon^2)^2} = 2\pi \frac{R^2}{R^2+\varepsilon^2} \leq 2\pi.$$

For a compactly supported continuous test function χ, substitute $w = \varepsilon\zeta$:

$$\frac{1}{2\pi}\int \chi(w) \frac{\varepsilon^2 i\,dw \wedge d\bar{w}}{(|w|^2+\varepsilon^2)^2} = \frac{1}{2\pi}\int \chi(\varepsilon\zeta) \frac{i\,d\zeta \wedge d\bar{\zeta}}{(1+|\zeta|^2)^2}.$$

The last density has total mass 2π. Split the integral into a large fixed disc and its uniformly small tail, then use dominated convergence on the disc. The limit is $\chi(0)$. Hence the forms divided by 2π converge weakly to δ_0. ◇

Lecture 30

Exercise 8.9: Linearization

Solution. Let $G(t) = (g_{j\bar{k}} + t\psi_{j\bar{k}})$. Jacobi's determinant formula gives

$$\frac{d}{dt} \det G(t) = \det G(t)\ \operatorname{tr}(G(t)^{-1}G'(t)).$$

Dividing by $\det G(t)$ and evaluating at zero yields

$$\left.\frac{d}{dt}\right|_0 \log\det(g_{j\bar{k}} + t\psi_{j\bar{k}}) = \operatorname{tr}(g^{-1}(\psi_{j\bar{k}})) = g^{j\bar{k}}\psi_{j\bar{k}} = \Delta_\omega \psi.$$

This is why the linearization of the logarithmic Monge–Ampère operator is the Laplacian of the current metric, not the fixed background Laplacian along the entire continuity path. ◇

Exercise 8.10: Complex dimension one

Solution. When $\dim_{\mathbb{C}} X = 1$, the Monge–Ampère equation has no nonlinear wedge power:

$$\omega + i\partial\bar{\partial}\varphi = e^F \omega.$$

Thus

$$i\partial\bar{\partial}\varphi = (e^F - 1)\omega,$$

which is a Poisson equation $\Delta_\omega \varphi = c(e^F - 1)$, with the constant depending only on the Laplacian convention. Its solvability condition is $\int_X (e^F - 1)\omega = 0$, exactly the prescribed volume condition. Elliptic theory then gives a solution unique modulo constants, fixed by $\sup \varphi = 0$.

In higher dimension the determinant couples all eigenvalues of the complex Hessian, and one needs separate C^0, trace, and fully nonlinear regularity estimates. In dimension one the equation is linear, so these genuinely nonlinear difficulties are absent. ◇

Exercise 8.11: First variation of volume

Solution. Differentiate under the integral and use $\dot{\omega}_{\varphi_t} = i\partial\bar{\partial}\dot{\varphi}_t$:

$$\frac{d}{dt}\int_X \omega_{\varphi_t}^n = n\int_X i\partial\bar{\partial}\dot{\varphi}_t \wedge \omega_{\varphi_t}^{n-1}.$$

Because $d\omega_{\varphi_t} = 0$, the integrand is exact:

$$\mathrm{i}\partial\bar\partial\dot\varphi_t \wedge \omega_{\varphi_t}^{n-1} = d\big(\mathrm{i}\bar\partial\dot\varphi_t \wedge \omega_{\varphi_t}^{n-1}\big)$$

up to the fixed sign convention. Stokes' theorem on compact X makes the integral zero. Hence the volume depends only on the Kähler cohomology class and is constant along the path. ◇

Lecture 31

Exercise 8.18: Nef test on a surface

Solution. On $X = \mathrm{Bl}_p\,\mathbb{P}^2$, the intersection relations are $H^2 = 1$, $H \cdot E = 0$, and $E^2 = -1$. For $D = aH - bE$,

$$D \cdot H = a, \qquad D \cdot E = b, \qquad D \cdot (H - E) = a - b.$$

The cone of curves is generated by E and $H - E$ (and $H = (H - E) + E$ adds no new inequality). Thus Kleiman's criterion gives

$$D \text{ nef} \iff a \geq b \geq 0.$$

Strict positivity on the two extremal curve classes gives $b > 0$ and $a - b > 0$; then $D^2 = a^2 - b^2 > 0$. The Nakai–Moishezon criterion therefore gives

$$D \text{ Kähler} \iff a > b > 0.$$

◇

Exercise 8.19: Compatibility of truncations

Solution. Let $V_j = V_{\theta_j}$ and $U_k = \bigcap_j \{u_j > V_j - k\}$. If $k < \ell$, then on the open set U_k one has

$$u_j^{(k)} = \max(u_j, V_j - k) = u_j = \max(u_j, V_j - \ell) = u_j^{(\ell)}$$

for every j. Bedford–Taylor products are local in the plurifine topology, in particular local on ordinary open sets: bounded psh potentials that agree on an open set have equal wedge products there. Hence

$$1_{U_k} \bigwedge_j (\theta_j + \mathrm{dd^c} u_j^{(k)}) = 1_{U_k} \bigwedge_j (\theta_j + \mathrm{dd^c} u_j^{(\ell)}).$$

This compatibility makes the increasing limit in the definition of the non-pluripolar product unambiguous. ◇

Exercise 8.21: A two-curve decomposition

Solution. Put $q_{ij} = C_i \cdot C_j$, $b_j = \alpha \cdot C_j$, and $Q = (q_{ij})$. The orthogonality equations are

$$Q\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}.$$

Negative definiteness gives $q_{11} < 0$, $q_{22} < 0$, and $\Delta = q_{11}q_{22} - q_{12}^2 > 0$. Therefore

$$a_1 = \frac{q_{22}b_1 - q_{12}b_2}{\Delta}, \qquad a_2 = \frac{q_{11}b_2 - q_{12}b_1}{\Delta}.$$

The negative part is effective exactly when

$$q_{22}b_1 - q_{12}b_2 \geq 0, \qquad q_{11}b_2 - q_{12}b_1 \geq 0.$$

When distinct irreducible curves have $q_{12} \geq 0$, these inequalities make explicit how sufficiently negative intersections of α with the curves produce positive coefficients in $N = a_1C_1 + a_2C_2$. ◇

Lecture 32

Exercise 8.23: Capstone calculation

Solution. The total transform of a degree-d curve has class dH, and multiplicity m at the blown-up point contributes mE. Hence the strict transform is

$$\widetilde{C} = dH - mE.$$

Using $H^2 = 1$, $H \cdot E = 0$, and $E^2 = -1$,

$$\widetilde{C} \cdot E = m, \qquad \widetilde{C}^2 = d^2 - m^2.$$

Since $K_X = -3H + E$, adjunction gives

$$p_a(\widetilde{C}) = 1 + \frac{1}{2}\widetilde{C} \cdot (\widetilde{C} + K_X) = 1 + \frac{1}{2}(d^2 - 3d + m - m^2).$$

Equivalently,

$$p_a(\widetilde{C}) = \frac{(d-1)(d-2)}{2} - \frac{m(m-1)}{2}.$$

This is the plane arithmetic genus minus the numerical correction $m(m-1)/2$ contributed by the multiplicity at the blown-up point; the adjunction calculation confirms all signs, especially $E^2 = -1$. ◇

Exercise 8.24: Final project outline

Solution (one complete model). This exercise admits many correct answers. Here is a complete outline following the pluripotential-theory route.

Main theorem. State and explain the finite-energy existence principle: if (X, ω) is compact Kähler and a measure μ of total mass $\int_X \omega^n$ has finite energy, then there is a normalized $u \in \mathcal{E}^1(X, \omega)$ with $\langle(\omega + \mathrm{dd^c}u)^n\rangle = \mu$, unique up to constants. The project should state the chosen definition of finite energy and may prove the result for measures with bounded density while giving the precise dependency chain of the variational argument in general.

Three prerequisite tools.

(1) Bedford–Taylor monotone convergence and the comparison principle, used first for bounded approximants and then for uniqueness.
(2) Non-pluripolar products and truncations $u_j = \max(u, -j)$, used to define the Monge–Ampère measure without charging the pole set.
(3) The Aubin–Mabuchi energy and compactness of normalized energy sublevels, used to maximize $E(u) - \int u\, d\mu$ and pass to a limit.

Normalization trap. Record both $\mathrm{d^c} = (4\pi\mathrm{i})^{-1}(\partial - \bar\partial)$ and $\int_X \mathrm{MA}(u) = \int_X \omega^n$. Adding a constant to u does not change $\mathrm{MA}(u)$, so impose $\sup_X u = 0$. Confusing the classical product $(\mathrm{dd^c} \log |s|^2)^n$ with the non-pluripolar bracket would incorrectly retain divisor mass on the pole set.

Explicit example. On $\mathbb{P}^1$, take a smooth positive density $\mu = e^F \omega_{\mathrm{FS}}$ of total mass one. The equation becomes the Poisson equation $\mathrm{dd^c}u = \mu - \omega_{\mathrm{FS}}$, solved by the Green kernel and normalized by $\sup u = 0$. Then approximate a measure with a mild singular density and show how the same truncation and energy language persists.

Five-page organization. Page 1 states the theorem and defines $\mathcal{E}^1$; page 2 develops truncations and non-pluripolar mass; page 3 presents the energy functional and compactness; page 4 derives the Euler–Lagrange equation and uniqueness by comparison; page 5 works out $\mathbb{P}^1$, audits the normalization, and lists the exact point where the general proof requires deeper compactness. This meets all requested components while keeping the main logical dependency visible. ◇